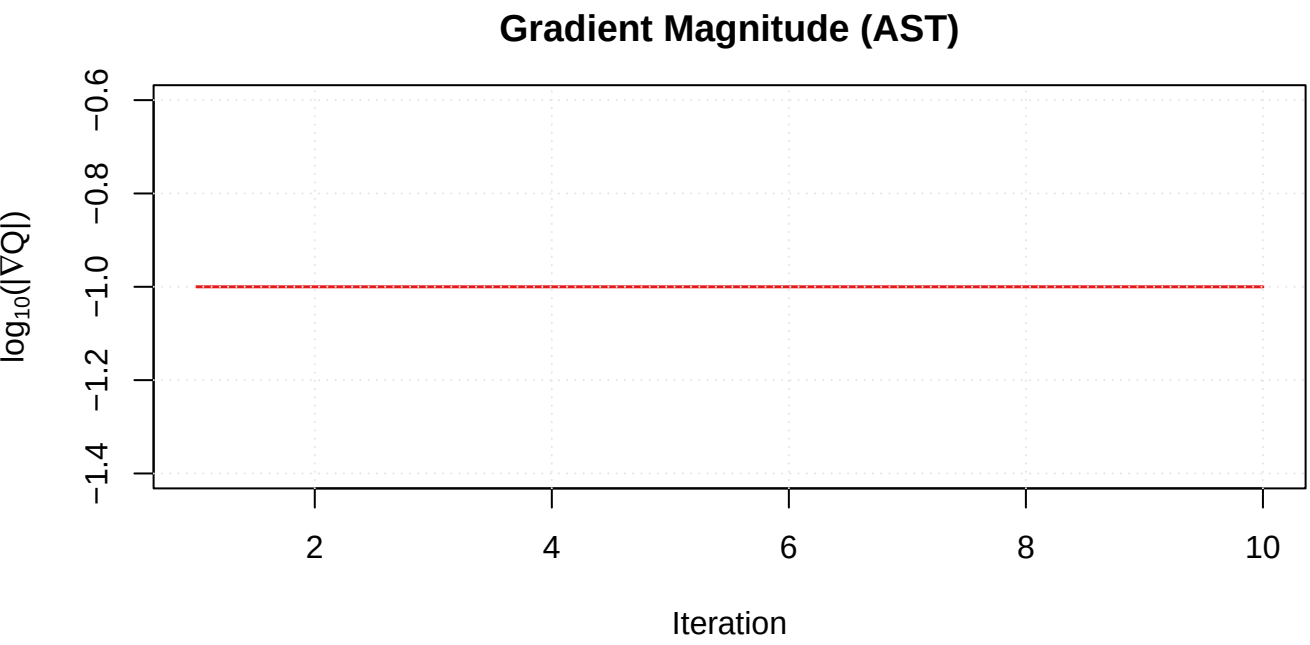
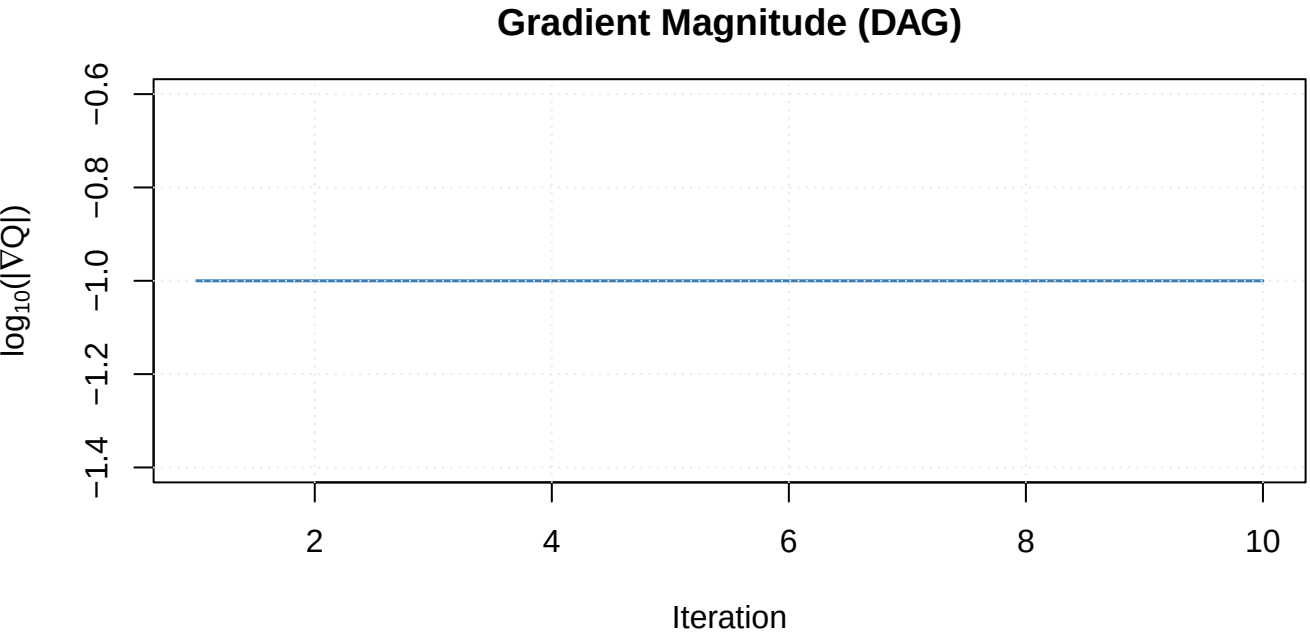
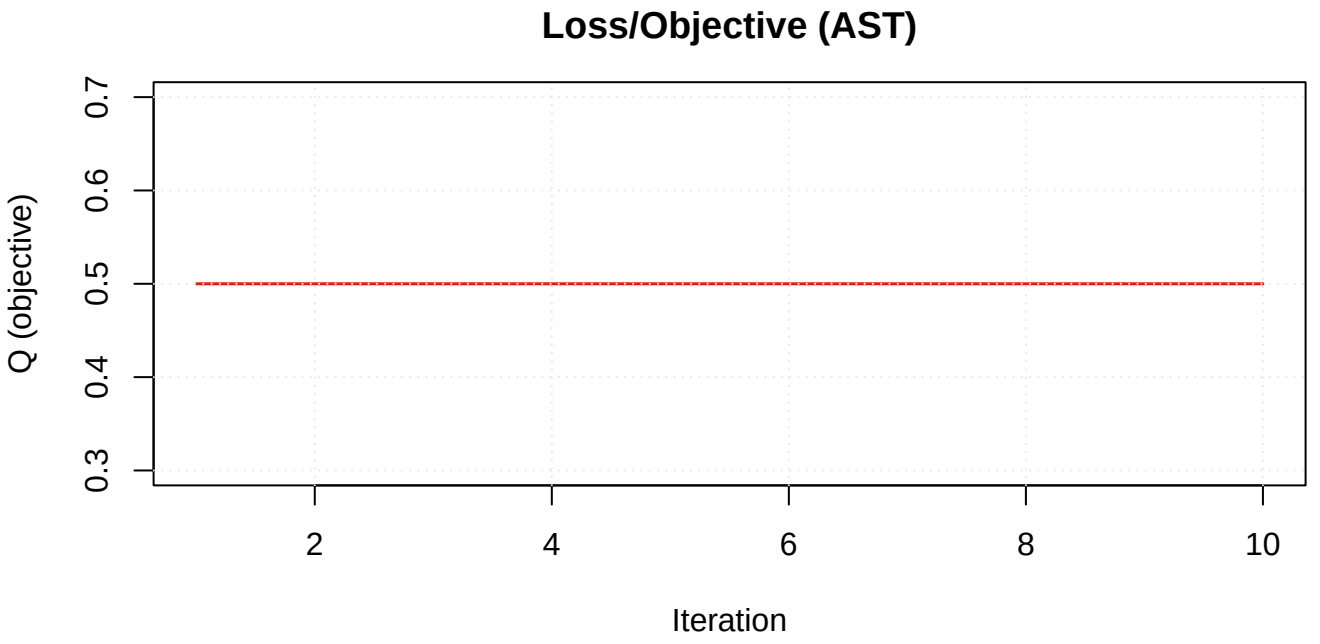
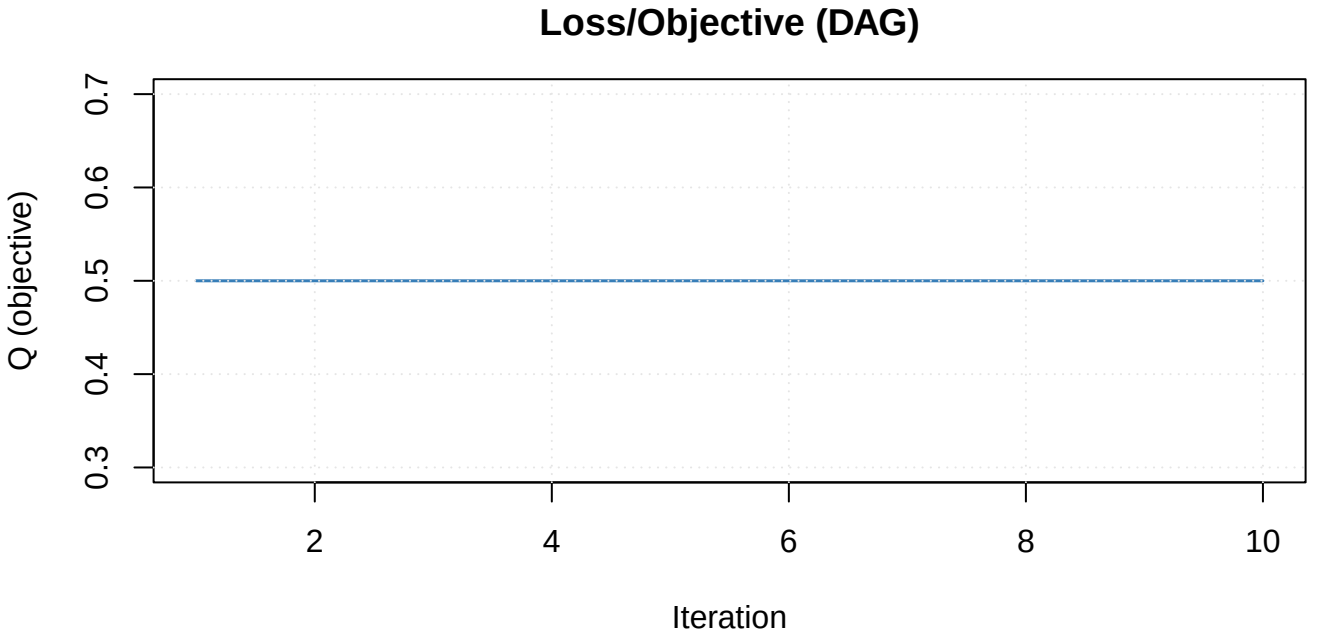
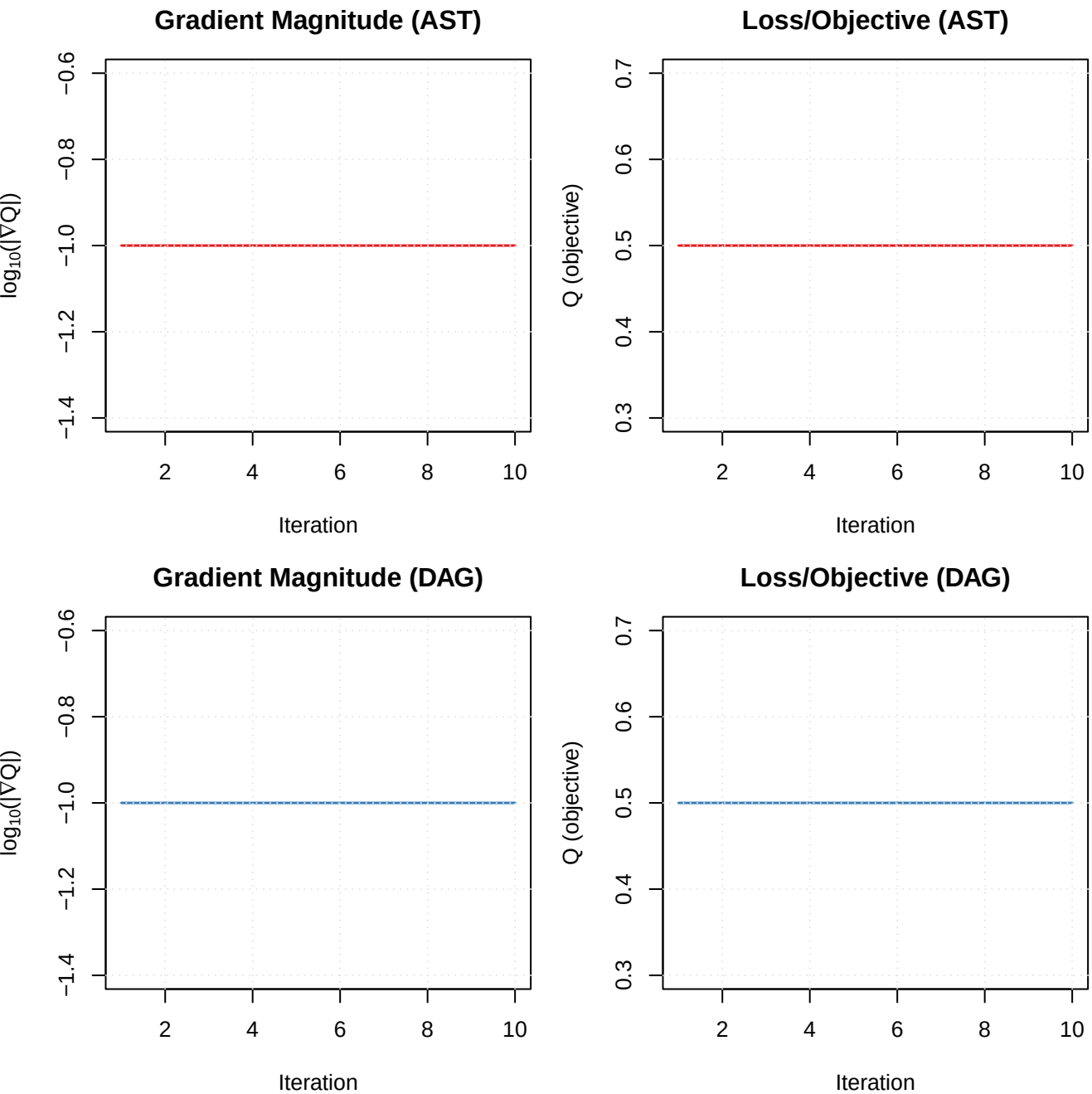
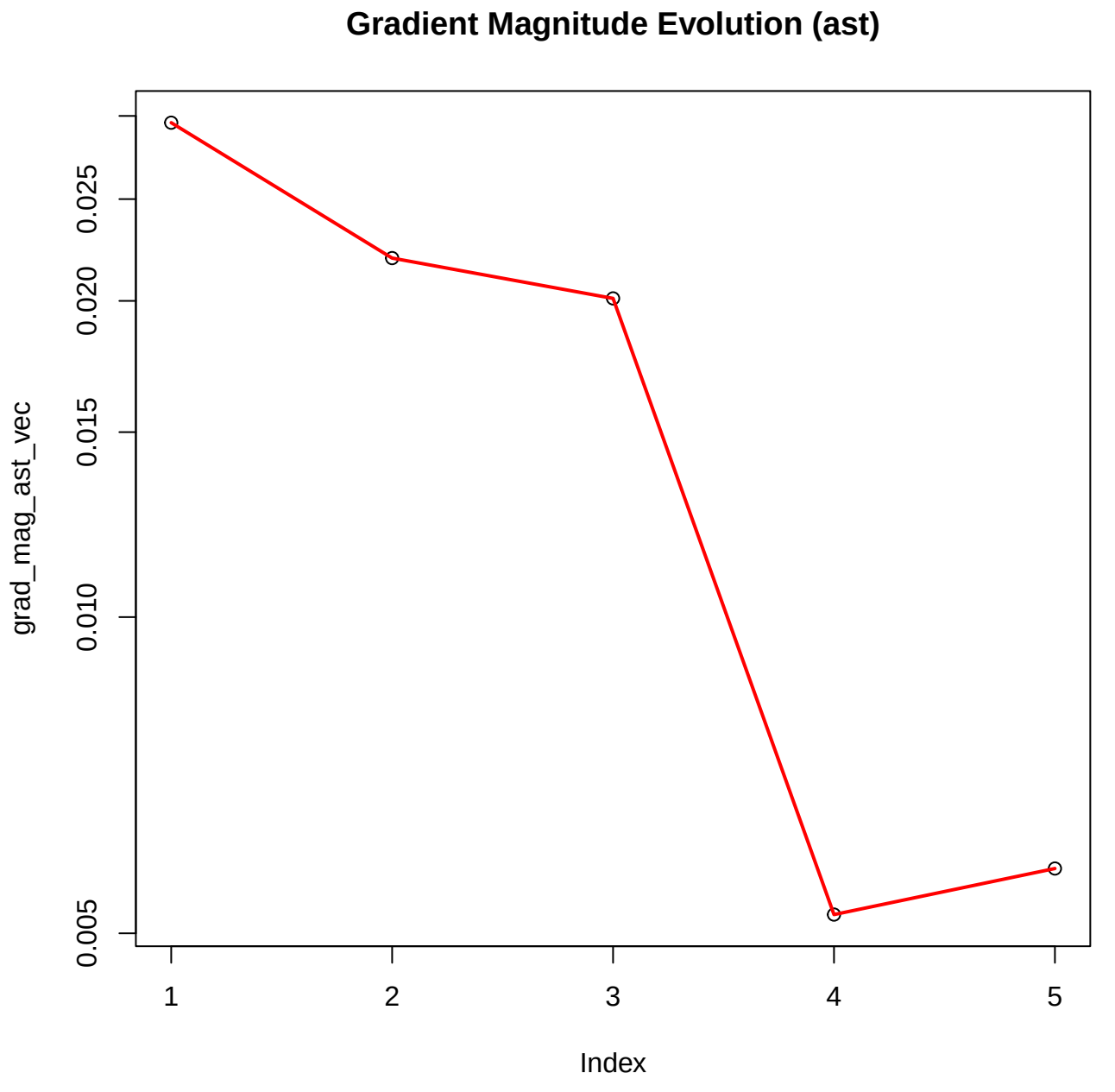

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

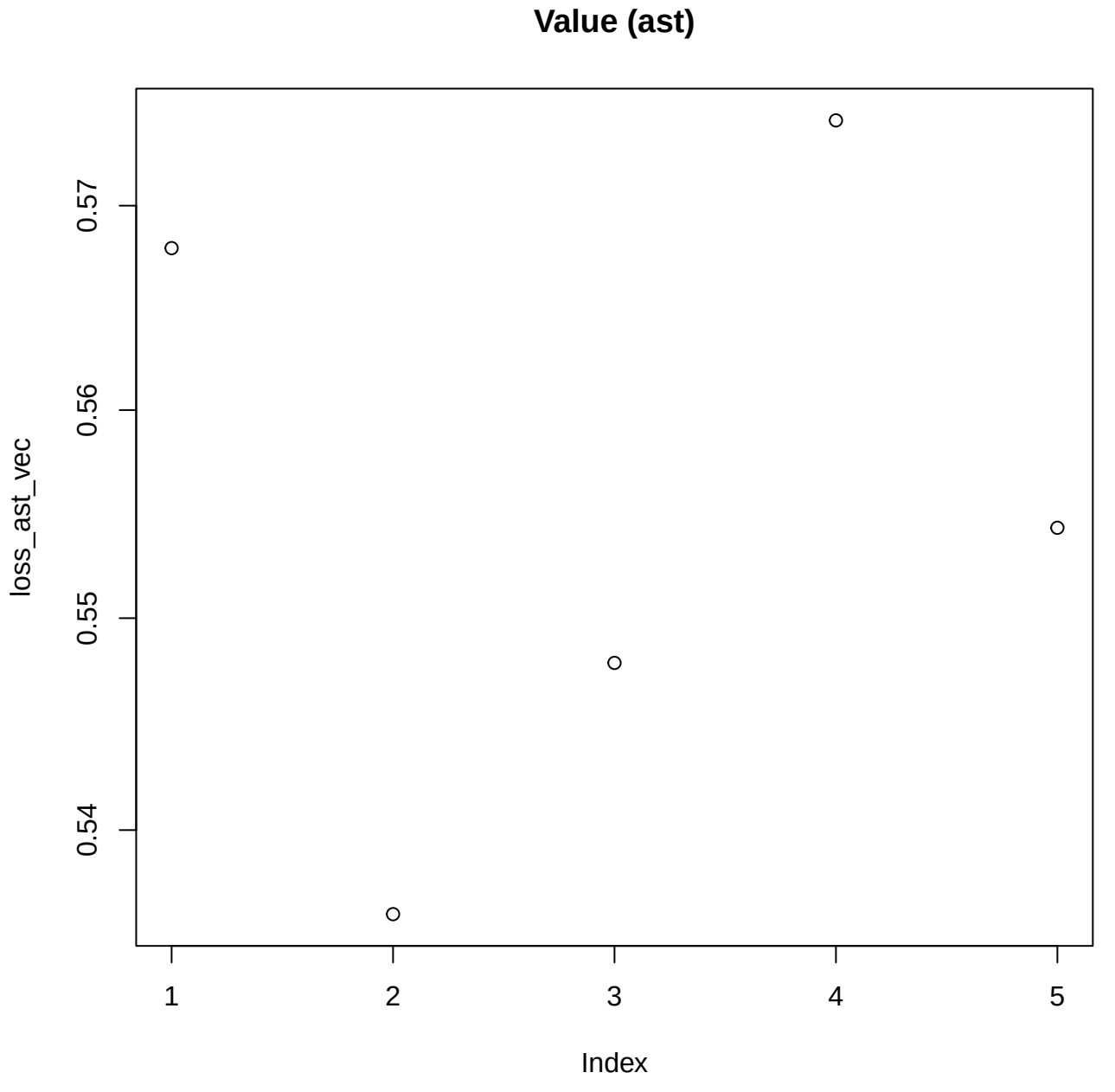

Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

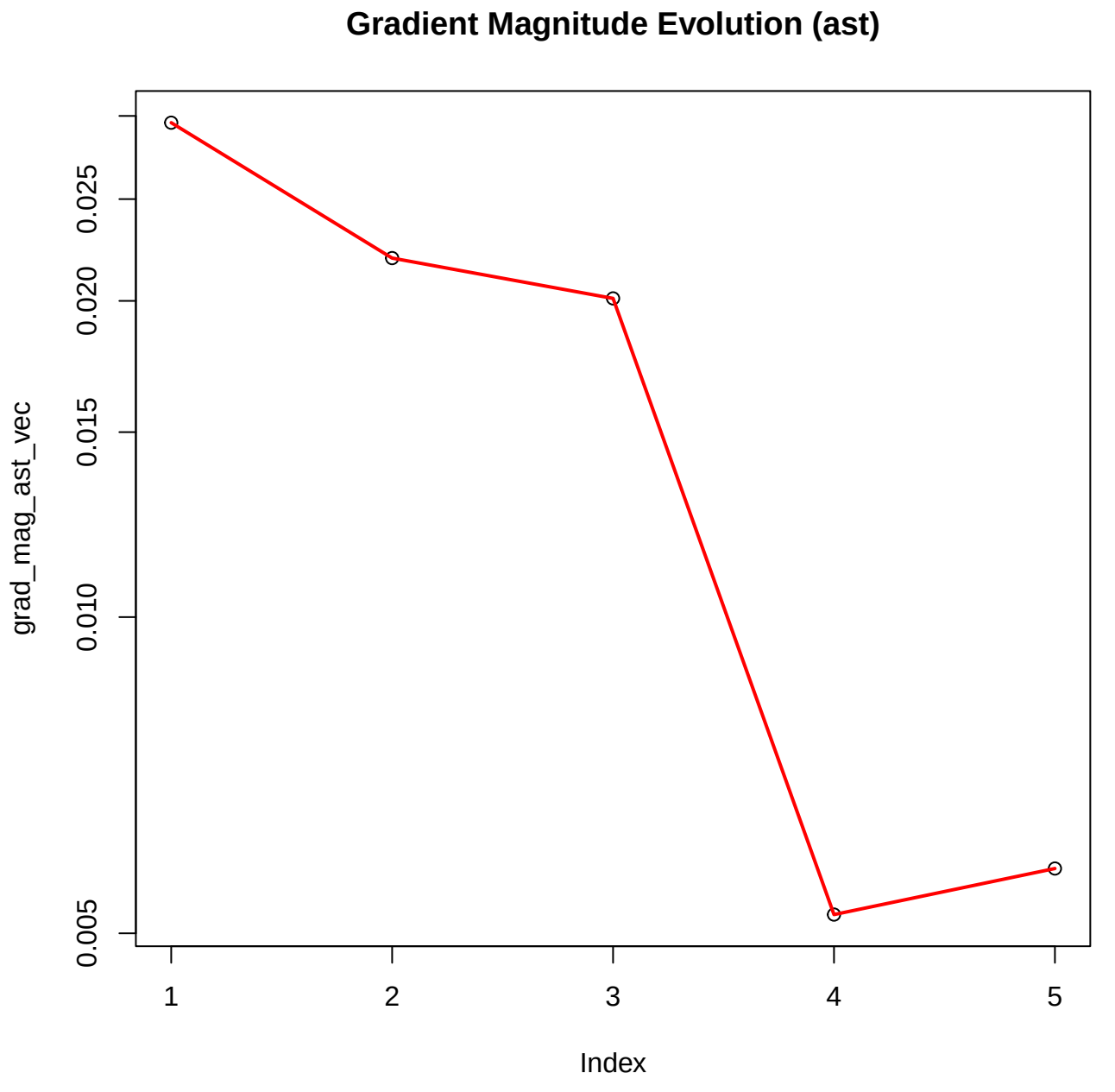

Loss/Objective (AST)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

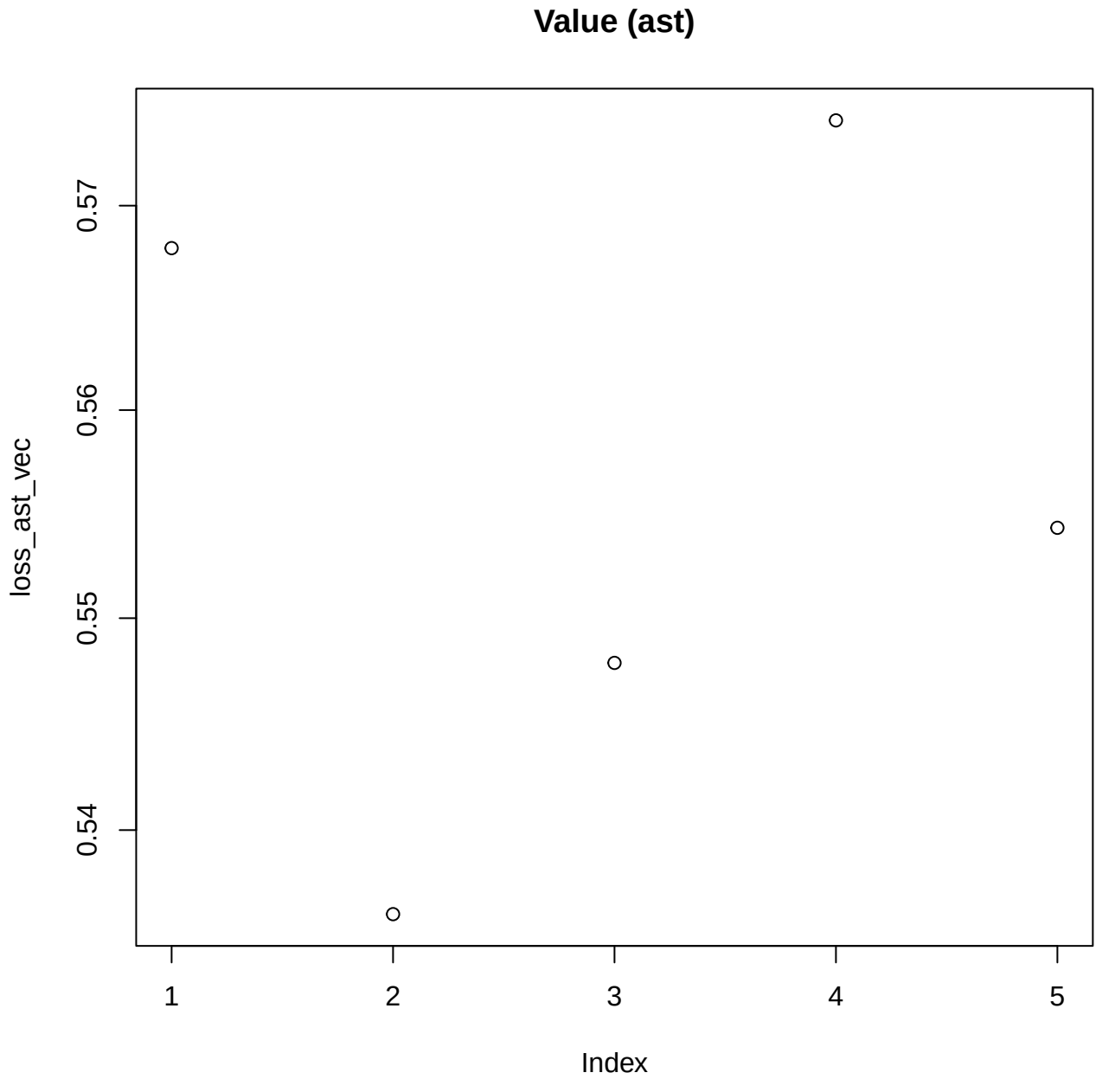

Loss/Objective (DAG)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

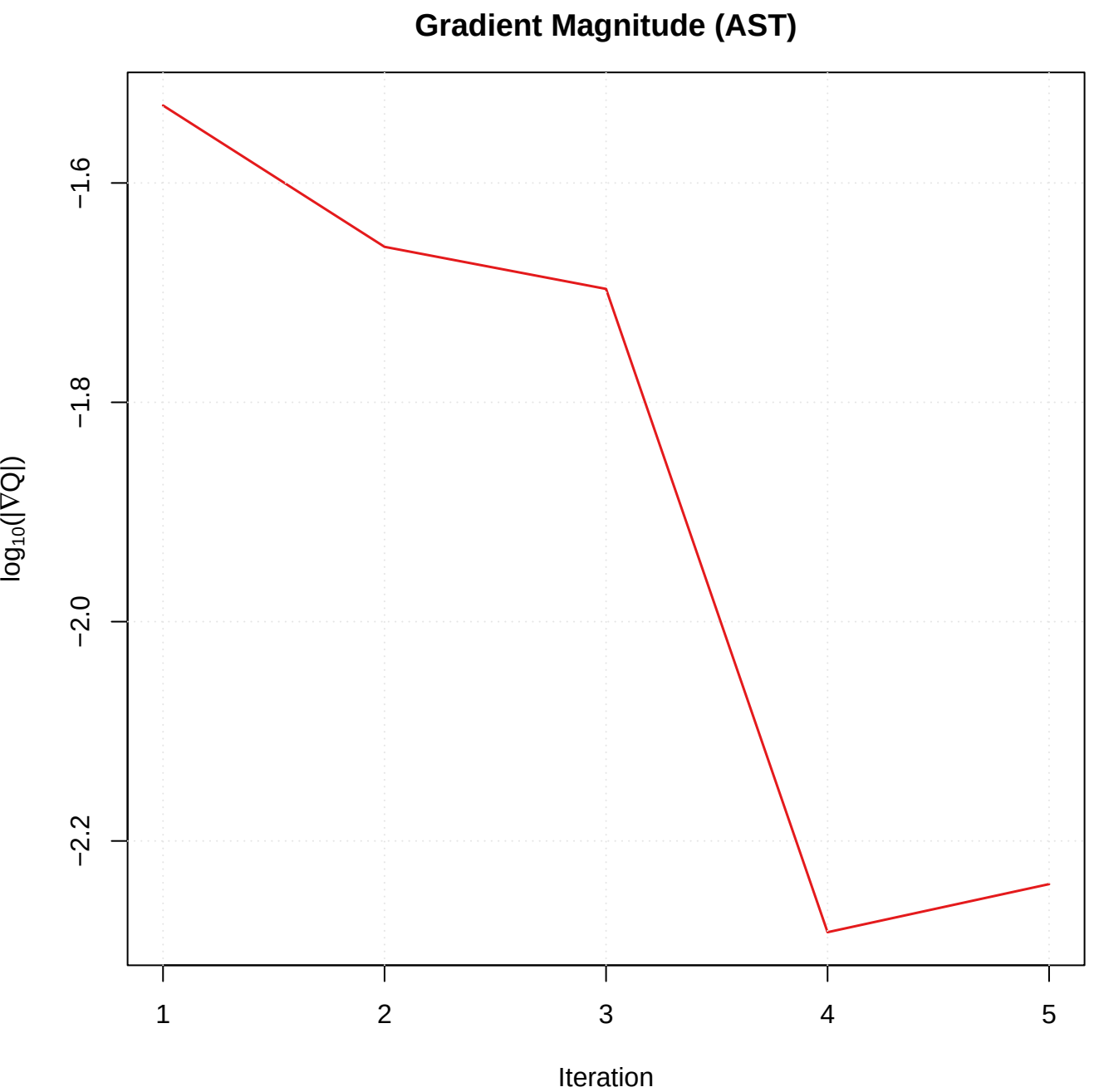

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (AST)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration
Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (DAG)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration

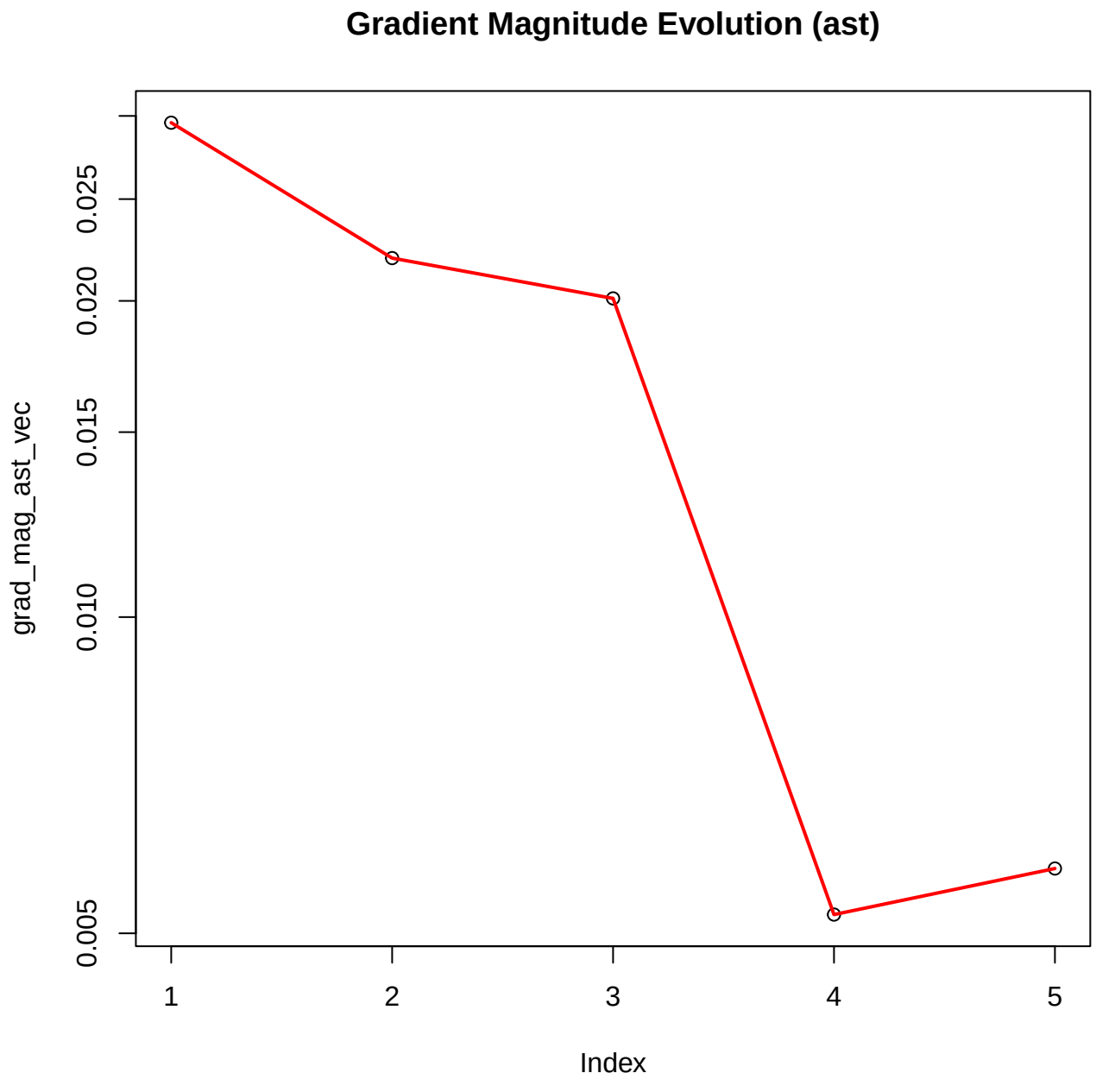

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

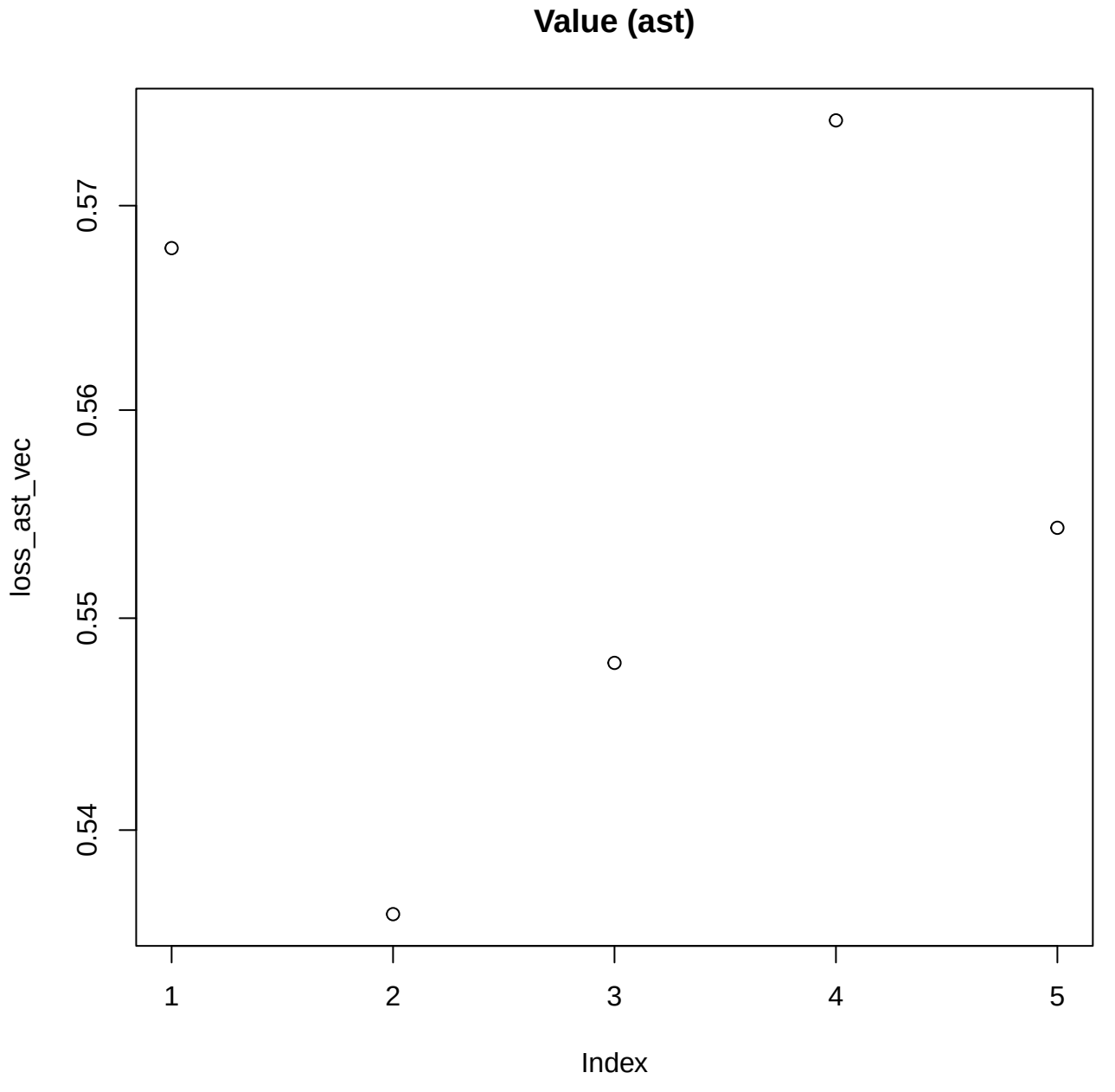

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

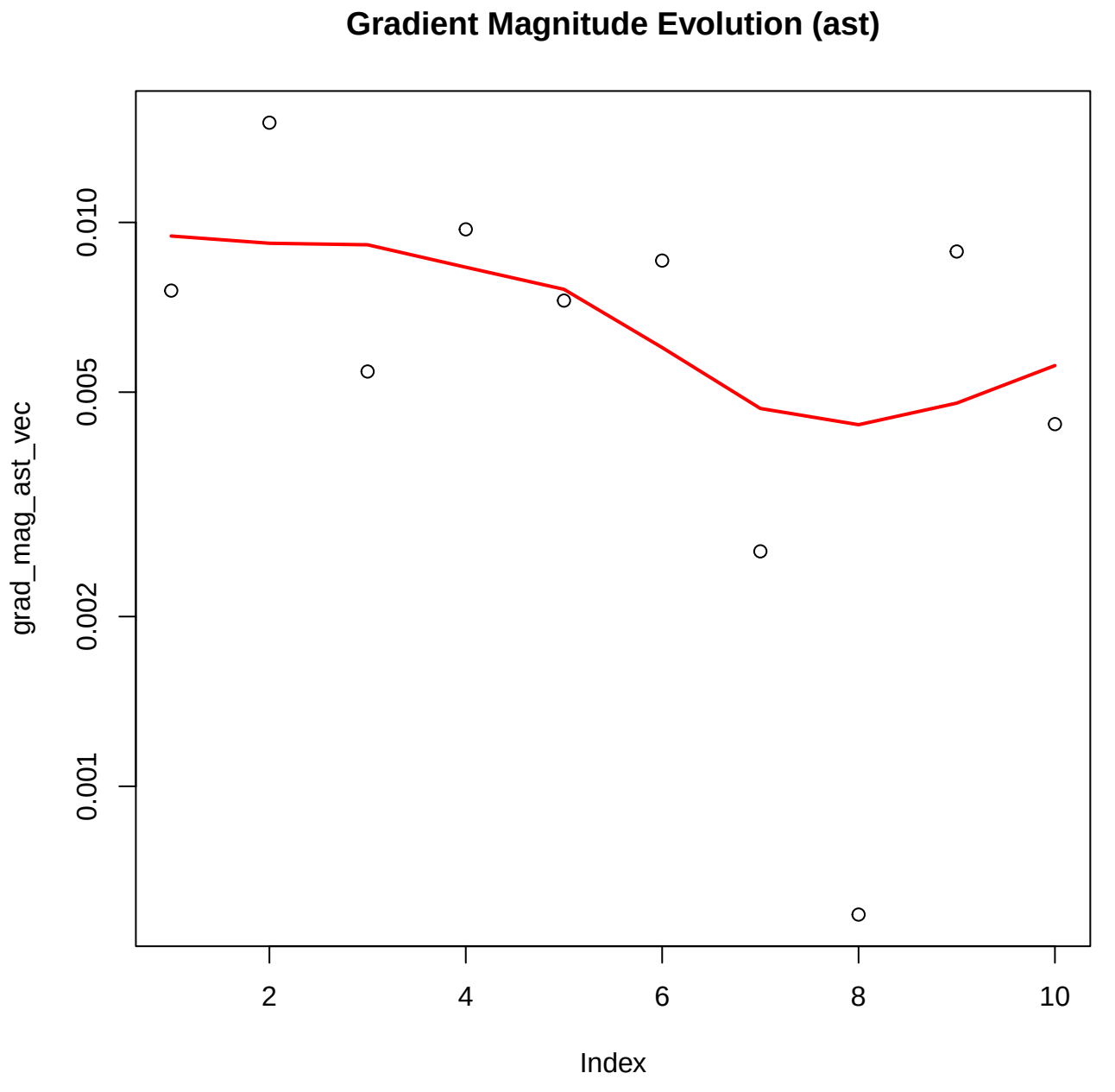

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

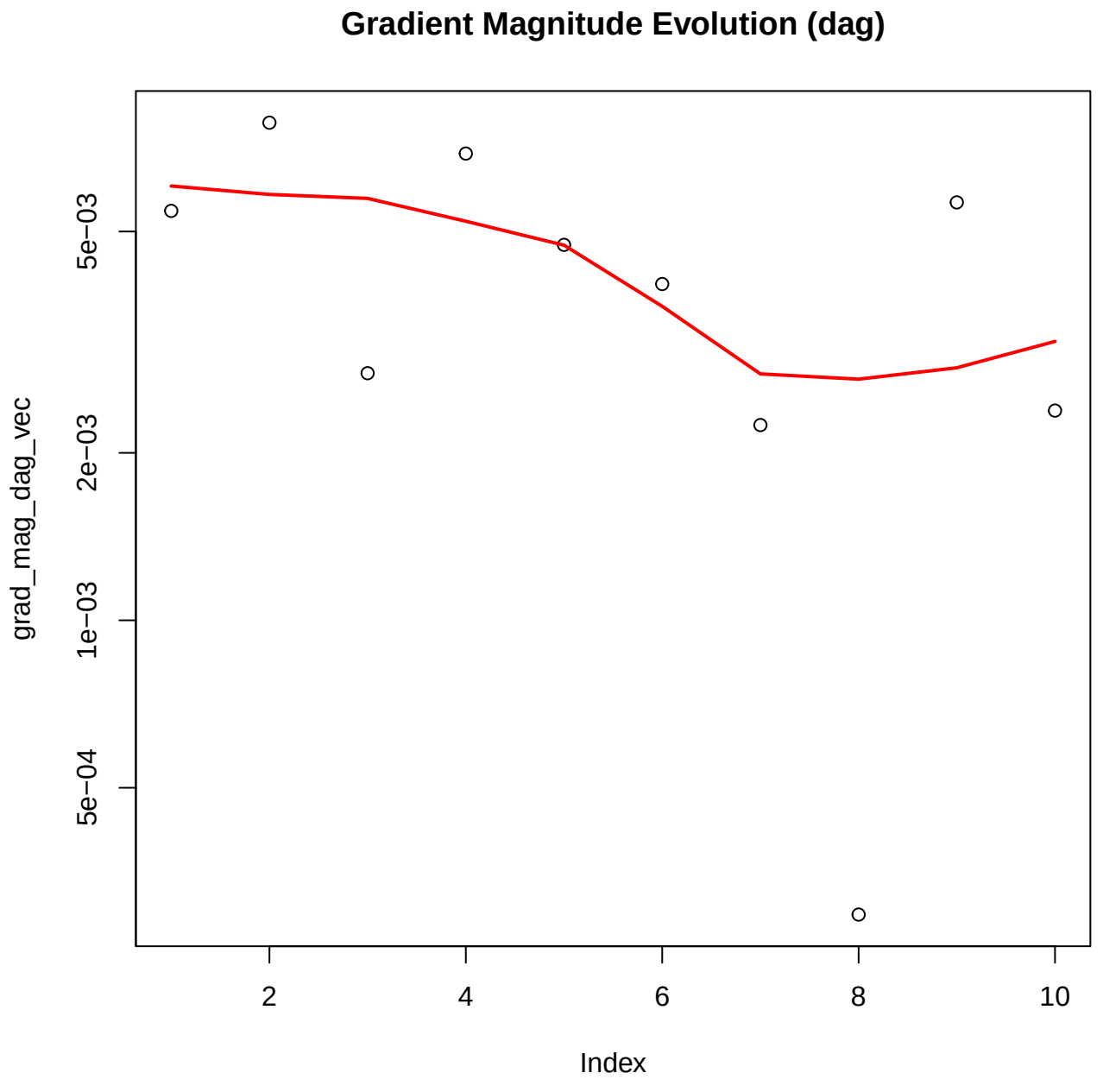

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

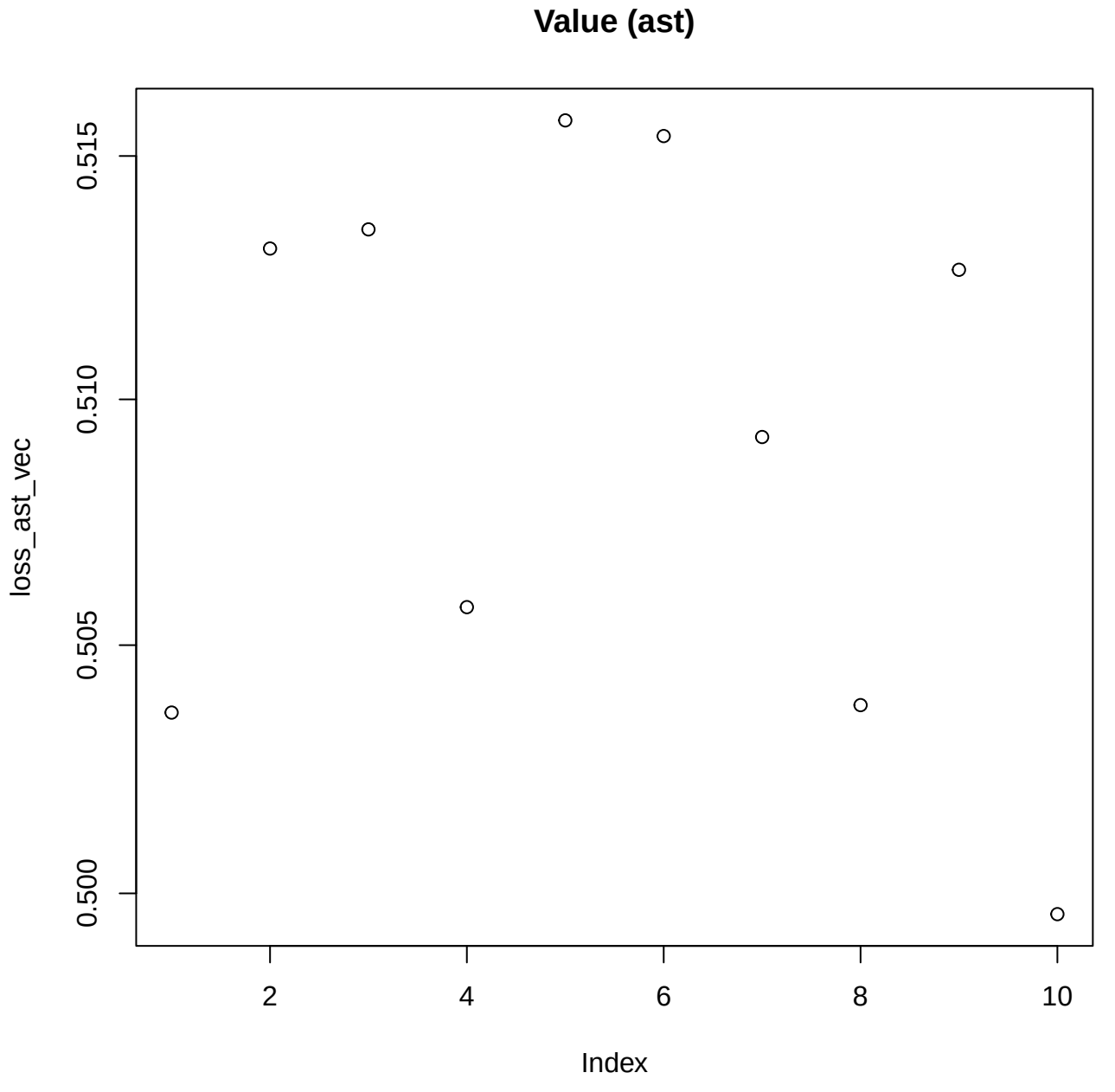

Gradient Magnitude (AST)
−1.6
−1.8
−2.0
−2.2
log10(|∇Q|)
1
2
3
4
5
Iteration

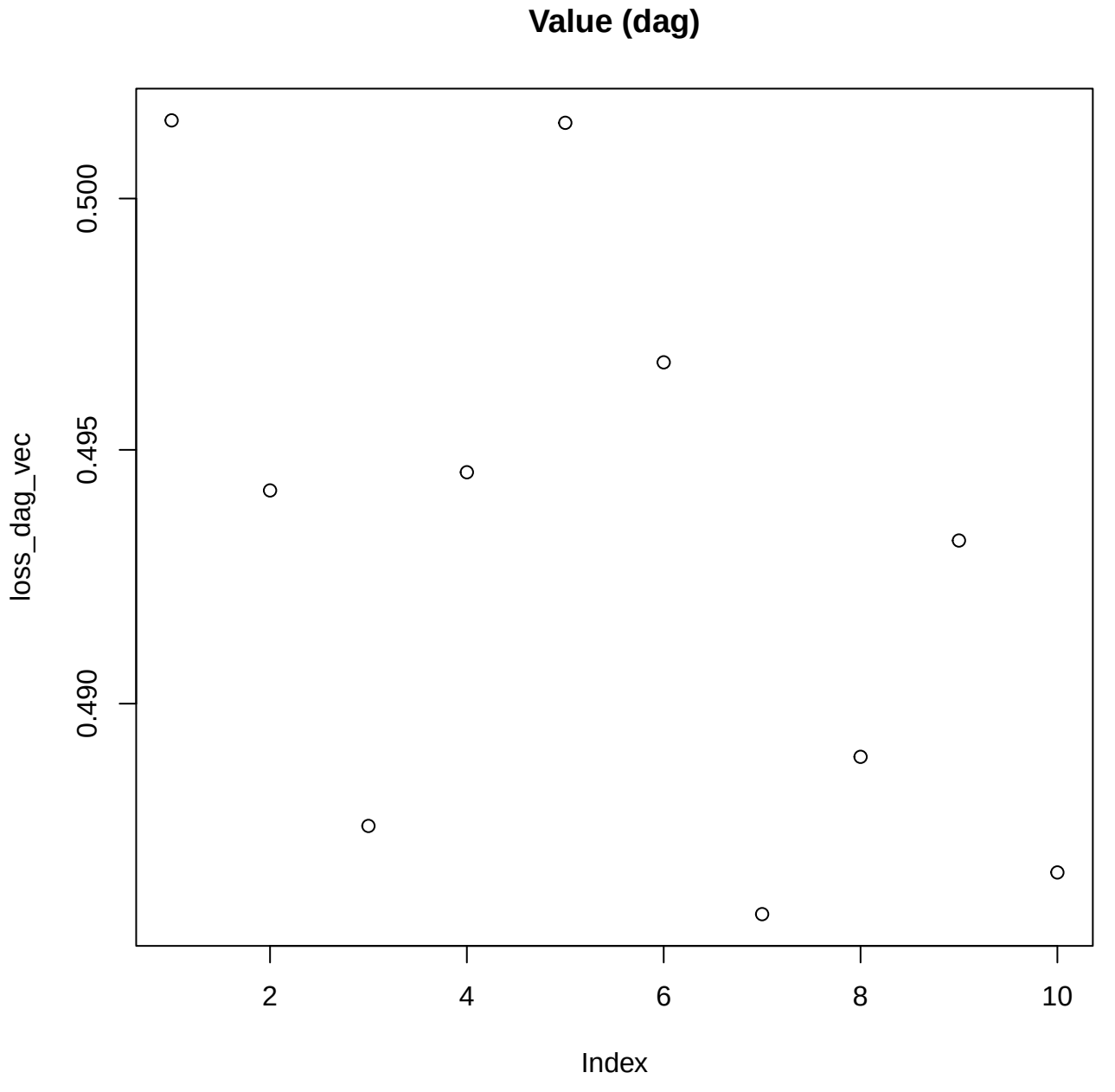

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

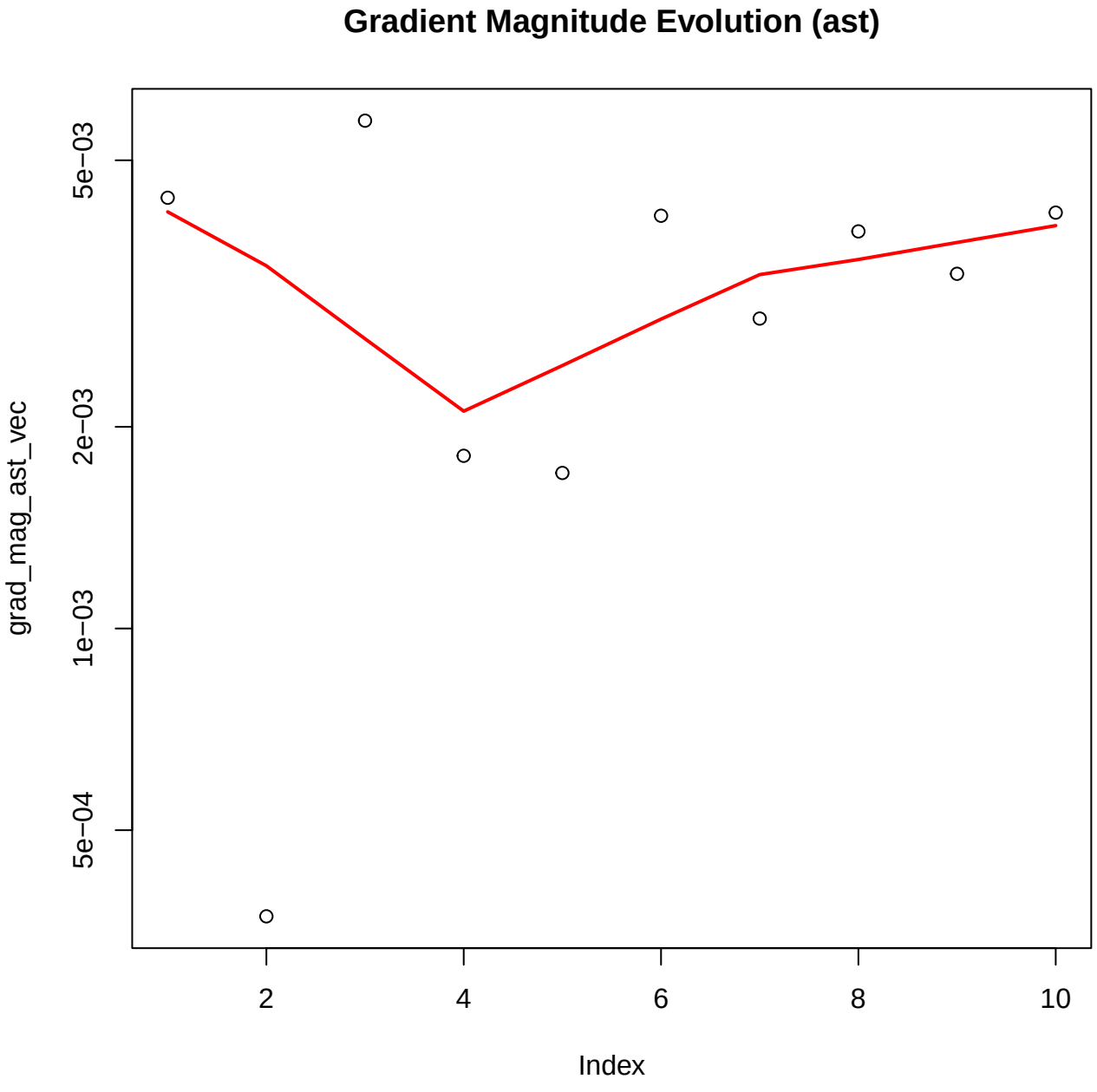

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

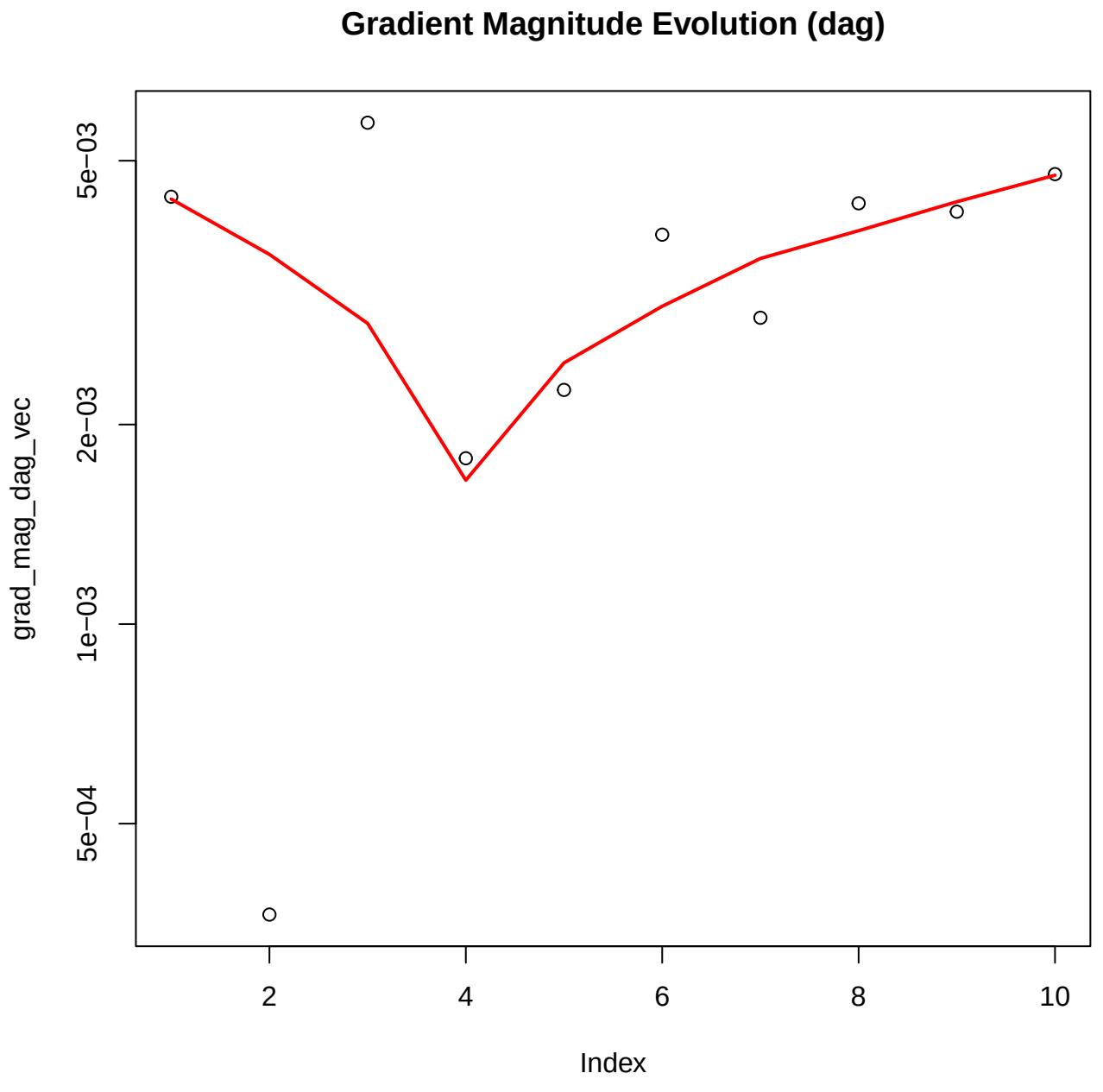

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.001
0.002
0.005
0.010
2
4
6
8
10
Index

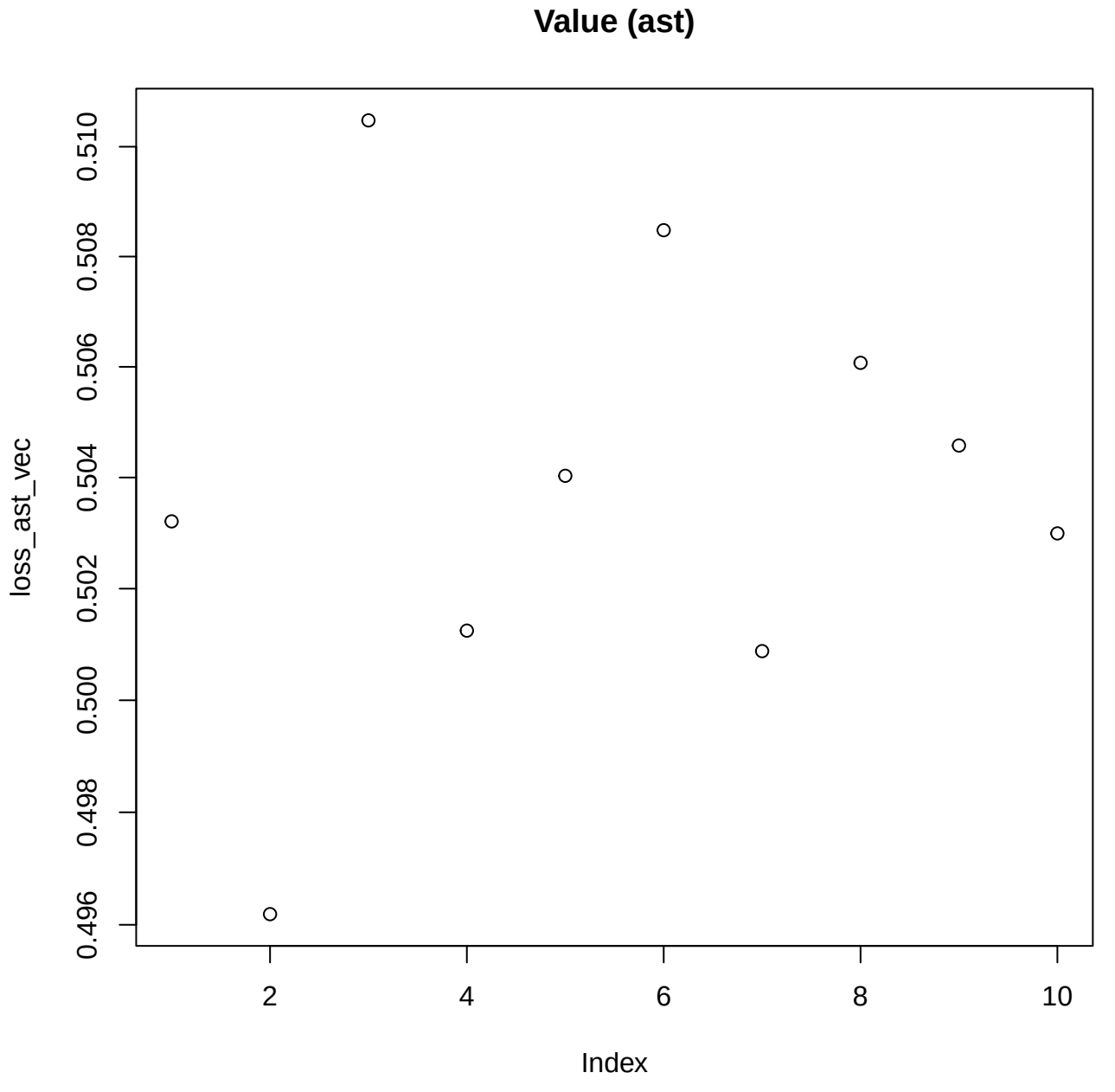

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

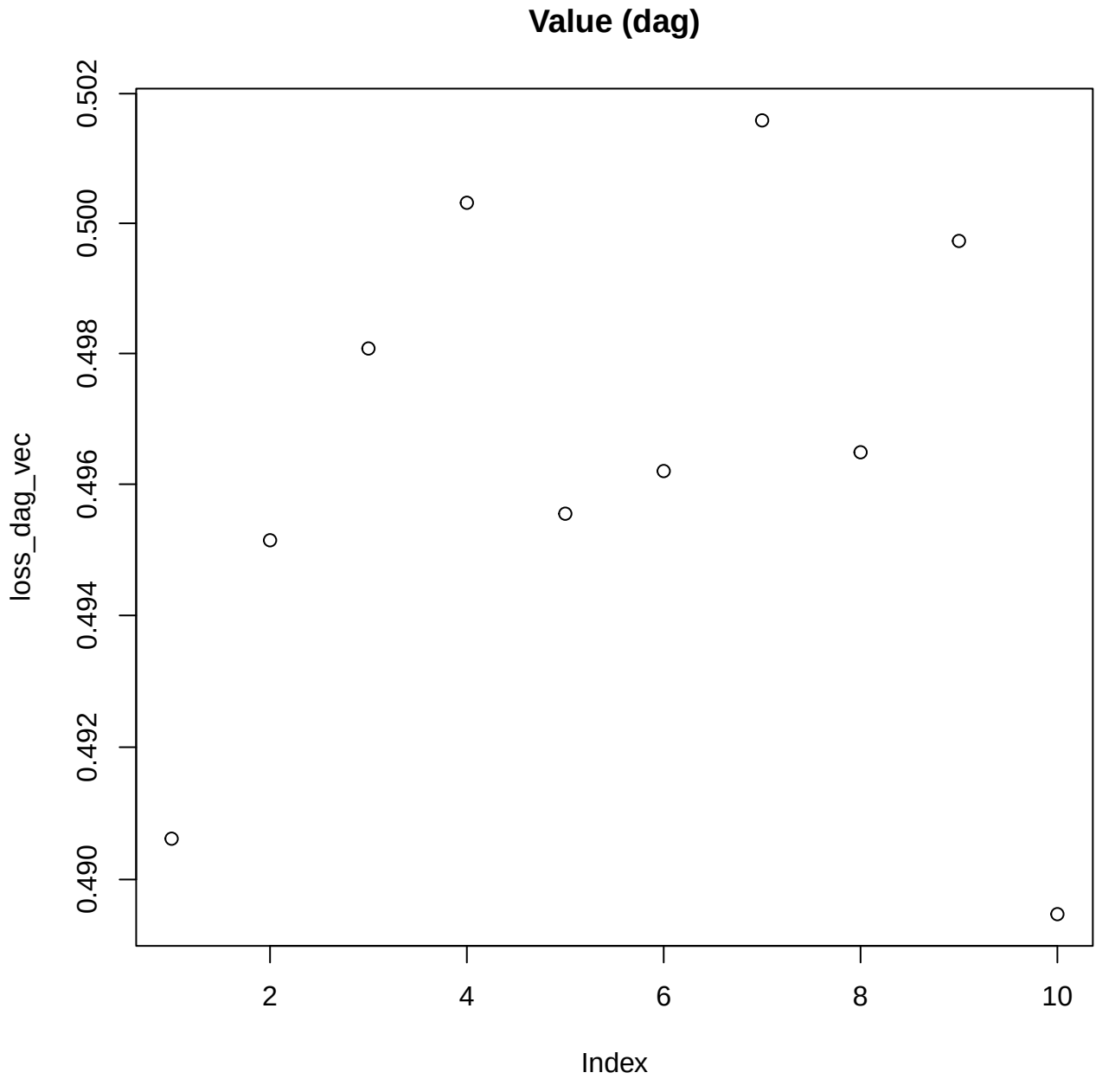

Value (ast)
loss_ast_vec
0.500
0.505
0.510
0.515
2
4
6
8
10
Index

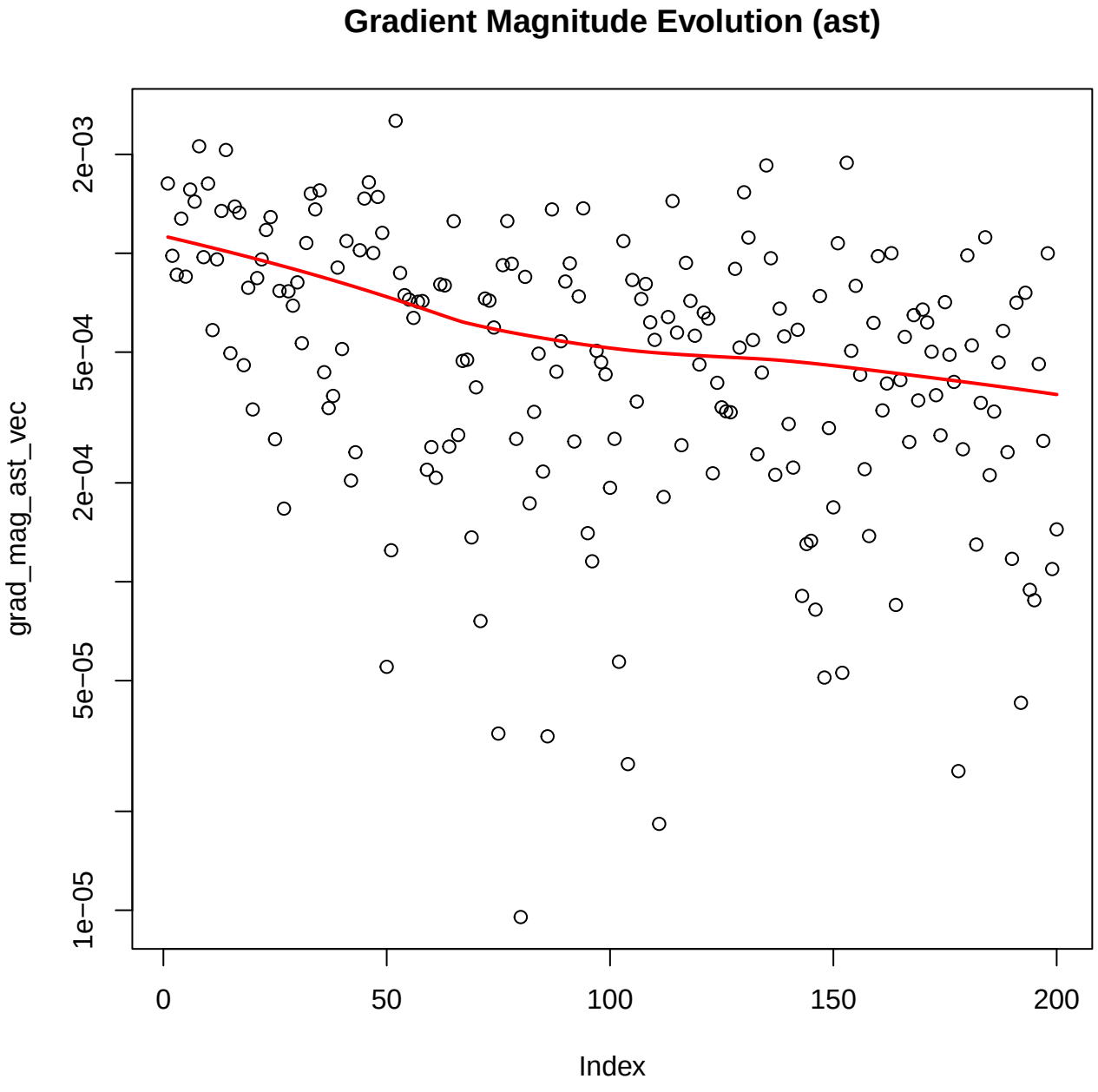

Value (dag)
loss_dag_vec
0.490
0.495
0.500
2
4
6
8
10
Index

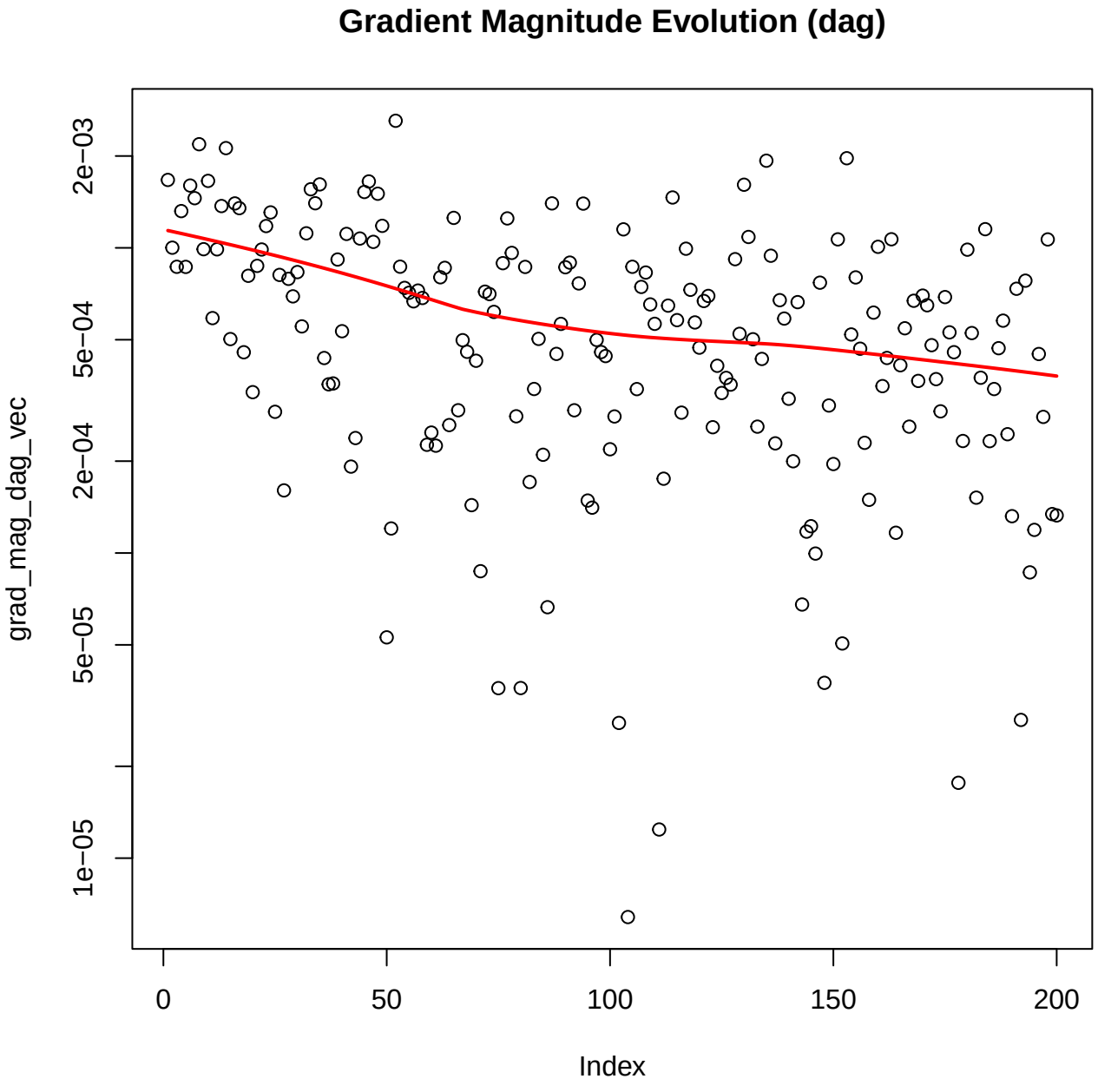

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

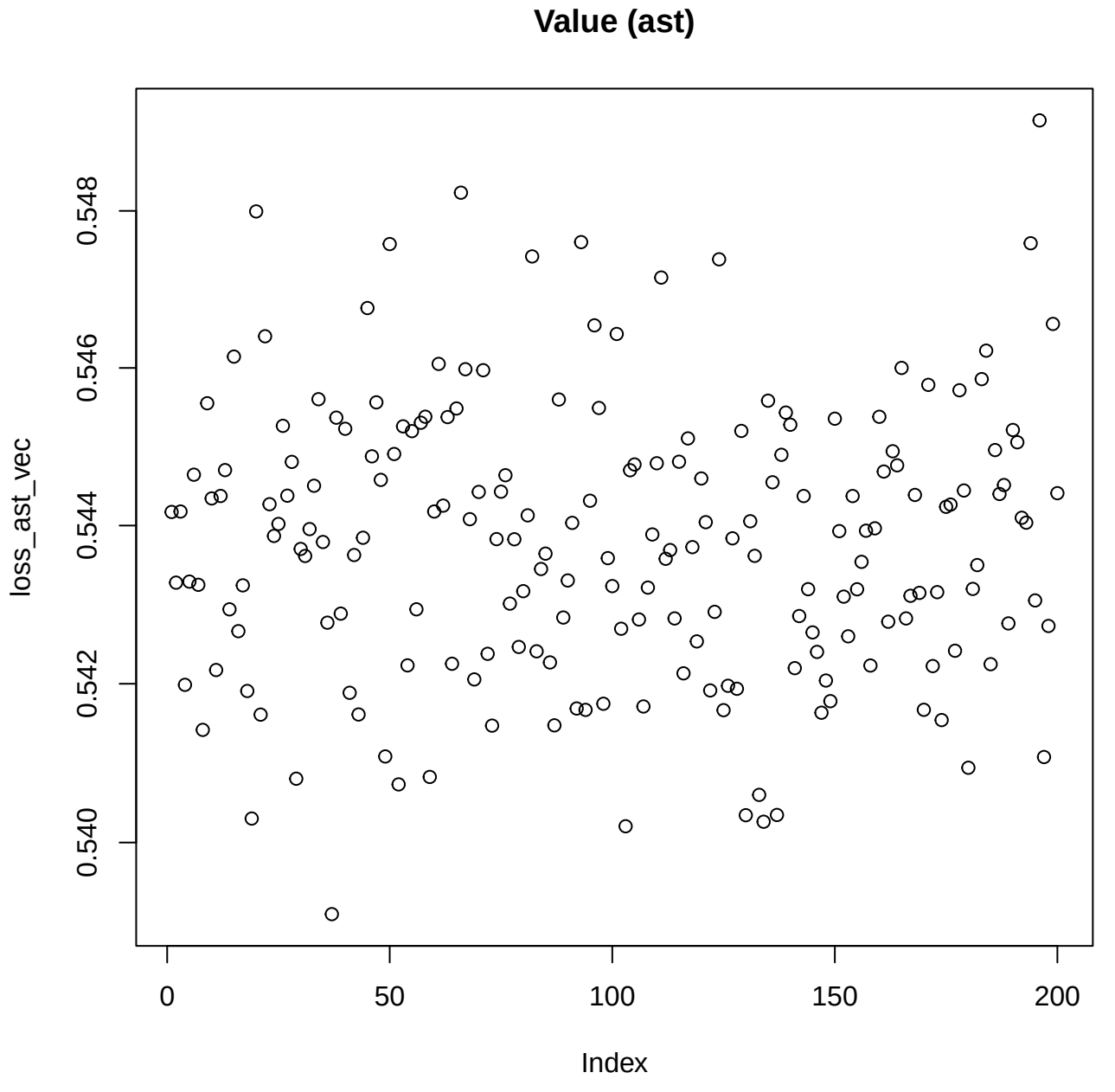

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

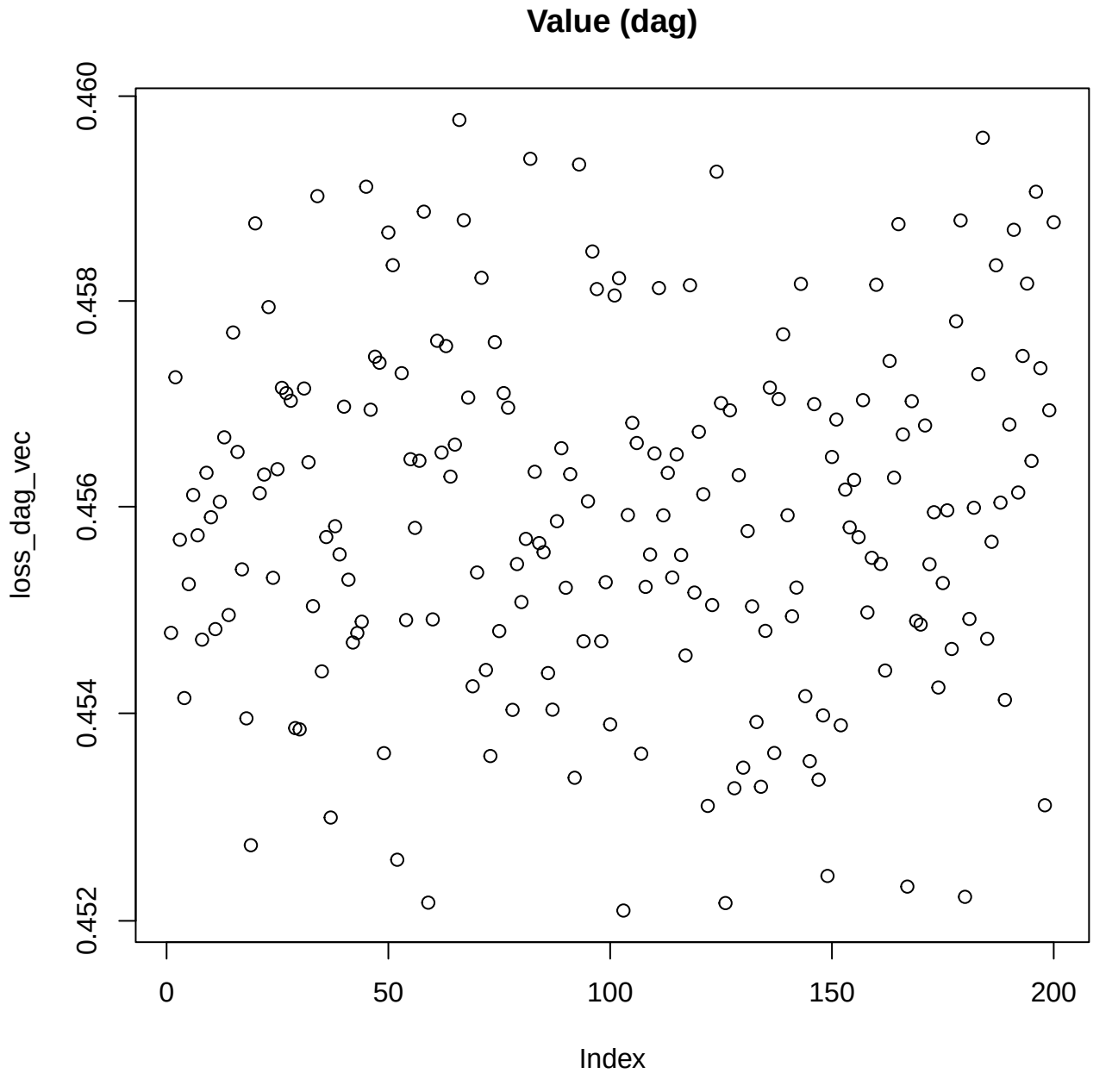

Value (ast)
loss_ast_vec
0.496
0.498
0.500
0.502
0.504
0.506
0.508
0.510
2
4
6
8
10
Index

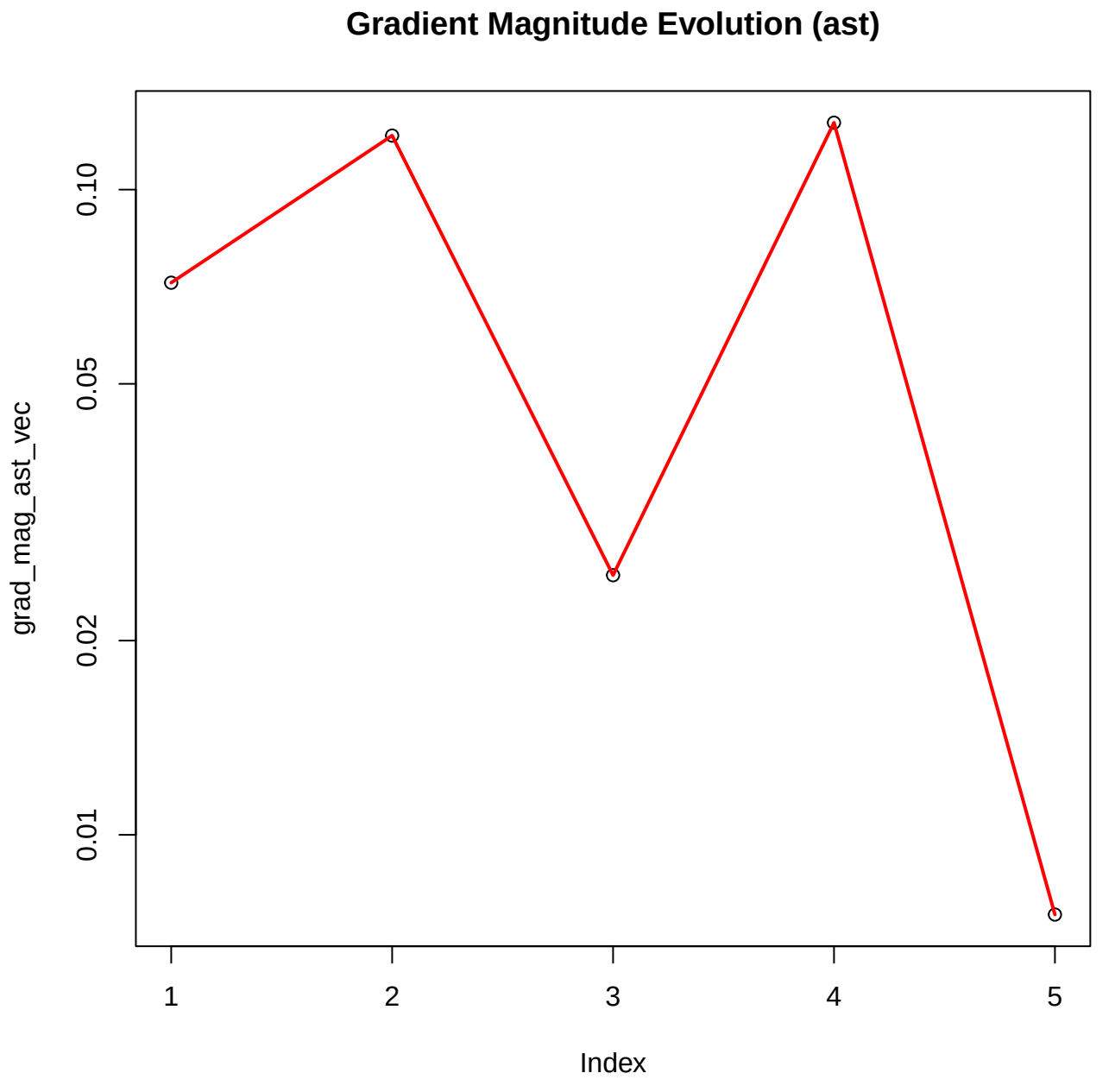

Value (dag)
loss_dag_vec
0.490
0.492
0.494
0.496
0.498
0.500
0.502
2
4
6
8
10
Index

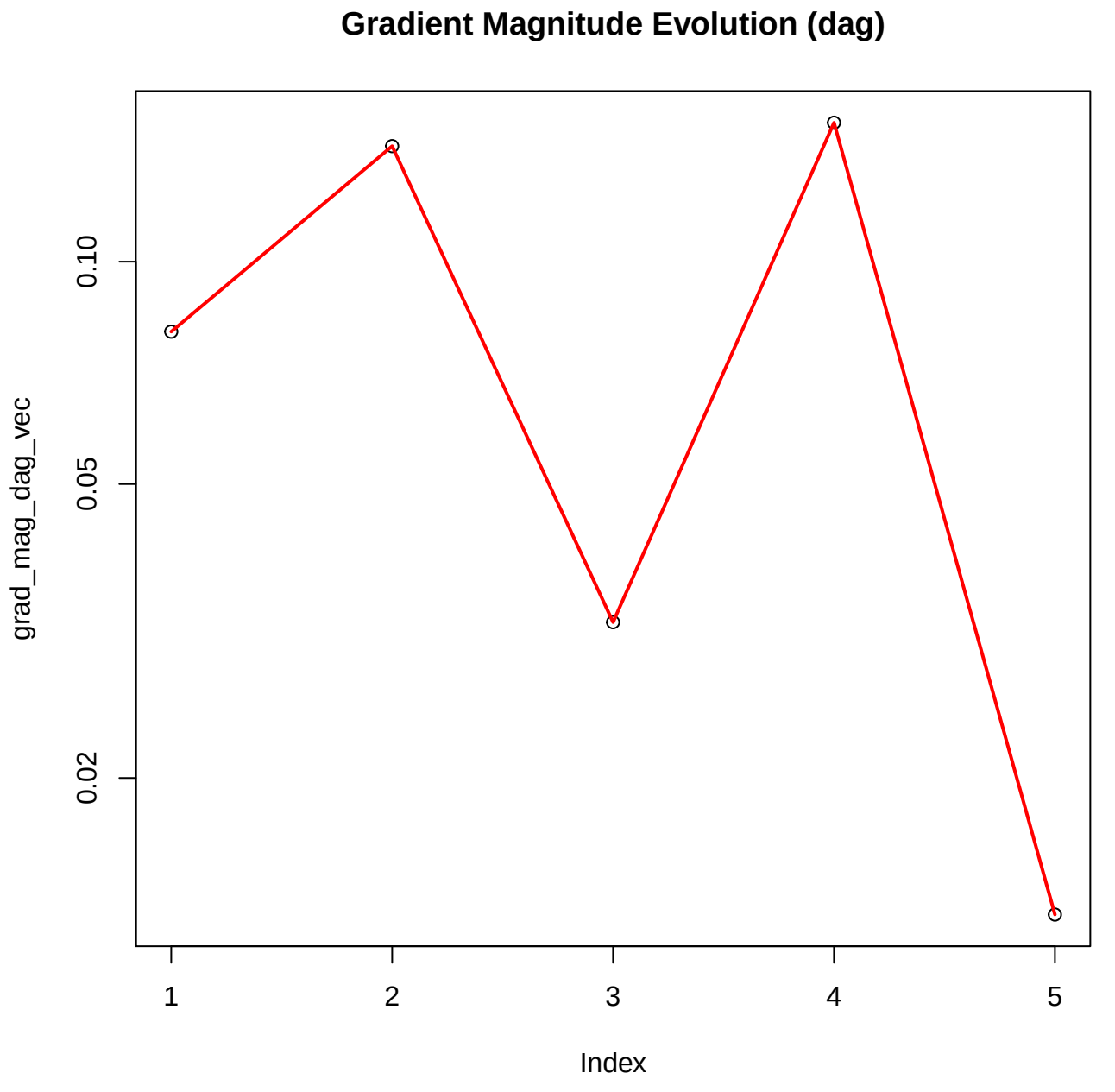

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

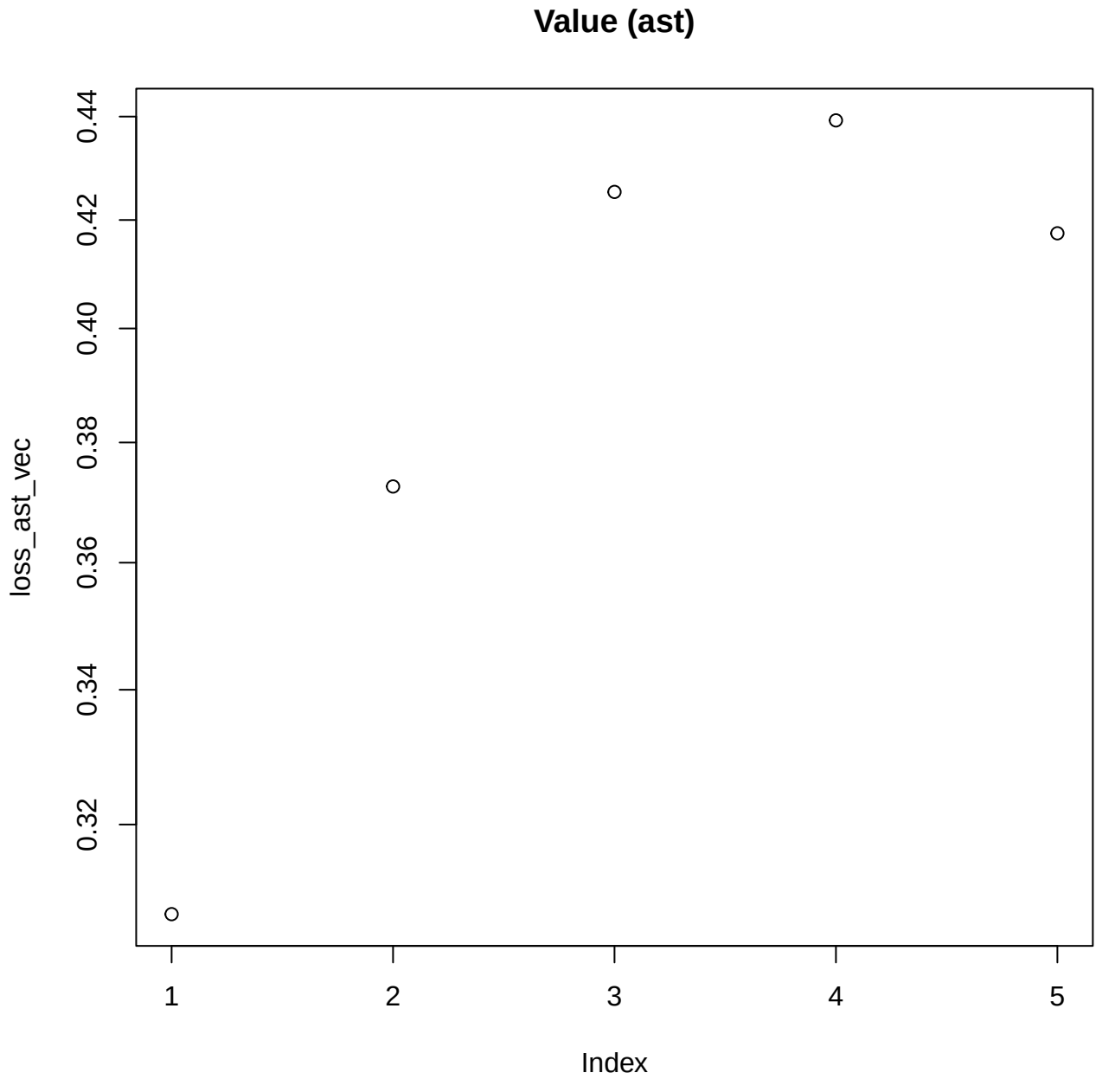

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

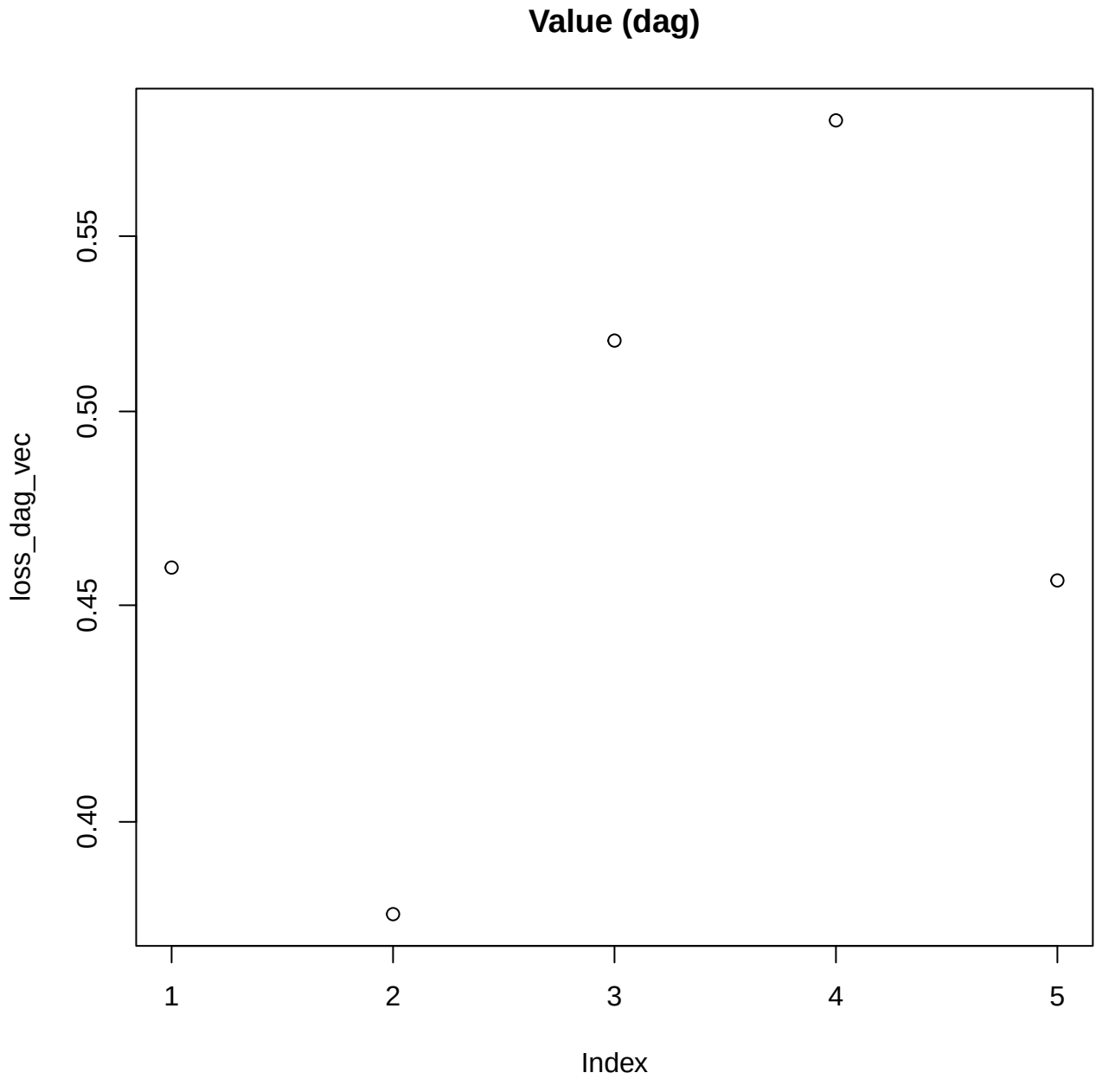

Value (ast)
loss_ast_vec
0.540
0.542
0.544
0.546
0.548
0
50
100
150
200
Index

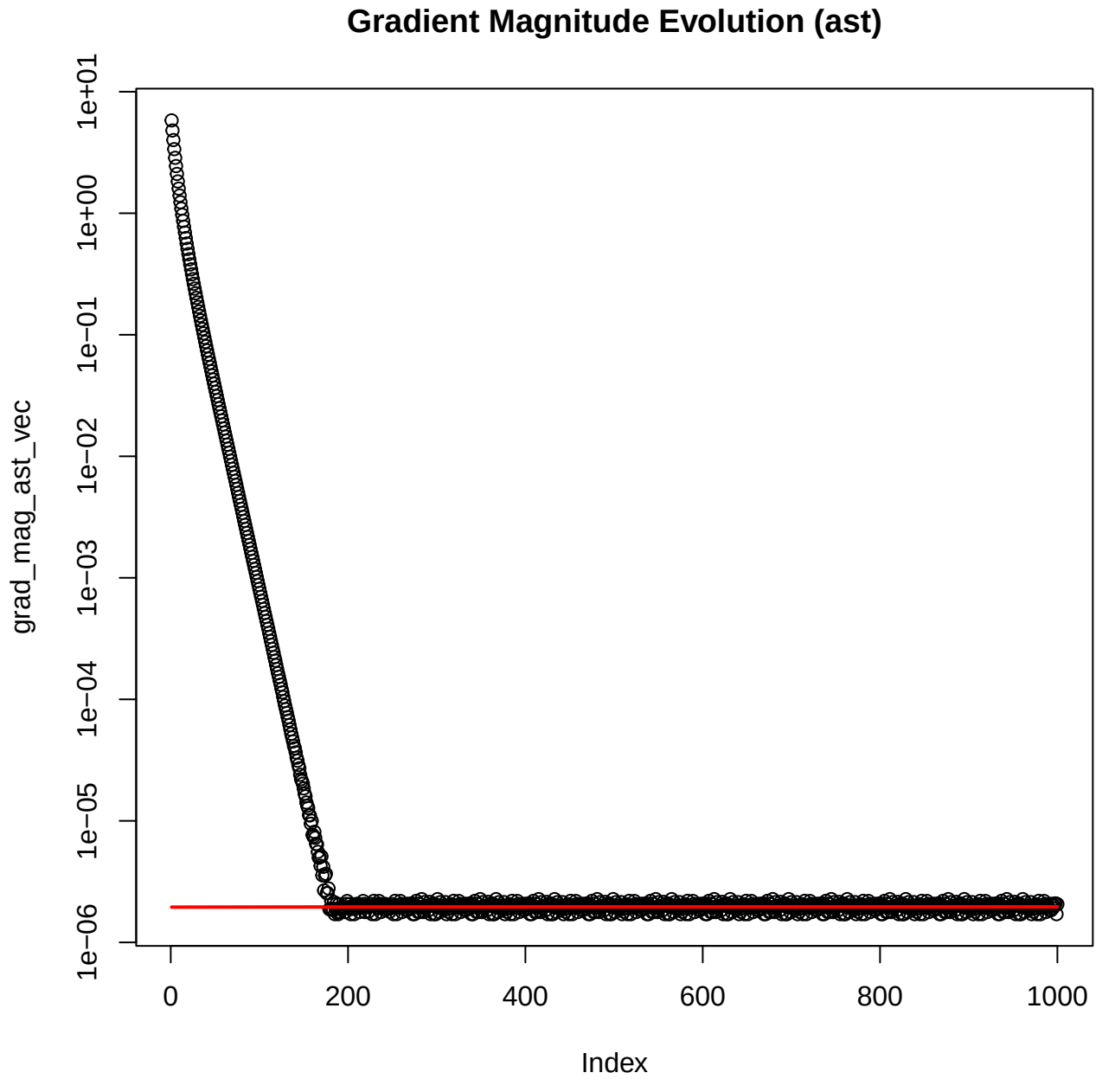

Value (dag)
loss_dag_vec
0.452
0.454
0.456
0.458
0.460
0
50
100
150
200
Index

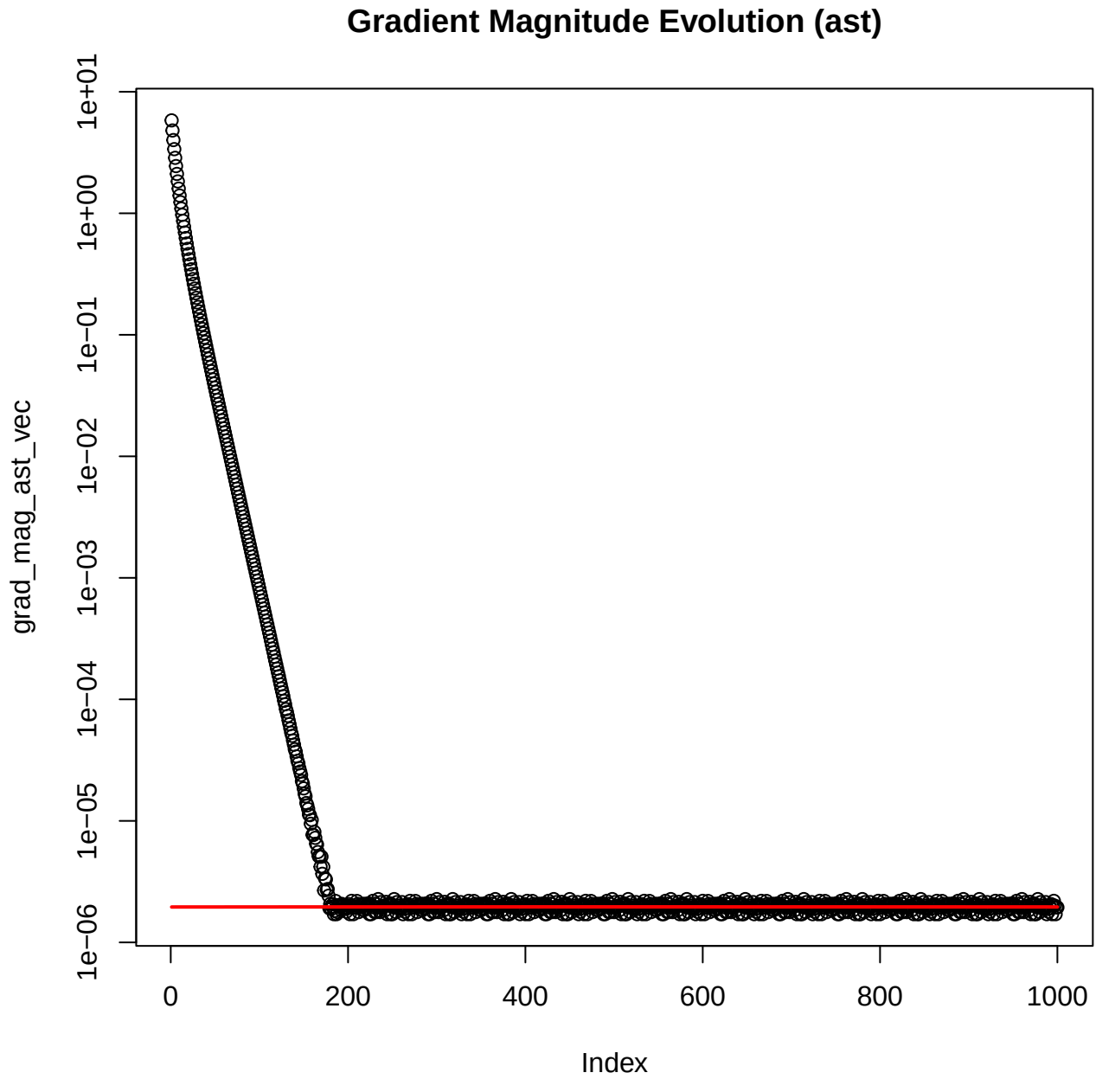

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
1
2
3
4
5
Index

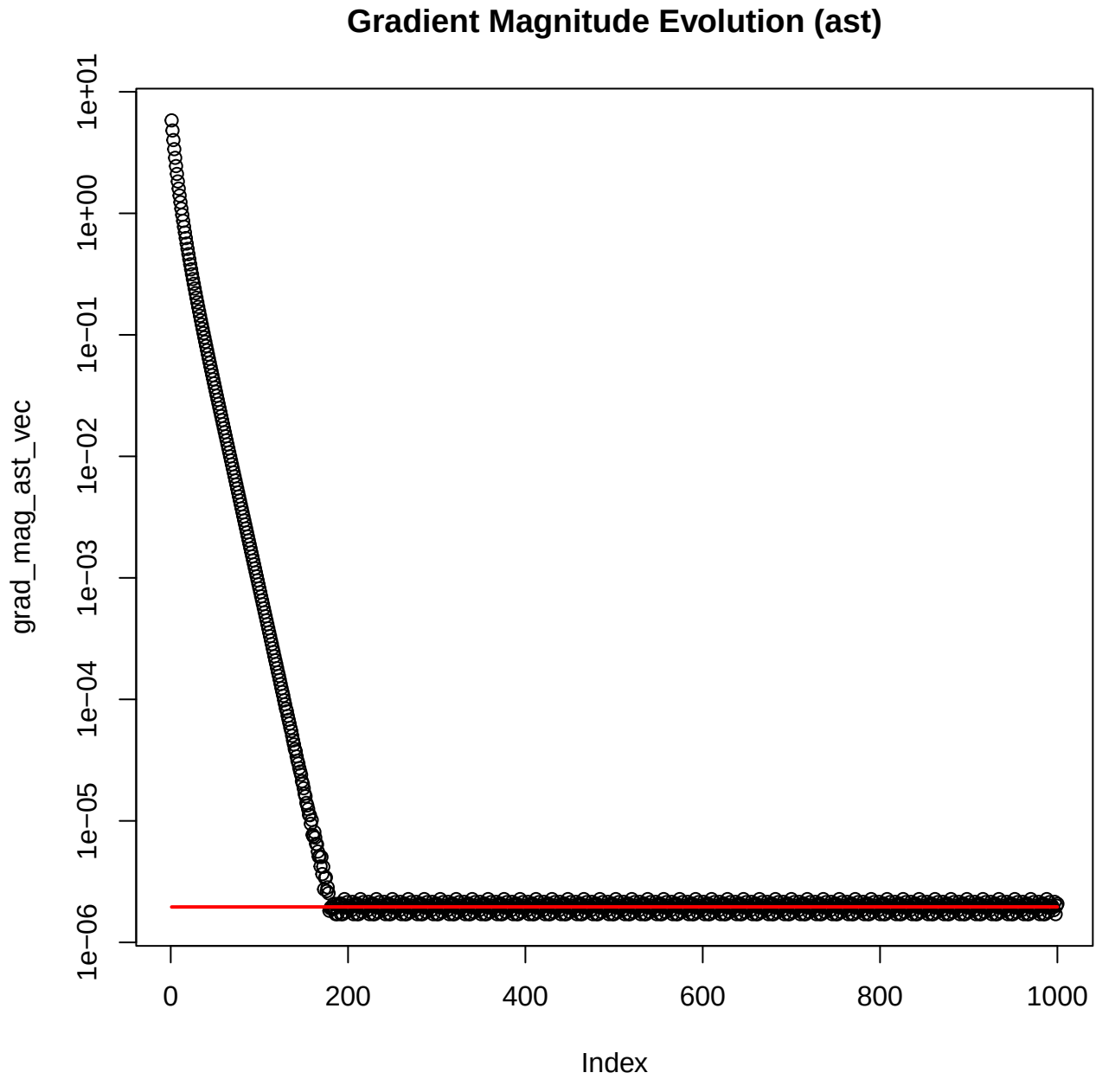

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
1
2
3
4
5
Index

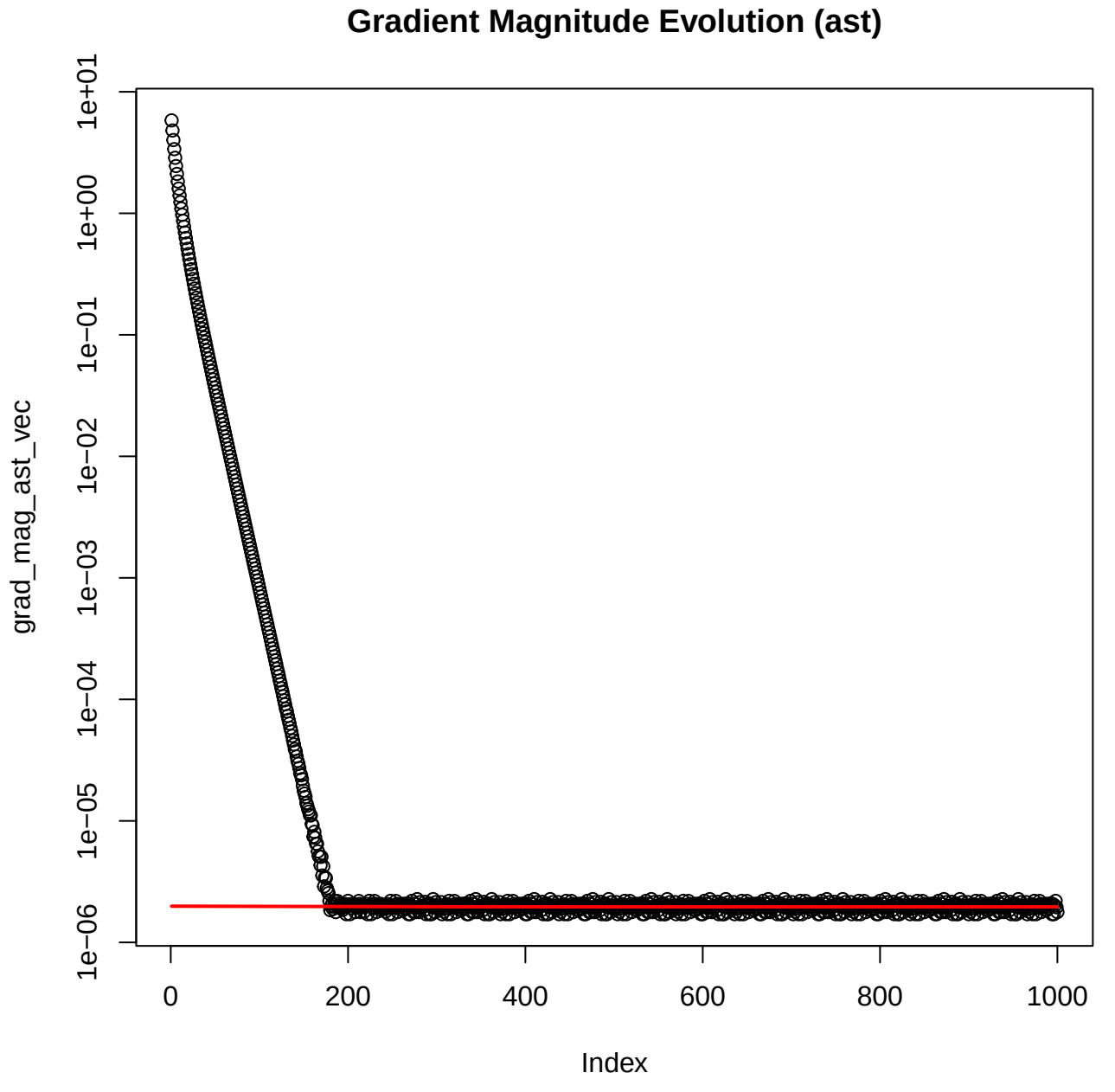

Value (ast)
loss_ast_vec
0.32
0.34
0.36
0.38
0.40
0.42
0.44
1
2
3
4
5
Index

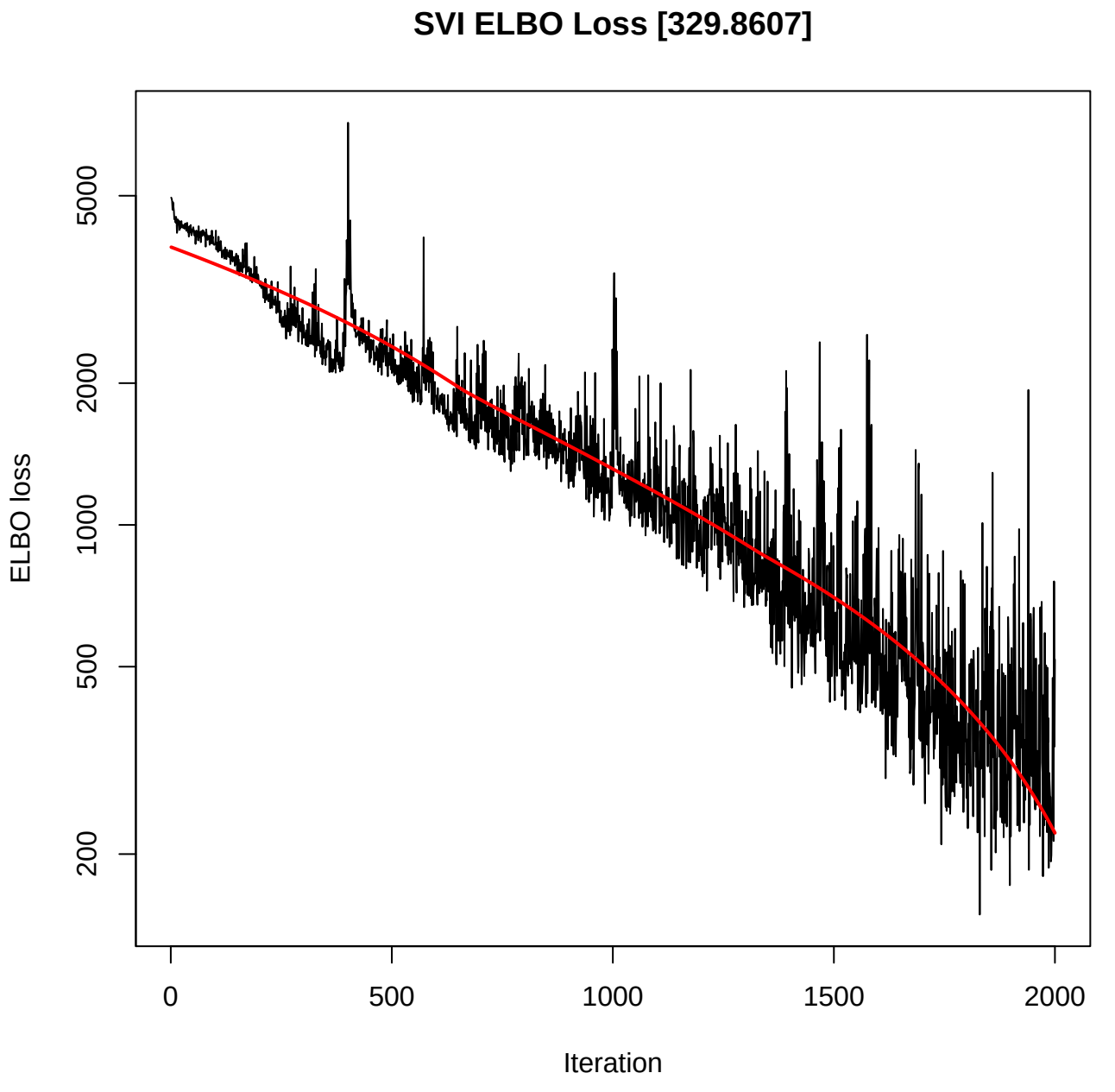

Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index

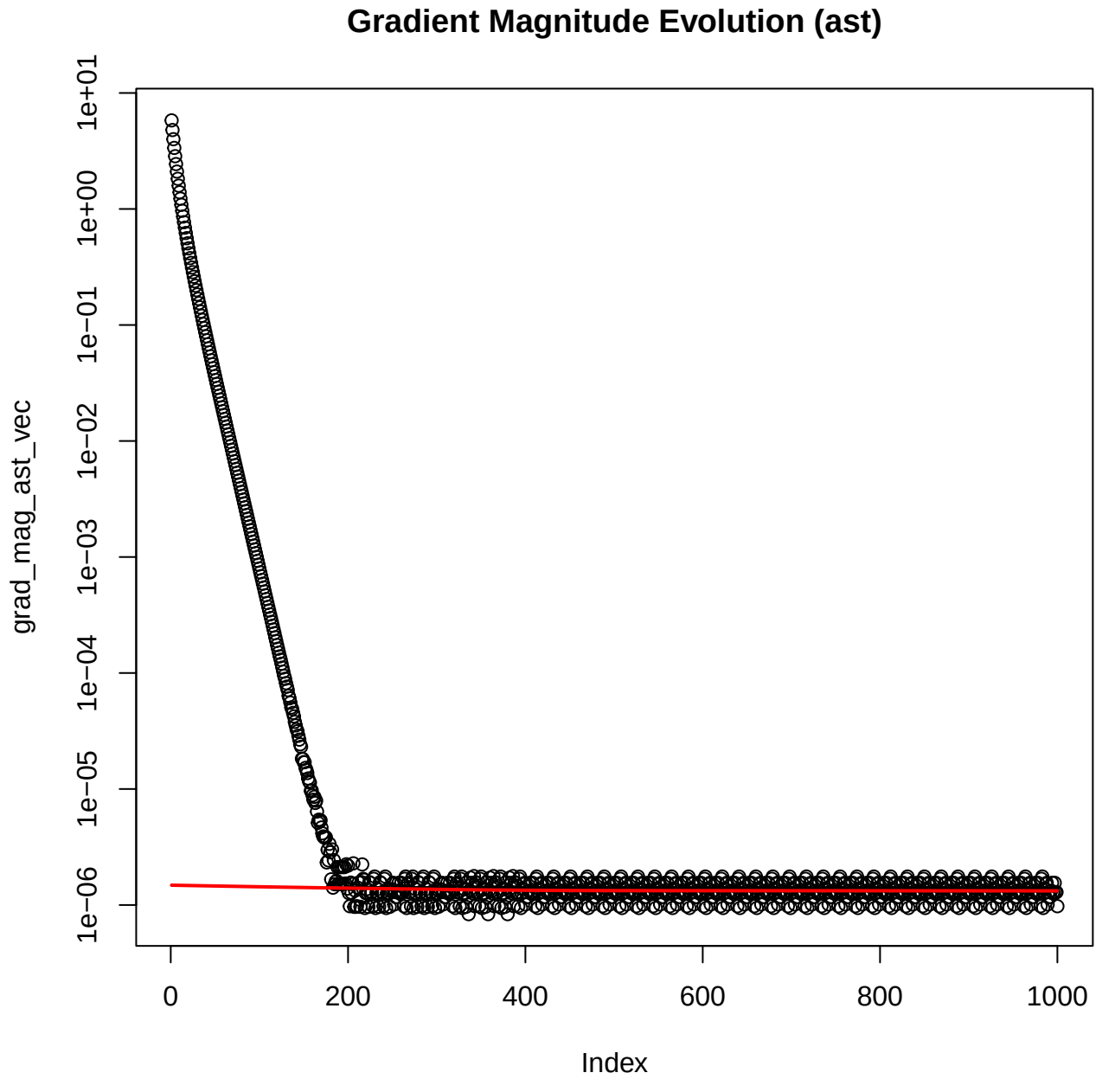

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

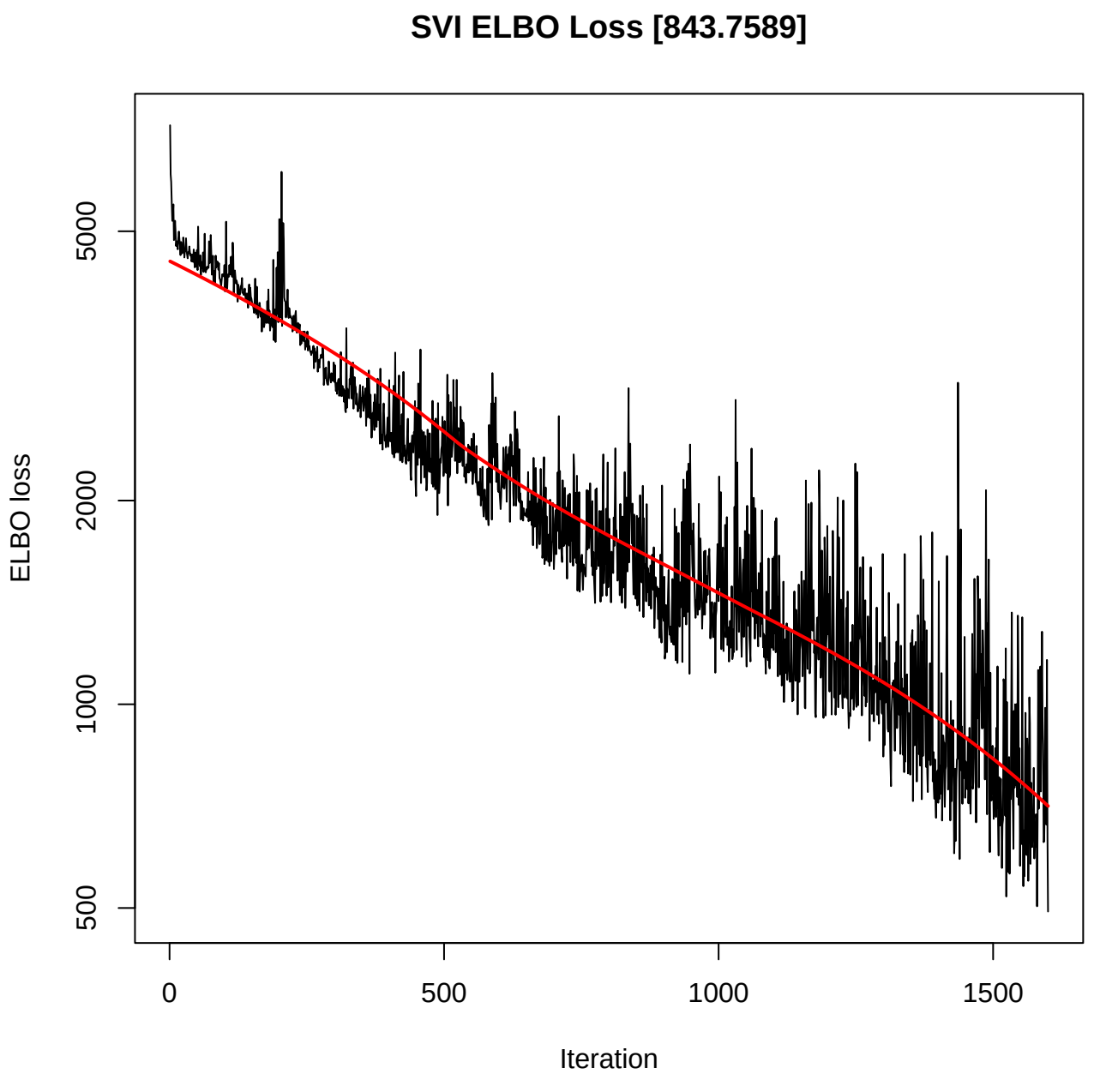

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

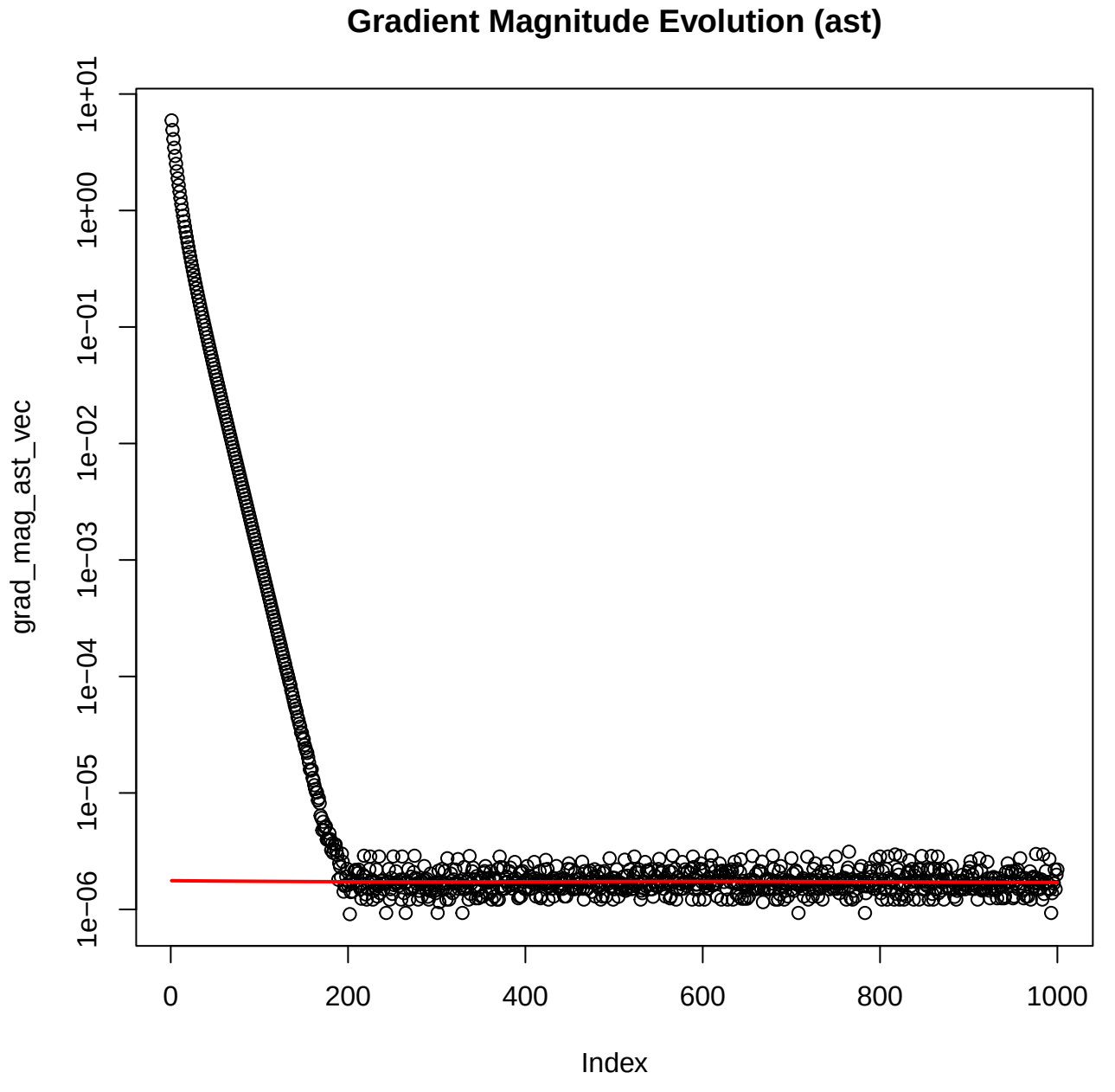

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

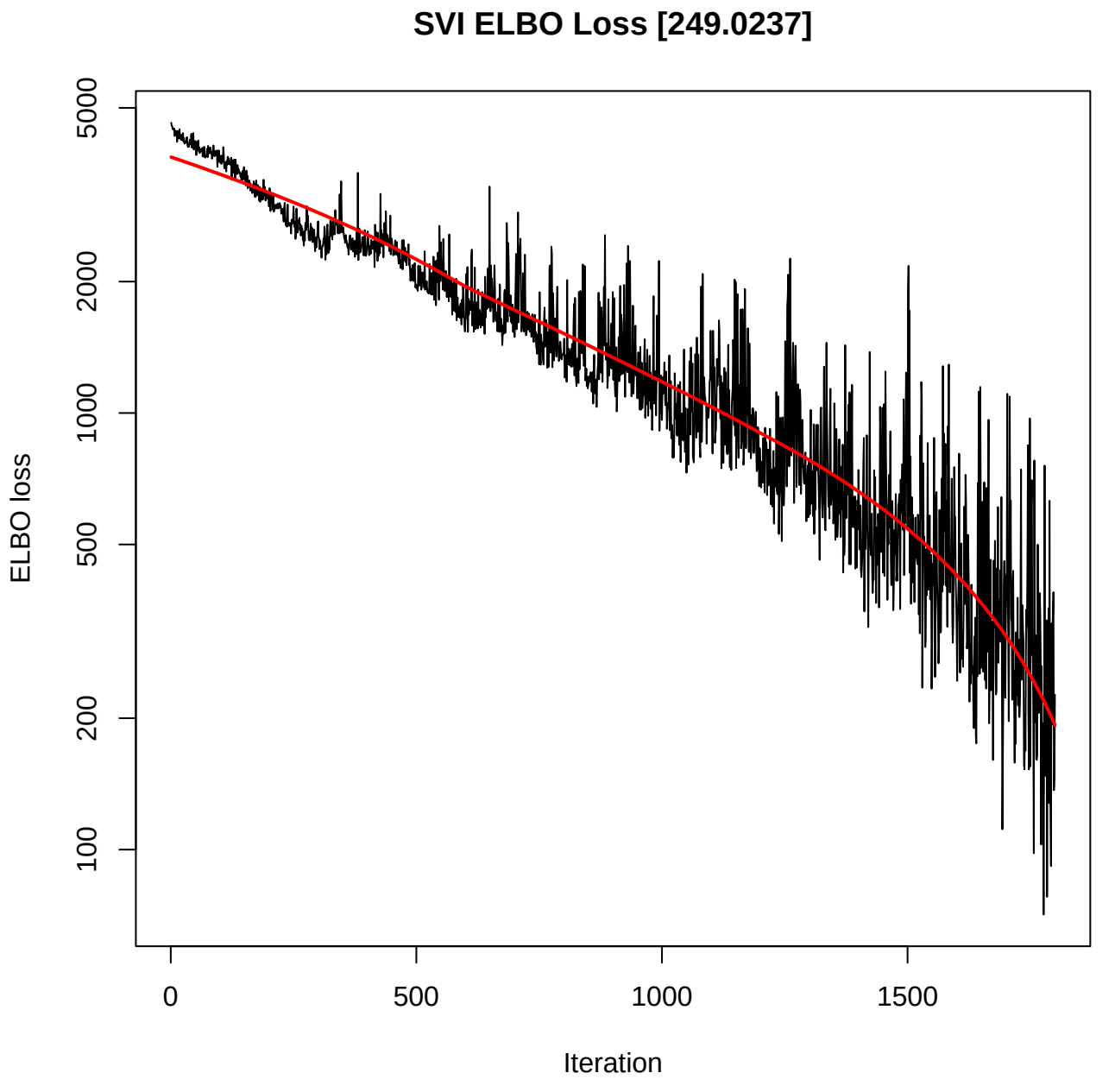

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

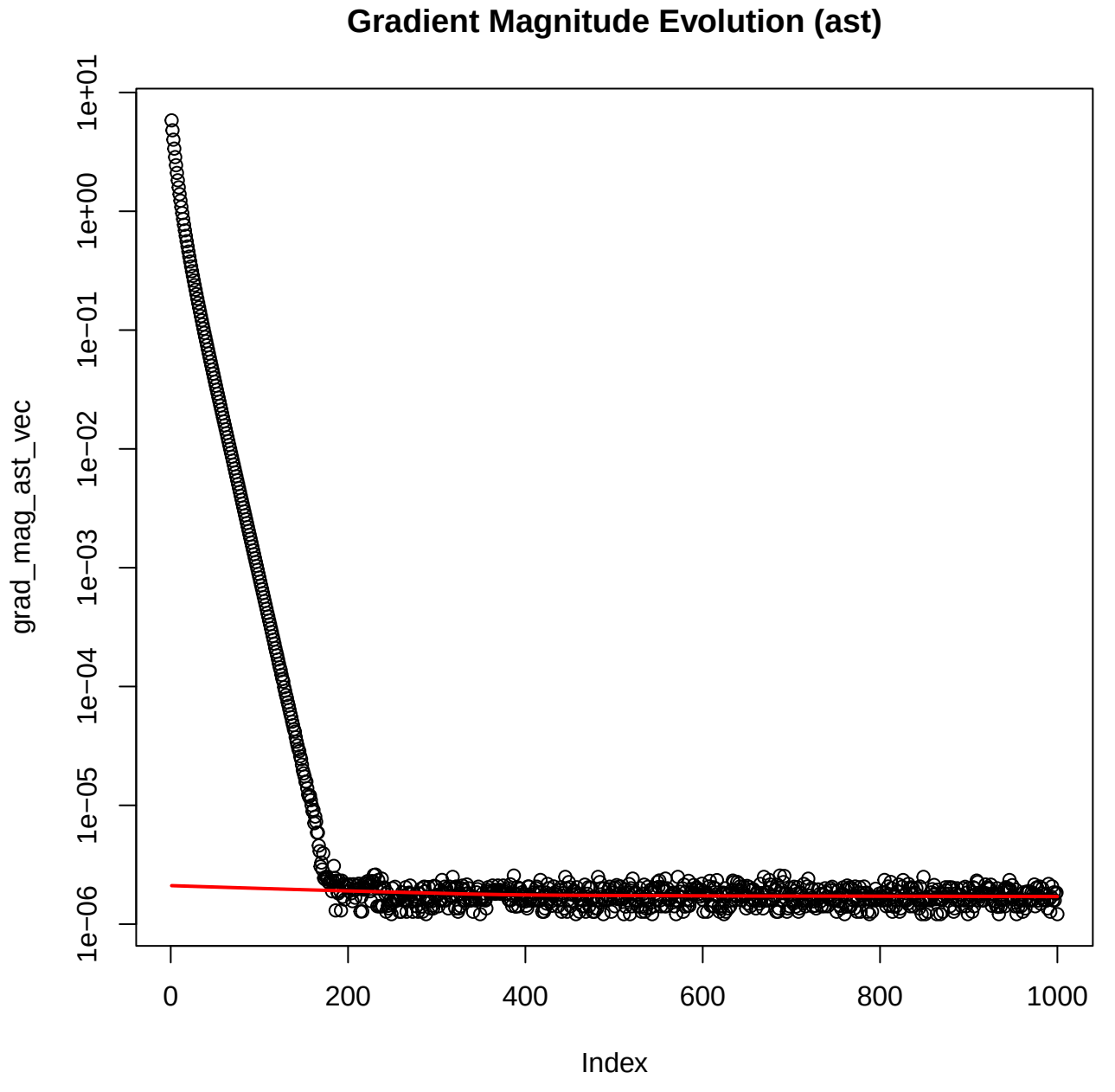

SVI ELBO Loss [329.8607]
ELBO loss
5000
2000
1000
500
200
0
500
1000
1500
2000
Iteration

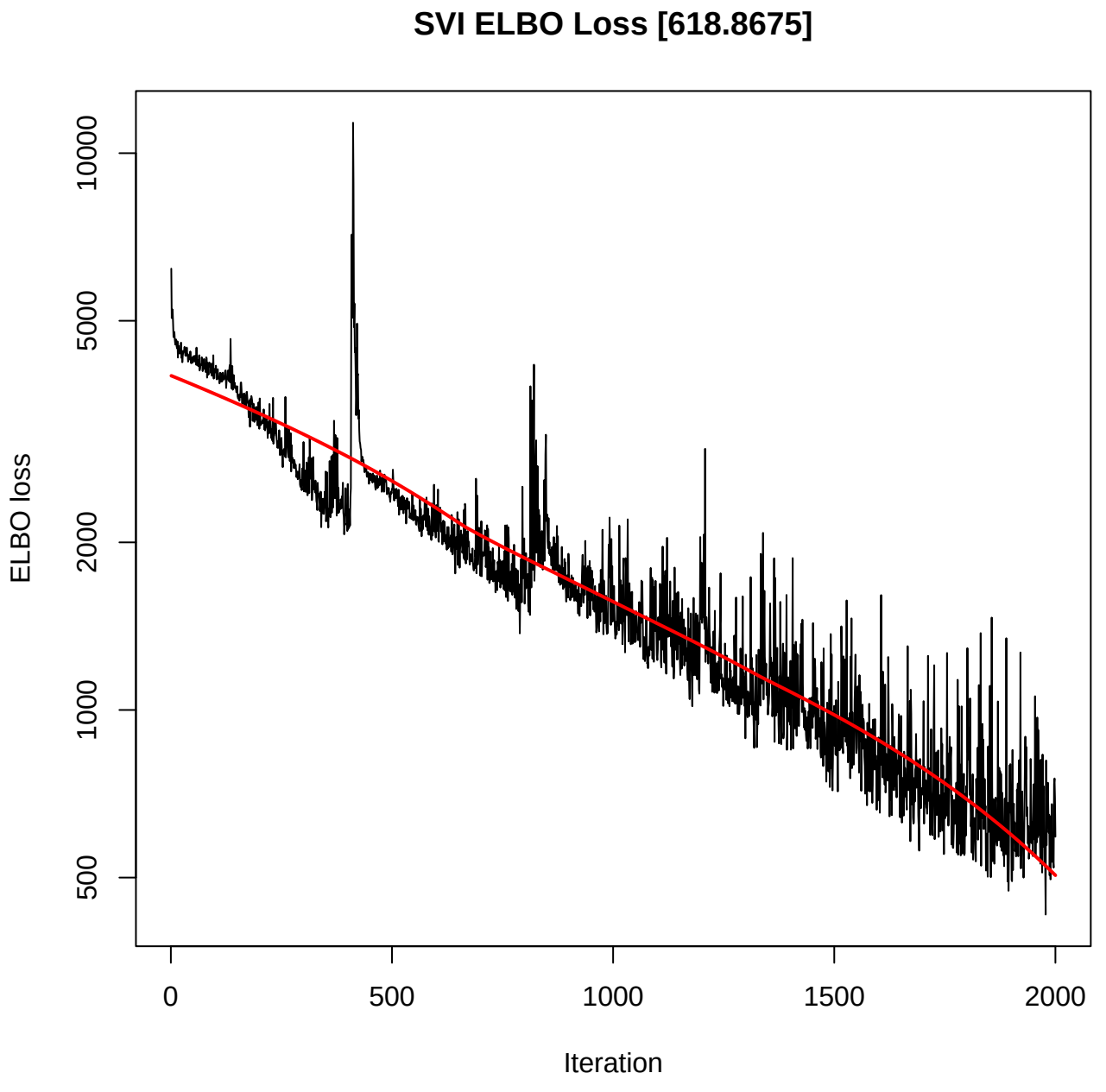

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

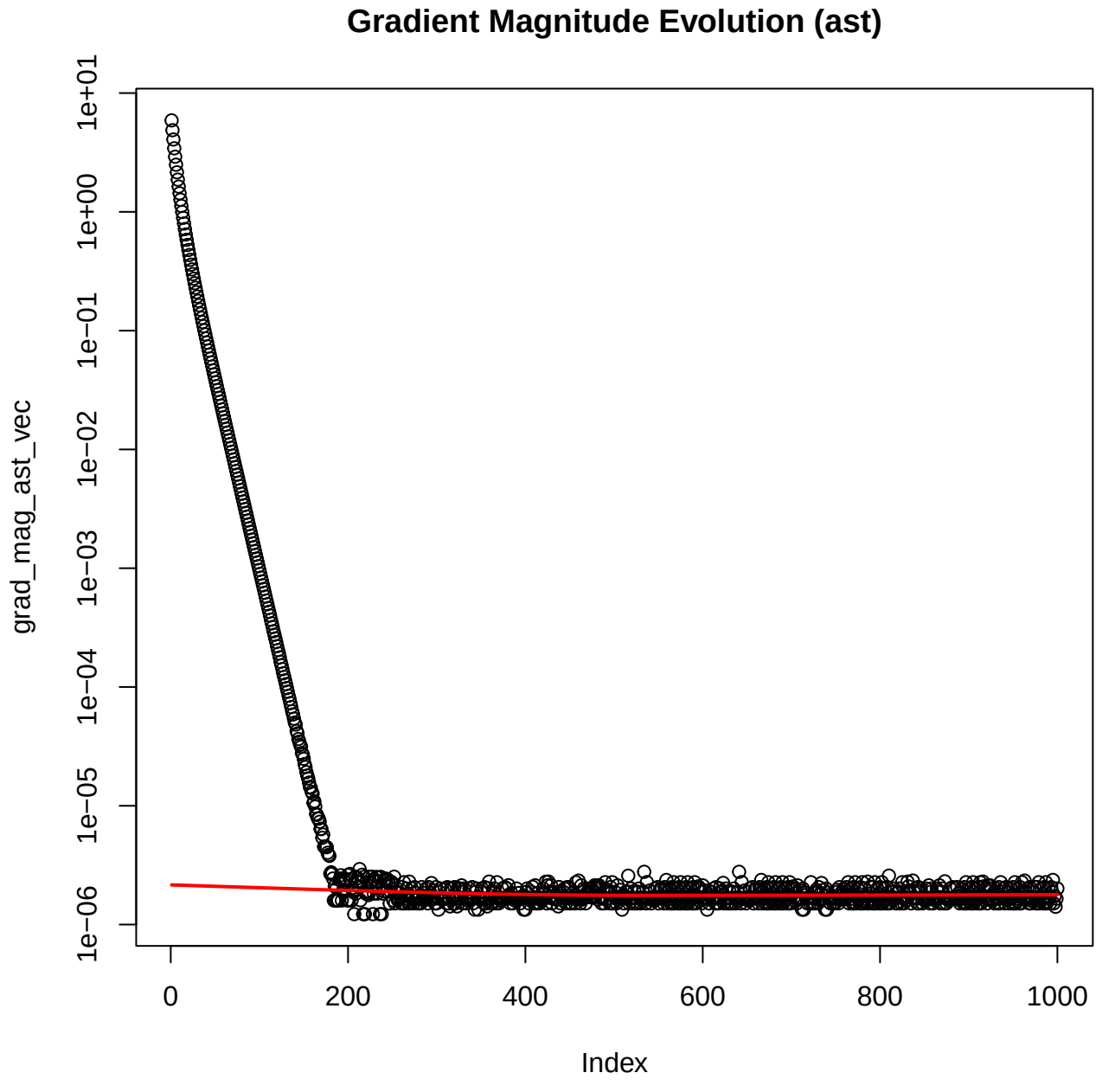

SVI ELBO Loss [843.7589]
ELBO loss
5000
2000
1000
500
0
500
1000
1500
Iteration

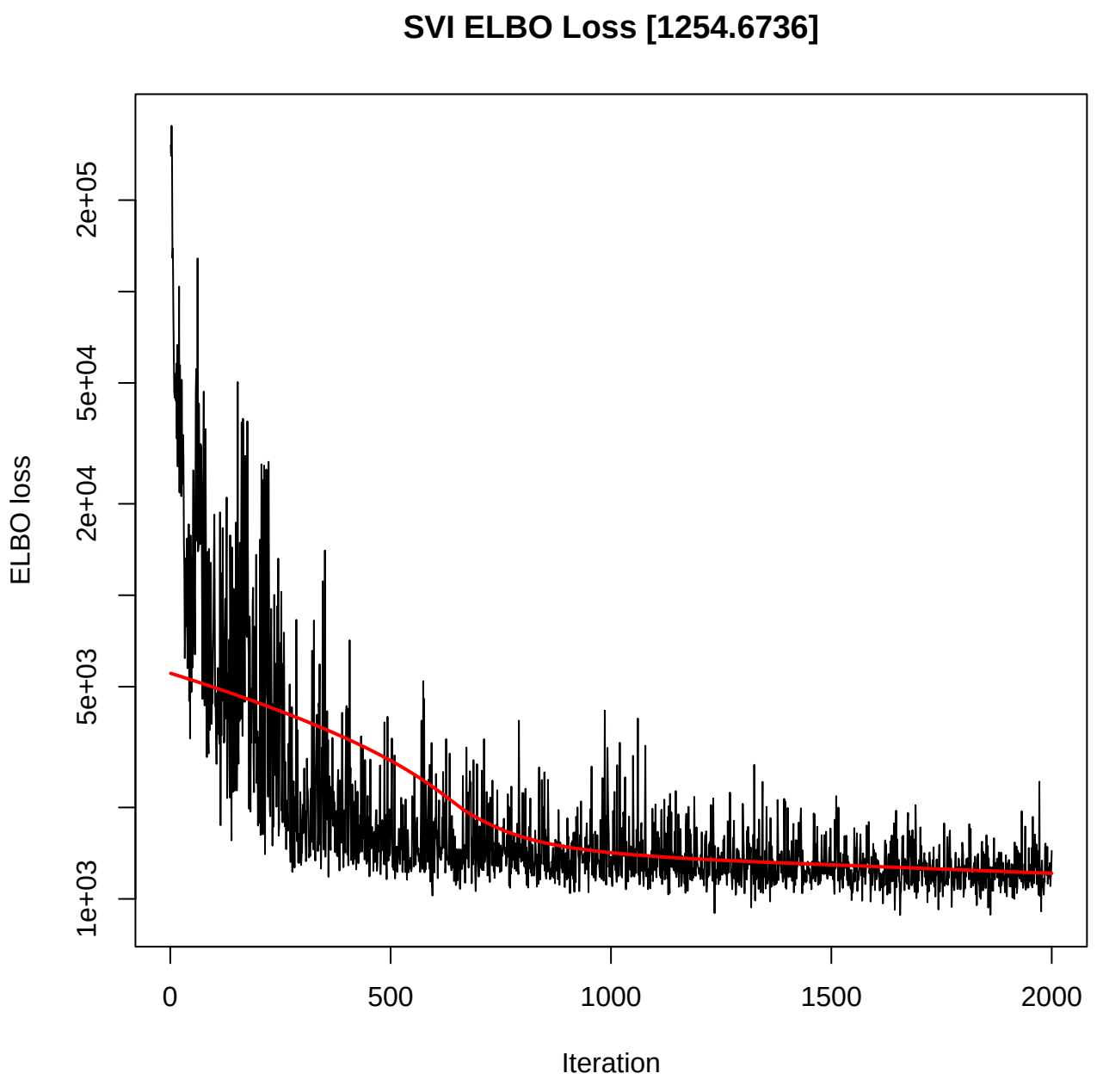

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

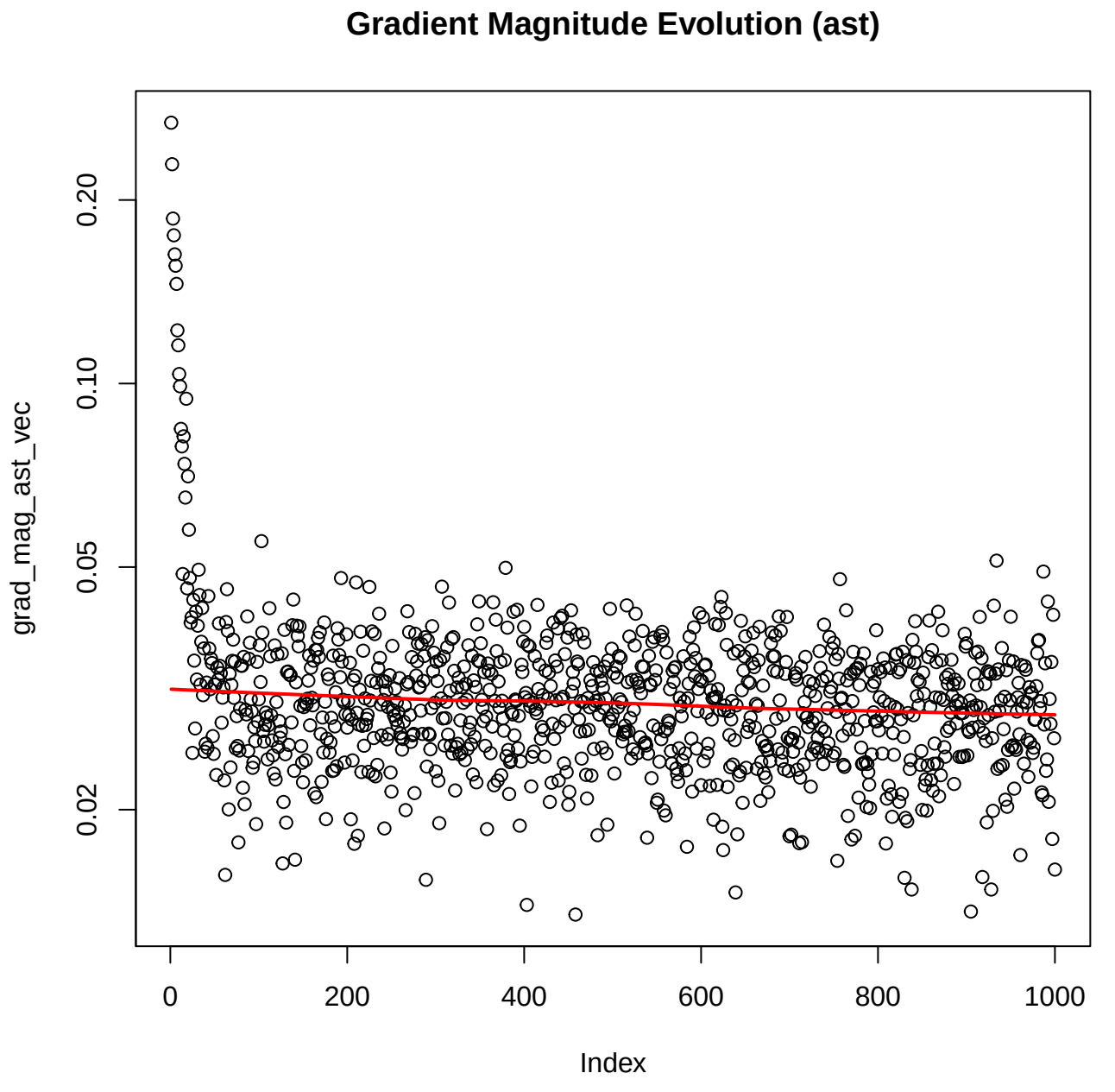

SVI ELBO Loss [249.0237]
ELBO loss
5000
2000
1000
500
200
100
0
500
1000
1500
Iteration

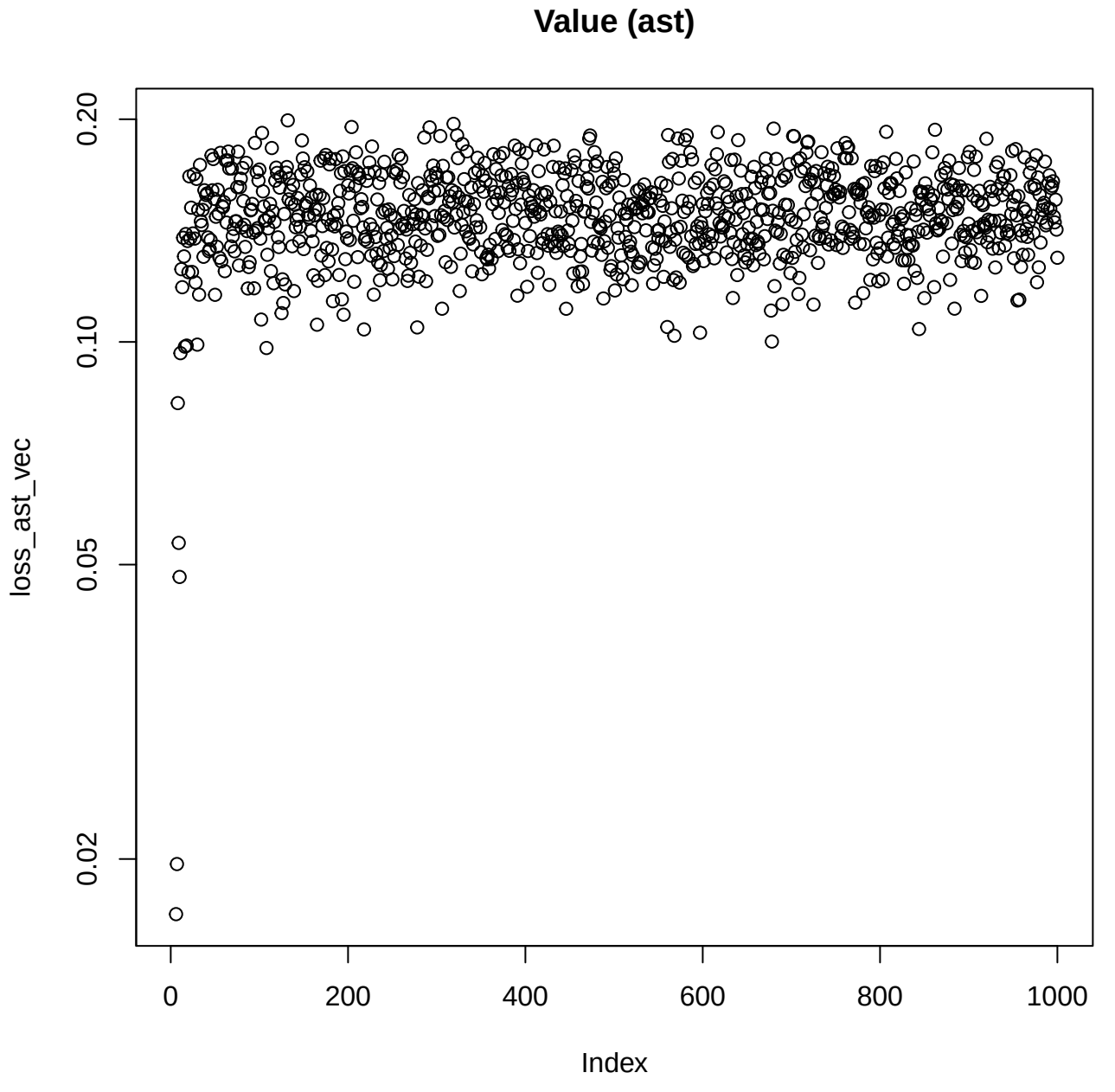

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

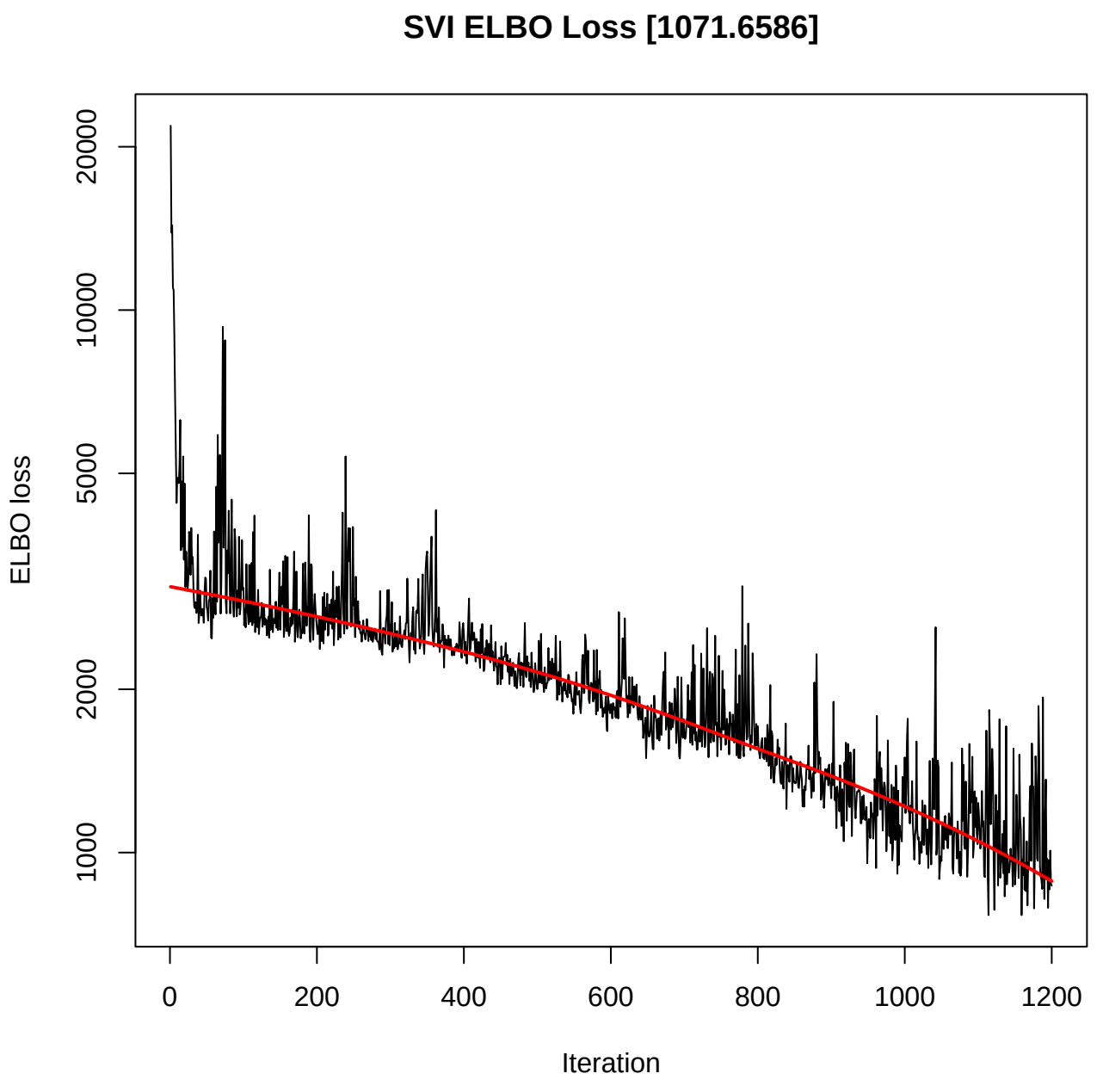

SVI ELBO Loss [618.8675]
ELBO loss
10000
5000
2000
1000
500
0
500
1000
1500
2000
Iteration

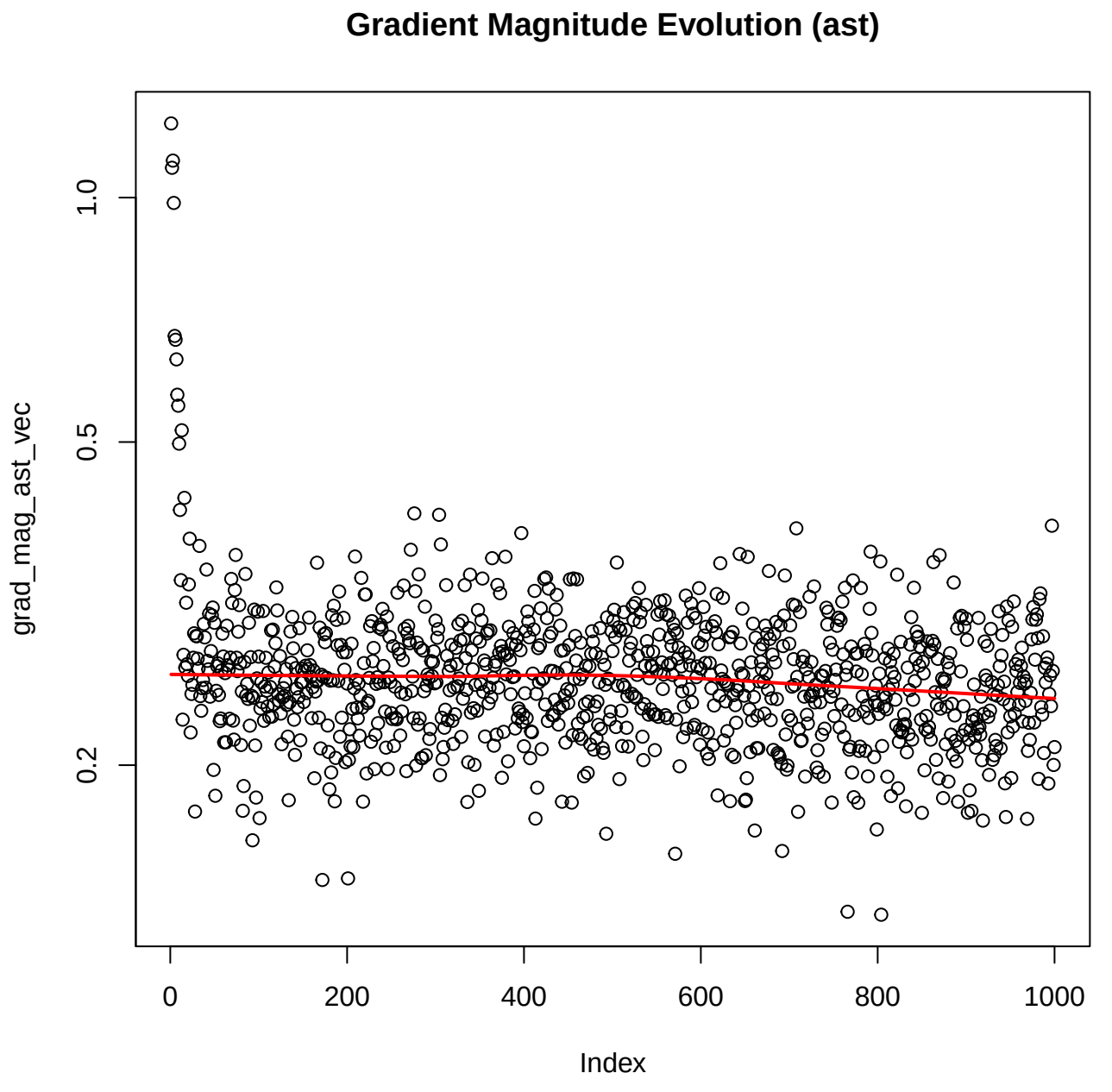

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index


SVI ELBO Loss [1254.6736]
ELBO loss
2e+05
5e+04
2e+04
5e+03
1e+03
0
500
1000
1500
2000
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.02
0.05
0.10
0.20
0
200
400
600
800
1000
Index


Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
0
200
400
600
800
1000
Index


SVI ELBO Loss [1071.6586]
ELBO loss
20000
10000
5000
2000
1000
0
200
400
600
800
1000
1200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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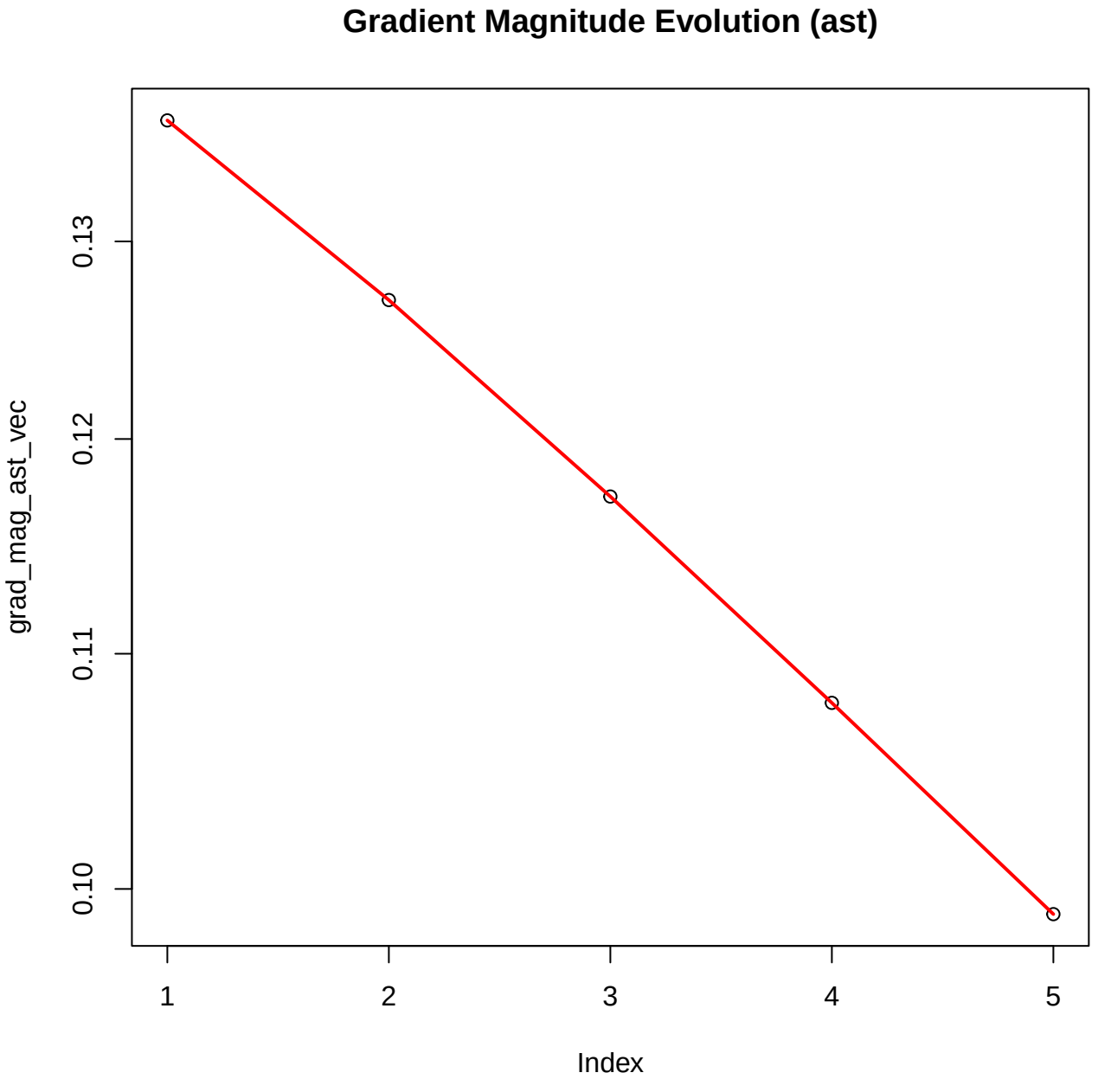
0.5
0.2
0
200
400
600
800
1000
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
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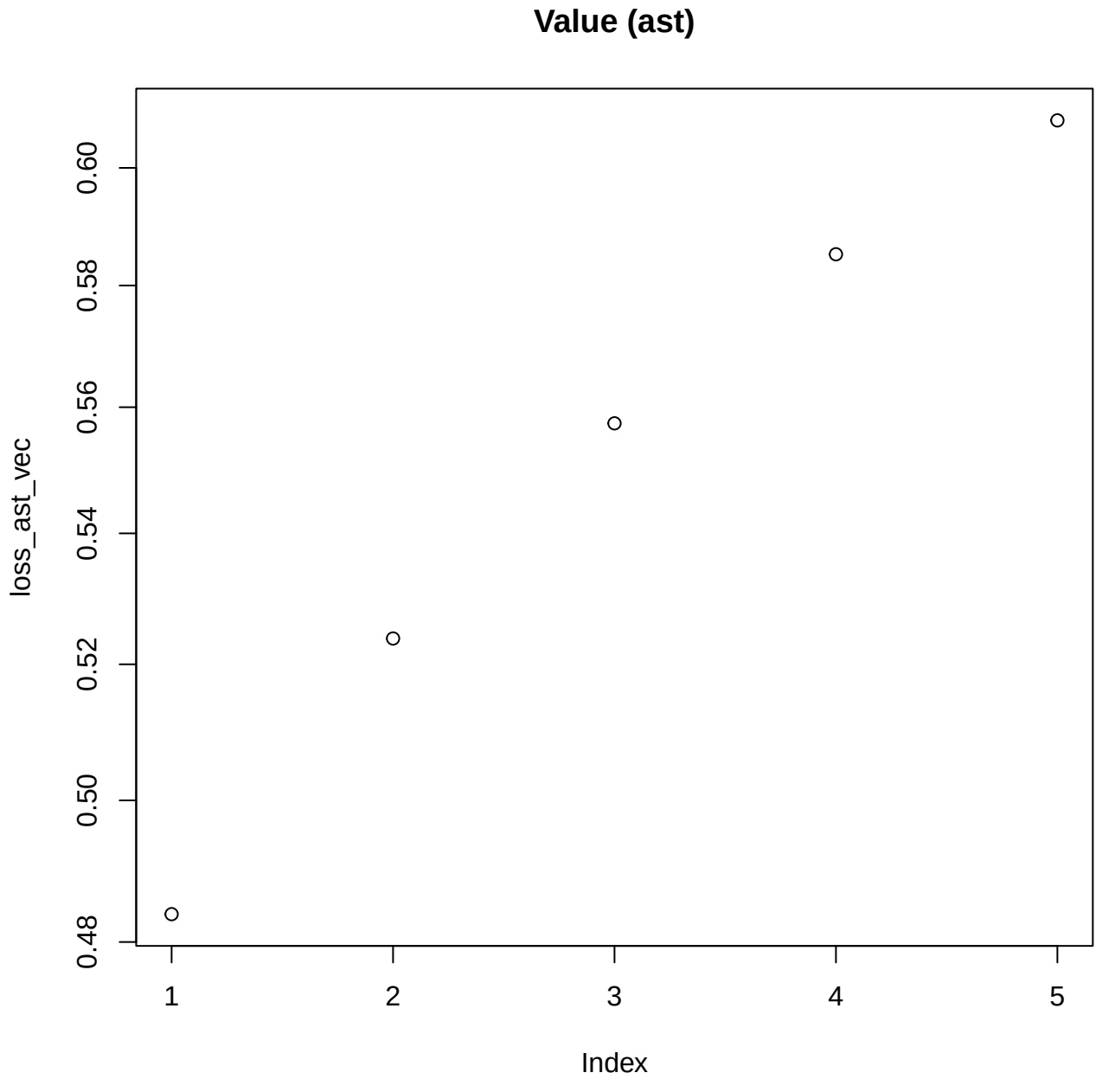
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.50
0.52
0.54
0.56
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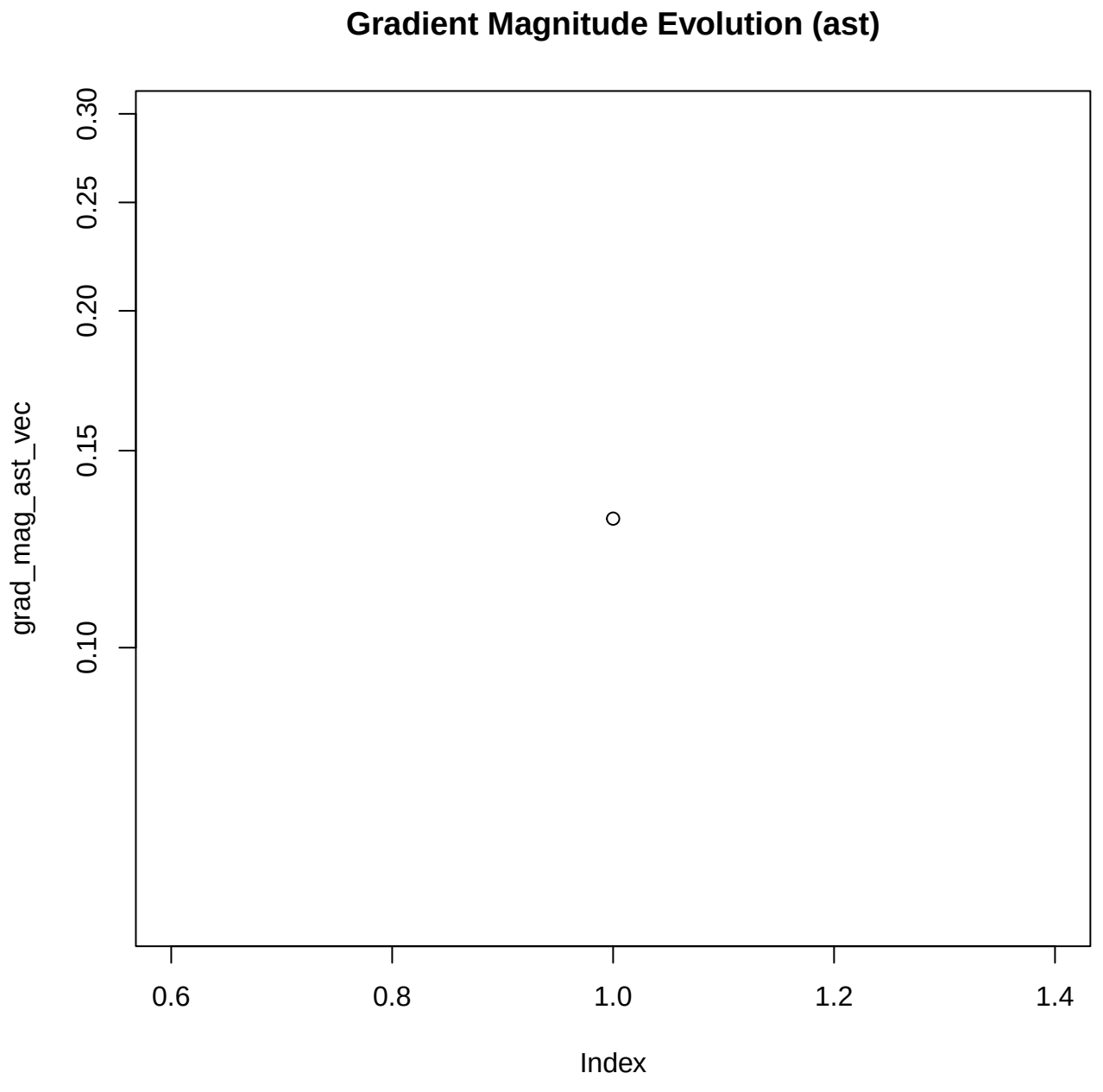
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.15
0.20
0.25
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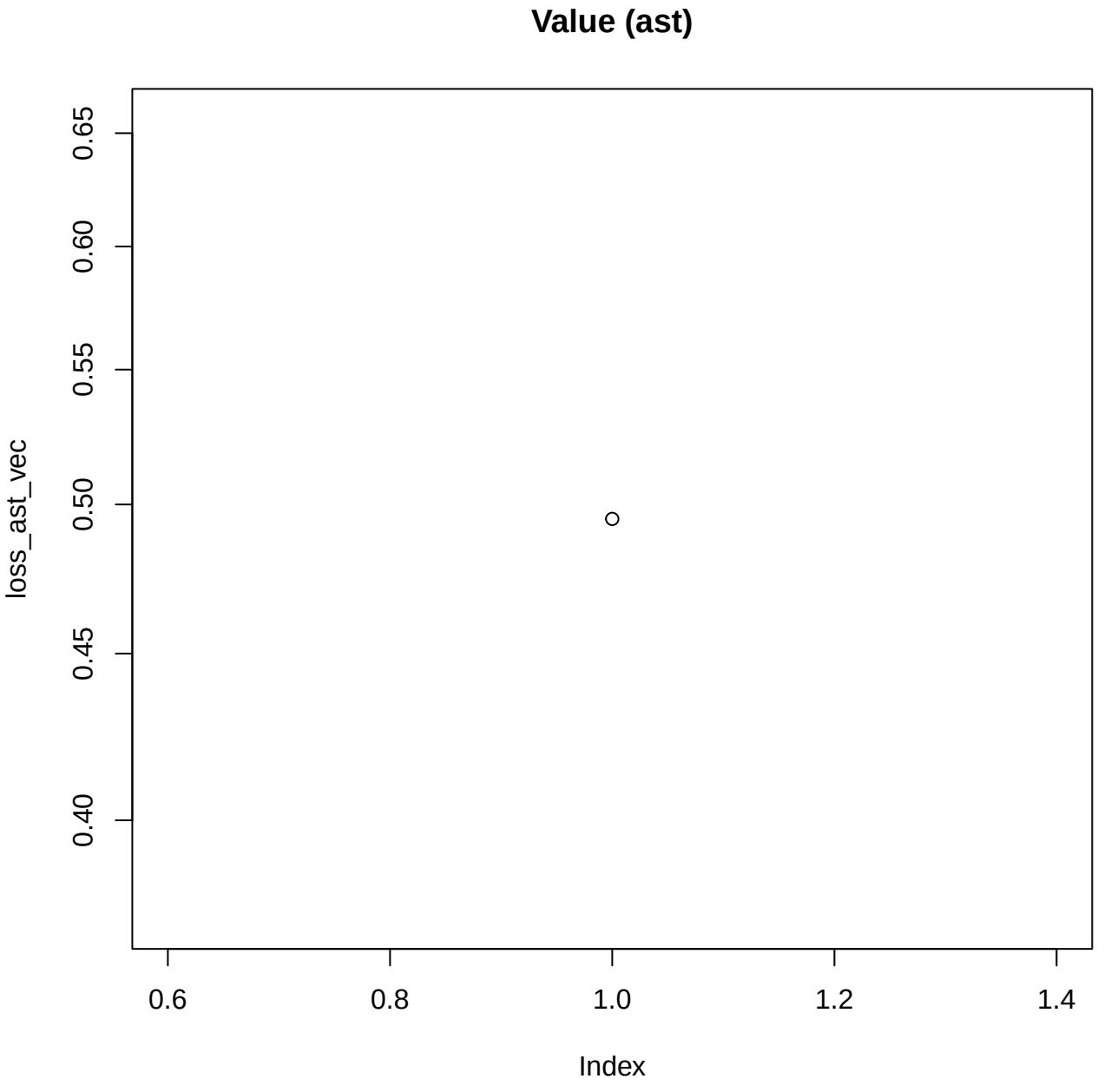
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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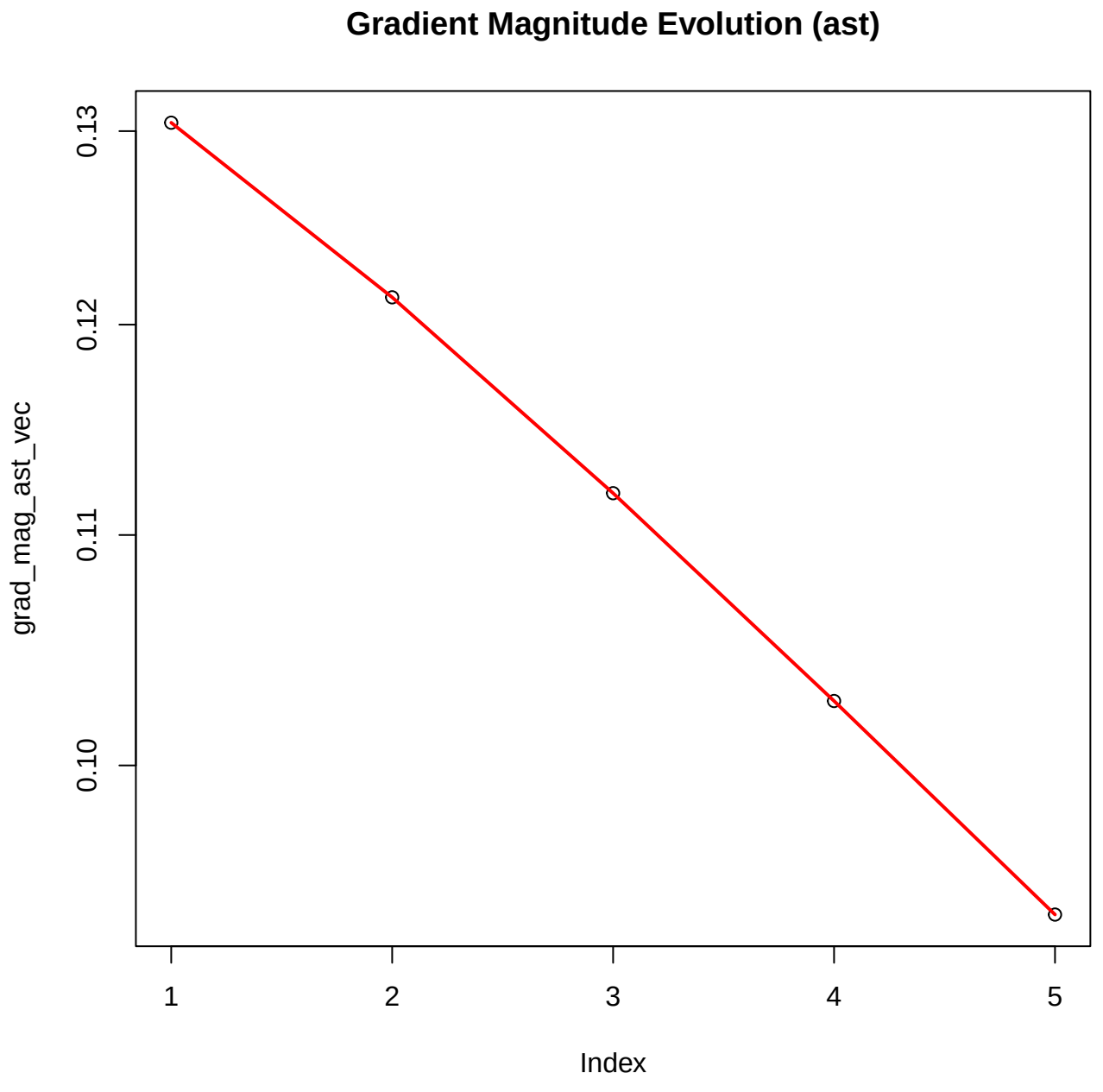
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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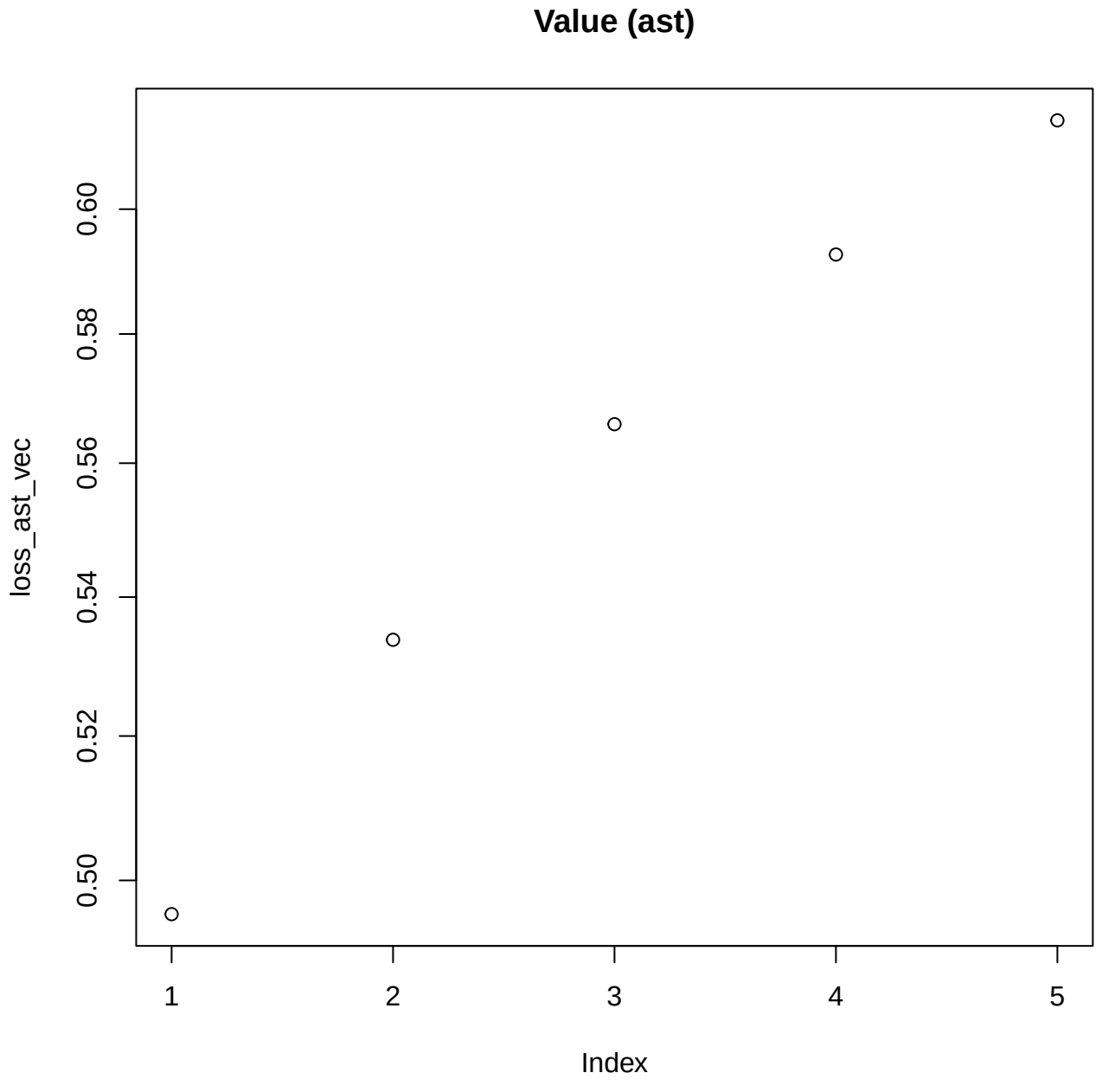
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
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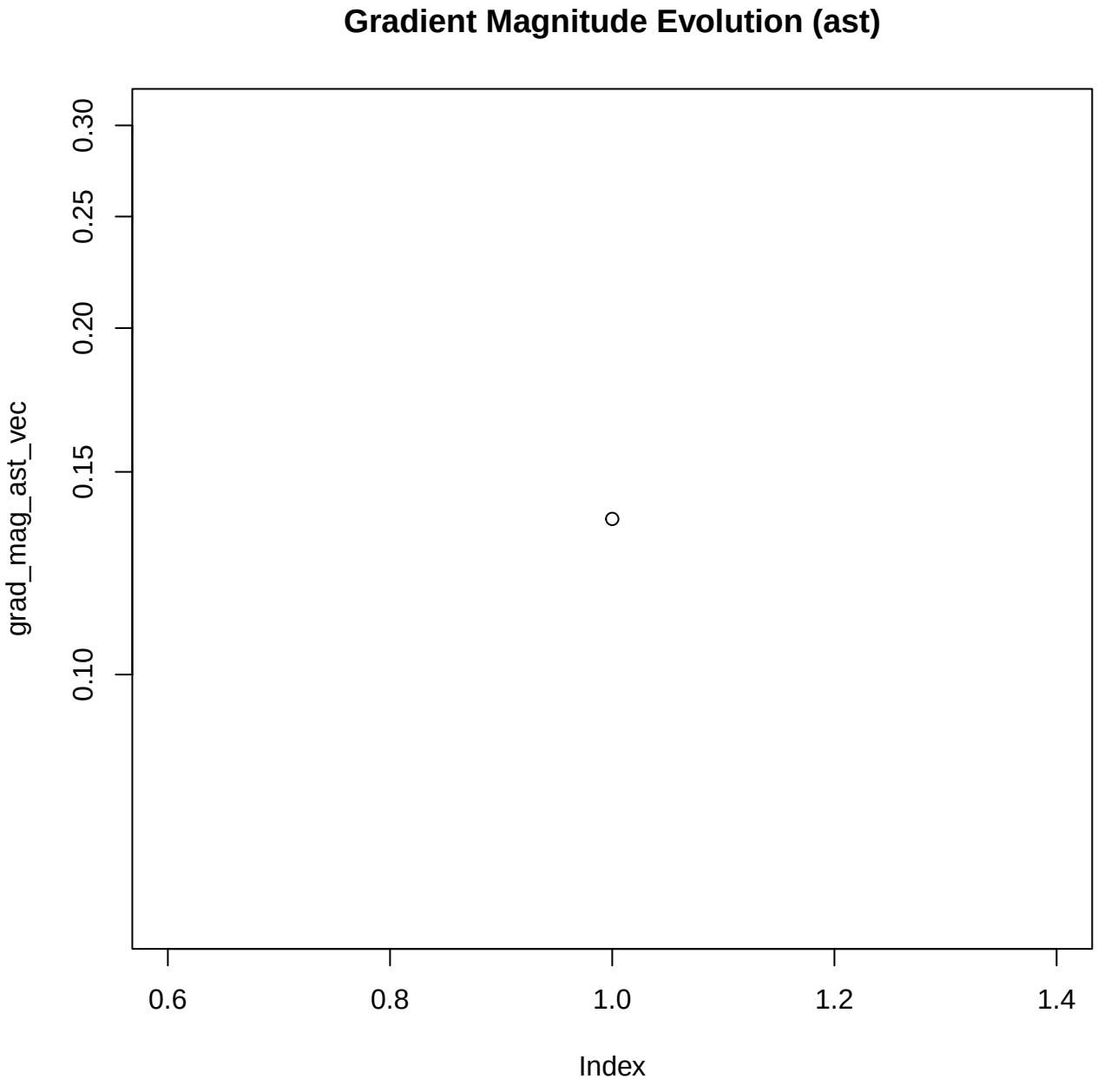
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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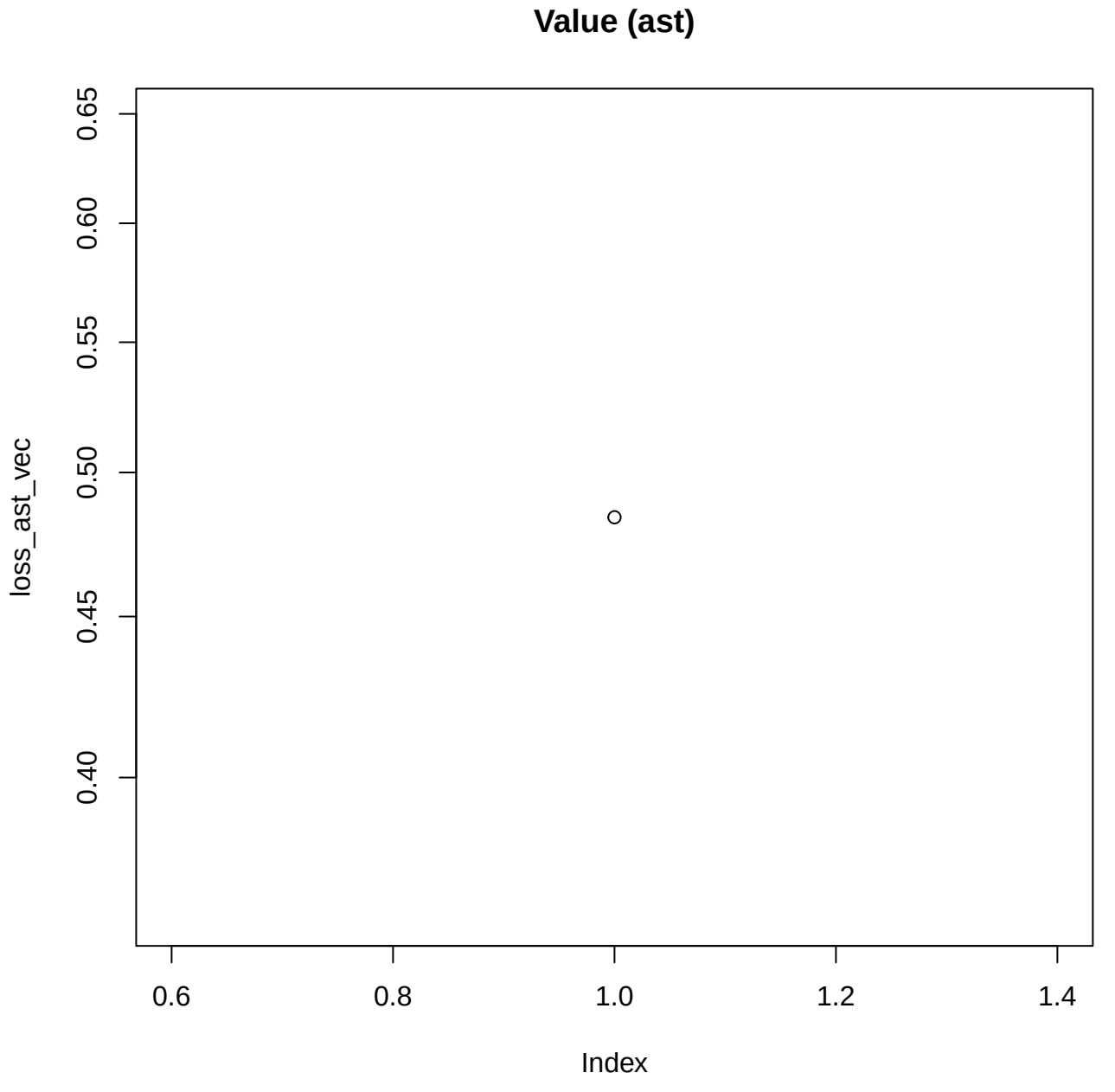
0.15
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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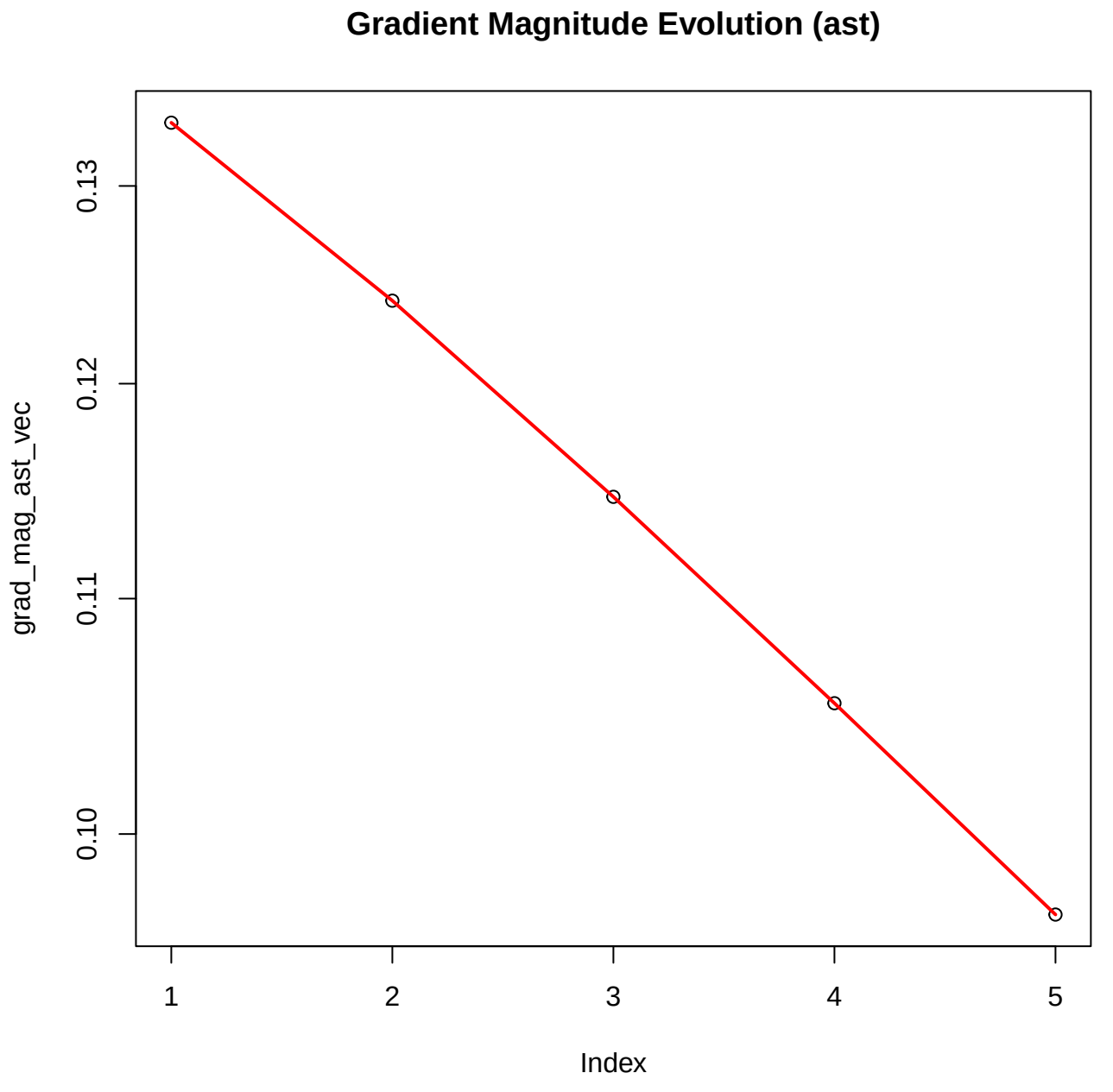
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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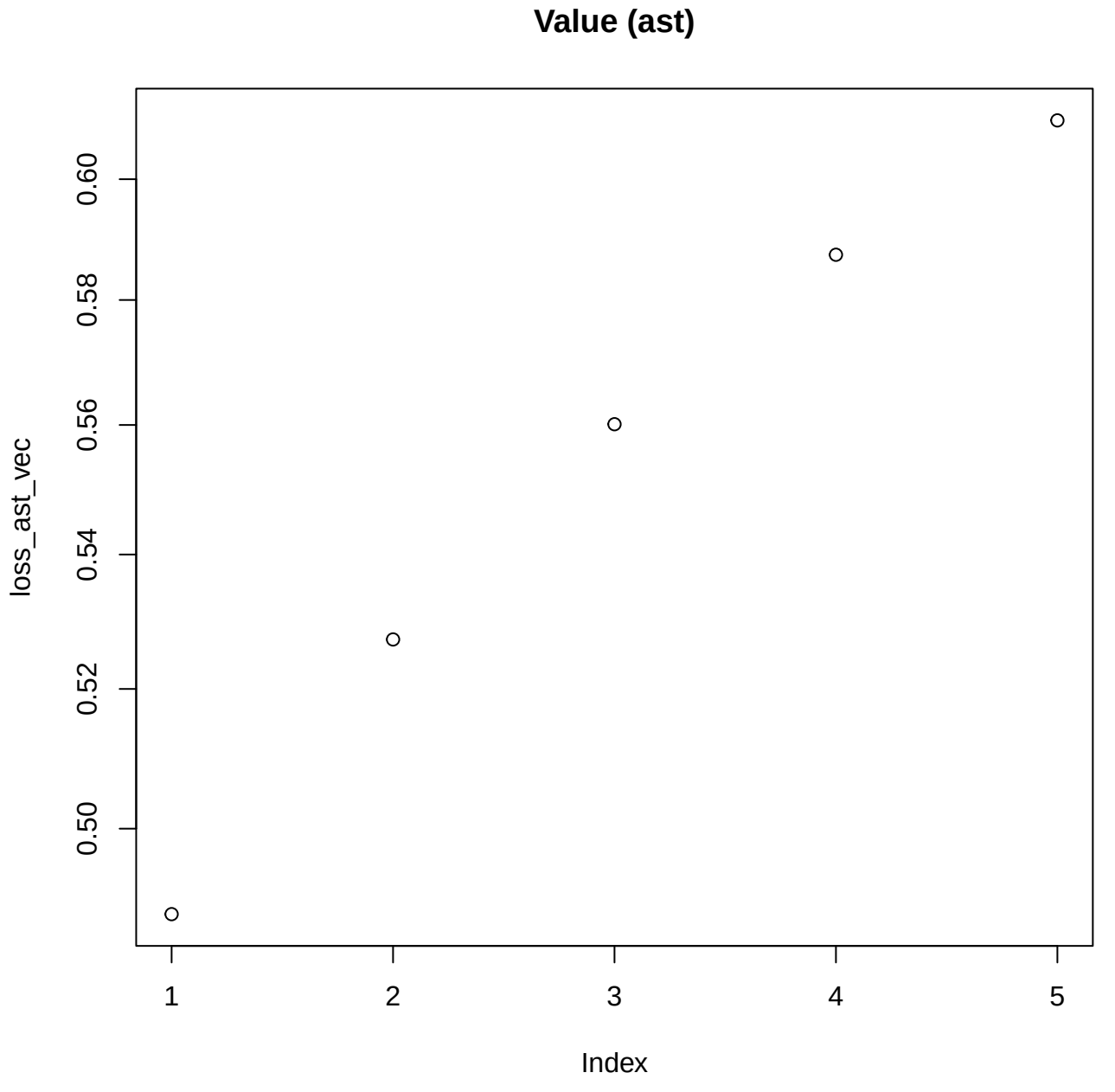
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.52
0.54
0.56
0.58
0.60
1
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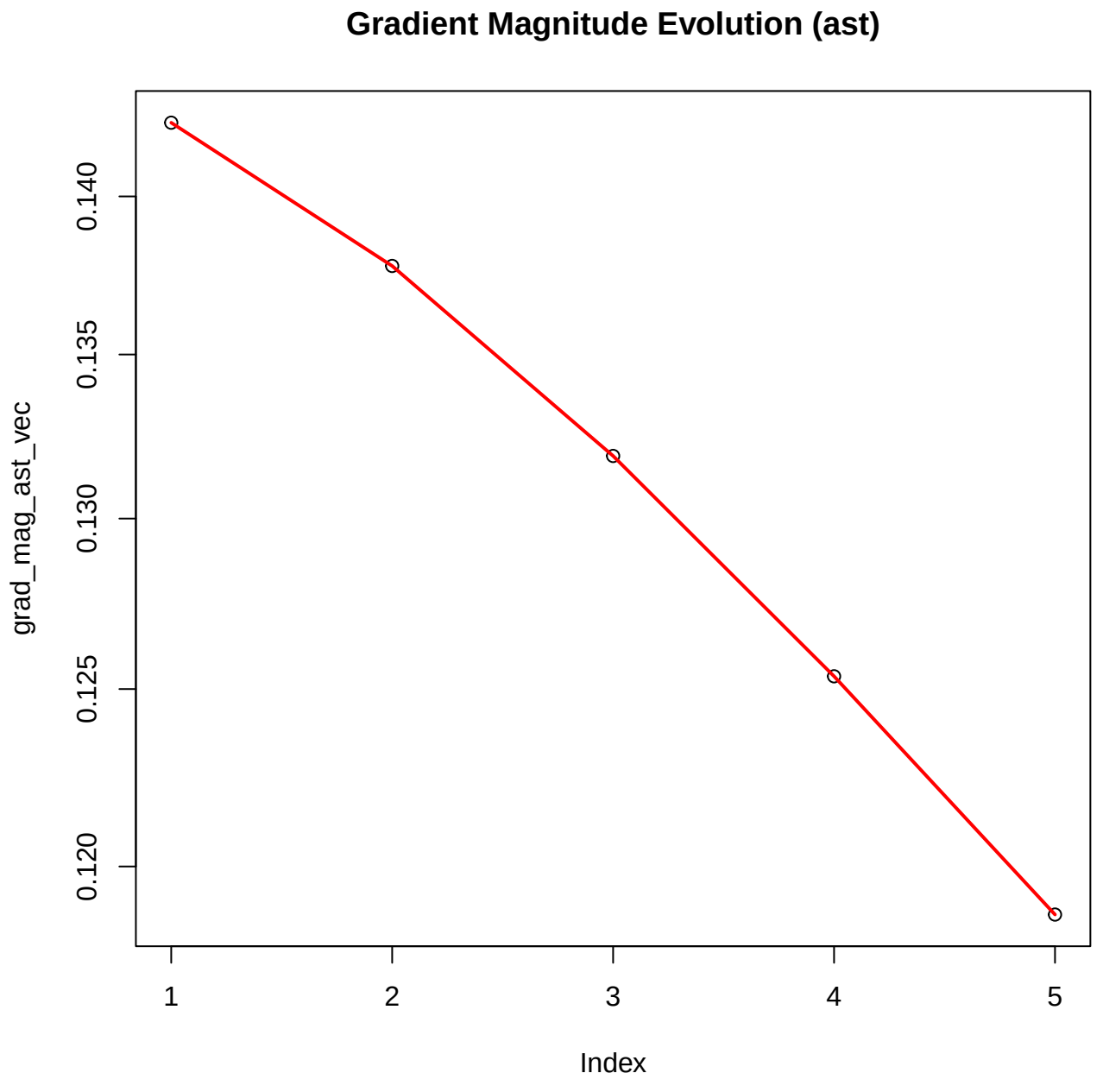
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.125
0.130
0.135
0.140
1
2
3
4
5
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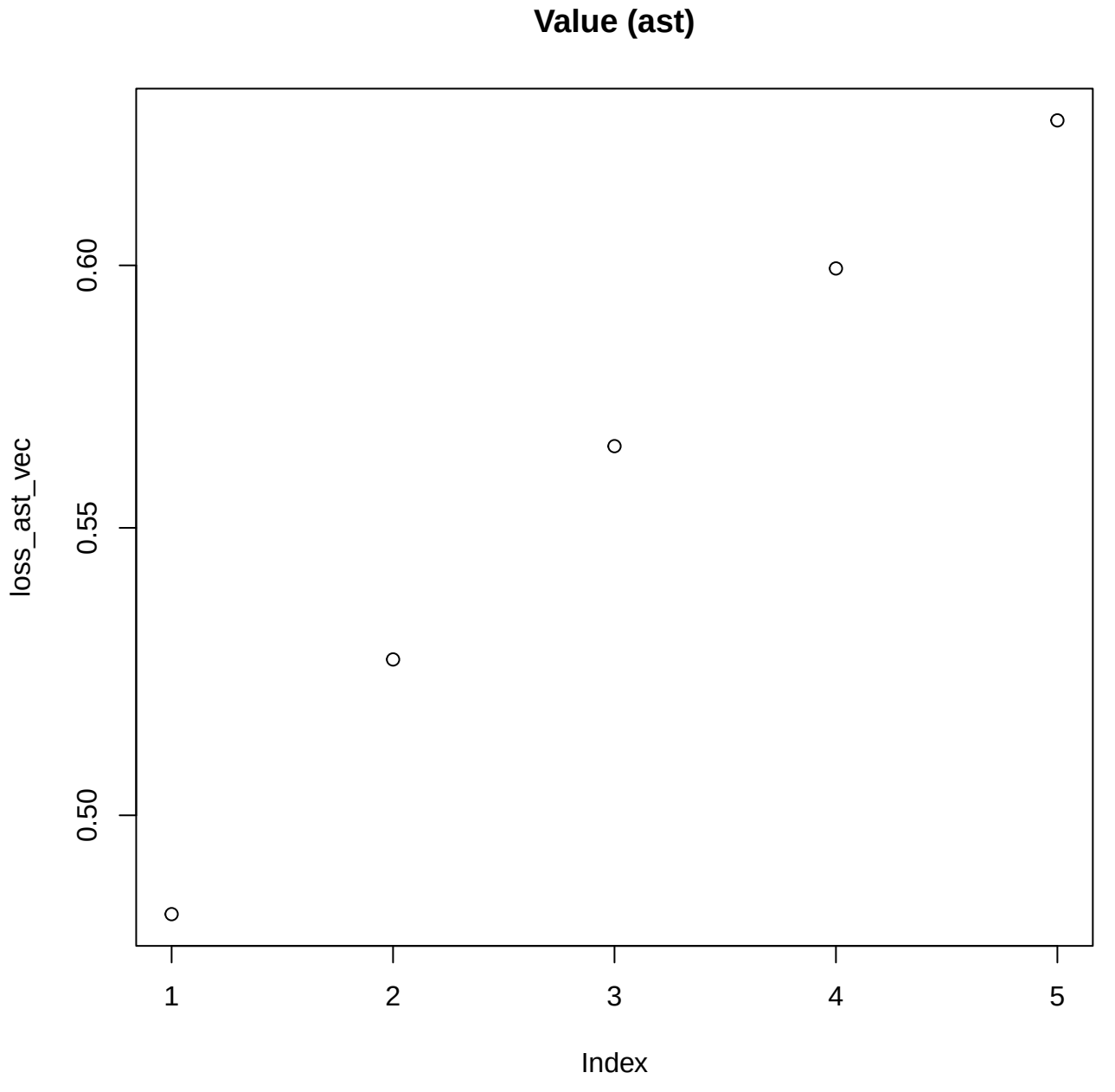
Value (ast)
loss_ast_vec
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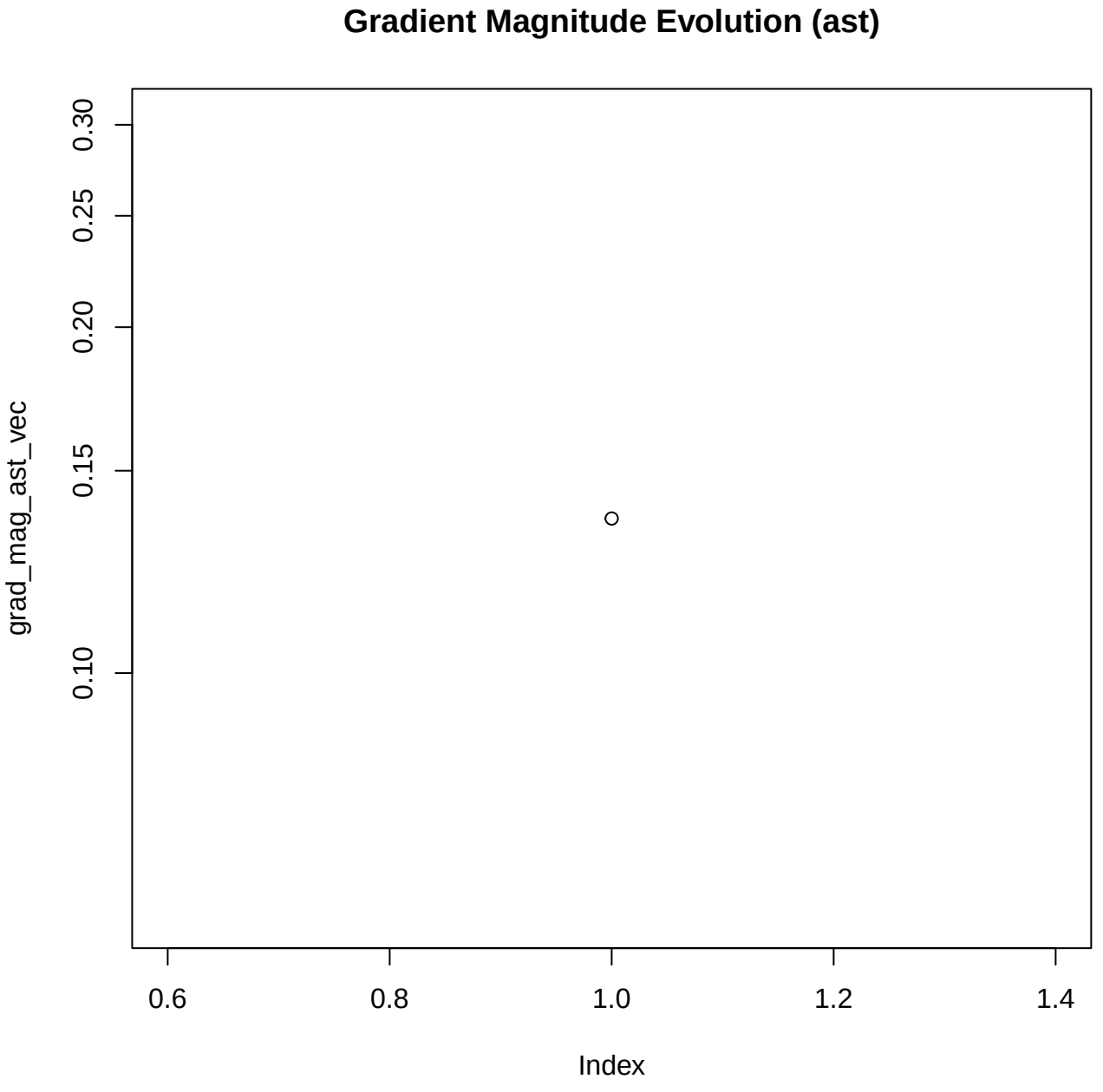
0.55
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.15
0.20
0.25
0.30
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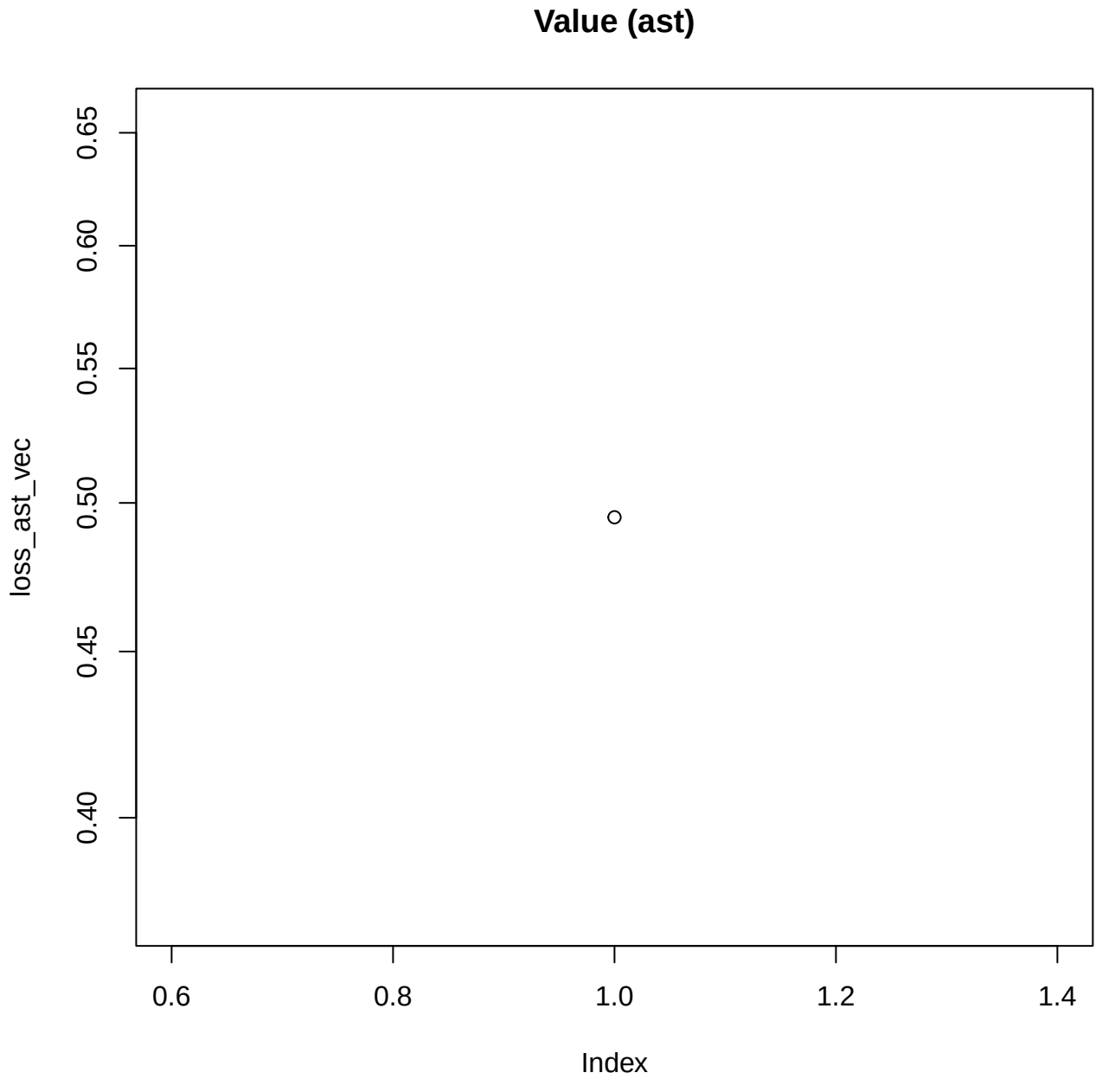
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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0.55
0.50
0.45
0.40
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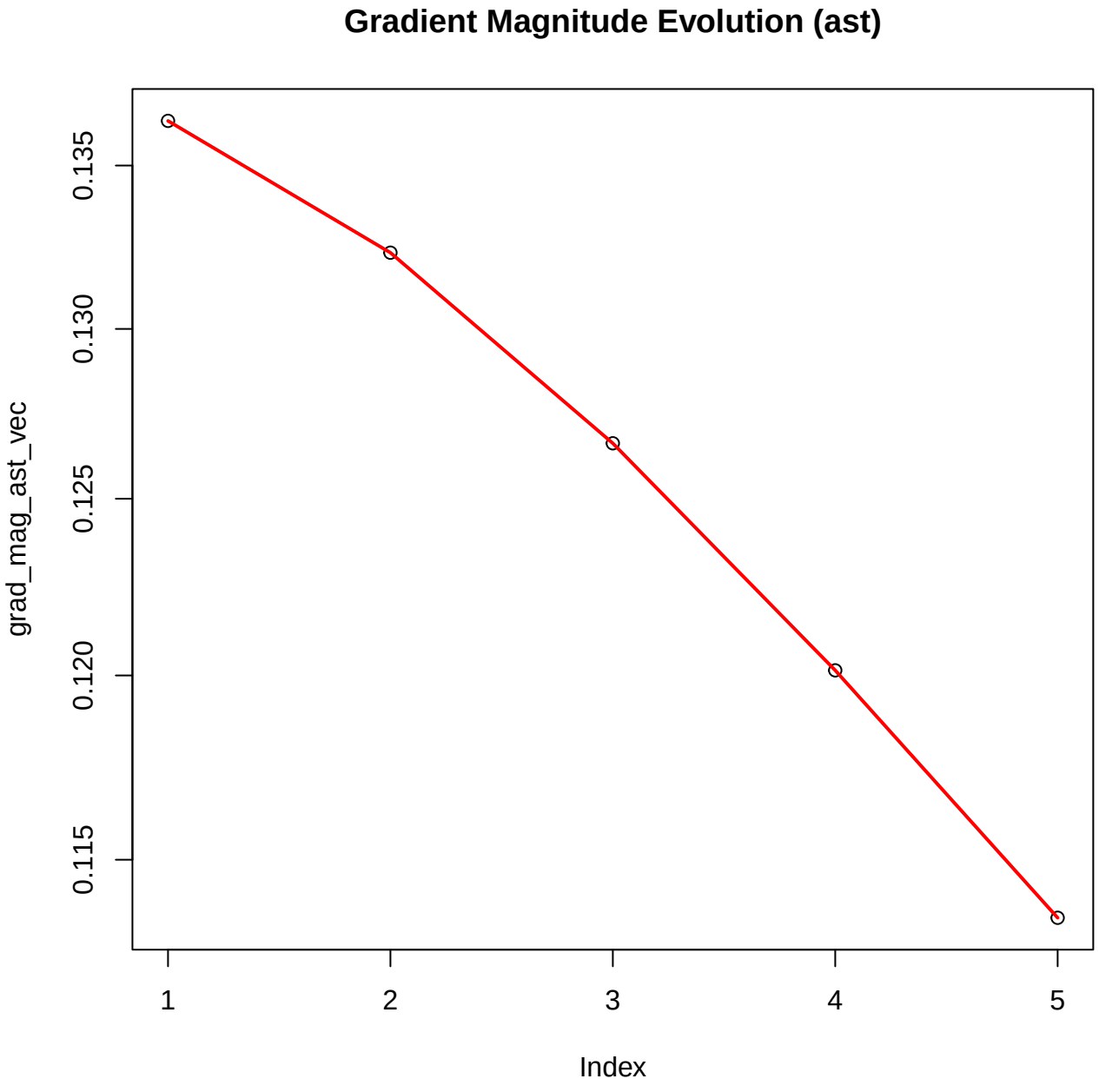
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.120
0.125
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0.135
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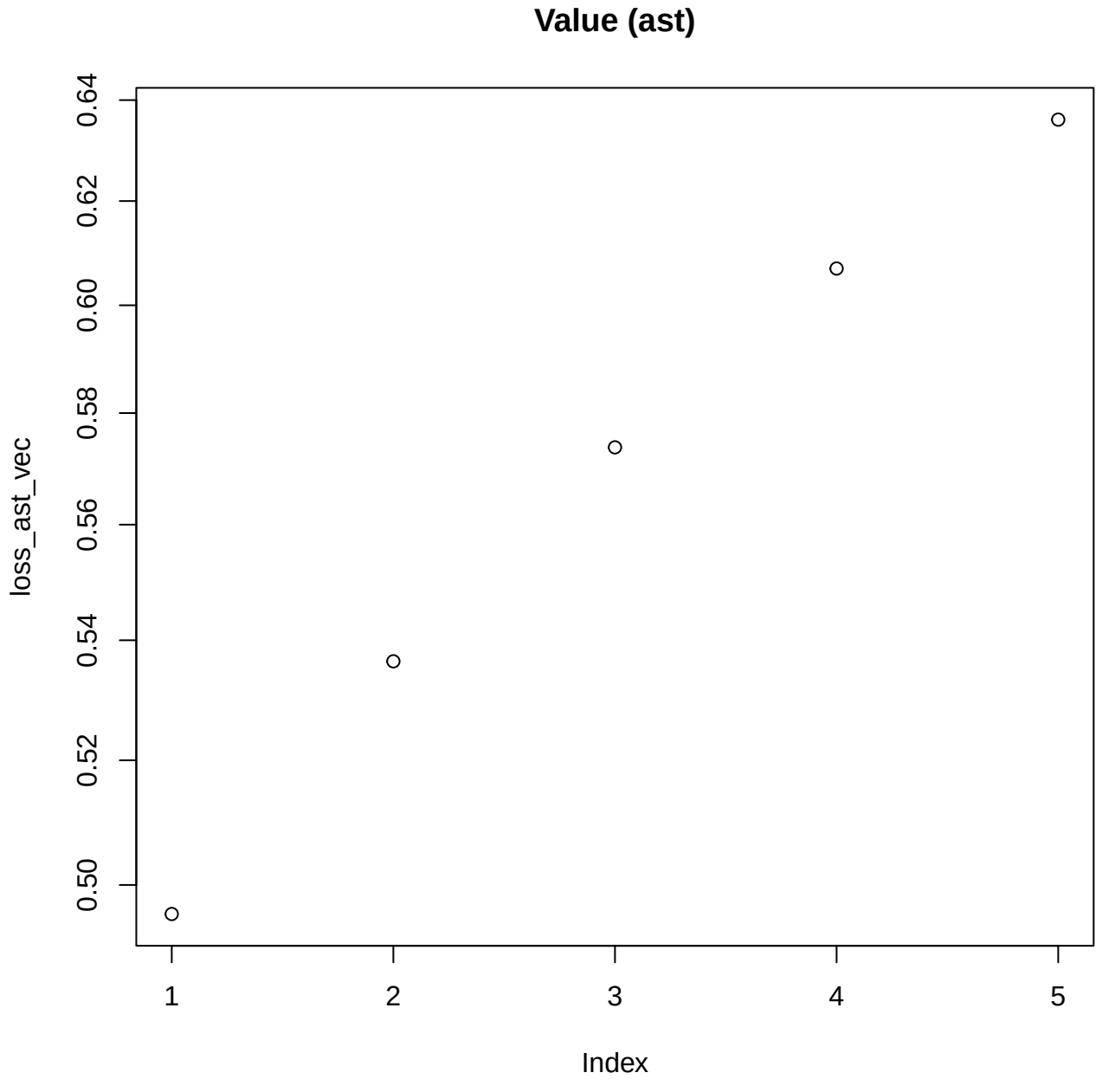
2
3
4
5
Index


Value (ast)
loss_ast_vec
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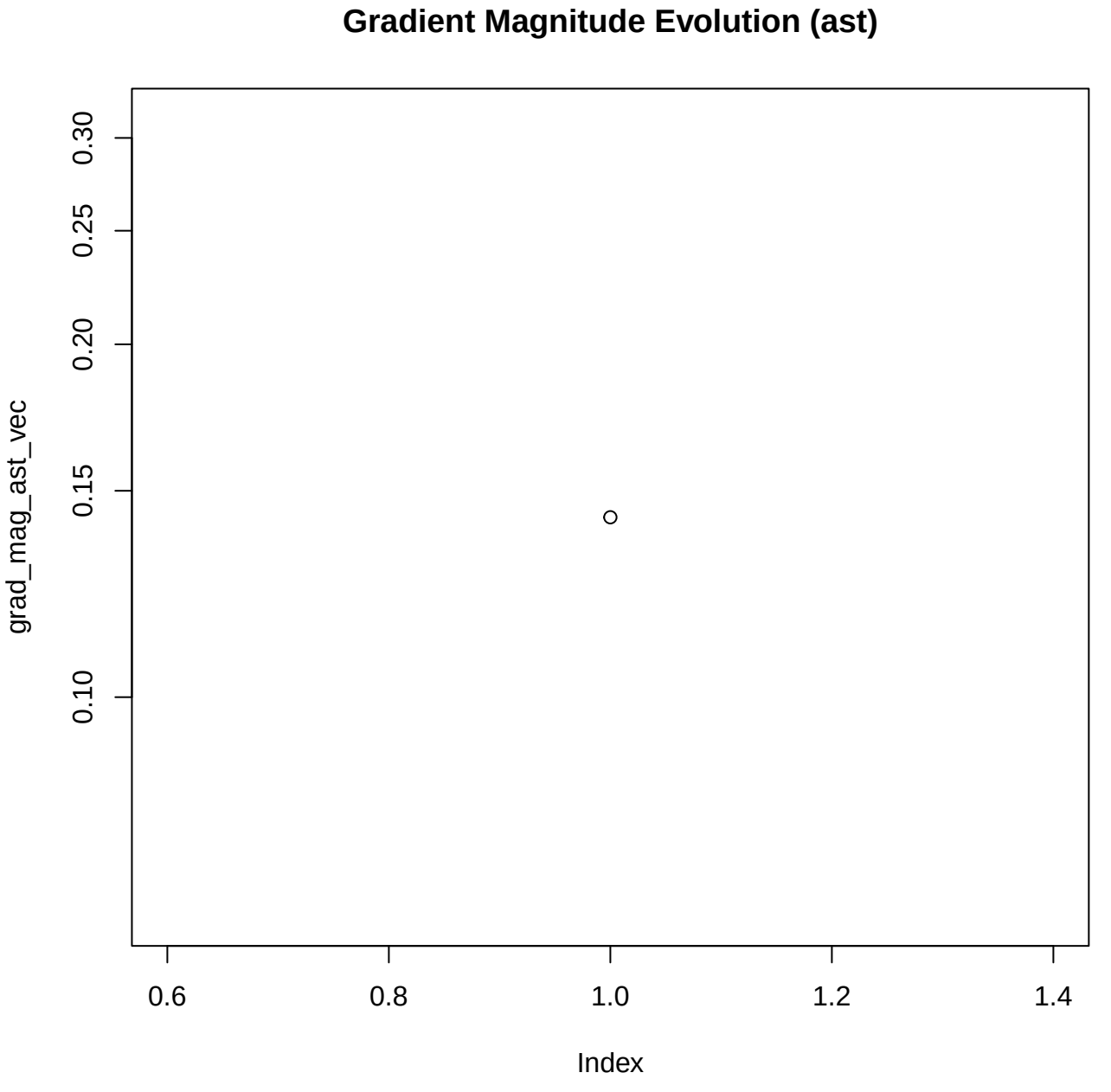
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.30
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0.8
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1.2
1.4
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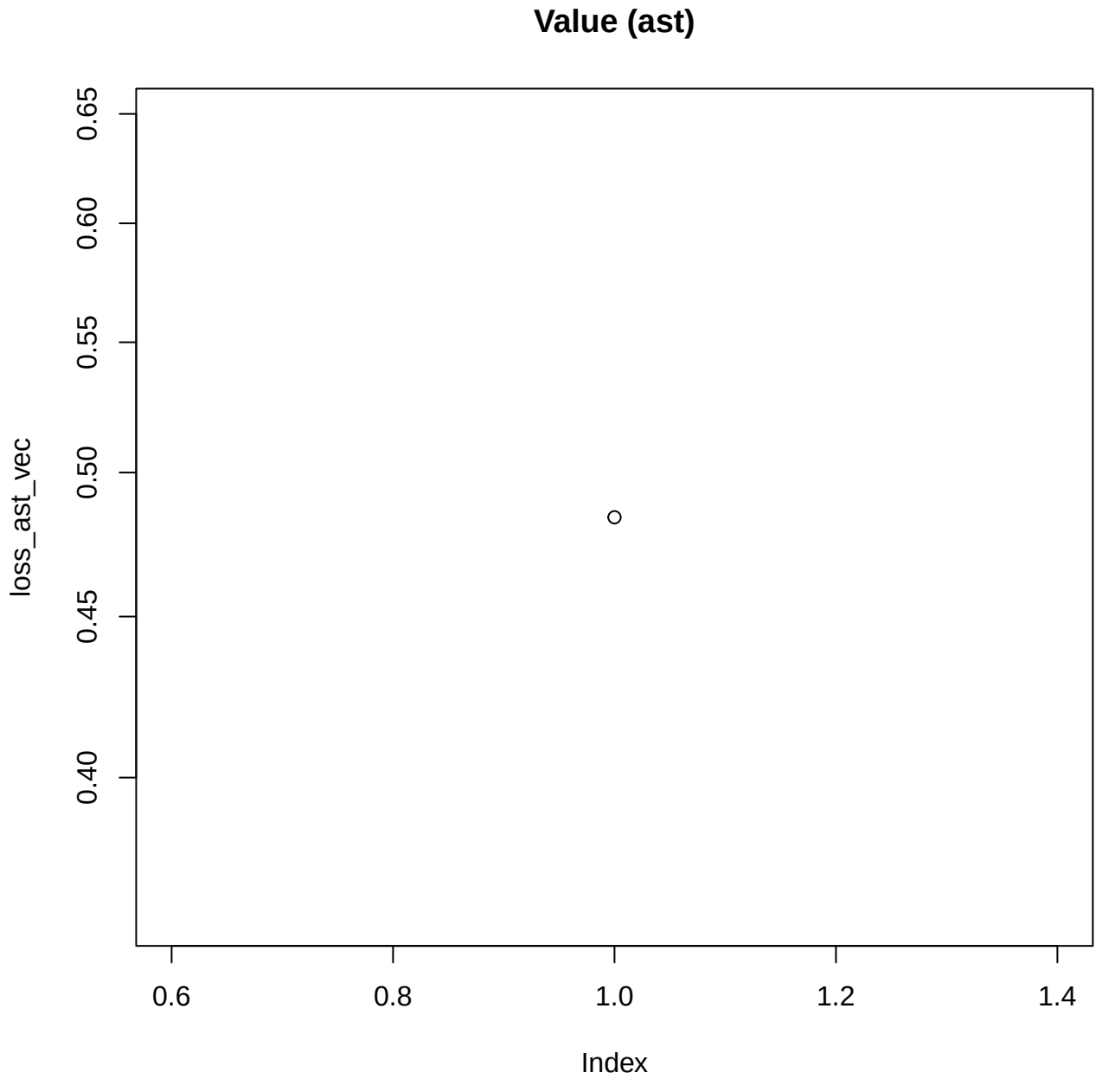
Value (ast)
loss_ast_vec
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0.8
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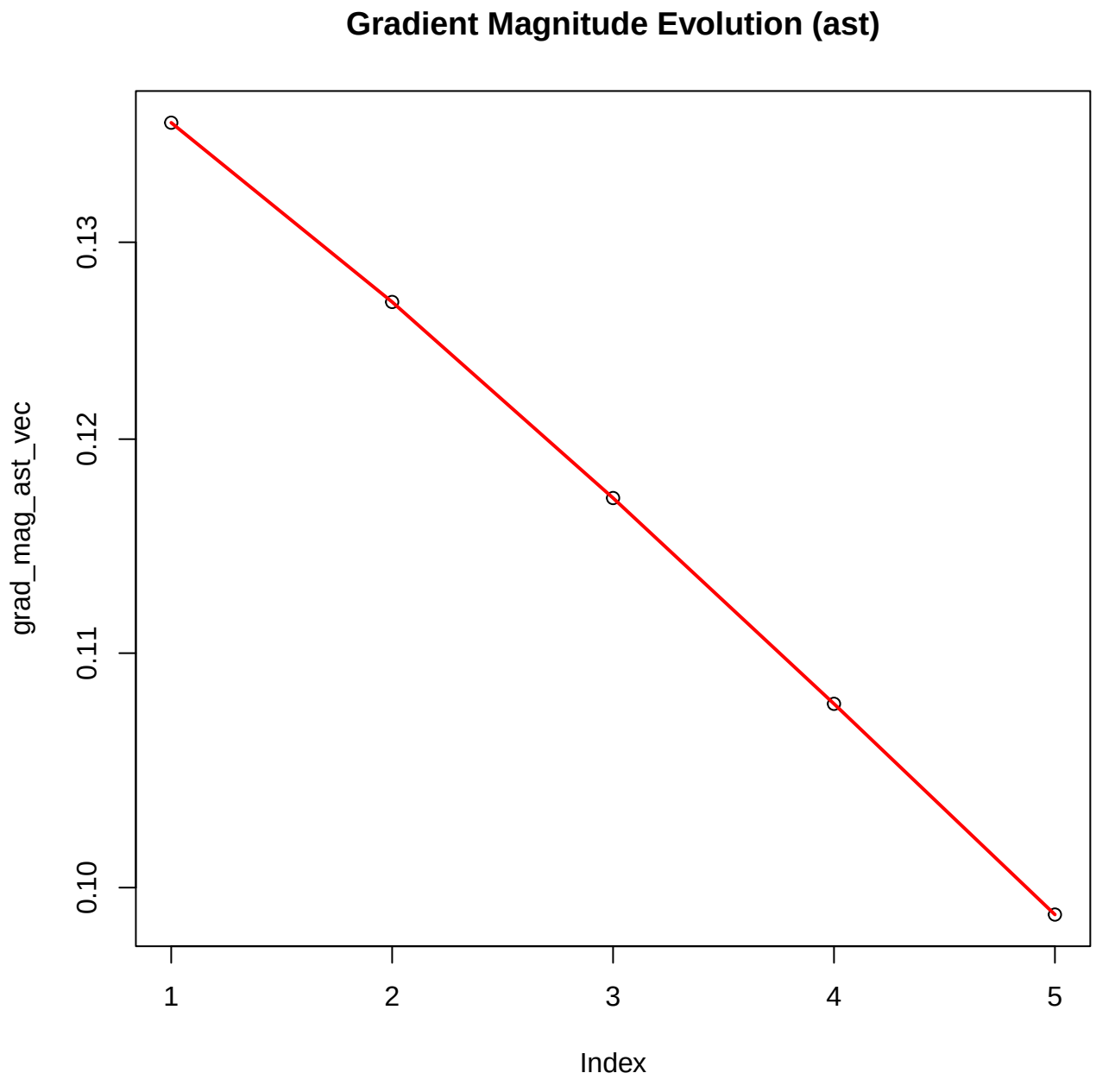
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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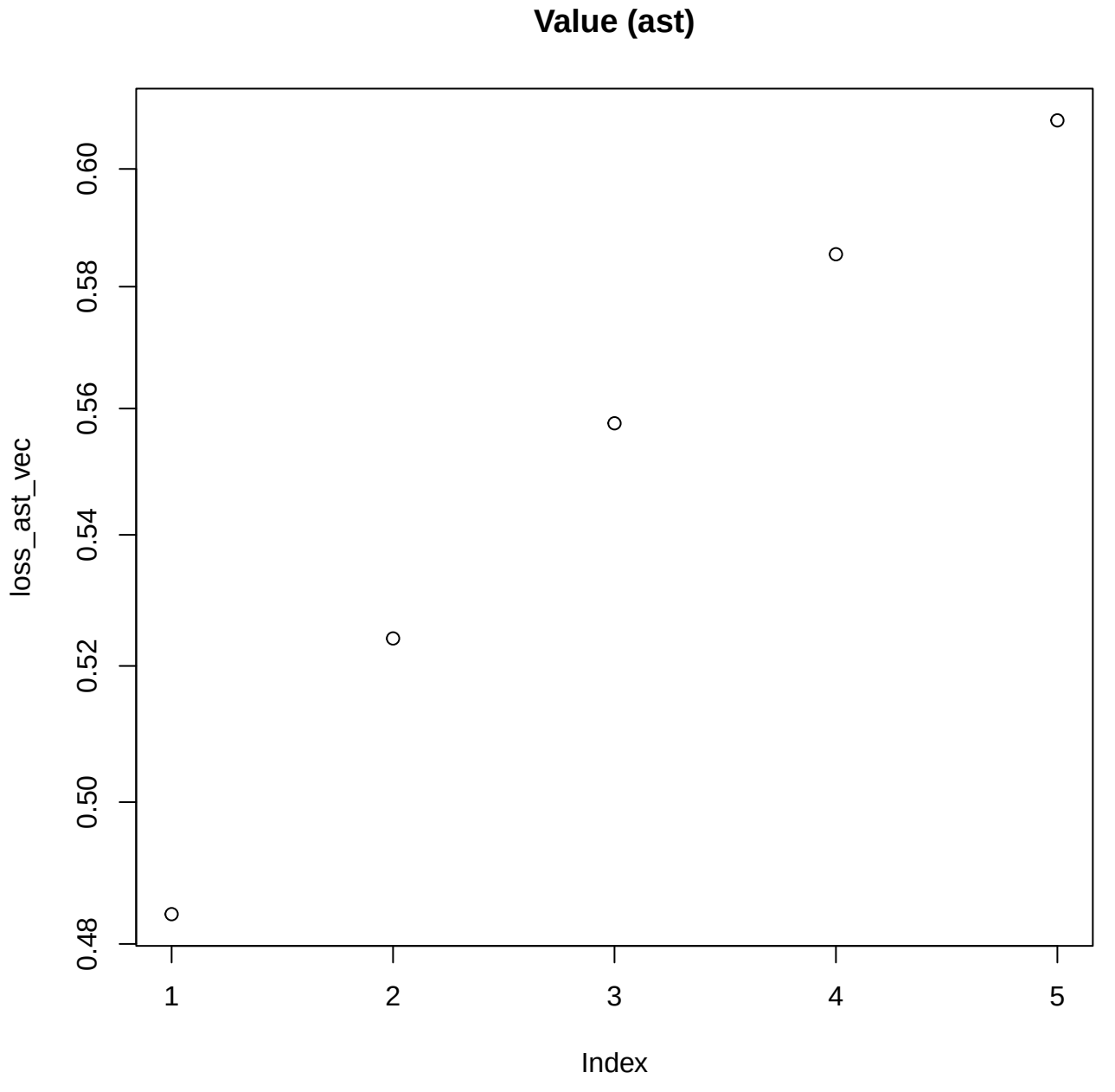
Value (ast)
loss_ast_vec
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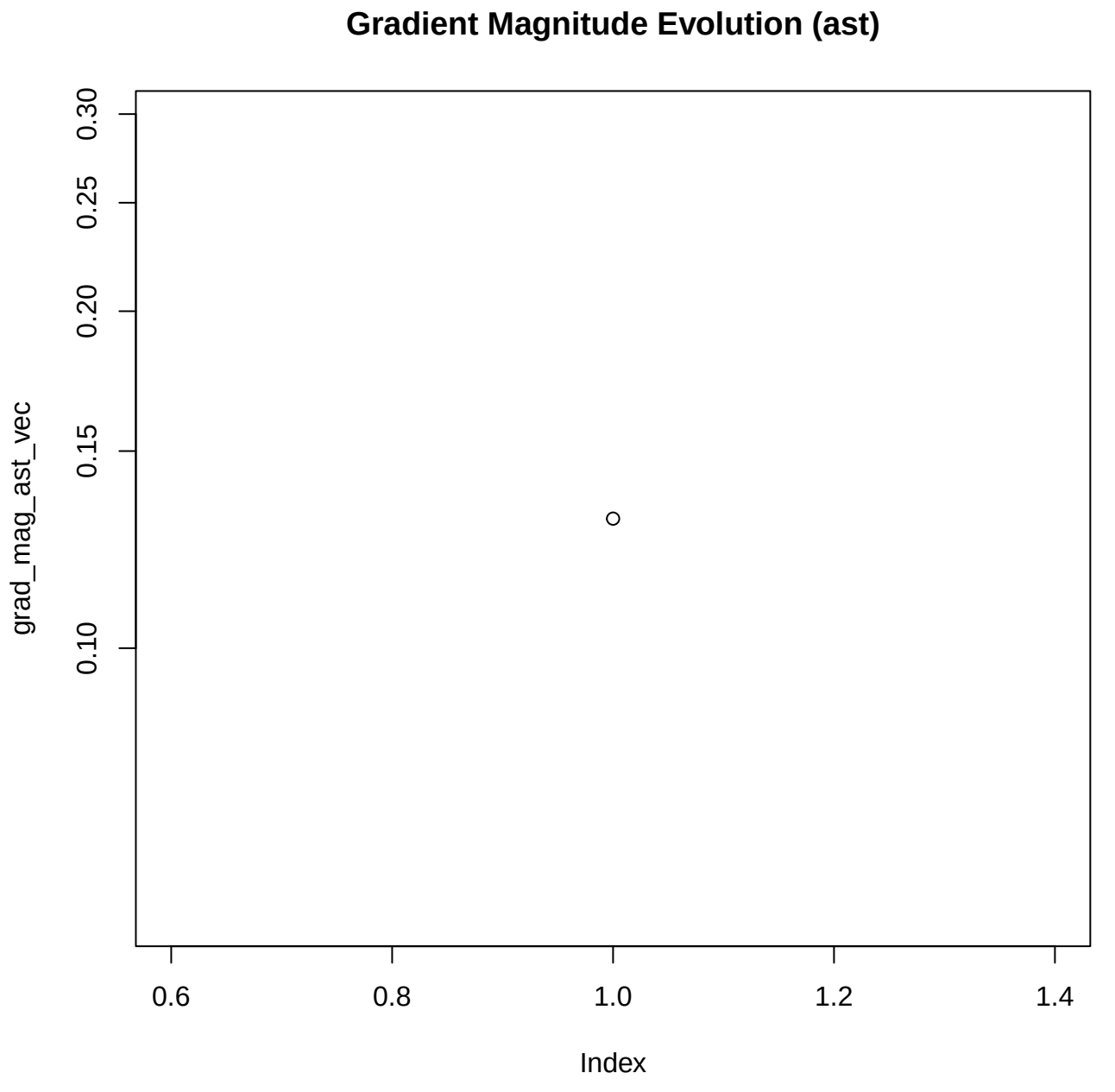
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.20
0.25
0.30
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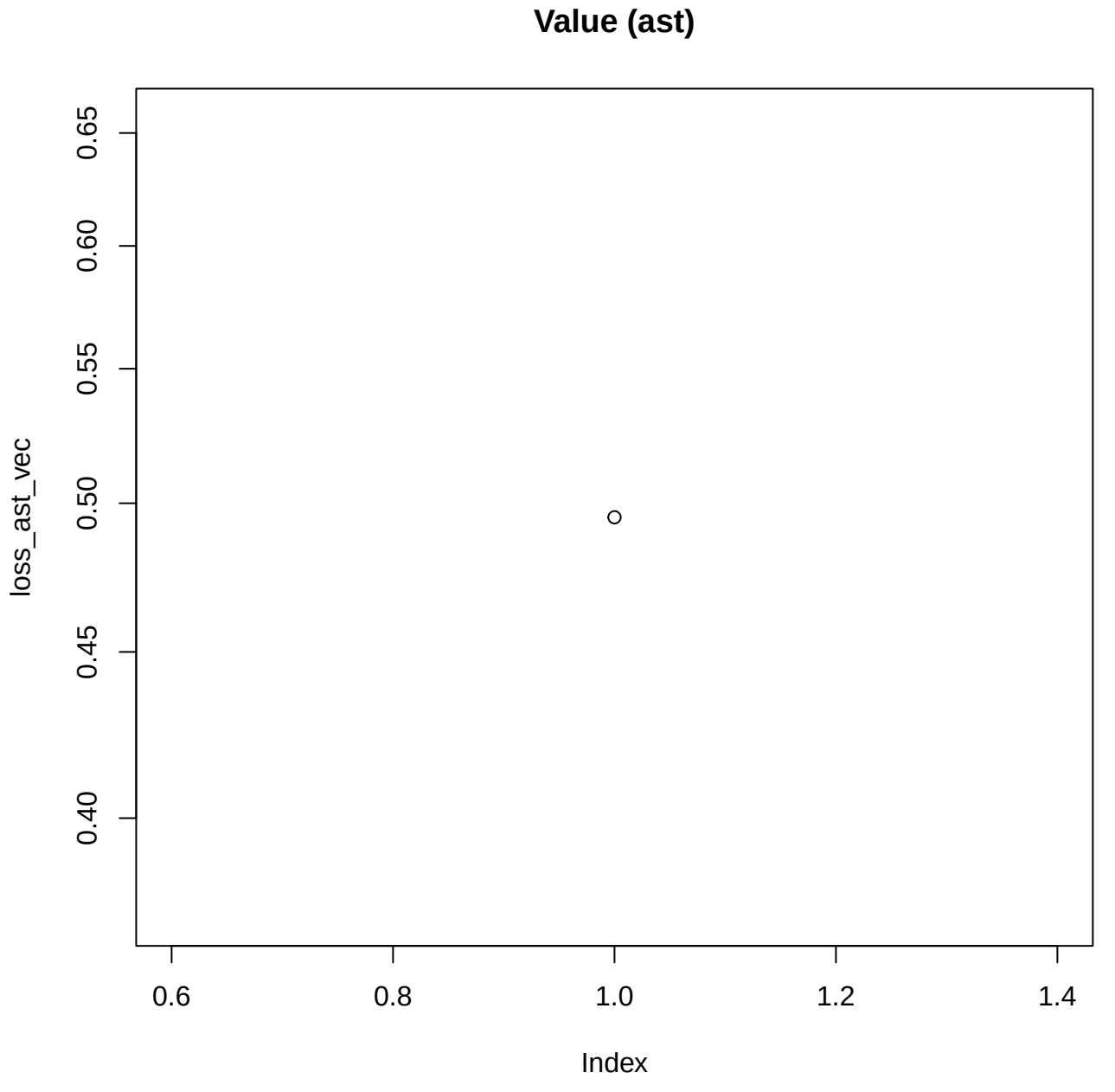
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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0.50
0.55
0.60
0.65
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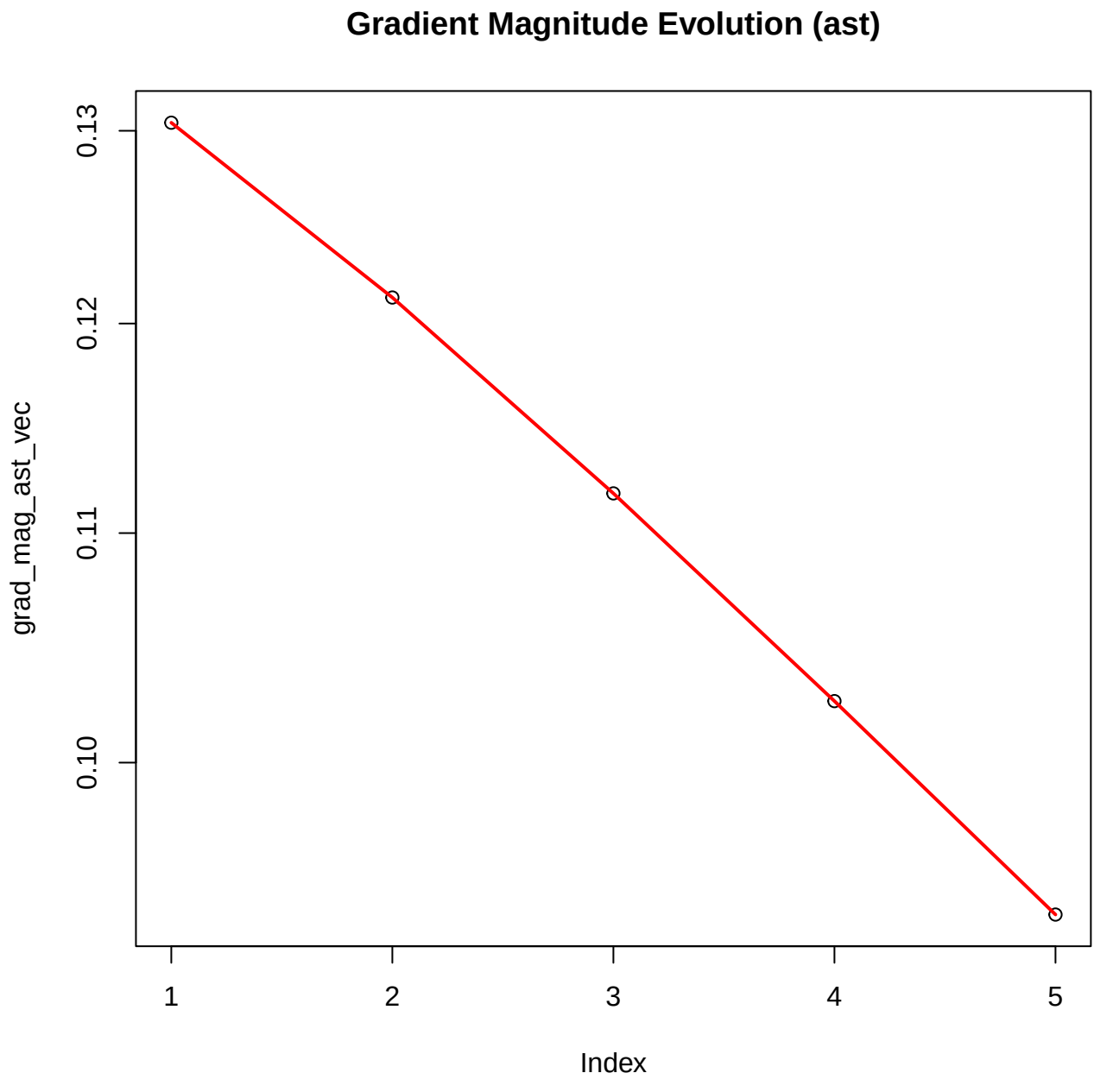
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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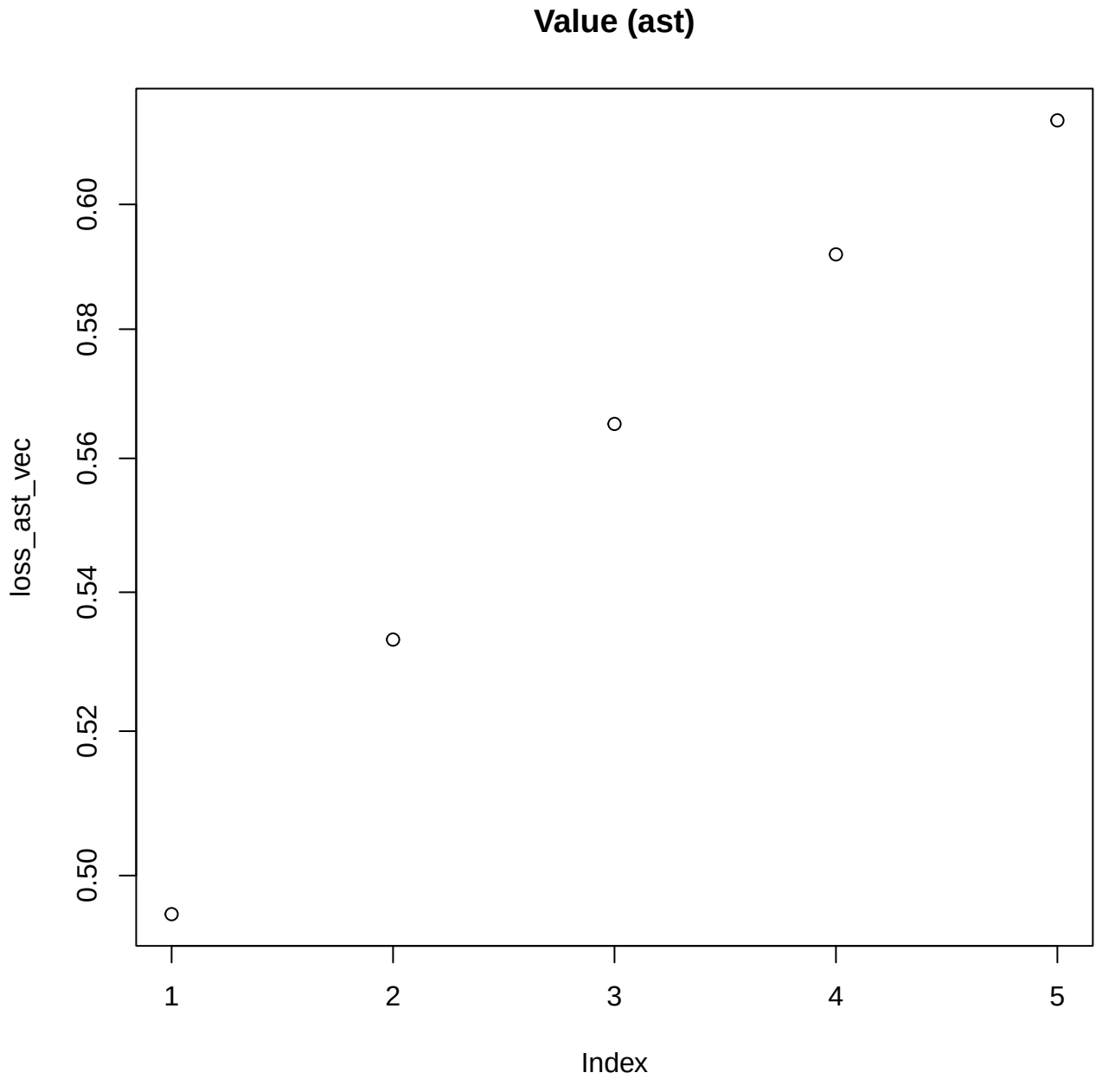
0.11
0.12
0.13
1
2
3
4
5
Index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loss_ast_vec
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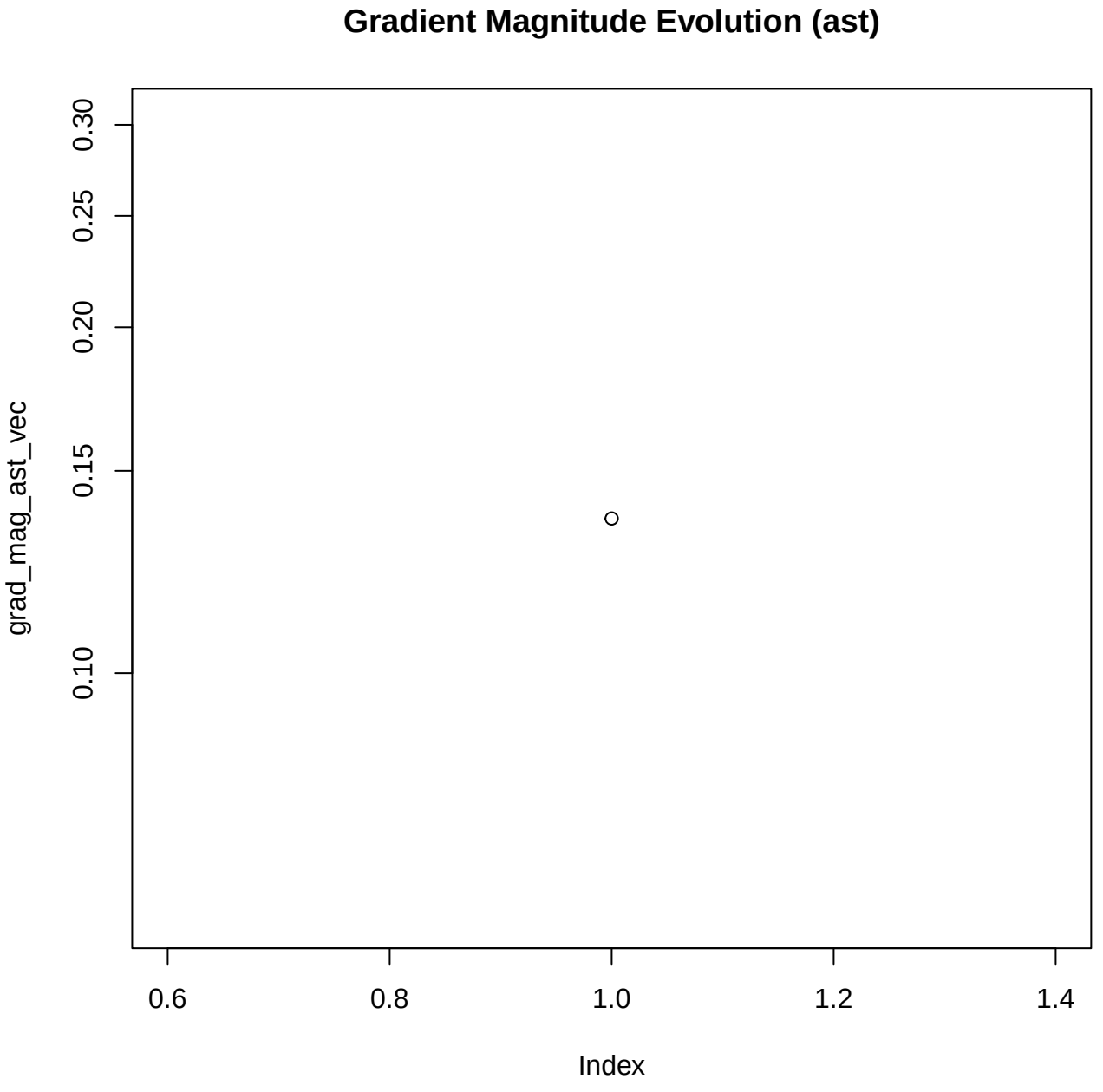
0.54
0.56
0.58
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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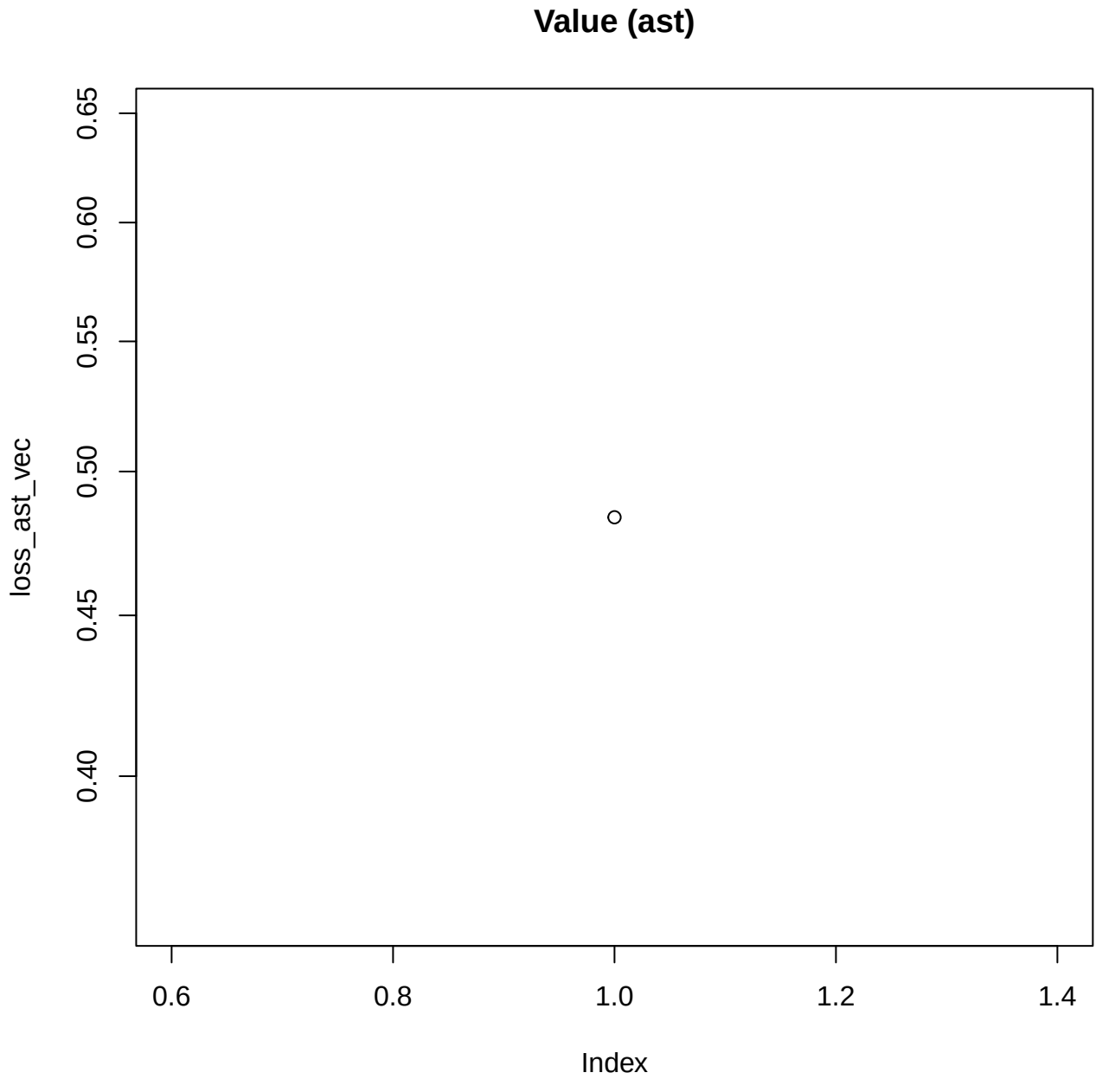
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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0.50
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0.40
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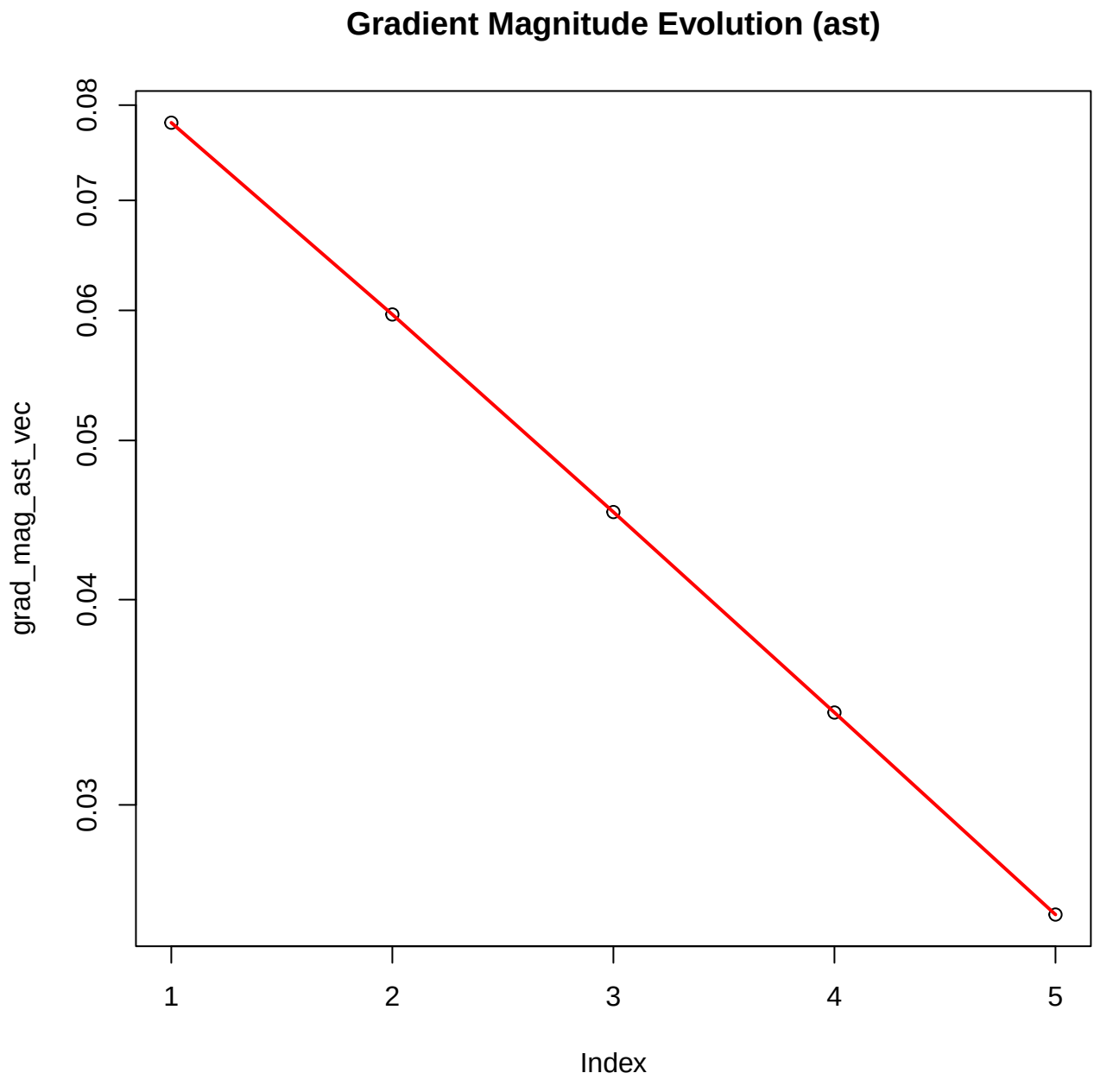
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
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0.03
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2
3
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5
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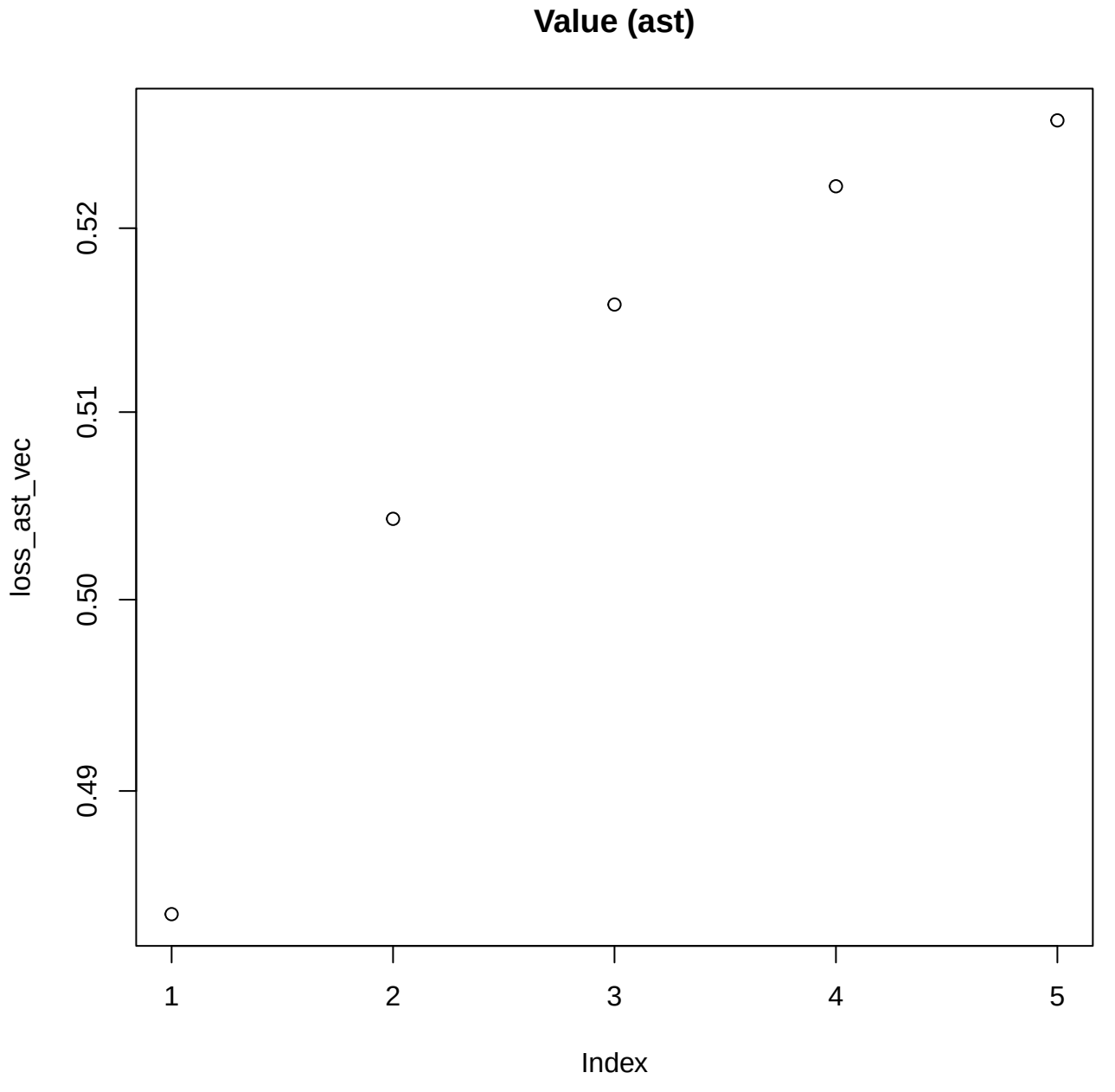
Value (ast)
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0.52
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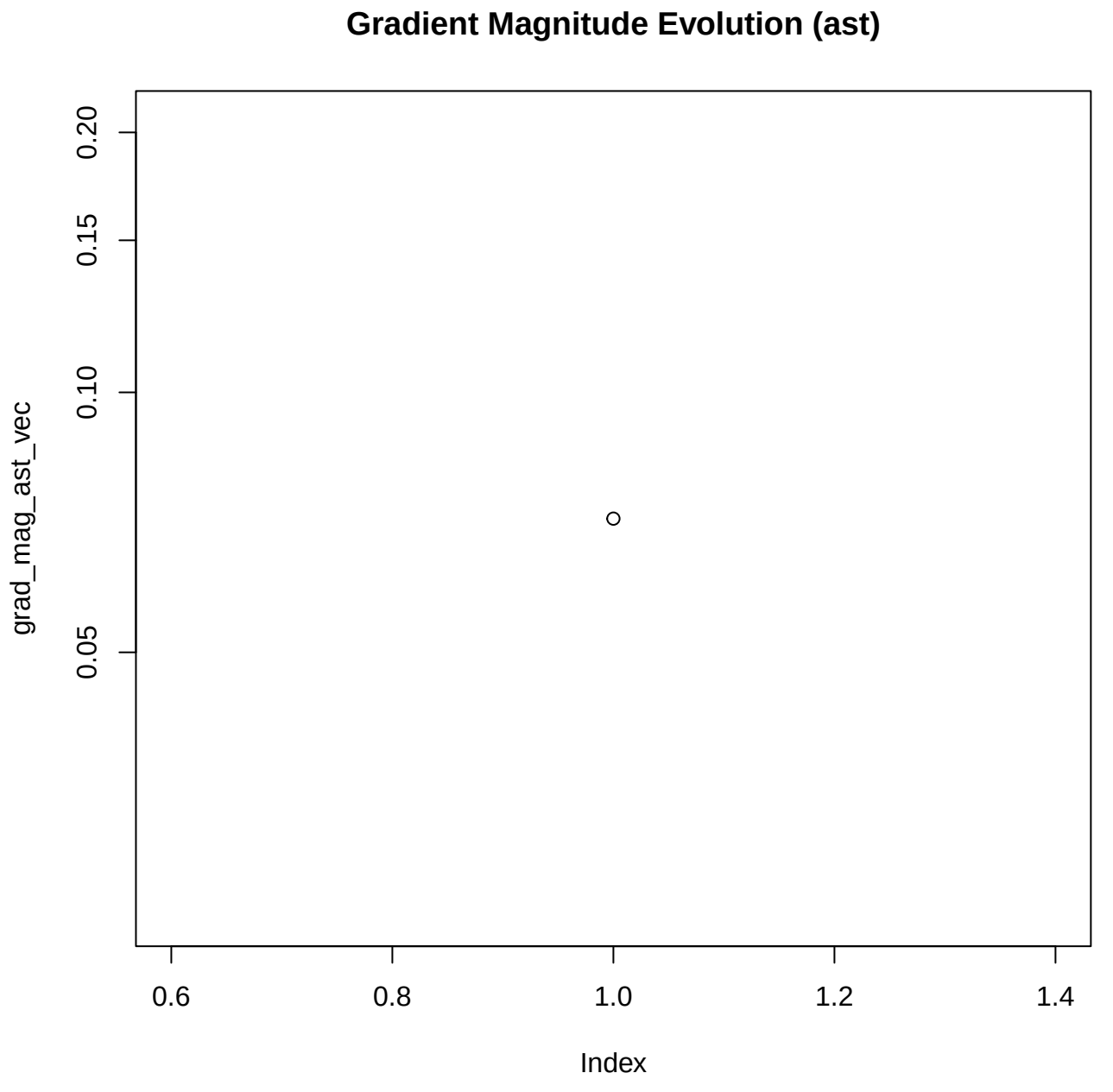
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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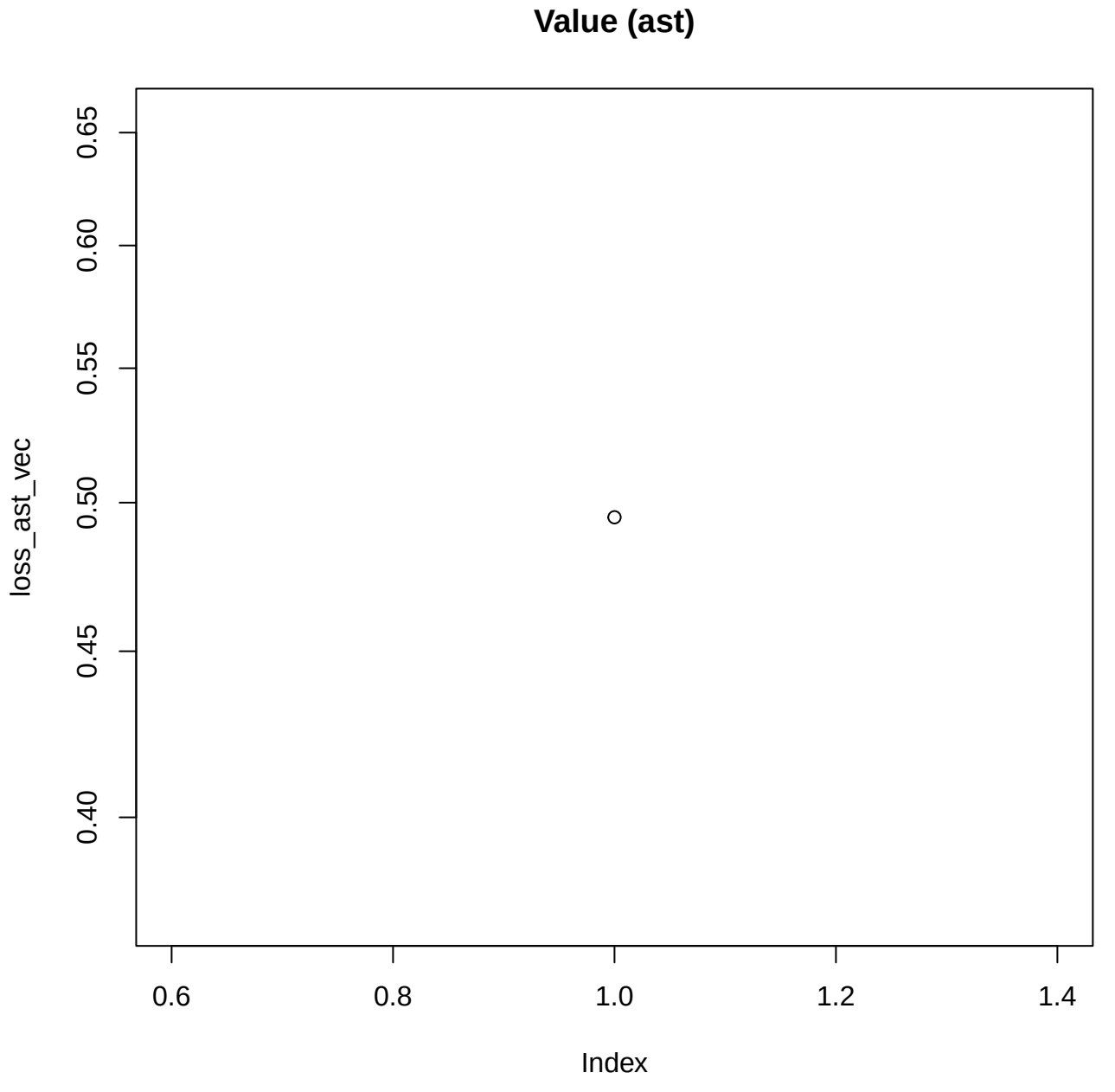
0.10
0.15
0.20
0.6
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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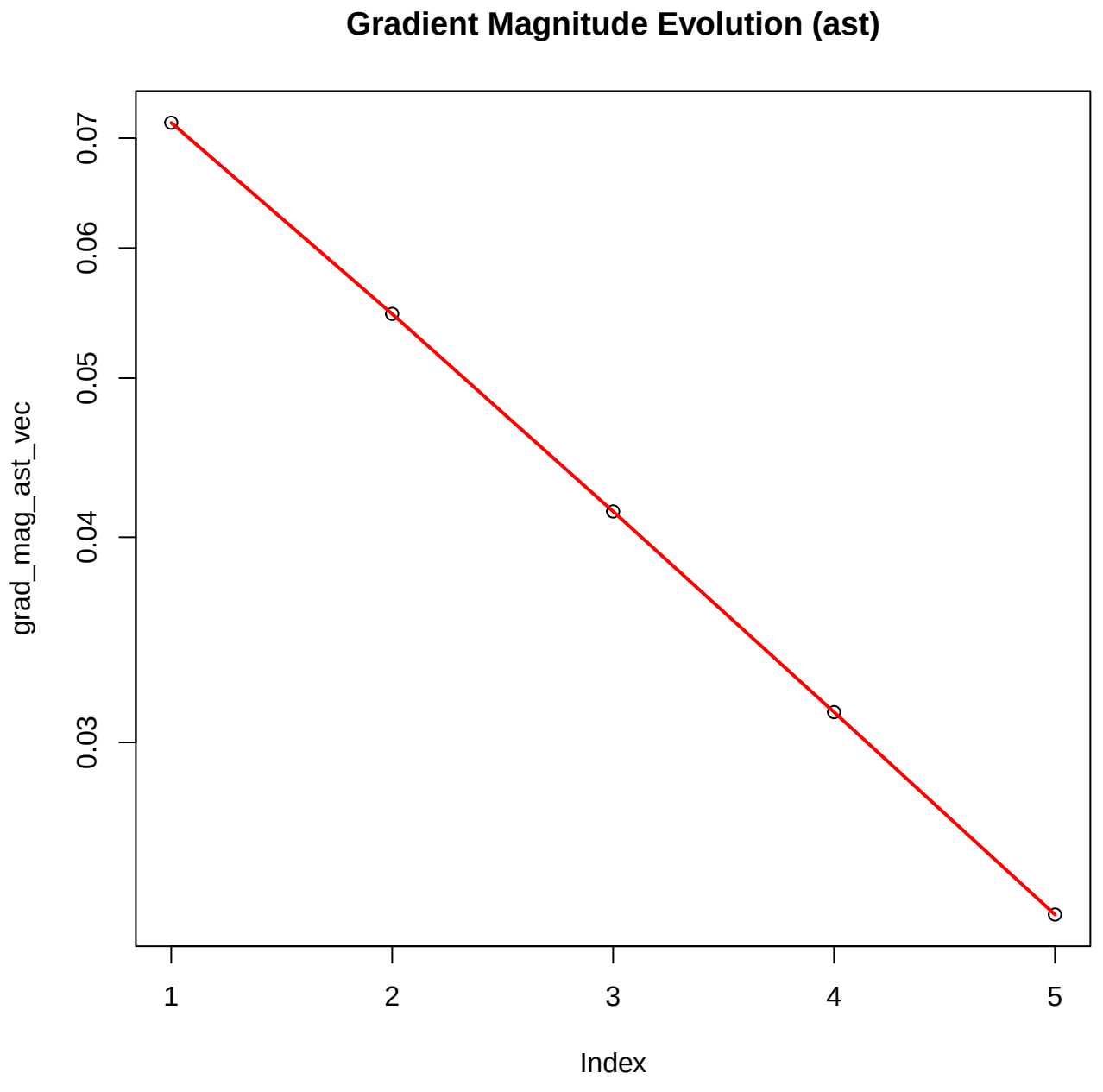
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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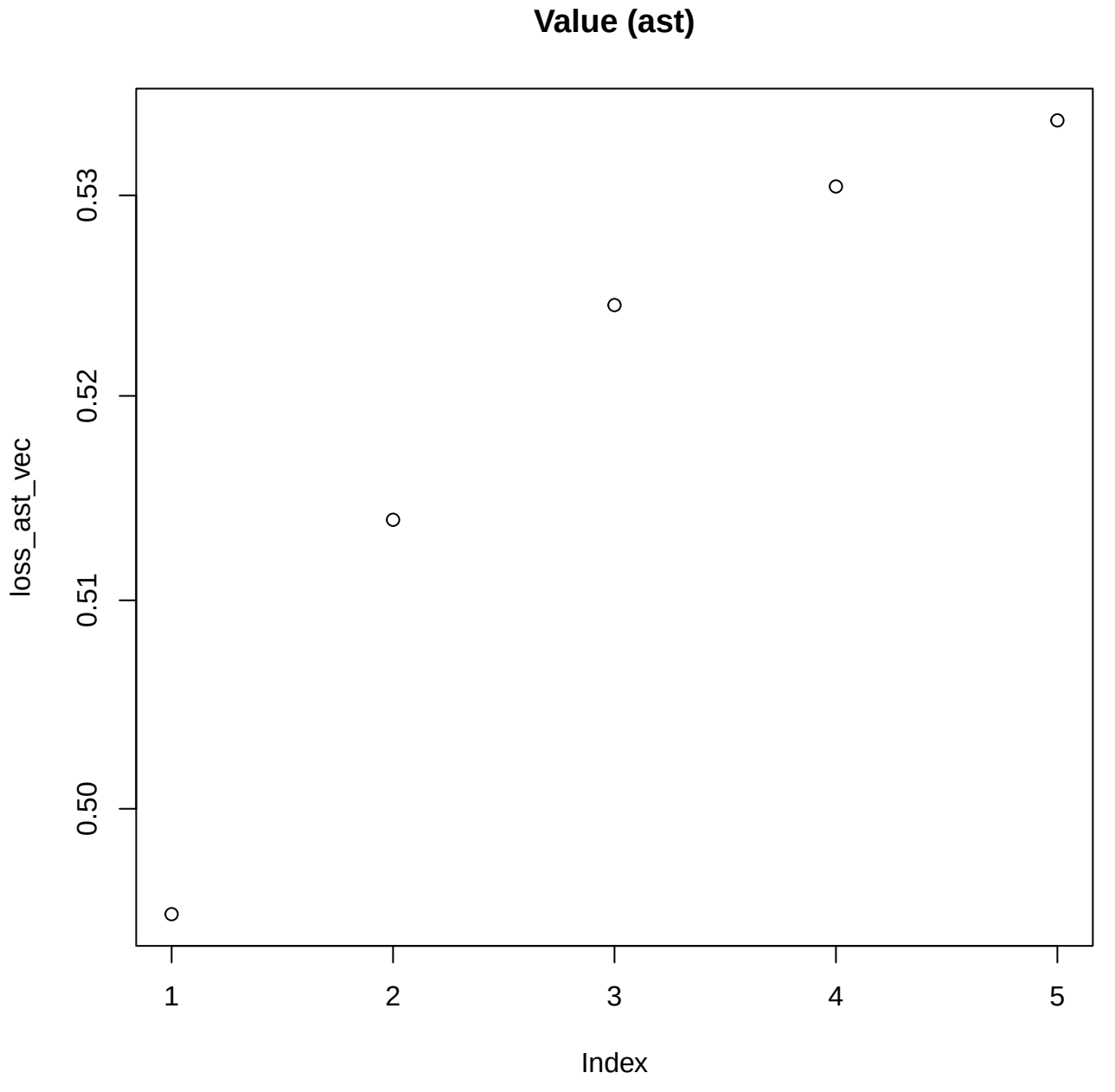
0.04
0.05
0.06
0.07
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.50
0.51
0.52
0.53
1
2
3
4
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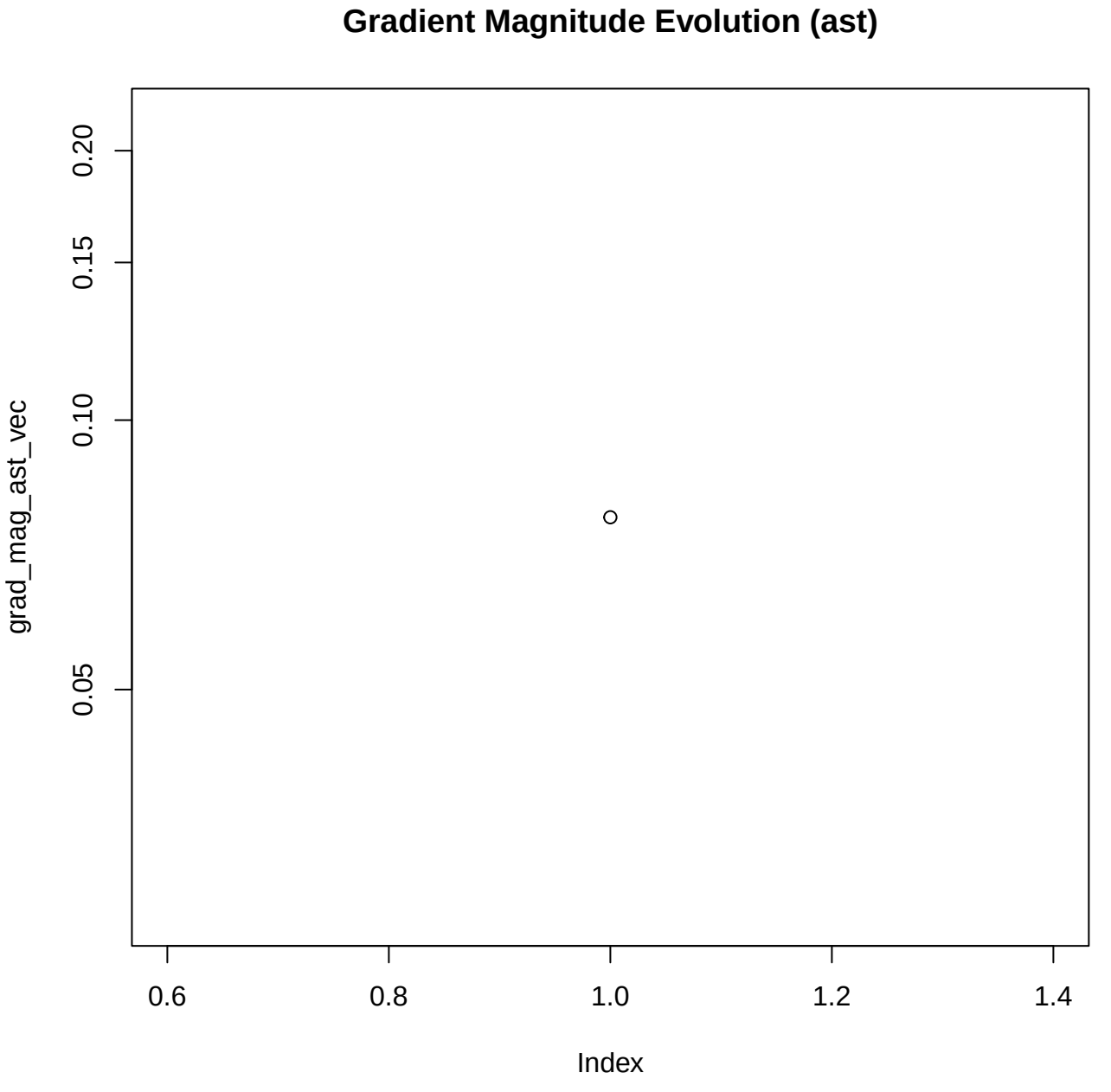
Gradient Magnitude Evolution (ast)
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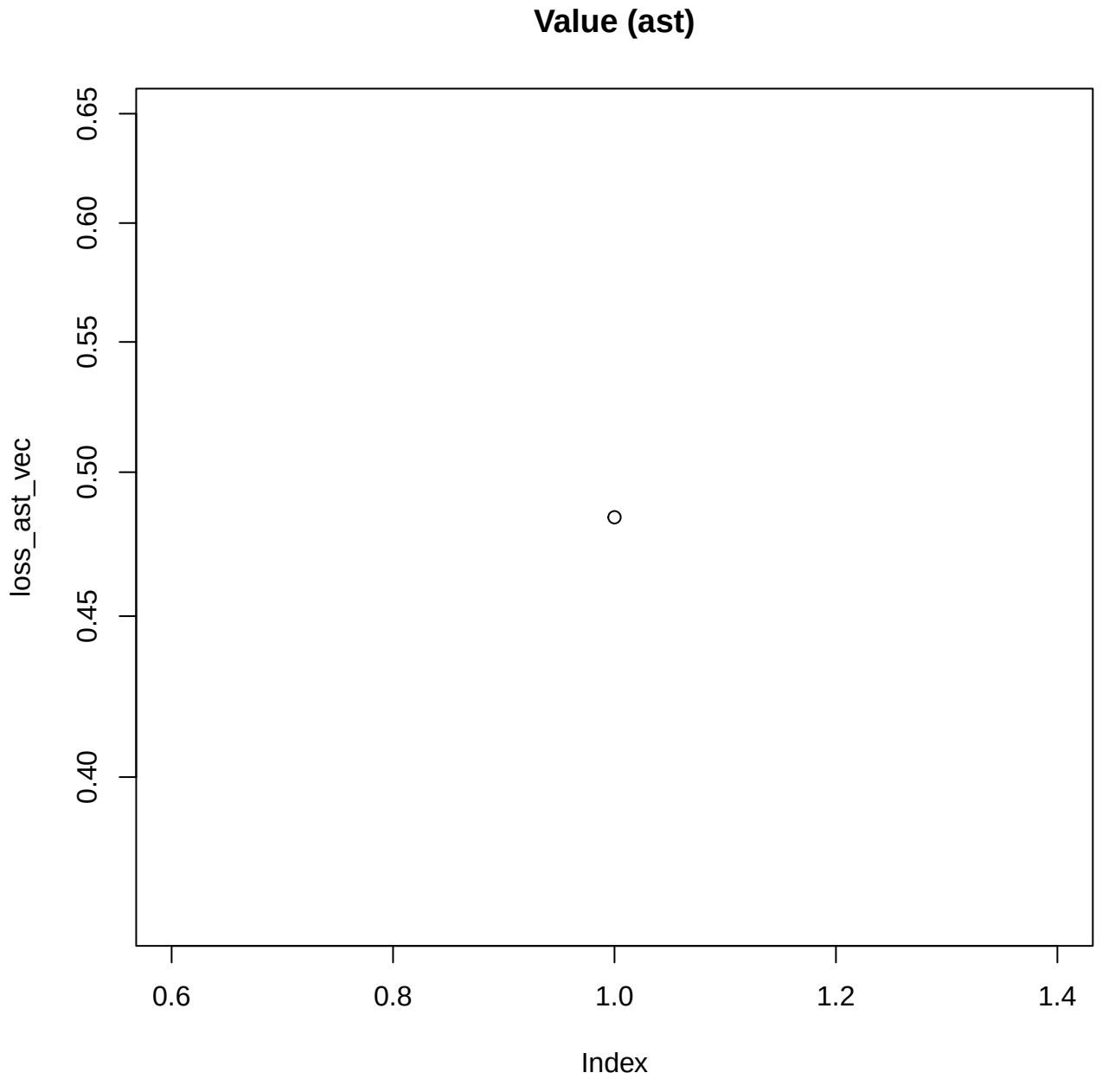
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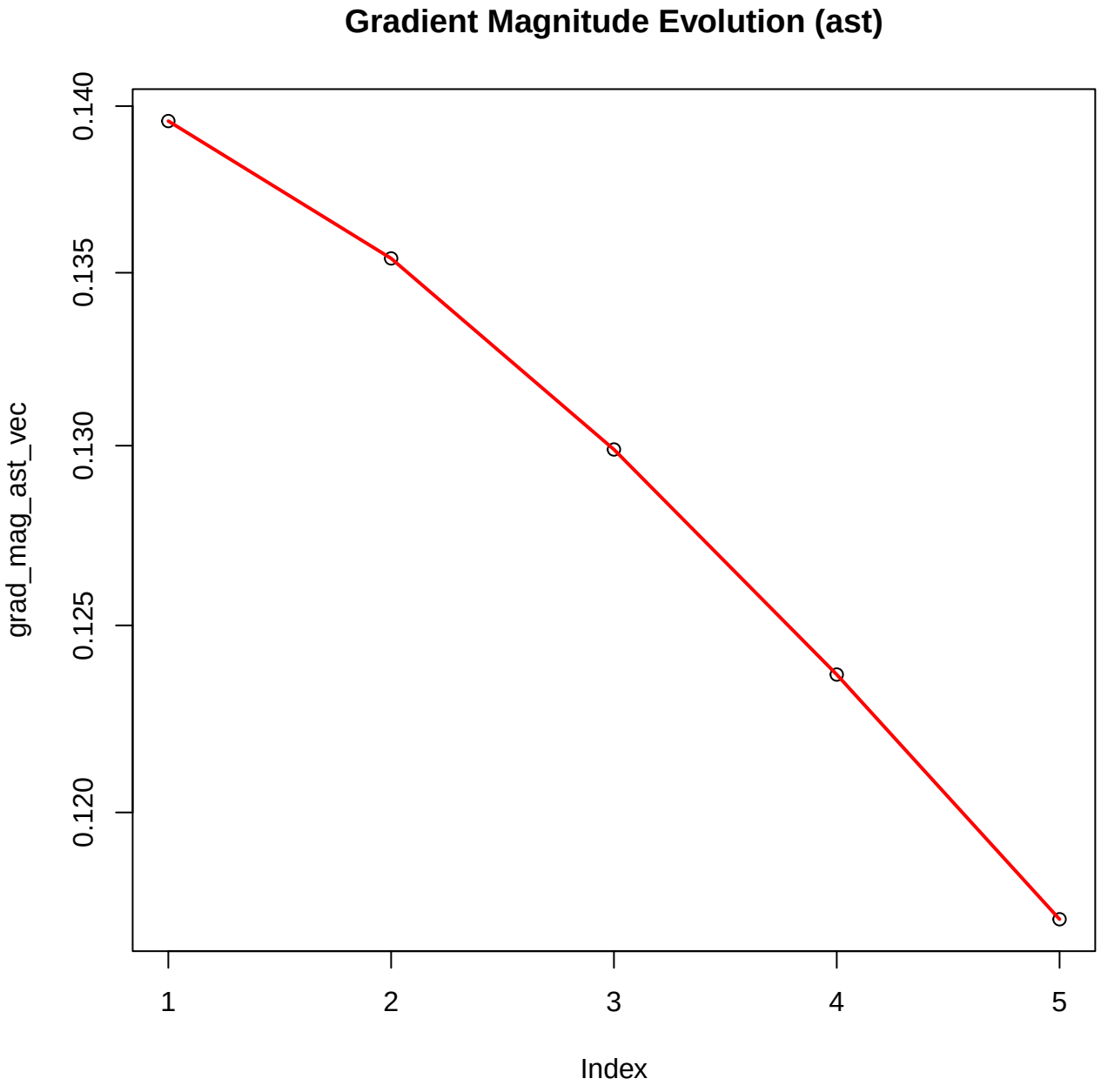
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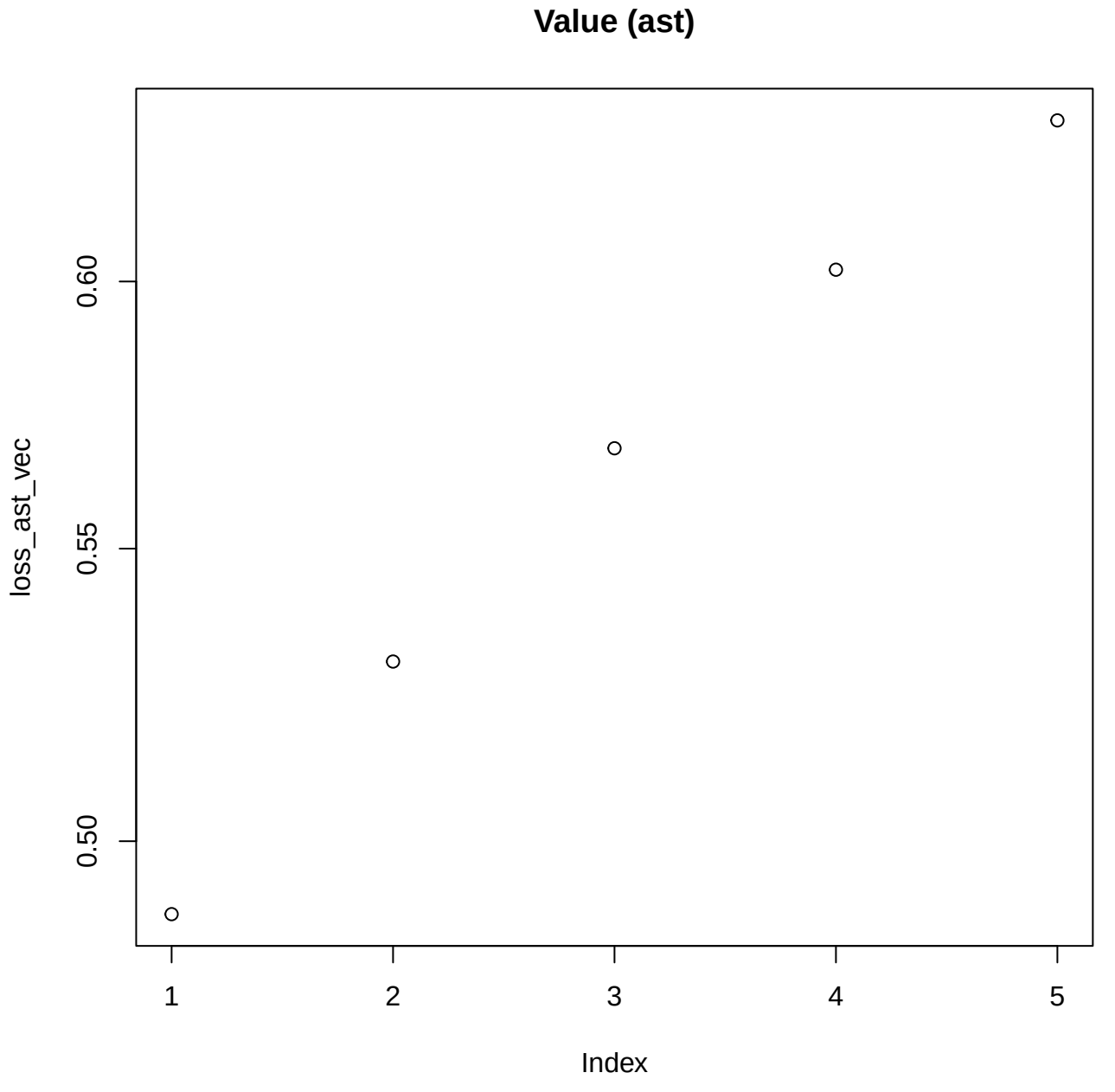
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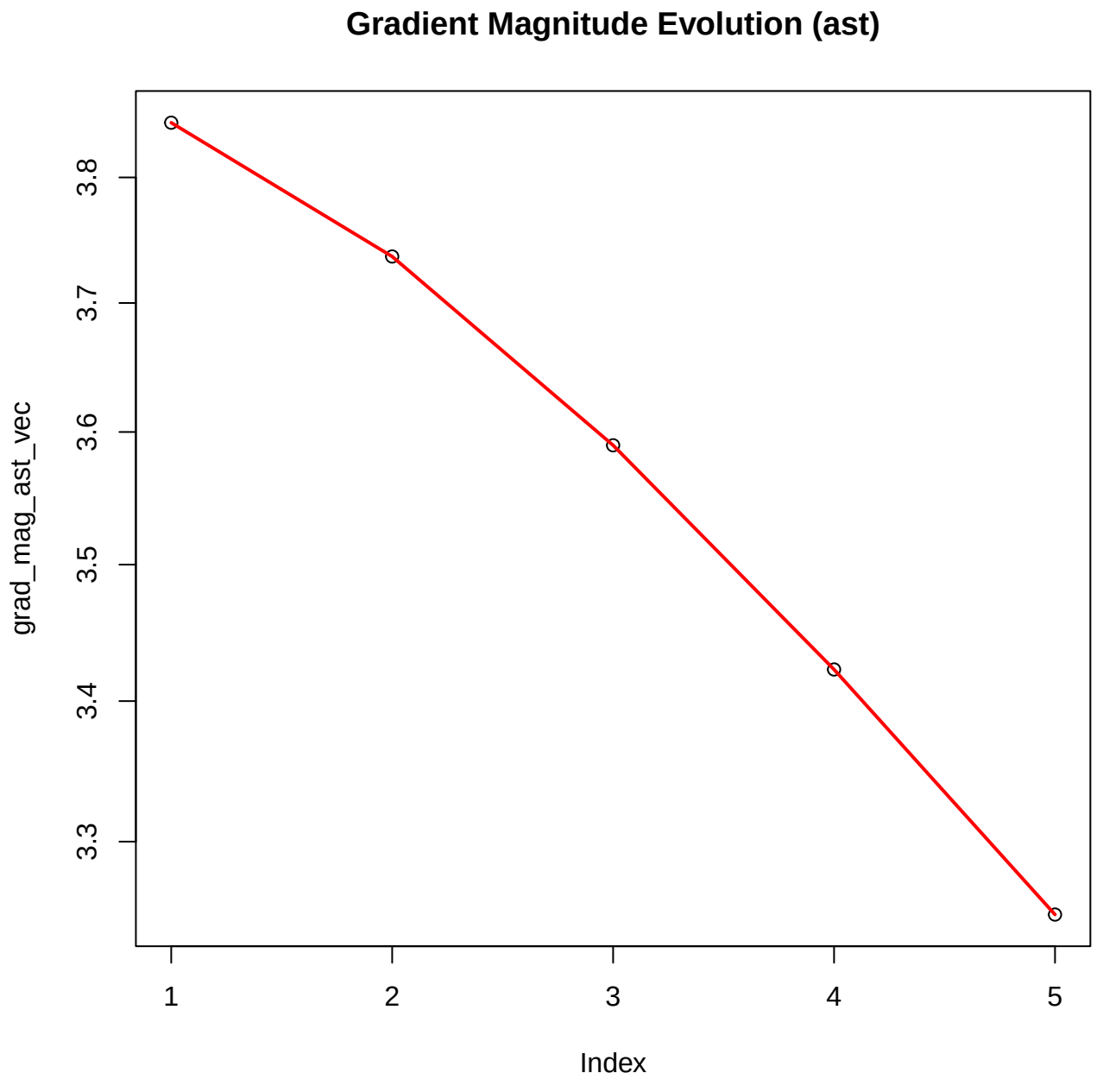
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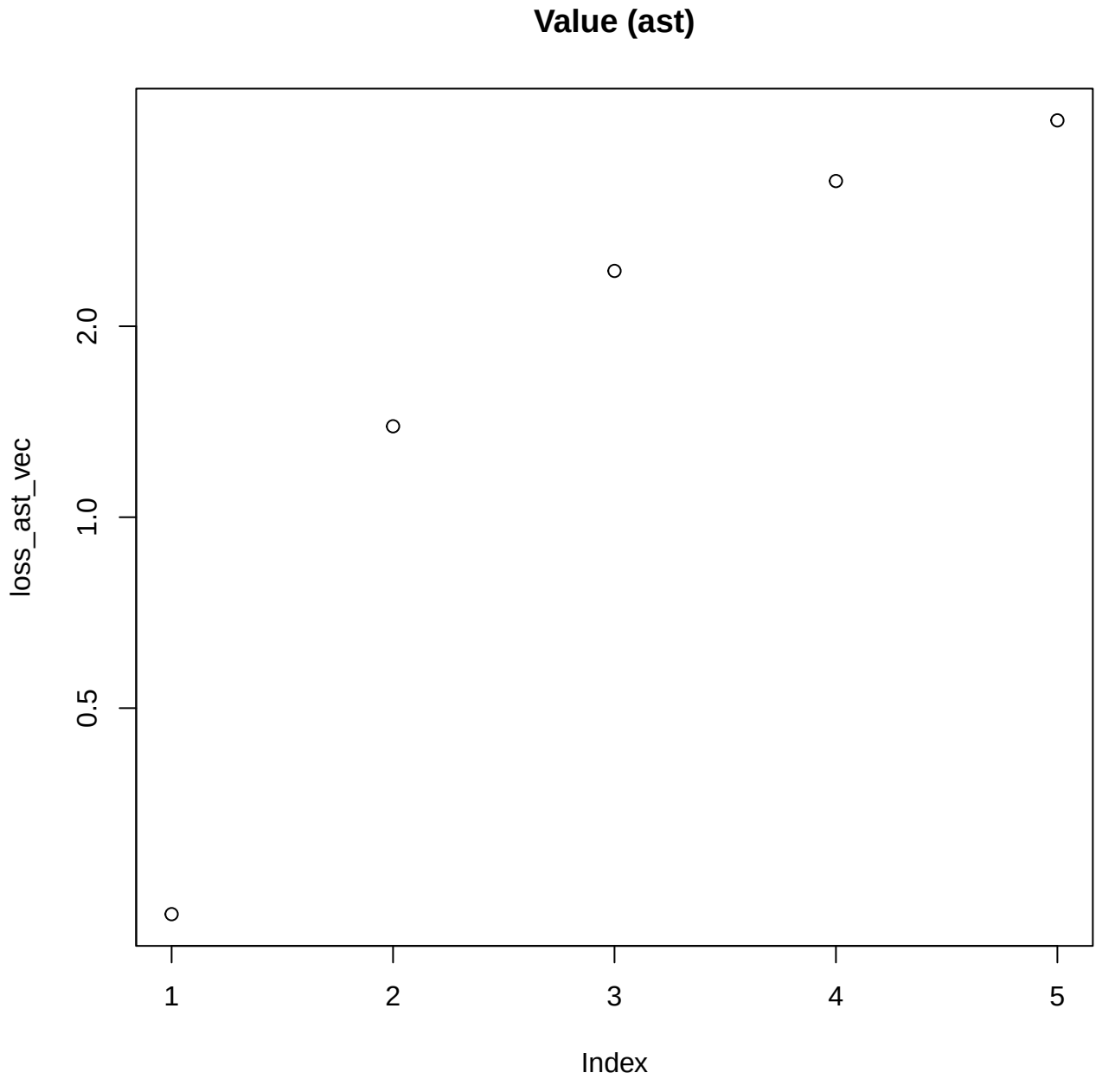
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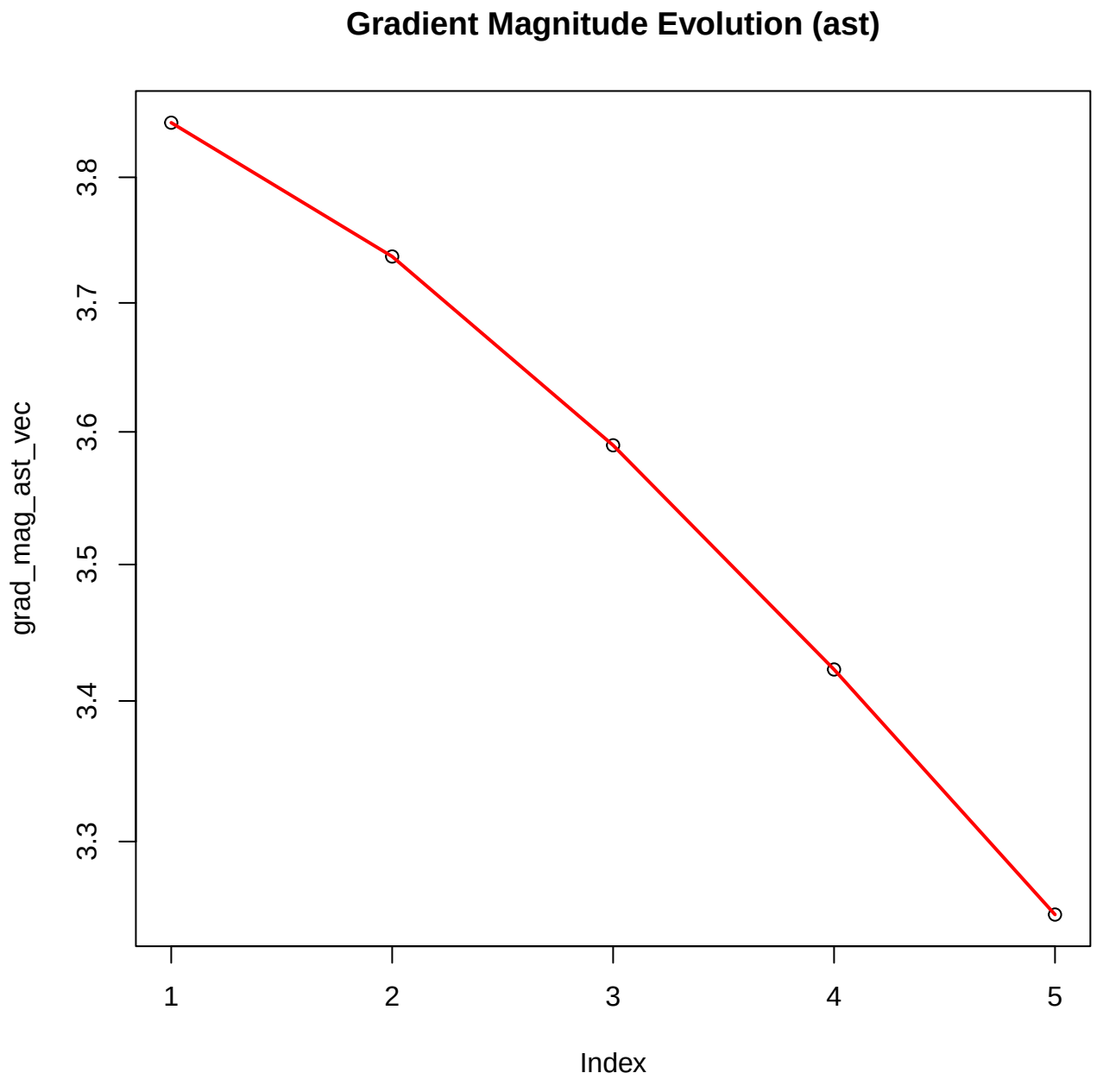
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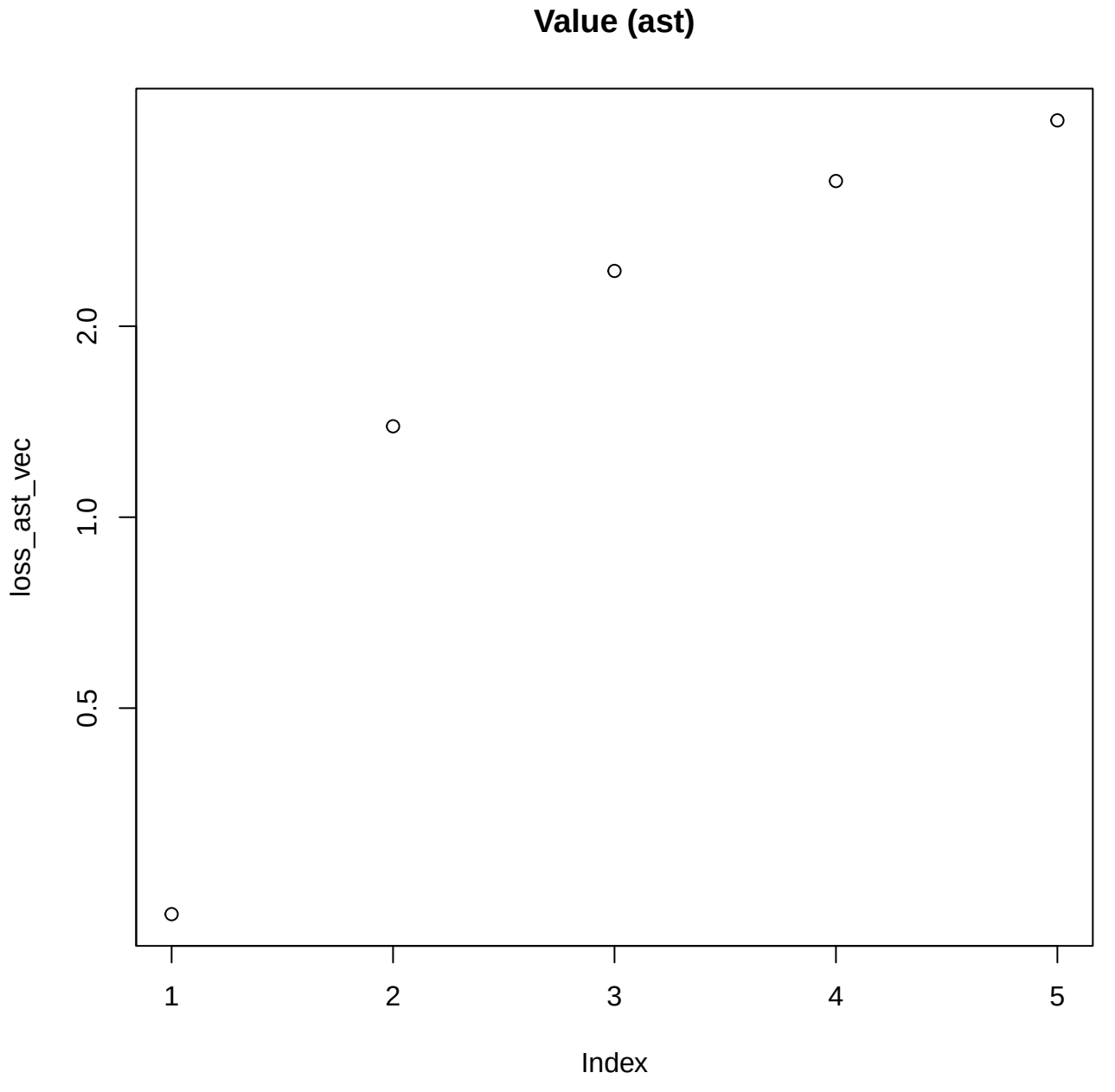
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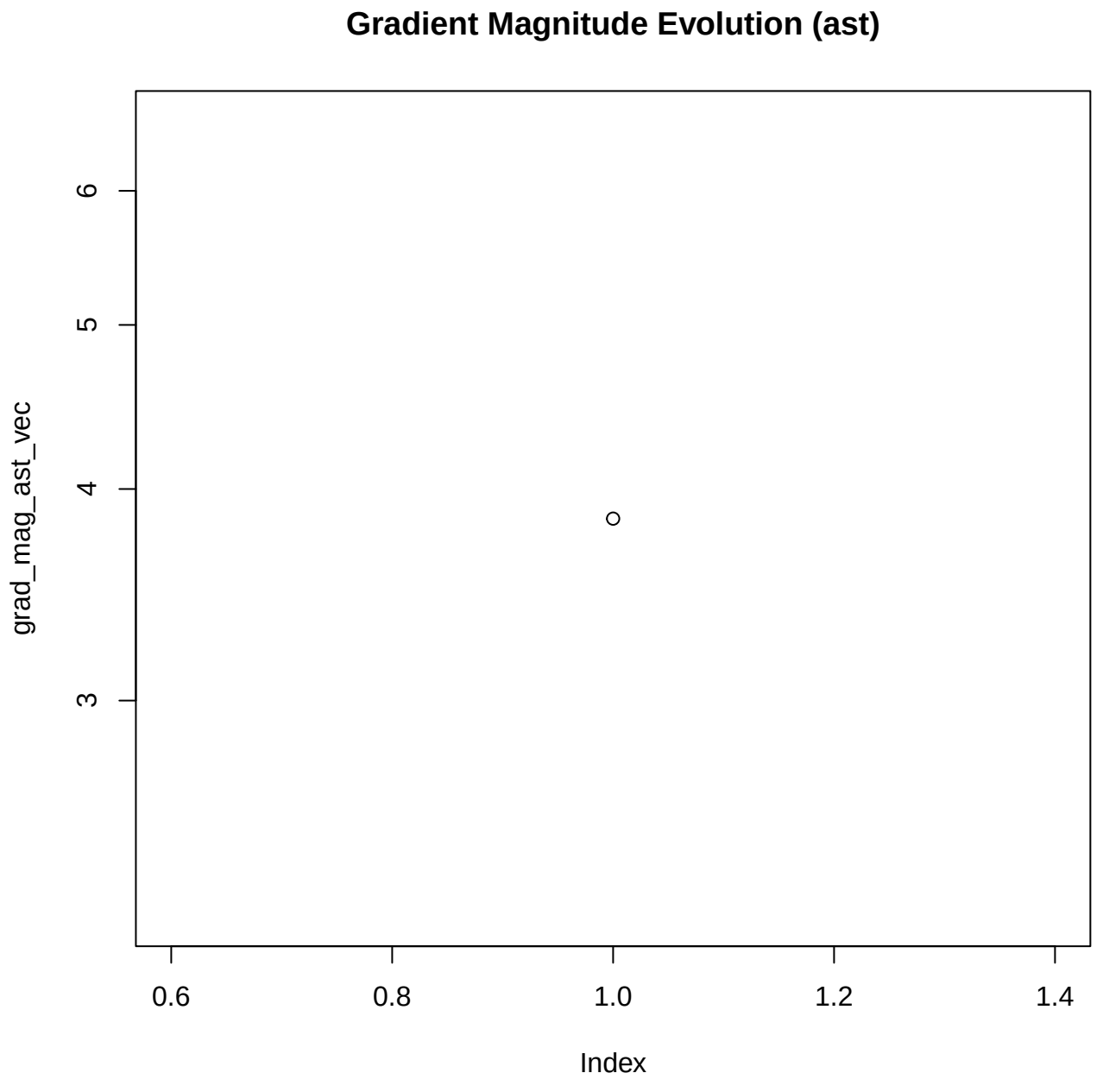
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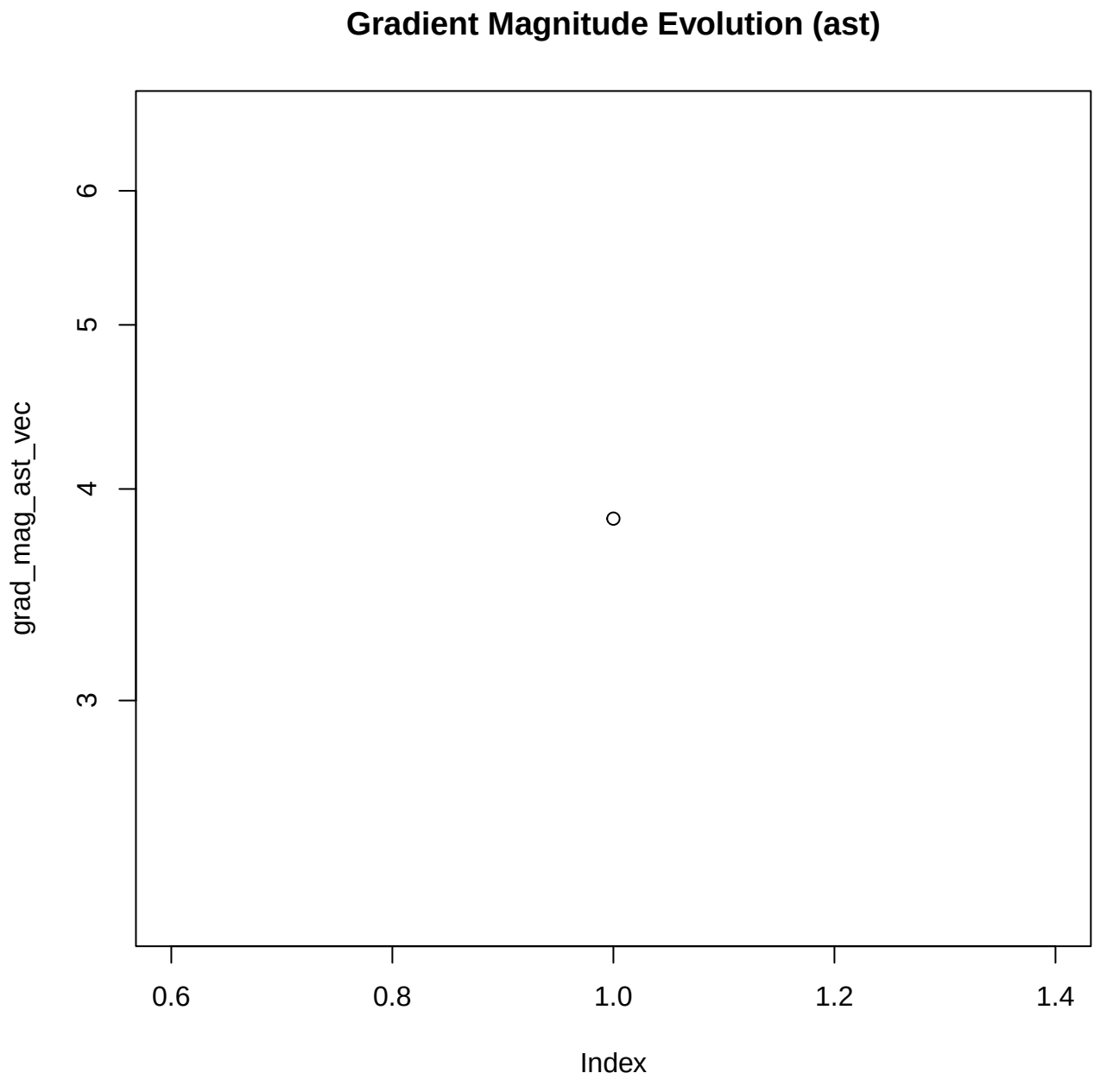
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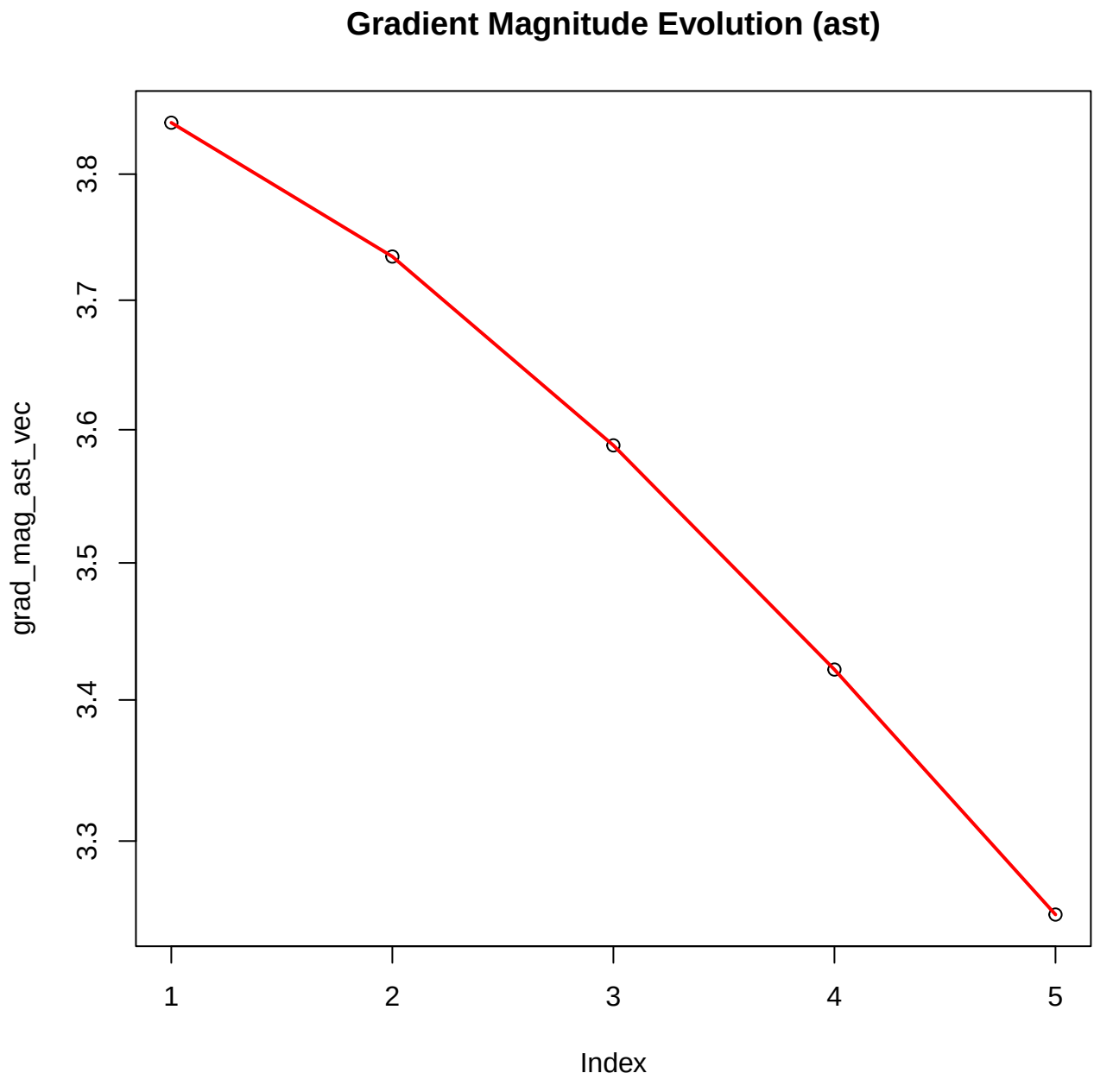
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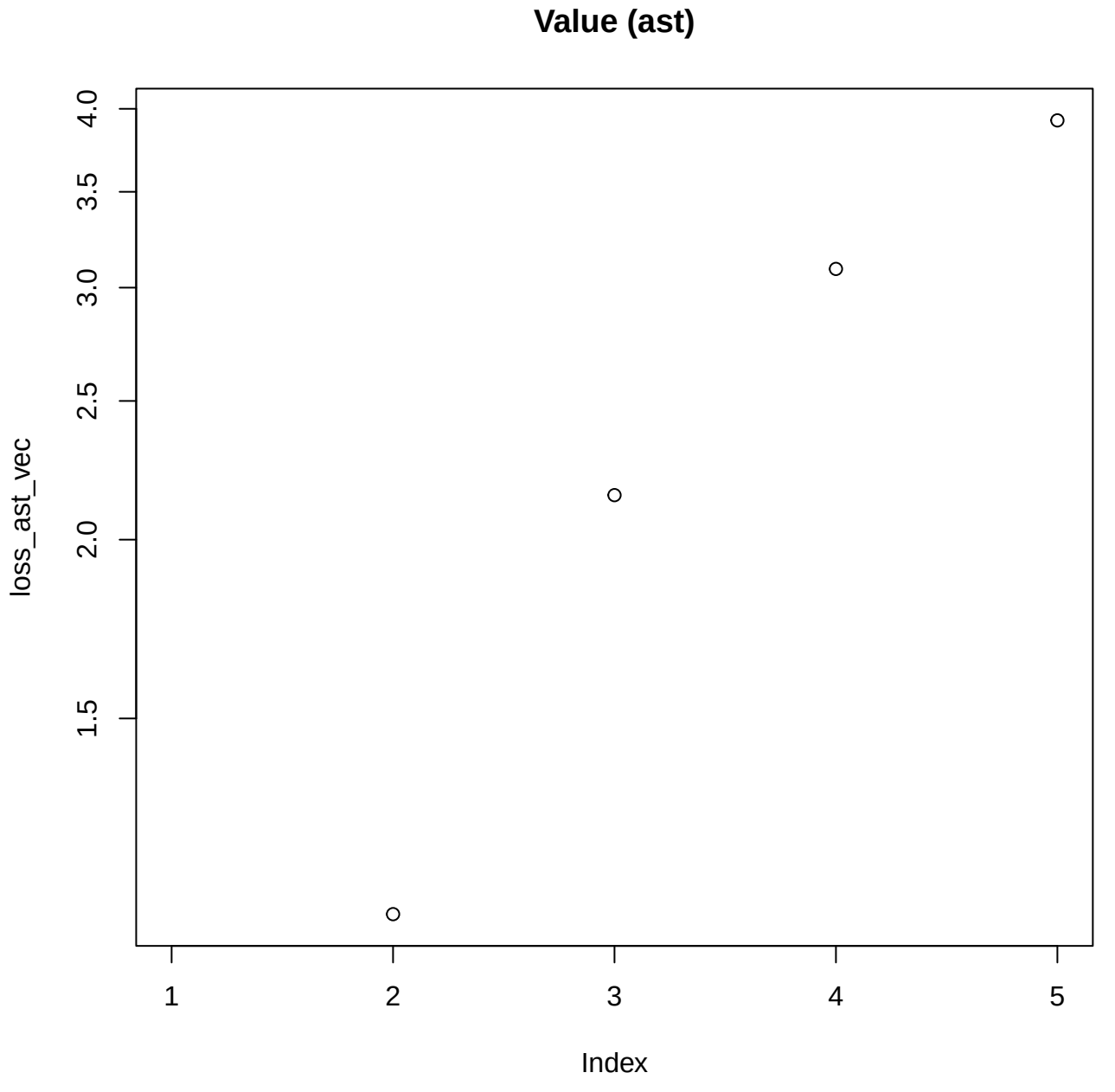
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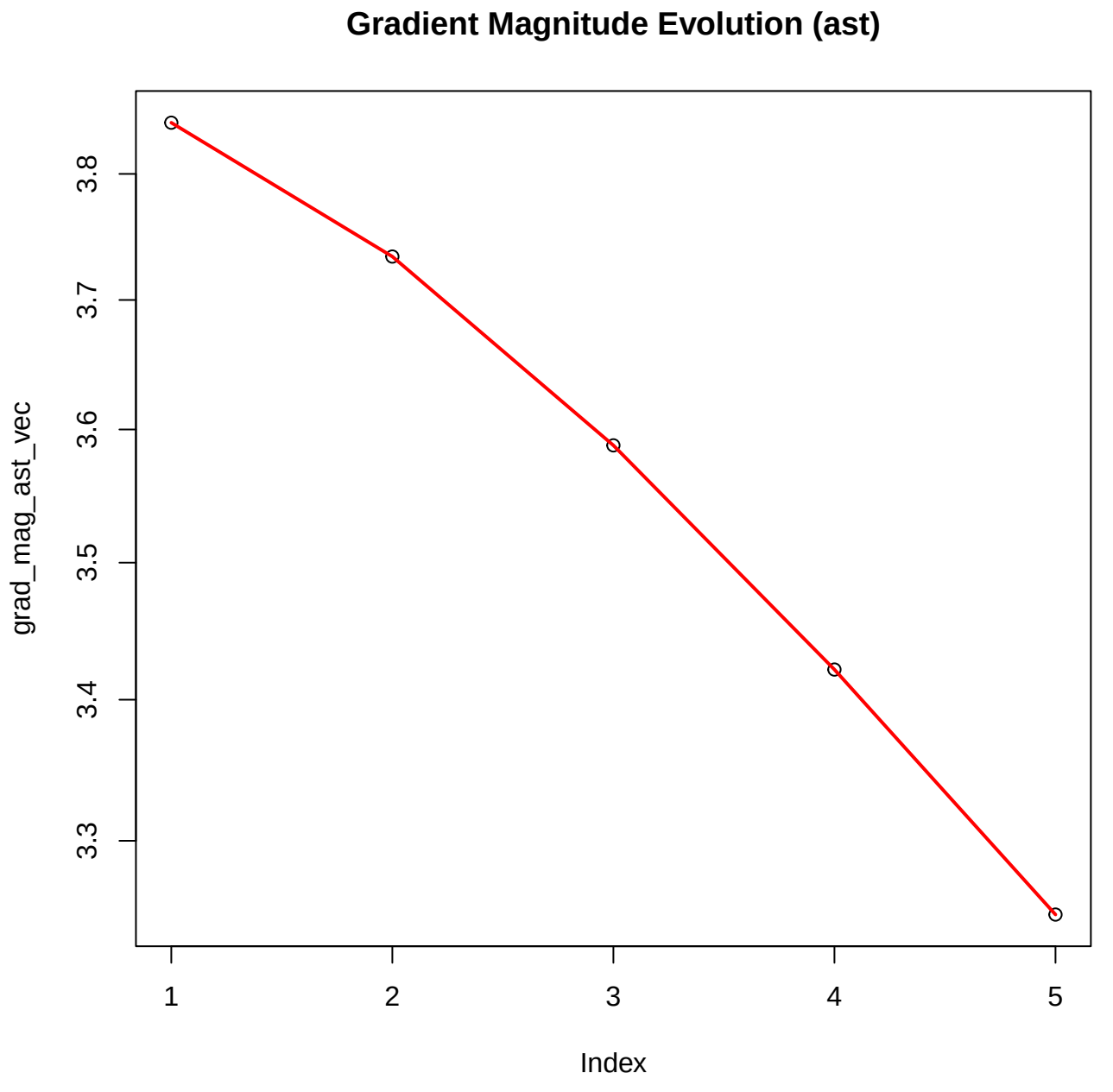
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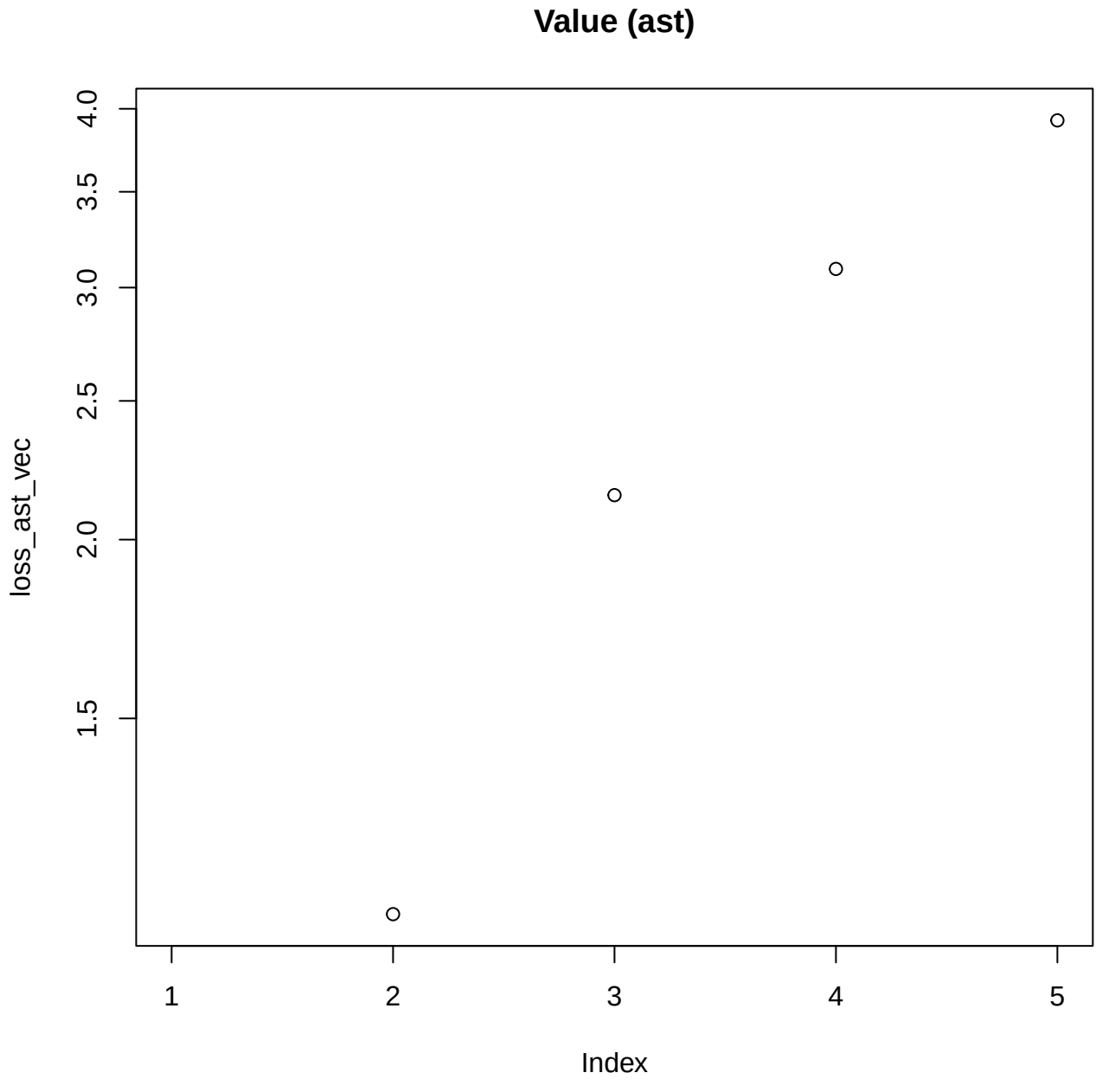
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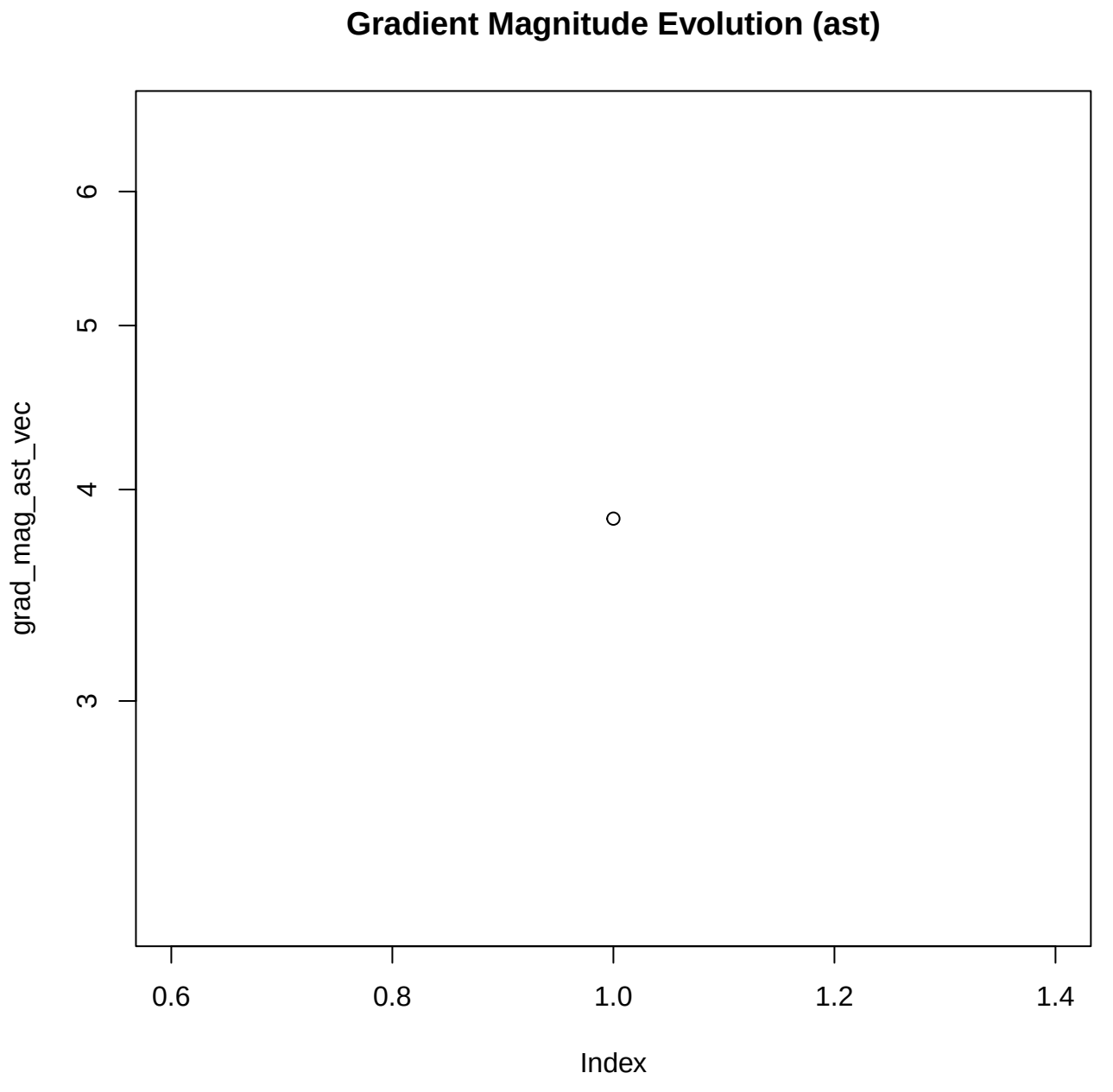
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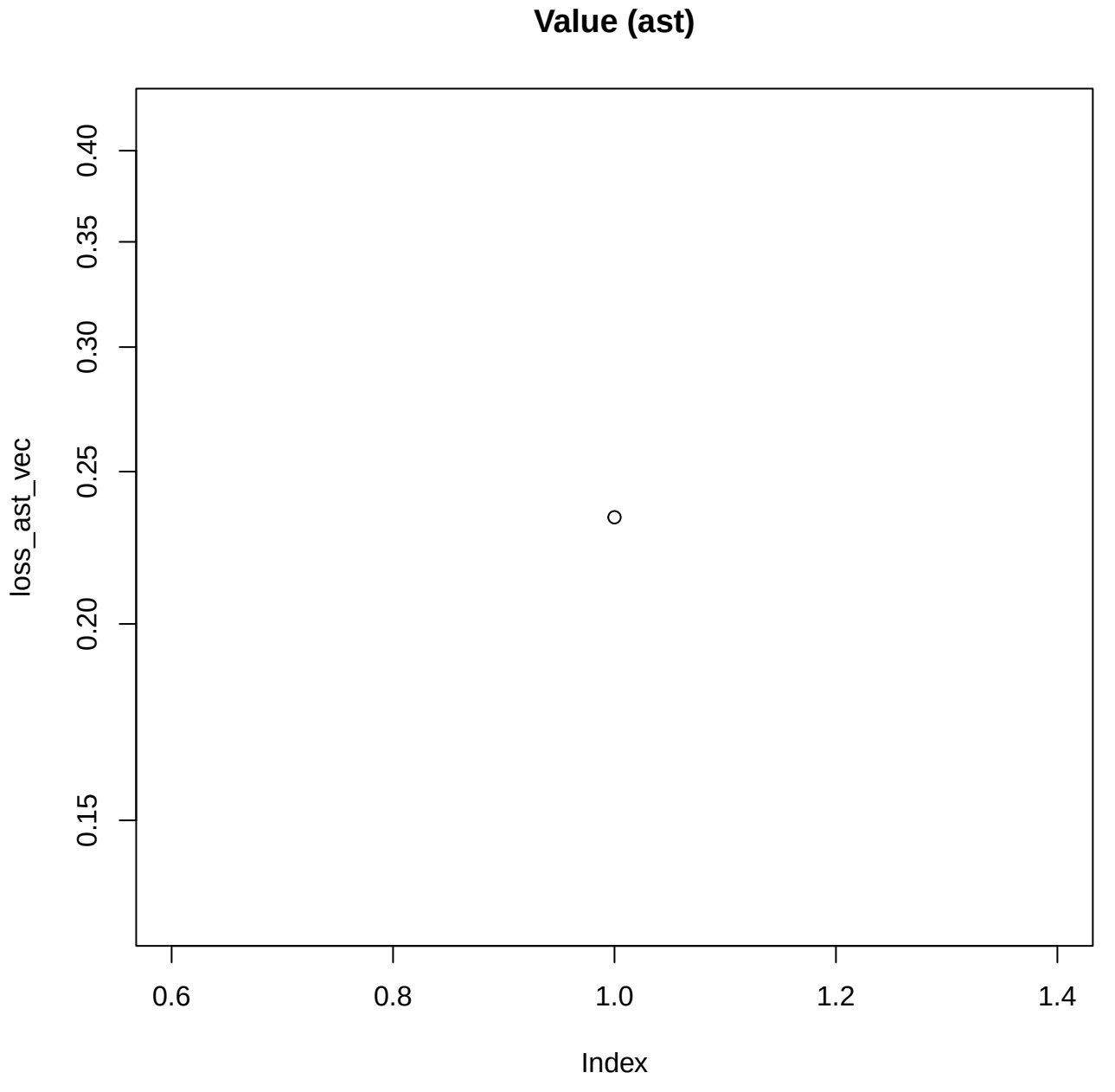
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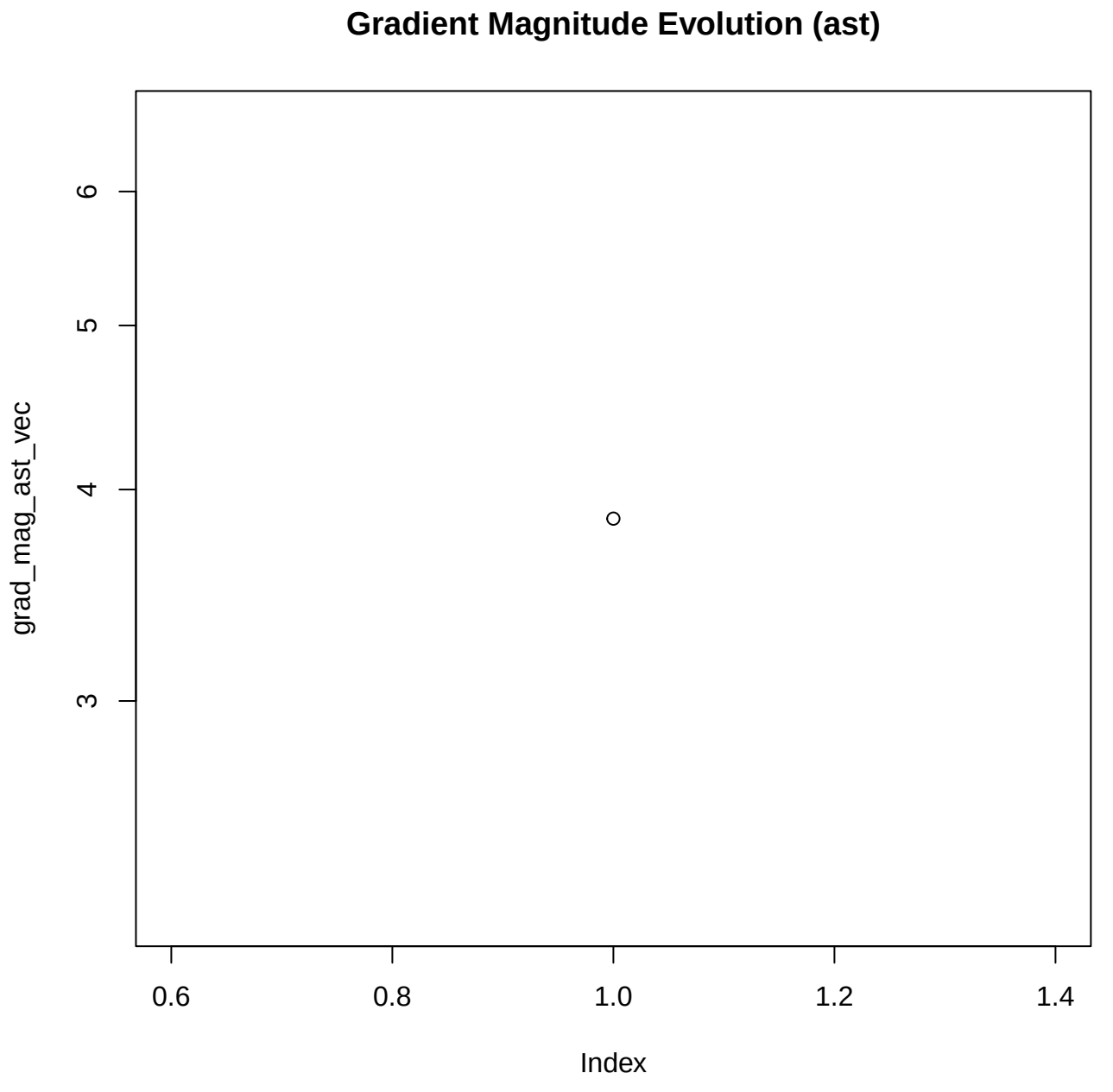
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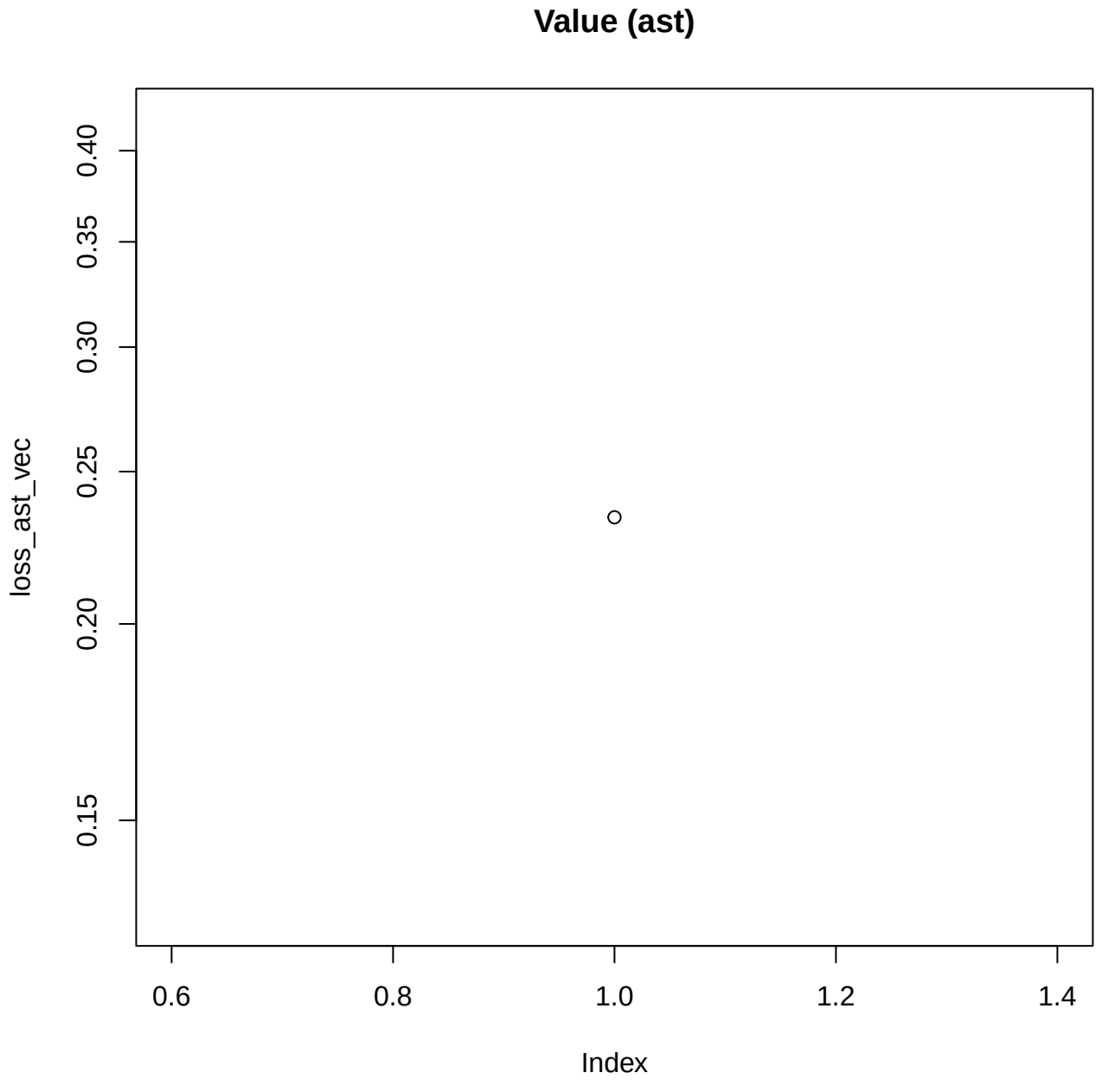
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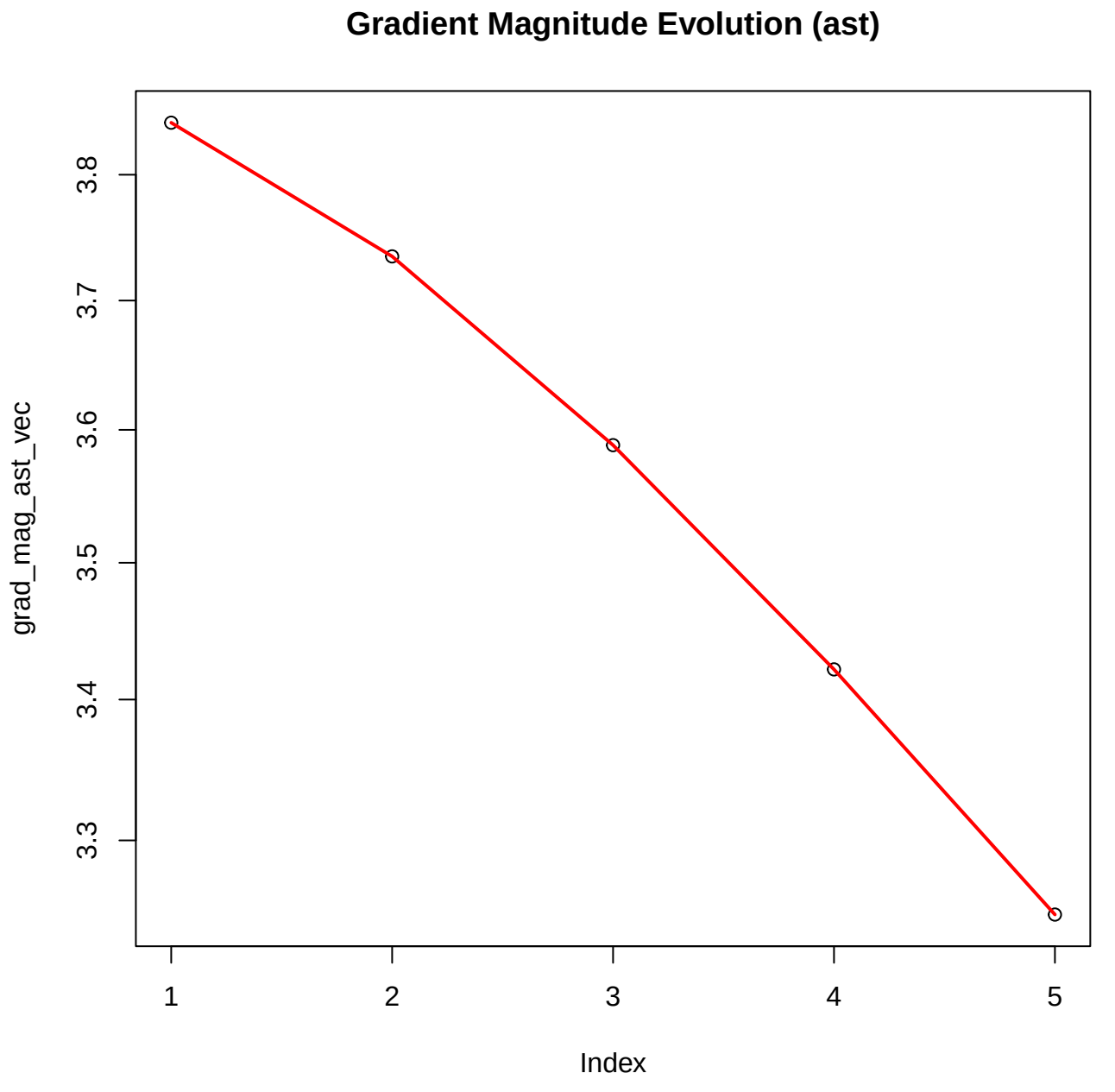
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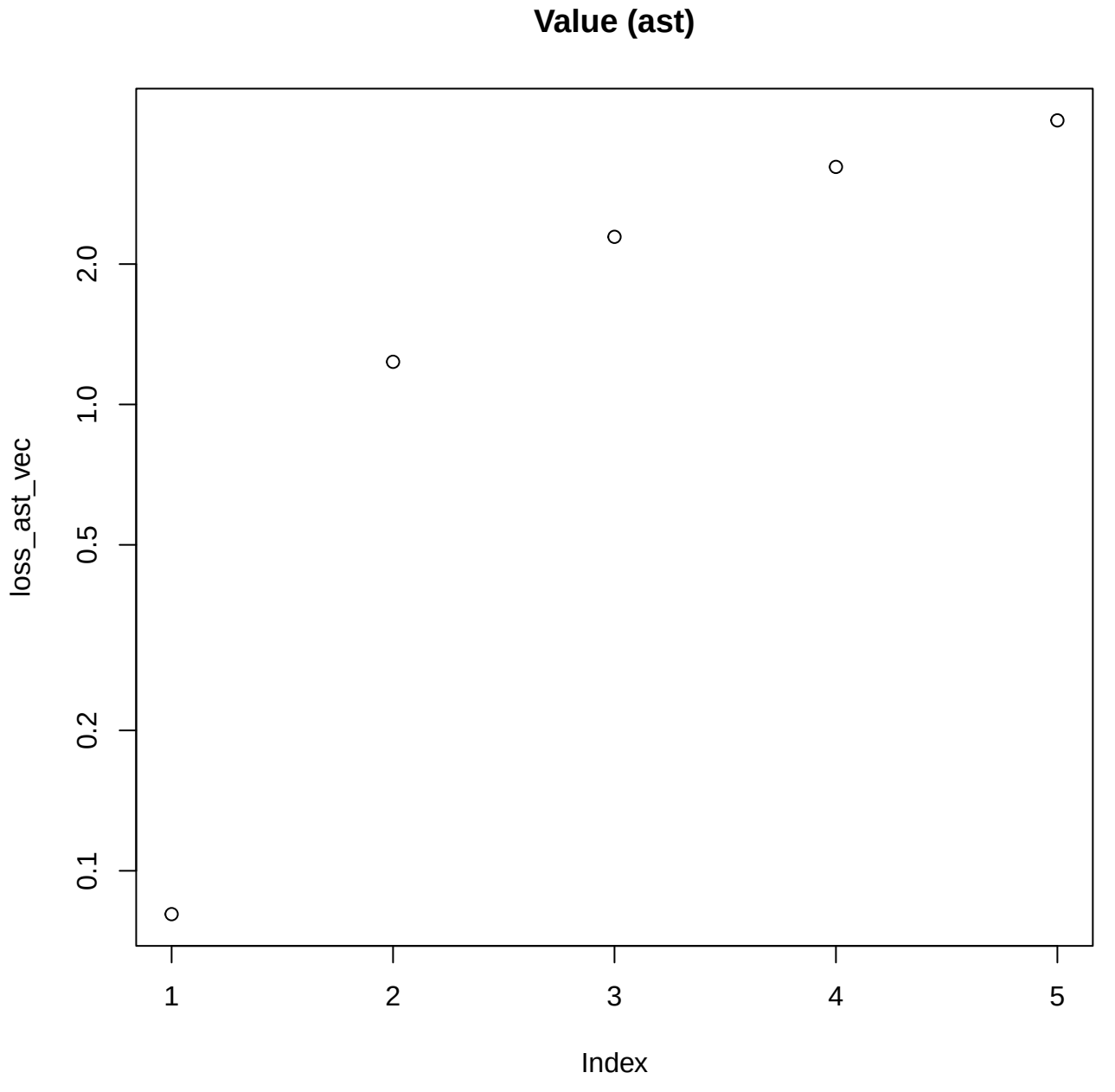
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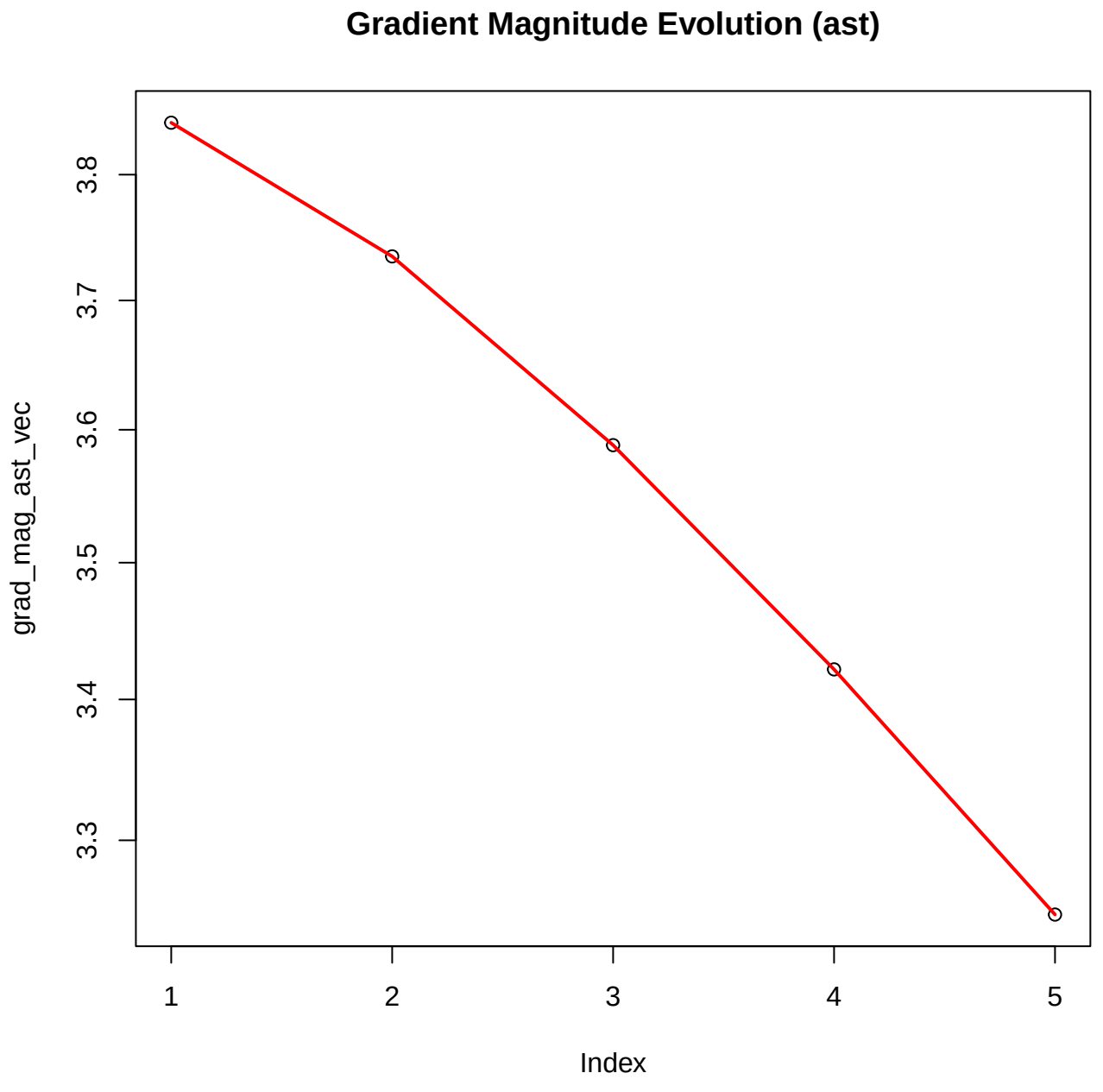
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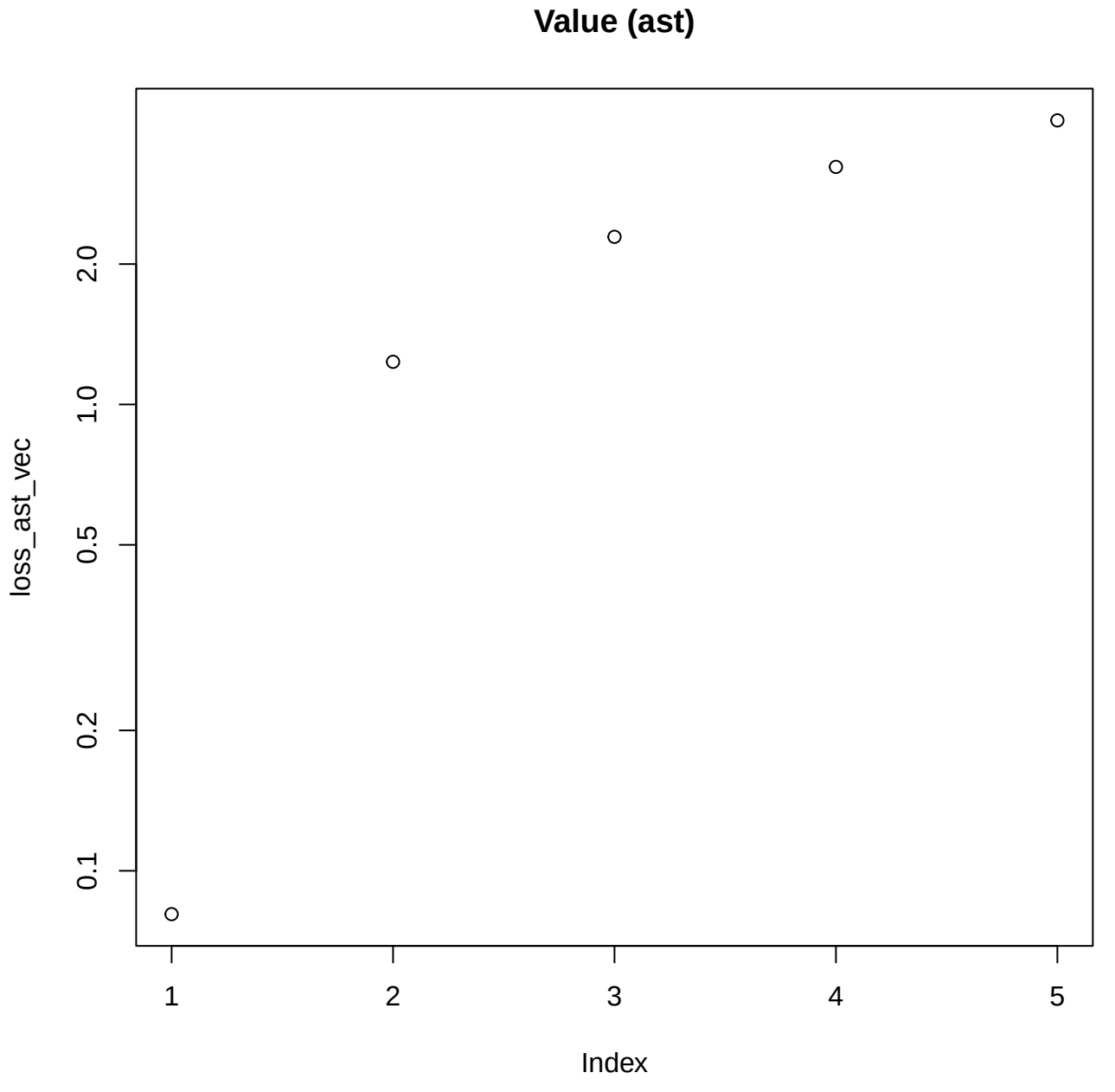
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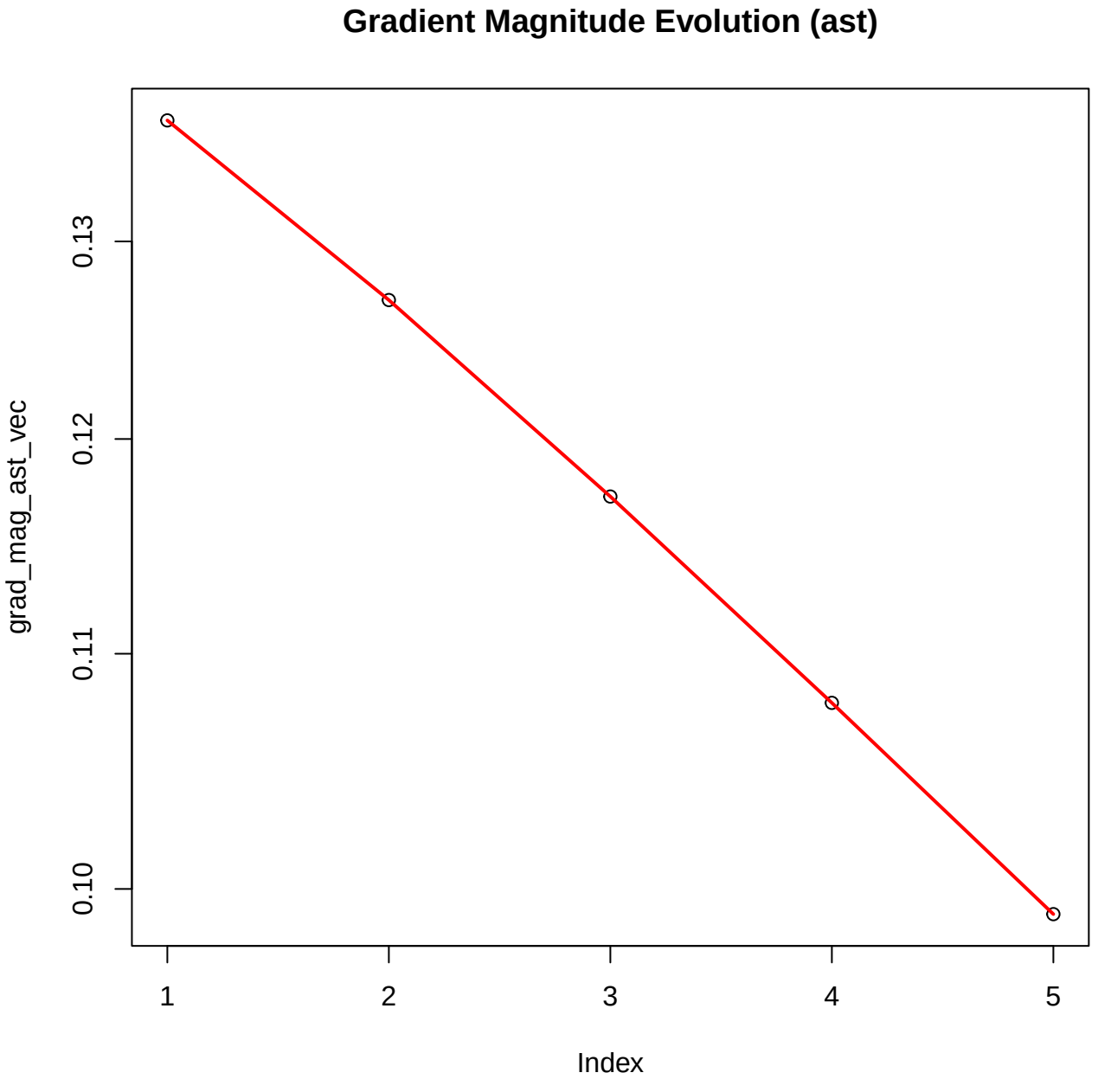
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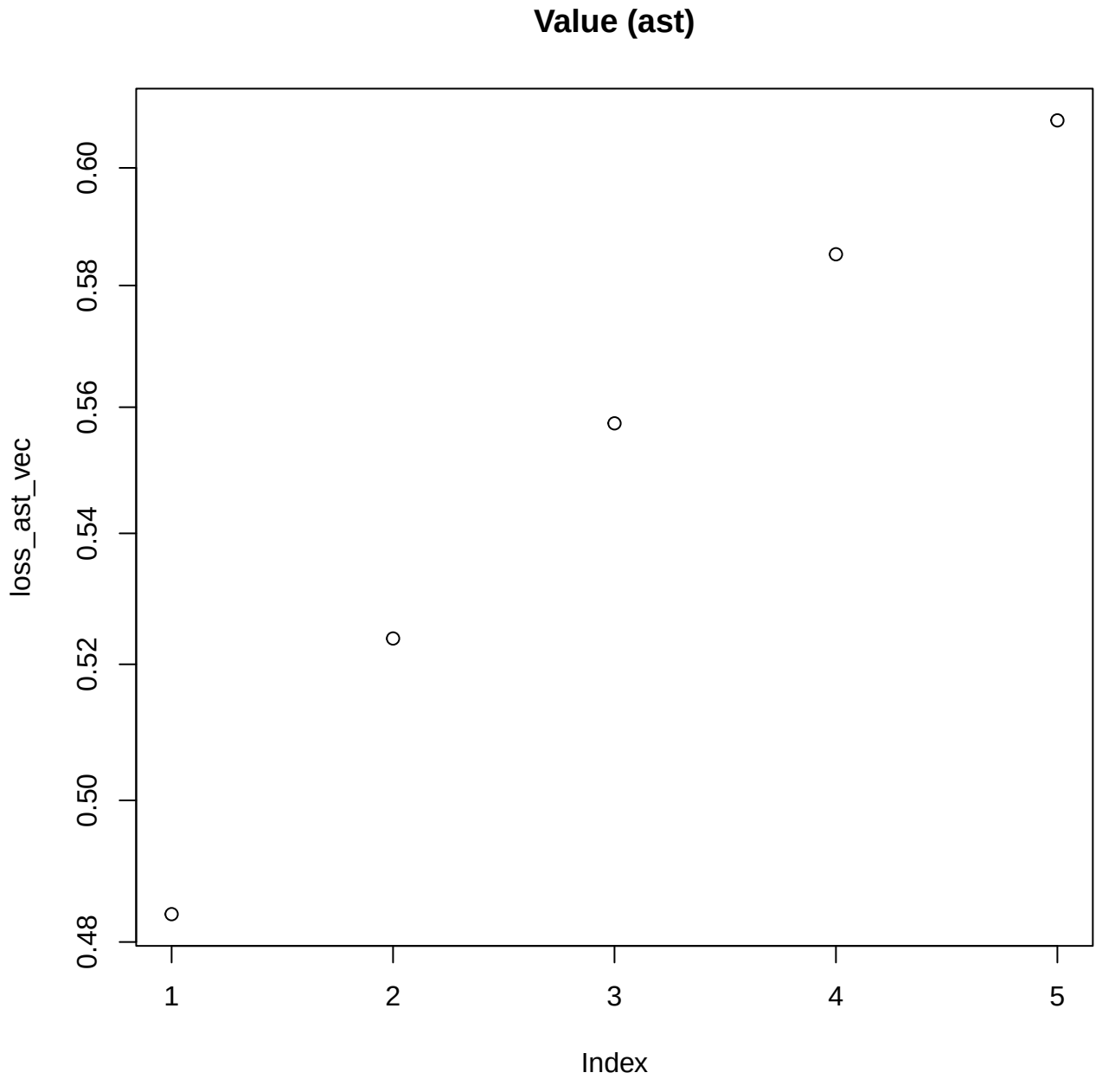
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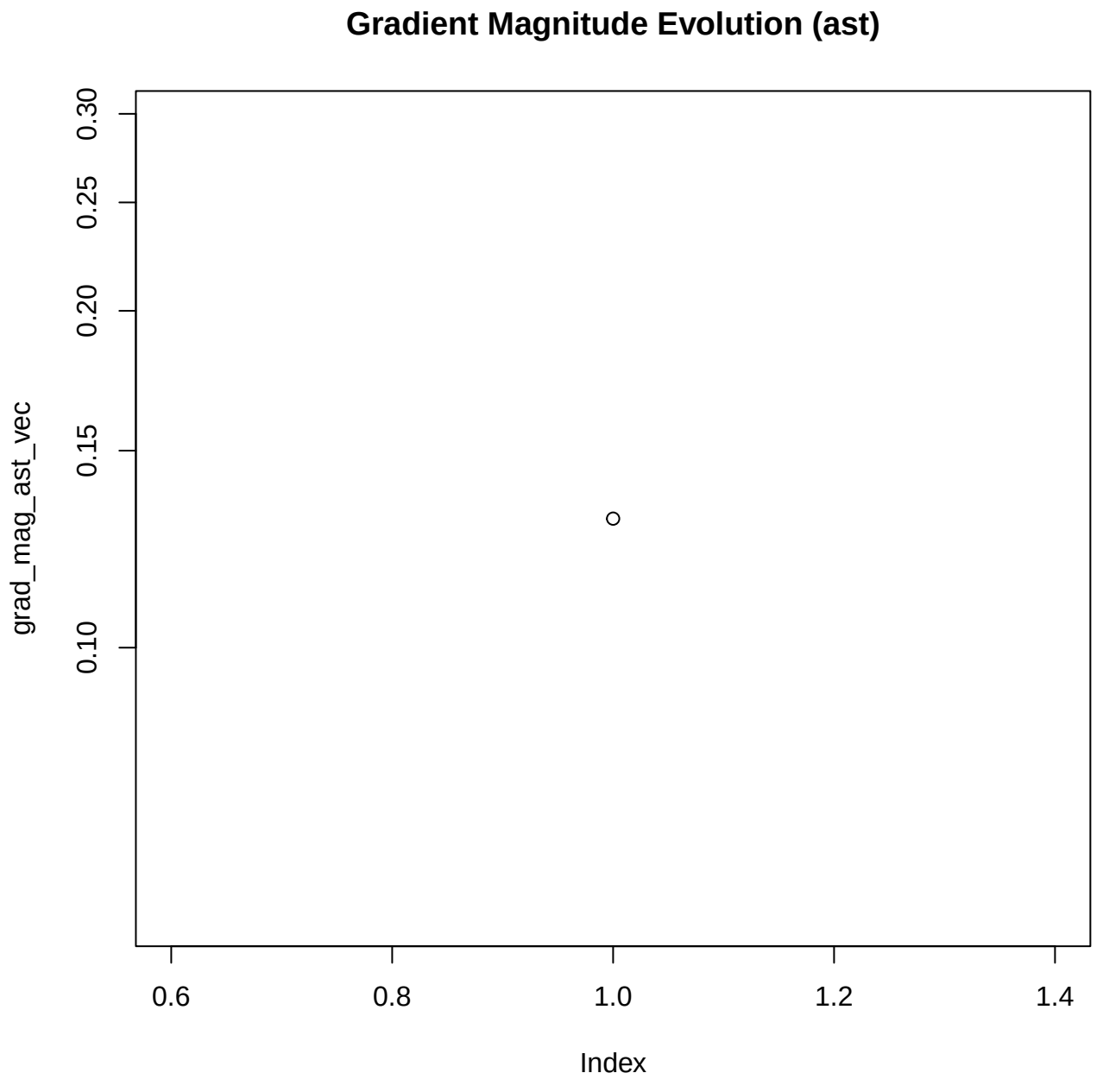
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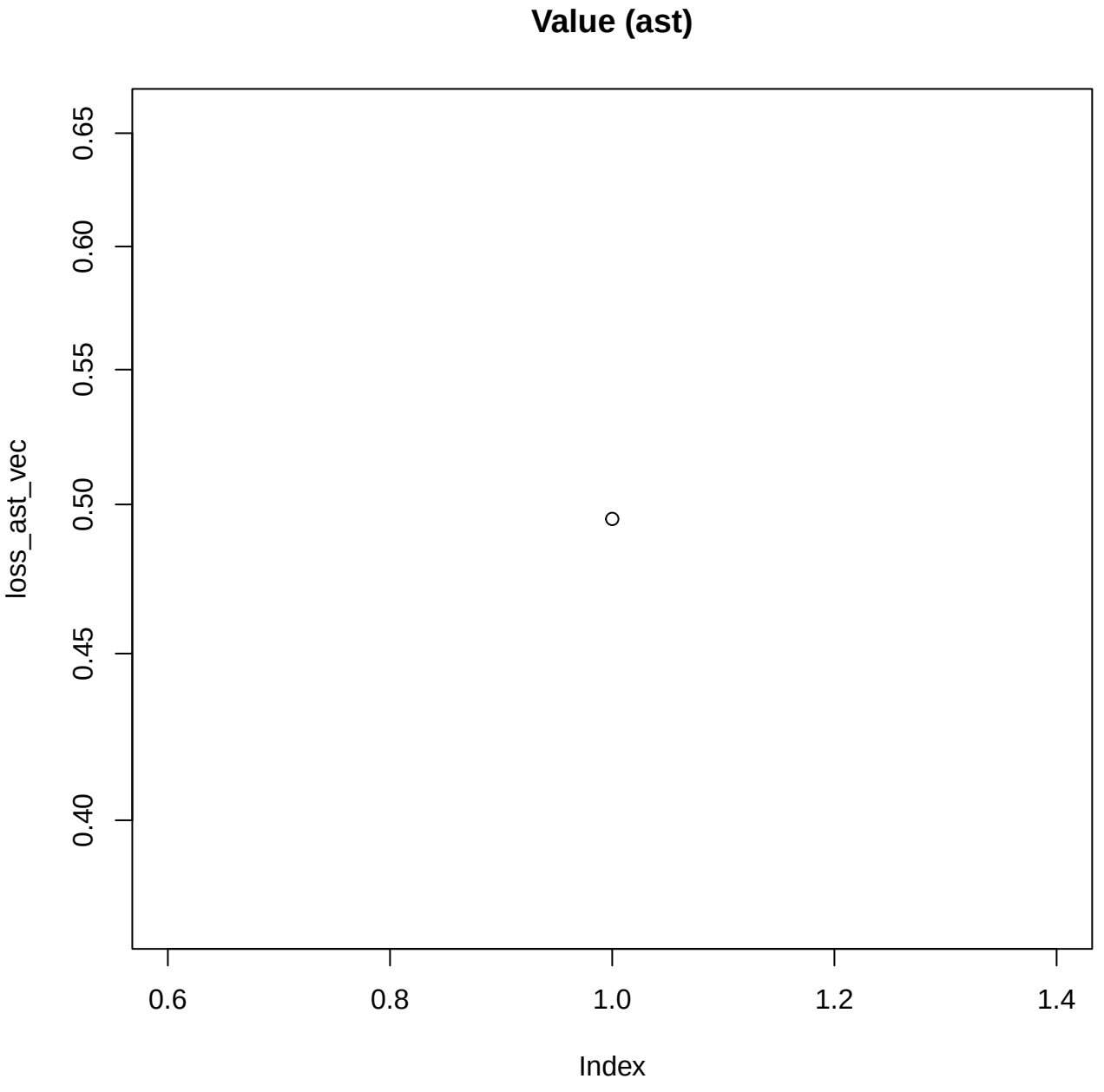
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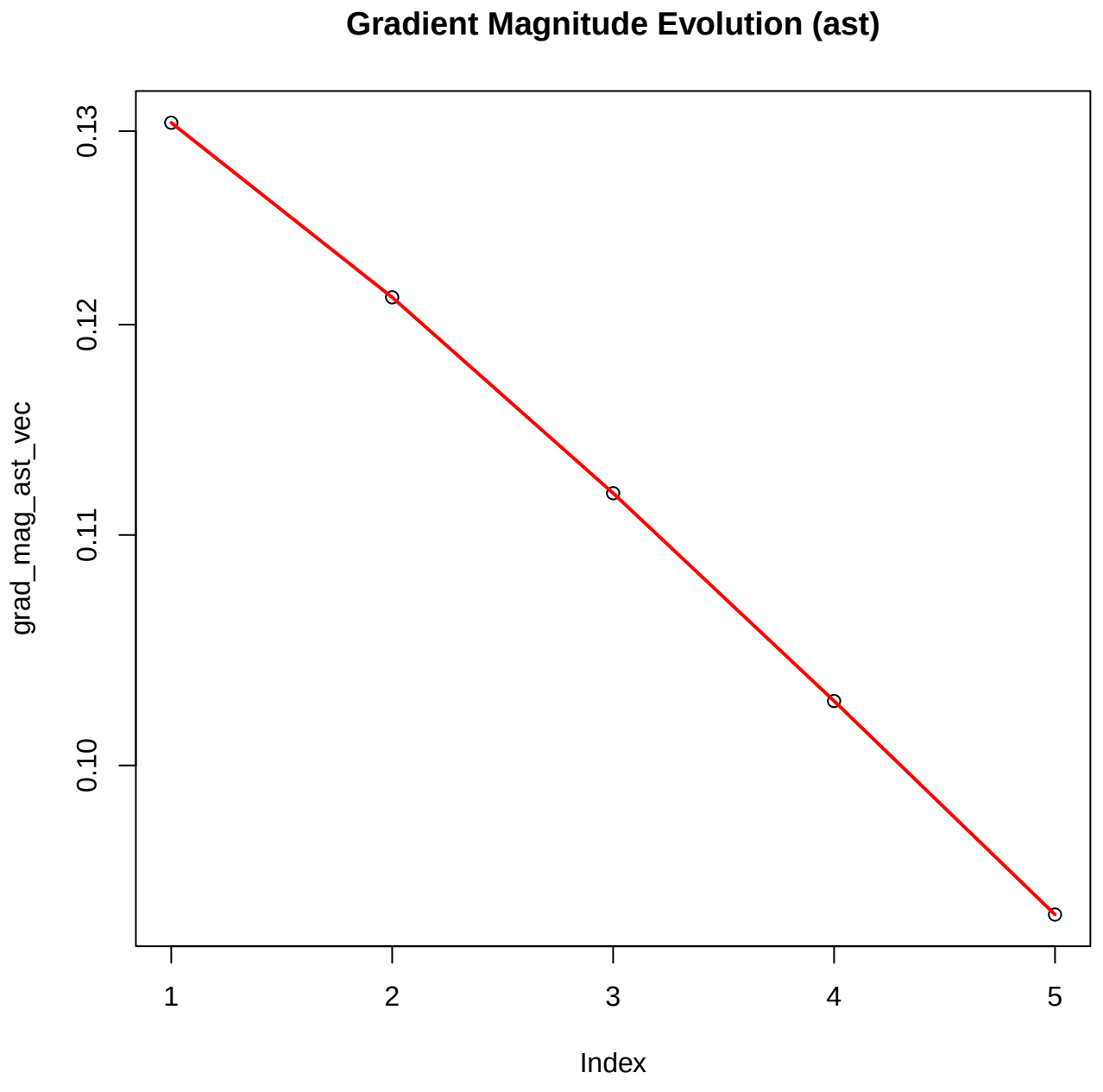
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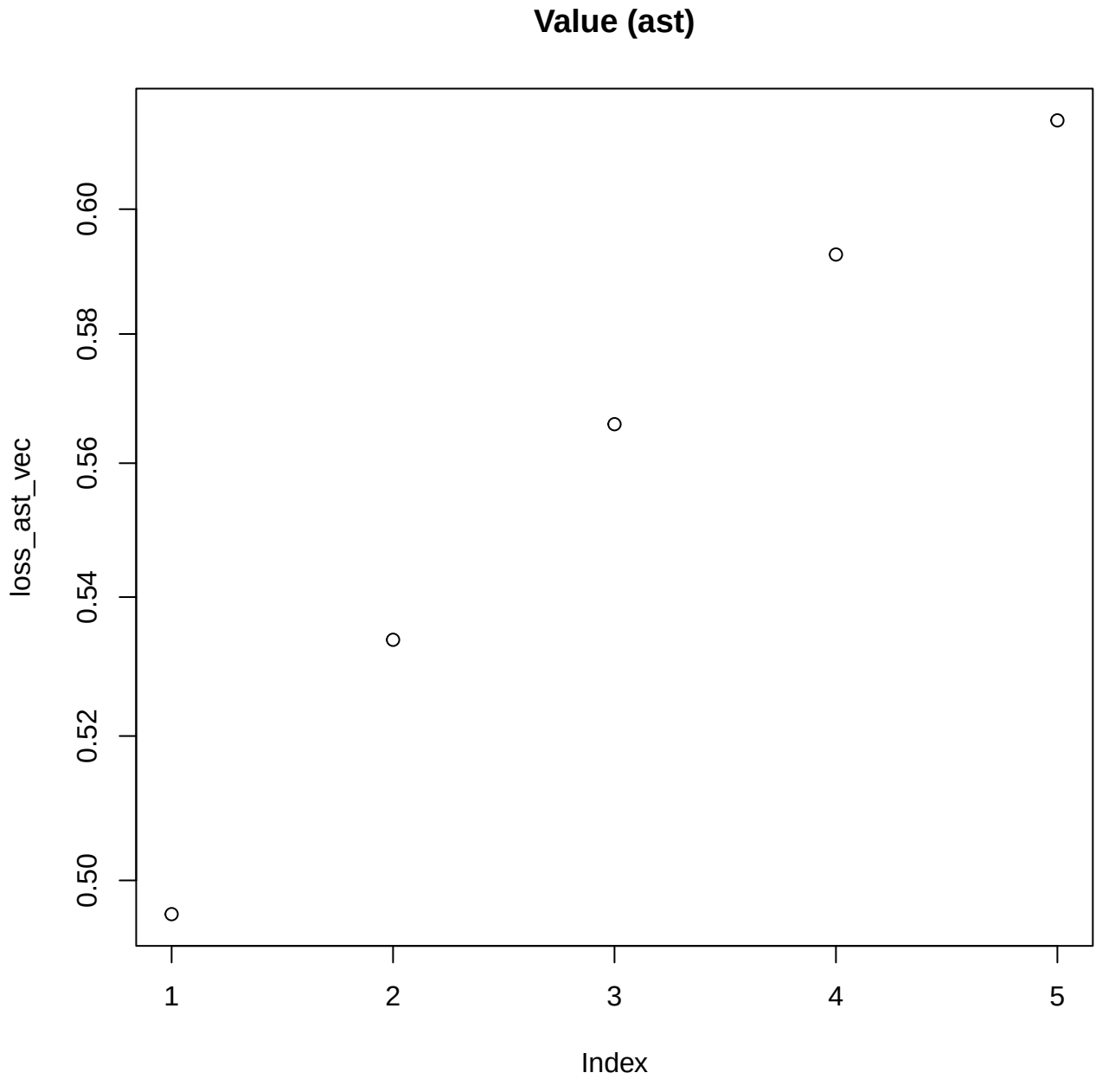
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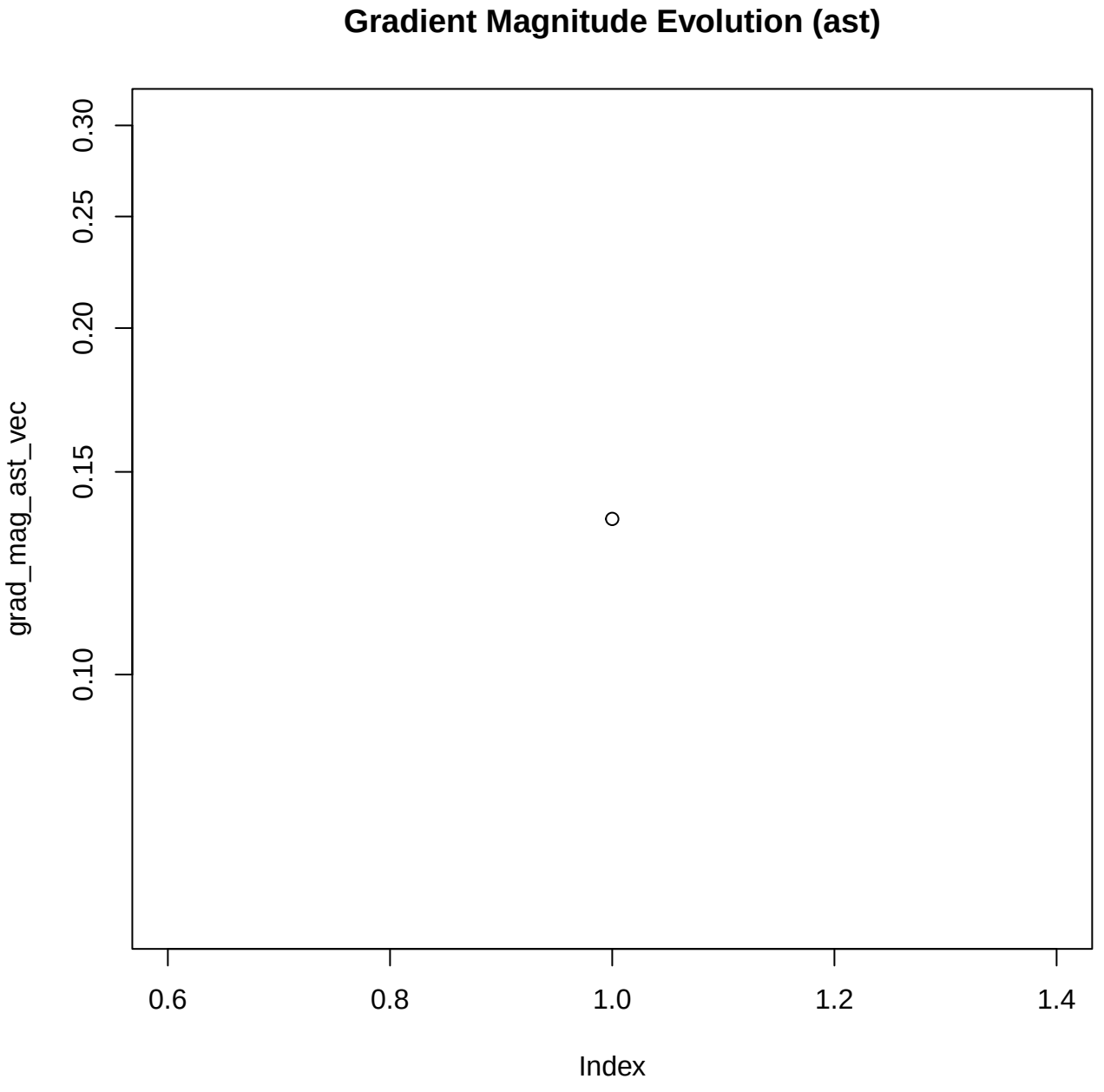
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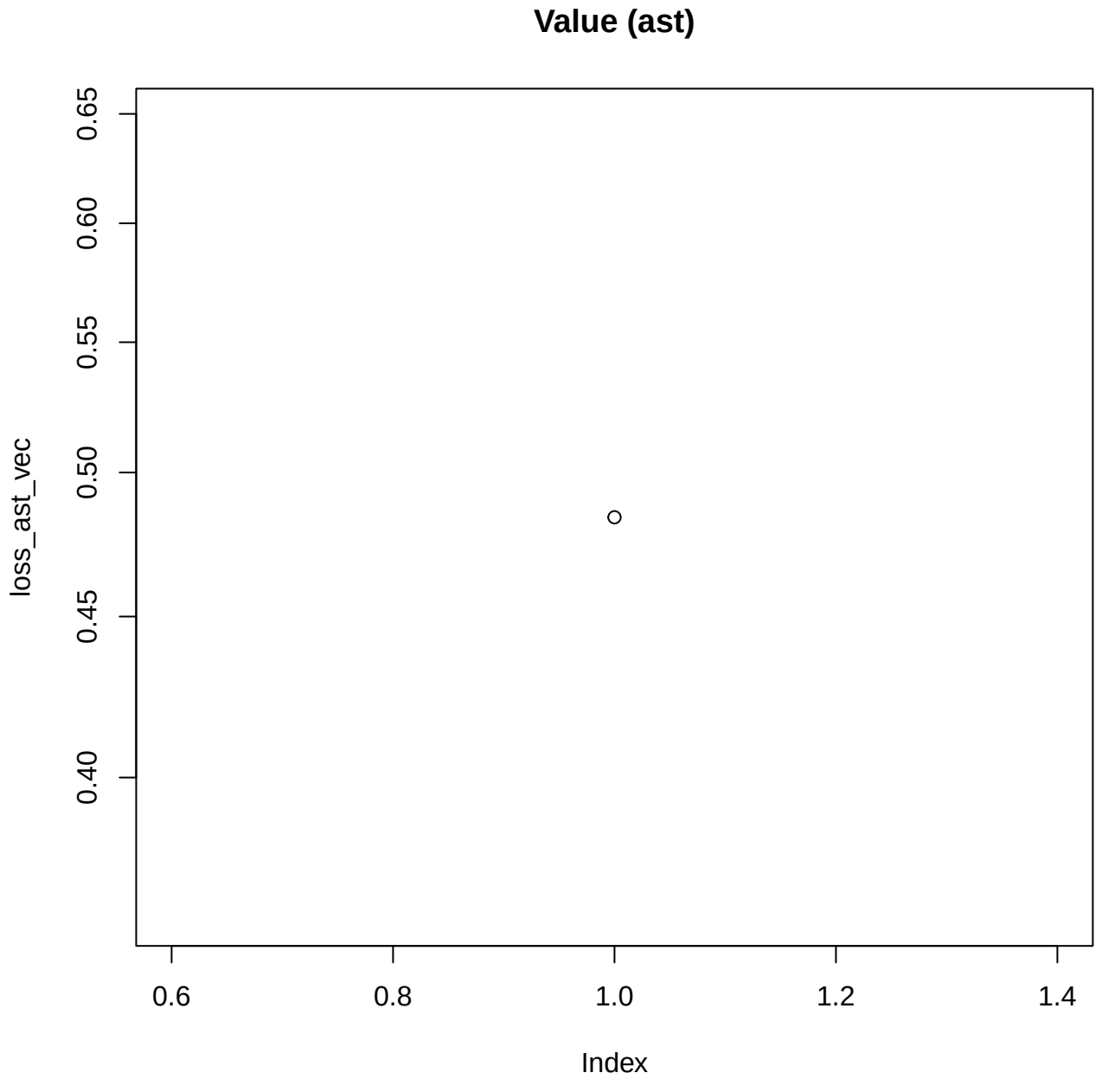
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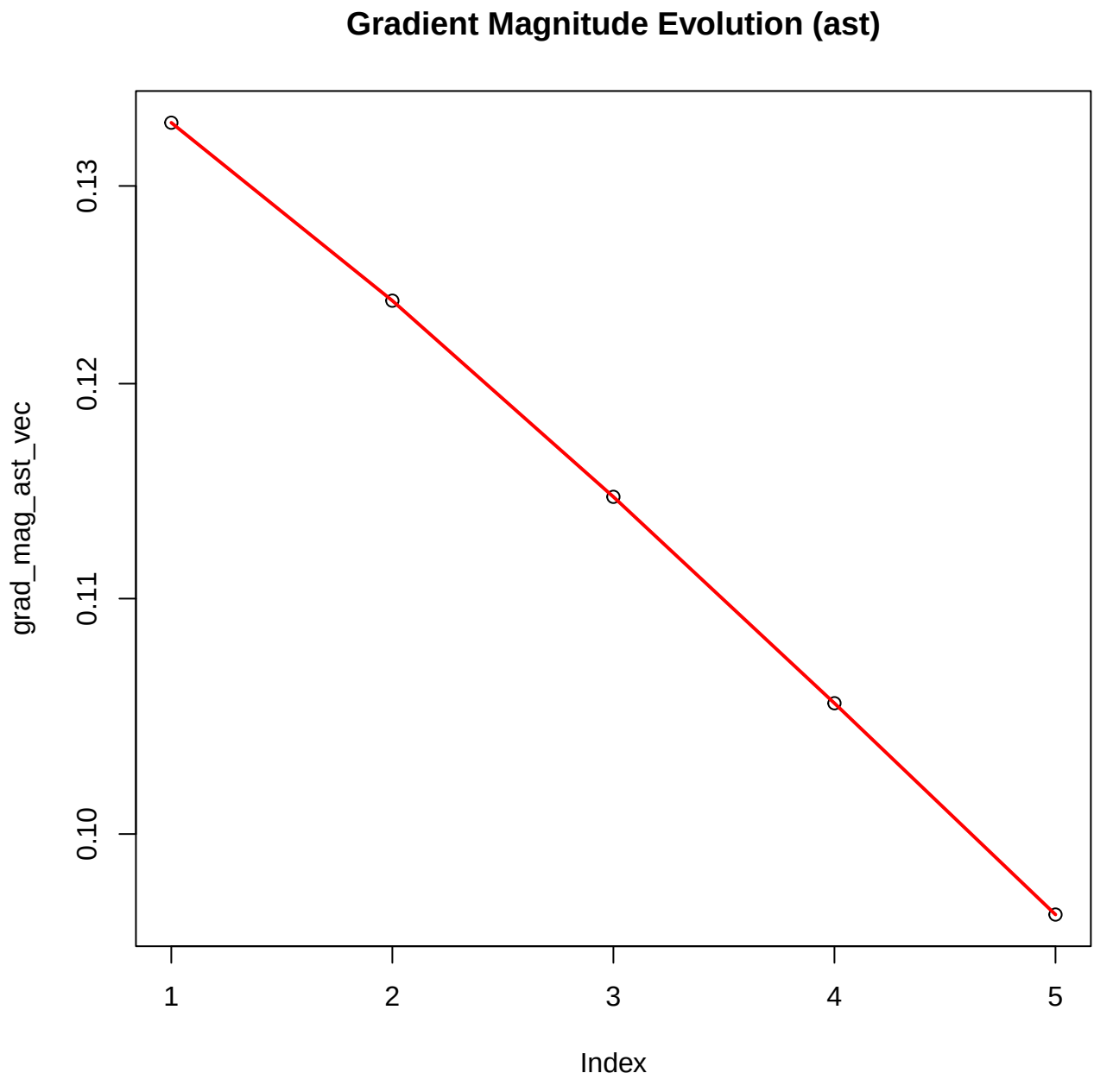
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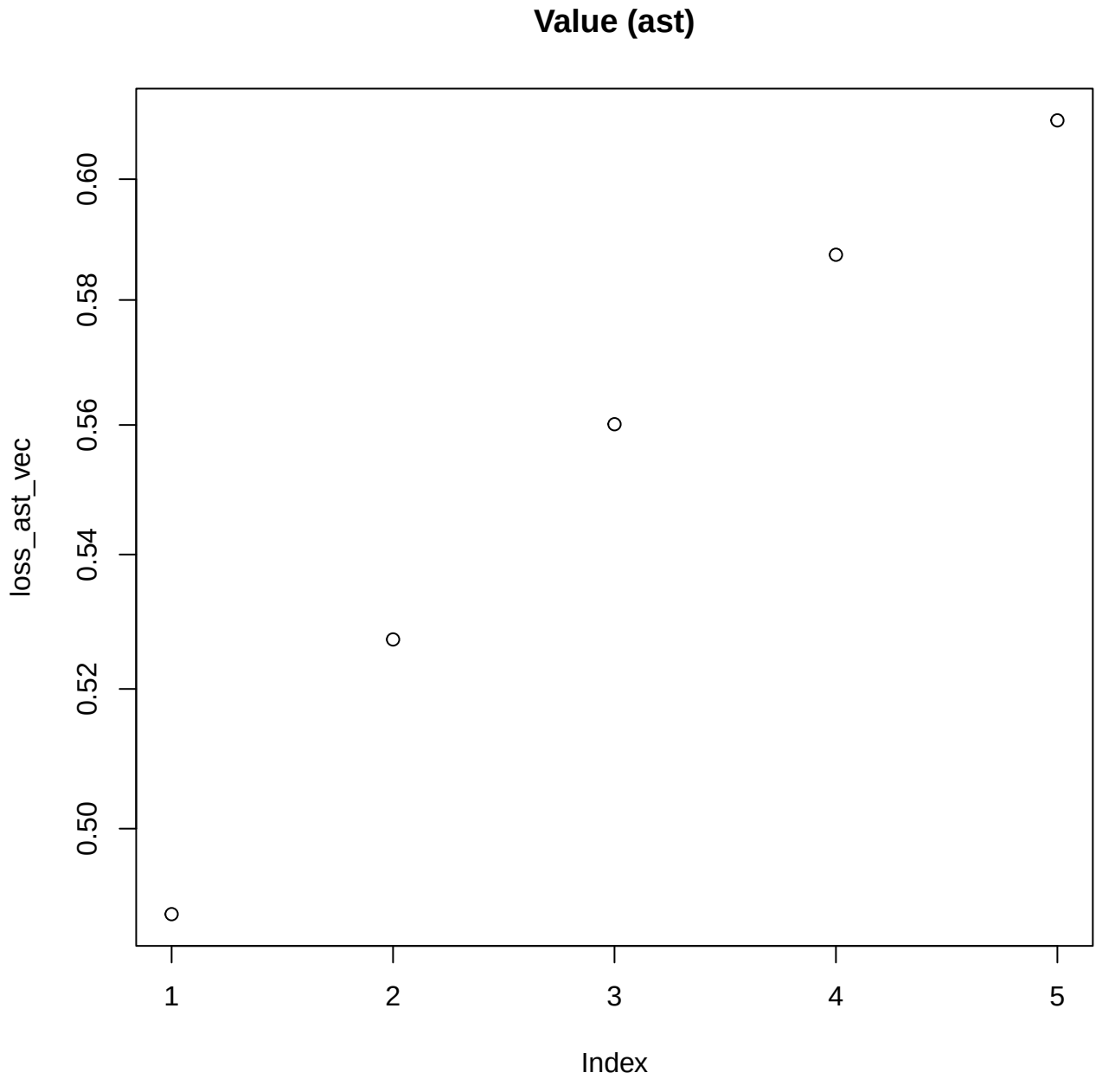
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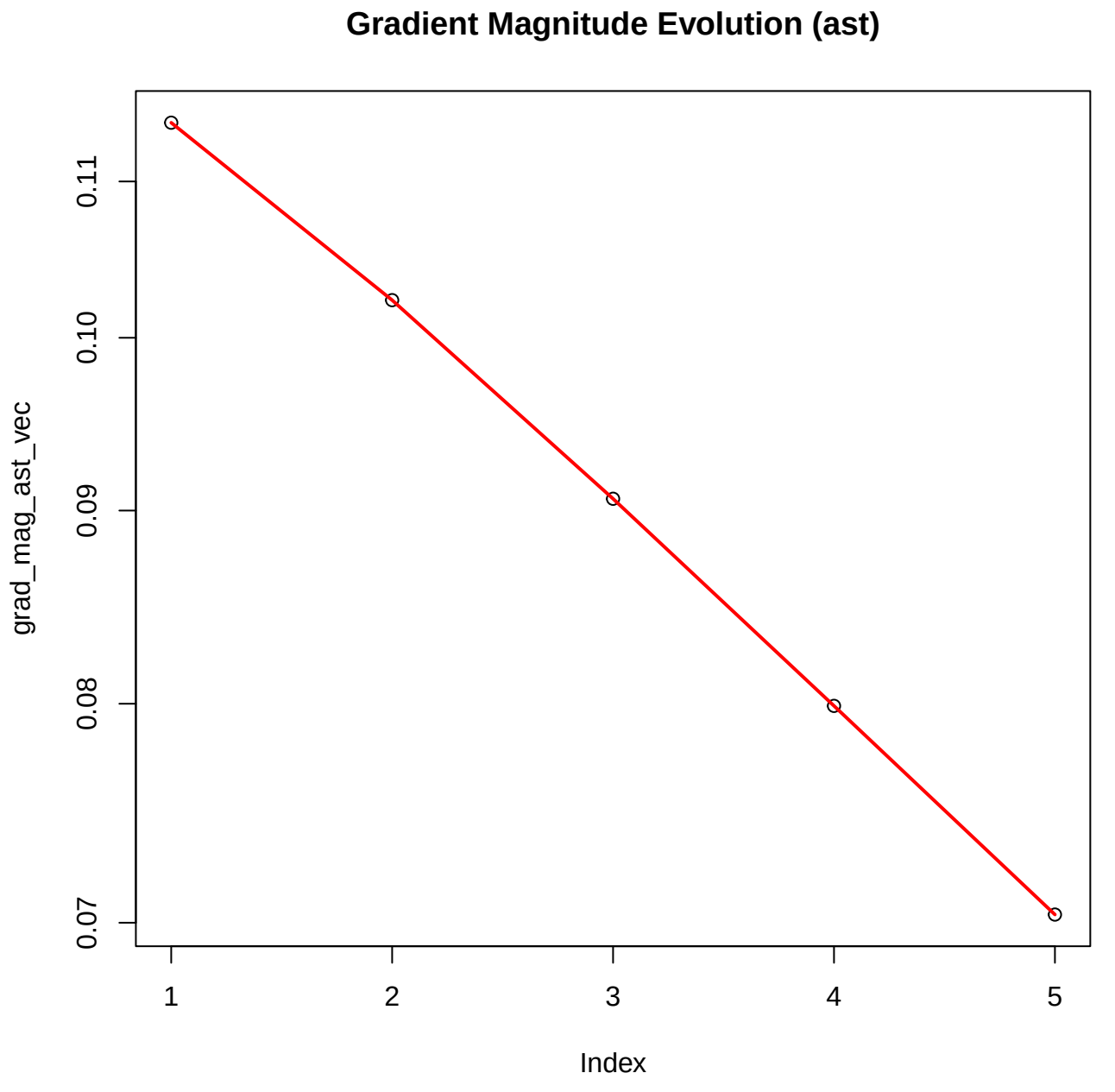
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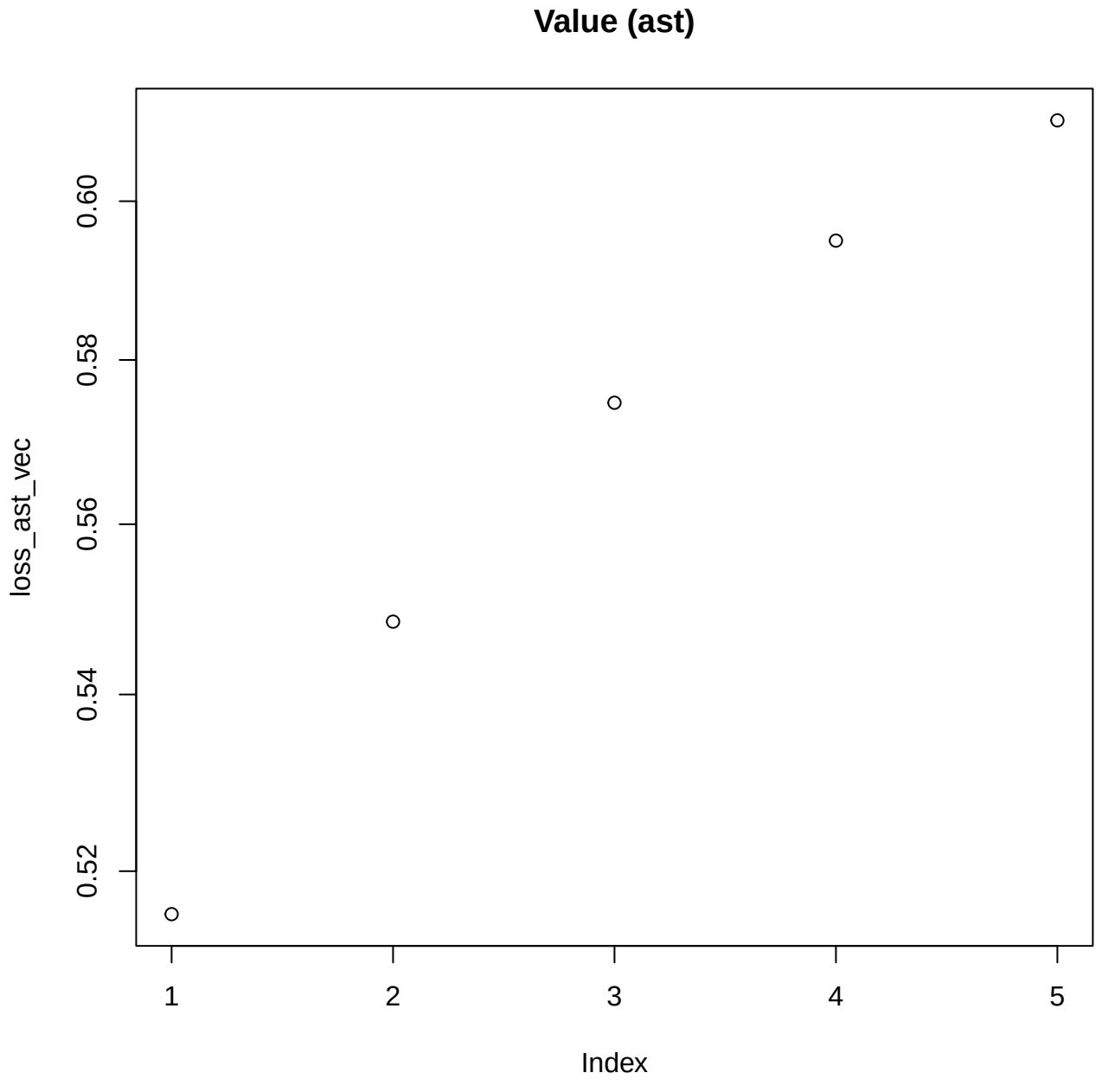
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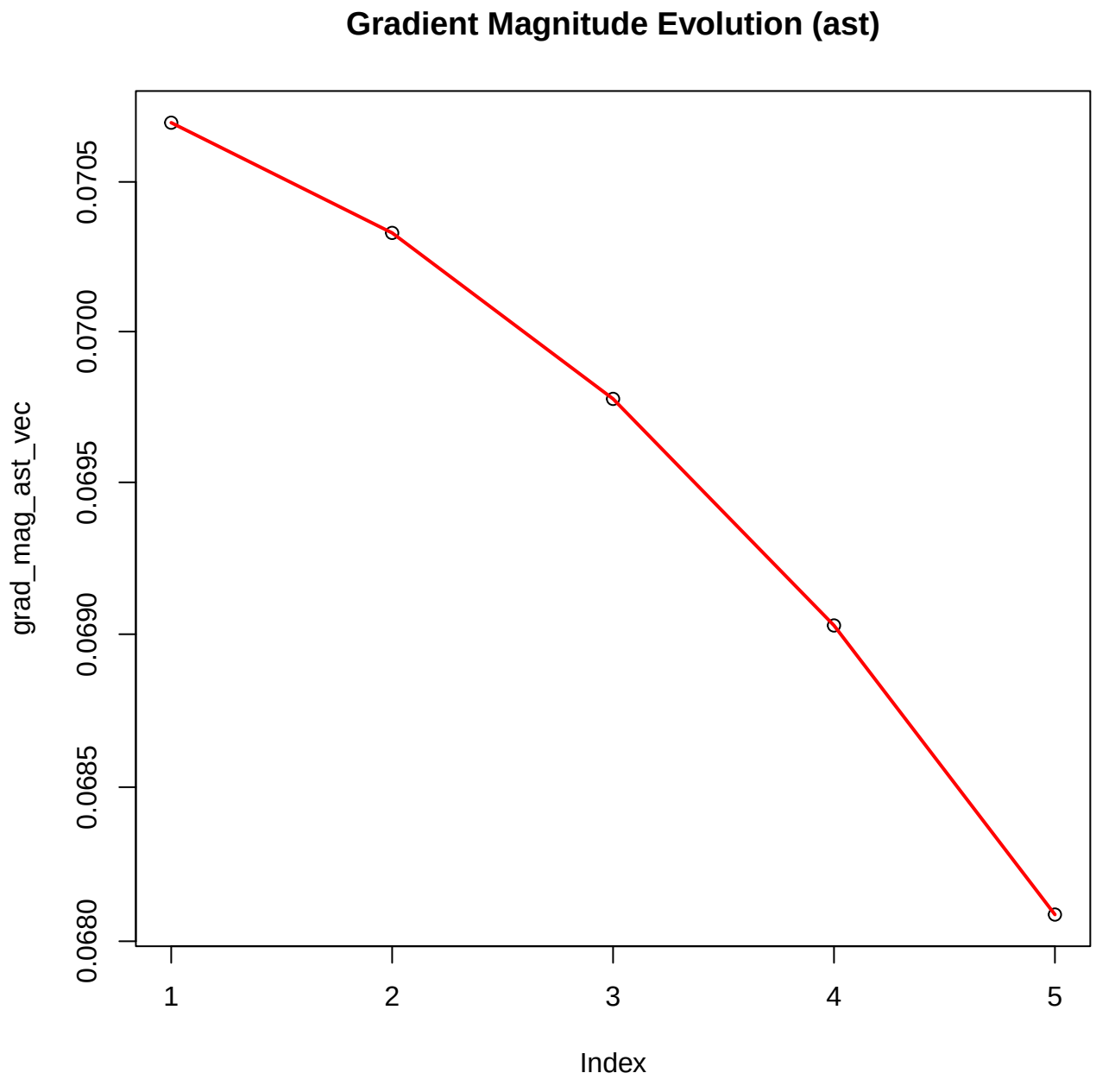
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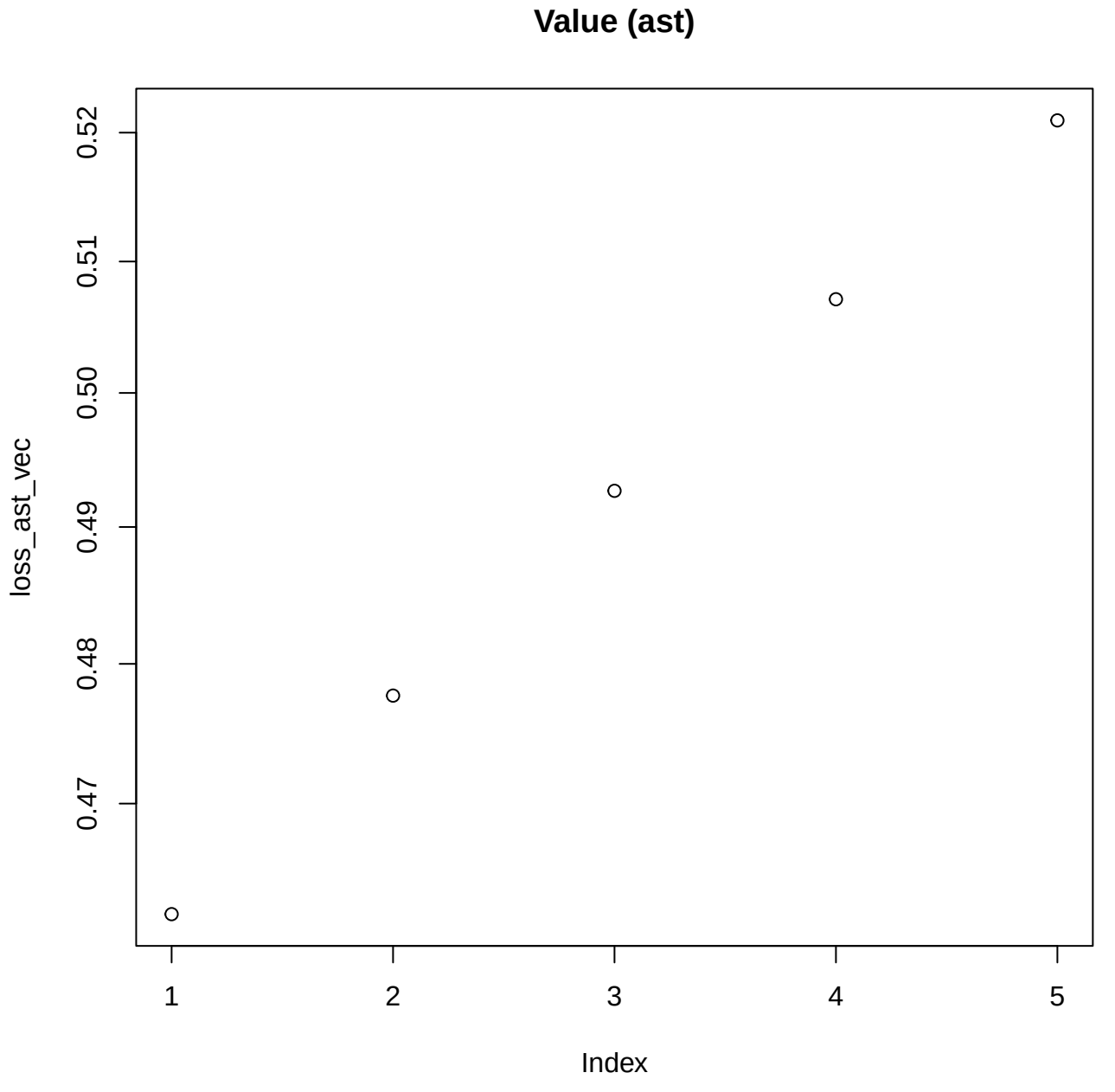
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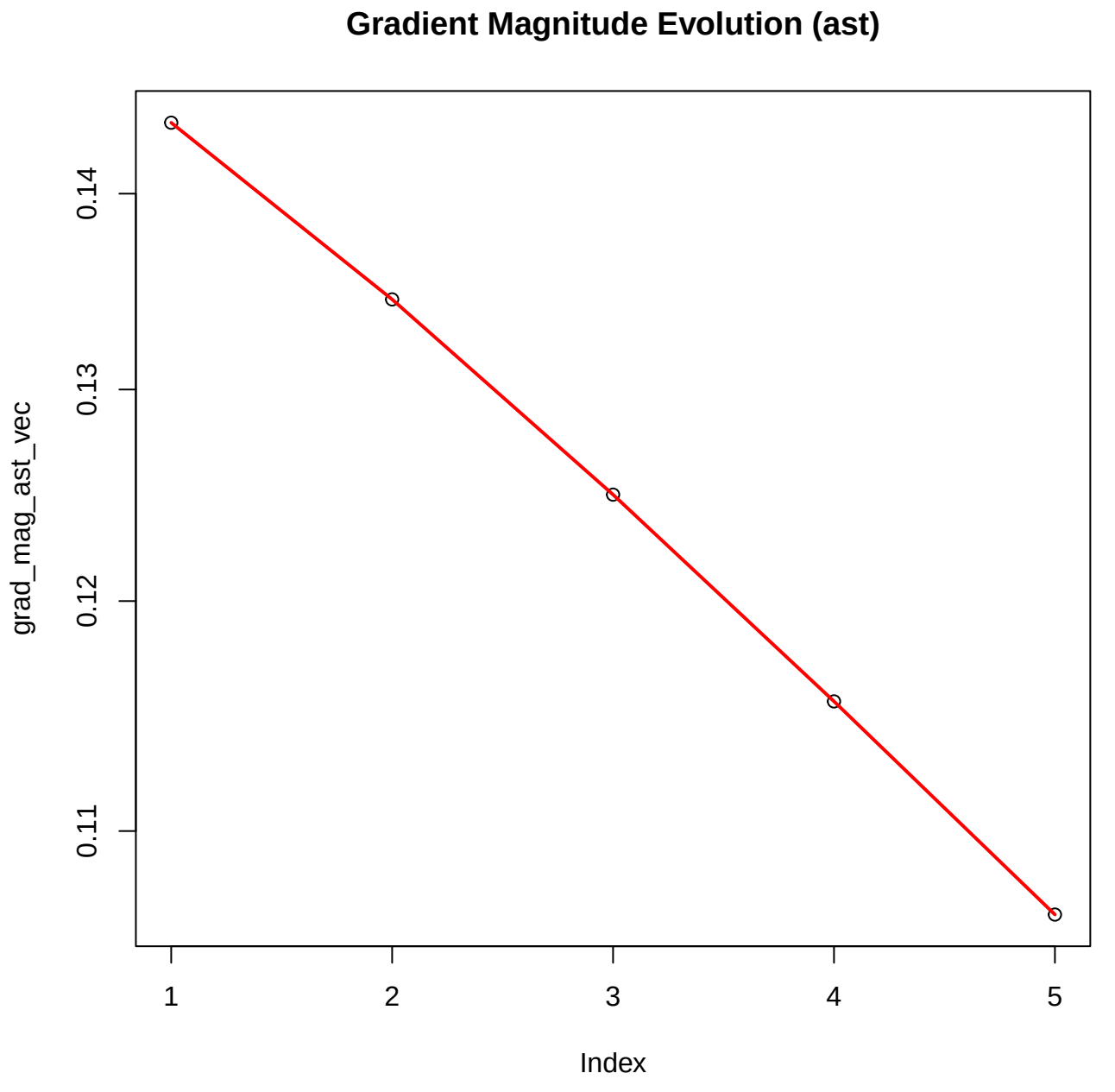
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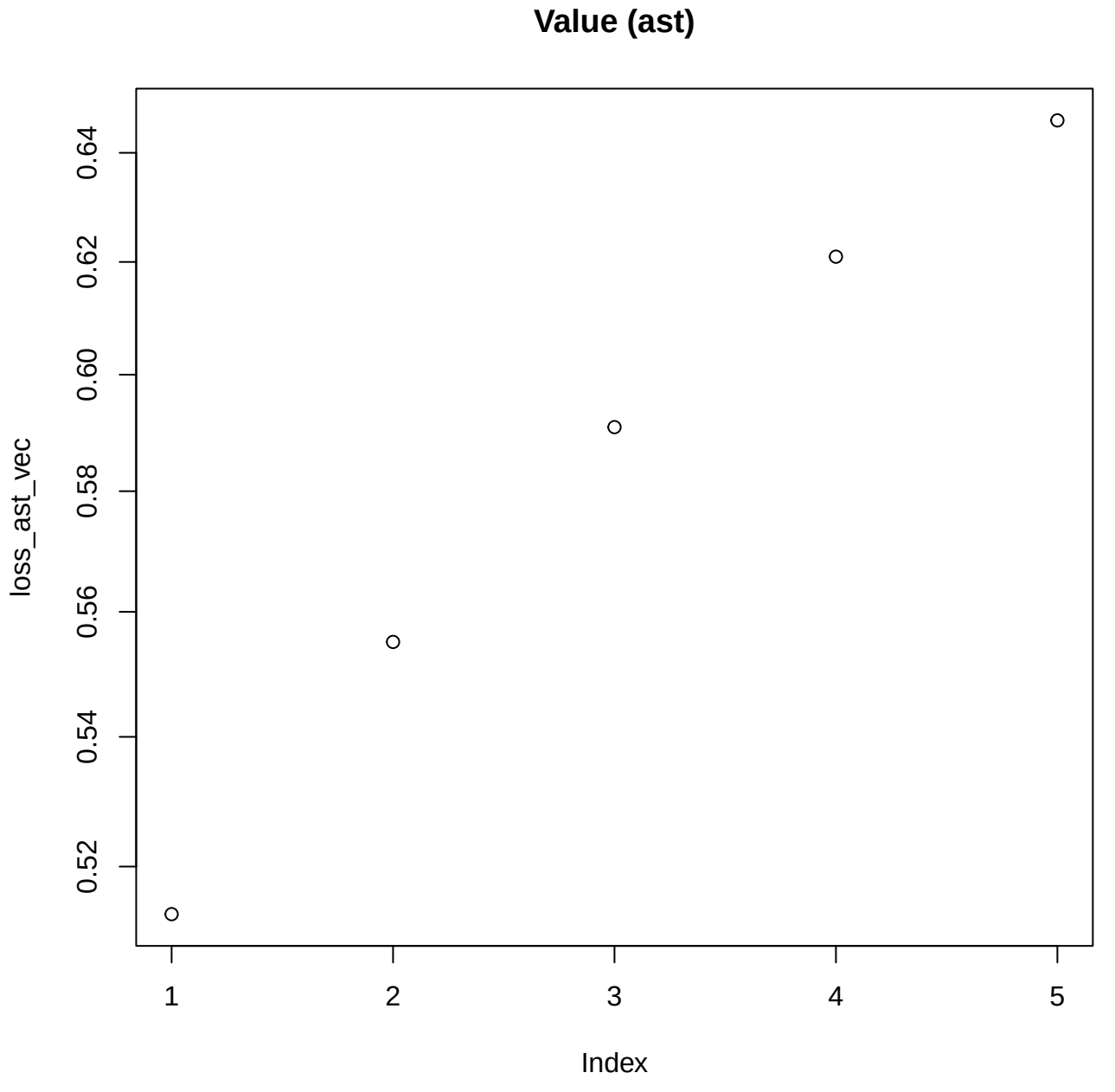
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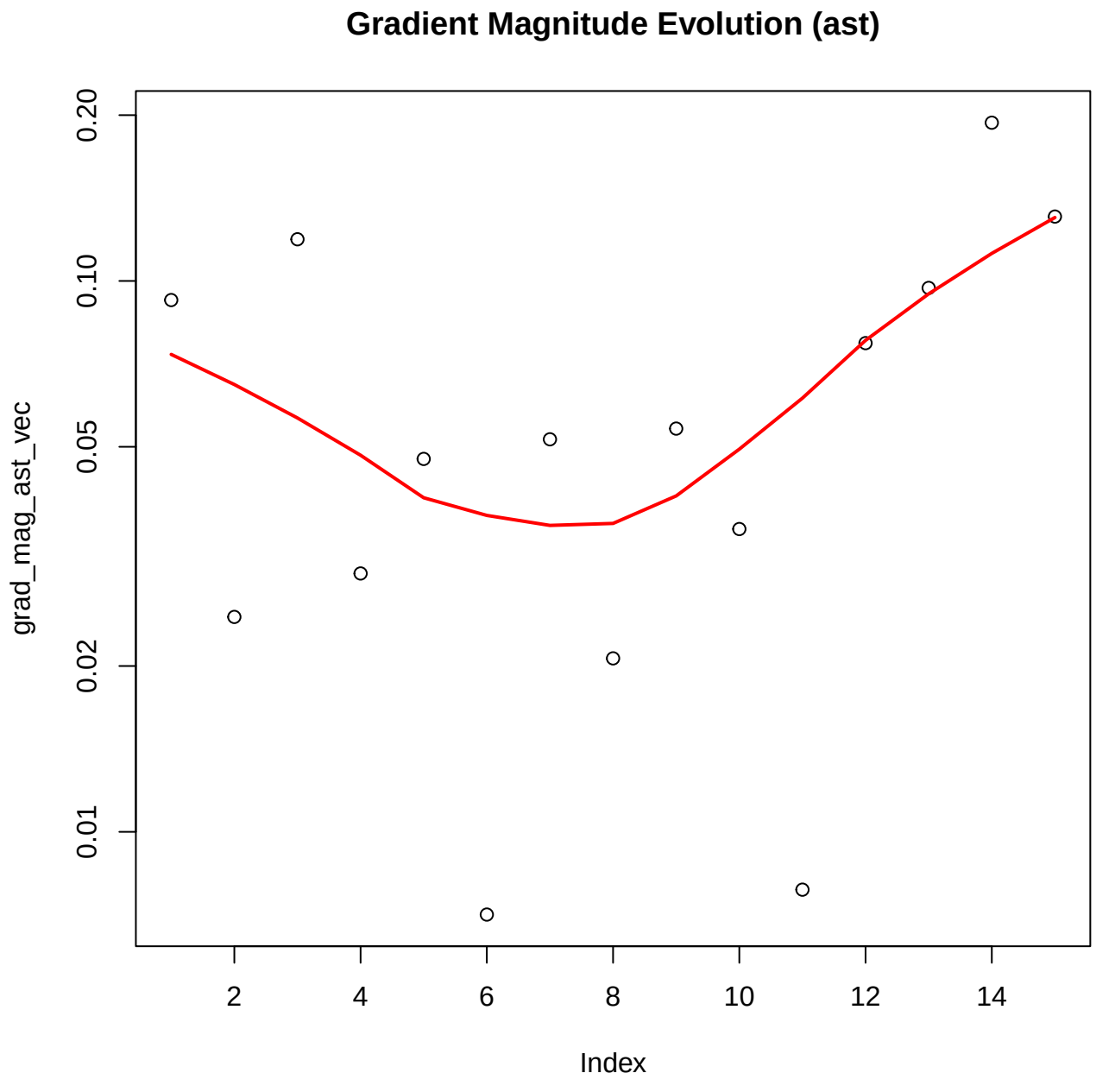
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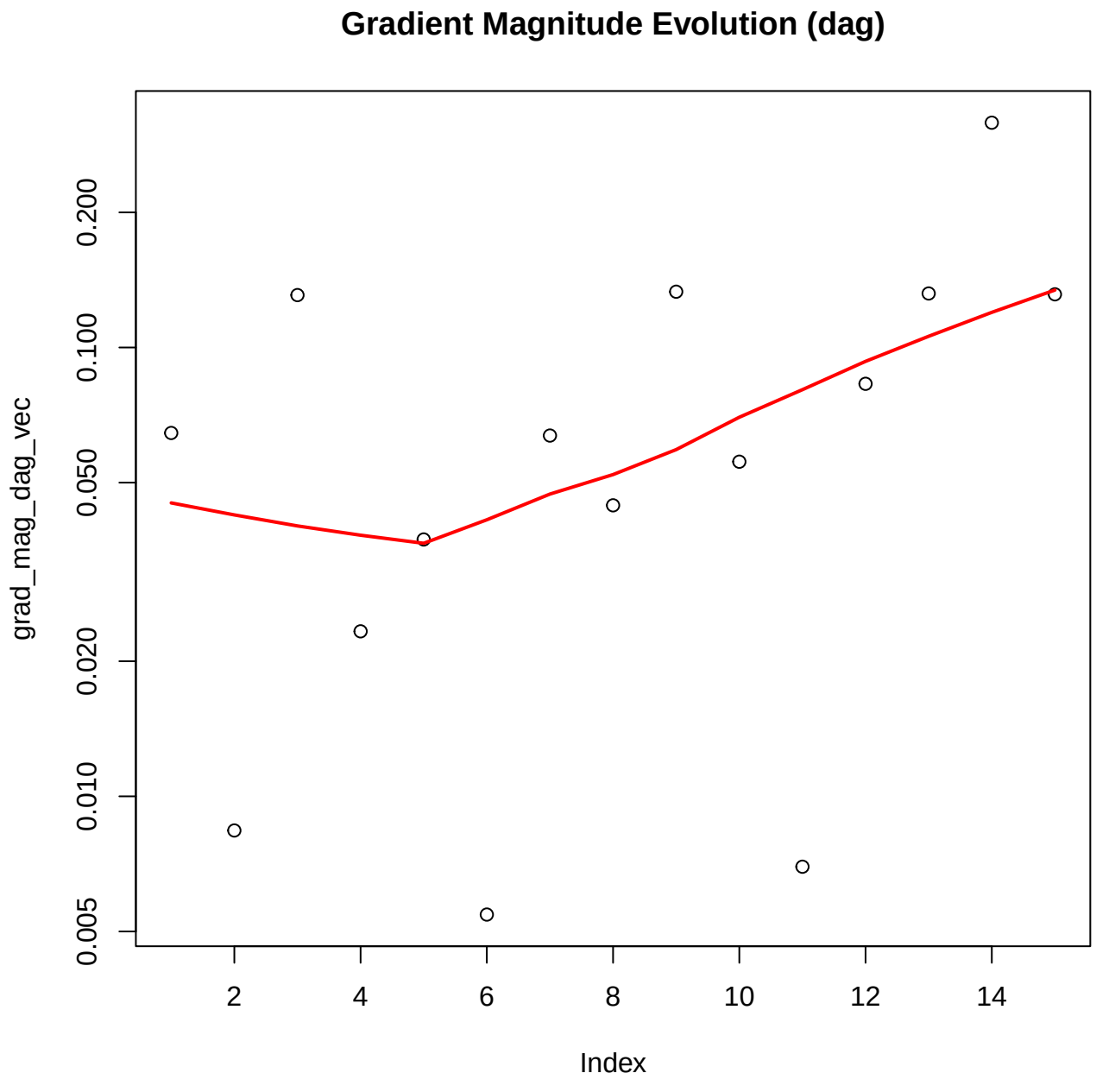
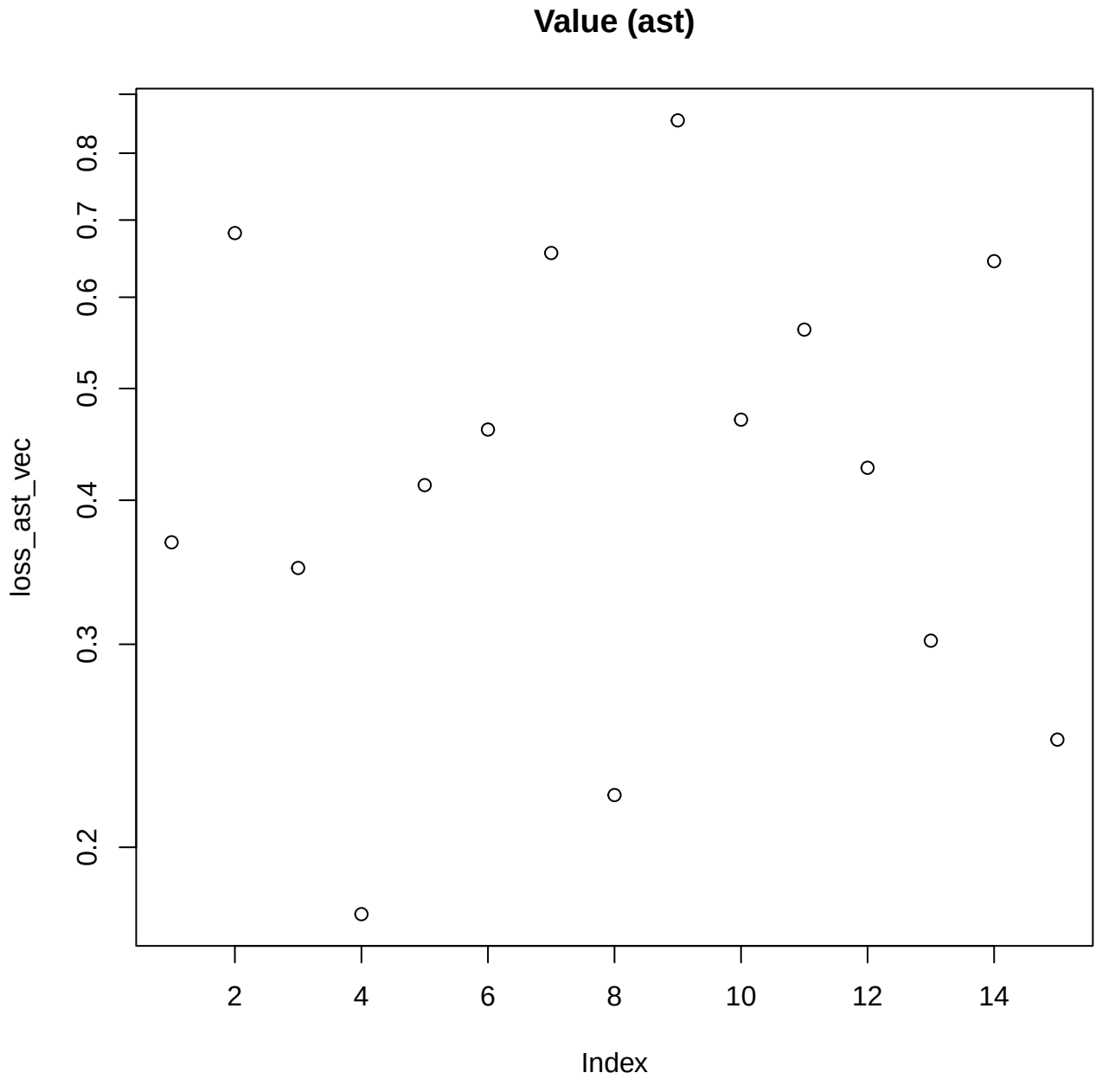
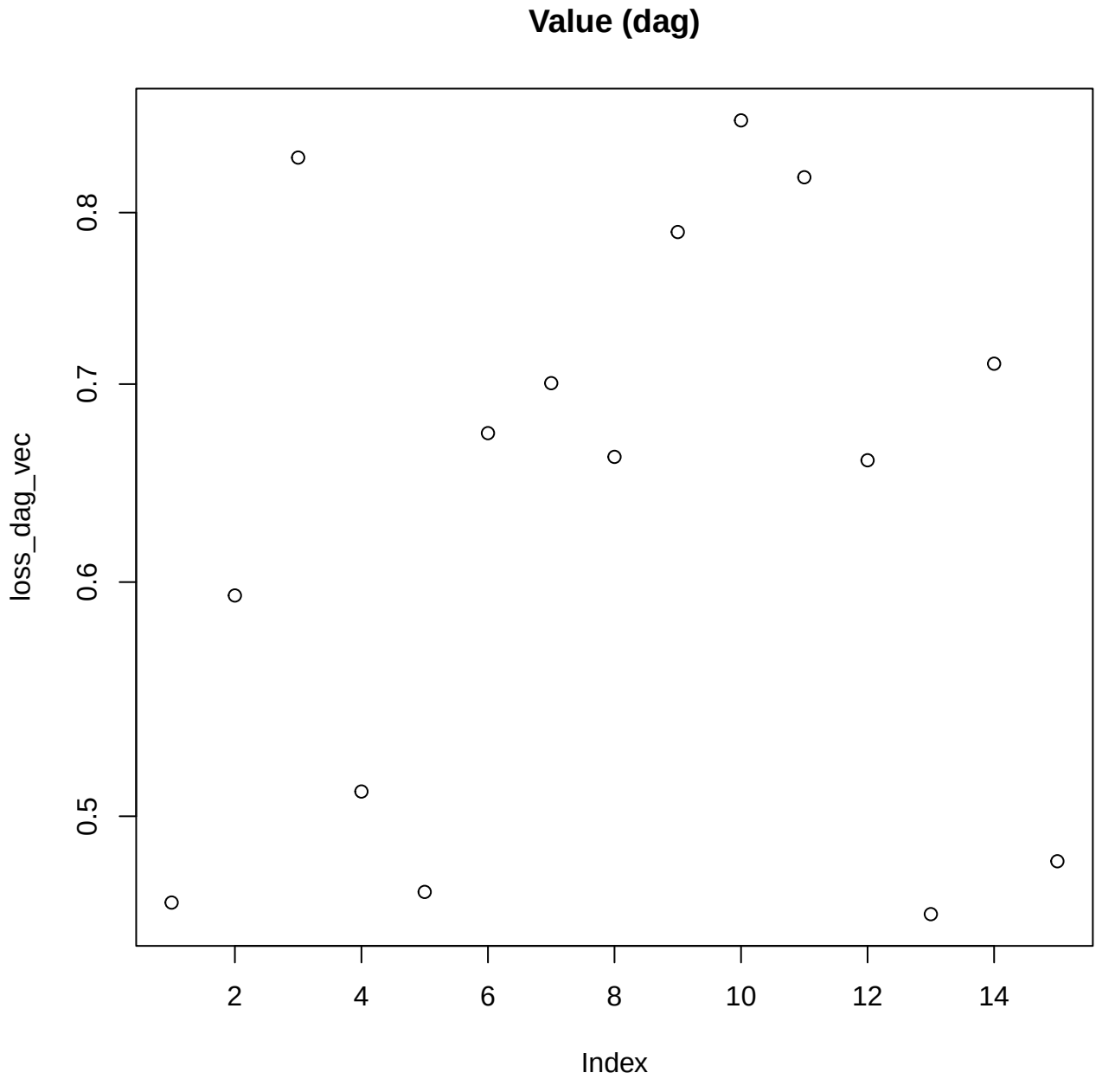
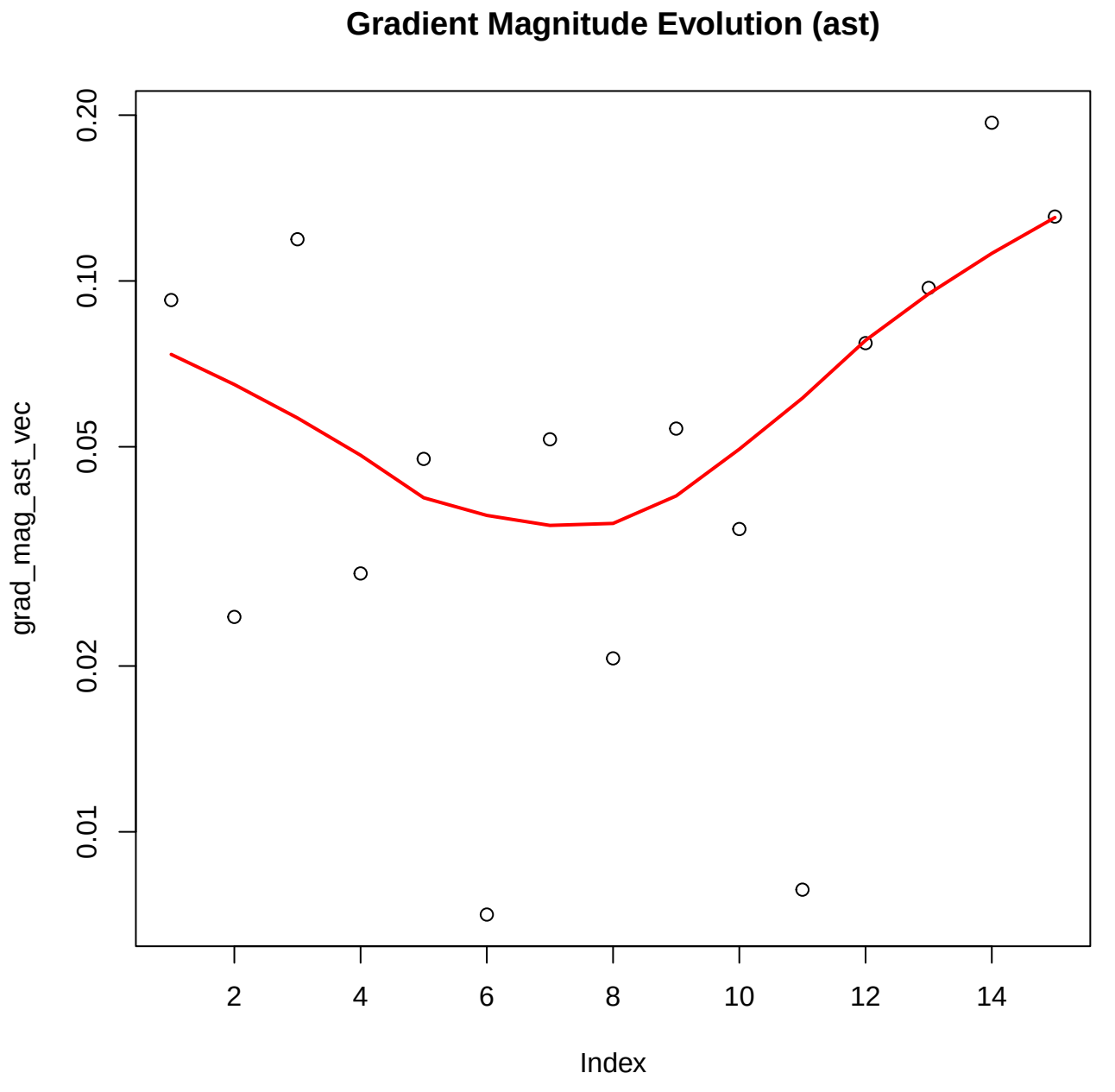
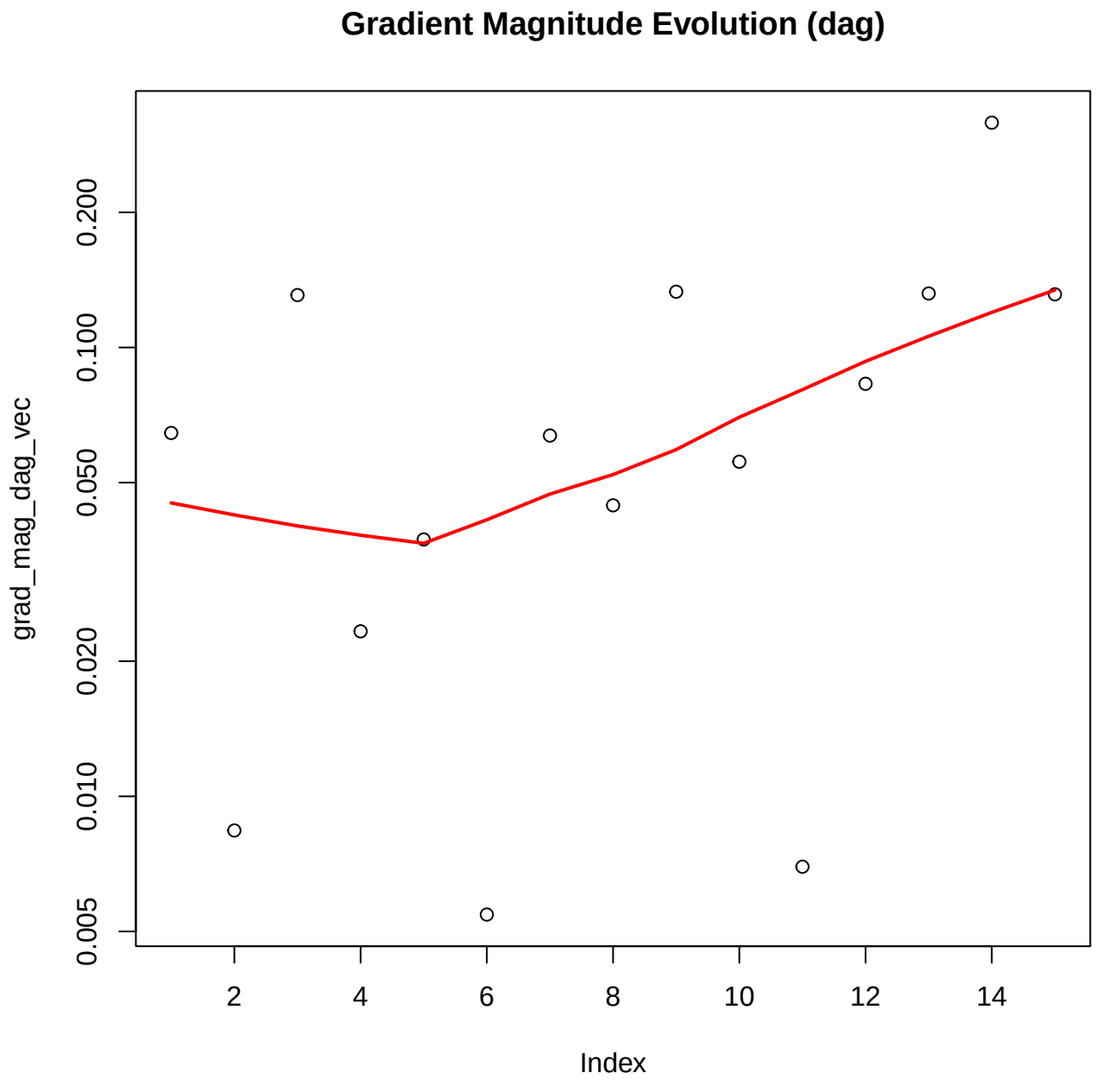
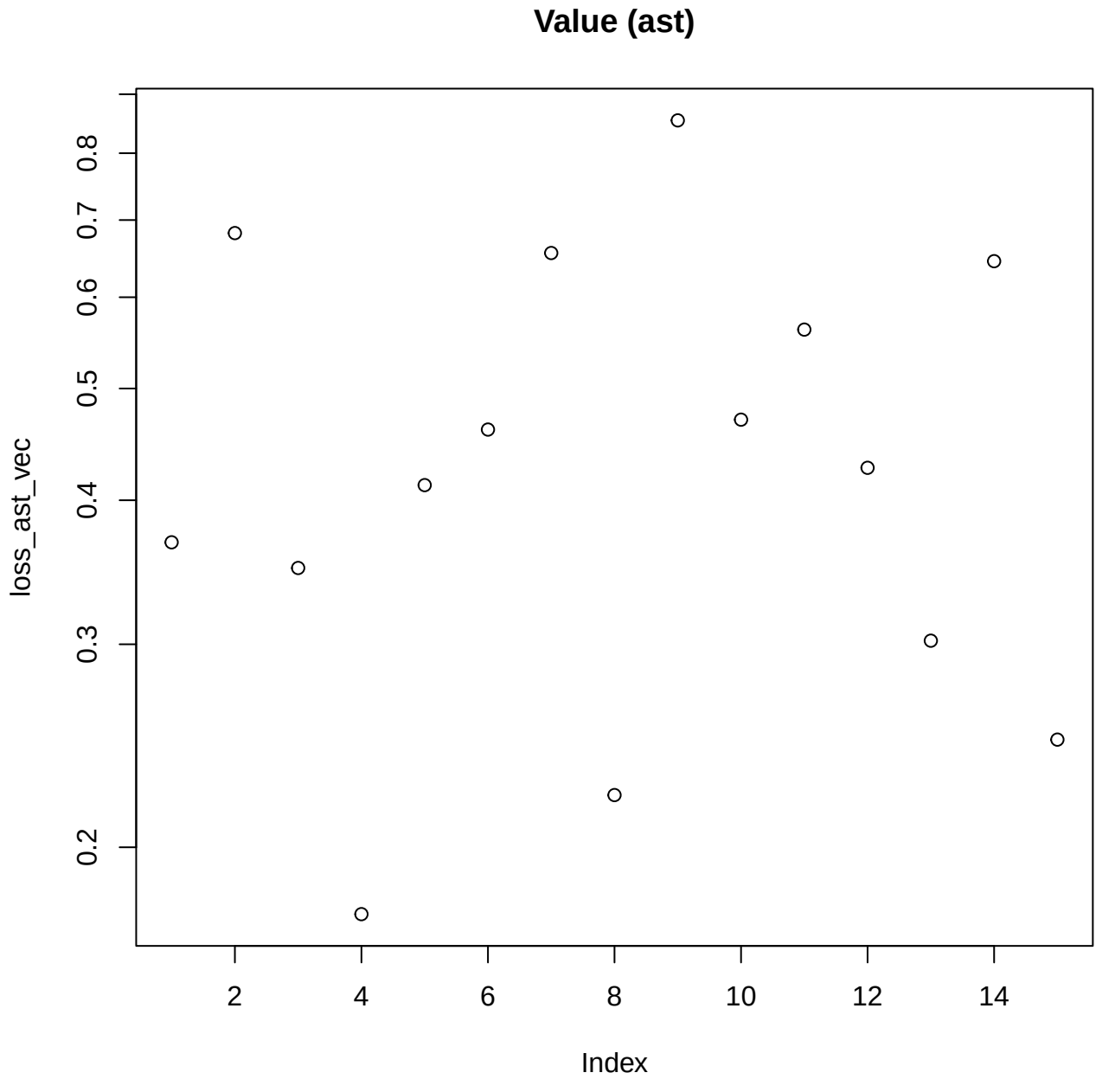

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
2
4
6
8
10
12
14
Index

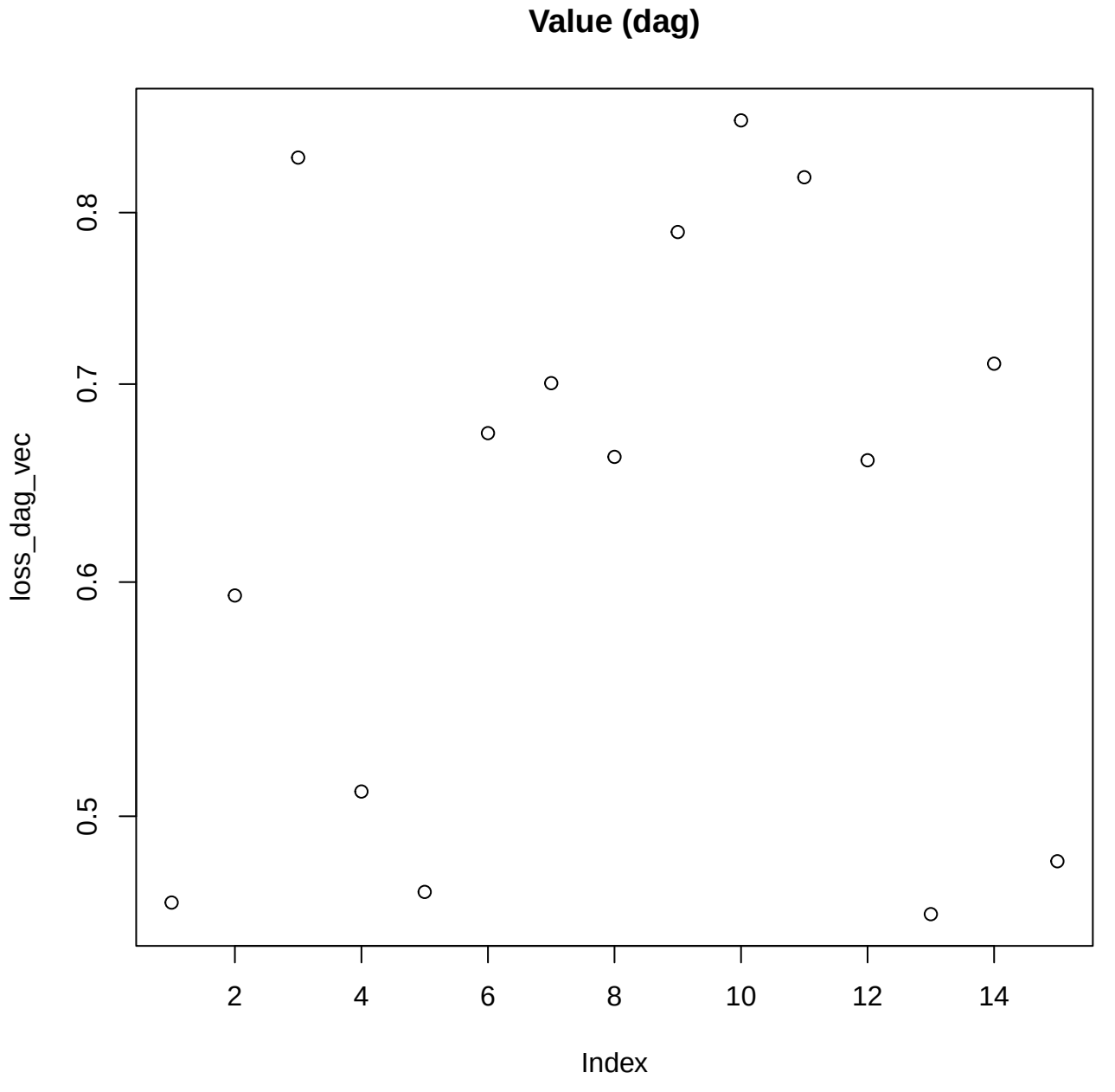

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

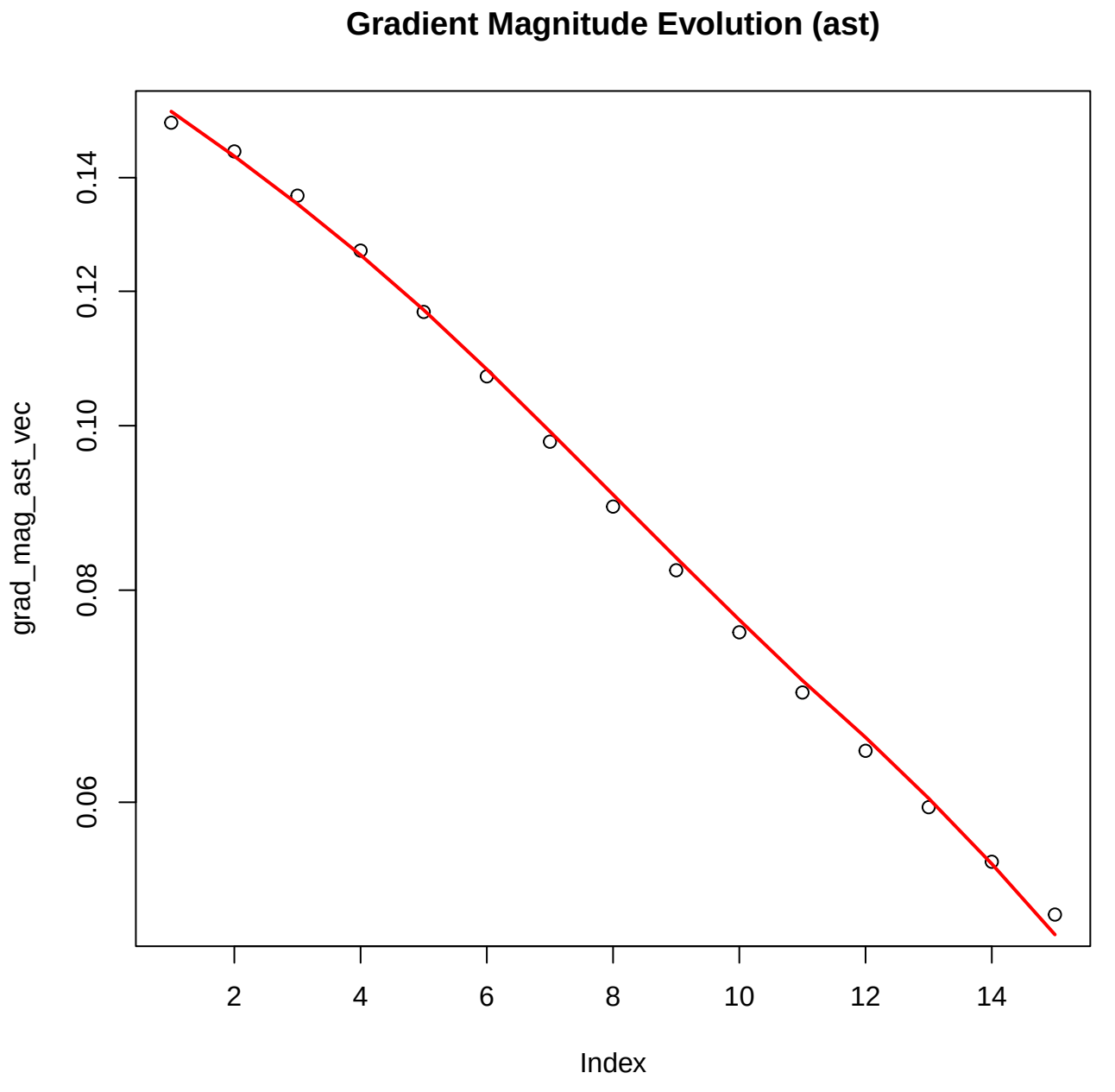

Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

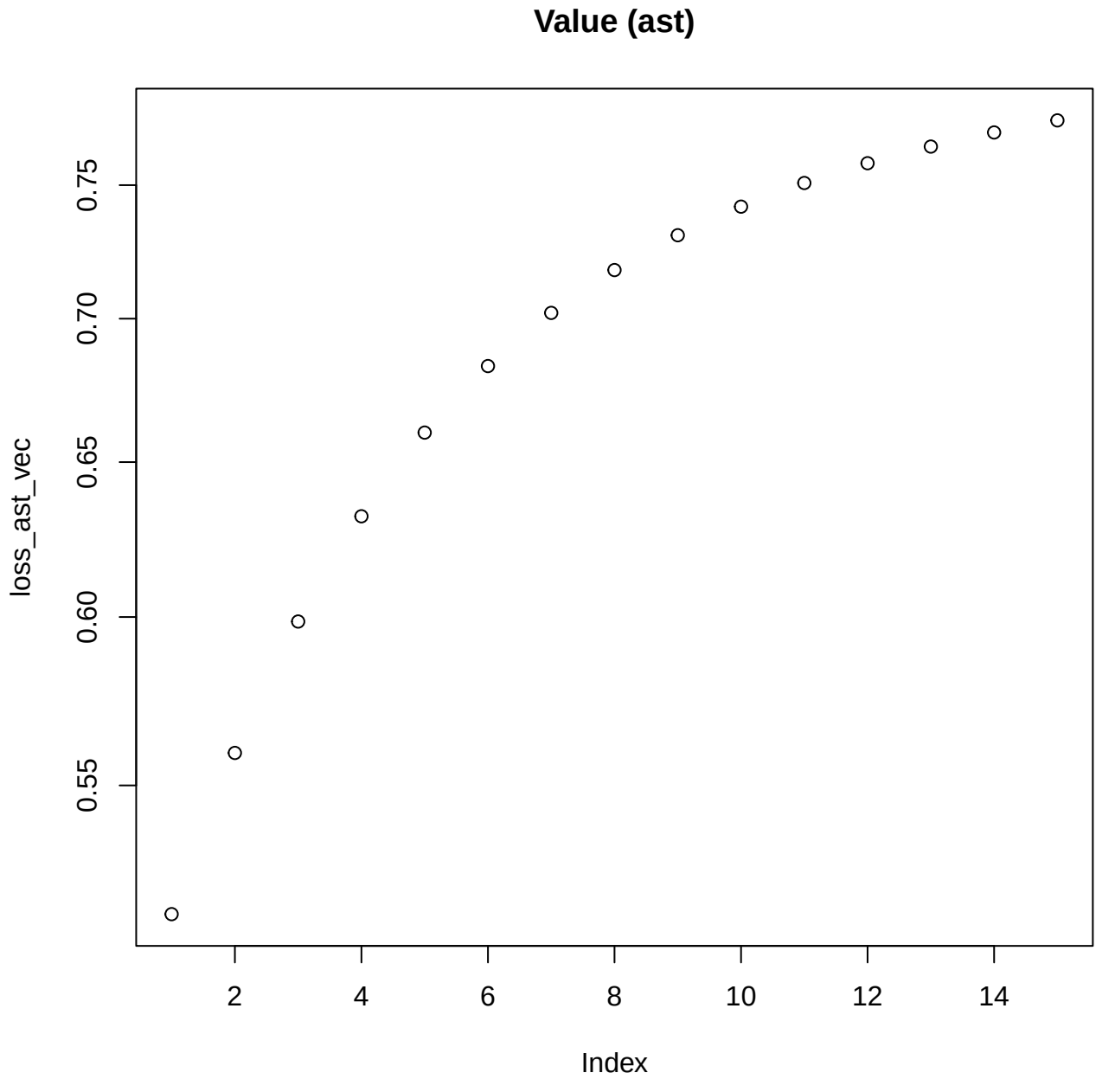

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
0.20
2
4
6
8
10
12
14
Index

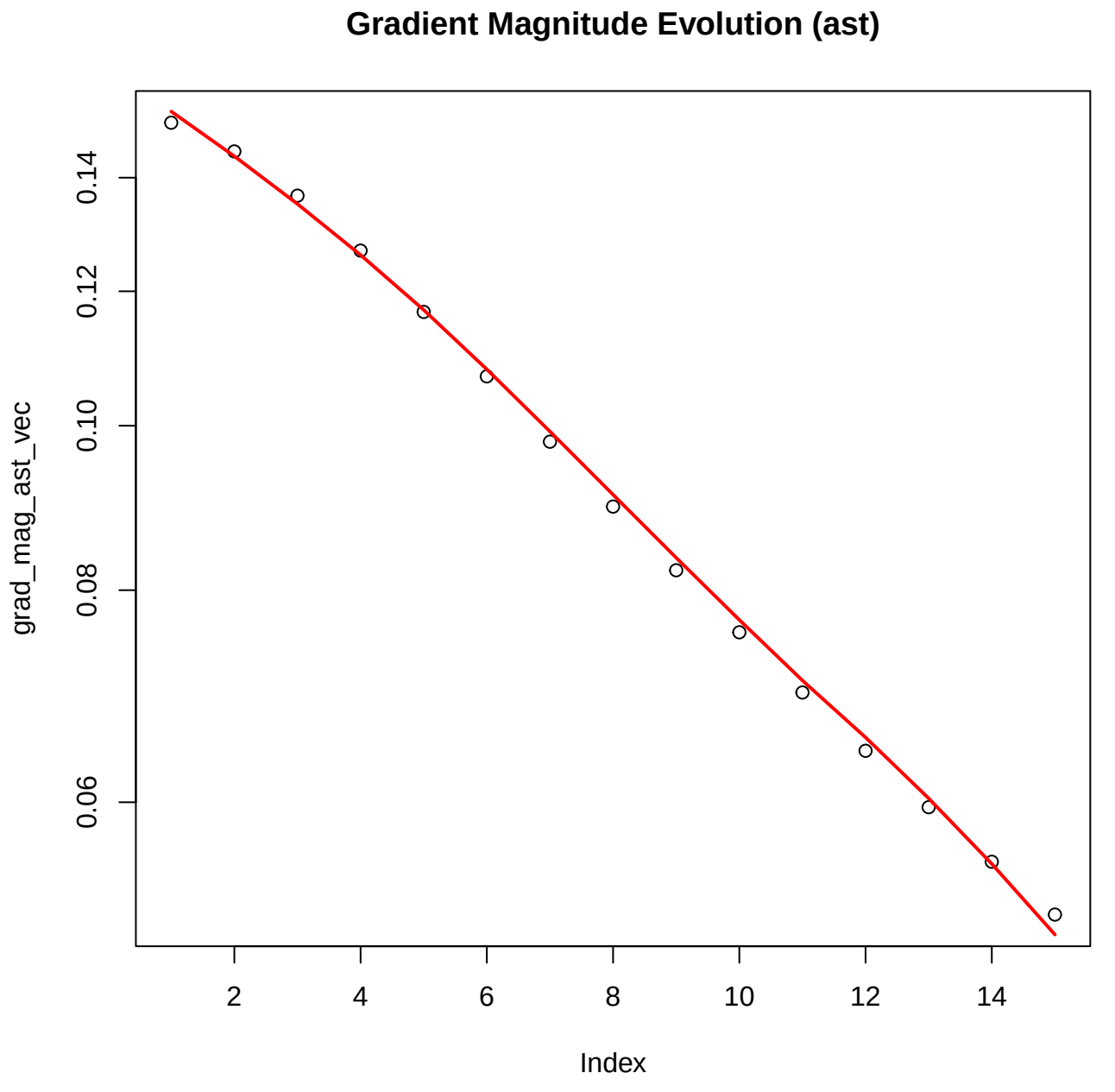

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
2
4
6
8
10
12
14
Index

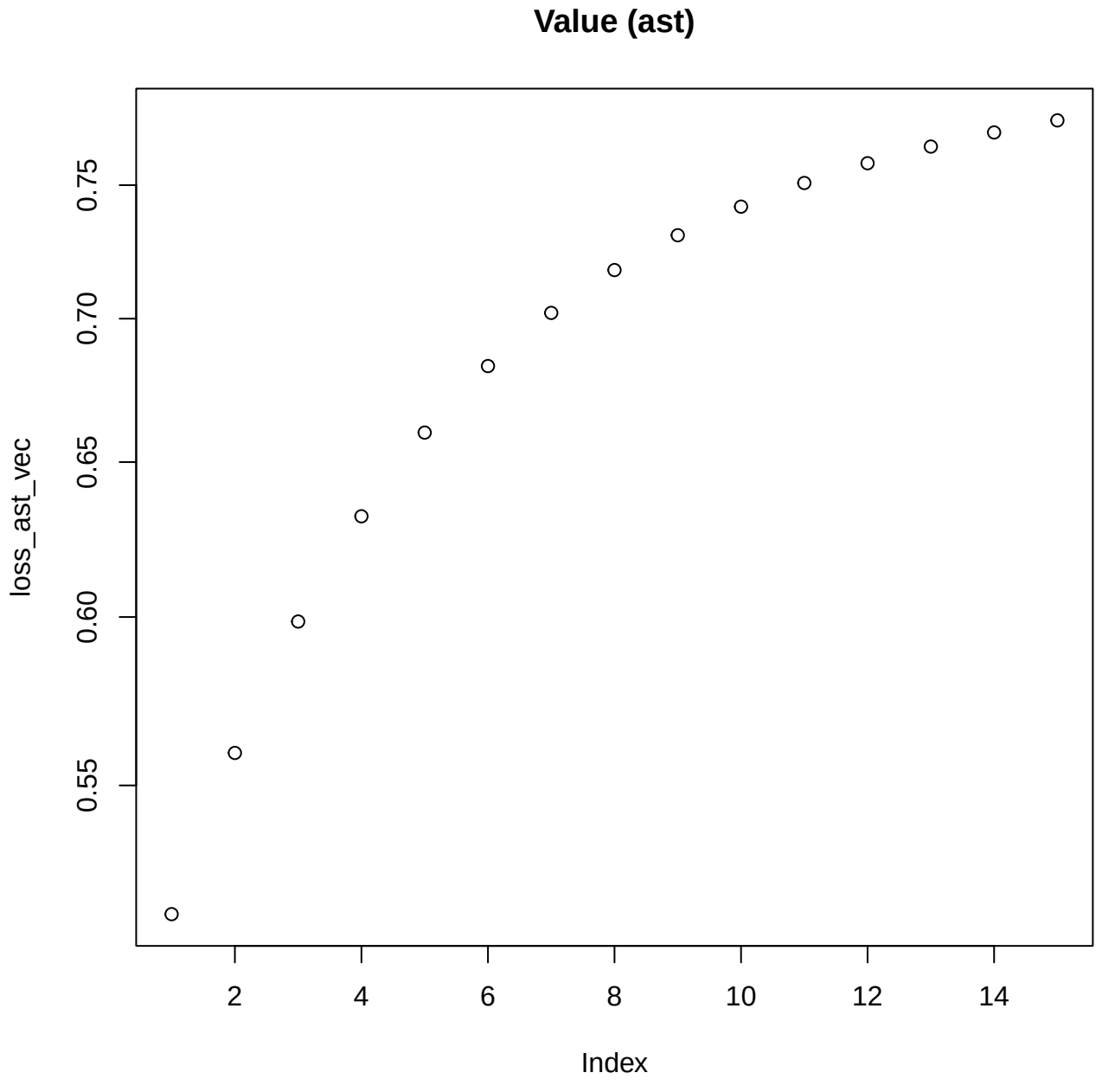

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

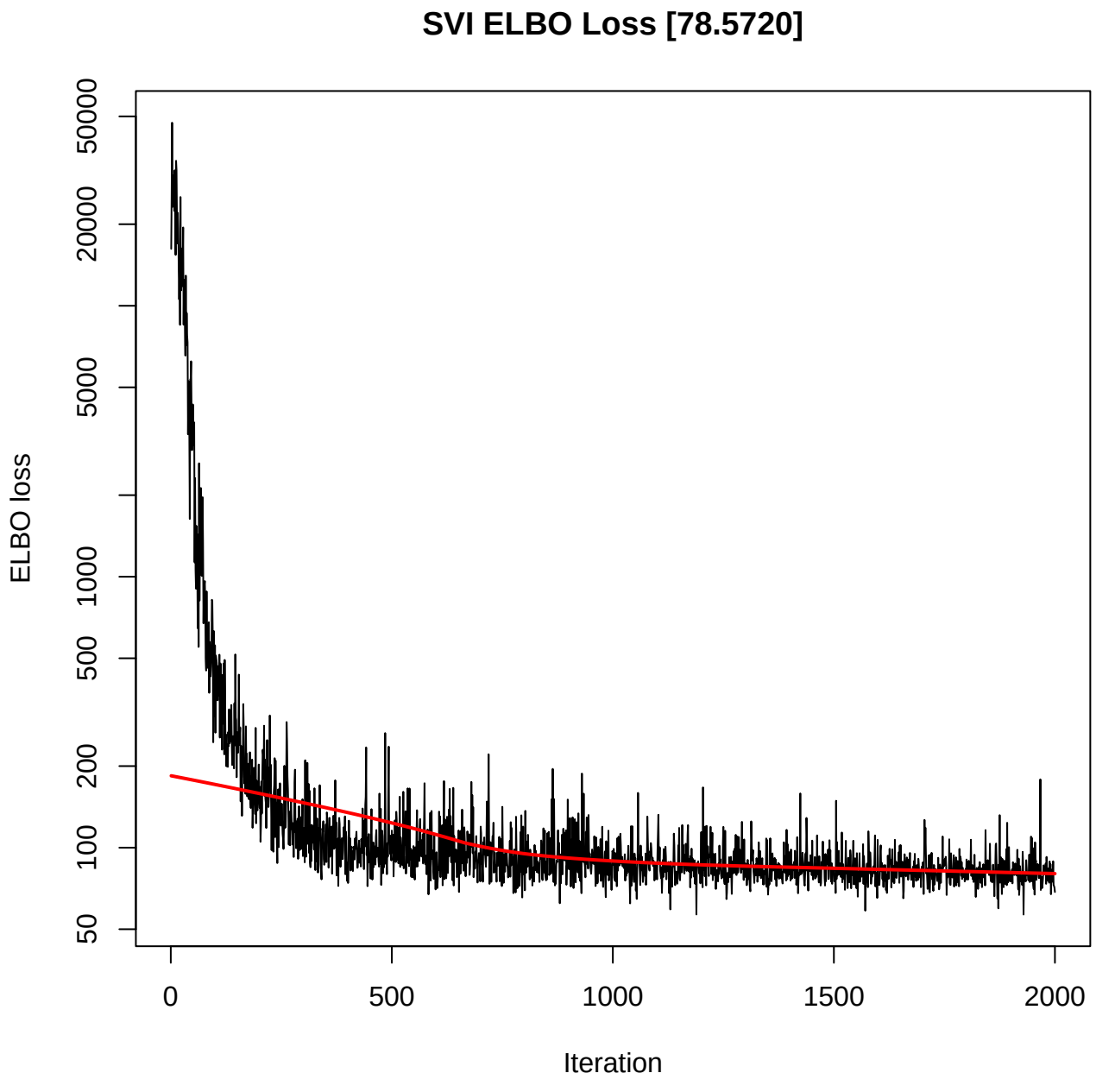

Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

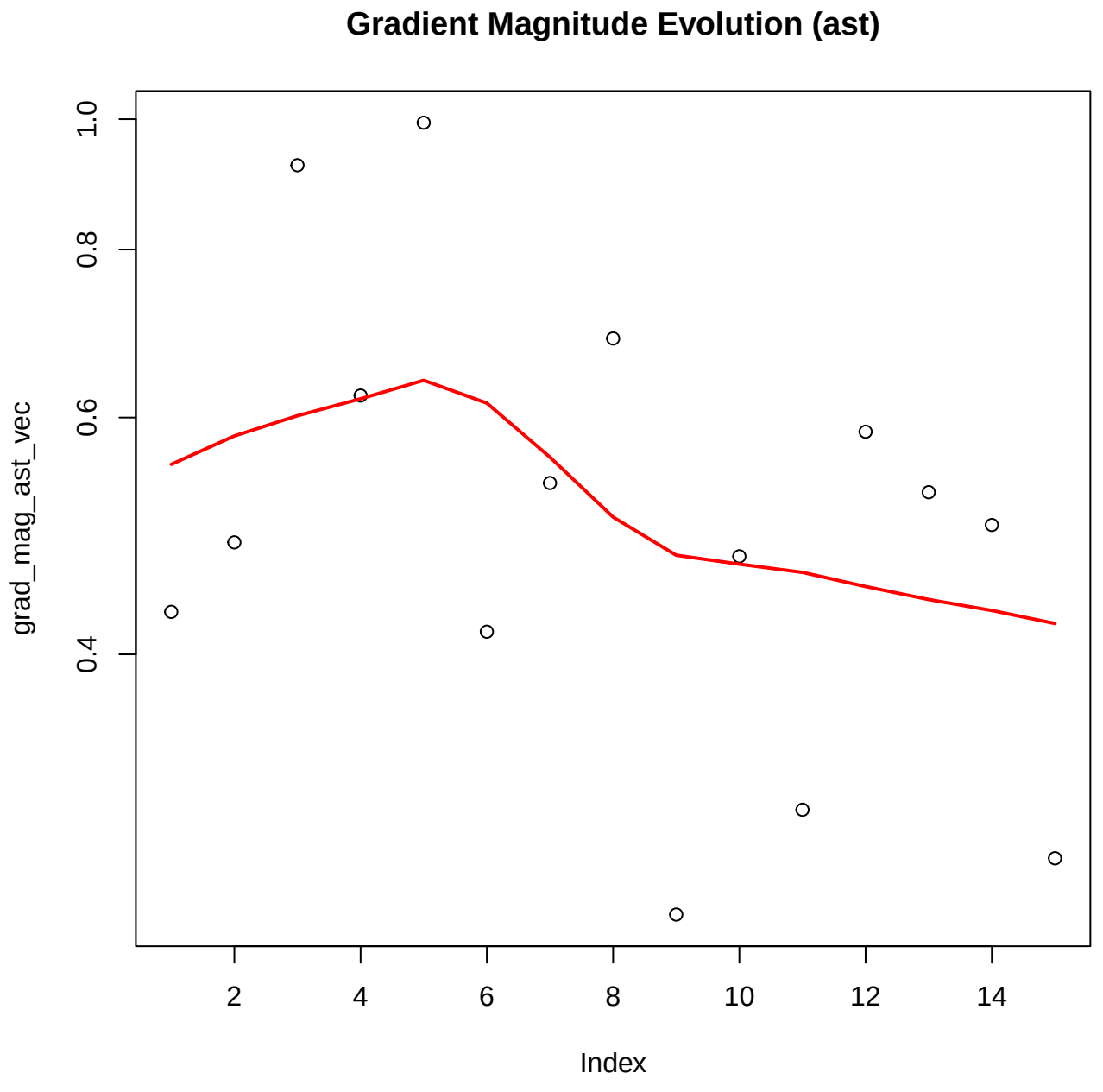

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.06
0.08
0.10
0.12
0.14
2
4
6
8
10
12
14
Index

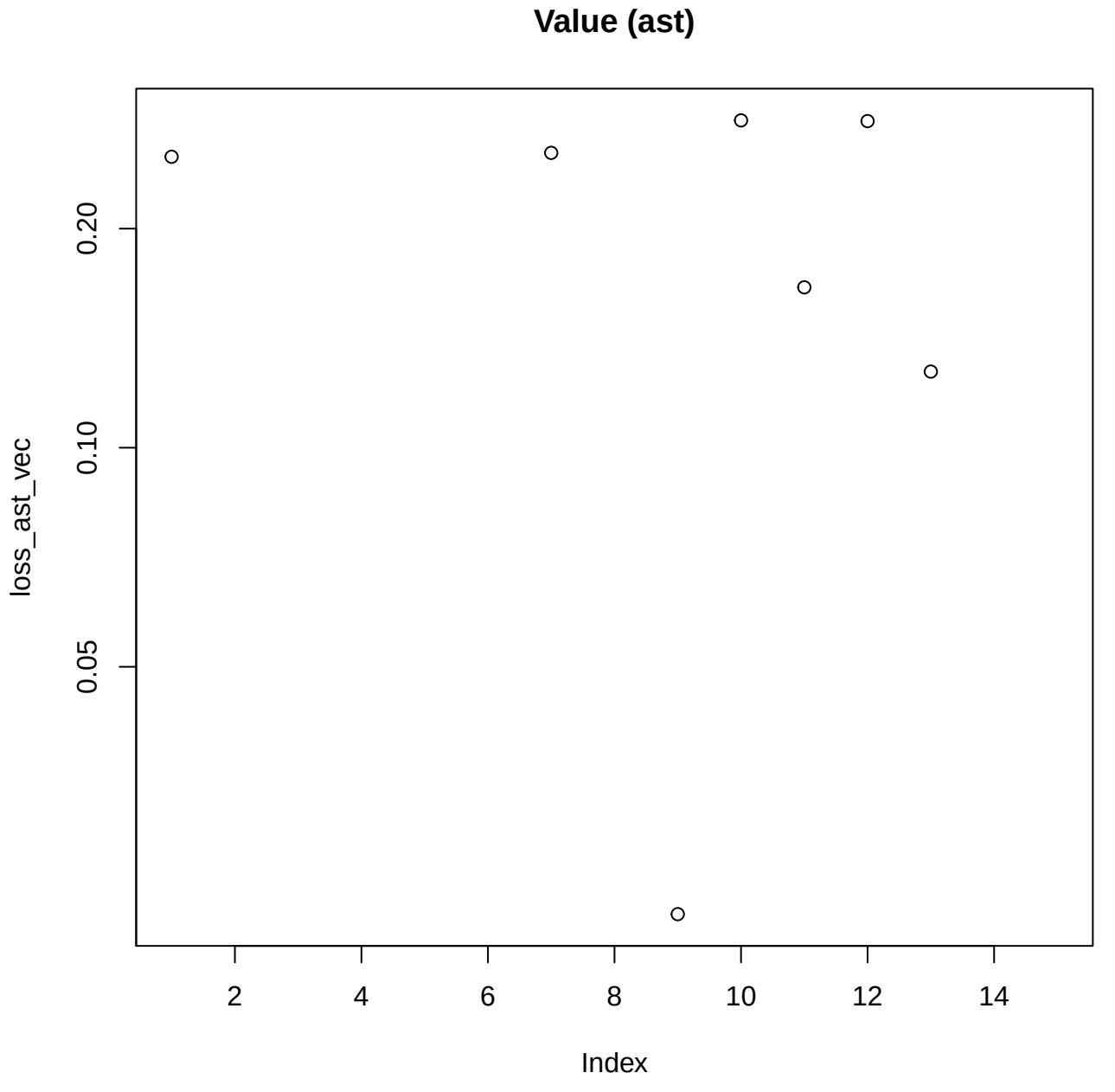

Value (ast)
loss_ast_vec
0.55
0.60
0.65
0.70
0.75
2
4
6
8
10
12
14
Index

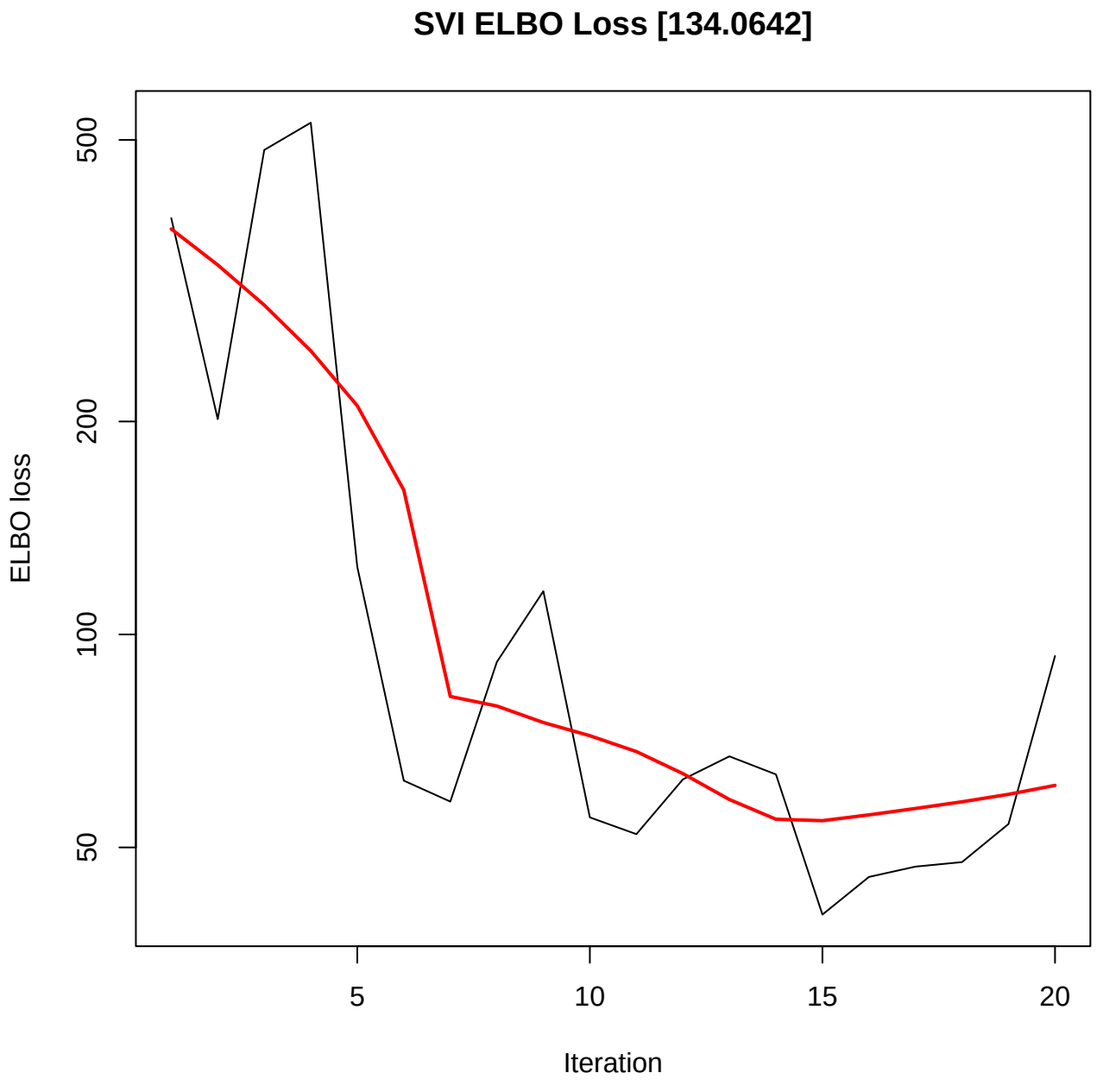

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.06
0.08
0.10
0.12
0.14
2
4
6
8
10
12
14
Index

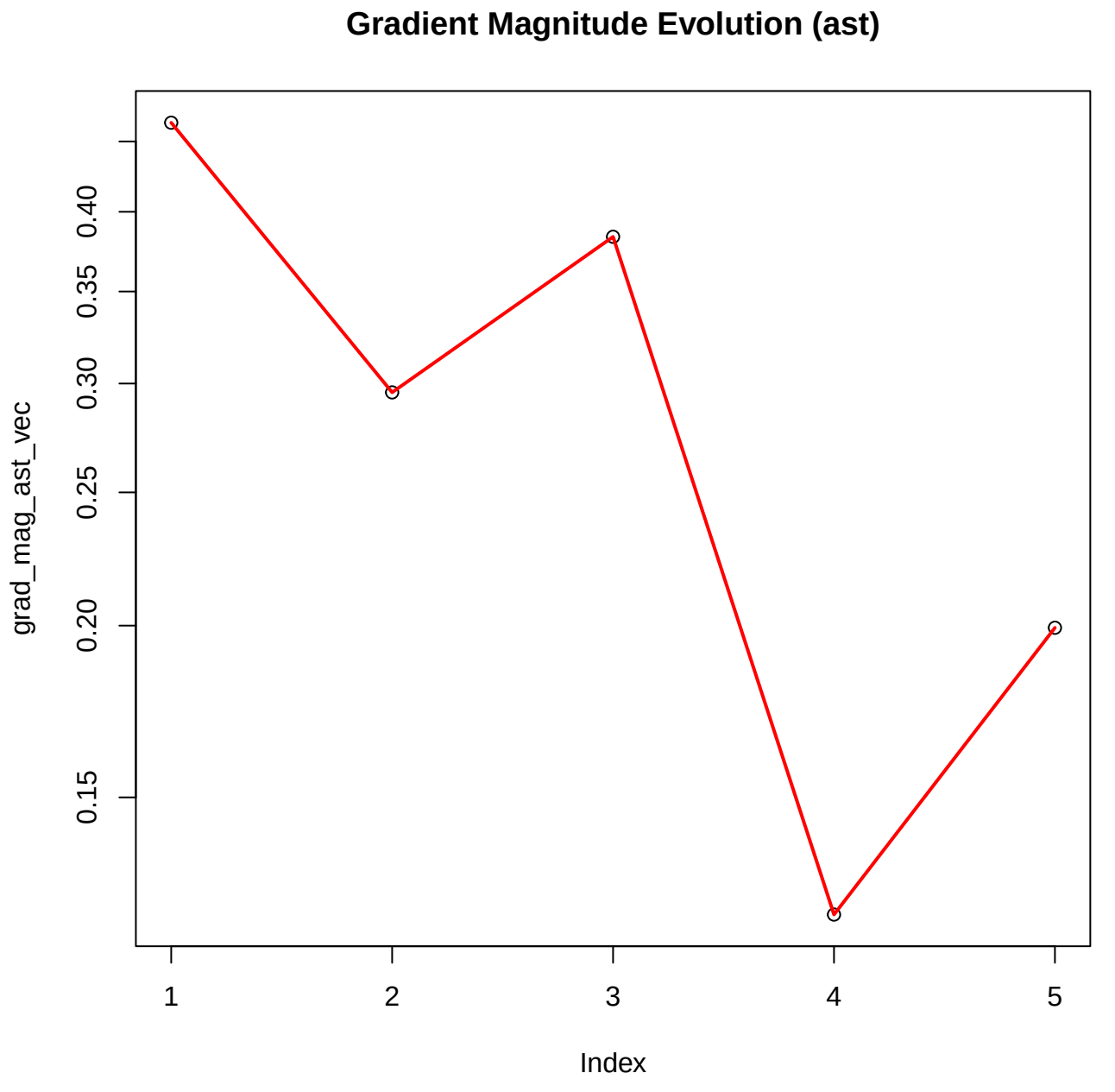

Value (ast)
loss_ast_vec
0.55
0.60
0.65
0.70
0.75
2
4
6
8
10
12
14
Index

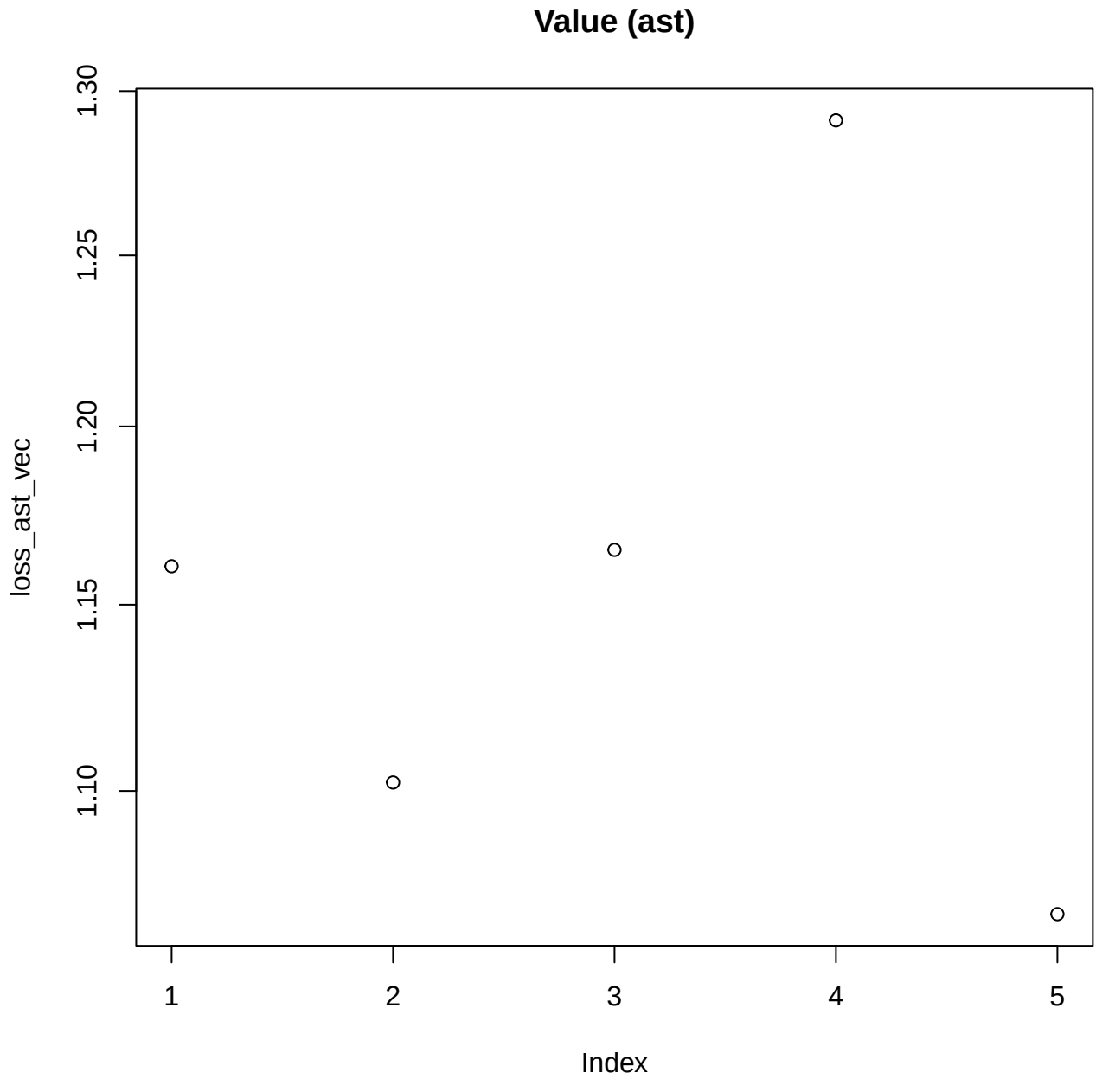

SVI ELBO Loss [78.5720]
ELBO loss
50000
20000
5000
1000
500
200
100
50
0
500
1000
1500
2000
Iteration

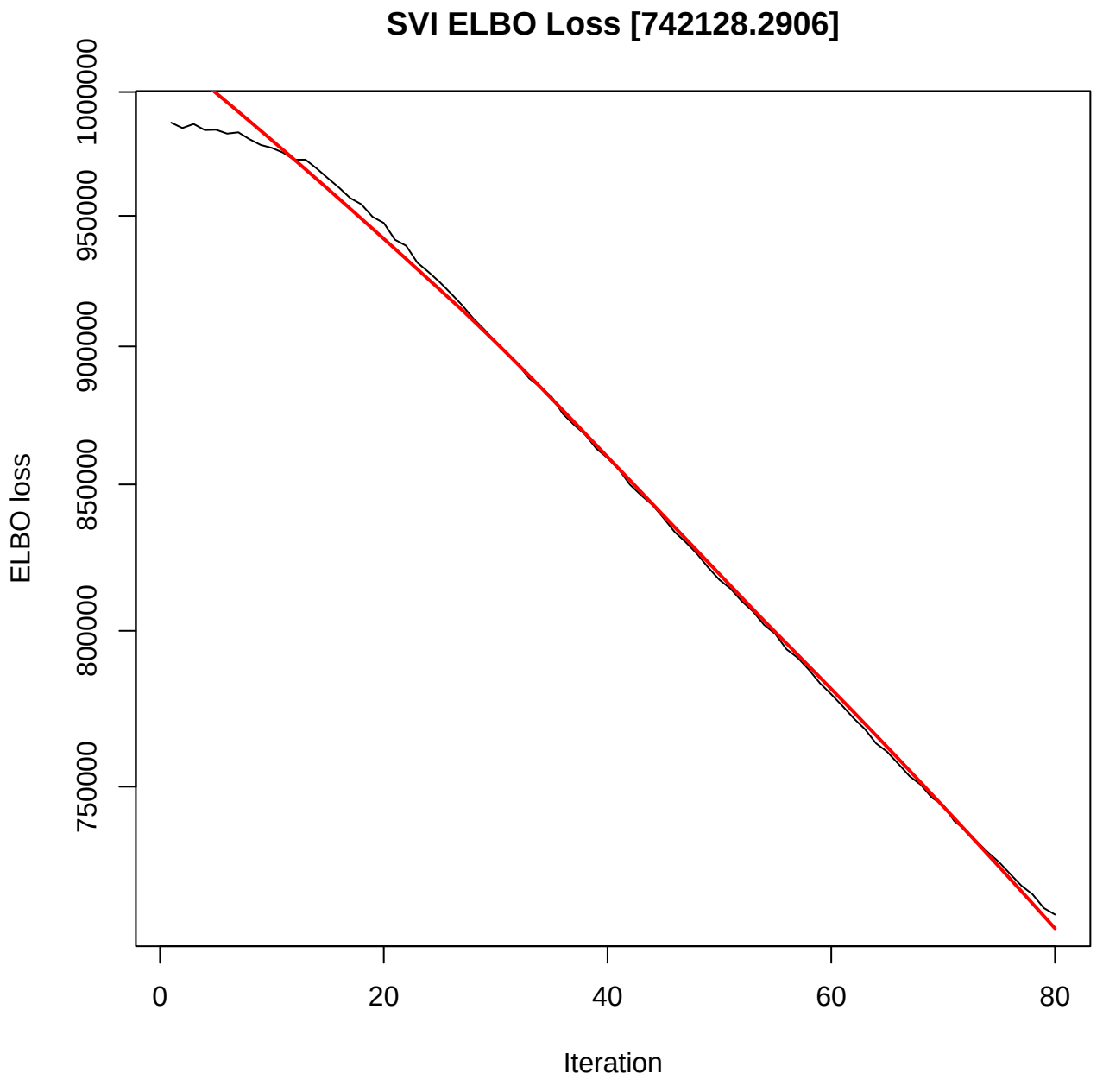

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1.0
0.8
0.6
0.4
2
4
6
8
10
12
14
Index

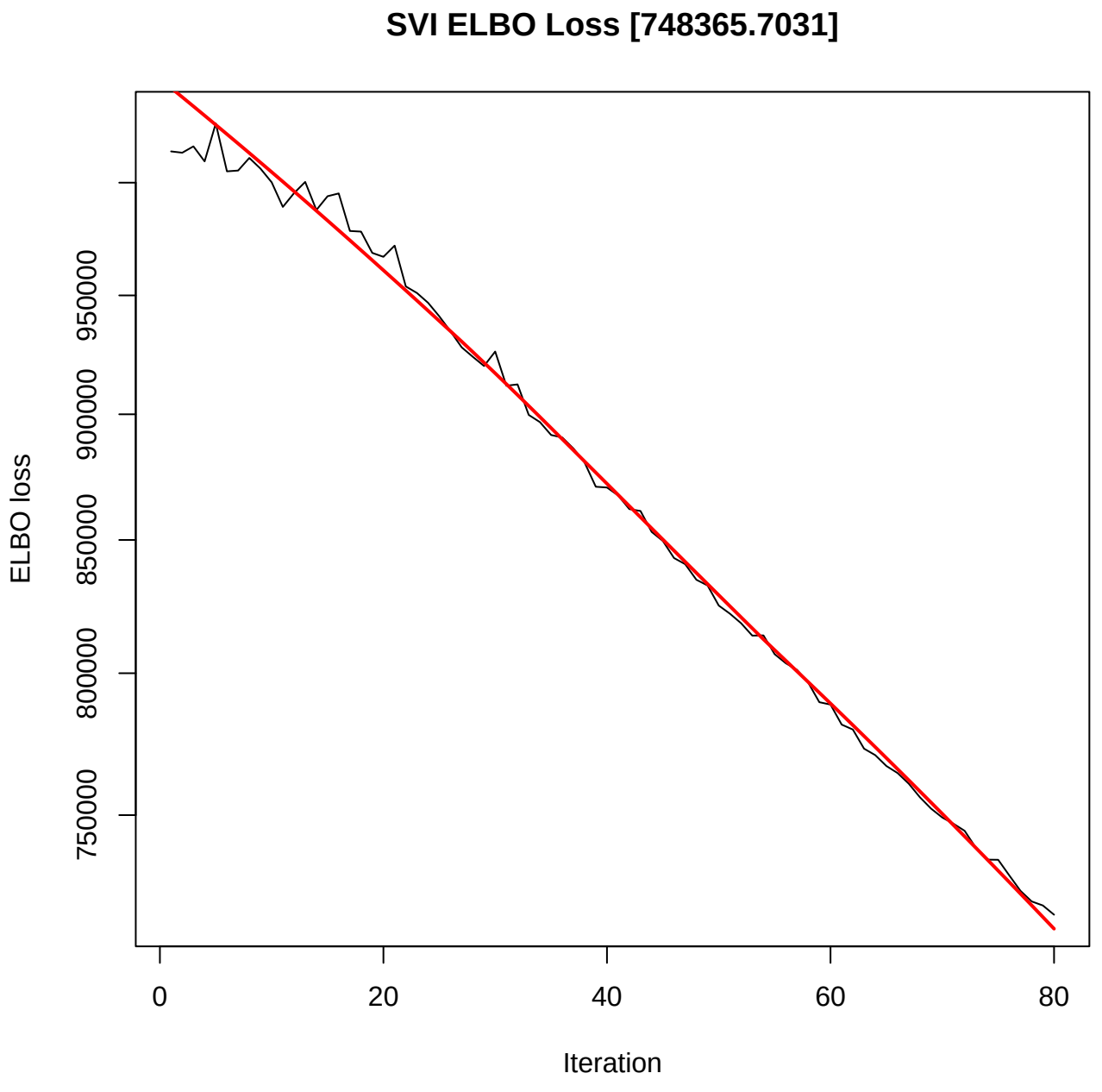

Value (ast)
loss_ast_vec
0.05
0.10
0.20
2
4
6
8
10
12
14
Index

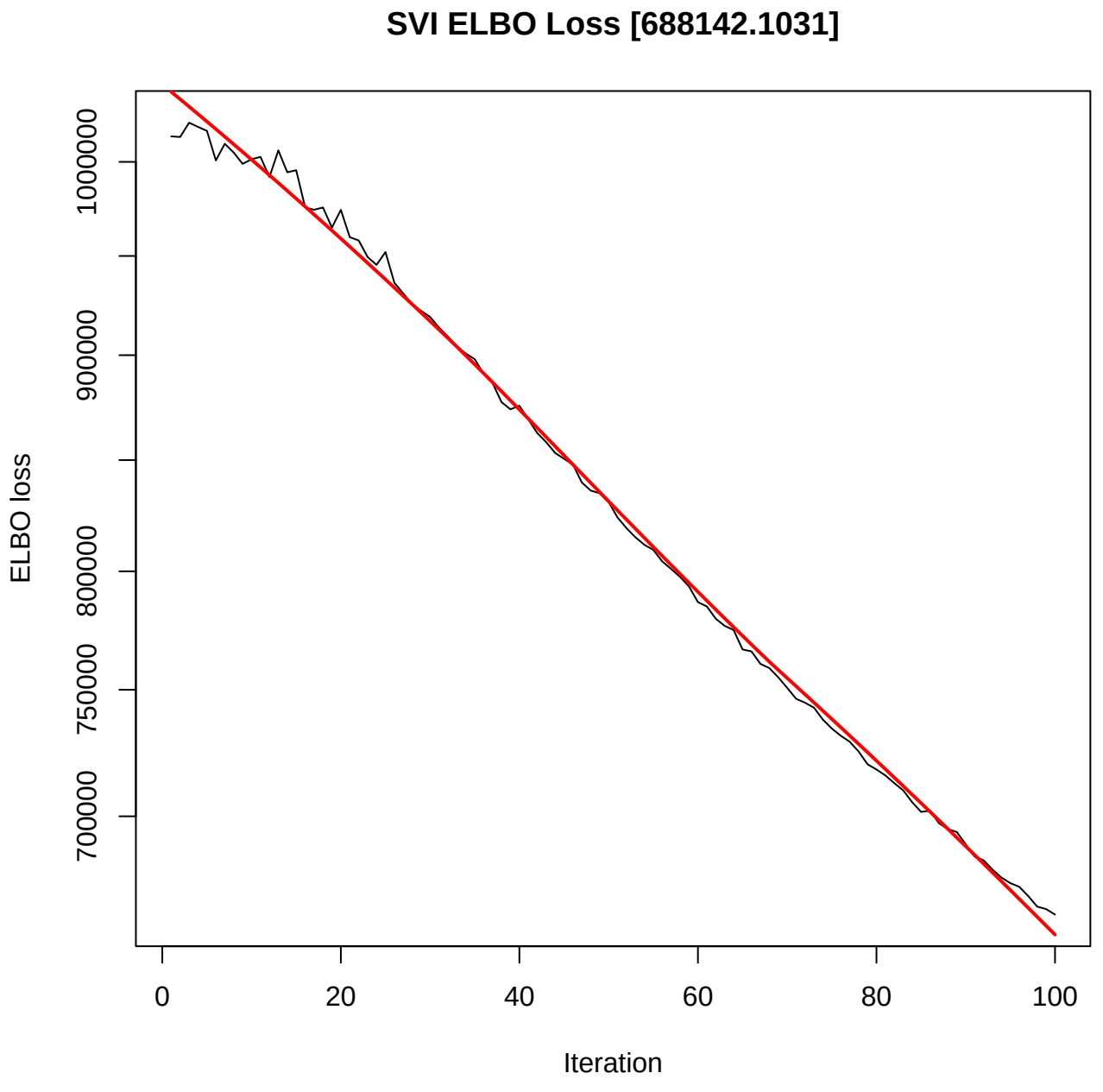

SVI ELBO Loss [134.0642]
ELBO loss
500
200
100
50
5
10
15
20
Iteration

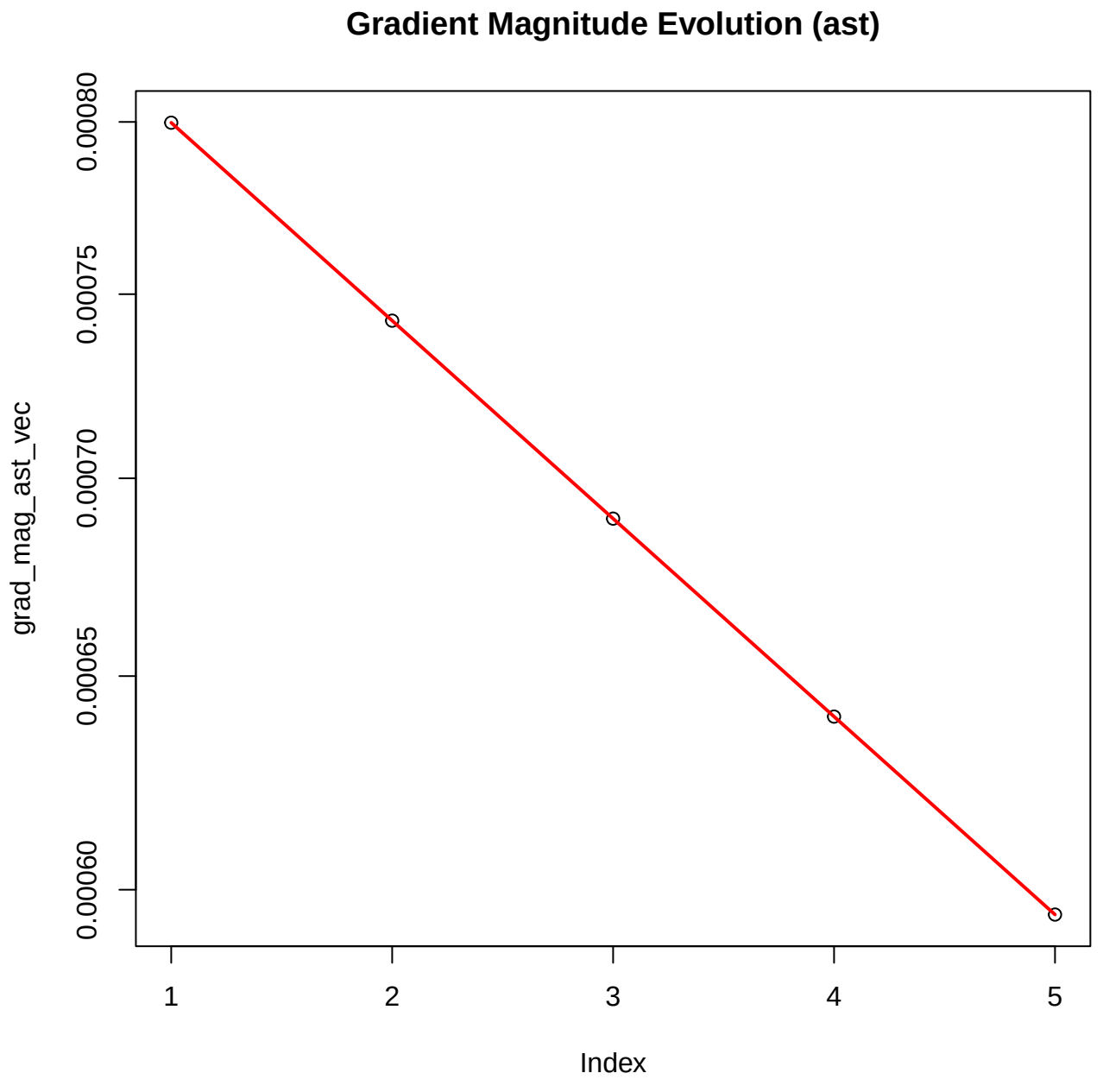

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.15
0.20
0.25
0.30
0.35
0.40
1
2
3
4
5
Index

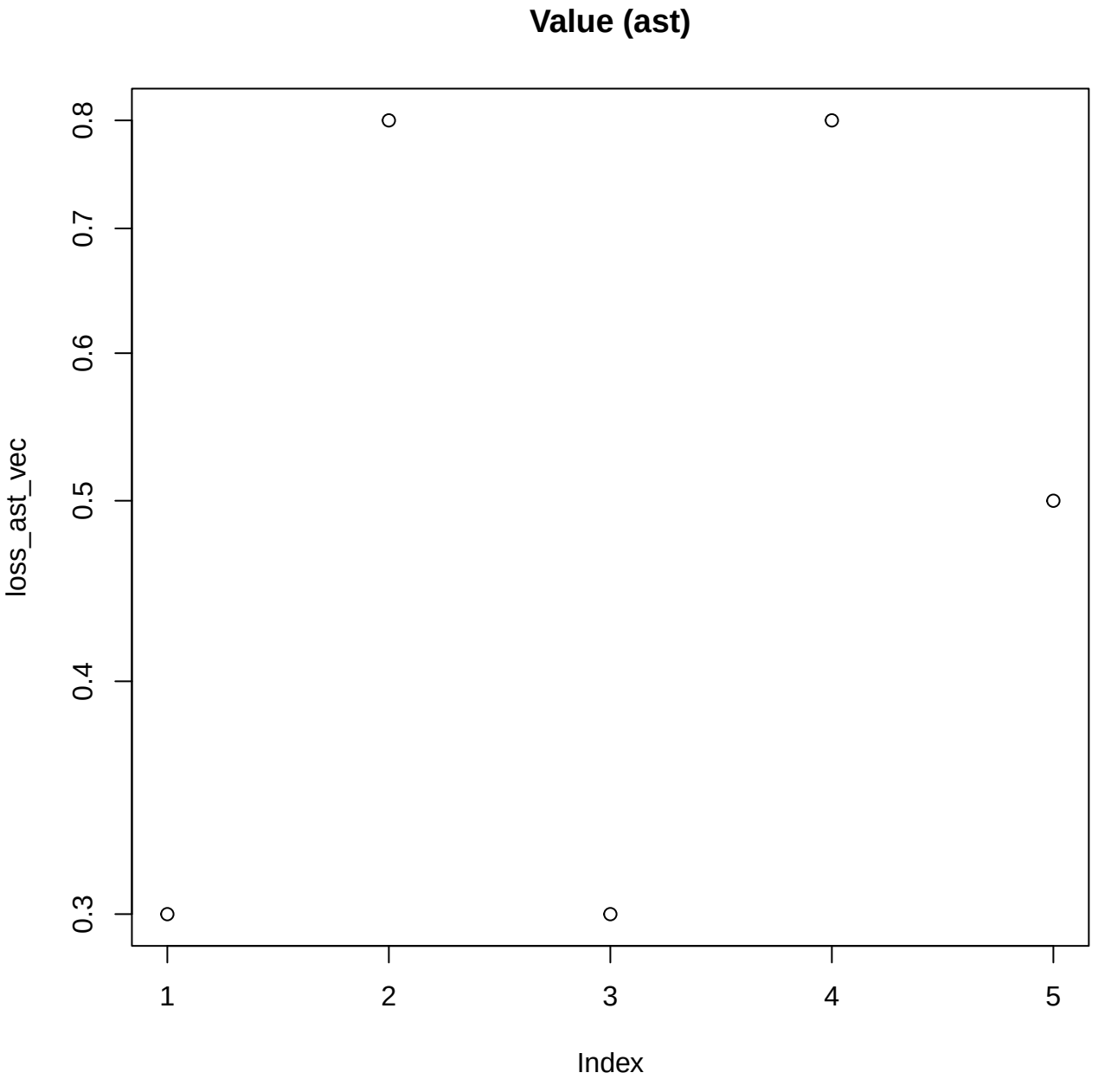

Value (ast)
loss_ast_vec
1.10
1.15
1.20
1.25
1.30
1
2
3
4
5
Index

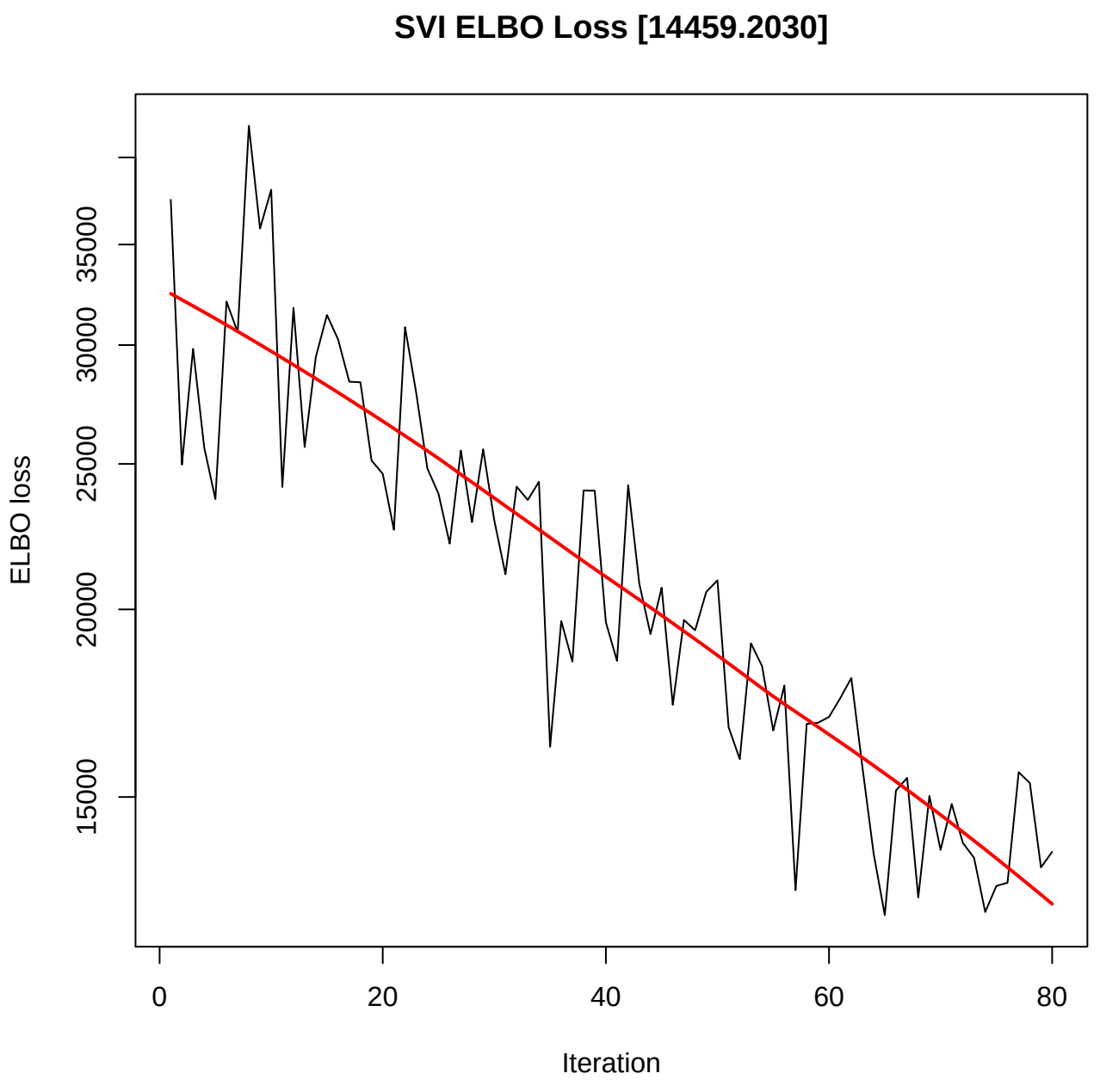

SVI ELBO Loss [742128.2906]
ELBO loss
1000000
950000
900000
850000
800000
750000
0
20
40
60
80
Iteration

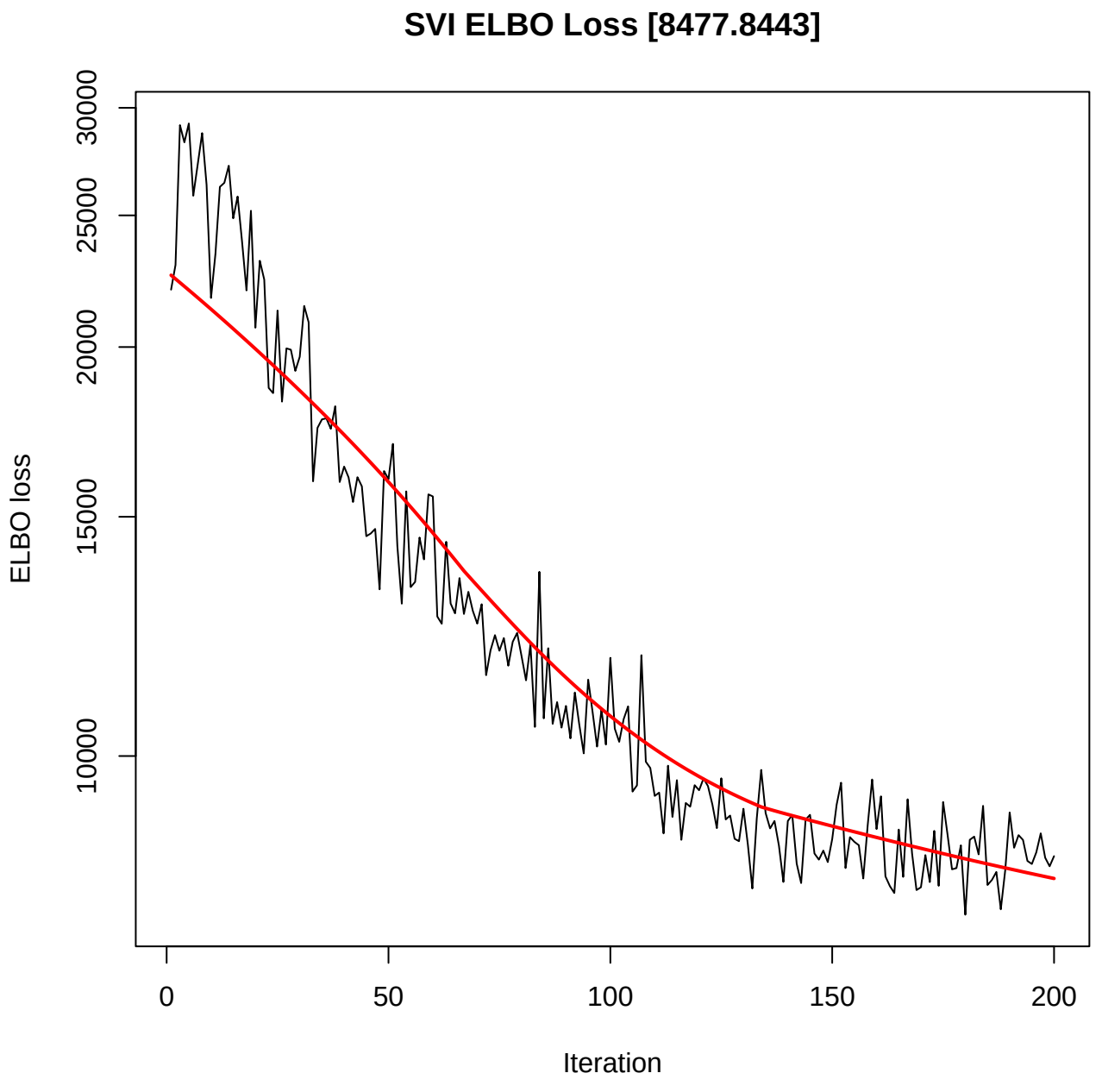

SVI ELBO Loss [748365.7031]
ELBO loss
750000
800000
850000
900000
950000
0
20
40
60
80
Iteration

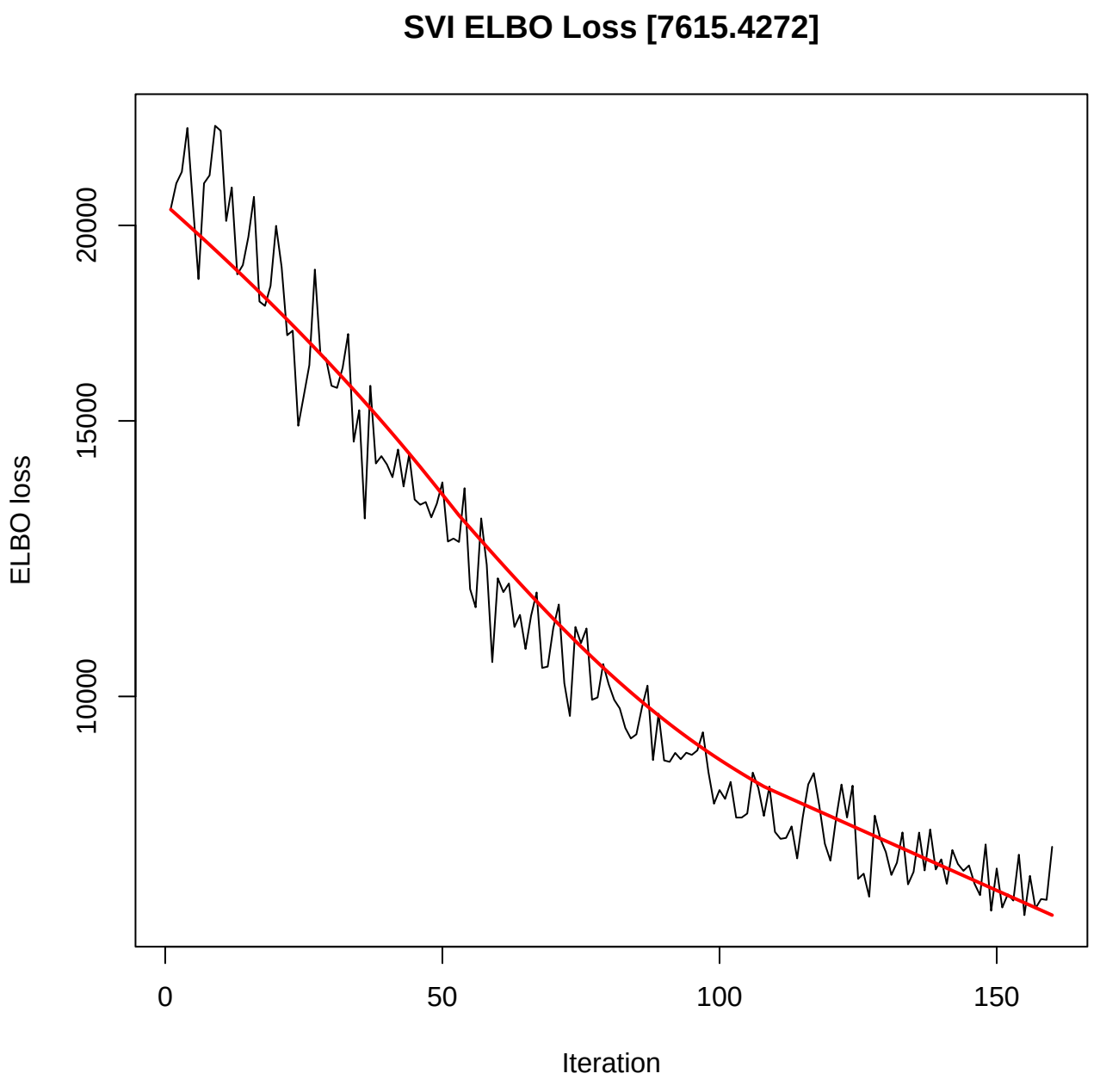

SVI ELBO Loss [688142.1031]
ELBO loss
1000000
900000
800000
750000
700000
0
20
40
60
80
100
Iteration

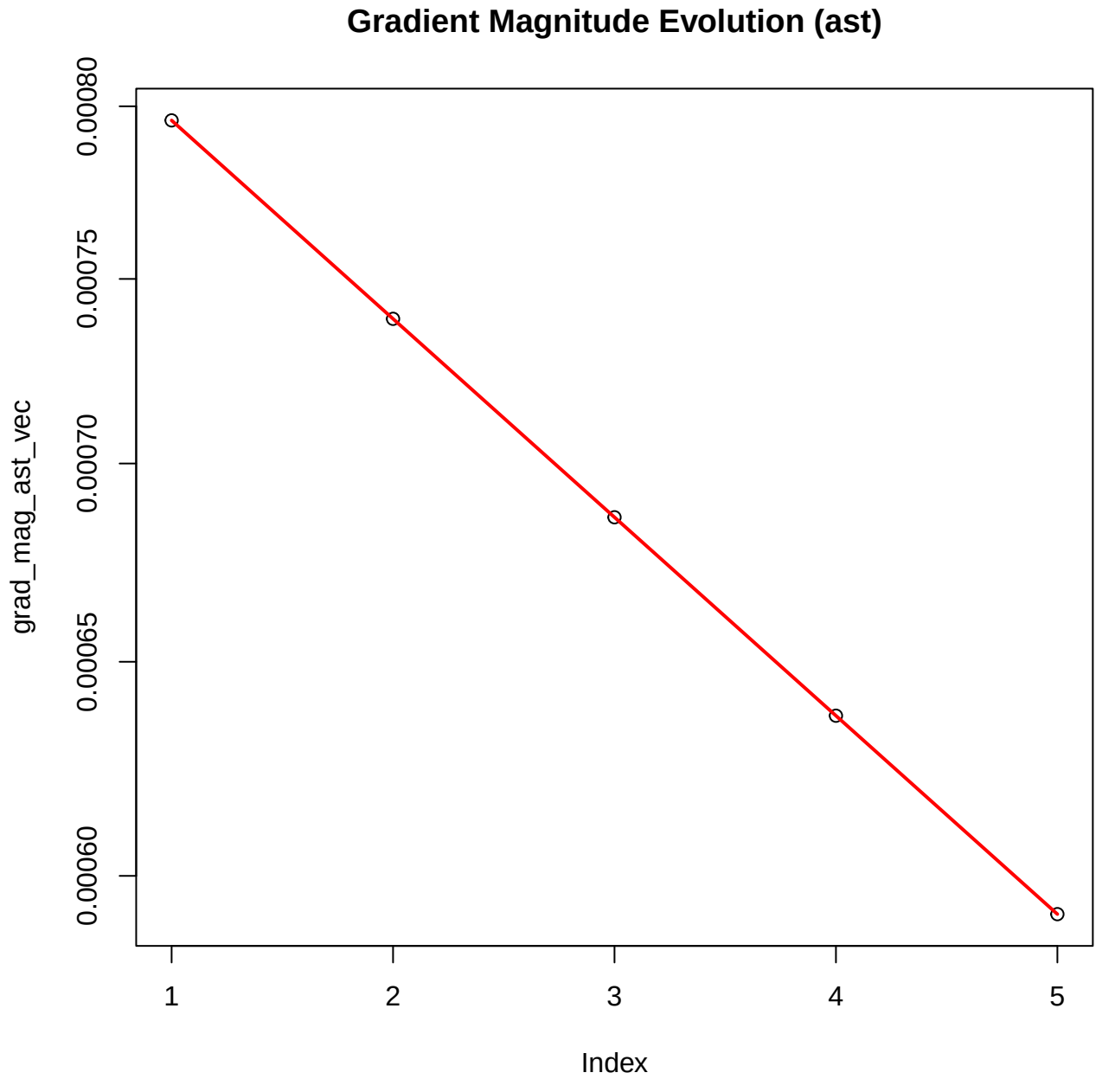

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

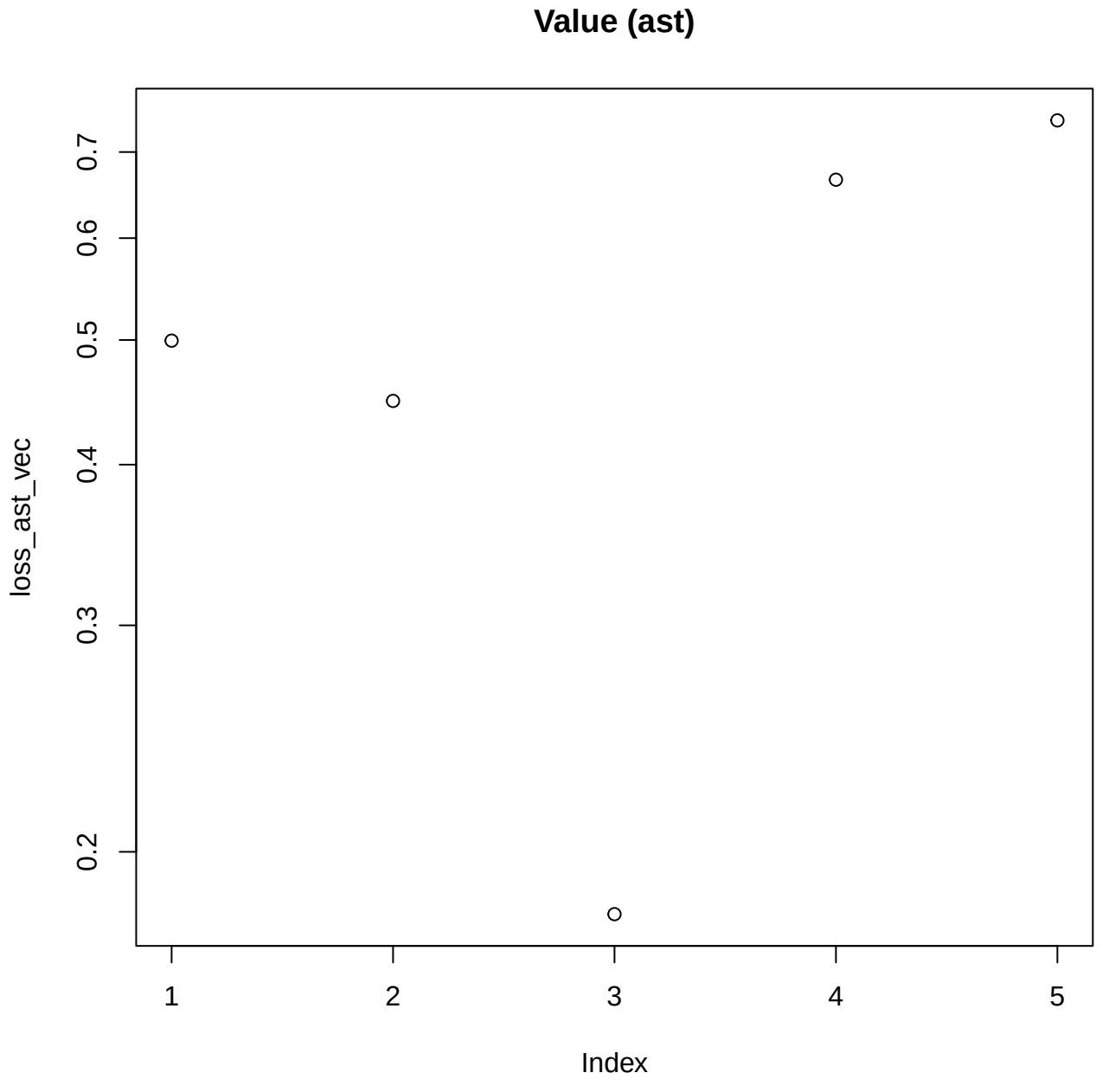

Value (ast)
loss_ast_vec
0.3
0.4
0.5
0.6
0.7
0.8
1
2
3
4
5
Index

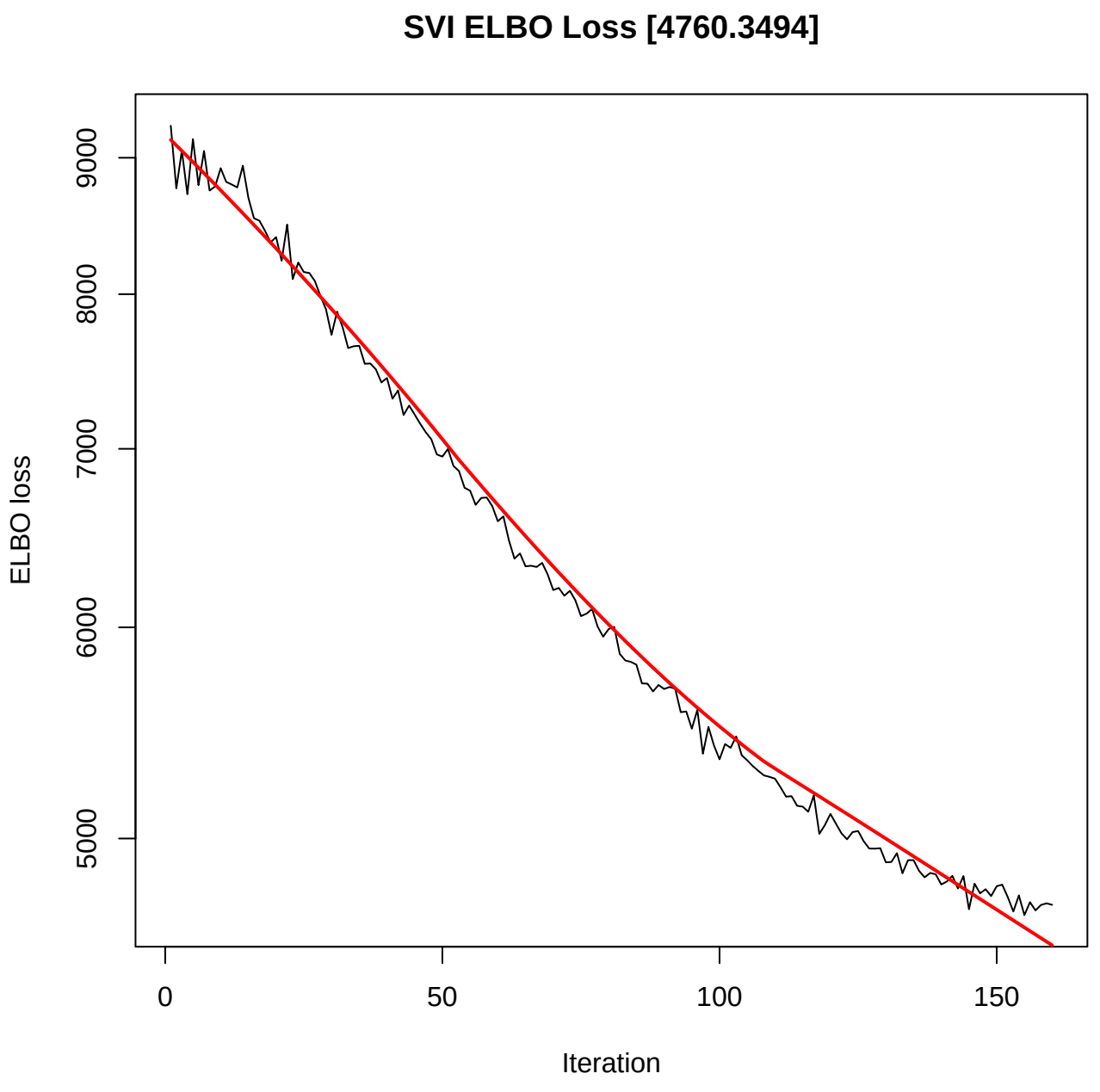

SVI ELBO Loss [14459.2030]
ELBO loss
15000
20000
25000
30000
35000
0
20
40
60
80
Iteration

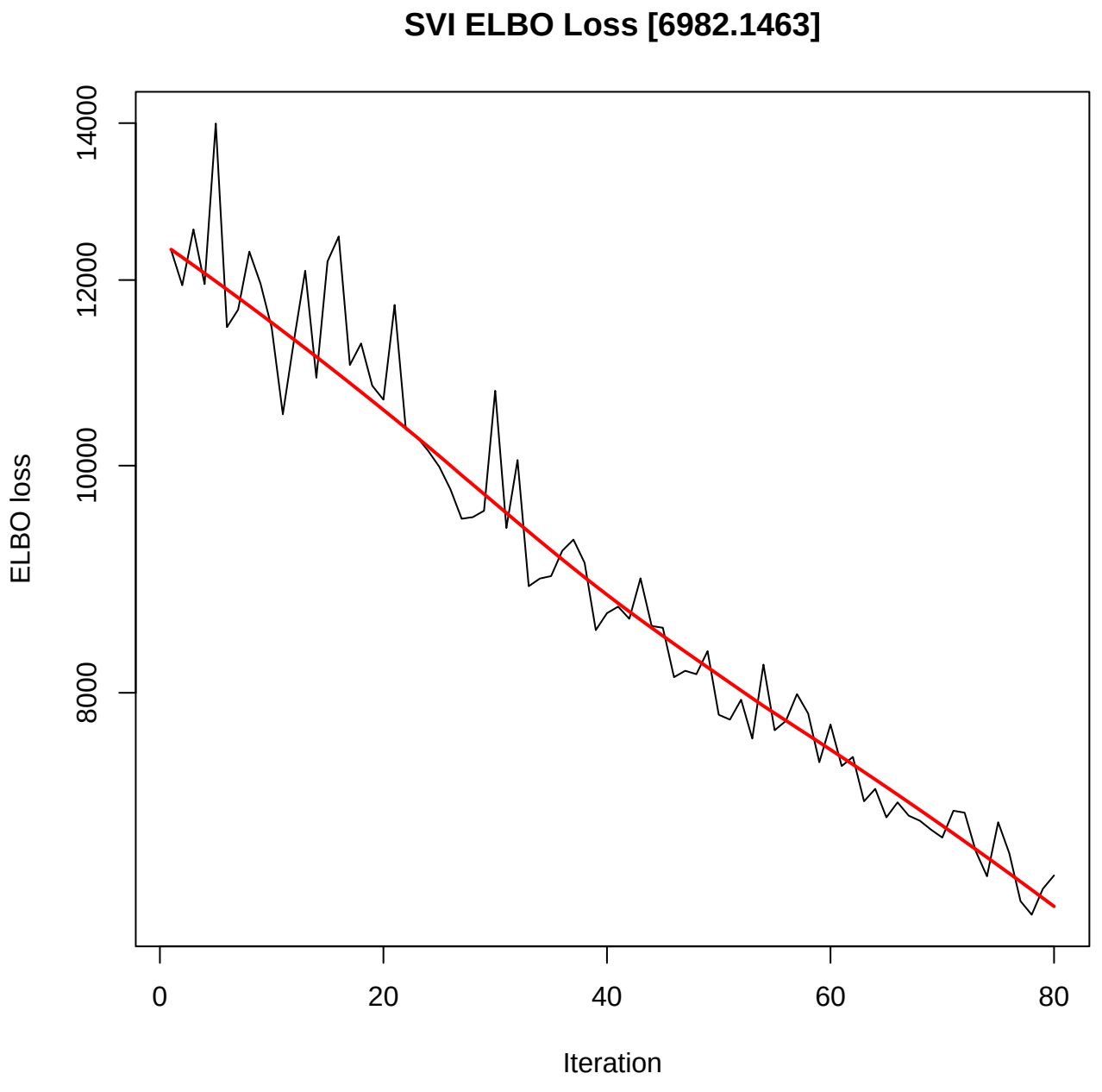

SVI ELBO Loss [8477.8443]
ELBO loss
30000
25000
20000
15000
10000
0
50
100
150
200
Iteration

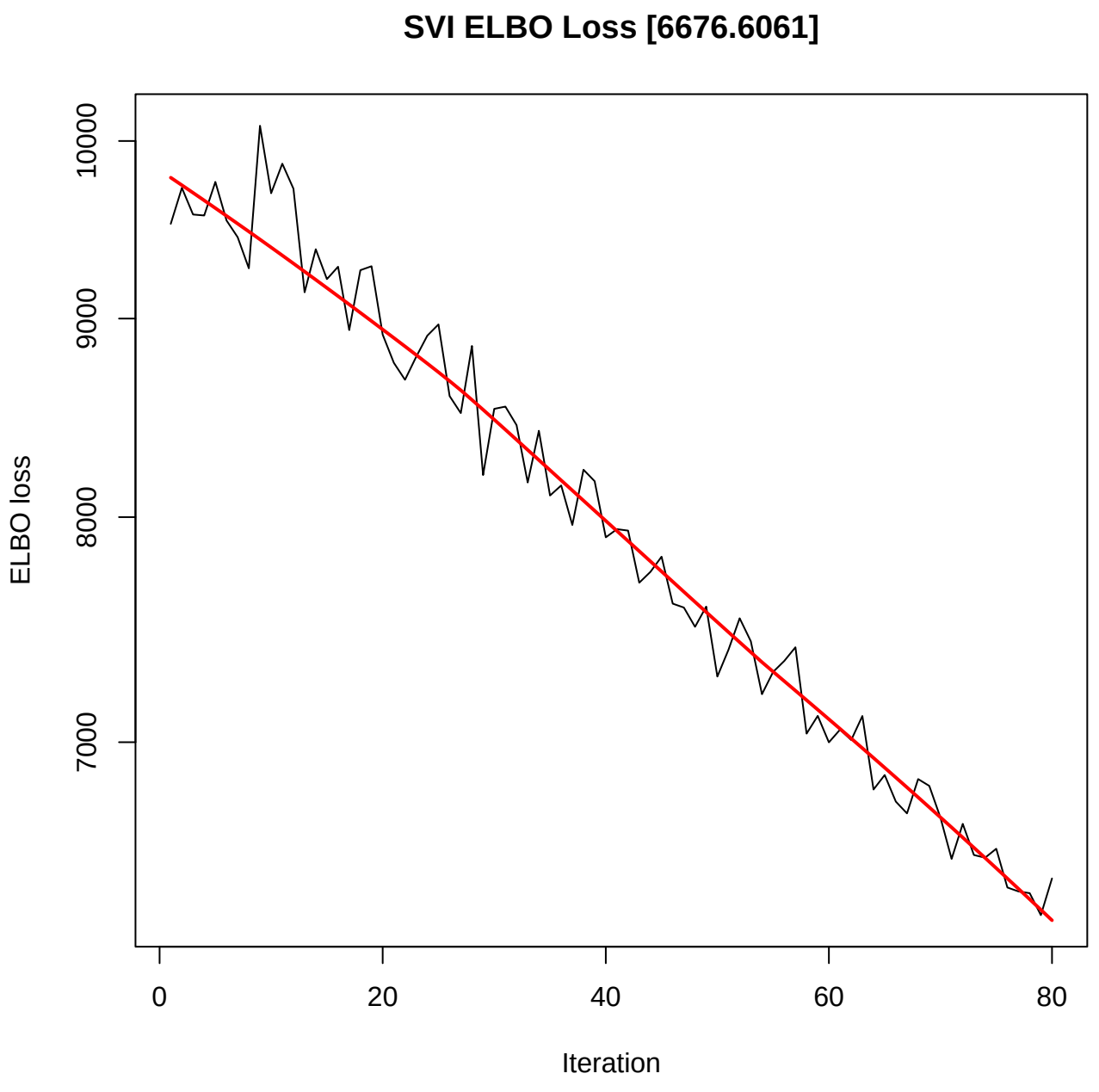

SVI ELBO Loss [7615.4272]
ELBO loss
10000
15000
20000
0
50
100
150
Iteration

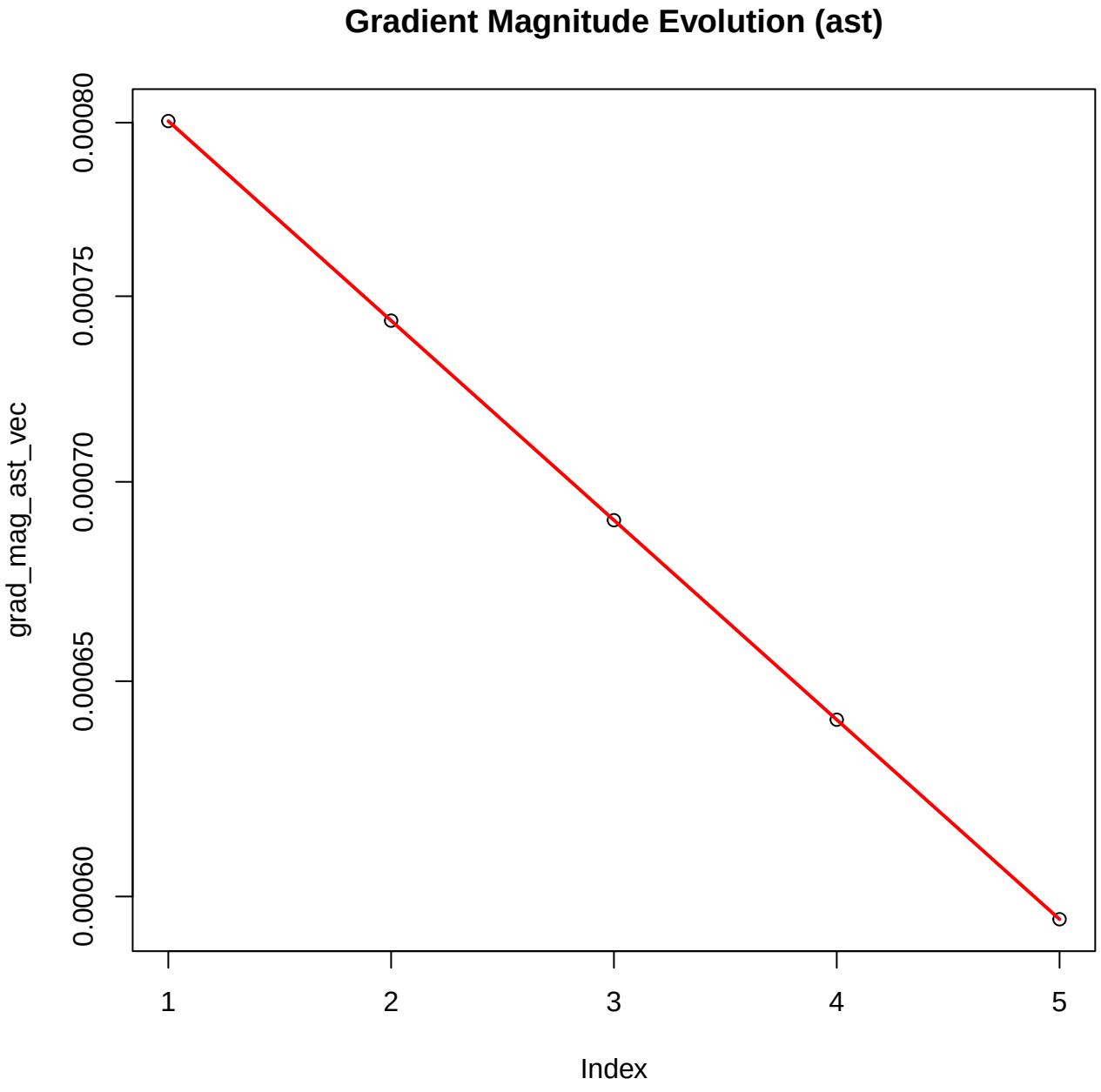

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

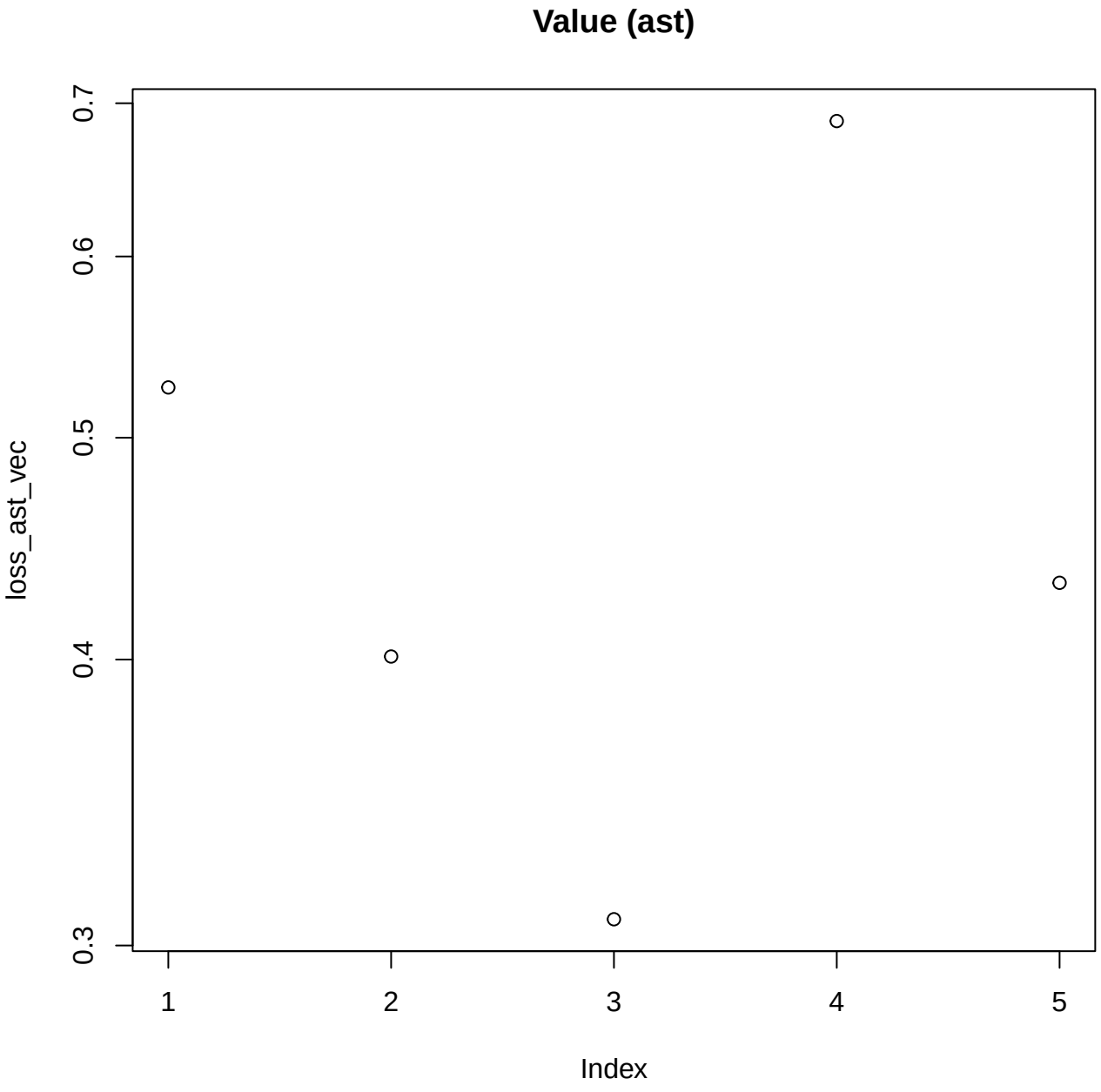

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
1
2
3
4
5
Index

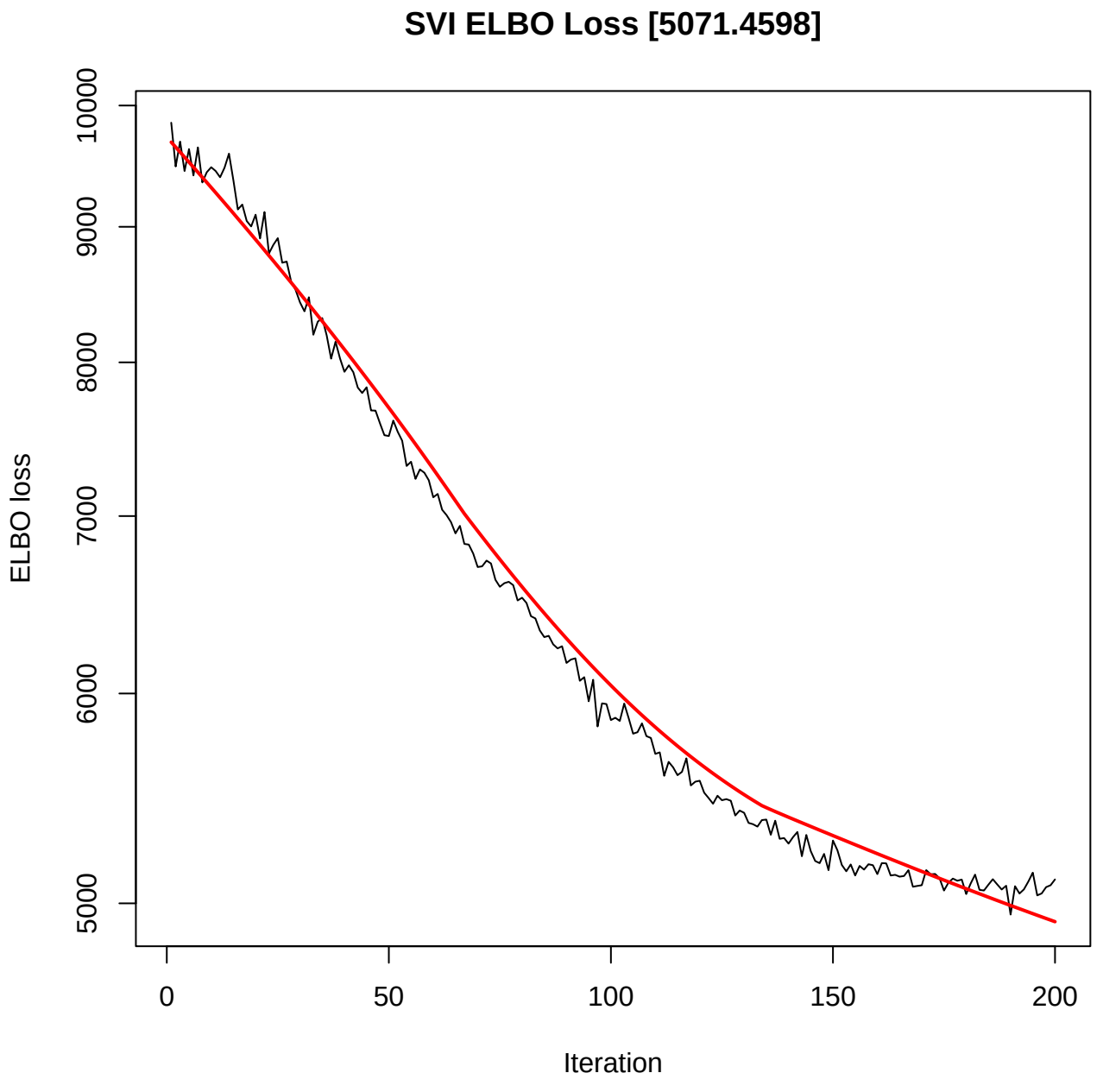

SVI ELBO Loss [4760.3494]
ELBO loss
9000
8000
7000
6000
5000
0
50
100
150
Iteration

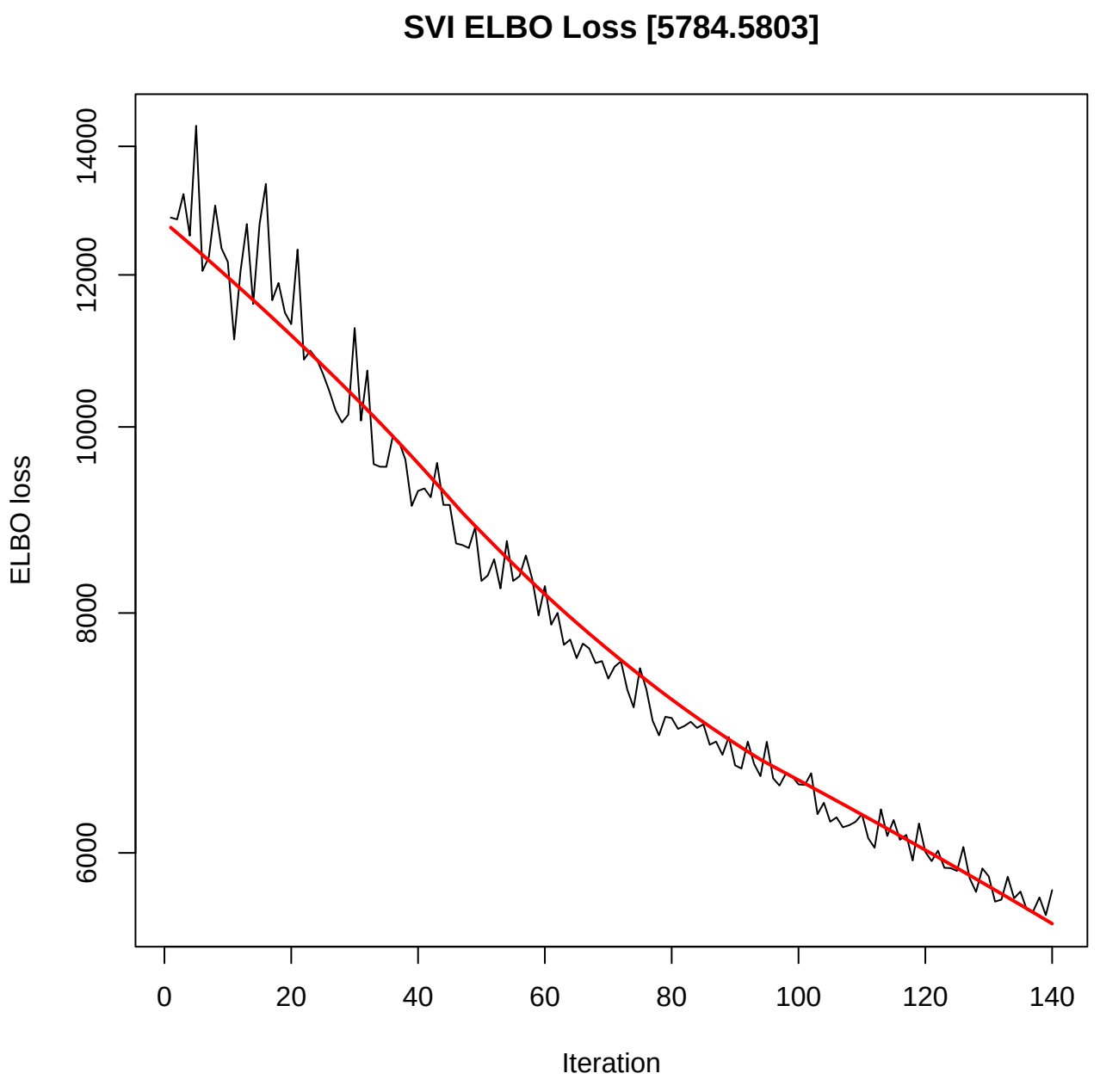

SVI ELBO Loss [6982.1463]
ELBO loss
14000
12000
10000
8000
0
20
40
60
80
Iteration

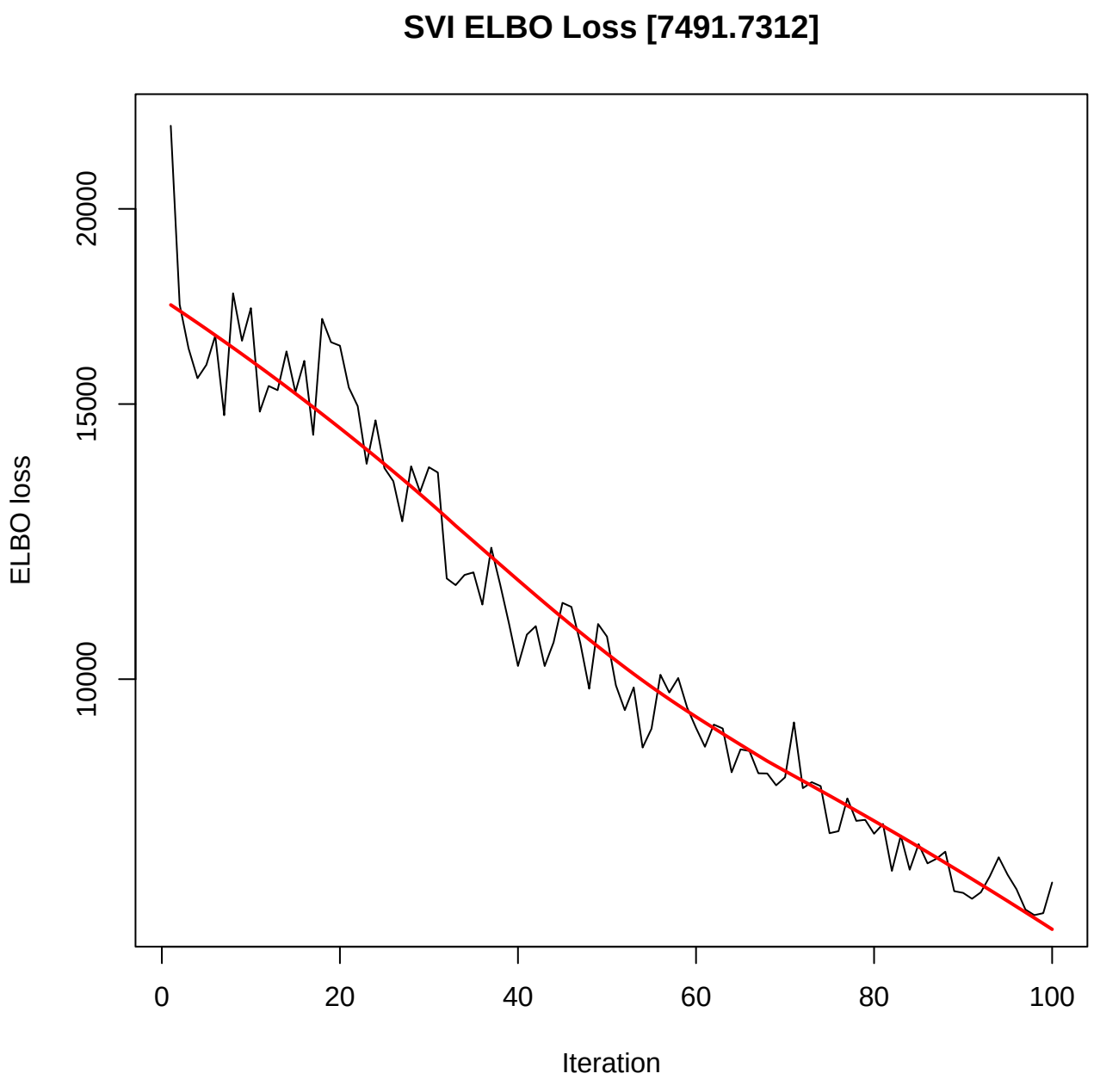

SVI ELBO Loss [6676.6061]
ELBO loss
10000
9000
8000
7000
0
20
40
60
80
Iteration

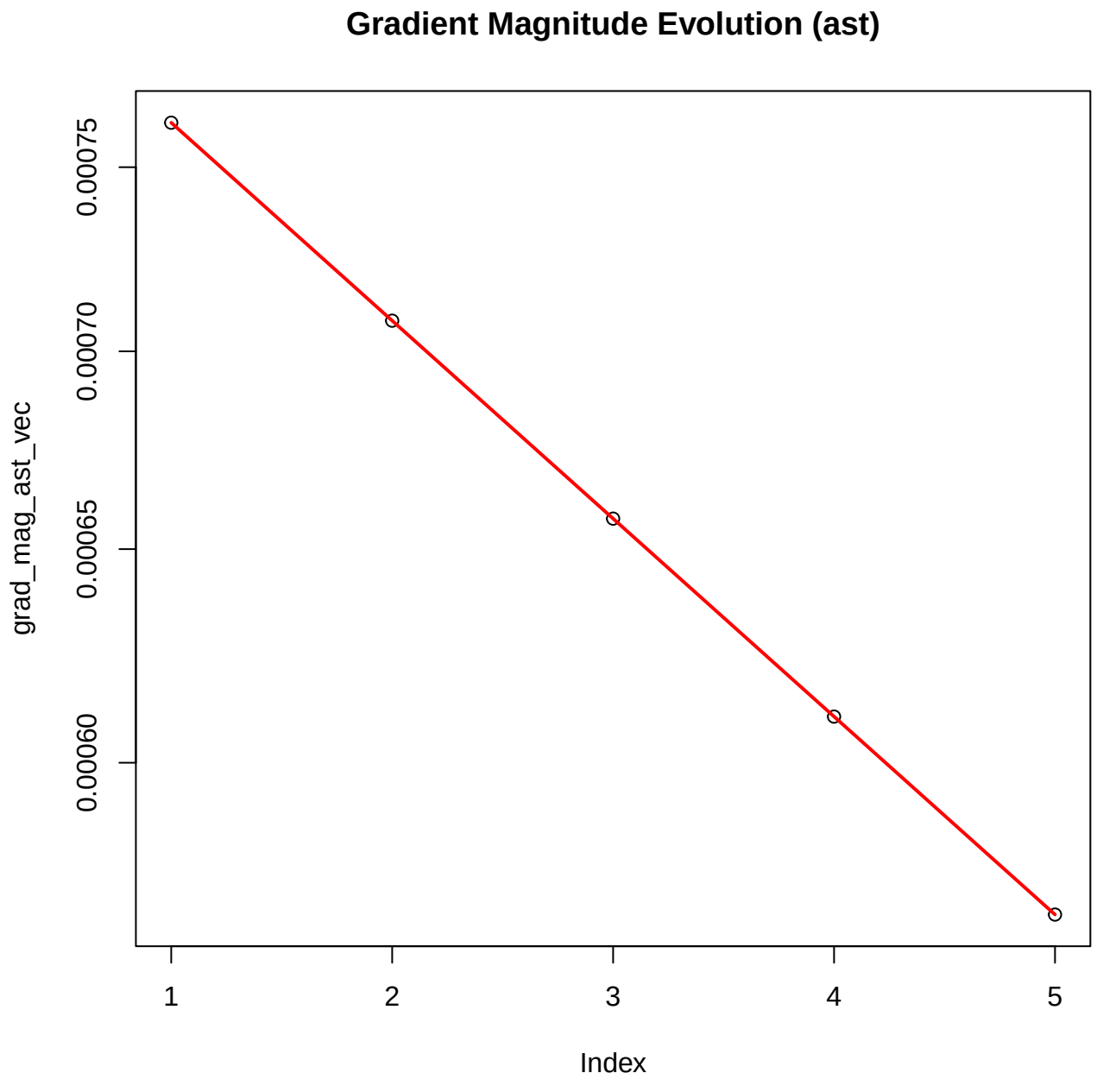

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

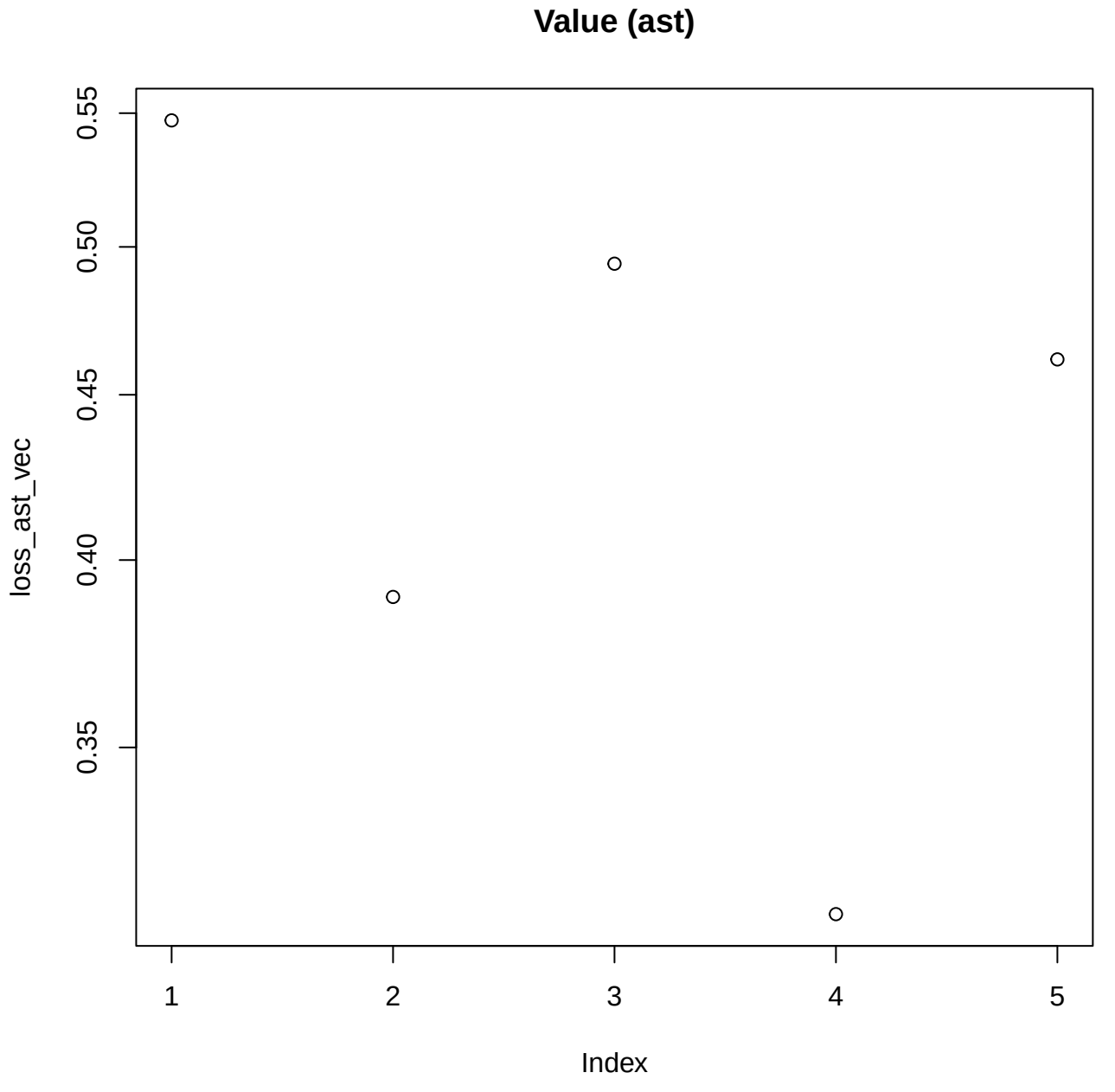

Value (ast)
loss_ast_vec
0.3
0.4
0.5
0.6
0.7
1
2
3
4
5
Index

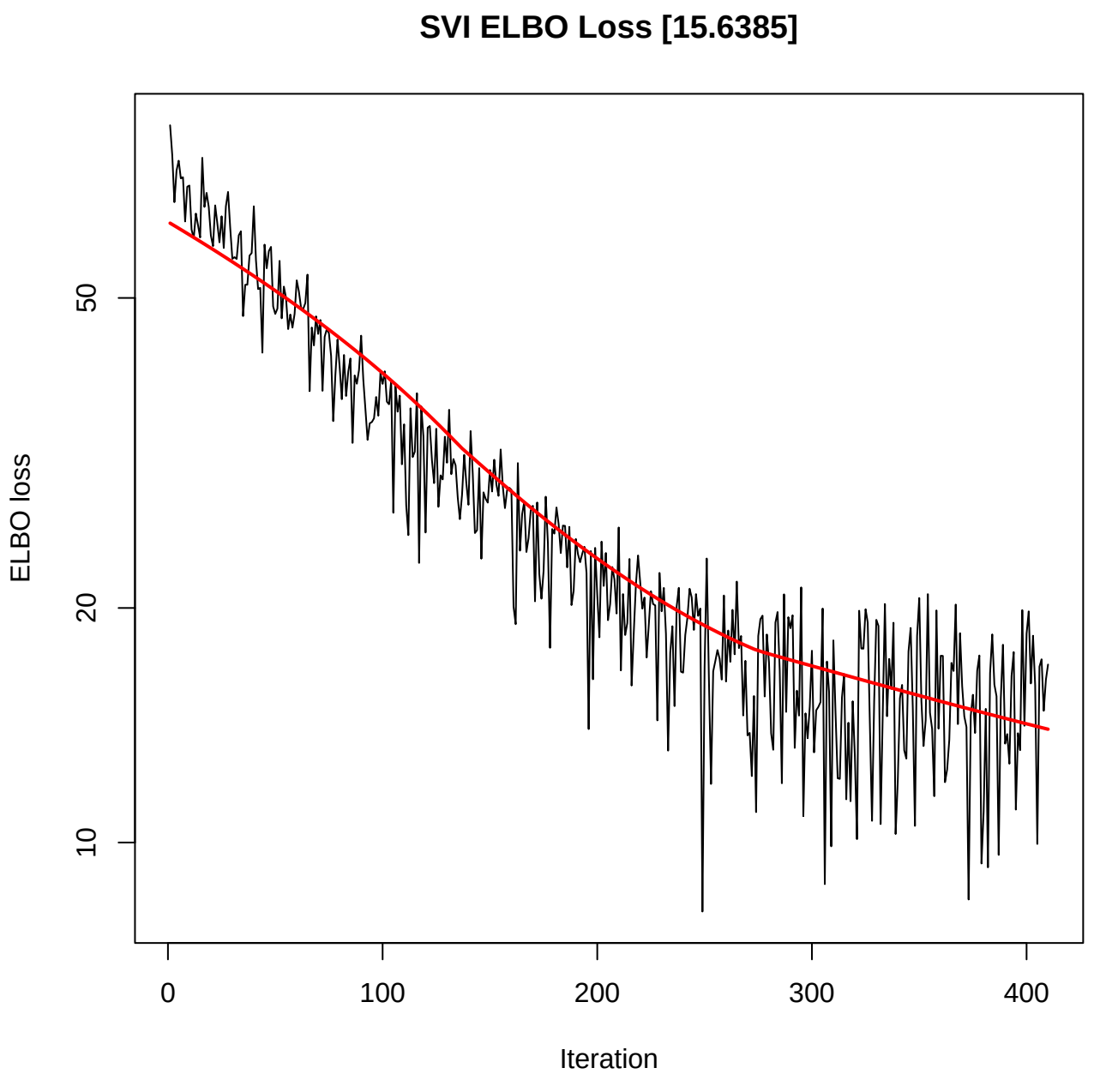

SVI ELBO Loss [5071.4598]
ELBO loss
10000
9000
8000
7000
6000
5000
0
50
100
150
200
Iteration

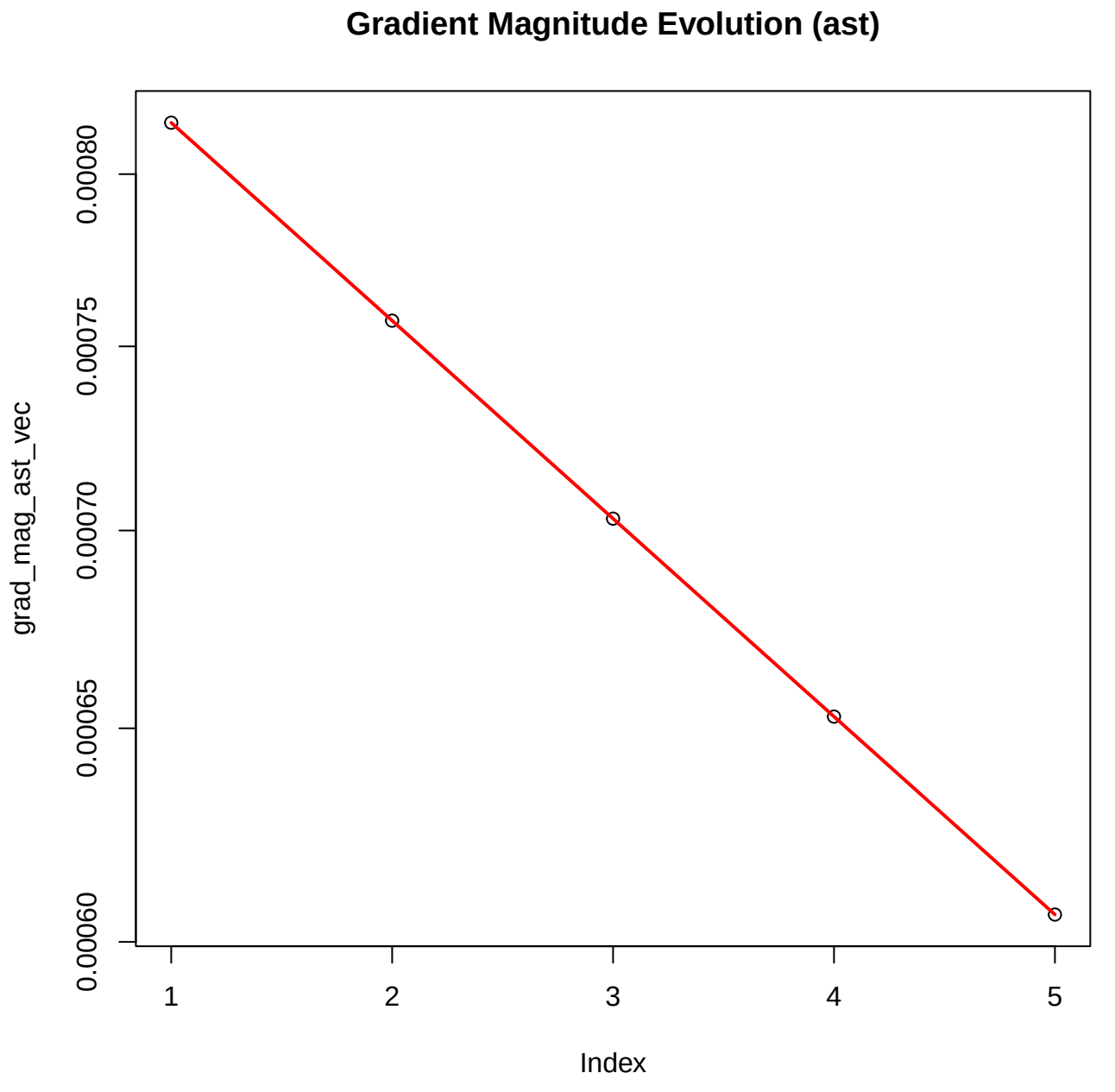

SVI ELBO Loss [5784.5803]
ELBO loss
14000
12000
10000
8000
6000
0
20
40
60
80
100
120
140
Iteration

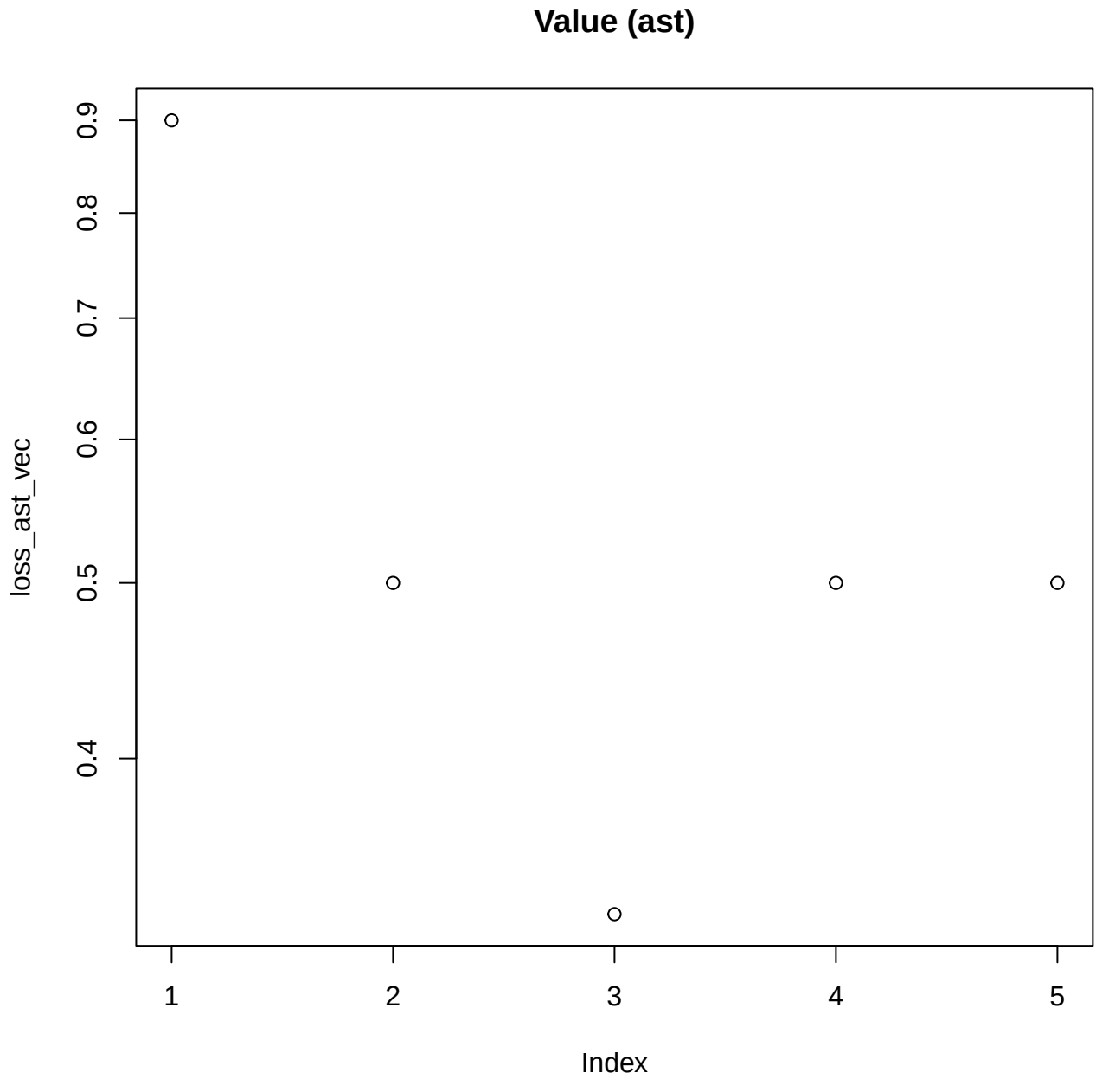

SVI ELBO Loss [7491.7312]
ELBO loss
10000
15000
20000
0
20
40
60
80
100
Iteration

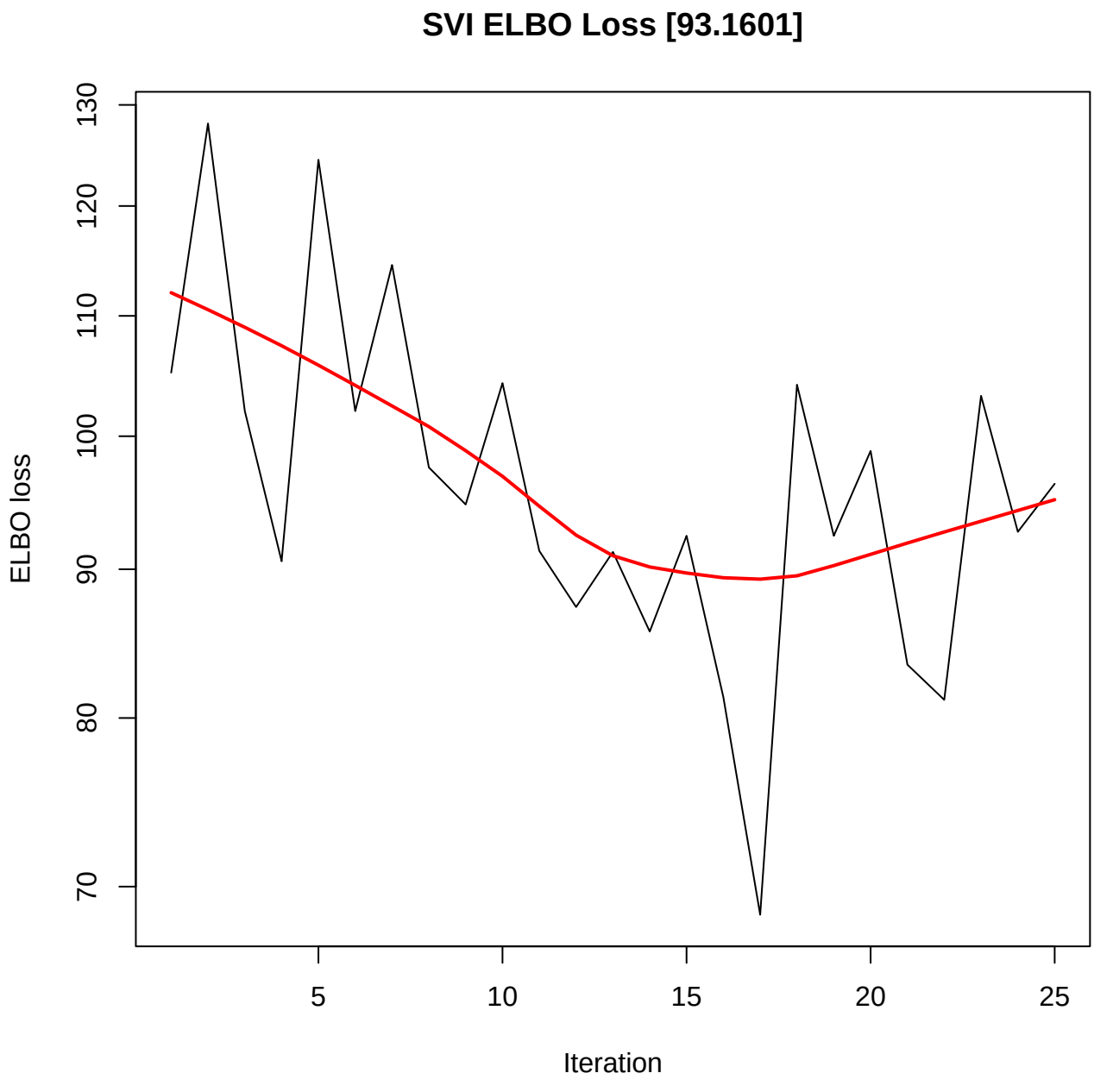

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

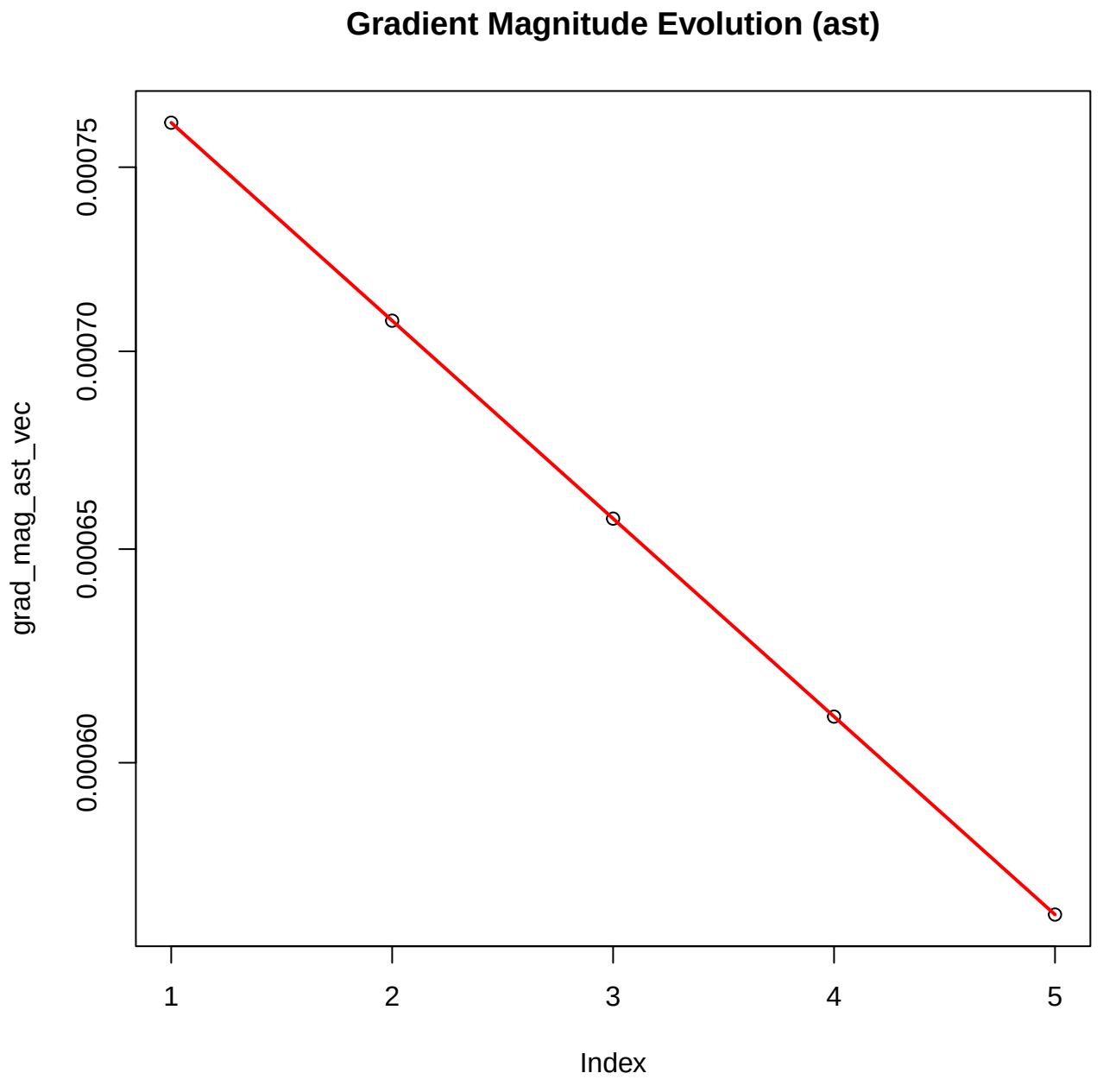

Value (ast)
loss_ast_vec
0.35
0.40
0.45
0.50
0.55
1
2
3
4
5
Index

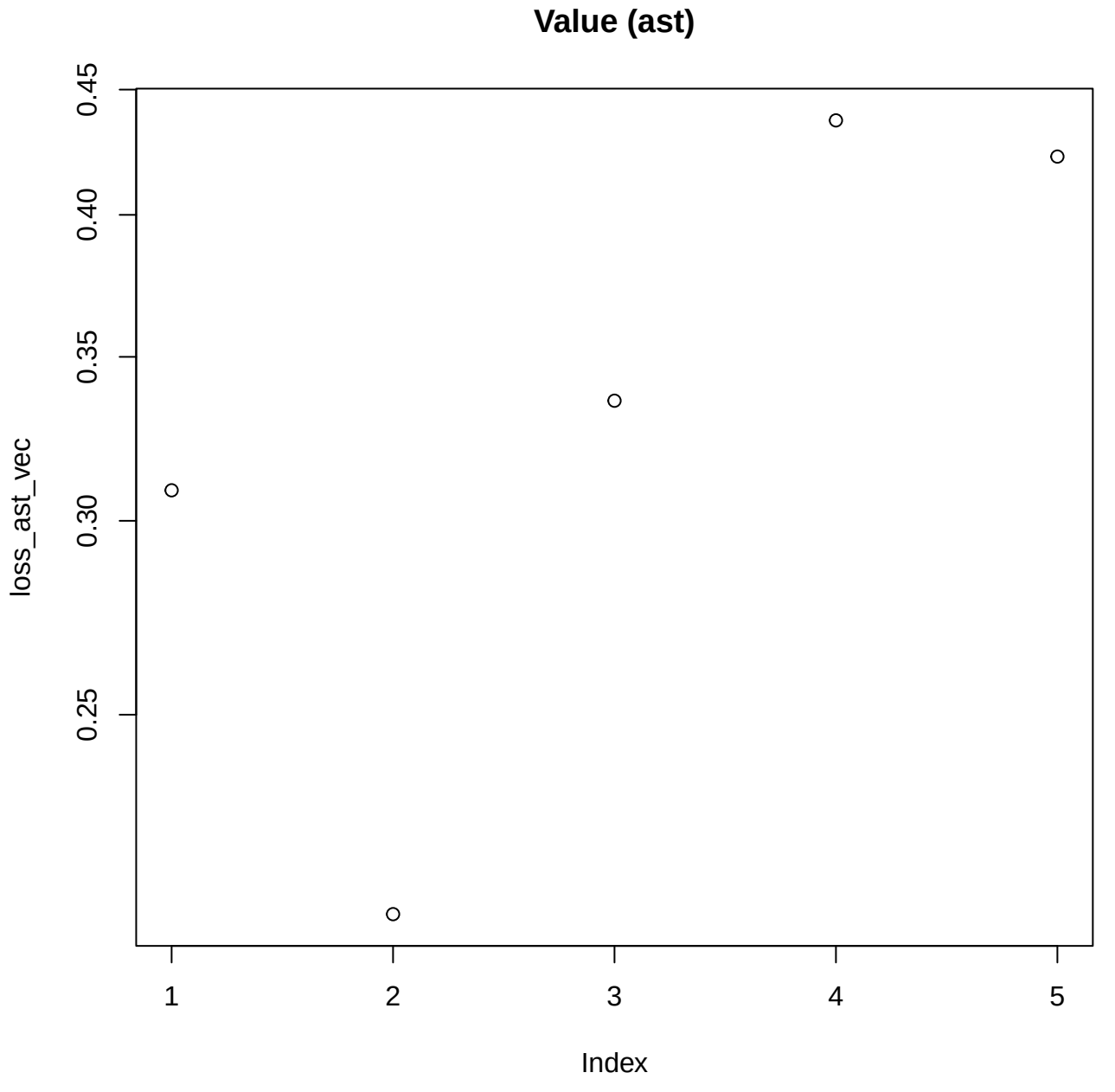

SVI ELBO Loss [15.6385]
ELBO loss
10
20
50
0
100
200
300
400
Iteration

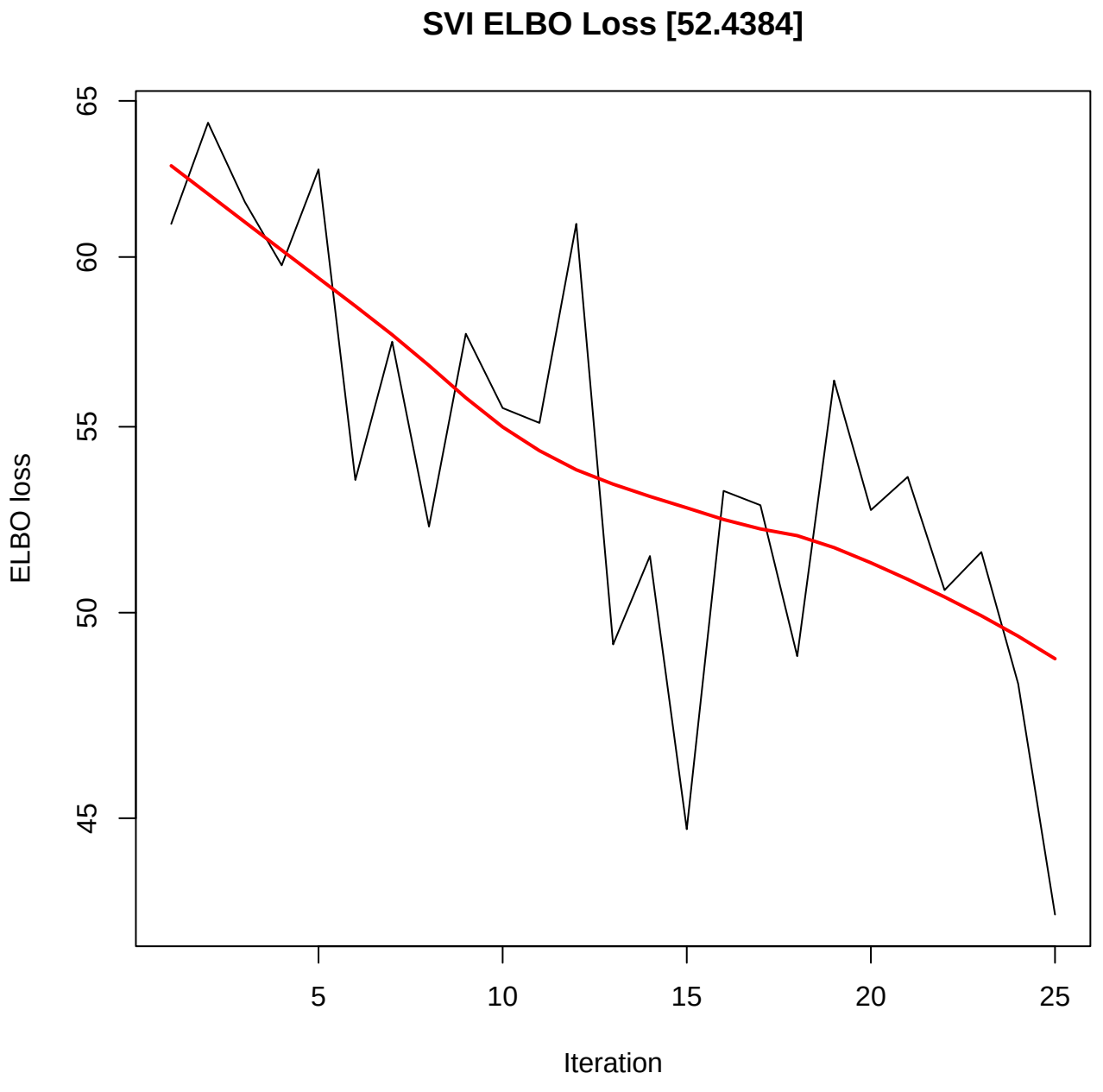

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

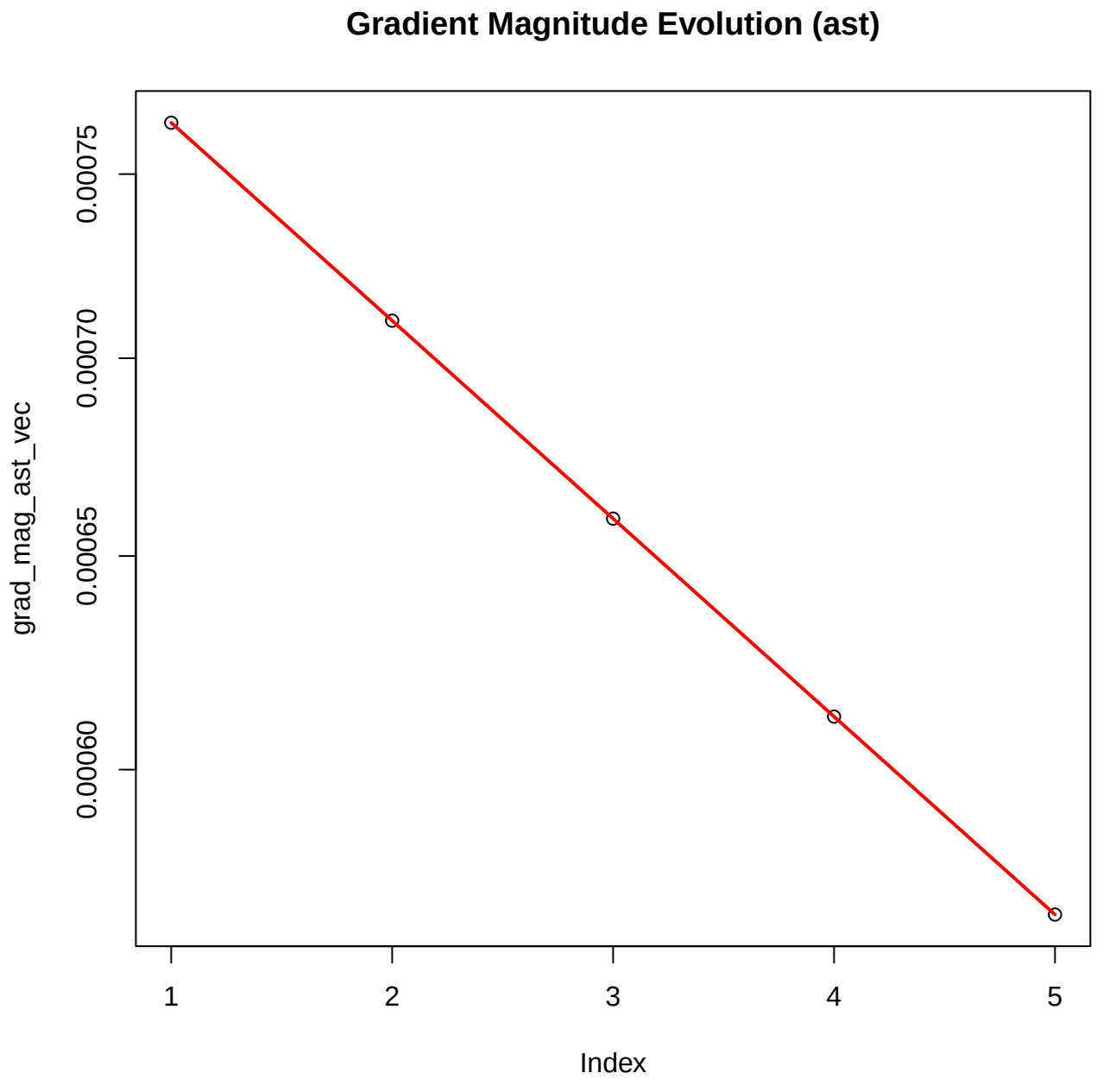

Value (ast)
loss_ast_vec
0.9
0.8
0.7
0.6
0.5
0.4
1
2
3
4
5
Index

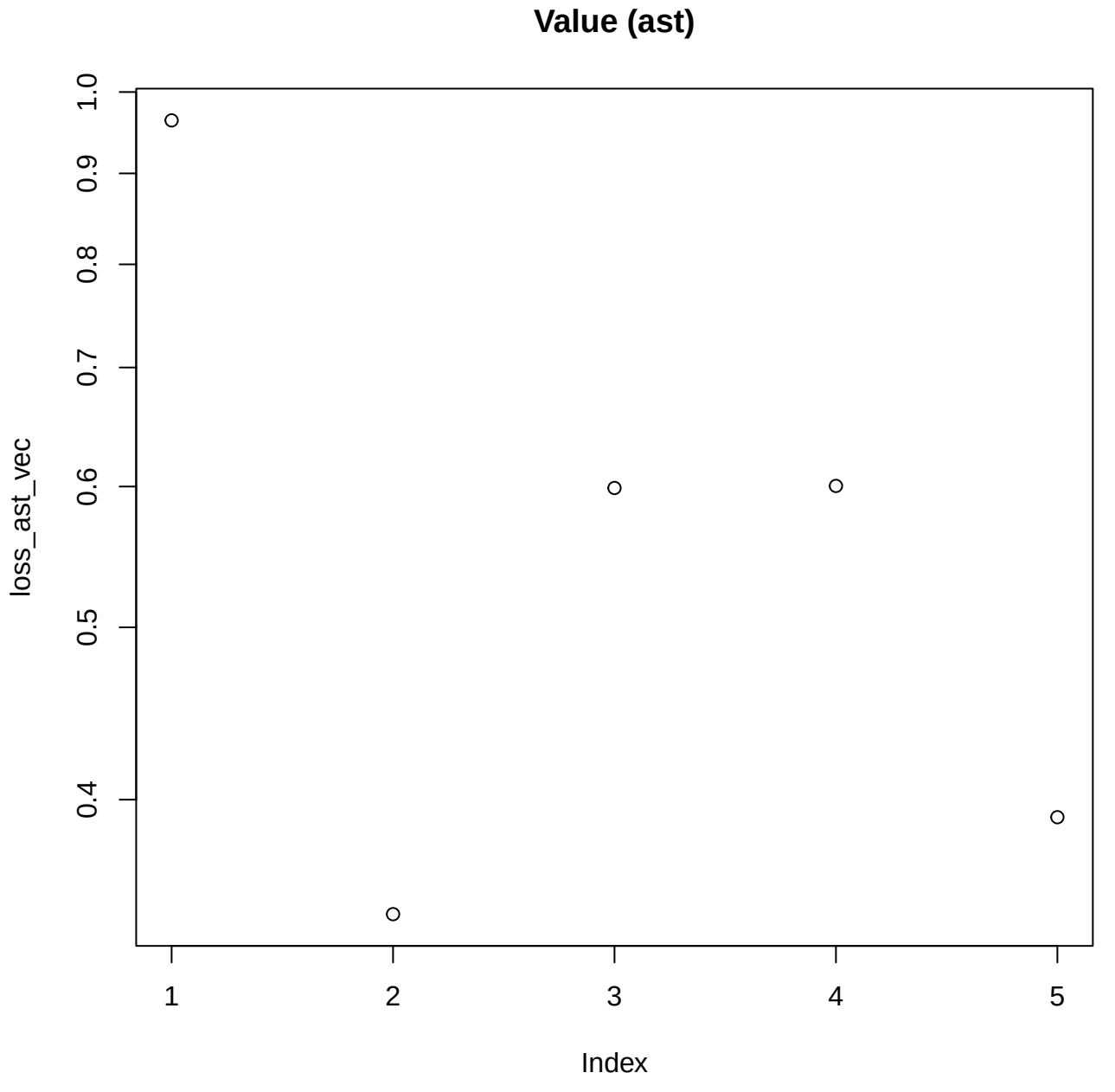

SVI ELBO Loss [93.1601]
ELBO loss
Iteration
130
120
110
100
90
80
70
5
10
15
20
25

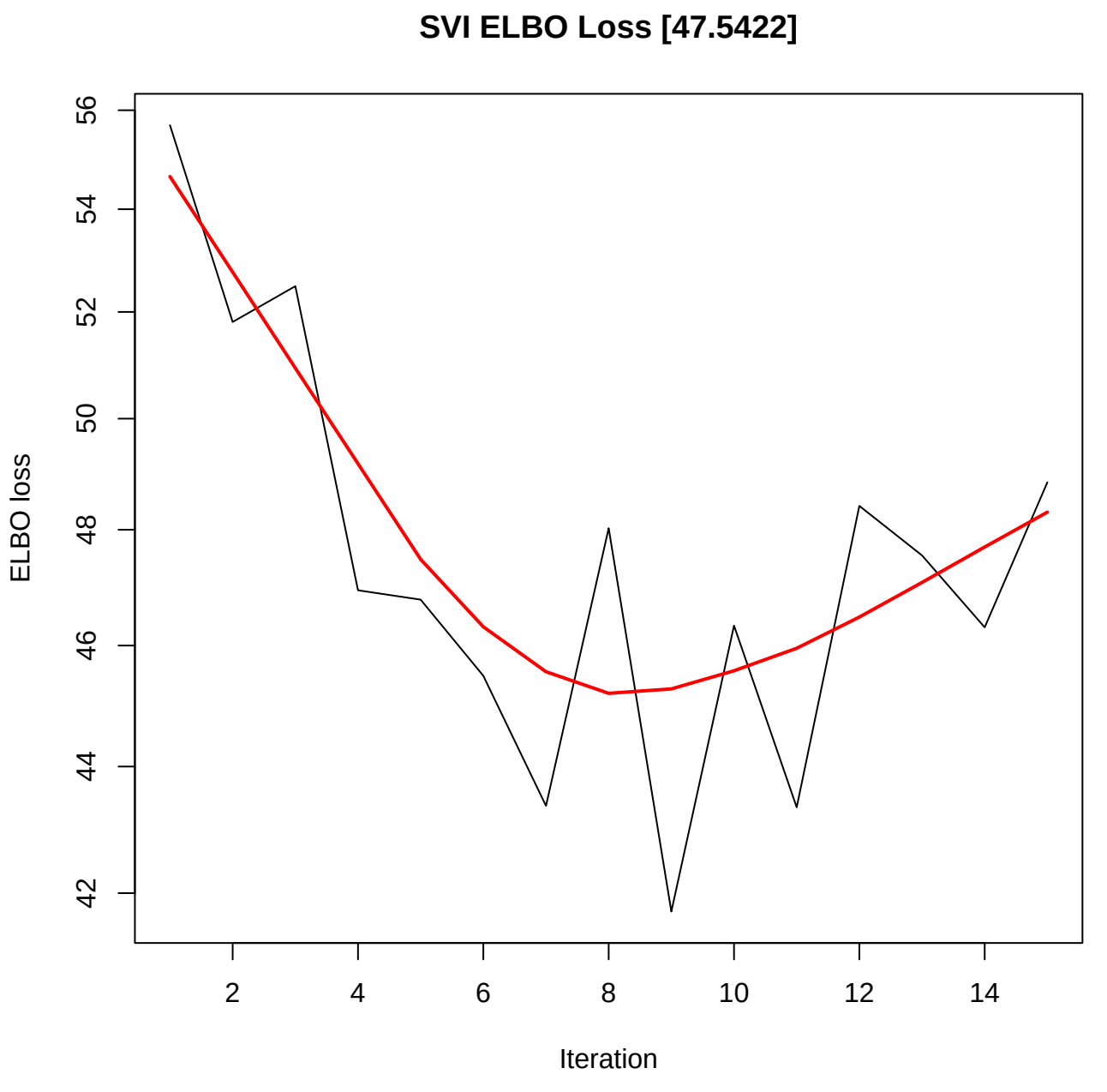

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

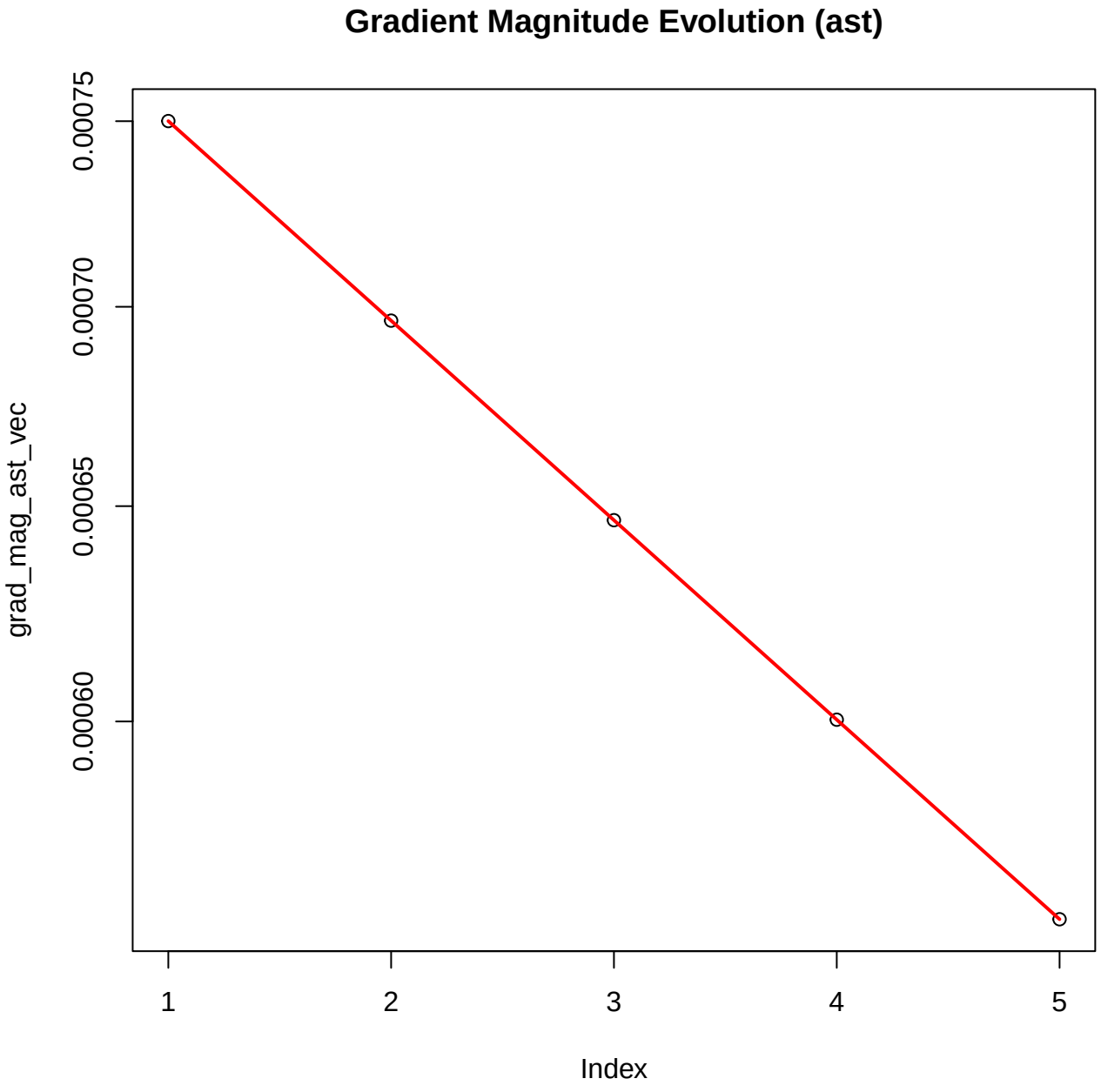

Value (ast)
loss_ast_vec
0.25
0.30
0.35
0.40
0.45
1
2
3
4
5
Index

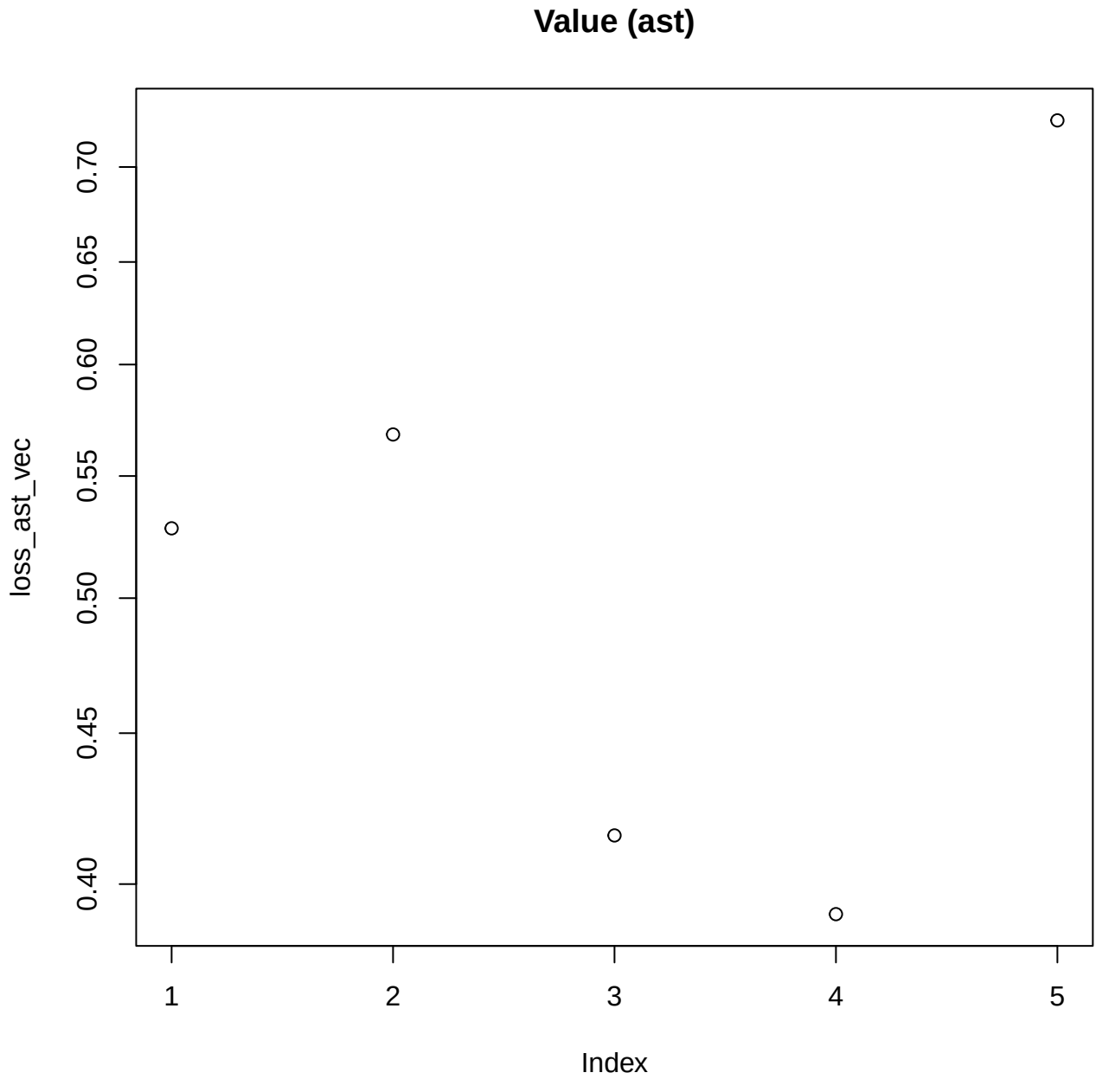

SVI ELBO Loss [52.4384]
ELBO loss
Iteration
65
60
55
50
45
5
10
15
20
25

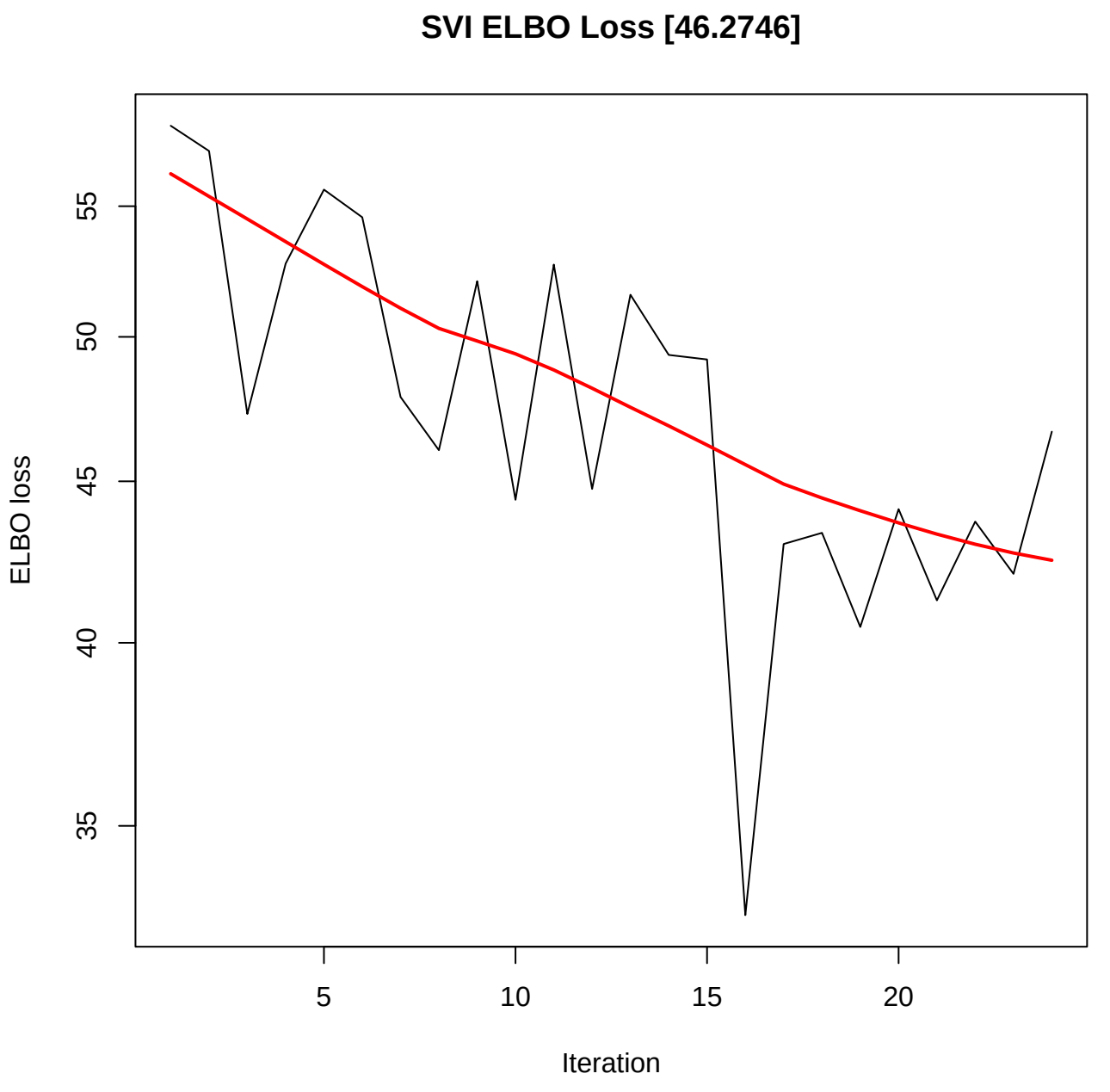

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

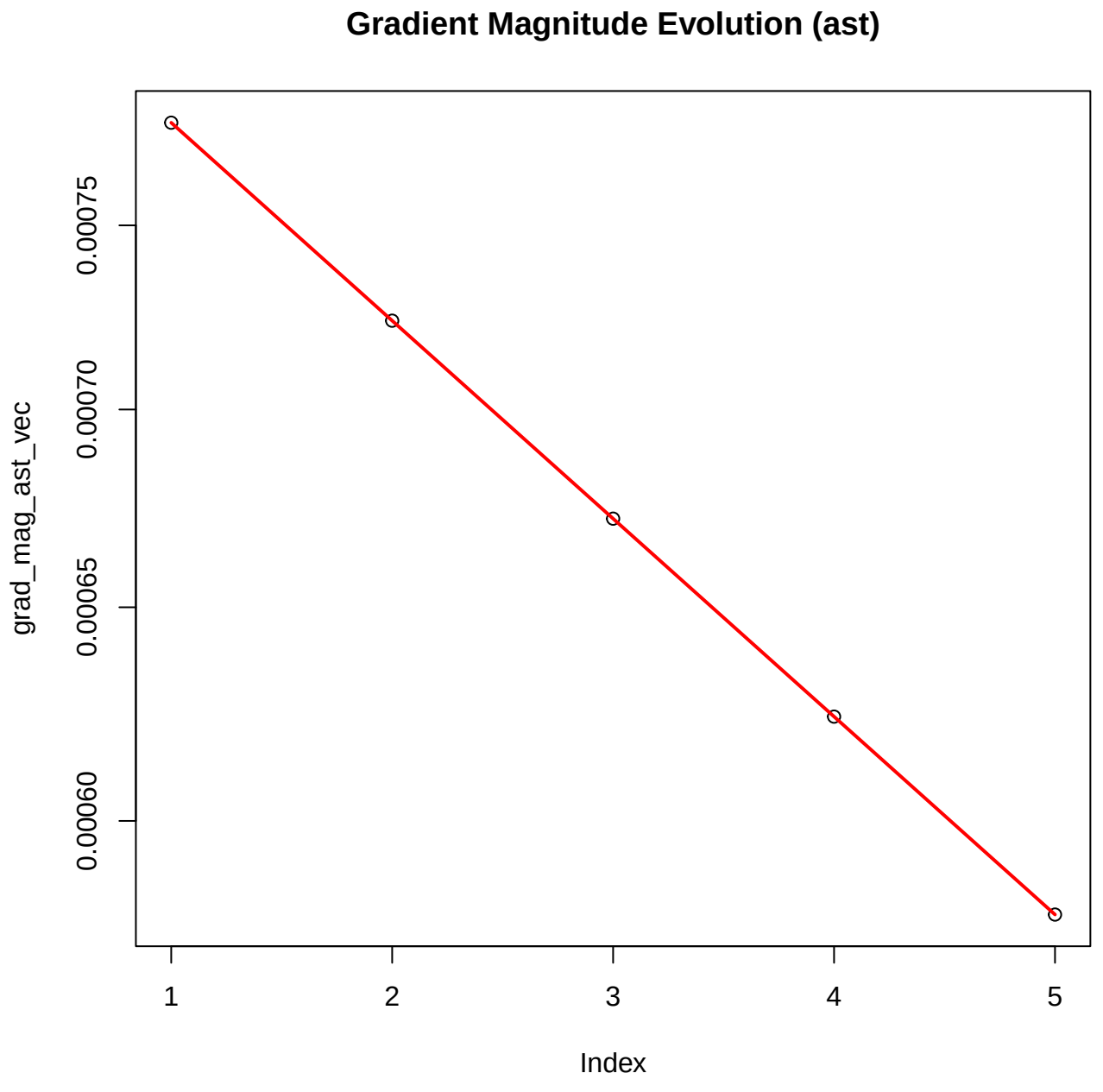

Value (ast)
loss_ast_vec
1.0
0.9
0.8
0.7
0.6
0.5
0.4
1
2
3
4
5
Index

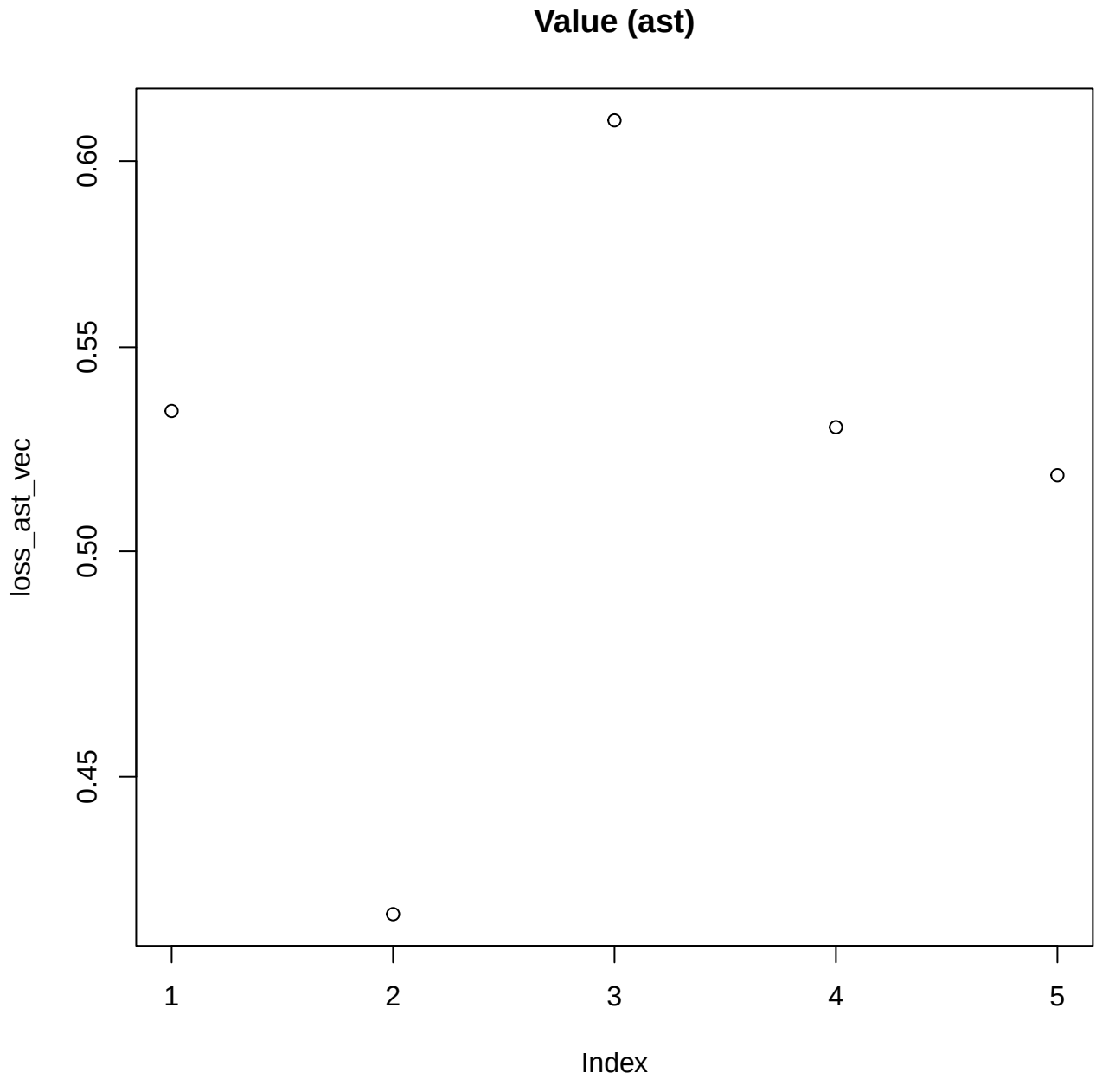

SVI ELBO Loss [47.5422]
ELBO loss
Iteration
56
54
52
50
48
46
44
42
2
4
6
8
10
12
14

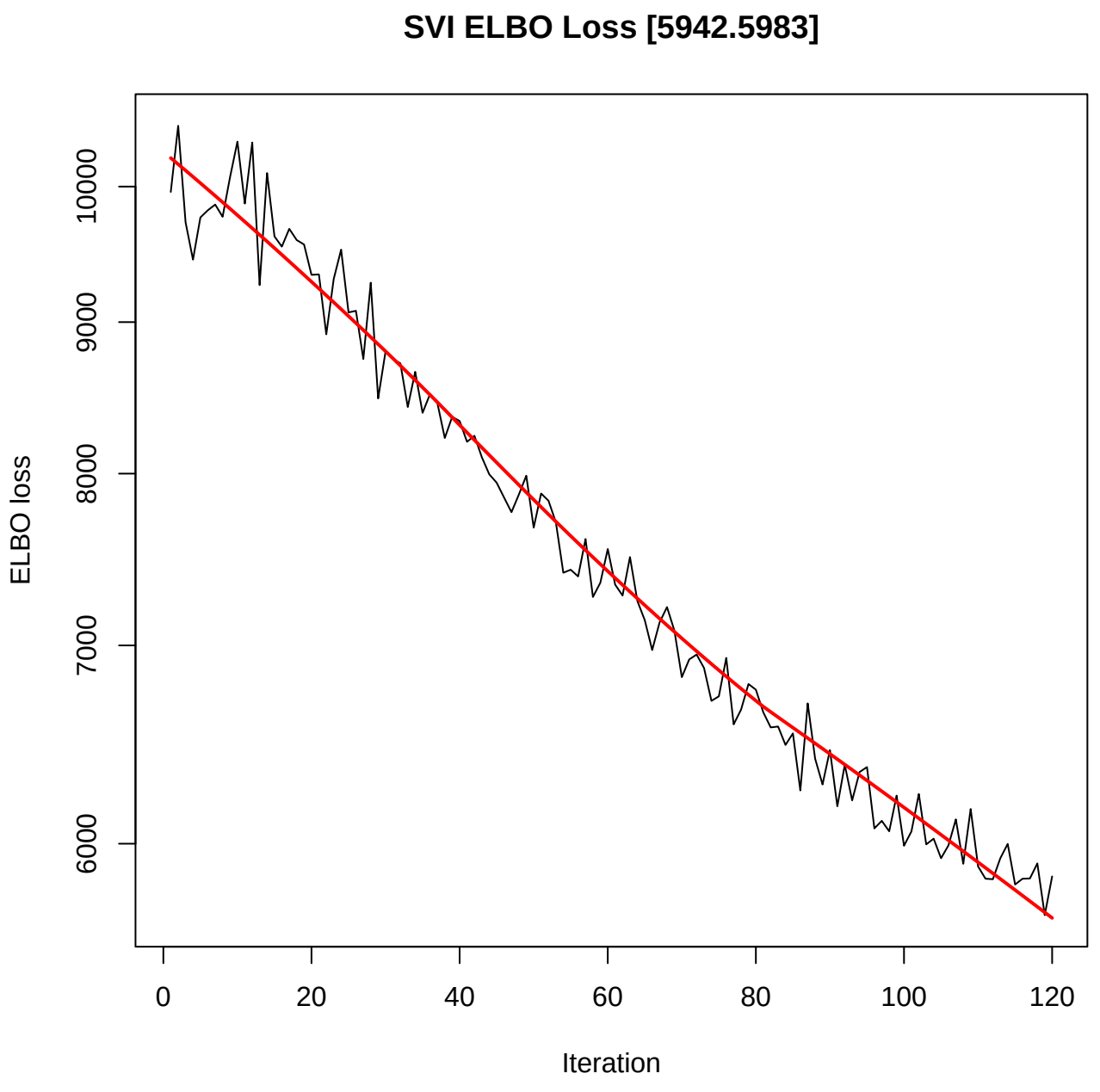

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

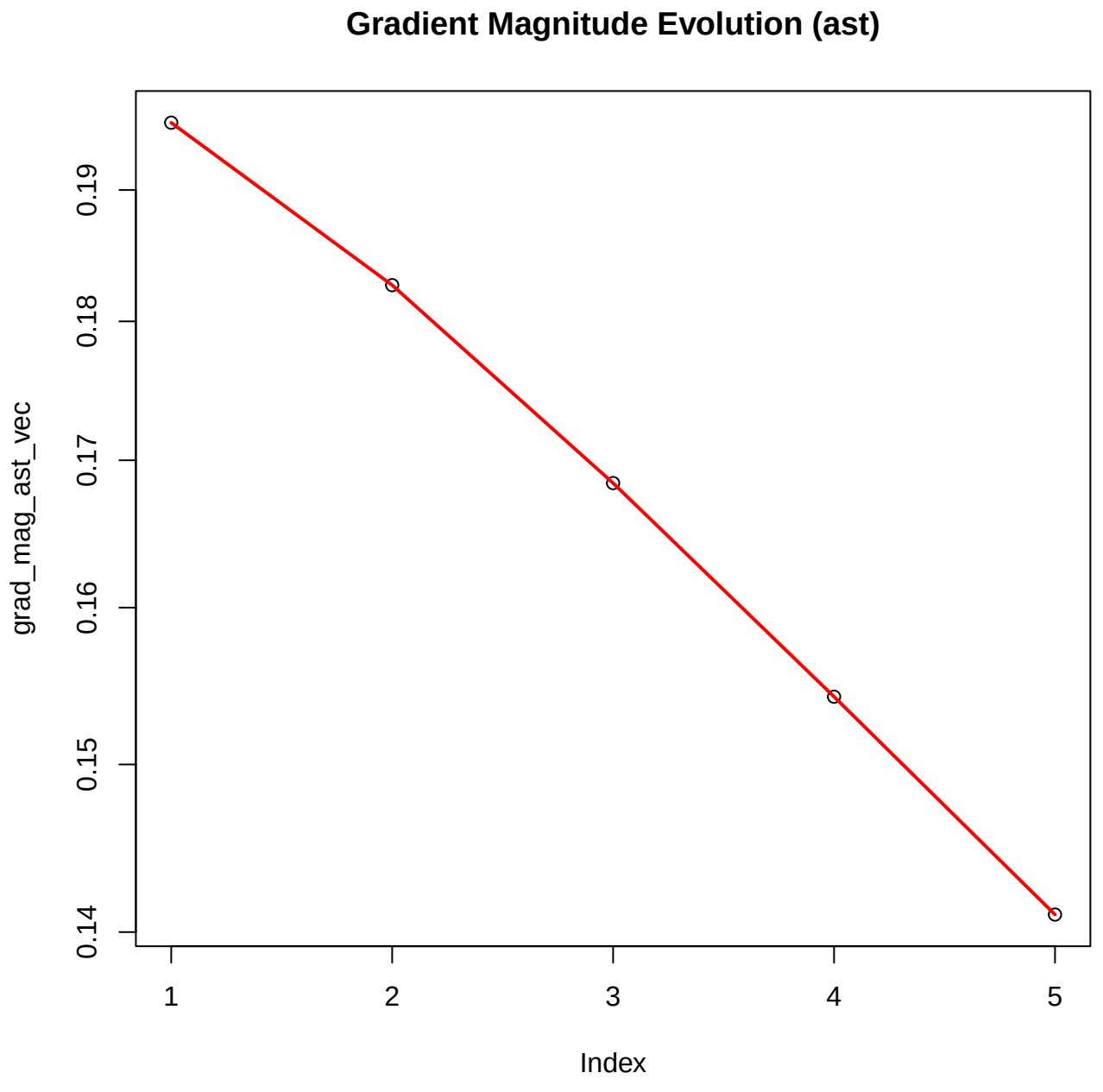

Value (ast)
loss_ast_vec
0.40
0.45
0.50
0.55
0.60
0.65
0.70
1
2
3
4
5
Index

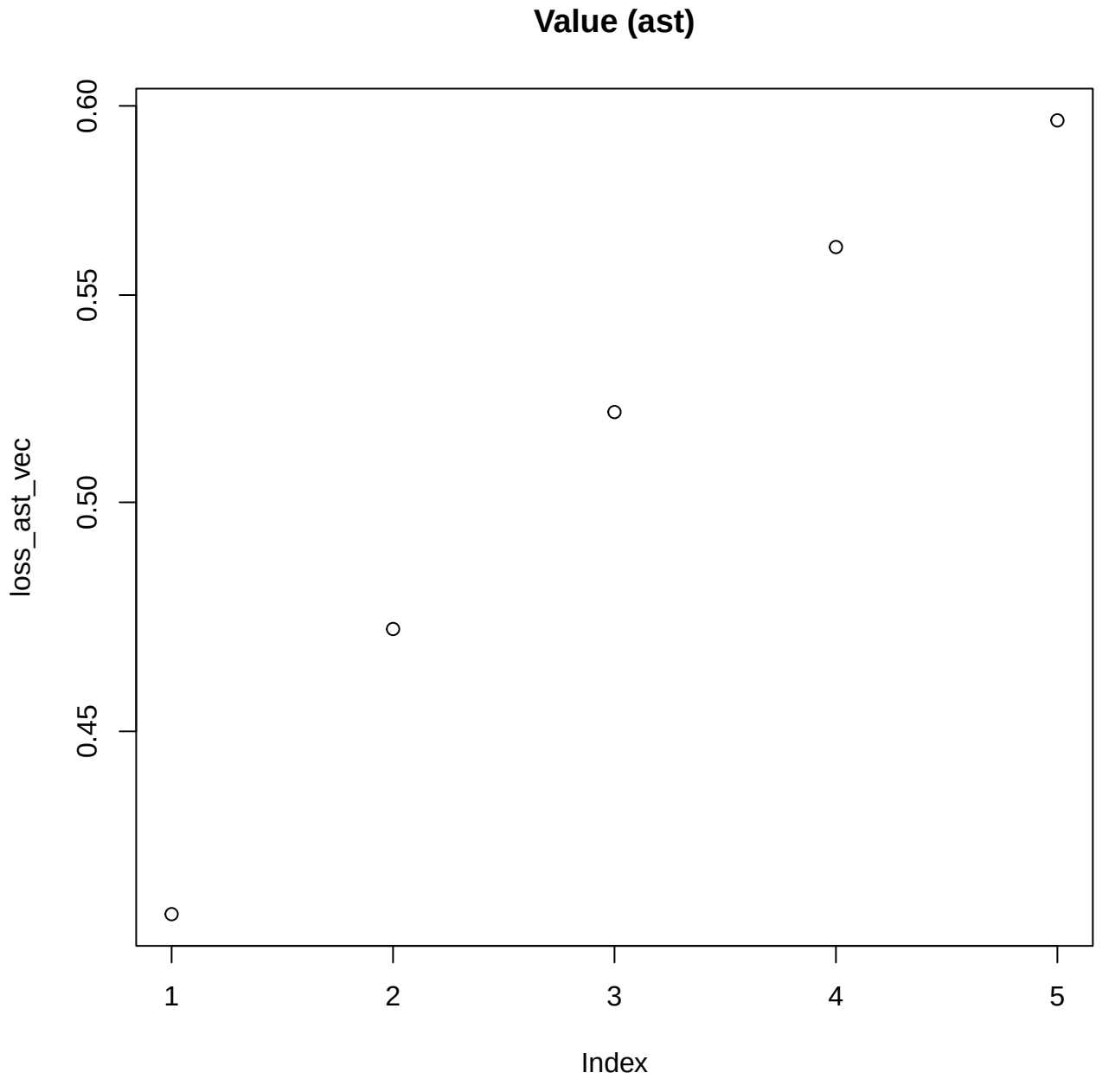

SVI ELBO Loss [46.2746]
ELBO loss
35
40
45
50
55
5
10
15
20
Iteration

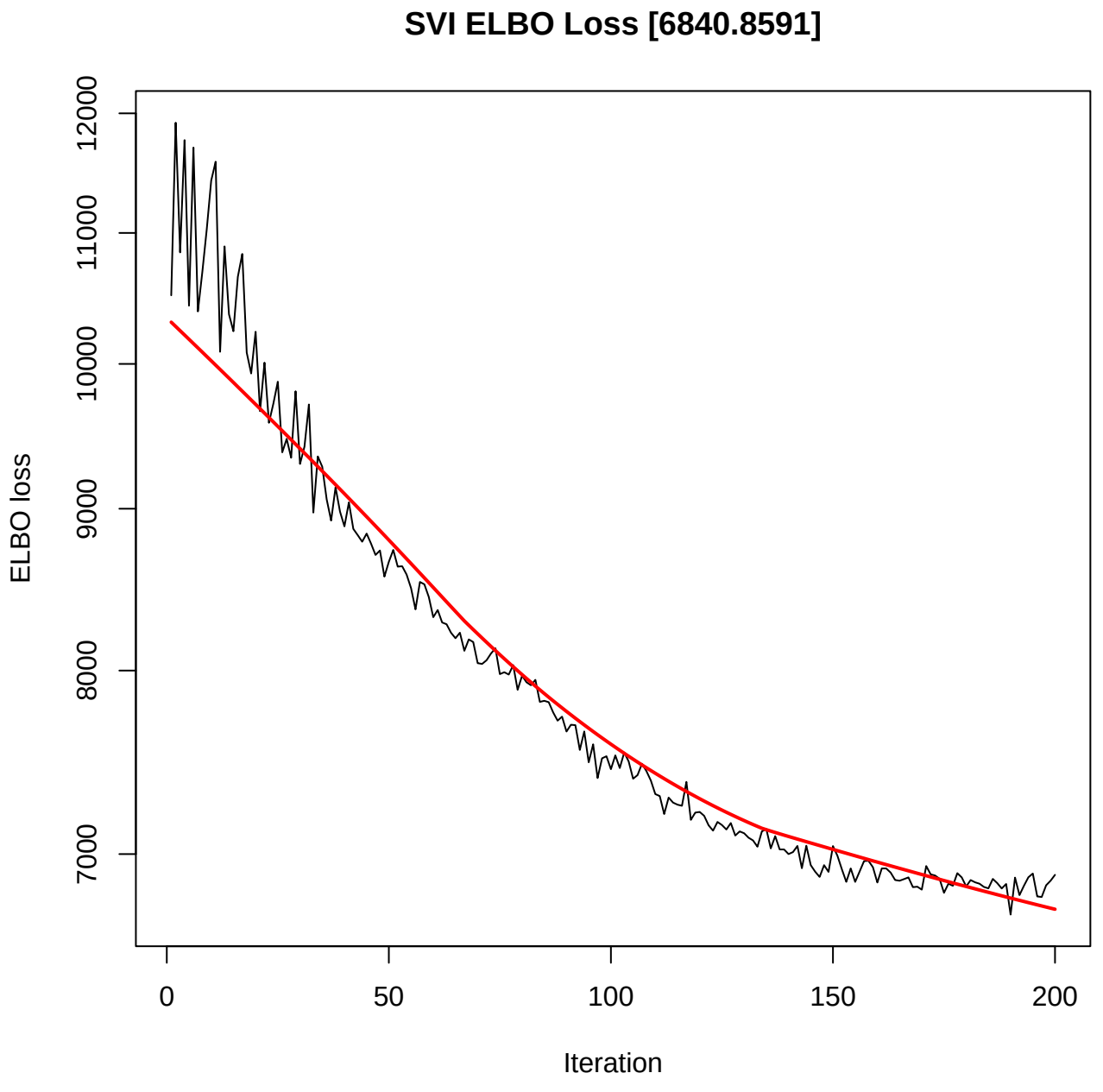

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

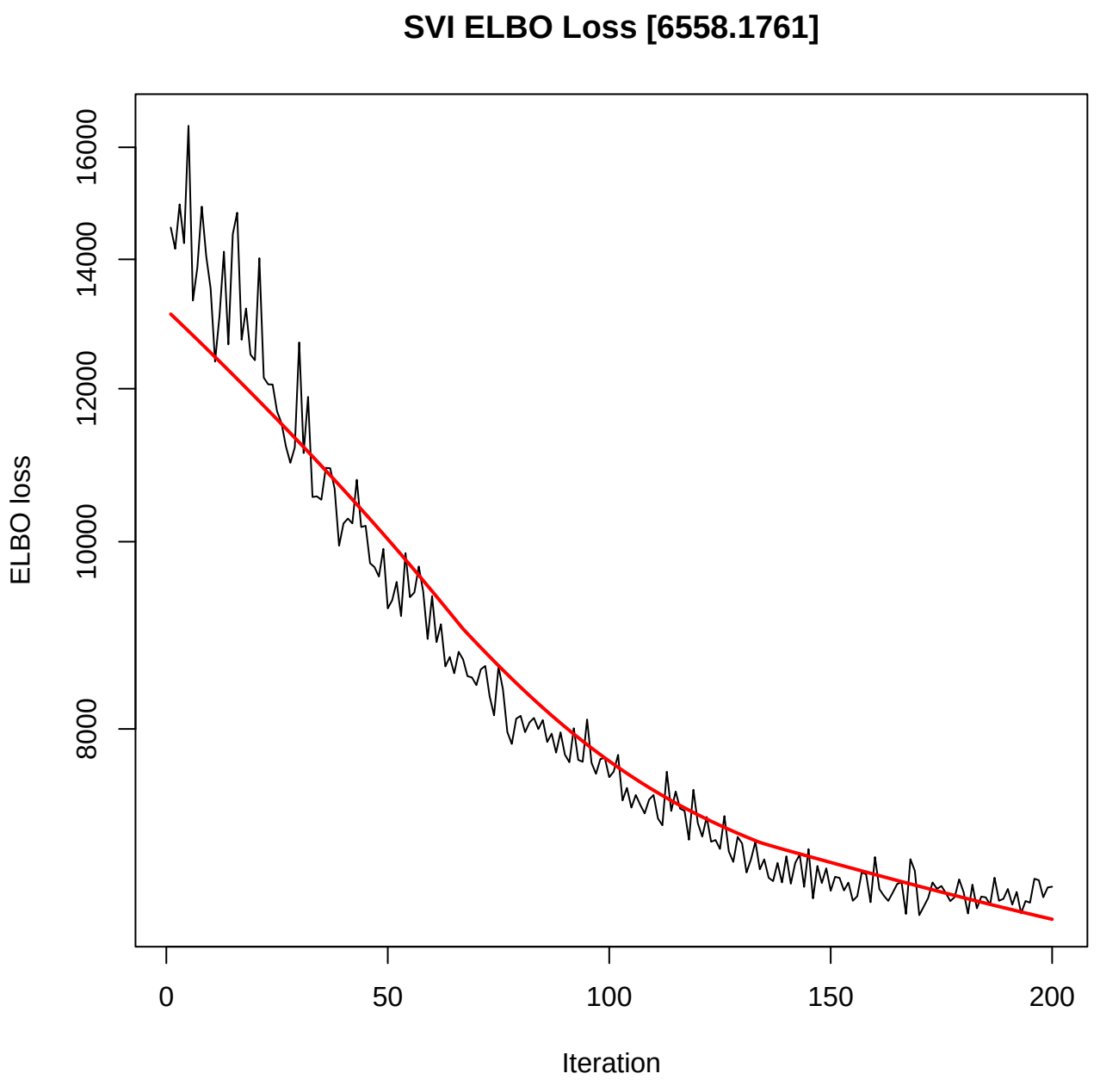

Value (ast)
loss_ast_vec
0.45
0.50
0.55
0.60
1
2
3
4
5
Index

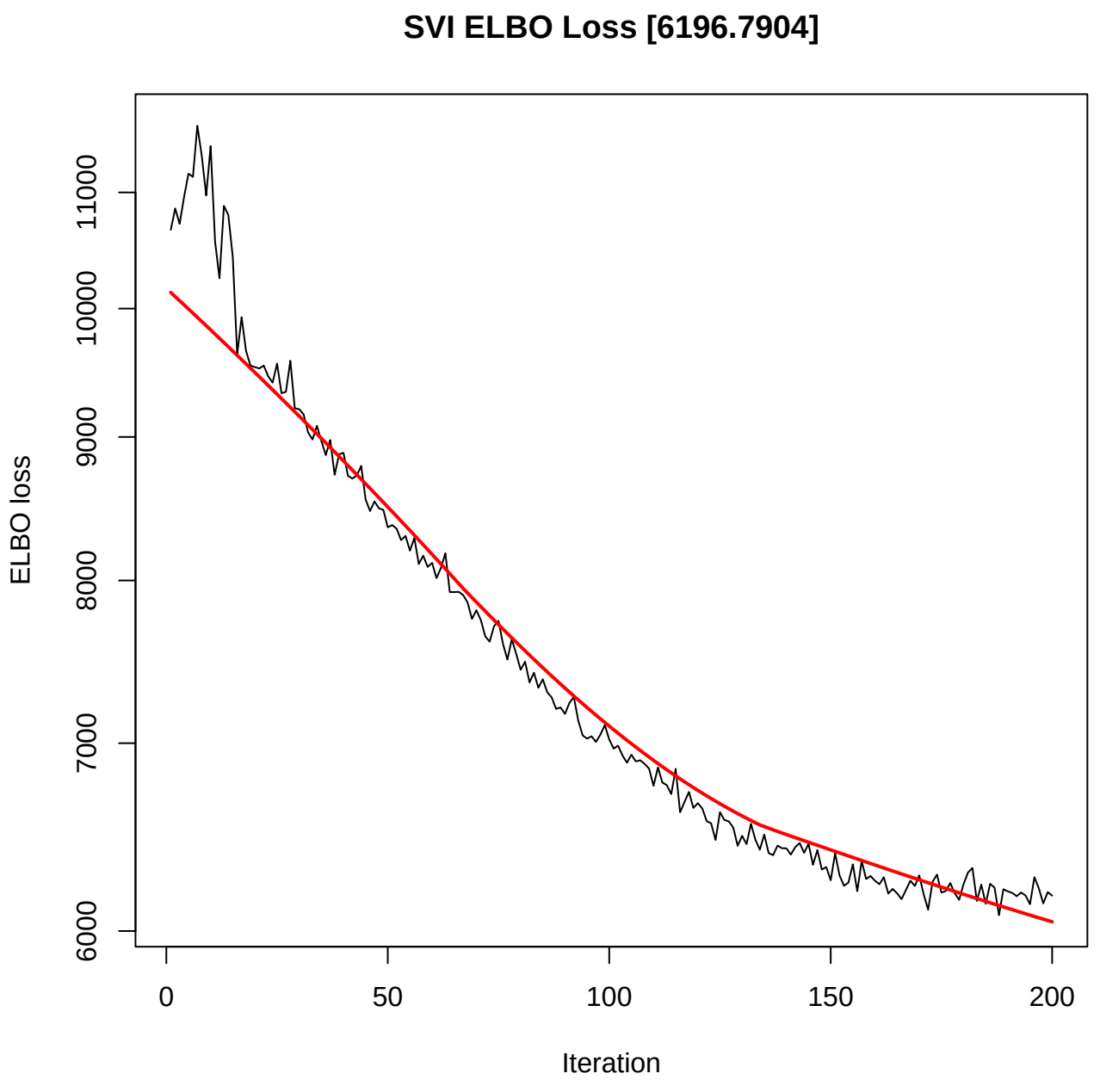

SVI ELBO Loss [5942.5983]
ELBO loss
10000
9000
8000
7000
6000
0
20
40
60
80
100
120
Iteration

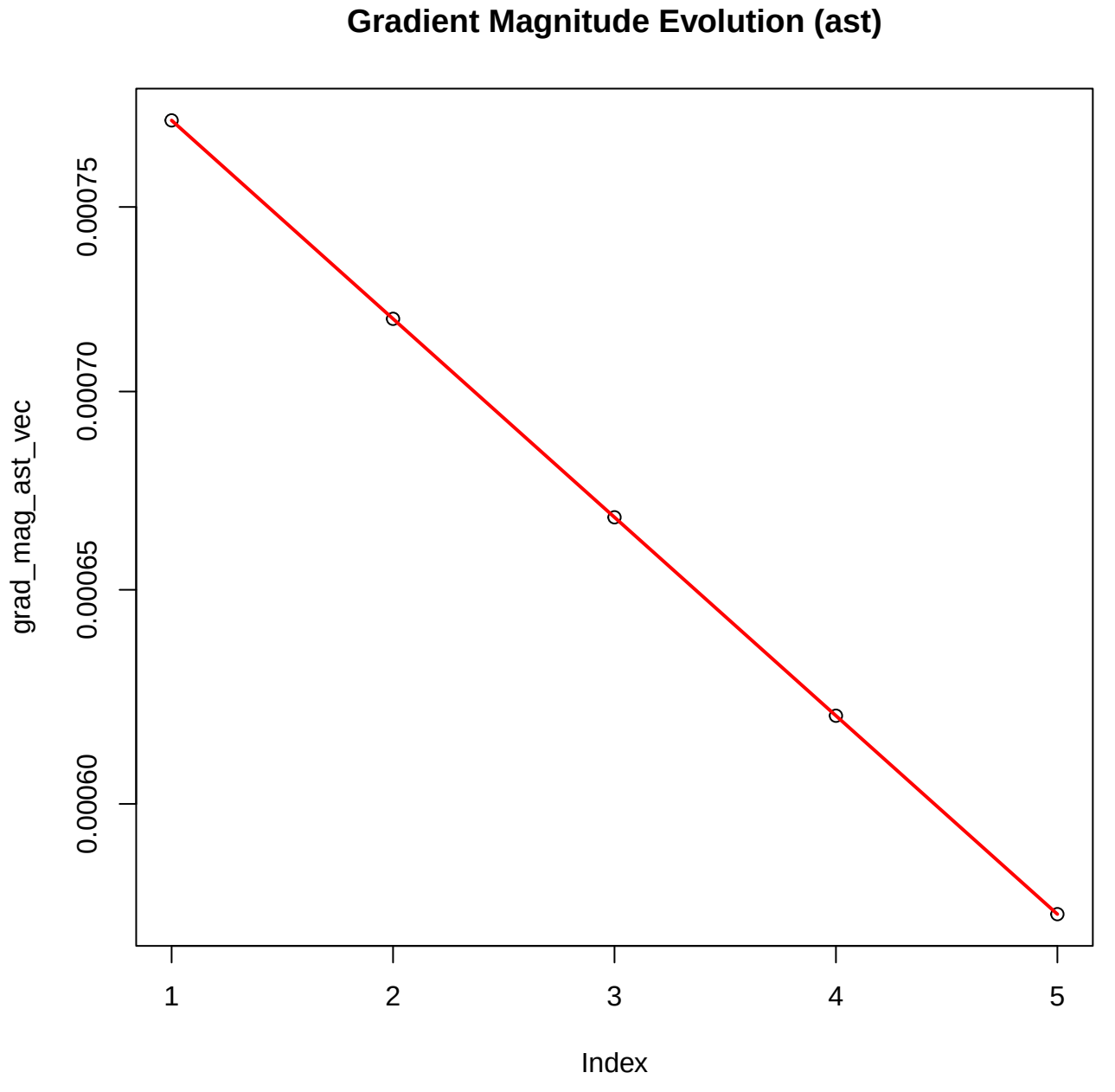

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.14
0.15
0.16
0.17
0.18
0.19
1
2
3
4
5
Index

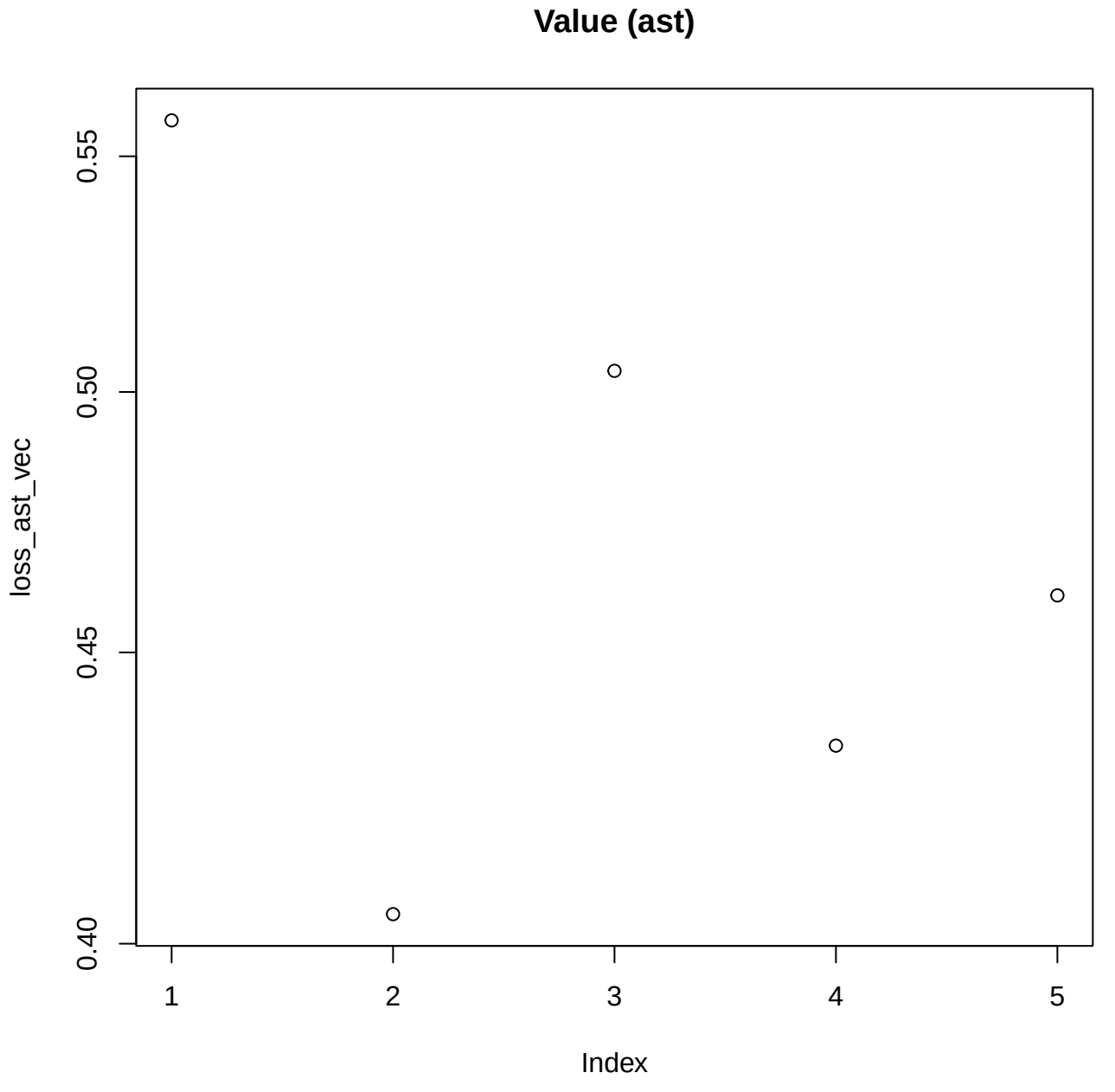

Value (ast)
loss_ast_vec
0.45
0.50
0.55
0.60
1
2
3
4
5
Index

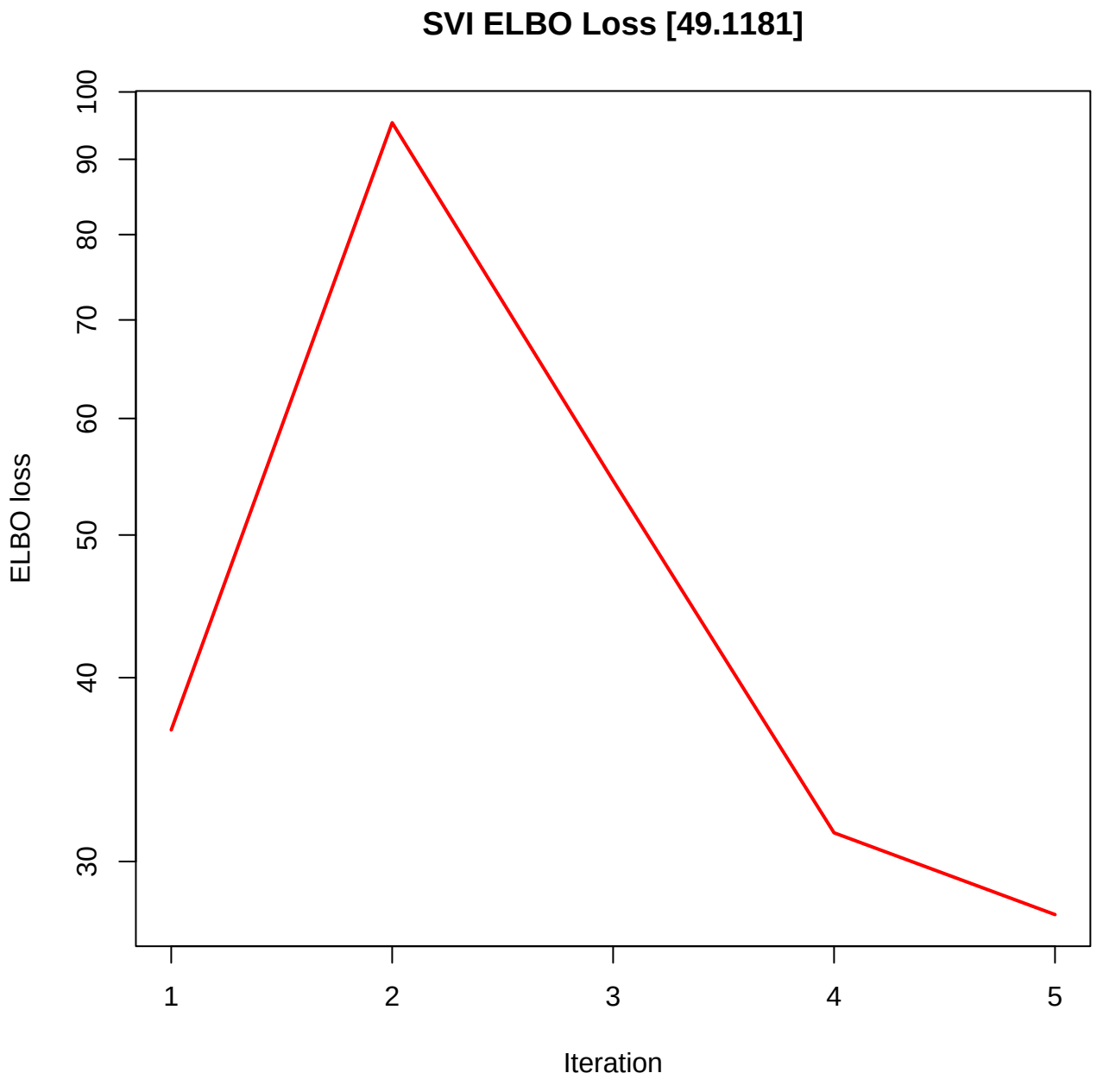

SVI ELBO Loss [6840.8591]
ELBO loss
12000
11000
10000
9000
8000
7000
0
50
100
150
200
Iteration

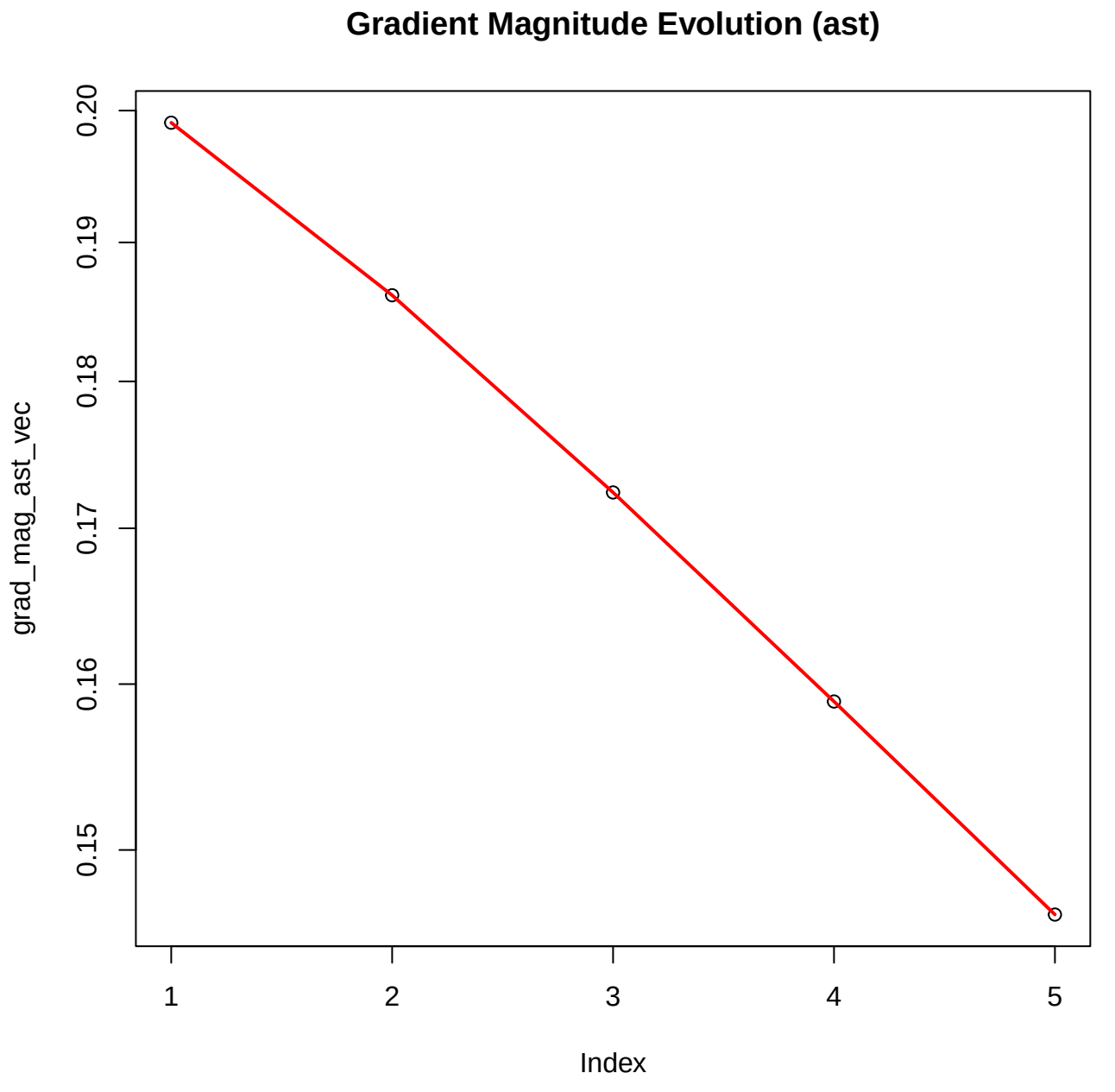

SVI ELBO Loss [6558.1761]
ELBO loss
8000
10000
12000
14000
16000
0
50
100
150
200
Iteration

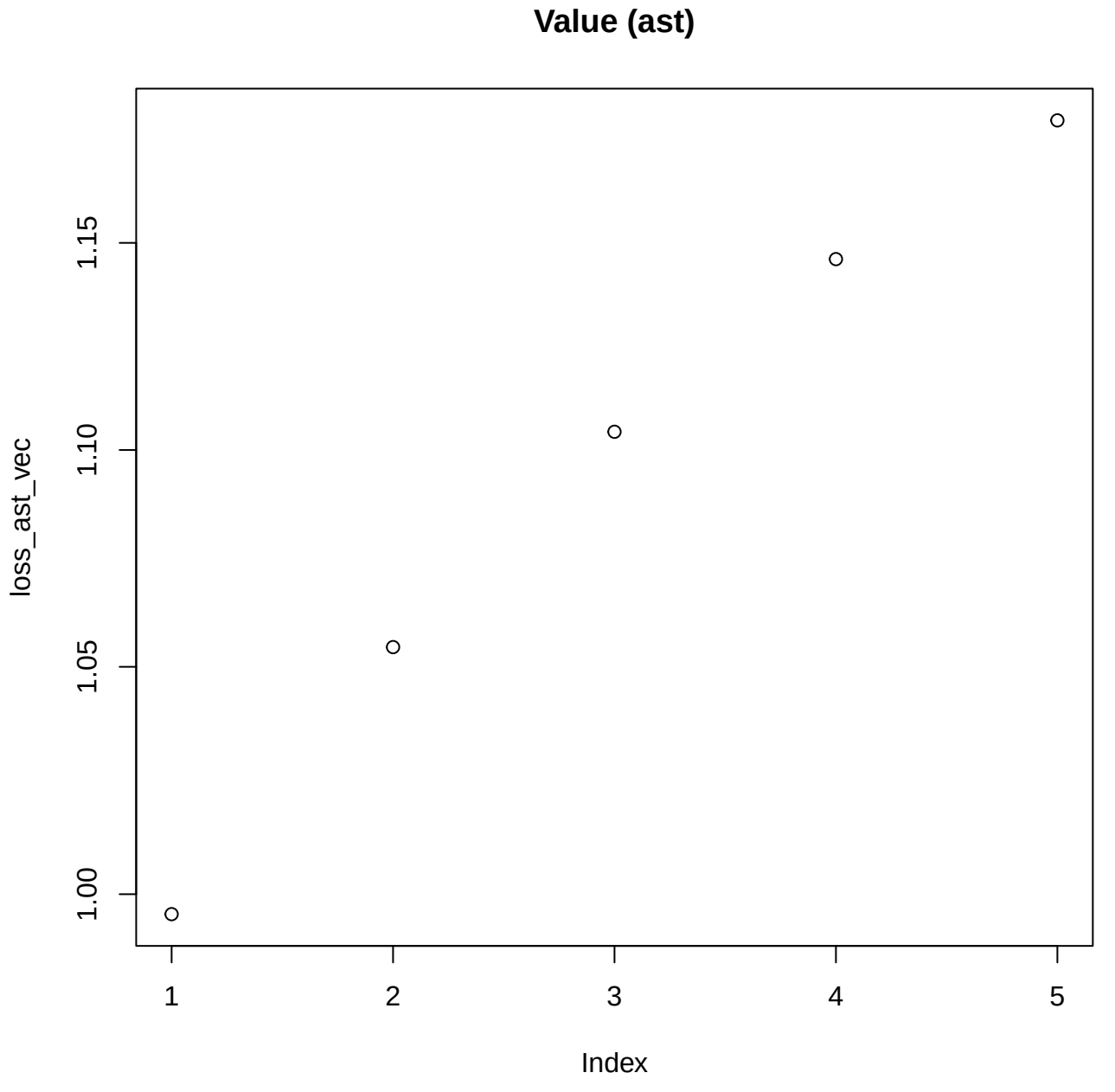

SVI ELBO Loss [6196.7904]
ELBO loss
6000
7000
8000
9000
10000
11000
0
50
100
150
200
Iteration

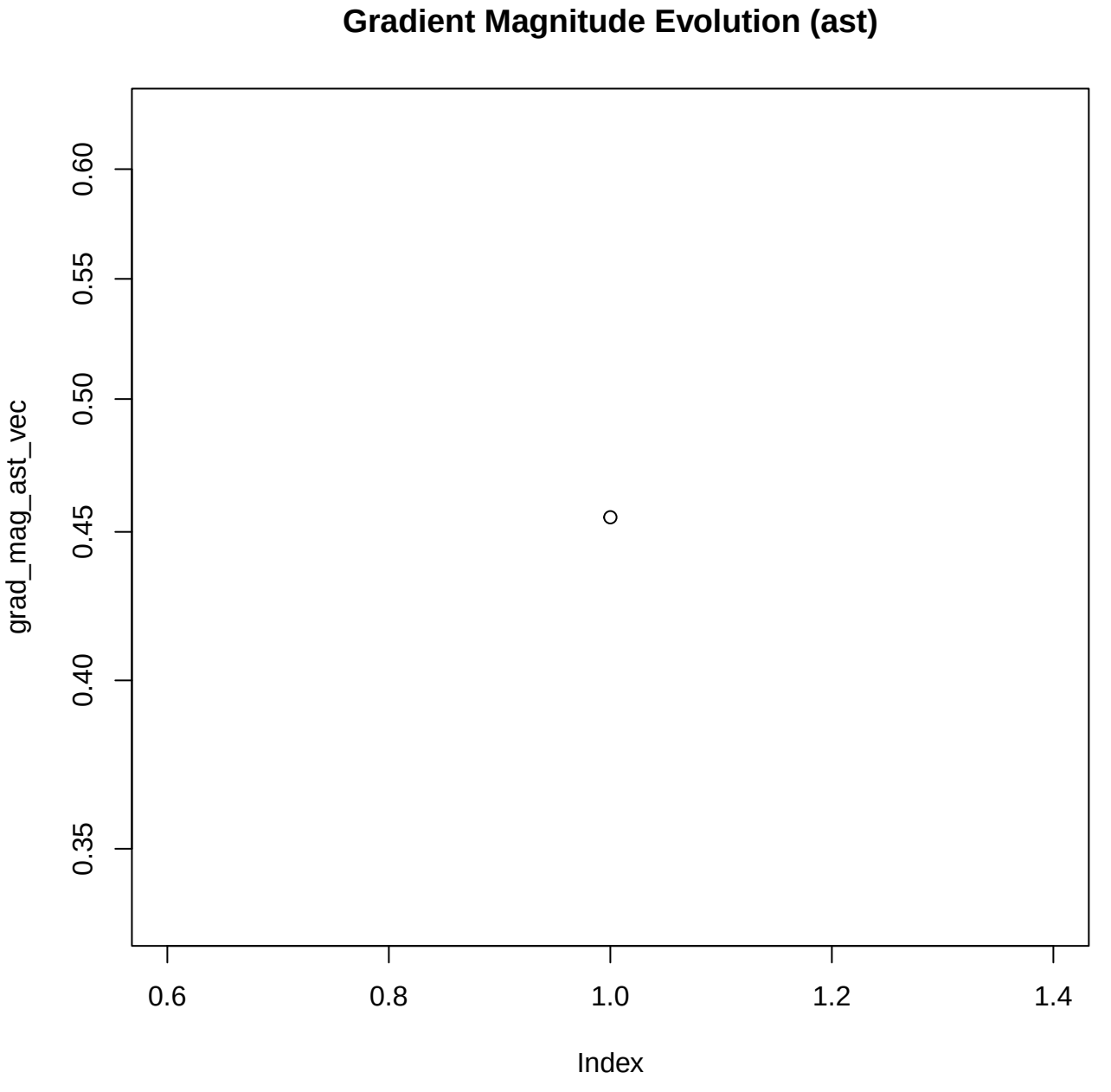

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

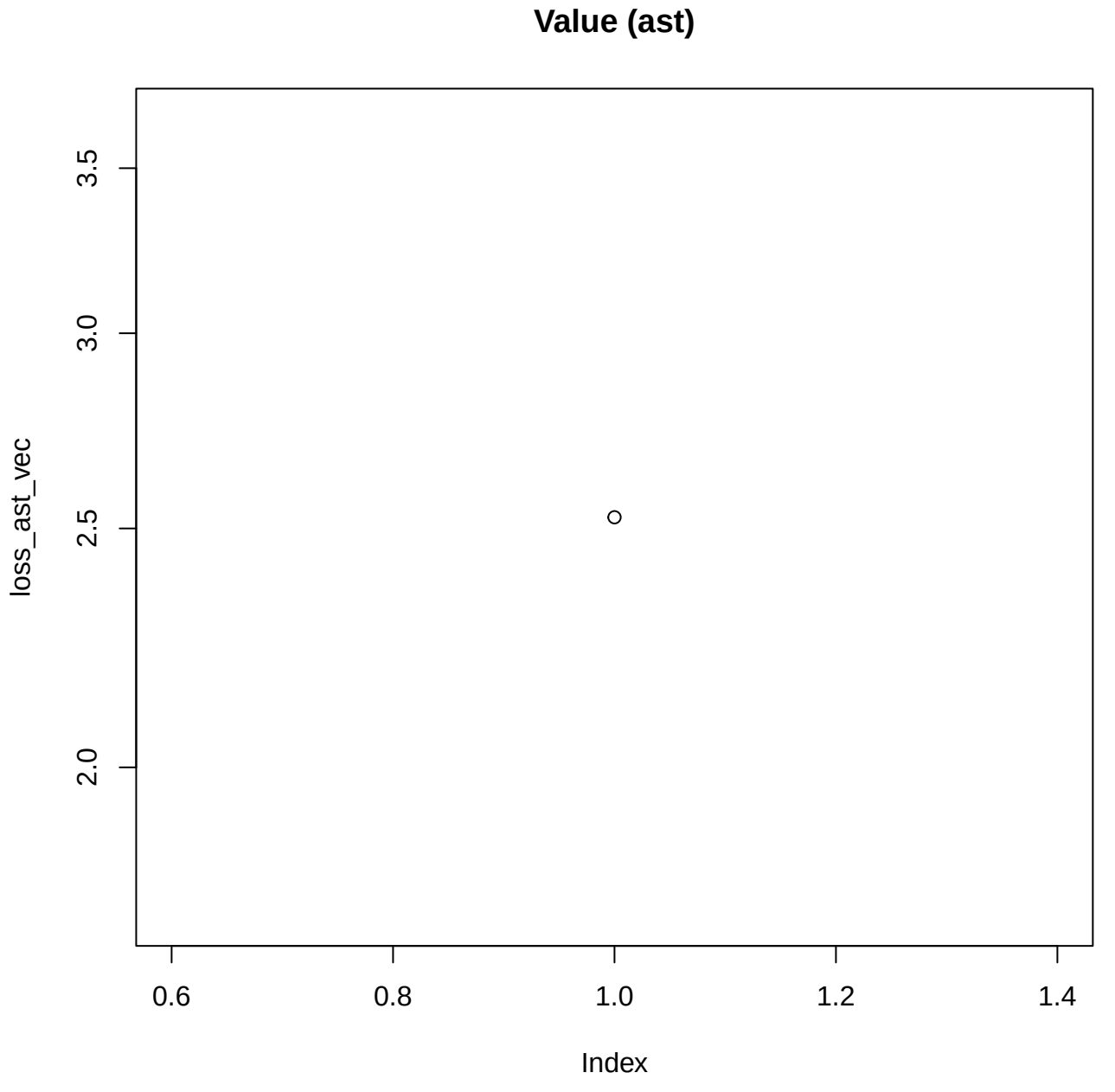

Value (ast)
loss_ast_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index

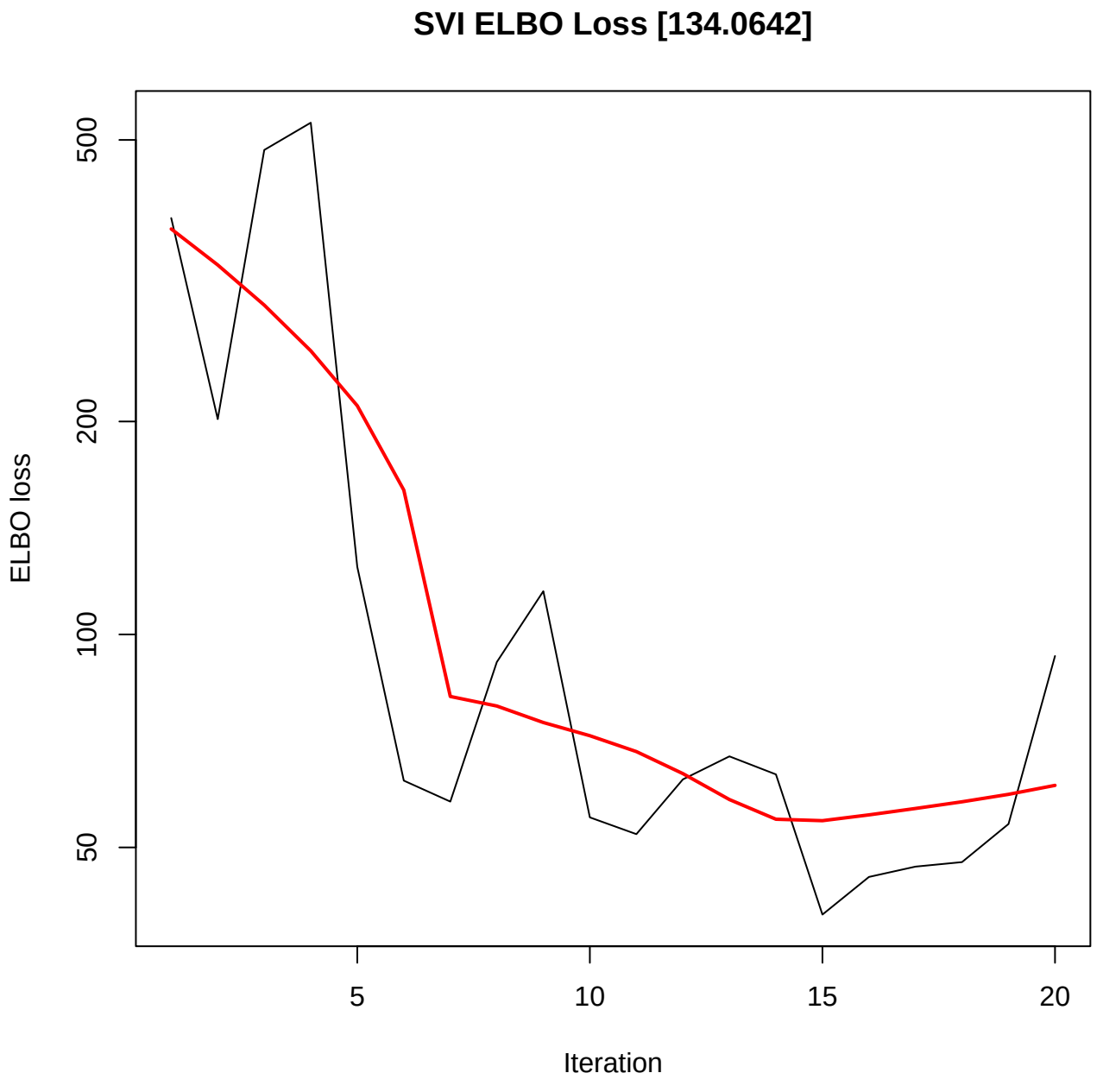

SVI ELBO Loss [49.1181]
ELBO loss
100
90
80
70
60
50
40
30
1
2
3
4
5
Iteration

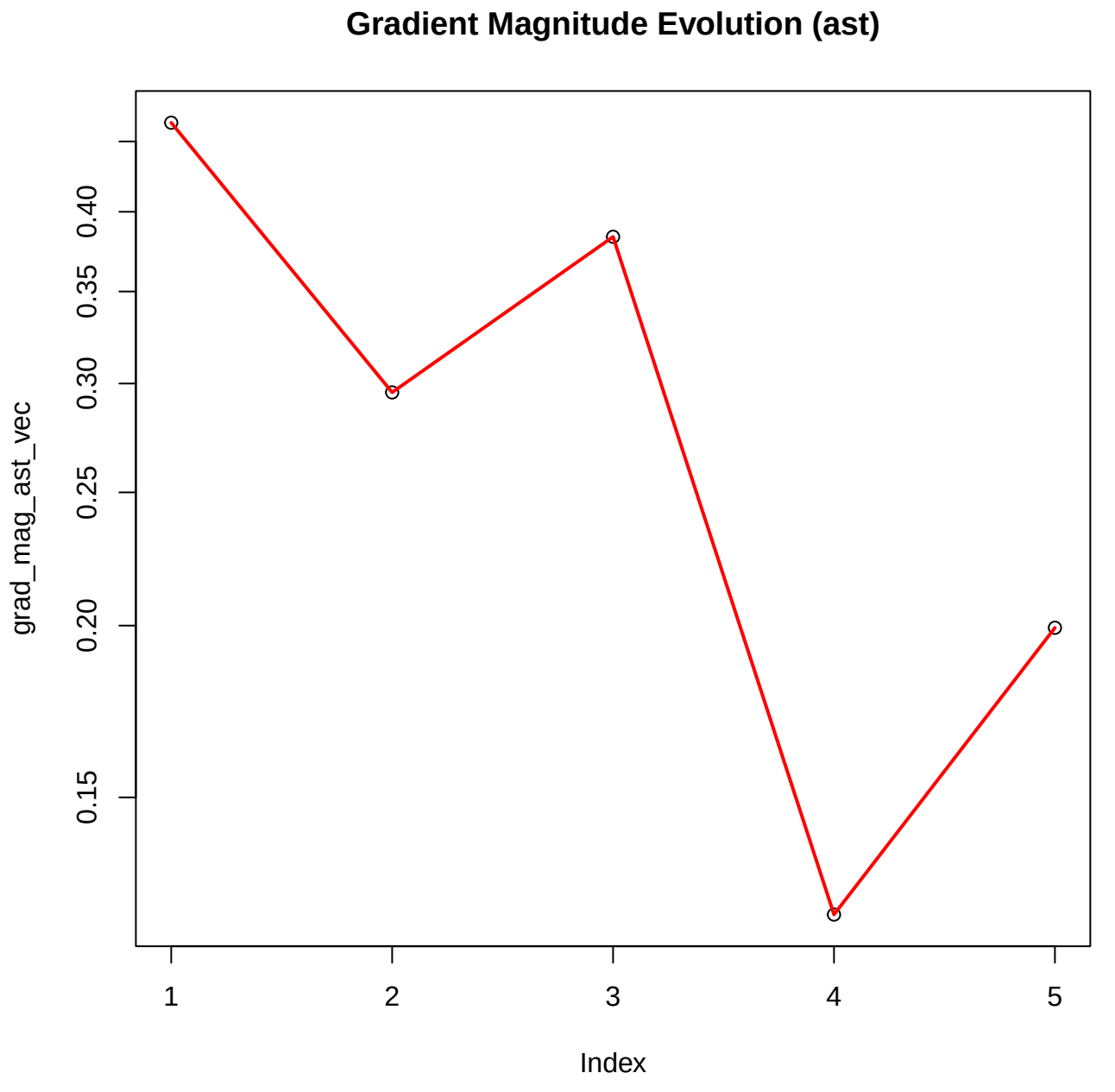

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.15
0.16
0.17
0.18
0.19
0.20
1
2
3
4
5
Index

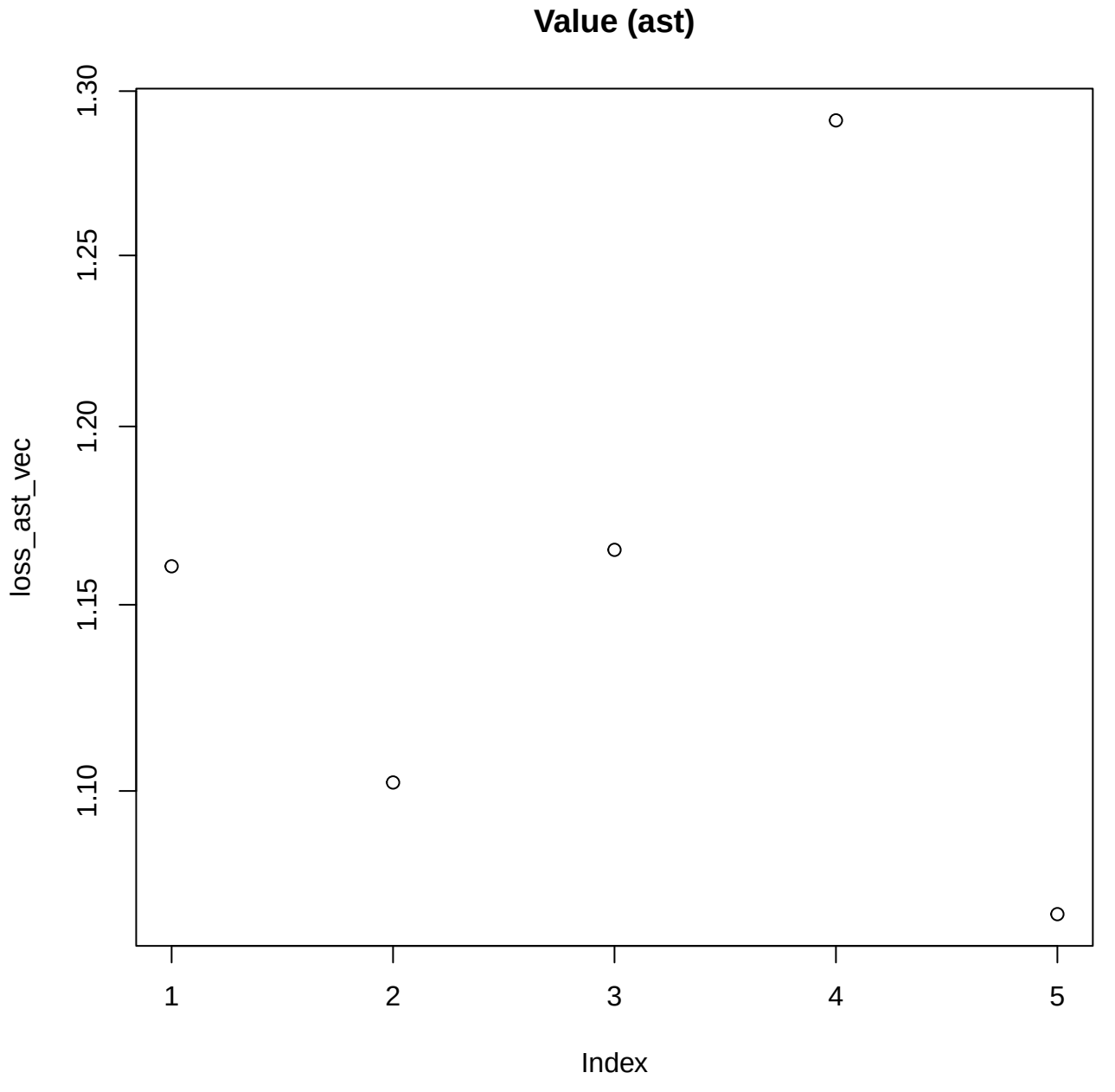

Value (ast)
loss_ast_vec
1.00
1.05
1.10
1.15
1
2
3
4
5
Index

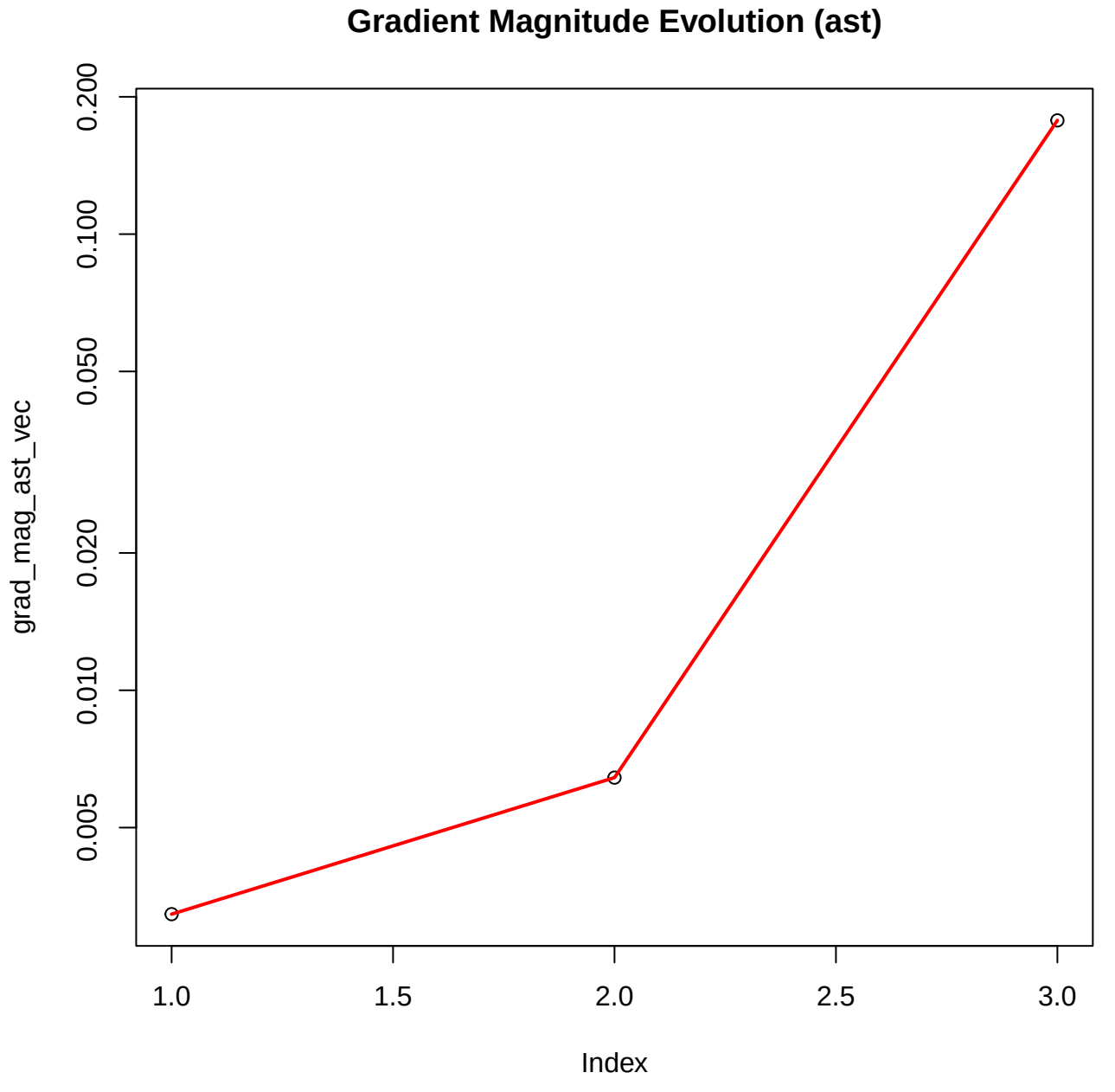

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.35
0.40
0.45
0.50
0.55
0.60
0.6
0.8
1.0
1.2
1.4
Index

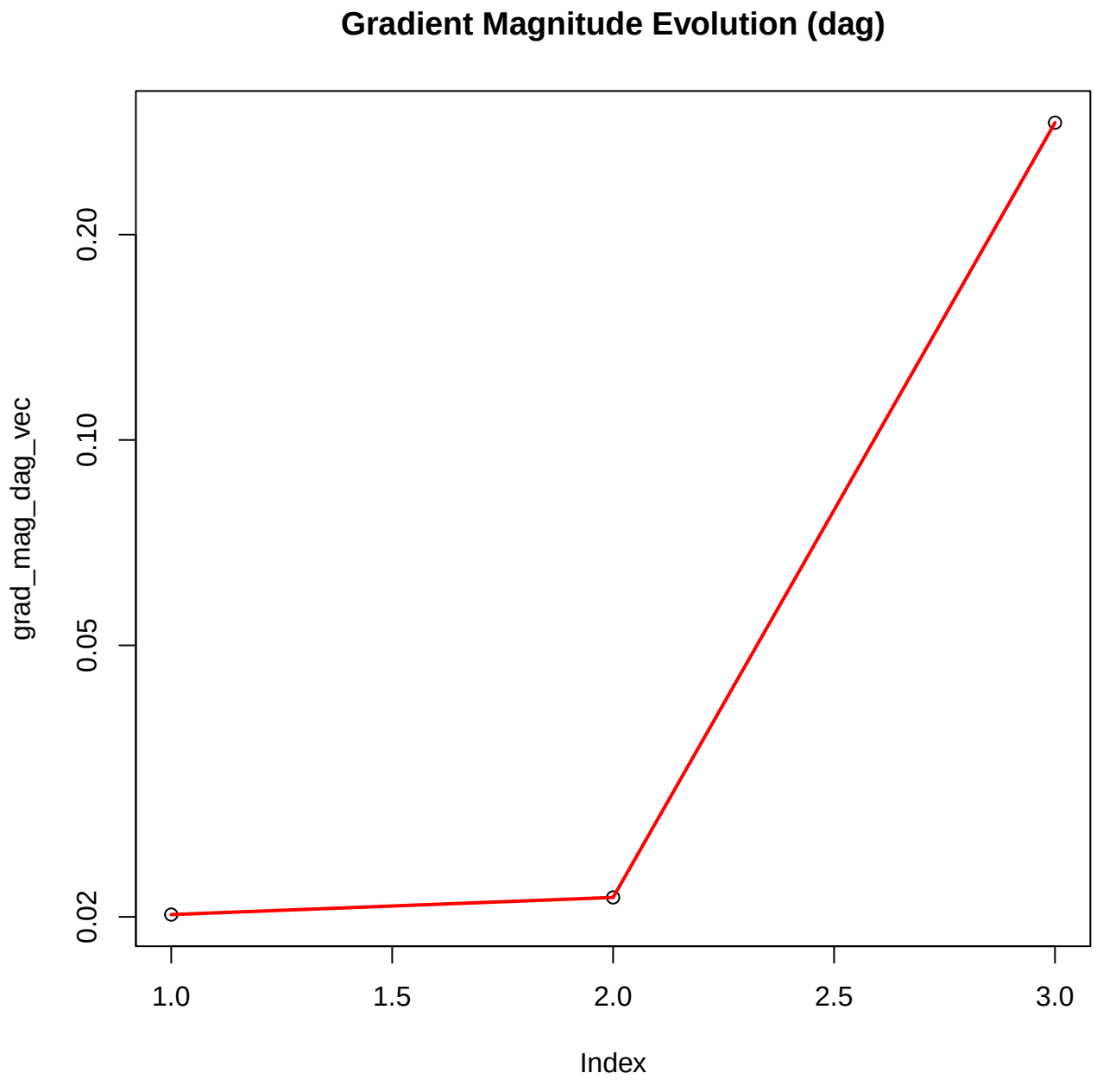

Value (ast)
loss_ast_vec
3.5
3.0
2.5
2.0
0.6
0.8
1.0
1.2
1.4
Index

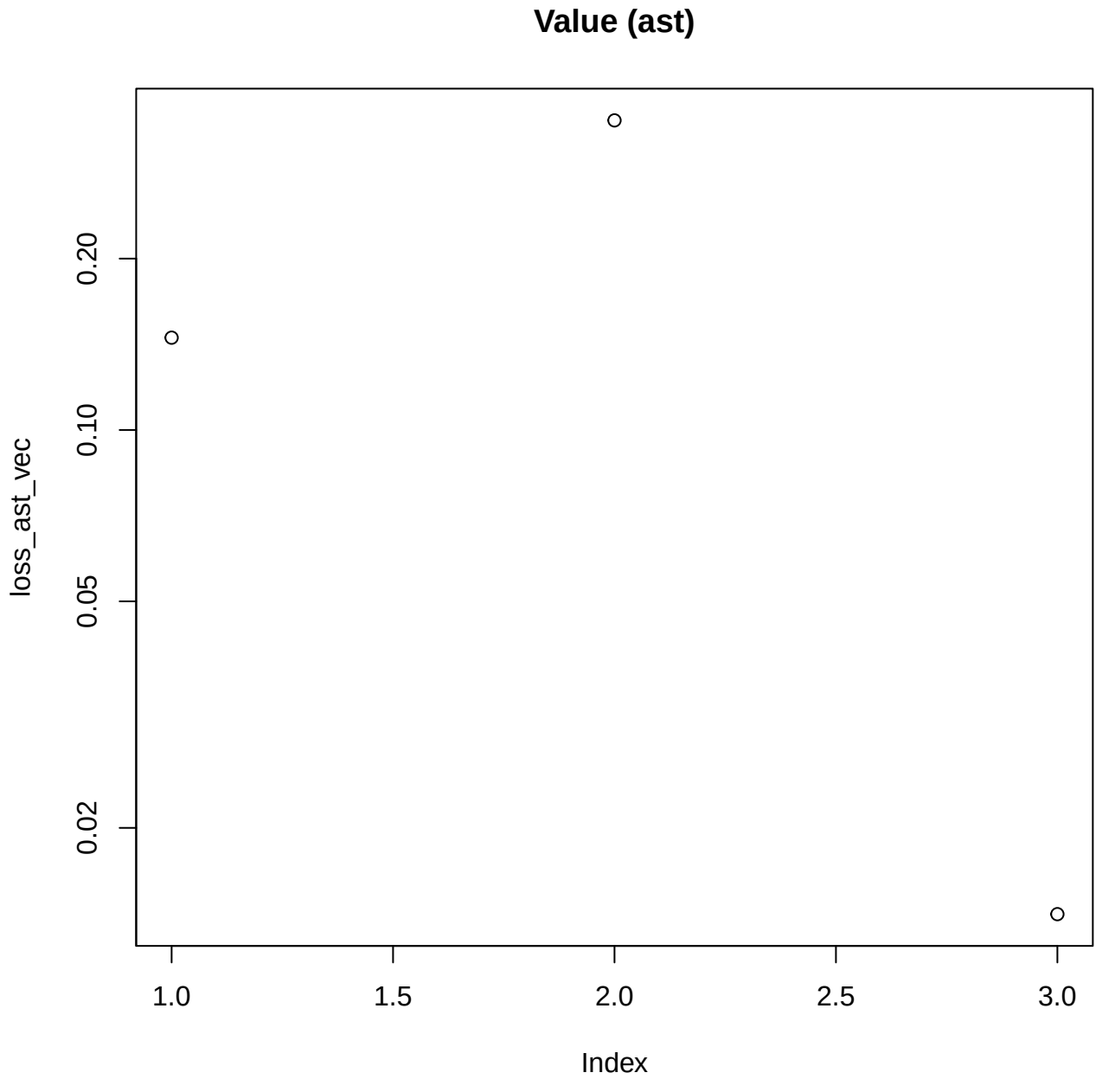

SVI ELBO Loss [134.0642]
ELBO loss
500
200
100
50
5
10
15
20
Iteration

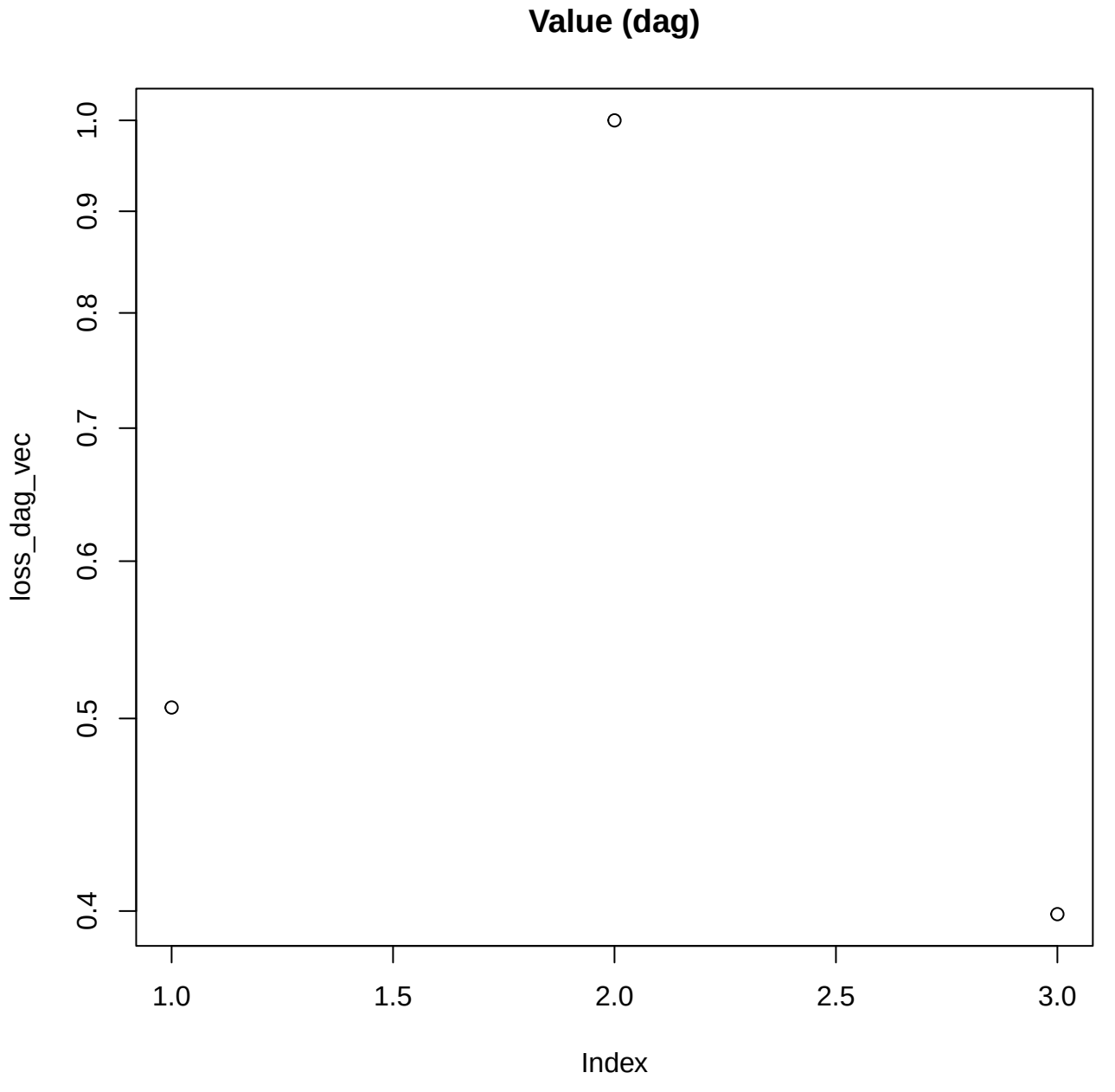

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.15
0.20
0.25
0.30
0.35
0.40
1
2
3
4
5
Index

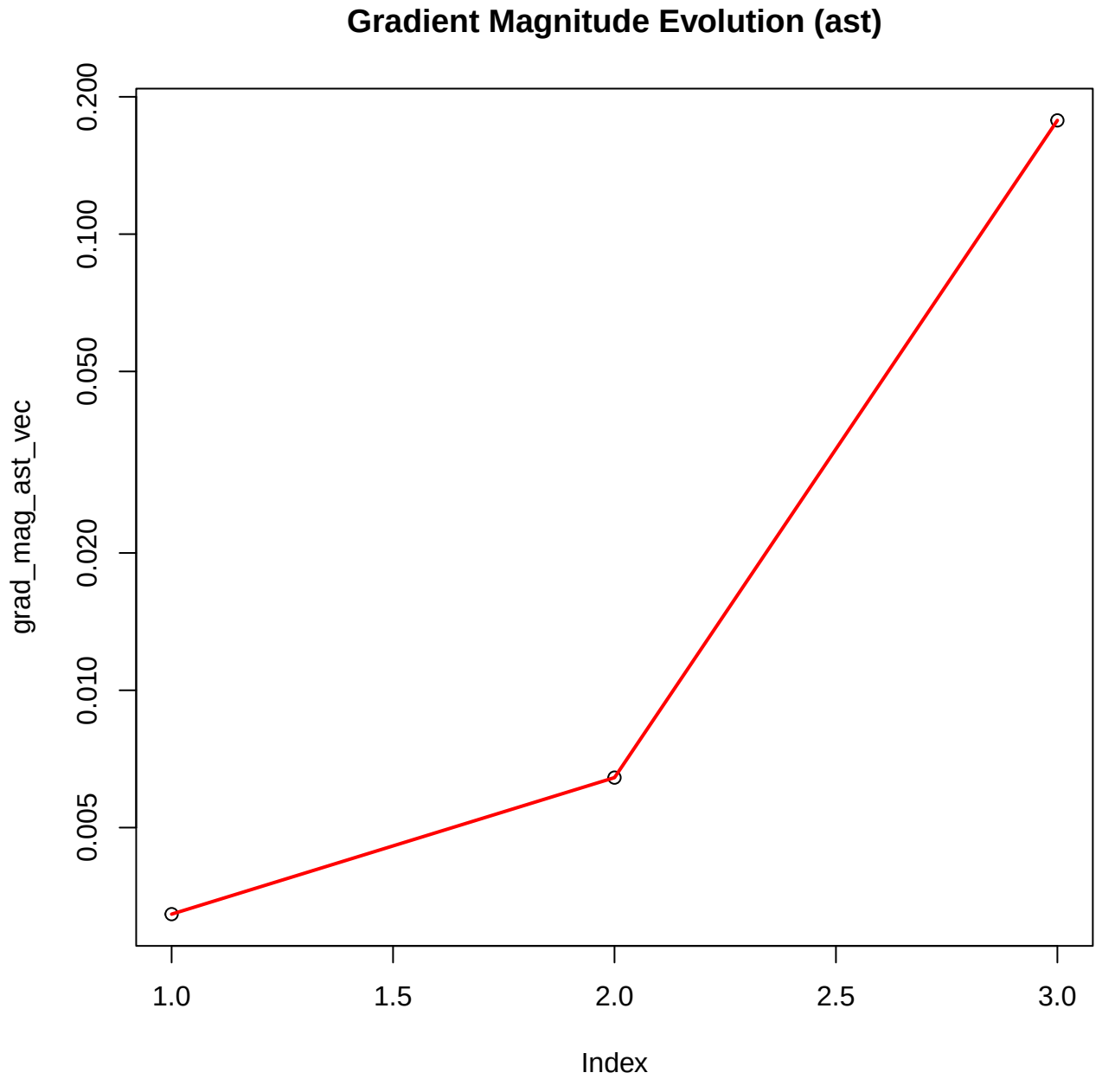

Value (ast)
loss_ast_vec
1.10
1.15
1.20
1.25
1.30
1
2
3
4
5
Index

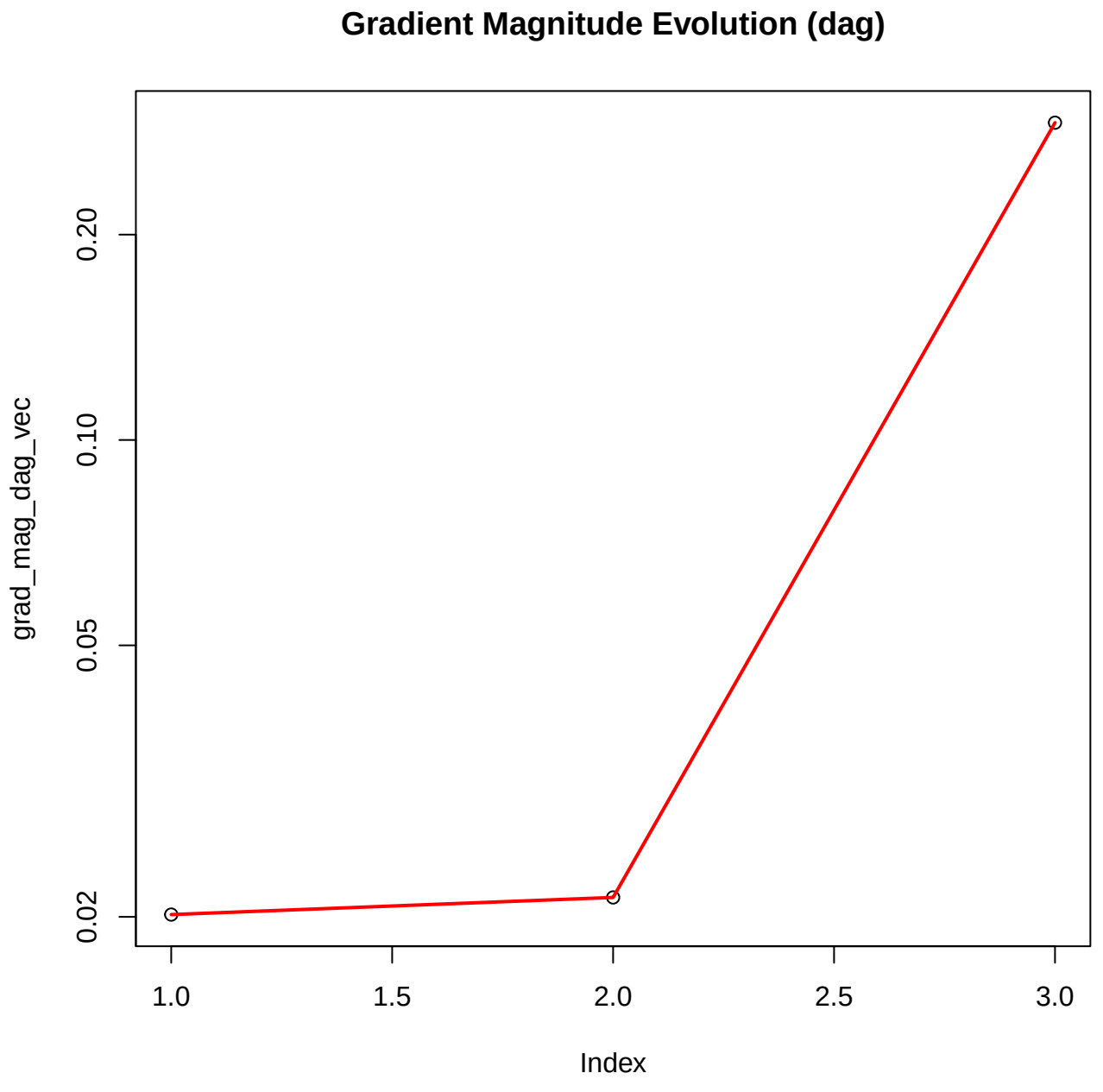

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

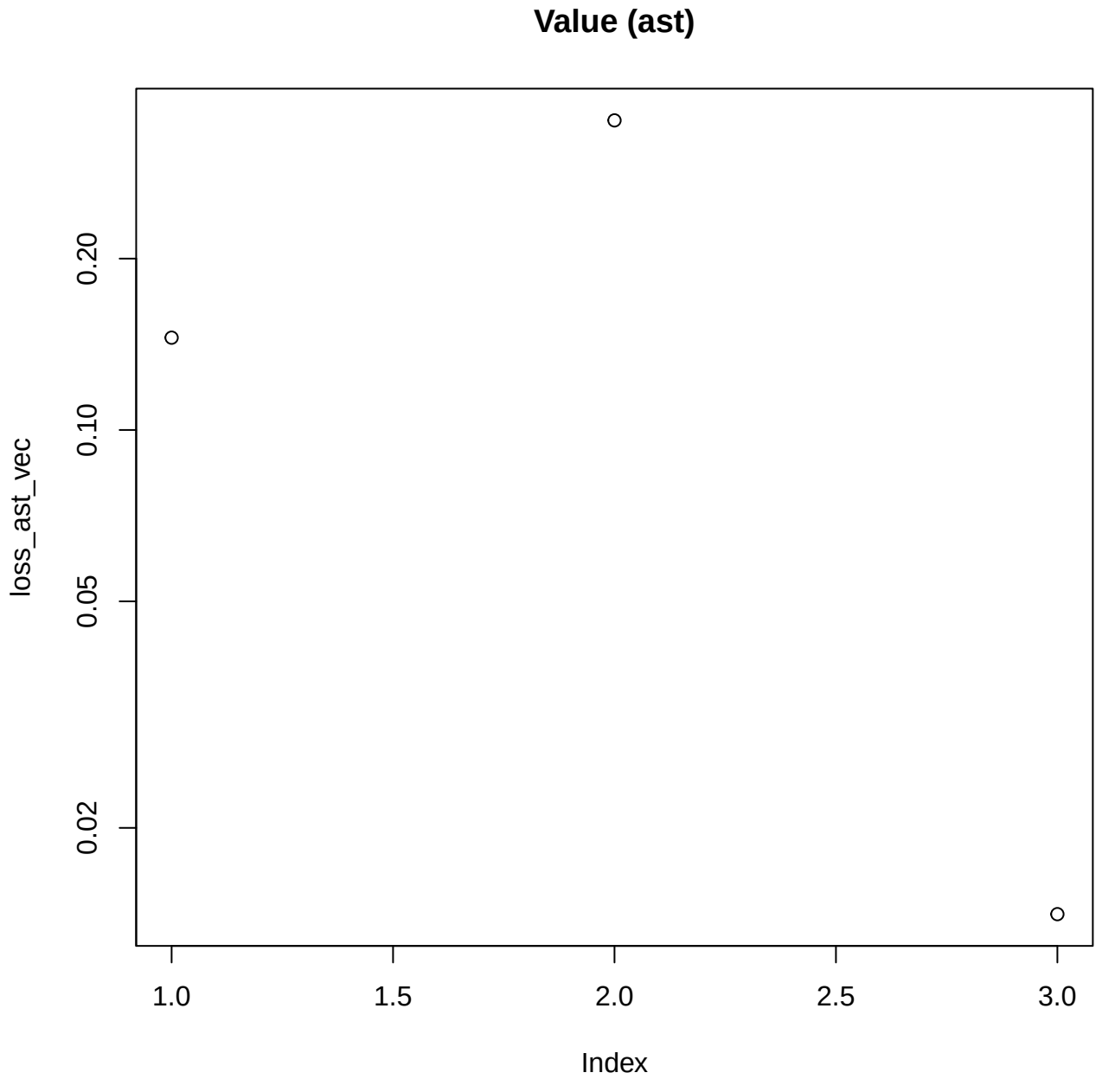

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

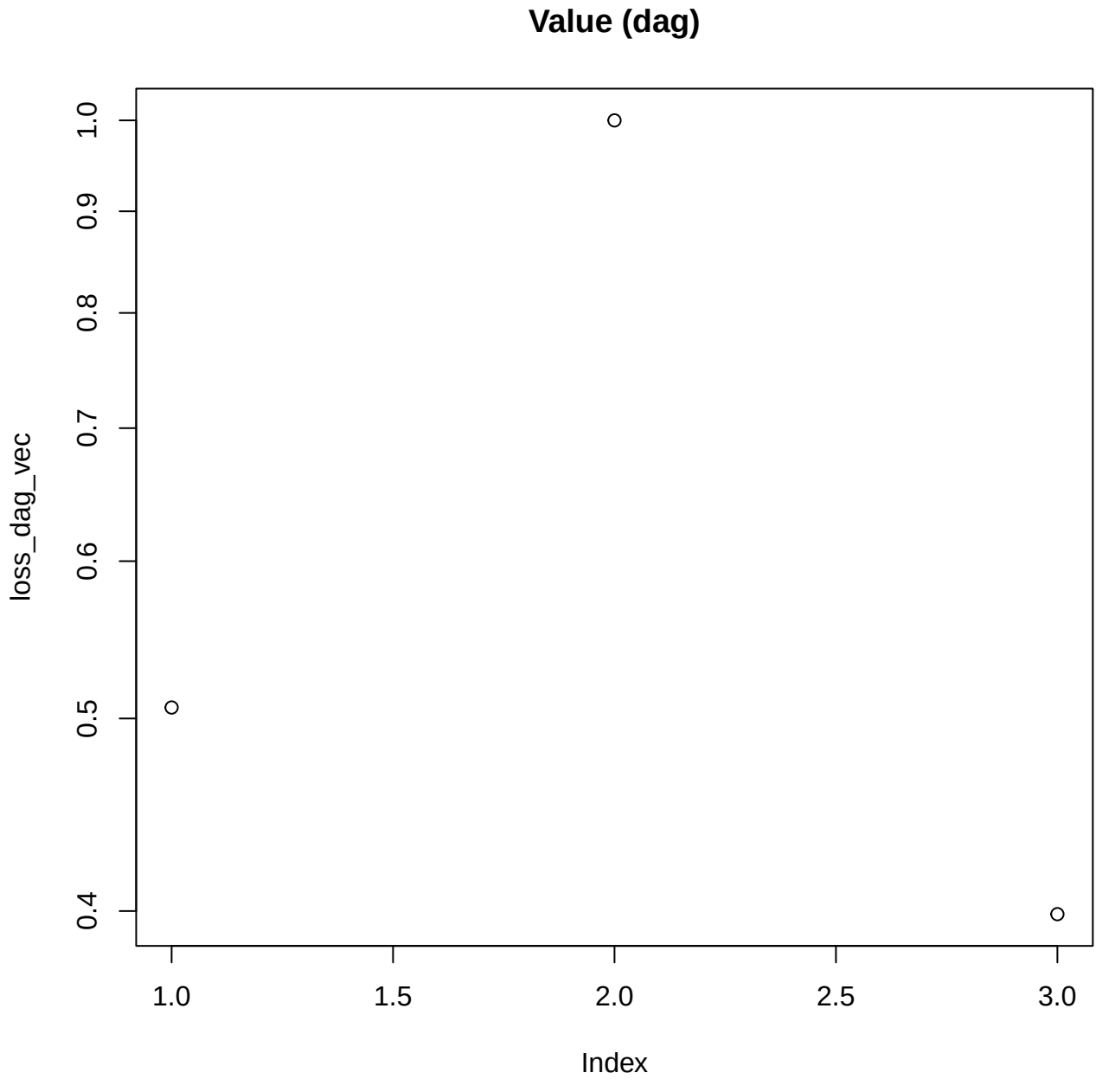

Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

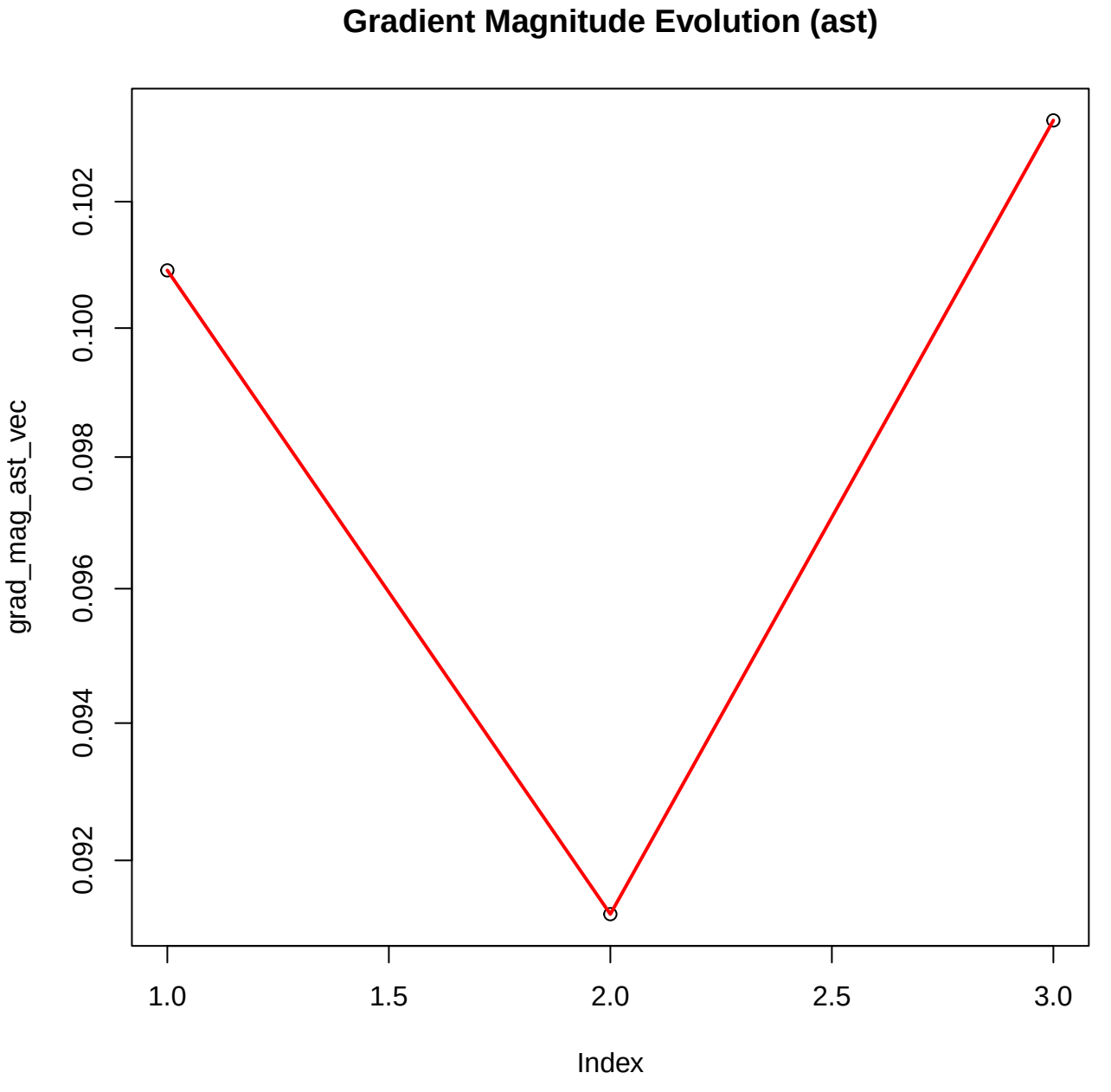

Value (dag)
loss_dag_vec
1.0
0.9
0.8
0.7
0.6
0.5
0.4
1.0
1.5
2.0
2.5
3.0
Index

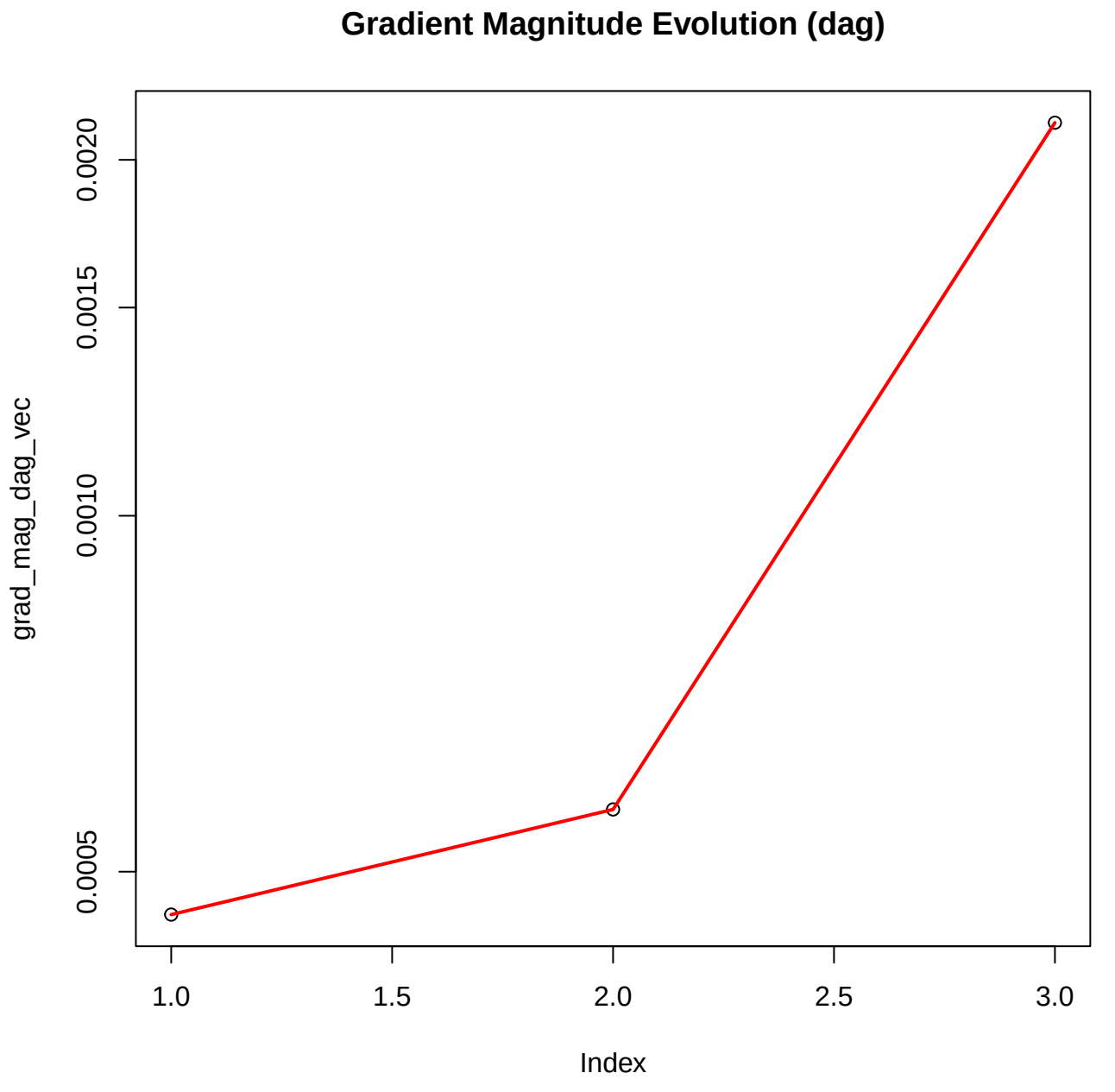

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

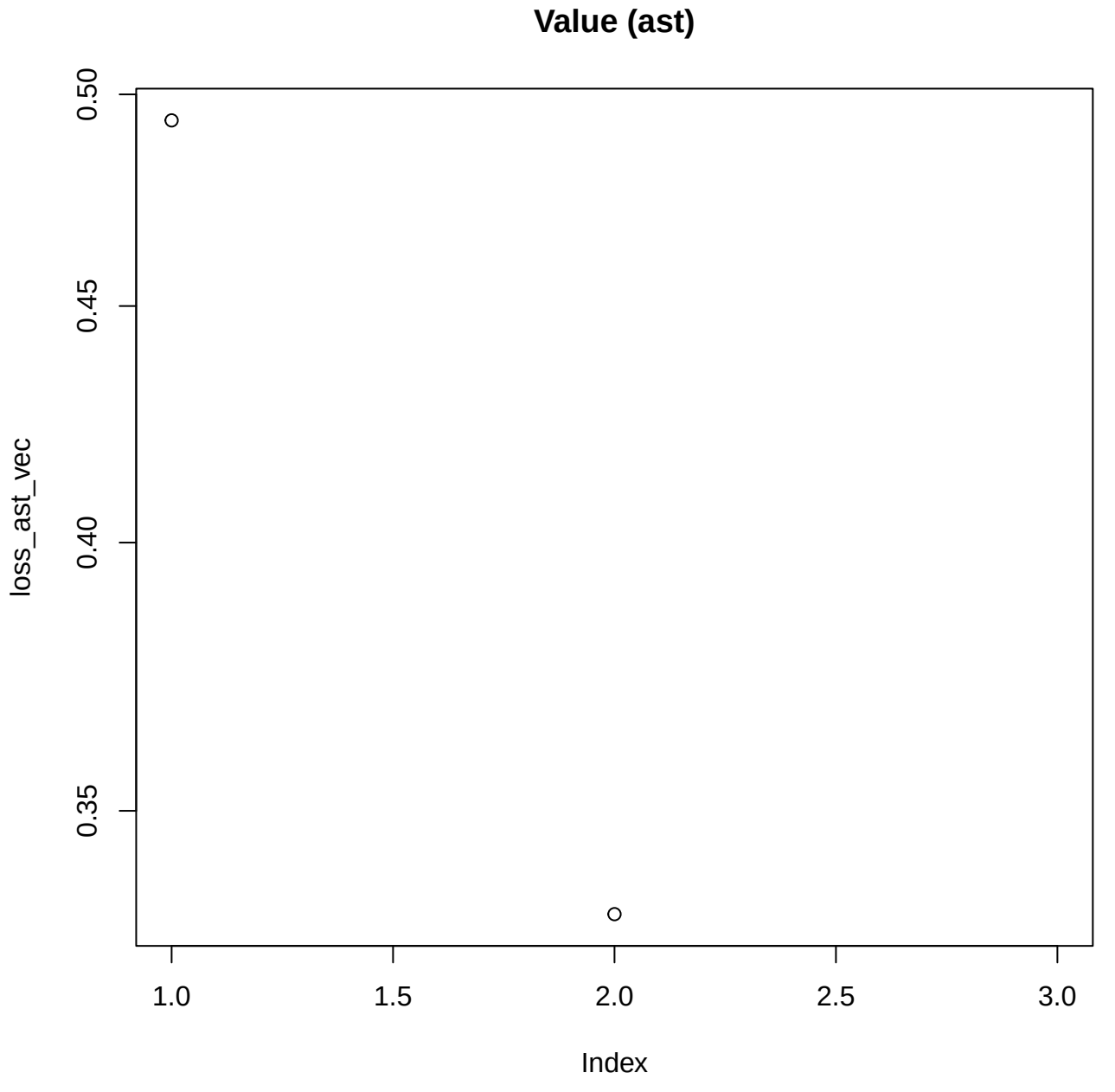

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

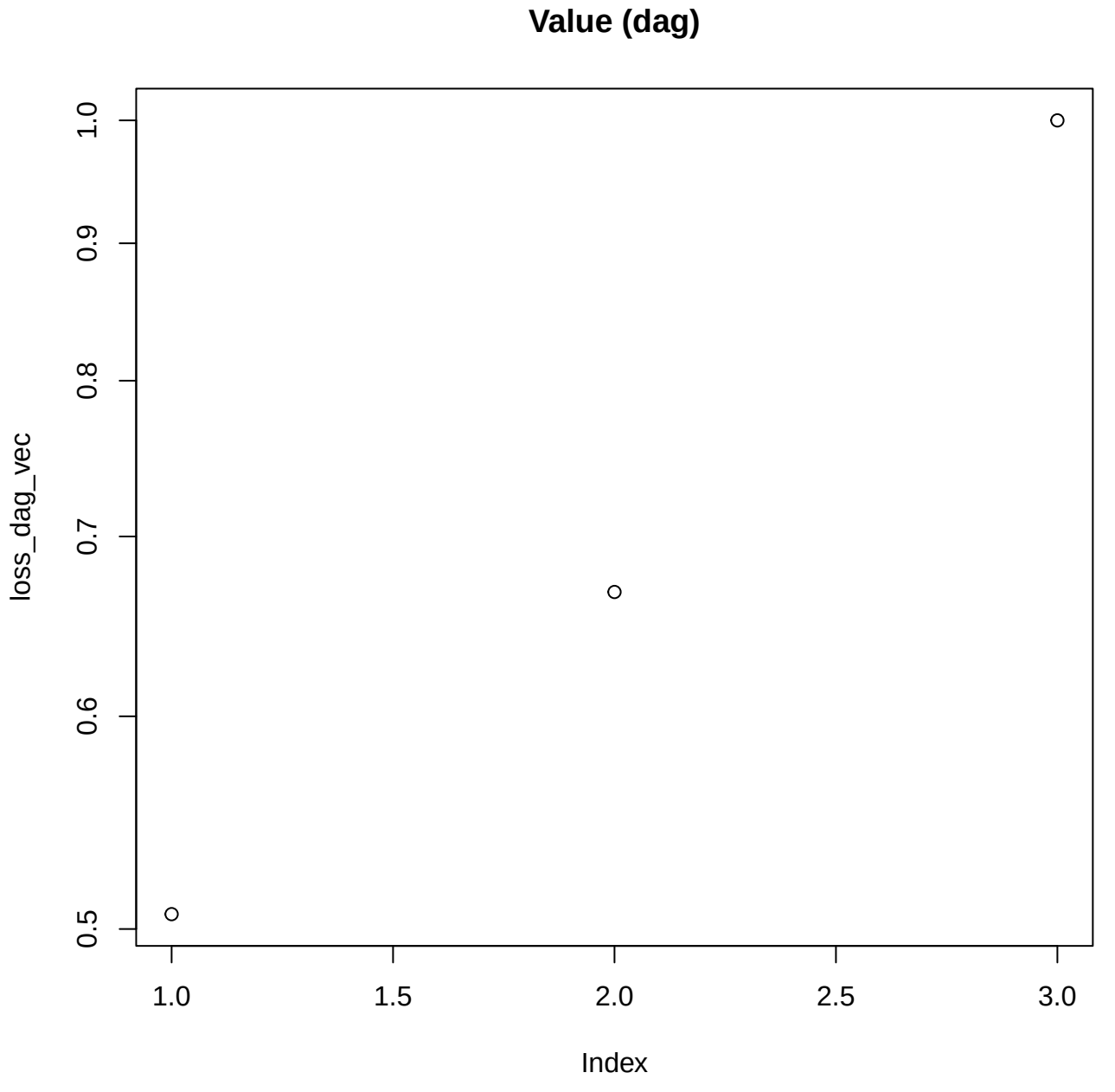

Value (ast)
loss_ast_vec
0.02
0.05
0.10
0.20
1.0
1.5
2.0
2.5
3.0
Index

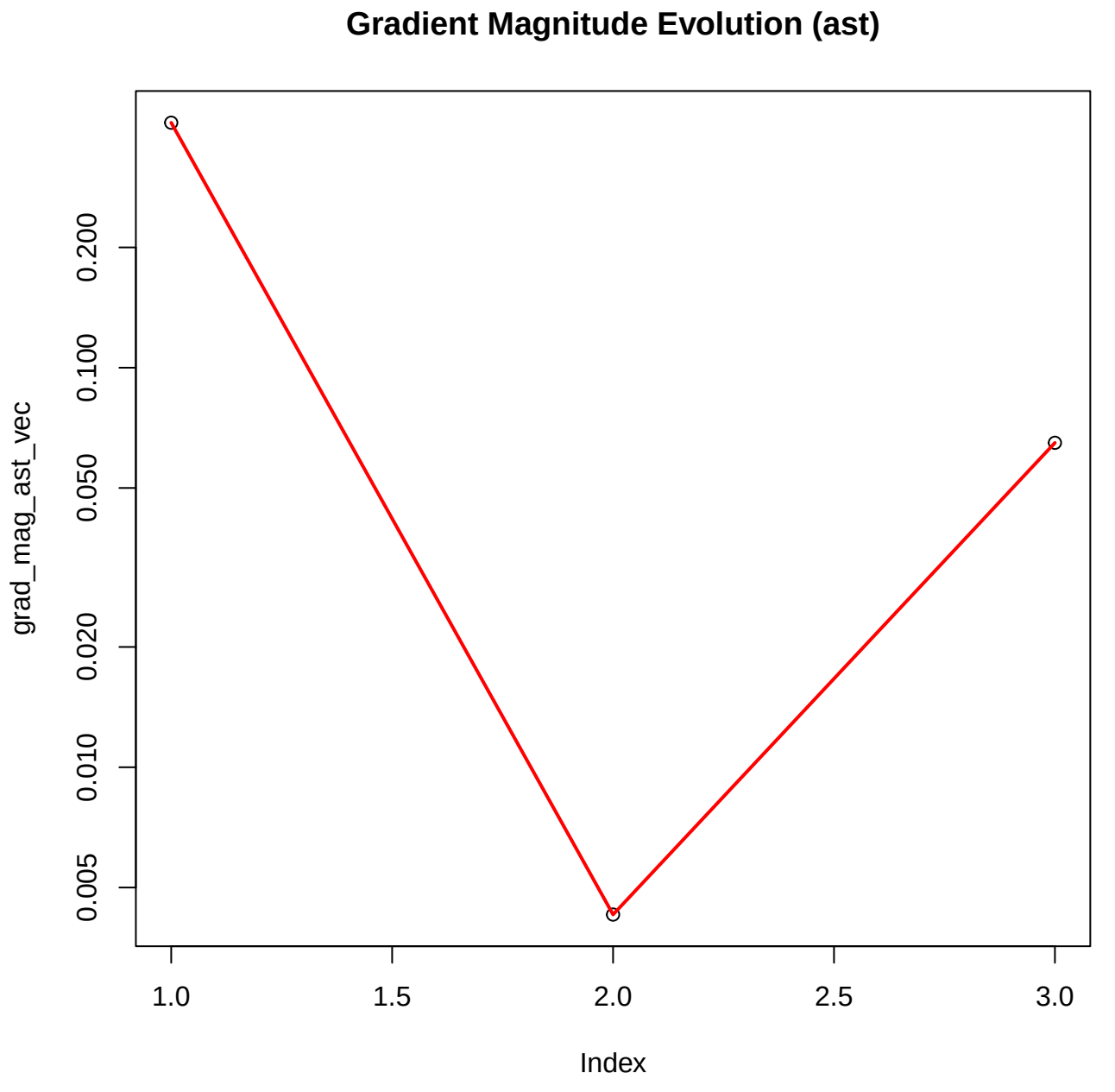

Value (dag)
loss_dag_vec
1.0
0.9
0.8
0.7
0.6
0.5
0.4
1.0
1.5
2.0
2.5
3.0
Index

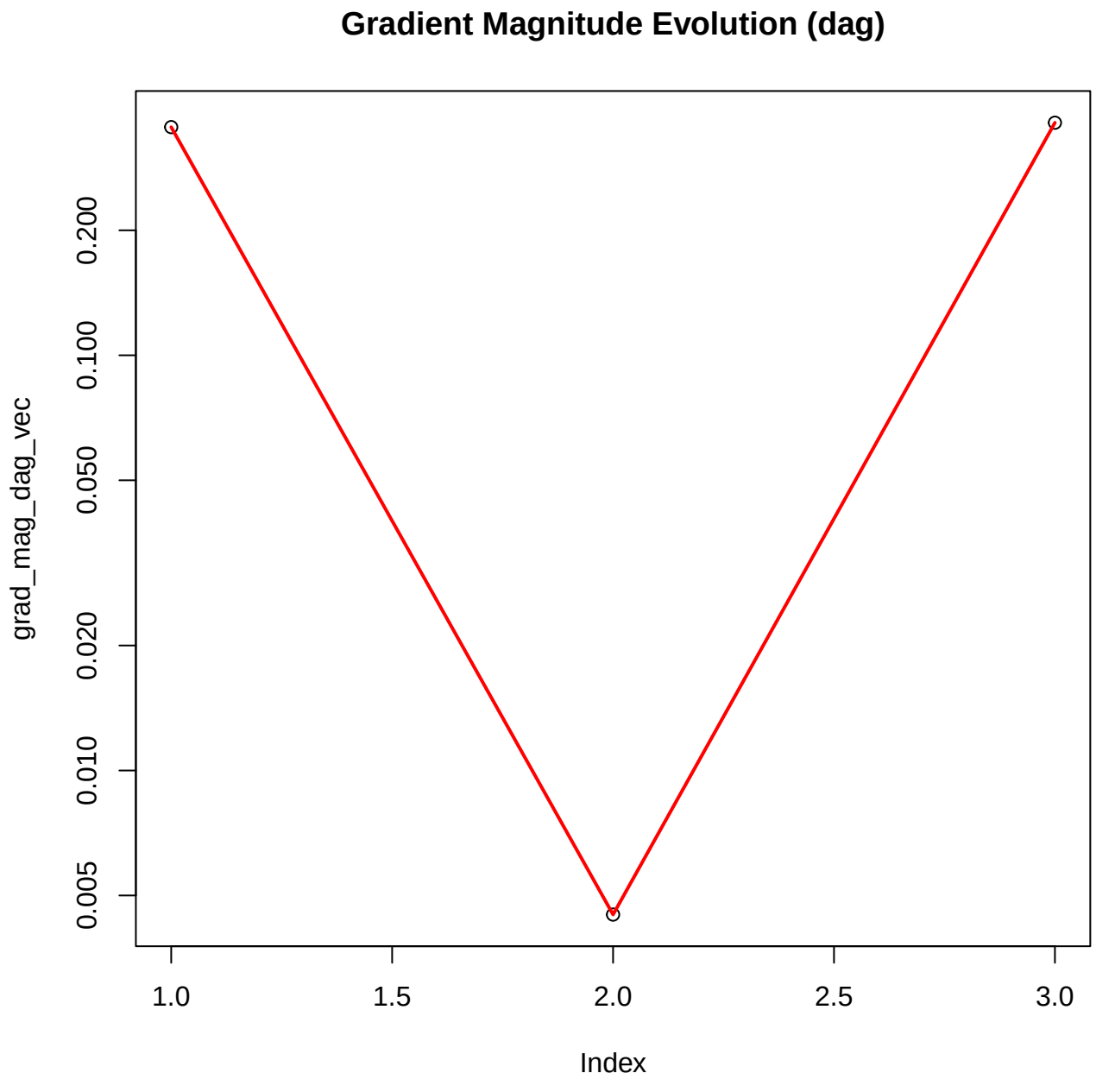

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.092
0.094
0.096
0.098
0.100
0.102
1.0
1.5
2.0
2.5
3.0
Index

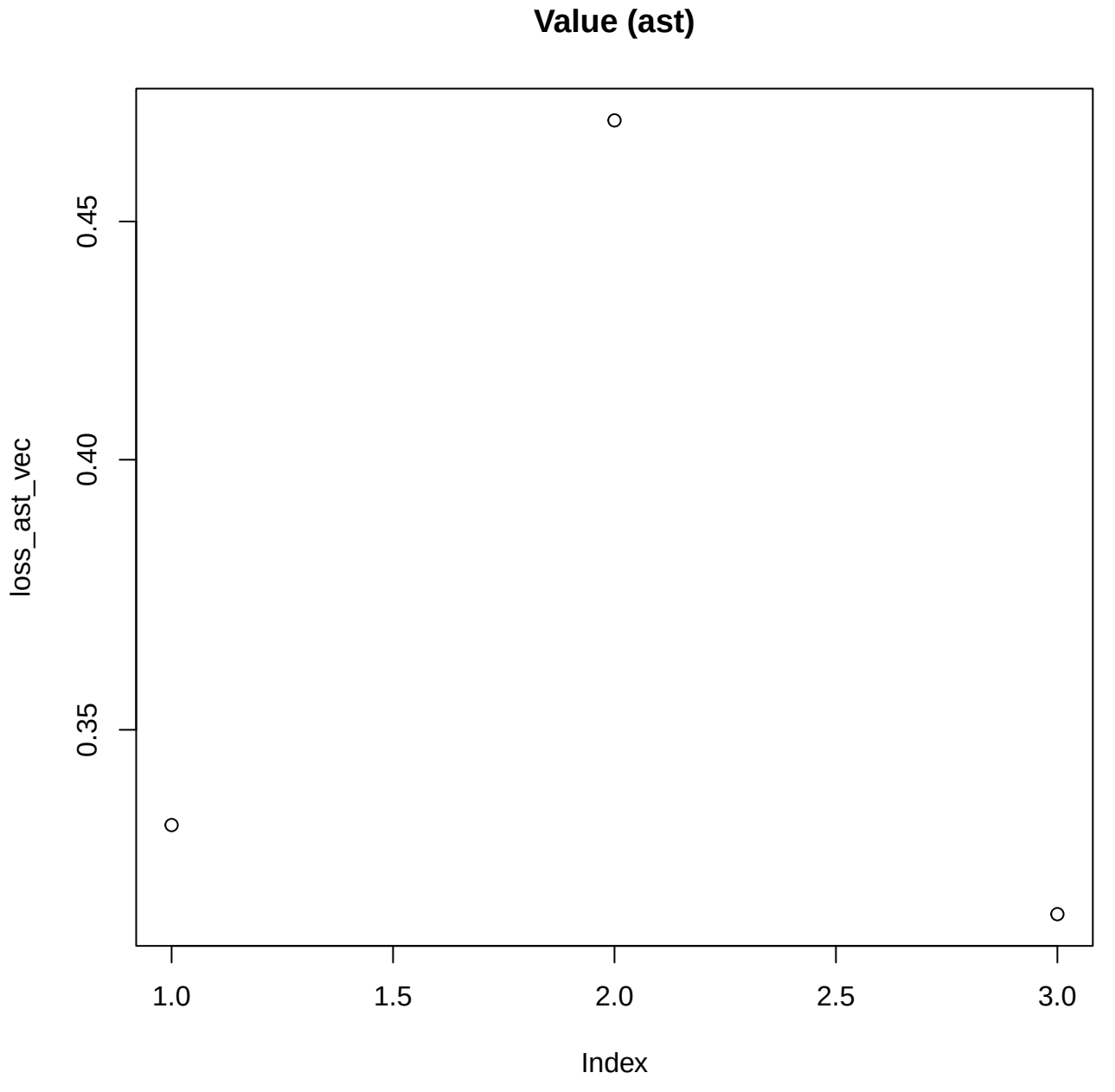

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.0005
0.0010
0.0015
0.0020
1.0
1.5
2.0
2.5
3.0
Index

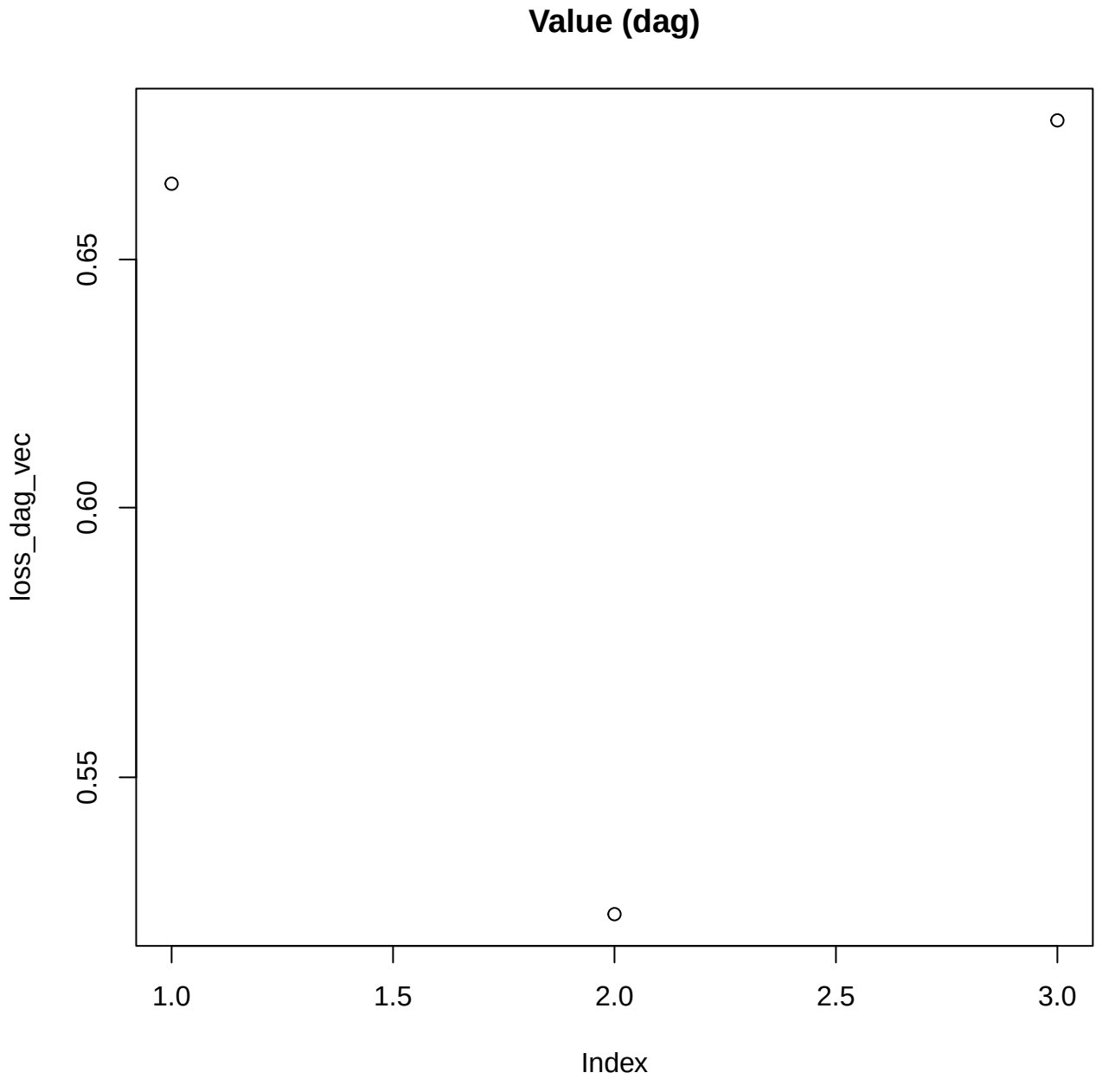

Value (ast)
loss_ast_vec
0.50
0.45
0.40
0.35
1.0
1.5
2.0
2.5
3.0
Index

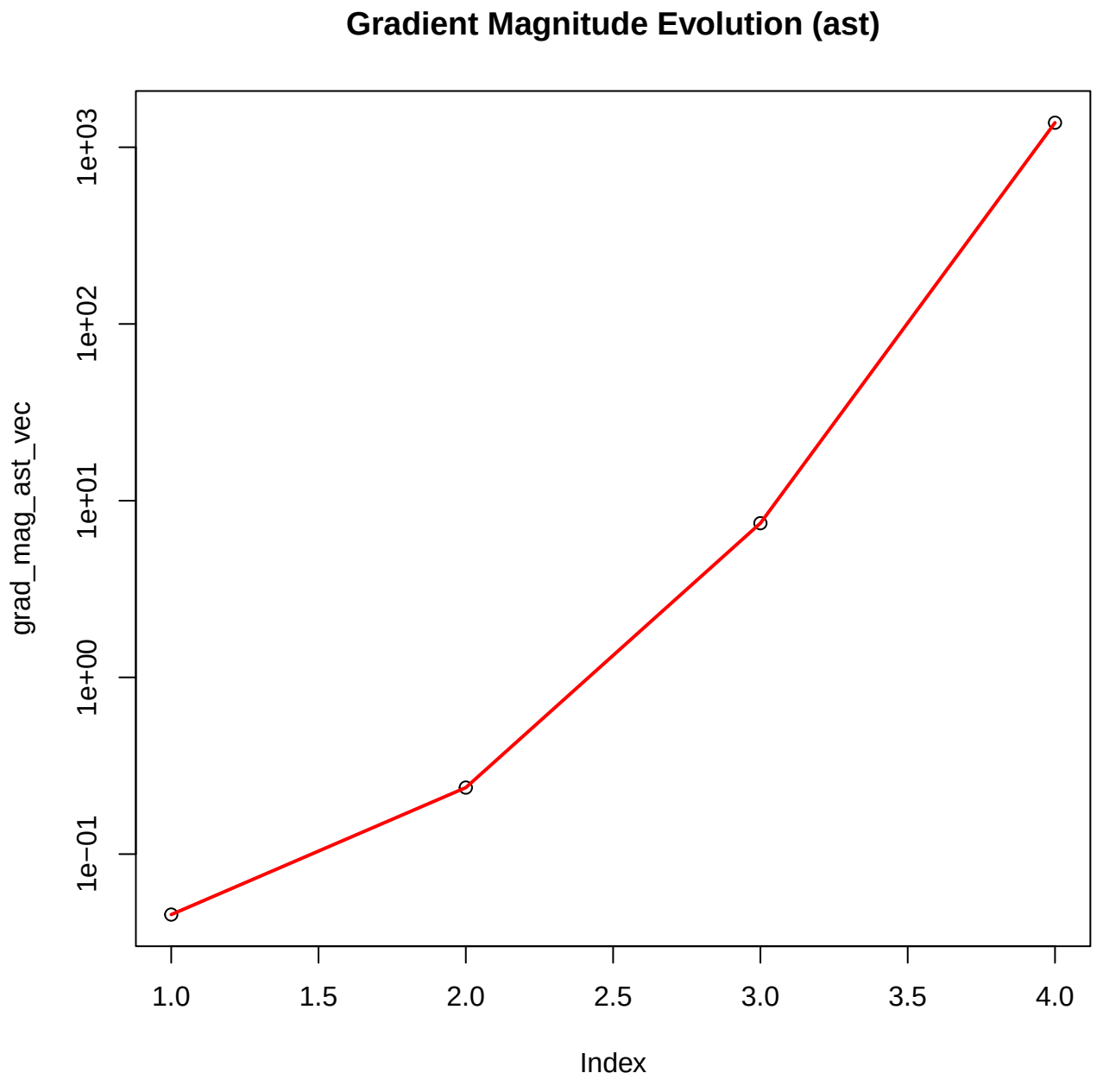

Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
0.9
1.0
1.0
1.5
2.0
2.5
3.0
Index

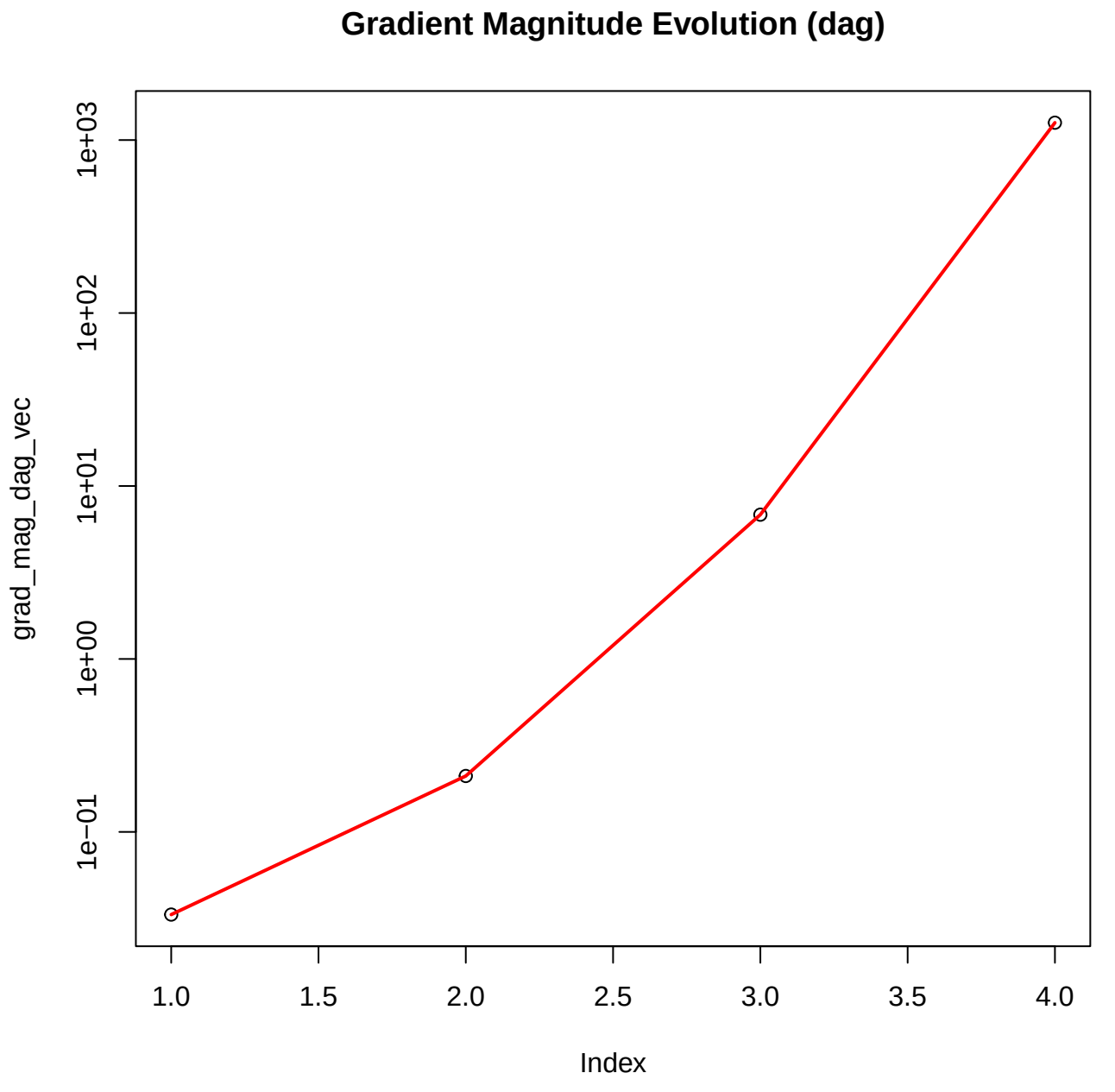

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

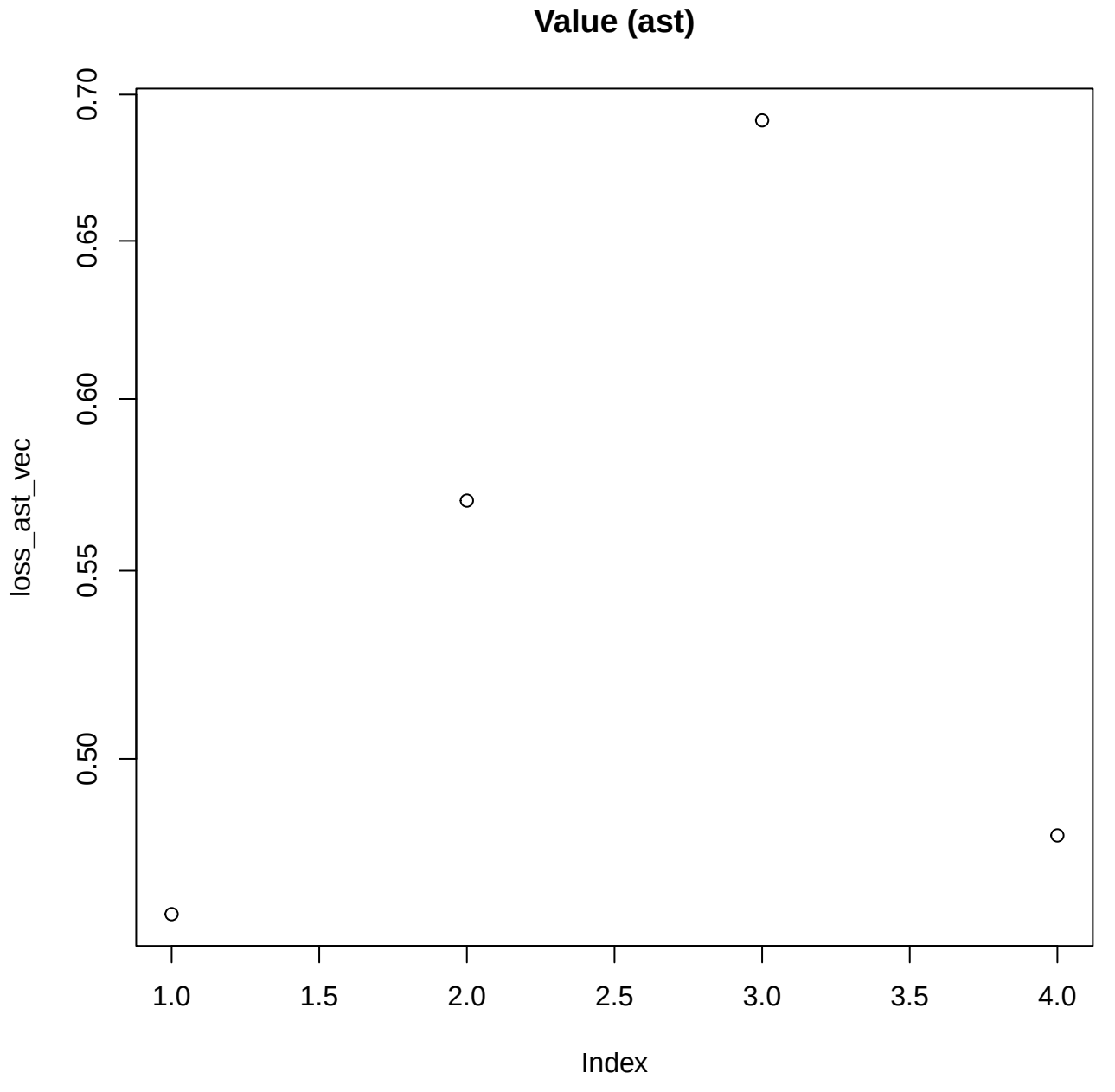

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
1.0
1.5
2.0
2.5
3.0
Index

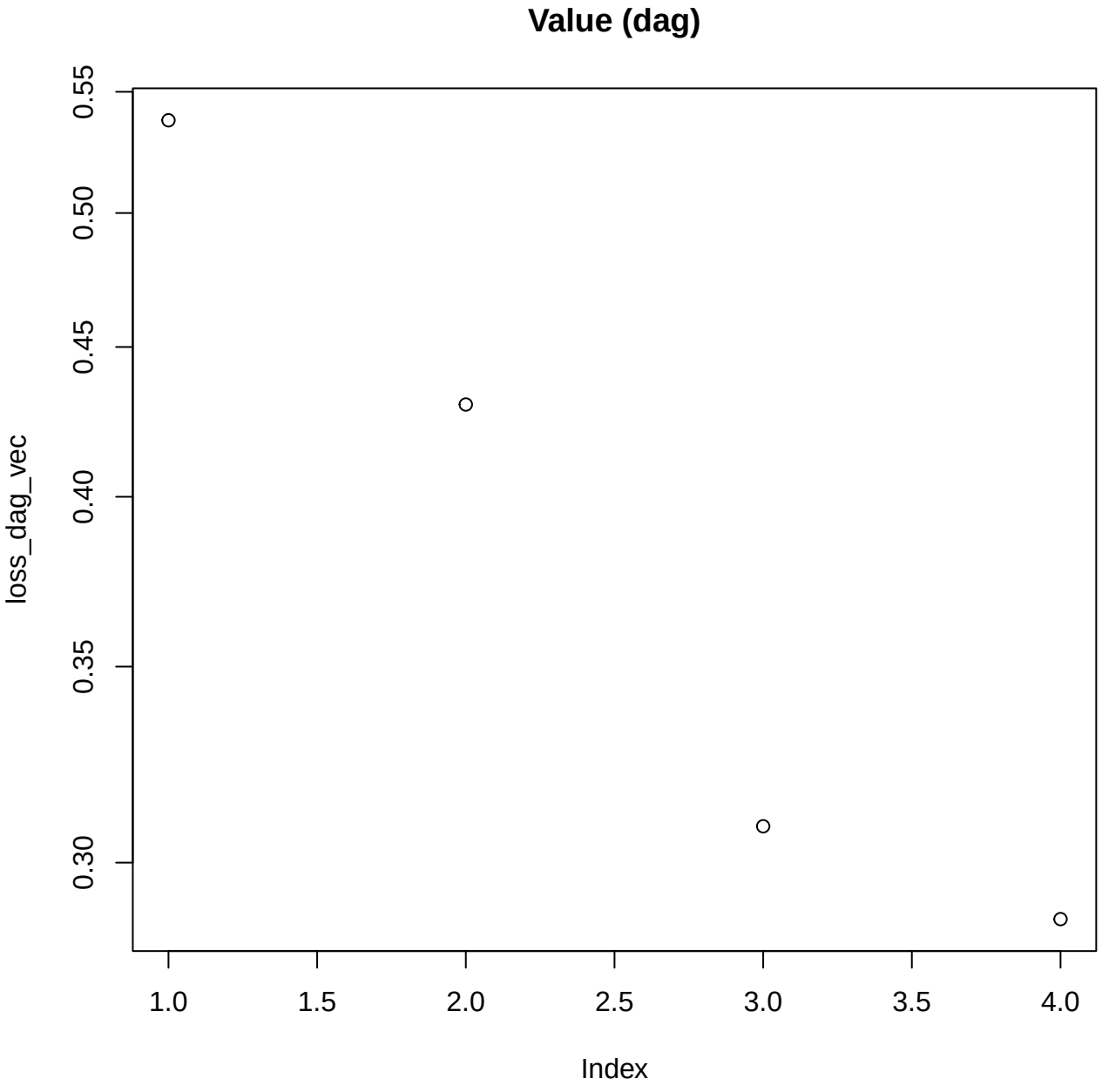

Value (ast)
loss_ast_vec
0.35
0.40
0.45
1.0
1.5
2.0
2.5
3.0
Index

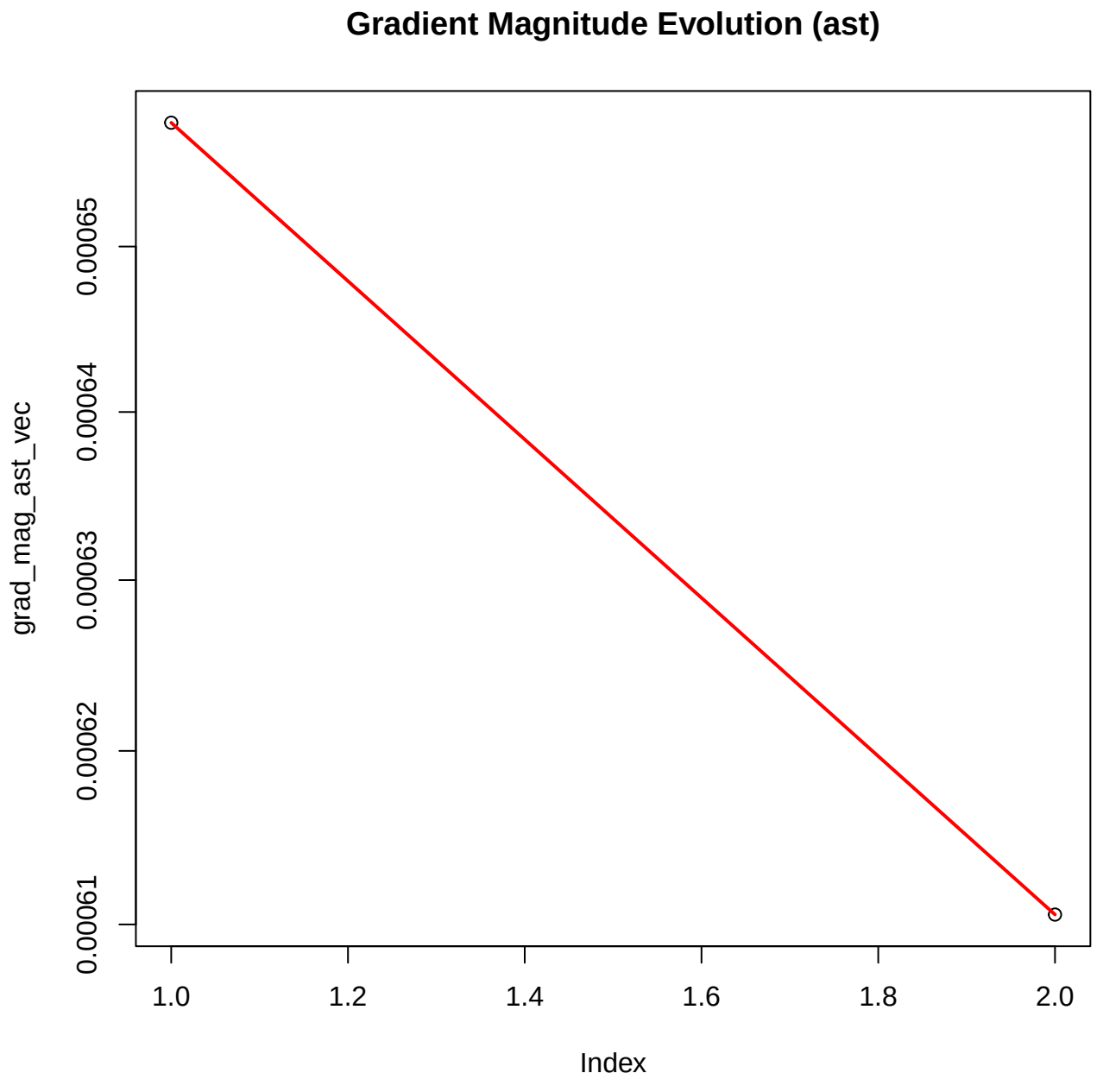

Value (dag)
loss_dag_vec
0.55
0.60
0.65
1.0
1.5
2.0
2.5
3.0
Index

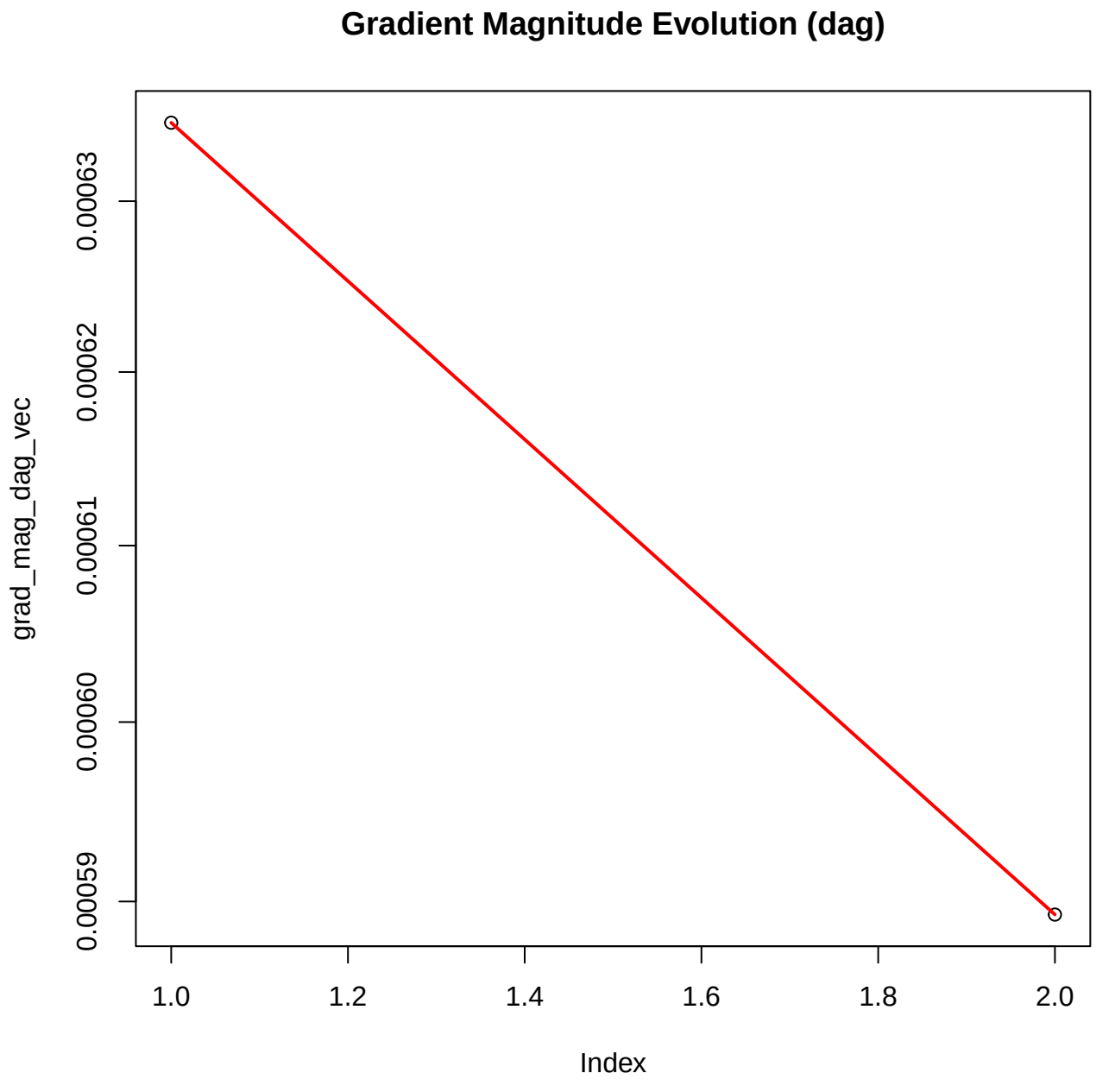

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−01
1e+00
1e+01
1e+02
1e+03
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

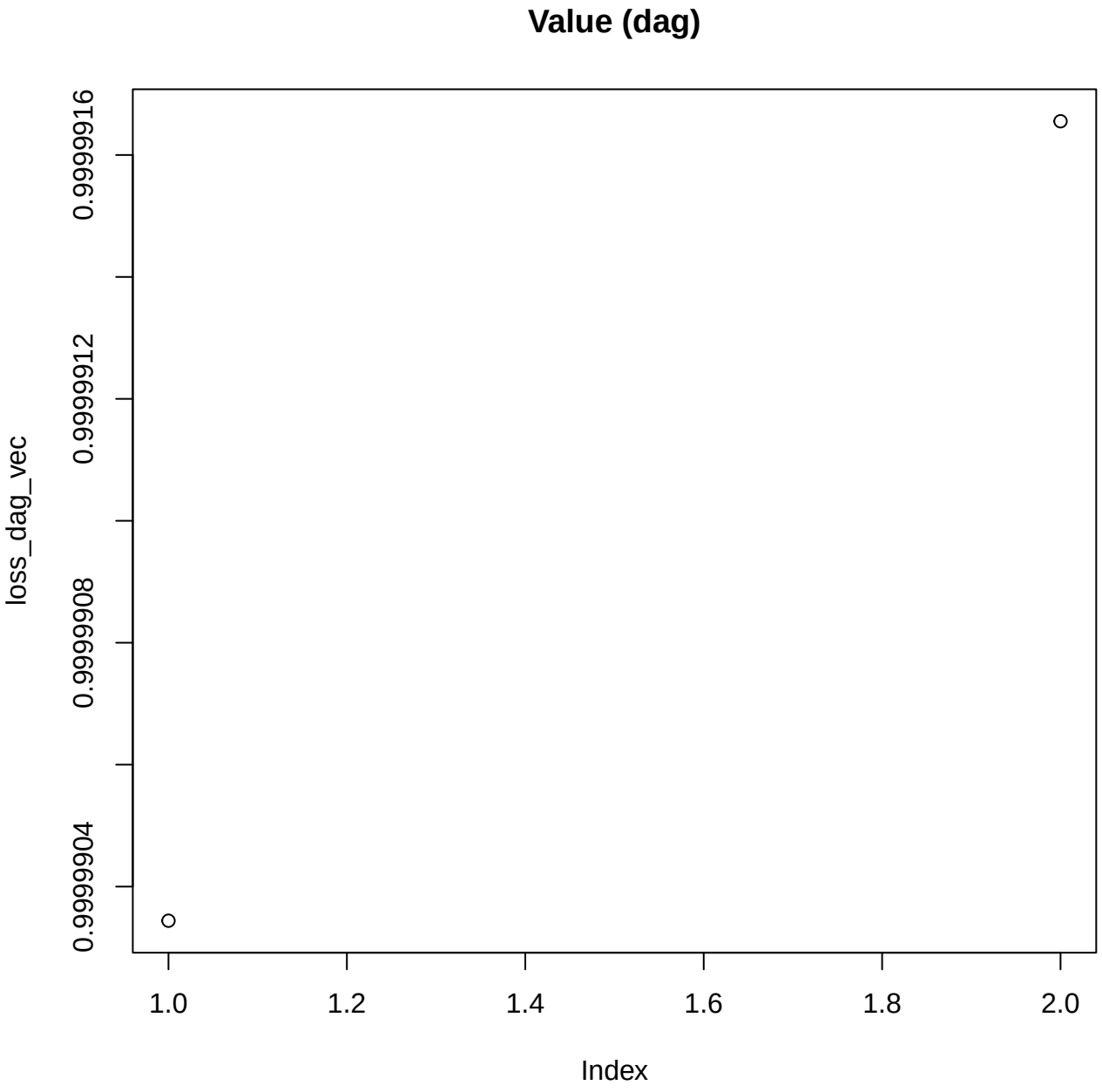

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
1e−01
1e+00
1e+01
1e+02
1e+03
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

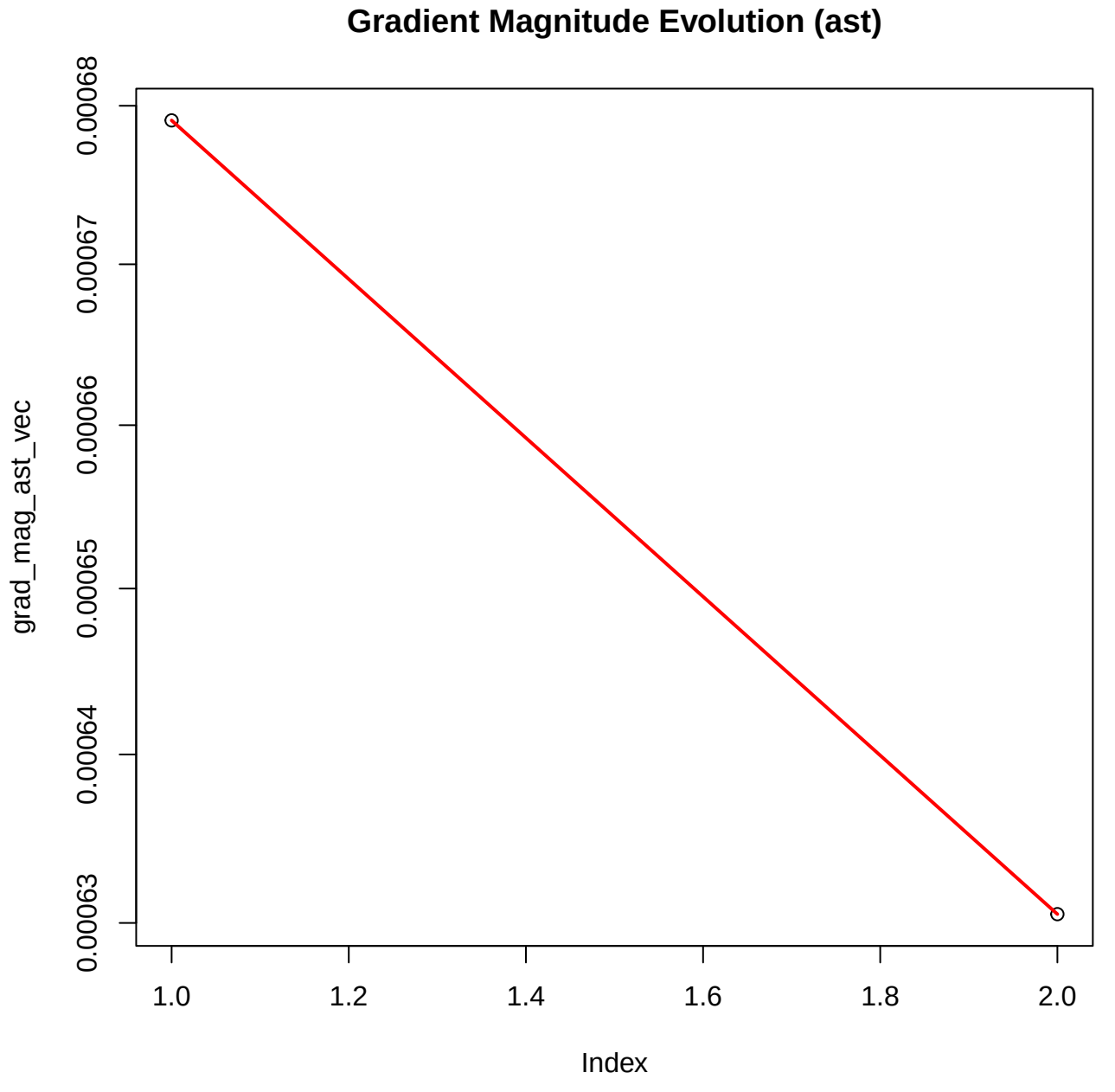

Value (ast)
loss_ast_vec
0.50
0.55
0.60
0.65
0.70
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

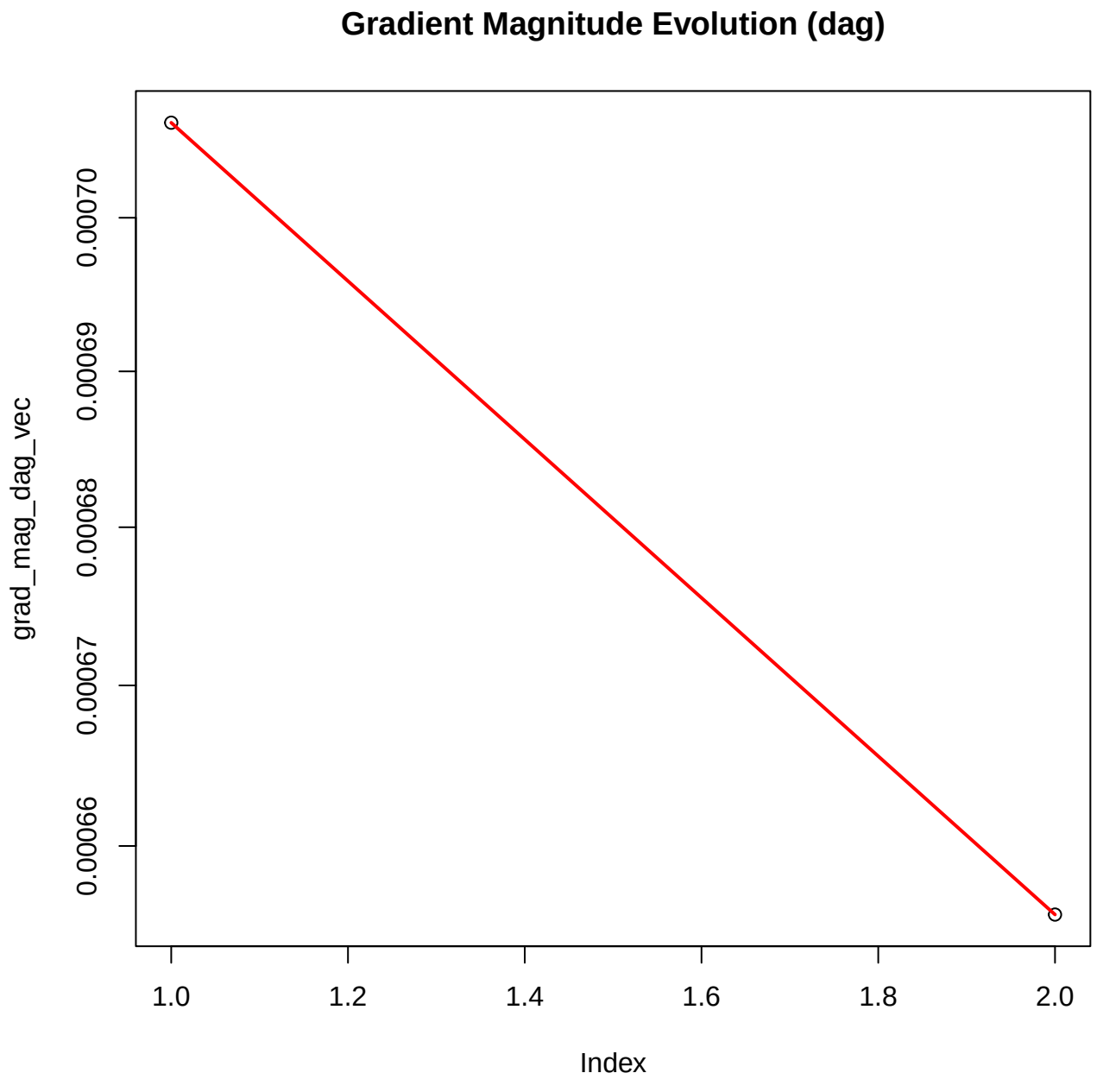

Value (dag)
loss_dag_vec
0.55
0.50
0.45
0.40
0.35
0.30
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Index

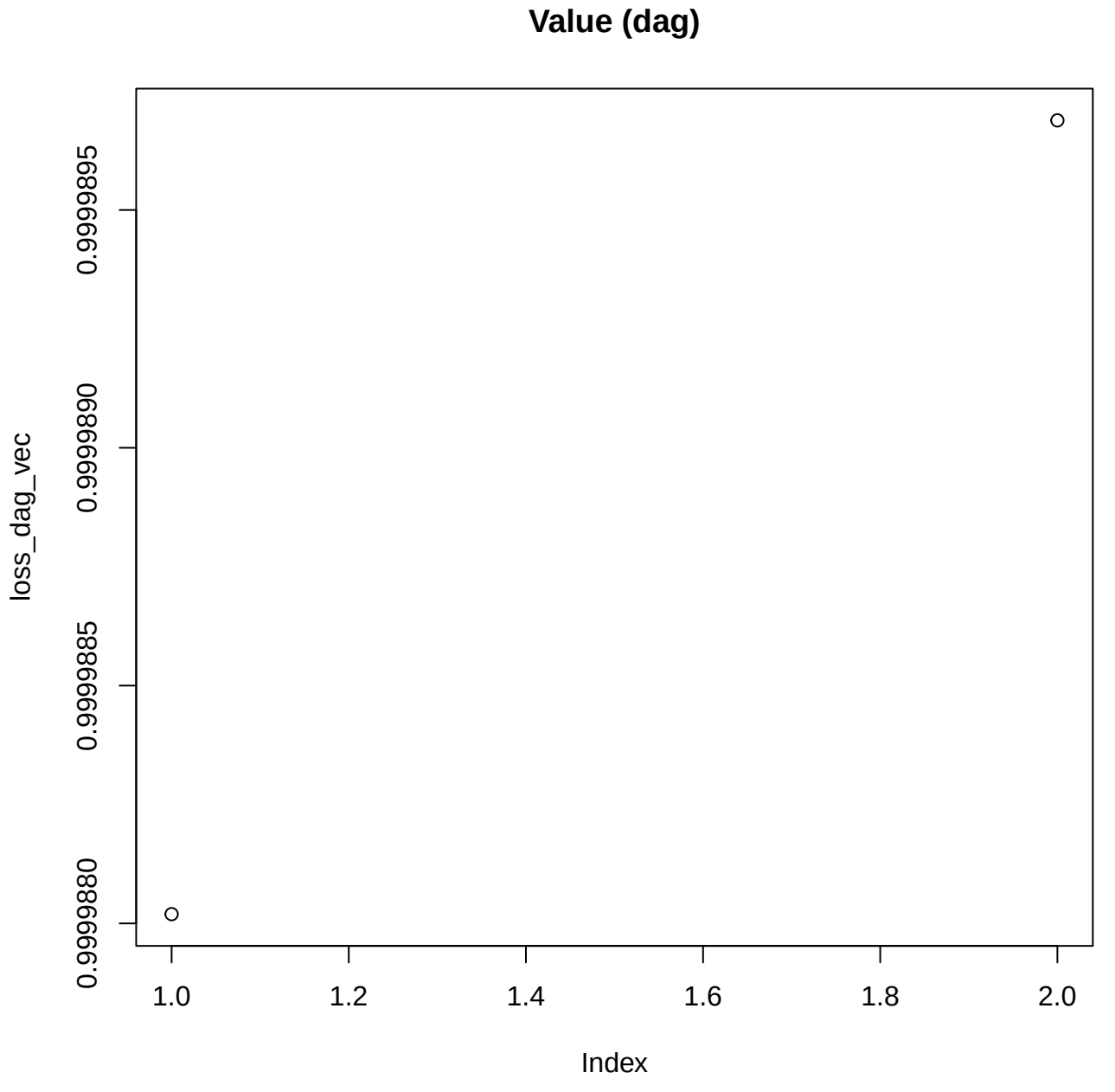

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00061
0.00062
0.00063
0.00064
0.00065
1.0
1.2
1.4
1.6
1.8
2.0
Index

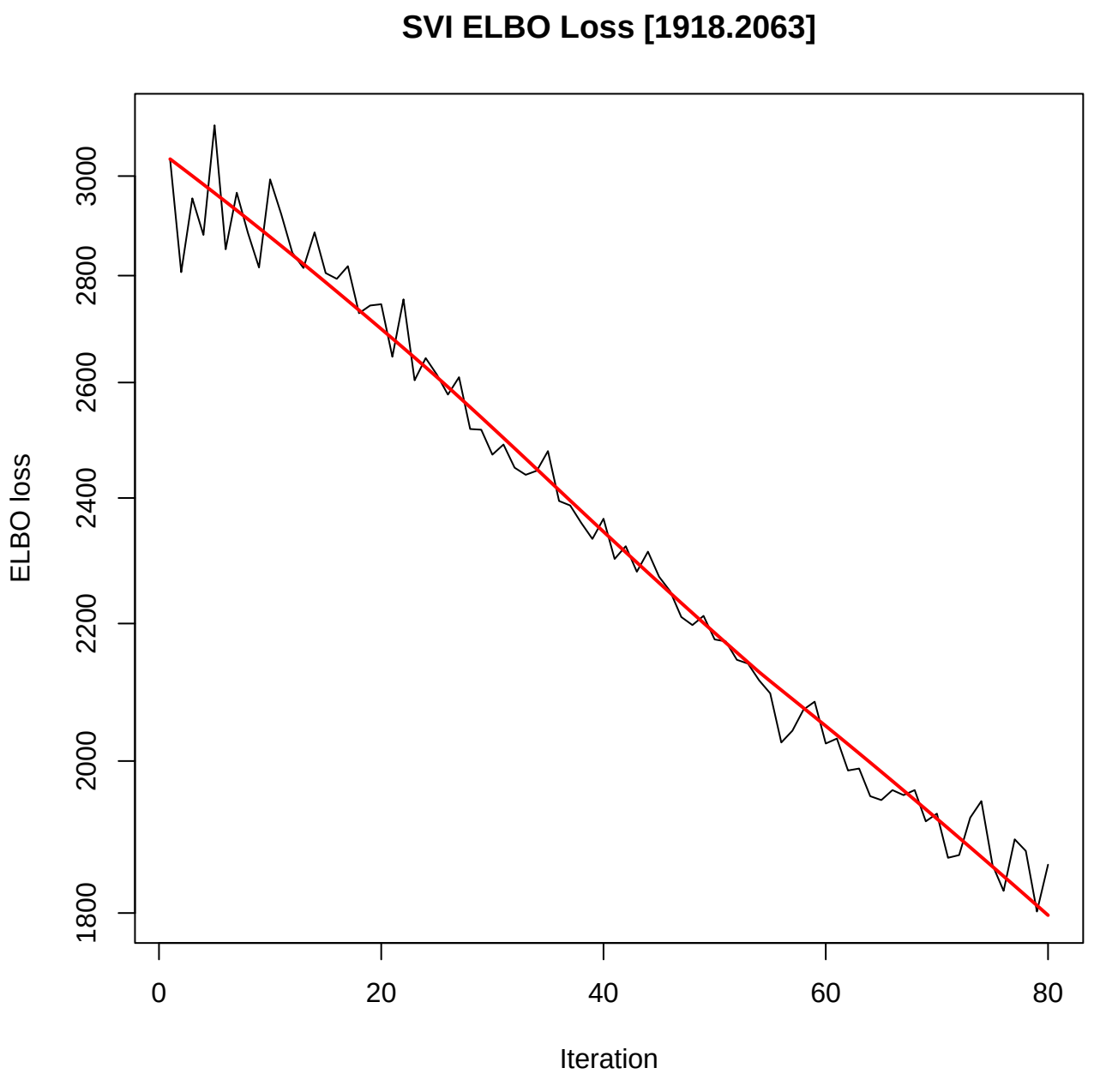

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.00059
0.00060
0.00061
0.00062
0.00063
1.0
1.2
1.4
1.6
1.8
2.0
Index

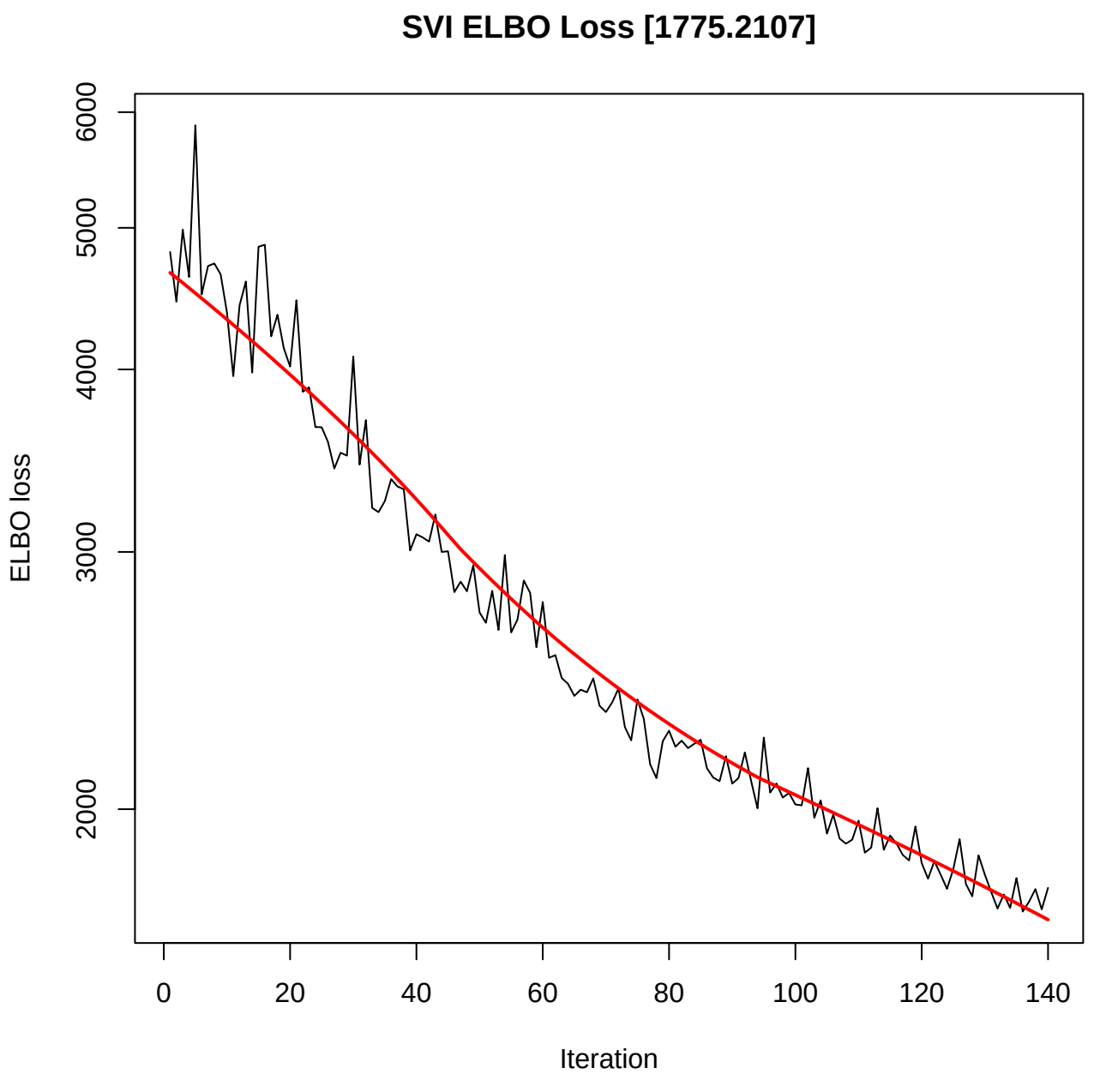

Value (dag)
loss_dag_vec
0.9999904
0.9999908
0.9999912
0.9999916
1.0
1.2
1.4
1.6
1.8
2.0
Index

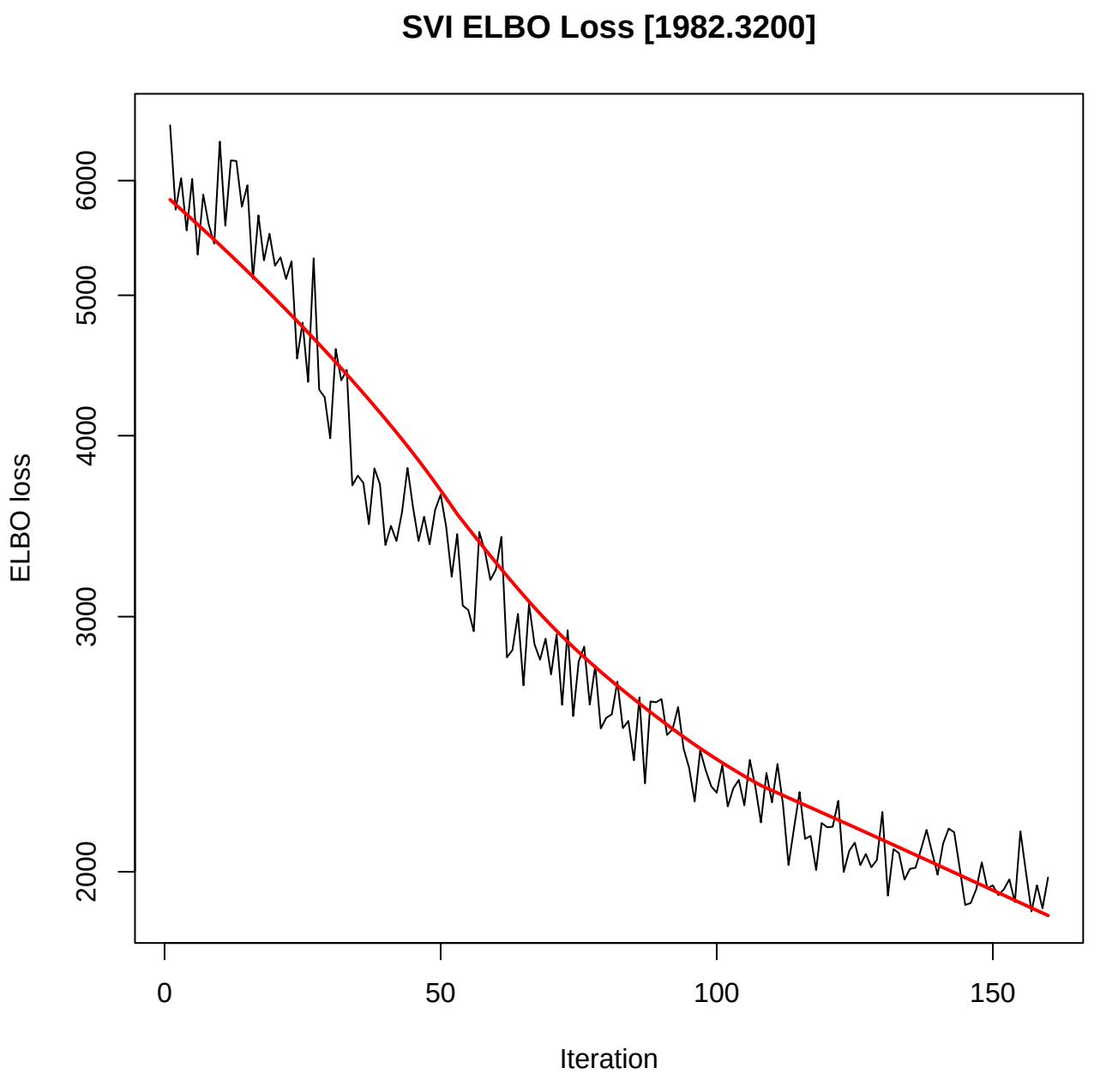

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00063
0.00064
0.00065
0.00066
0.00067
0.00068
1.0
1.2
1.4
1.6
1.8
2.0
Index

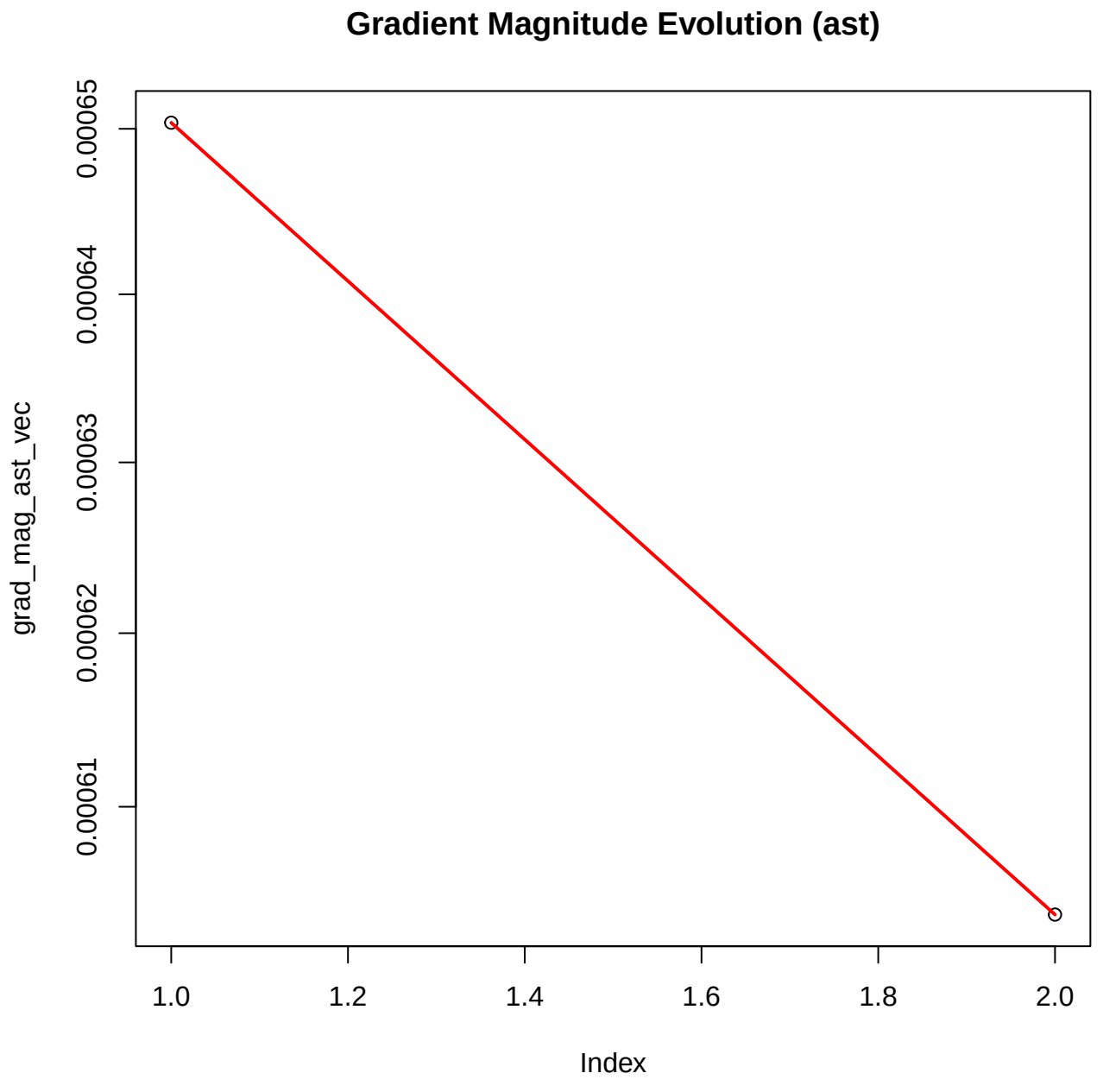

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.00066
0.00067
0.00068
0.00069
0.00070
1.0
1.2
1.4
1.6
1.8
2.0
Index

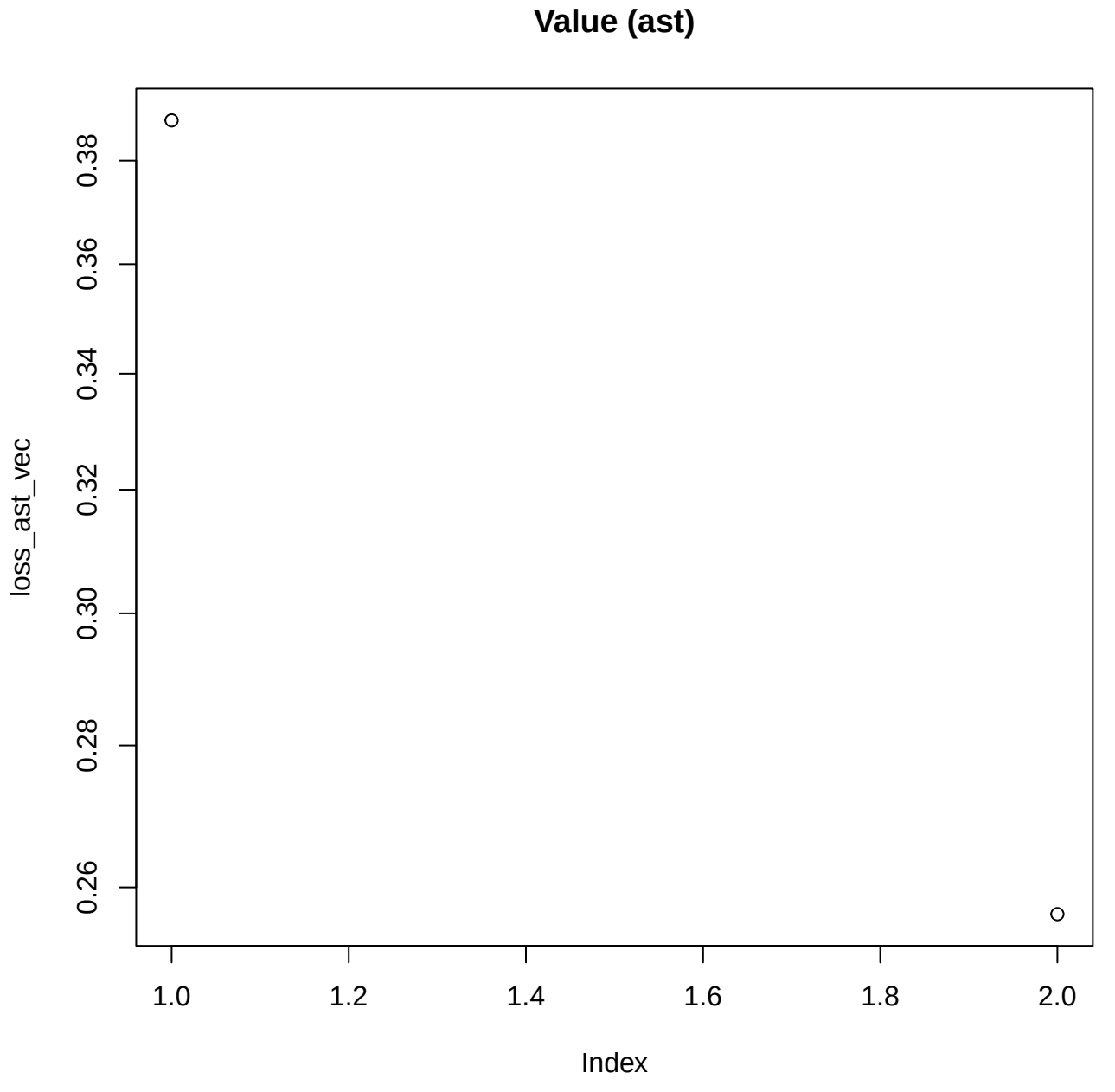

Value (dag)
loss_dag_vec
0.9999880
0.9999885
0.9999890
0.9999895
1.0
1.2
1.4
1.6
1.8
2.0
Index

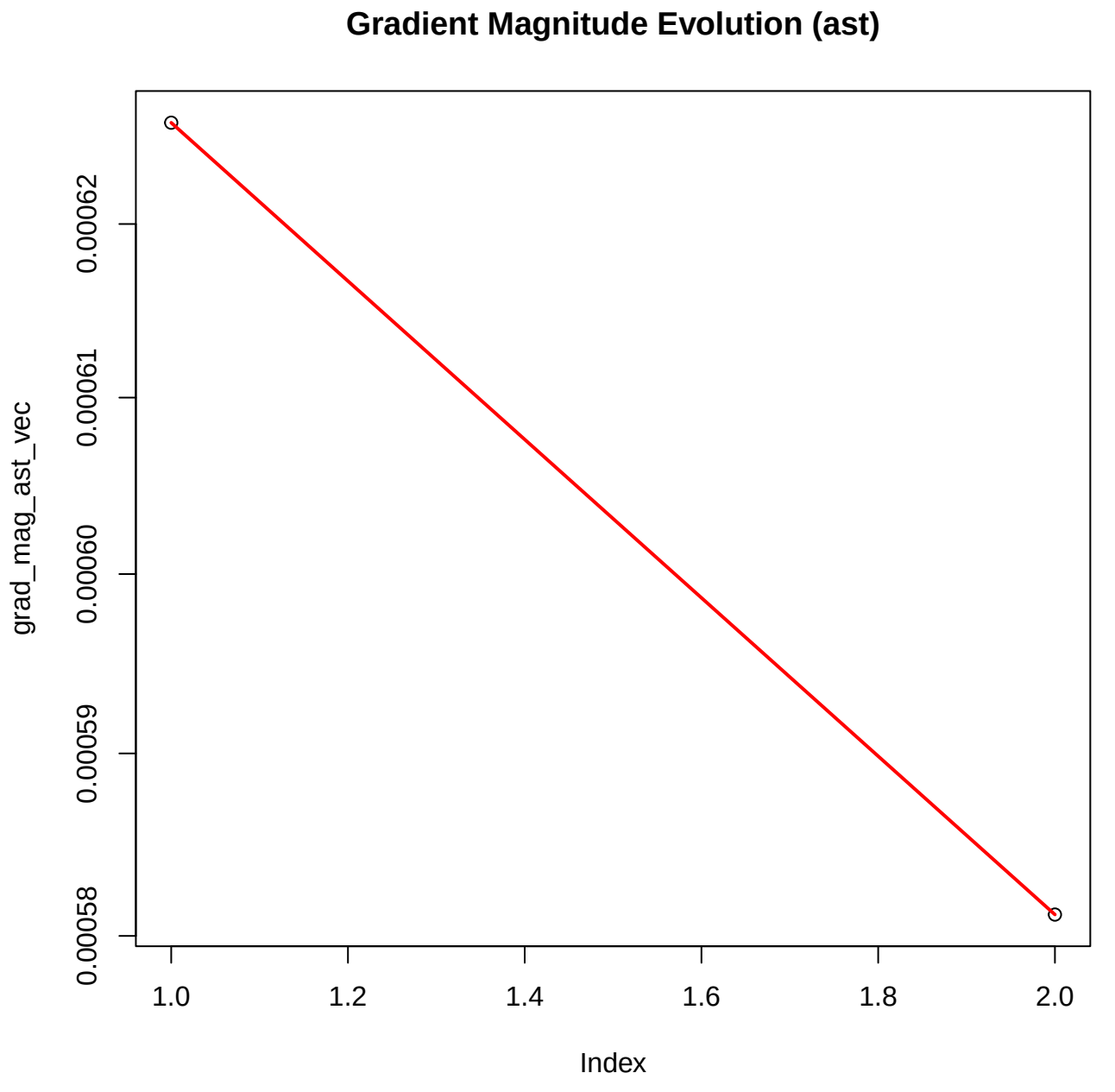

SVI ELBO Loss [1918.2063]
ELBO loss
1800
2000
2200
2400
2600
2800
3000
0
20
40
60
80
Iteration

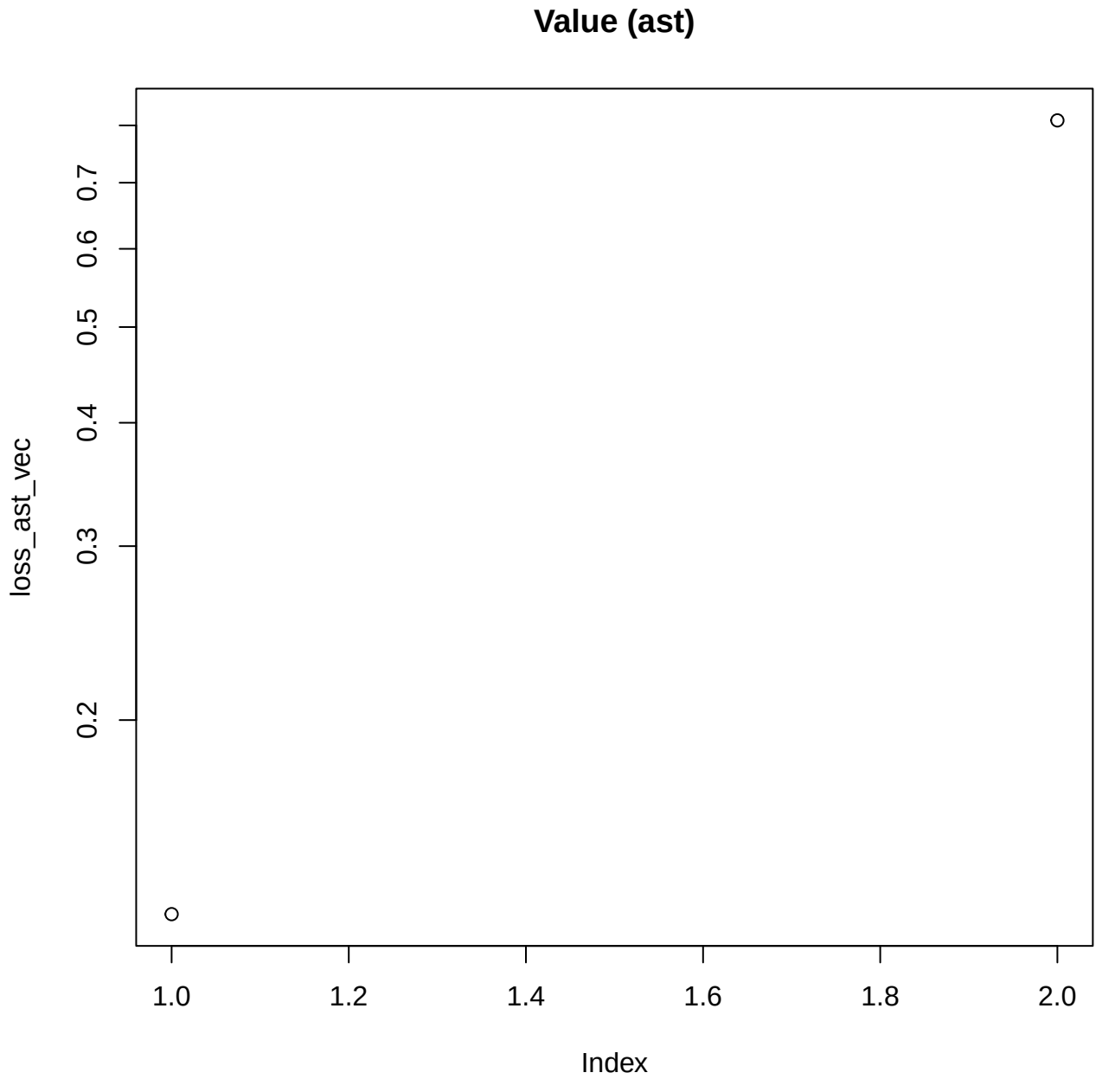

SVI ELBO Loss [1775.2107]
ELBO loss
6000
5000
4000
3000
2000
0
20
40
60
80
100
120
140
Iteration

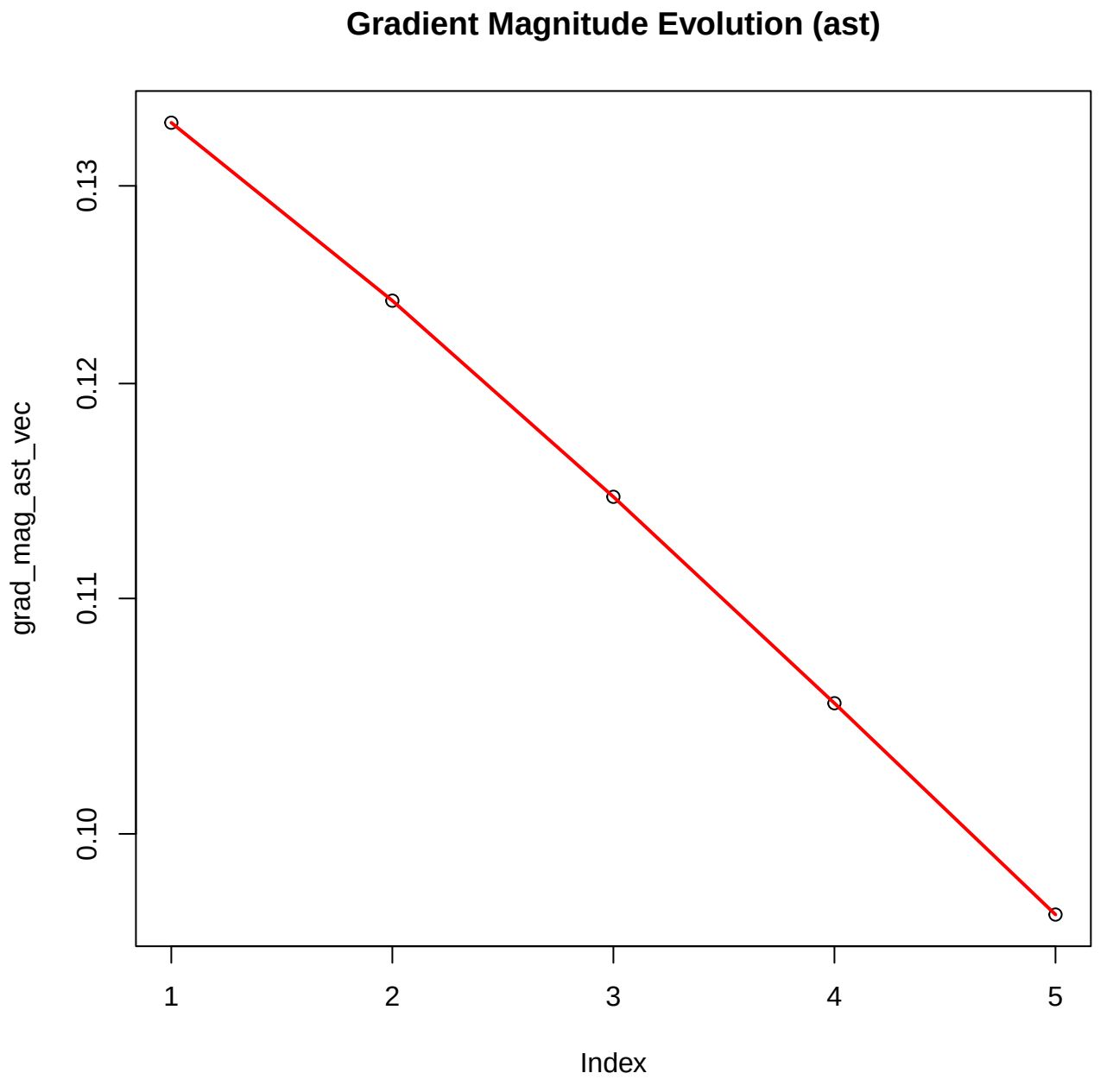

SVI ELBO Loss [1982.3200]
ELBO loss
6000
5000
4000
3000
2000
0
50
100
150
Iteration

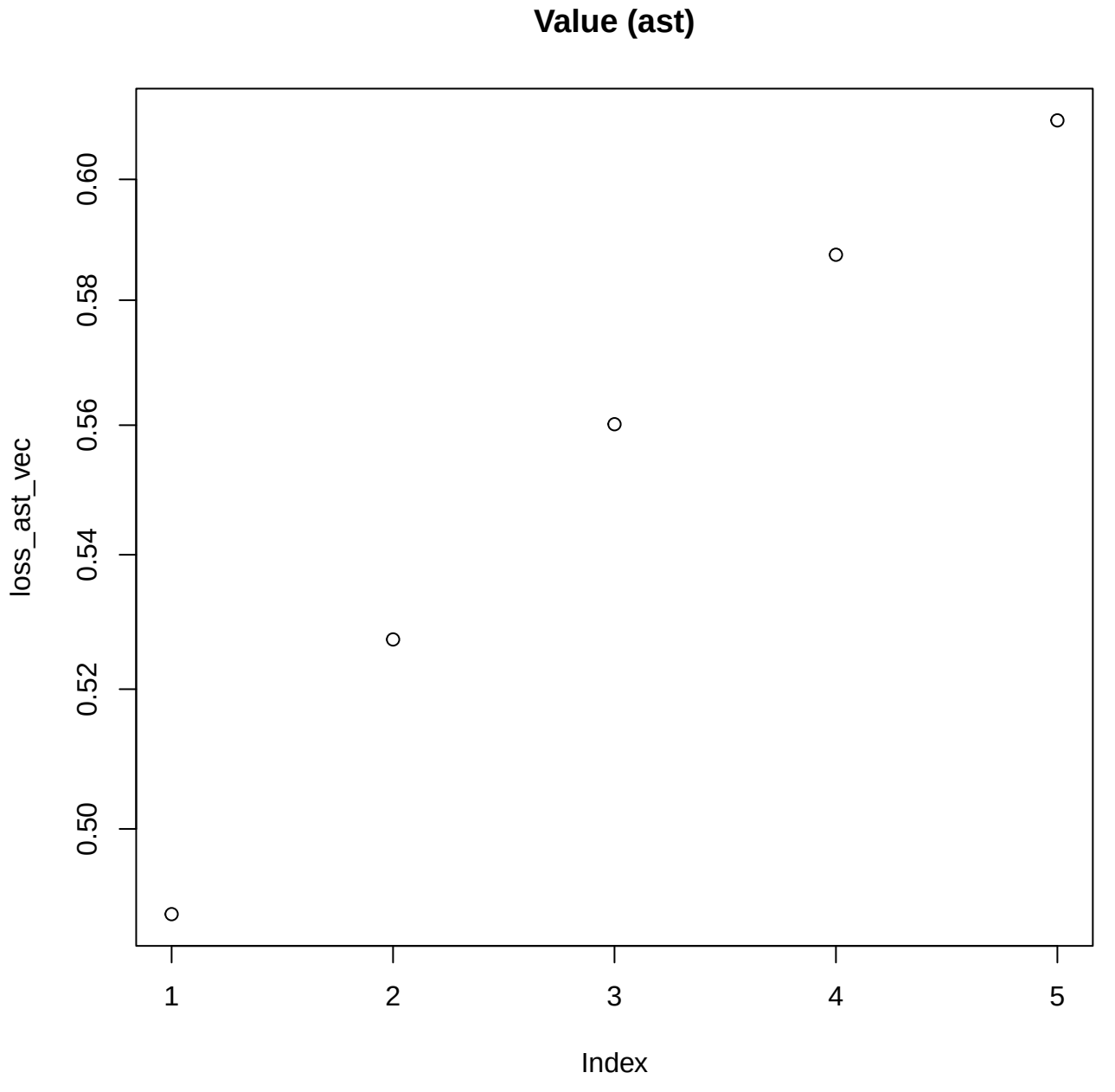

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00061
0.00062
0.00063
0.00064
0.00065
1.0
1.2
1.4
1.6
1.8
2.0
Index

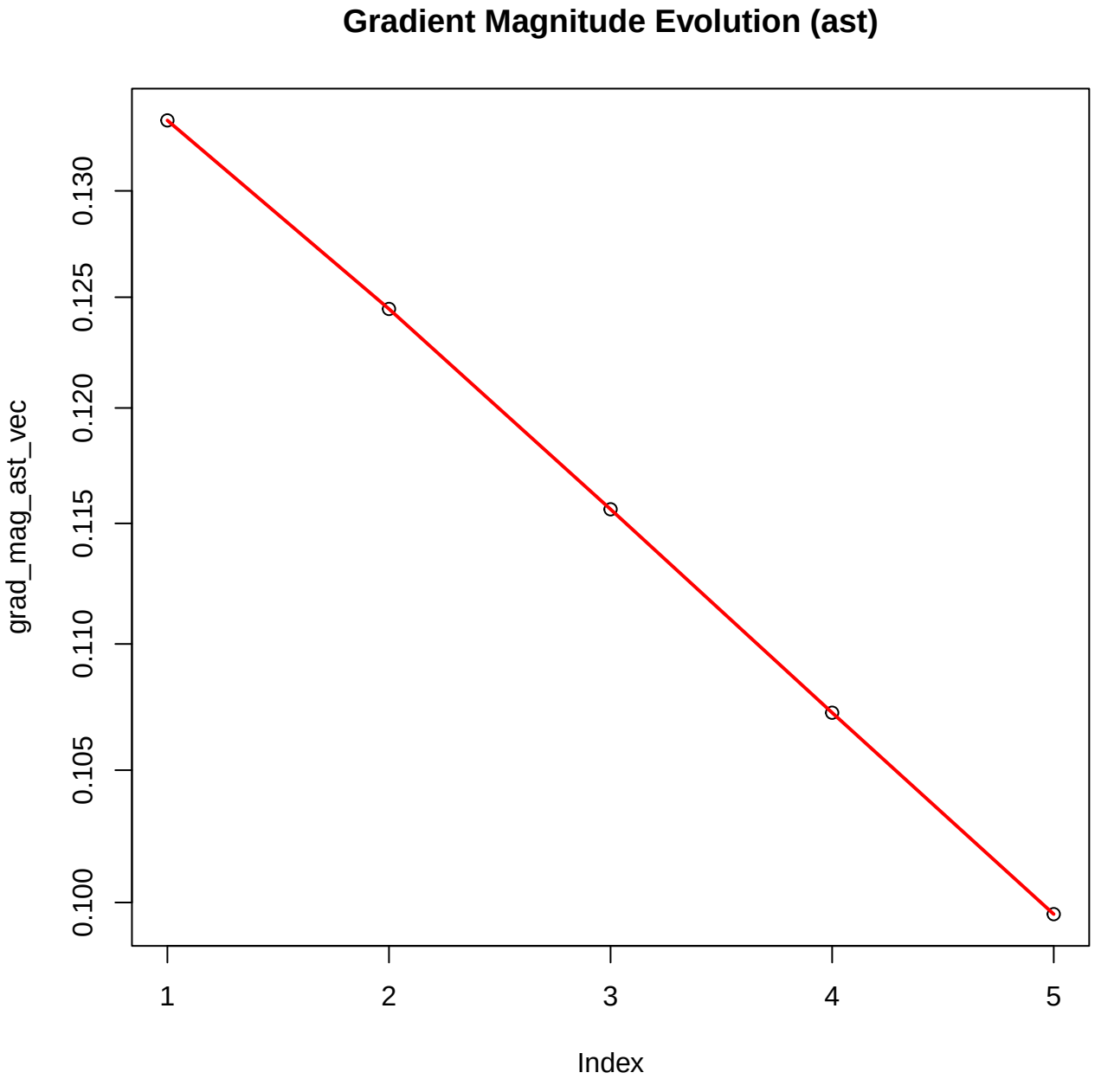

Value (ast)
loss_ast_vec
0.26
0.28
0.30
0.32
0.34
0.36
0.38
1.0
1.2
1.4
1.6
1.8
2.0
Index

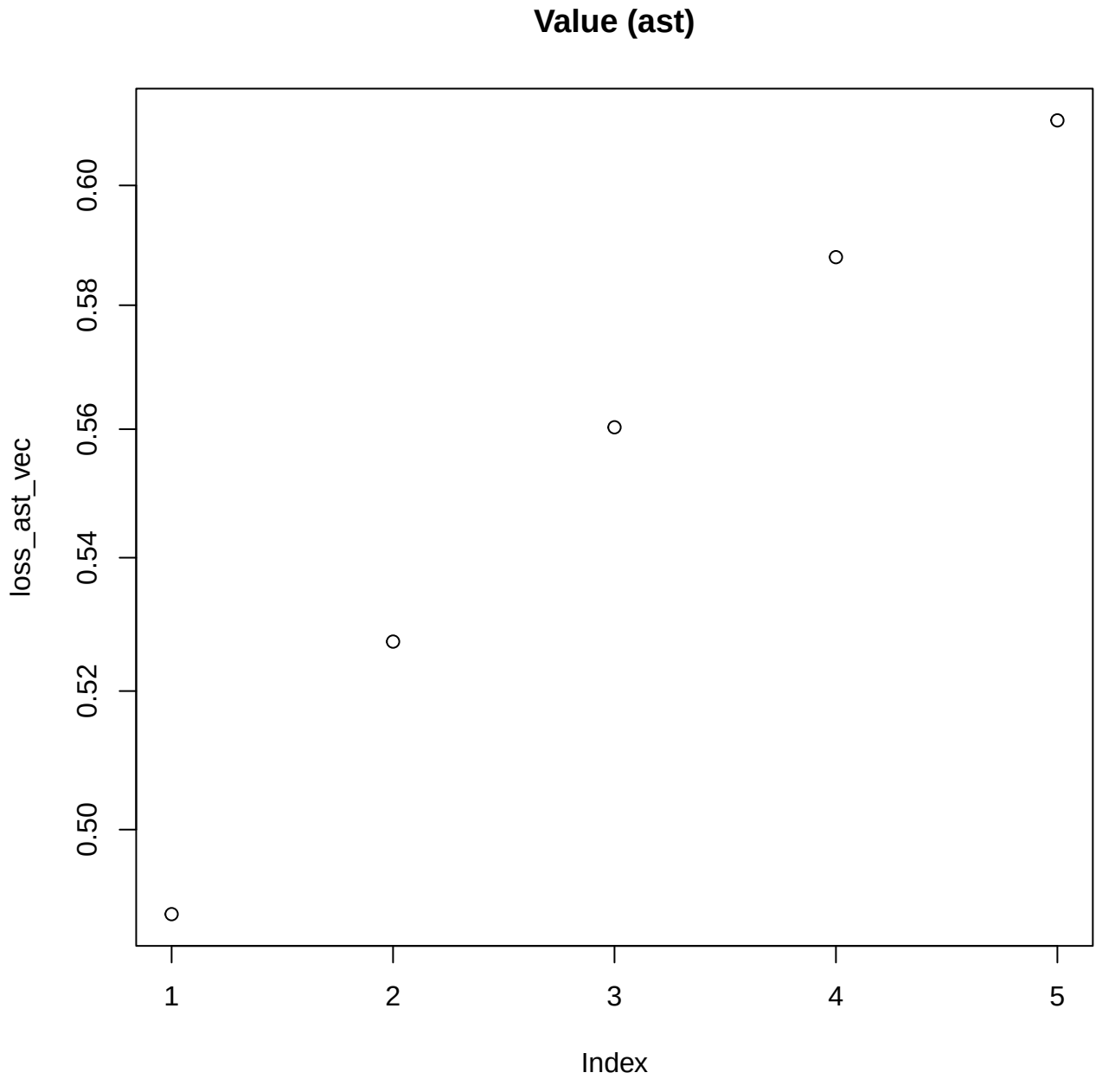

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00058
0.00059
0.00060
0.00061
0.00062
1.0
1.2
1.4
1.6
1.8
2.0
Index

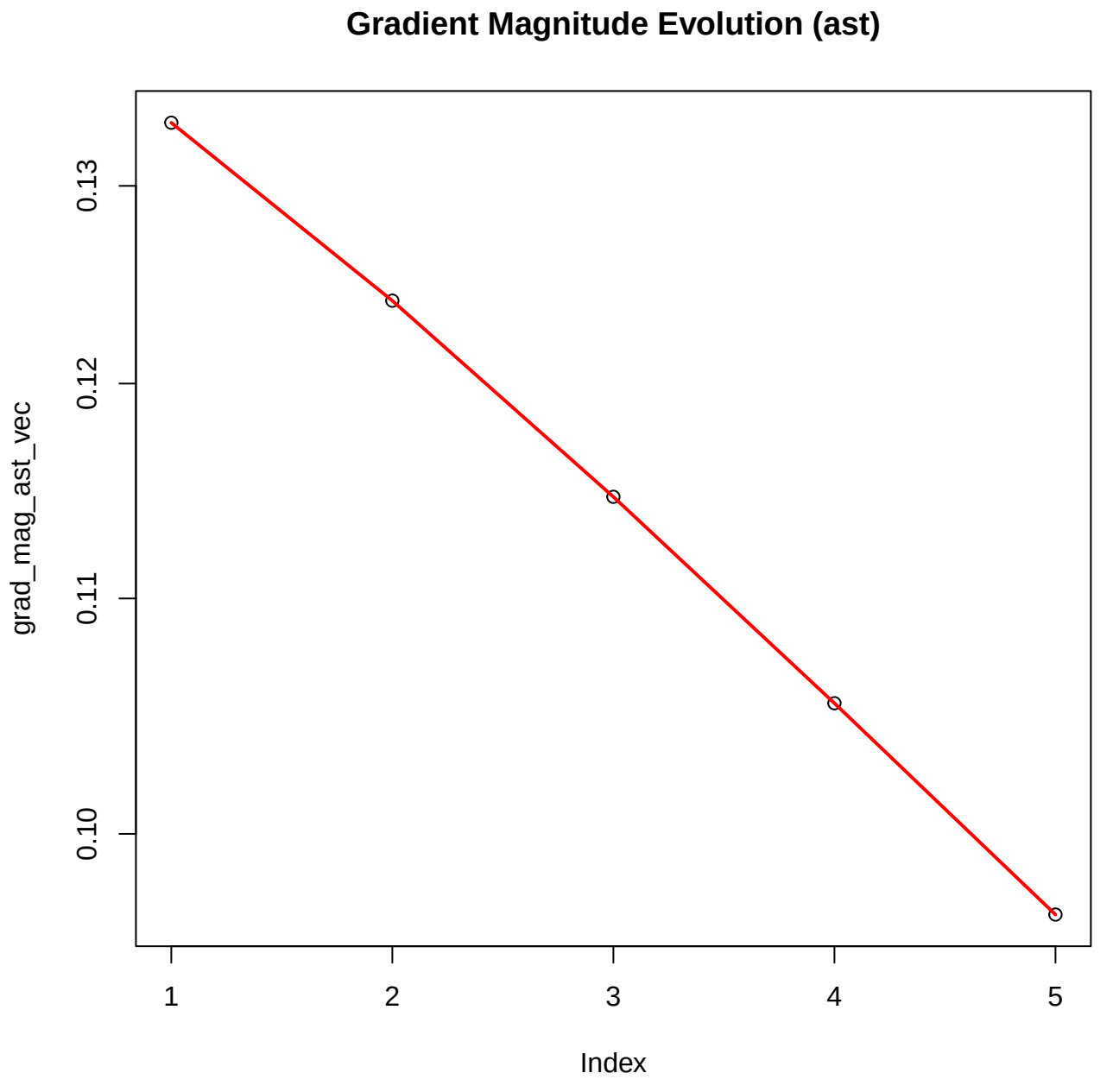

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
1.0
1.2
1.4
1.6
1.8
2.0
Index

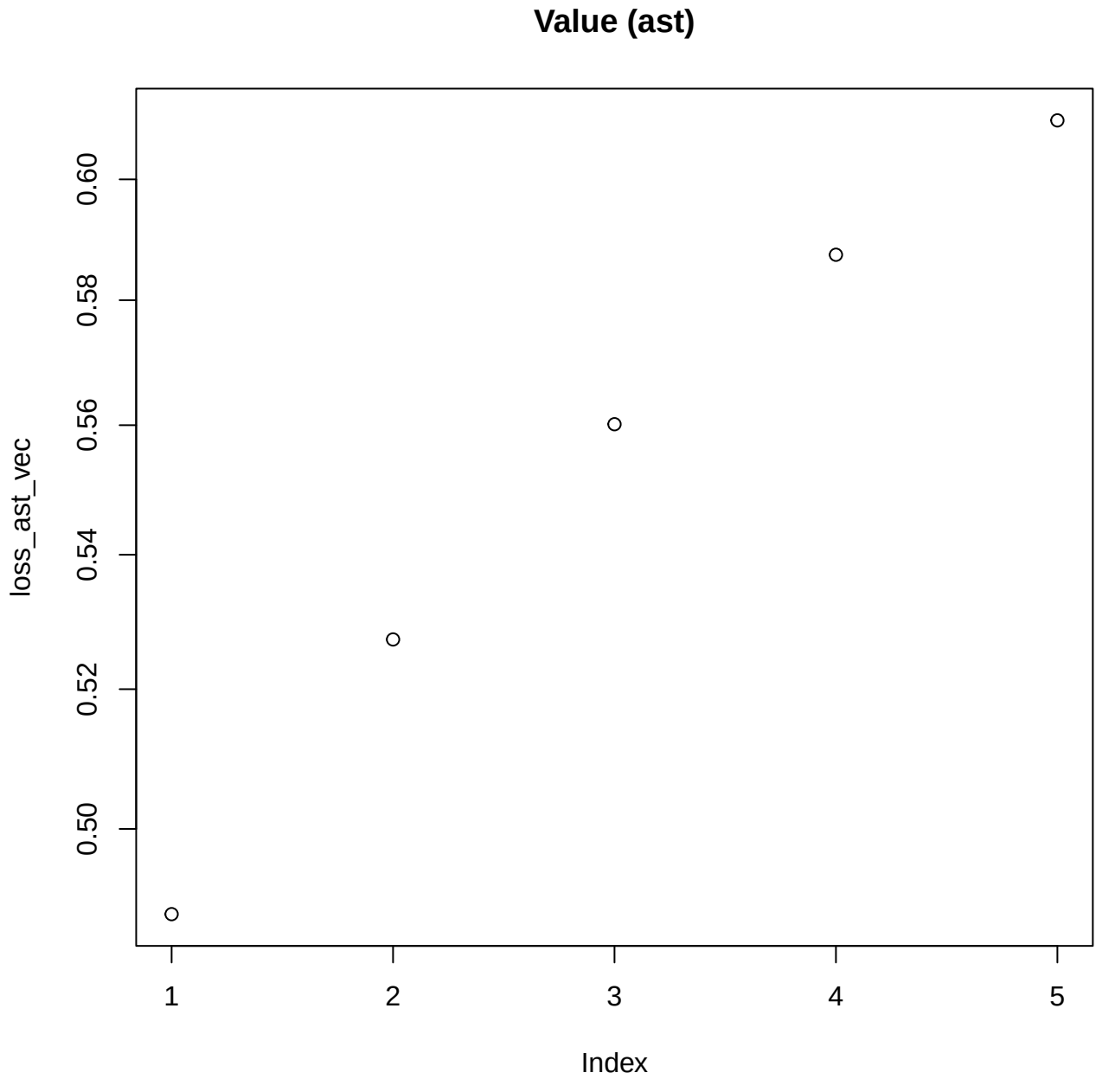

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index

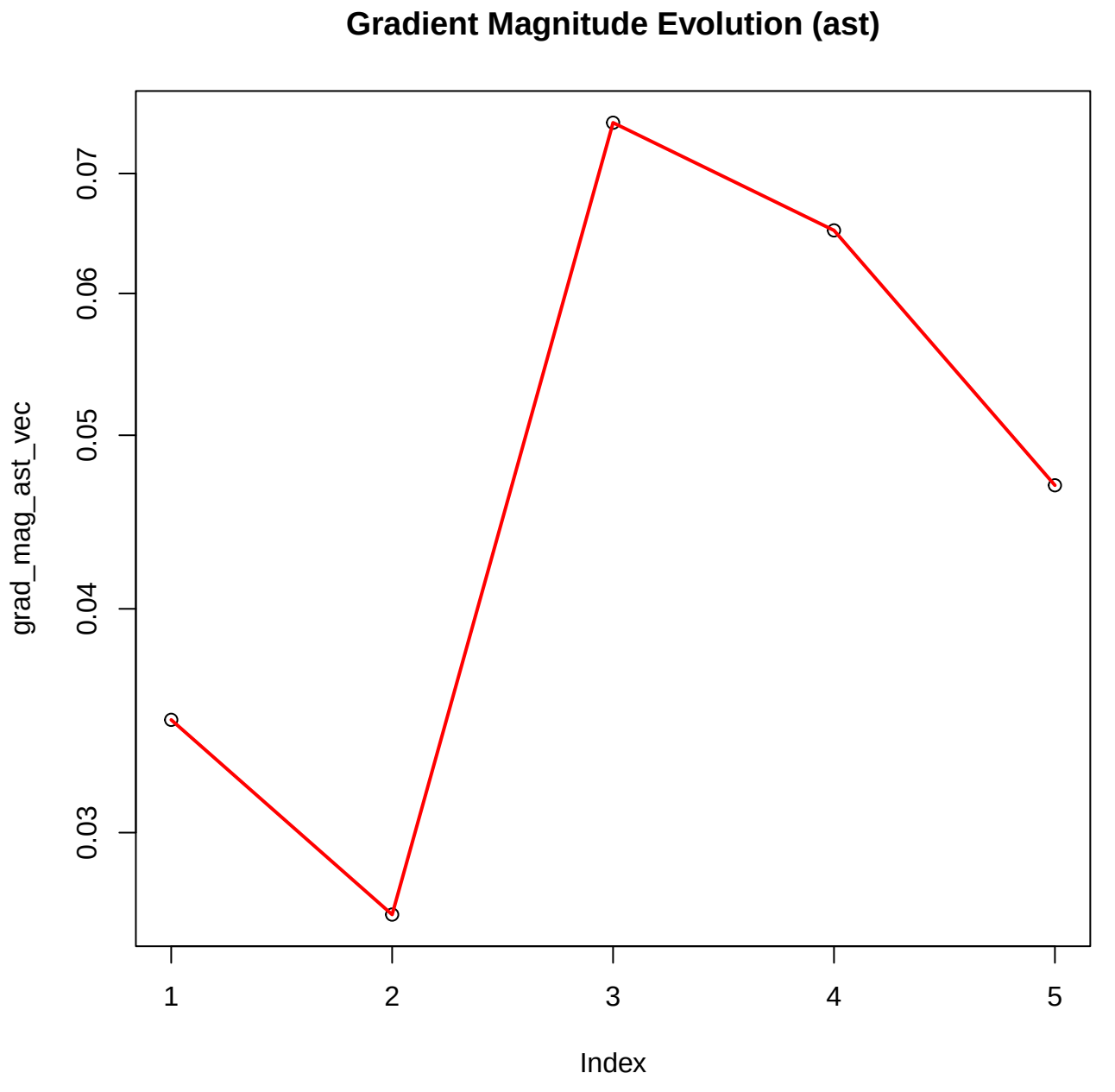

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

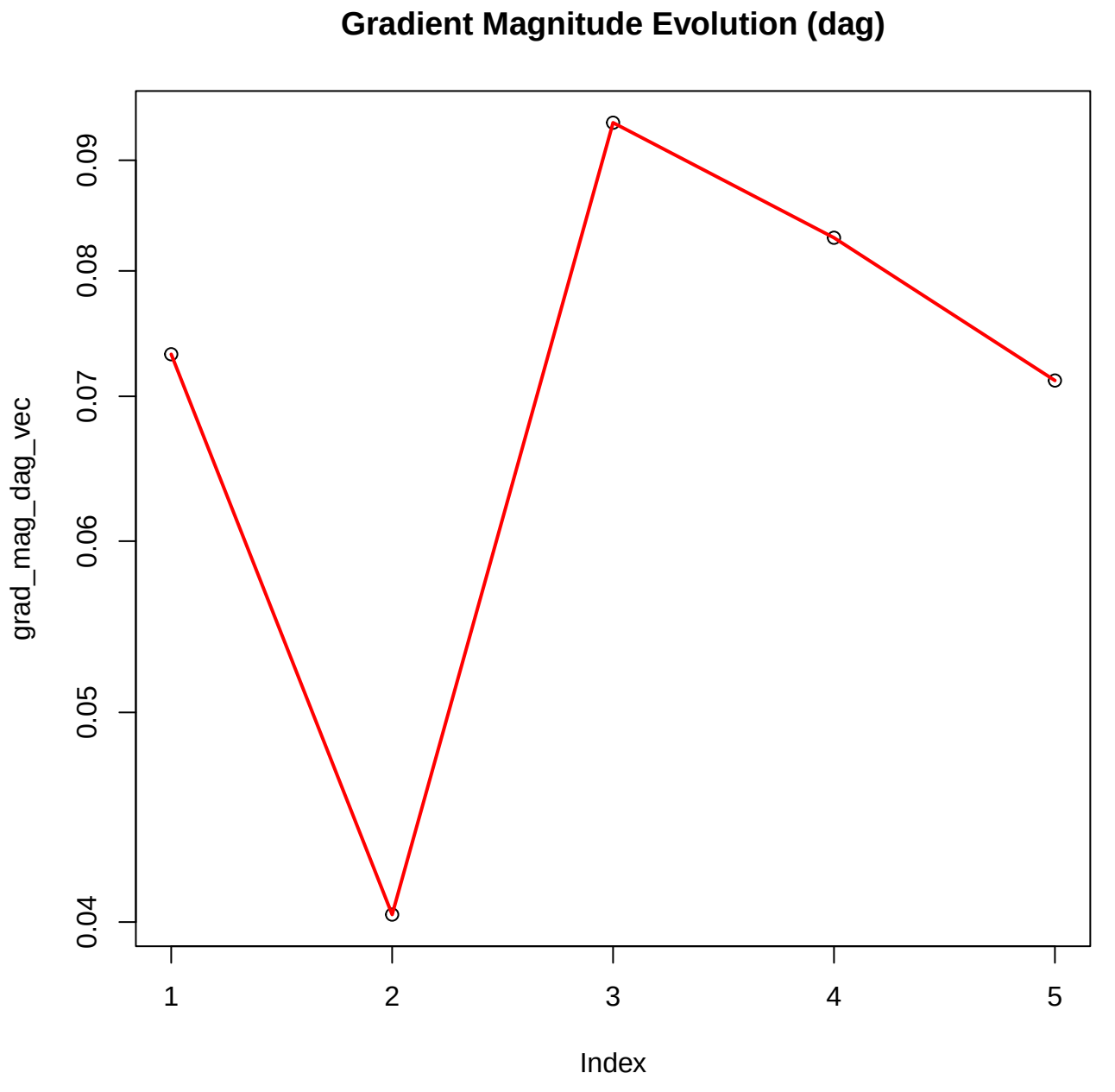

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.100
0.105
0.110
0.115
0.120
0.125
0.130
1
2
3
4
5
Index

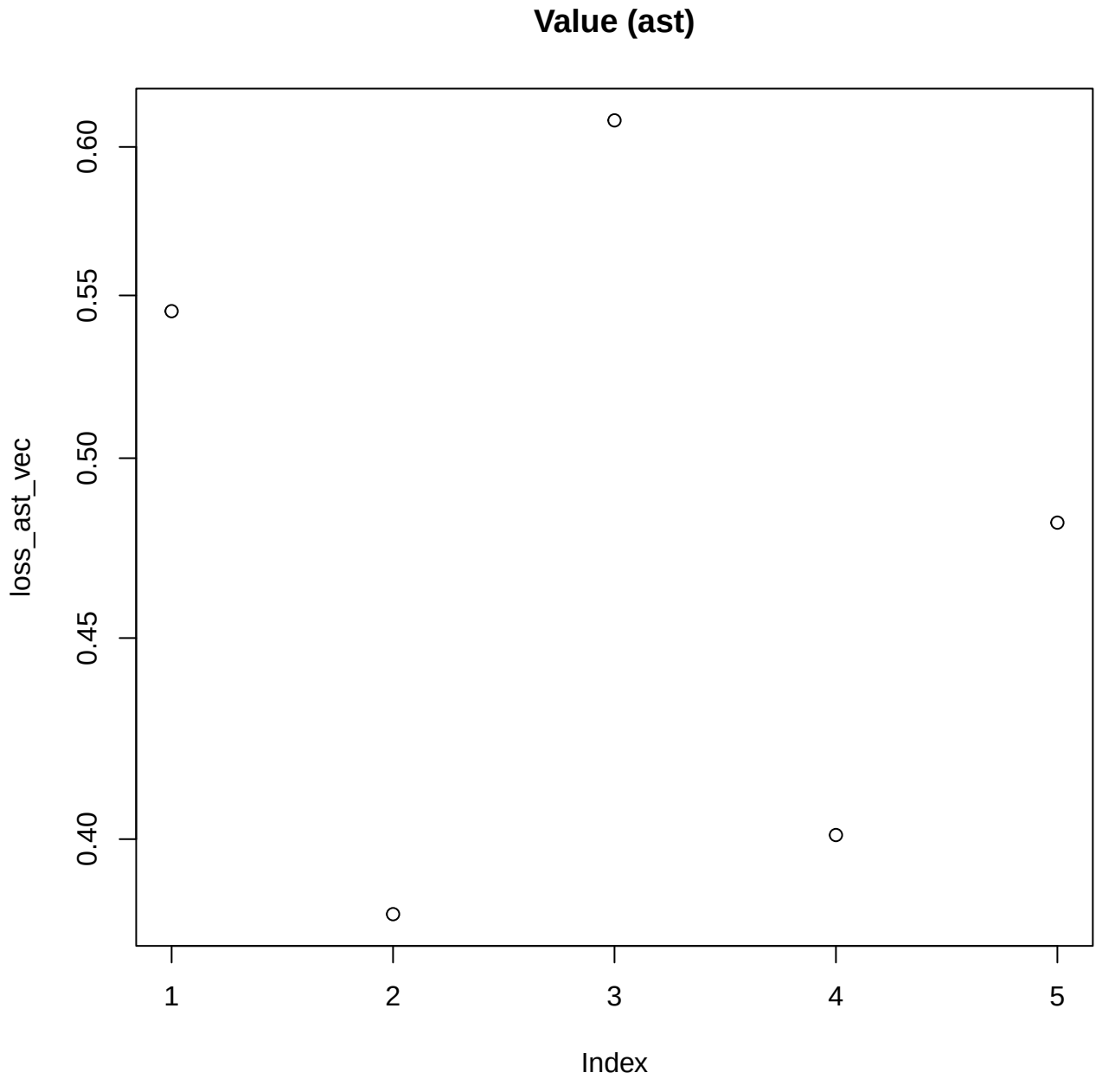

Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index

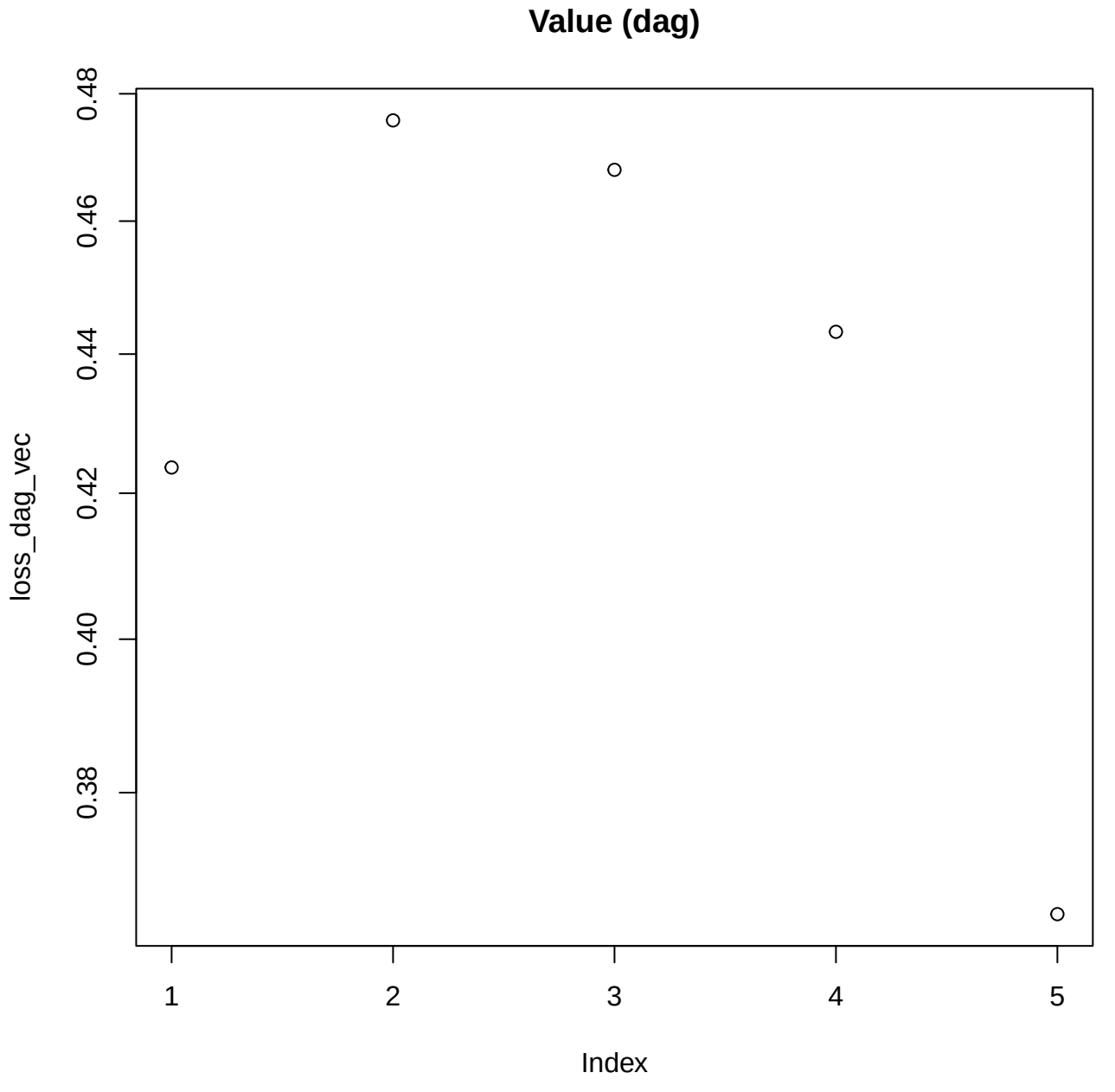

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.03
0.04
0.05
0.06
0.07
1
2
3
4
5
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.04
0.05
0.06
0.07
0.08
0.09
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.40
0.45
0.50
0.55
0.60
1
2
3
4
5
Index


Value (dag)
loss_dag_vec
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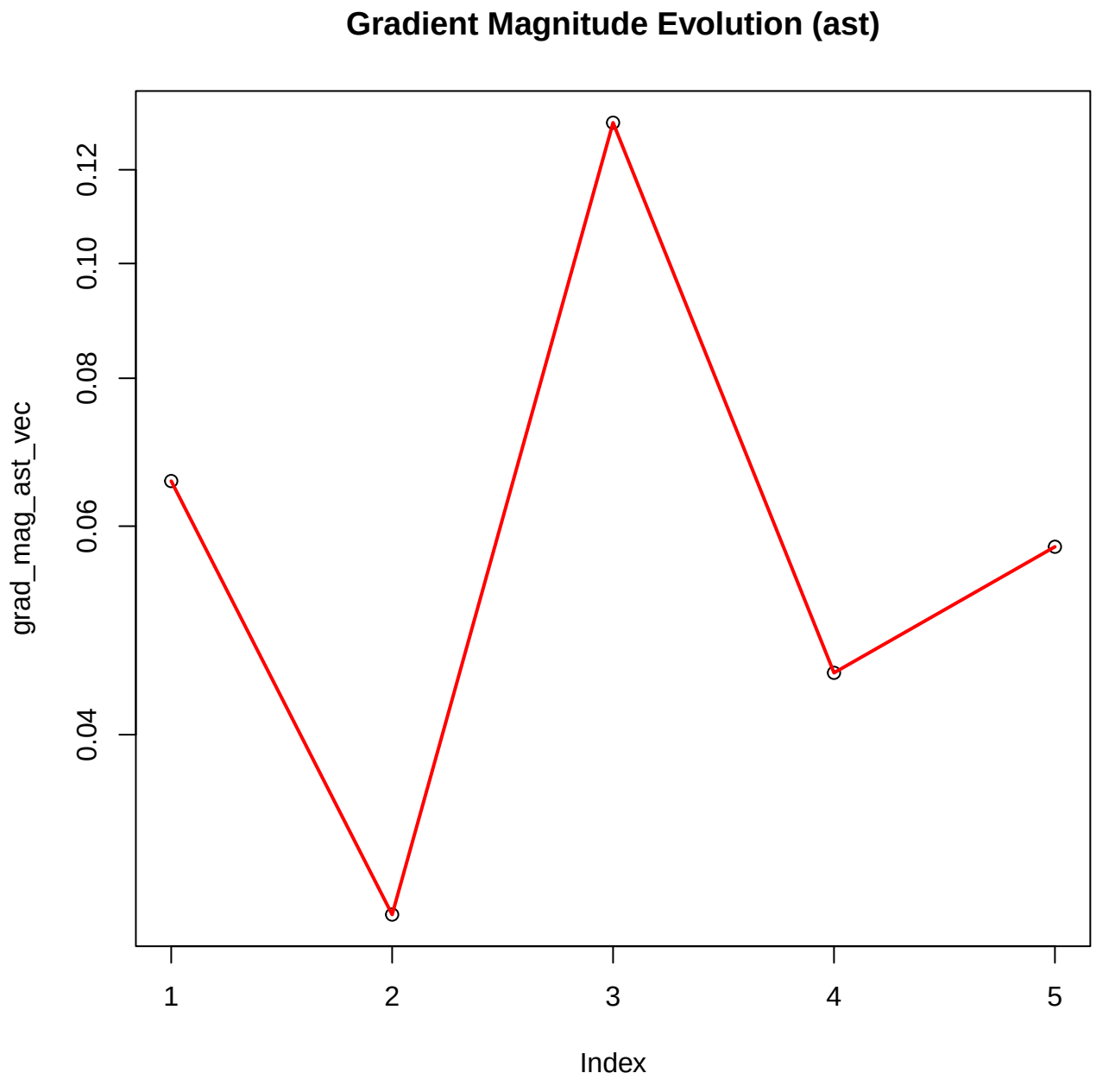
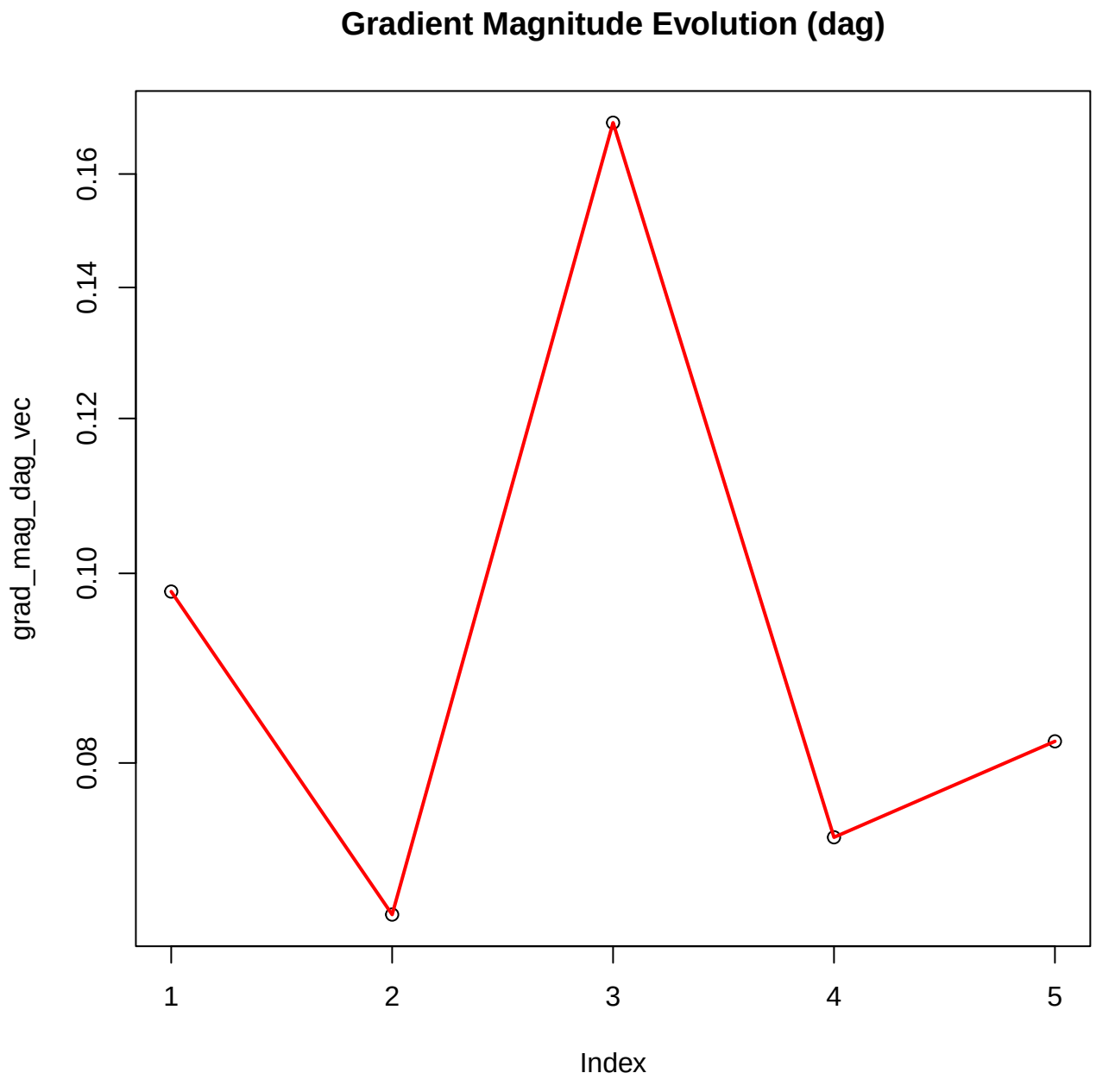
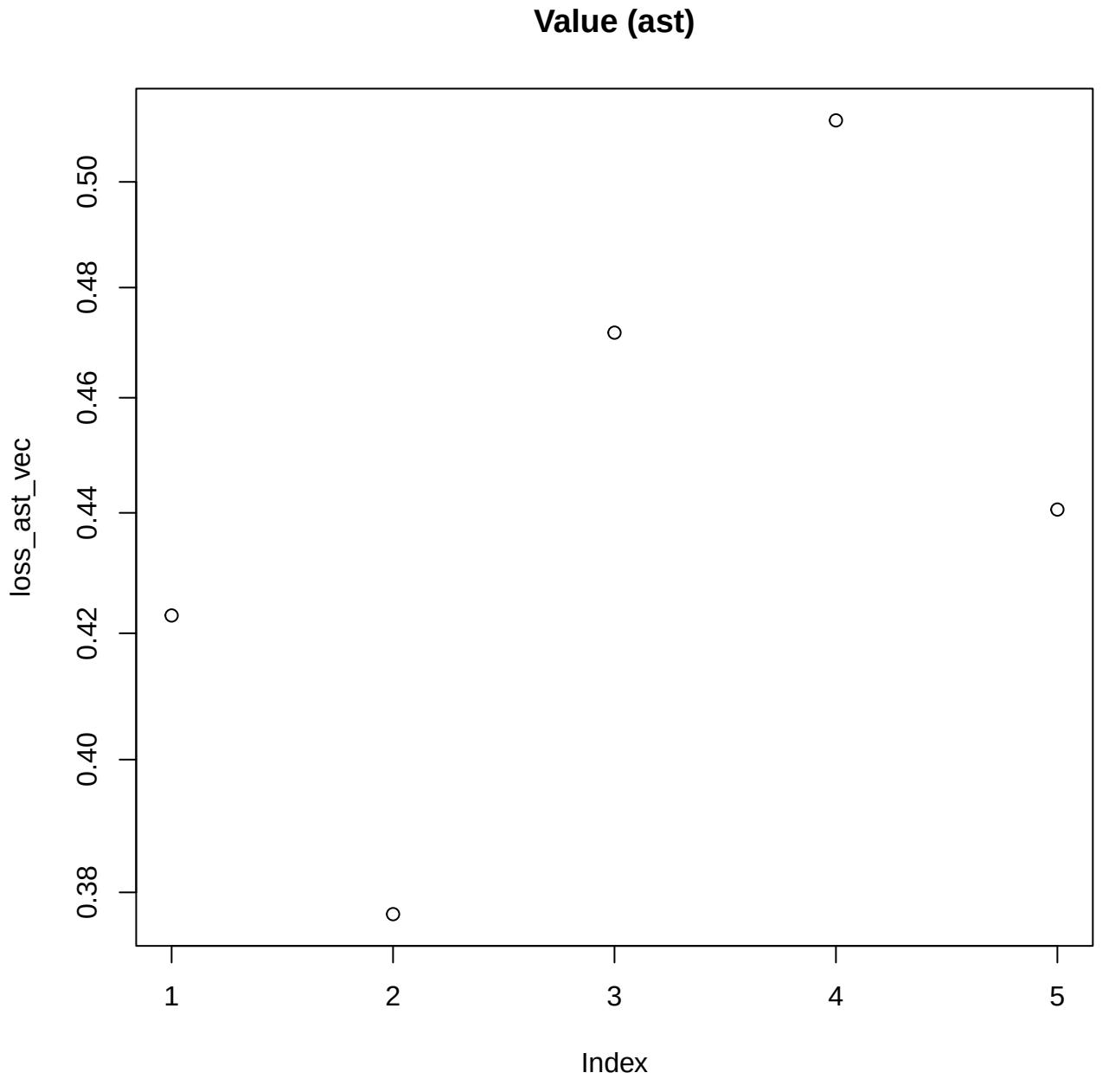
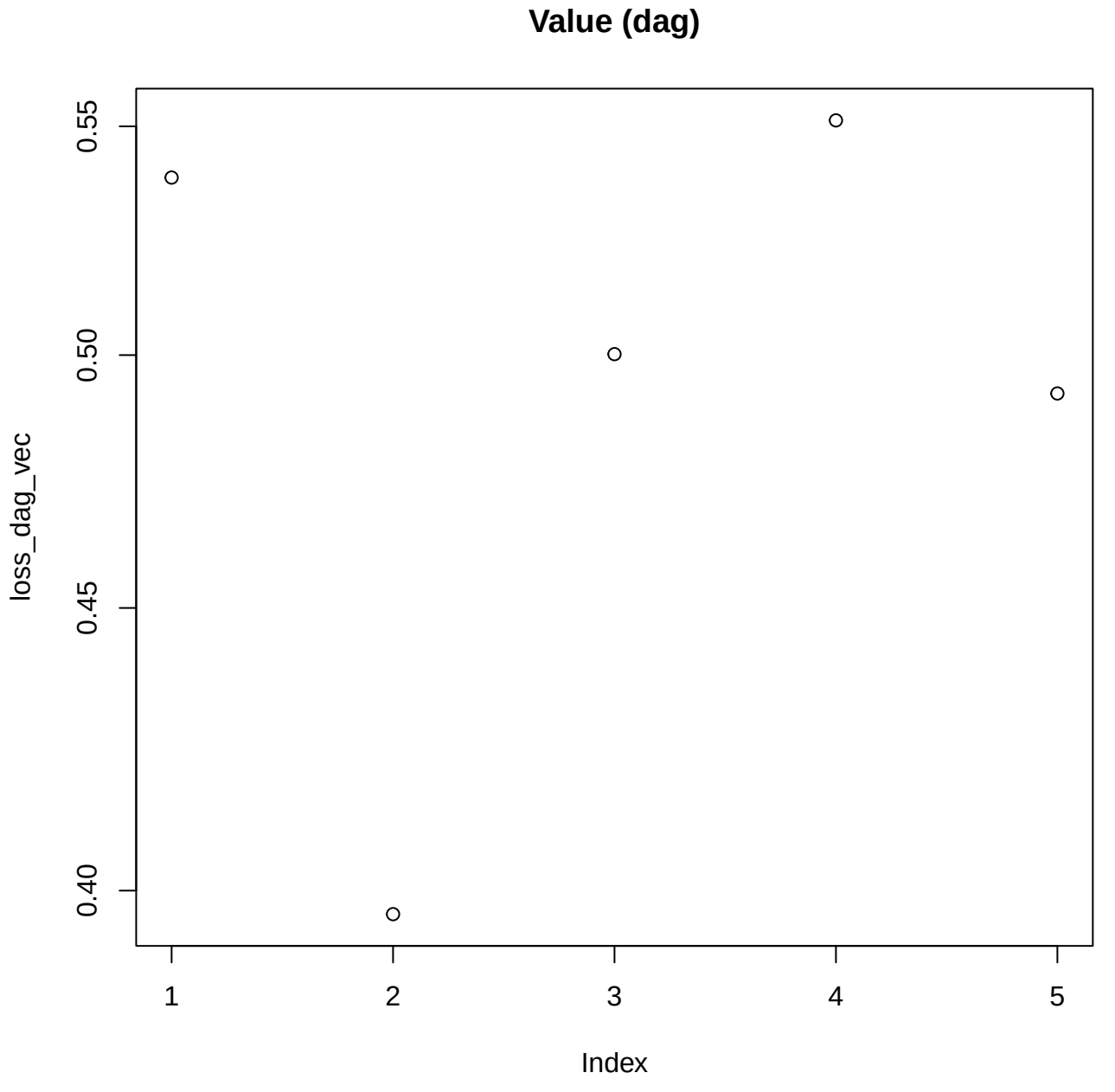
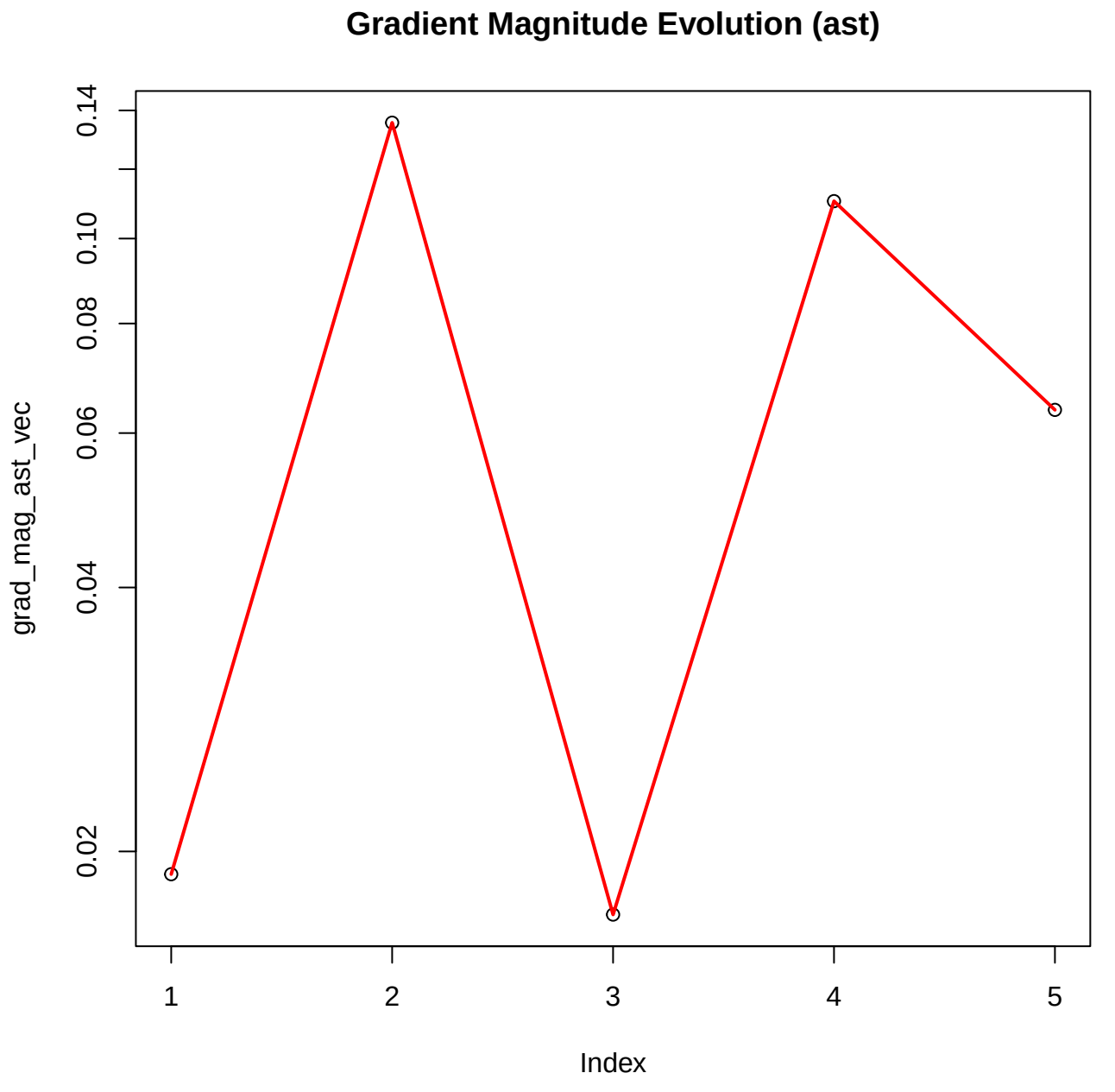
0.40
0.42
0.44
0.46
0.48
1
2
3
4
5
Index

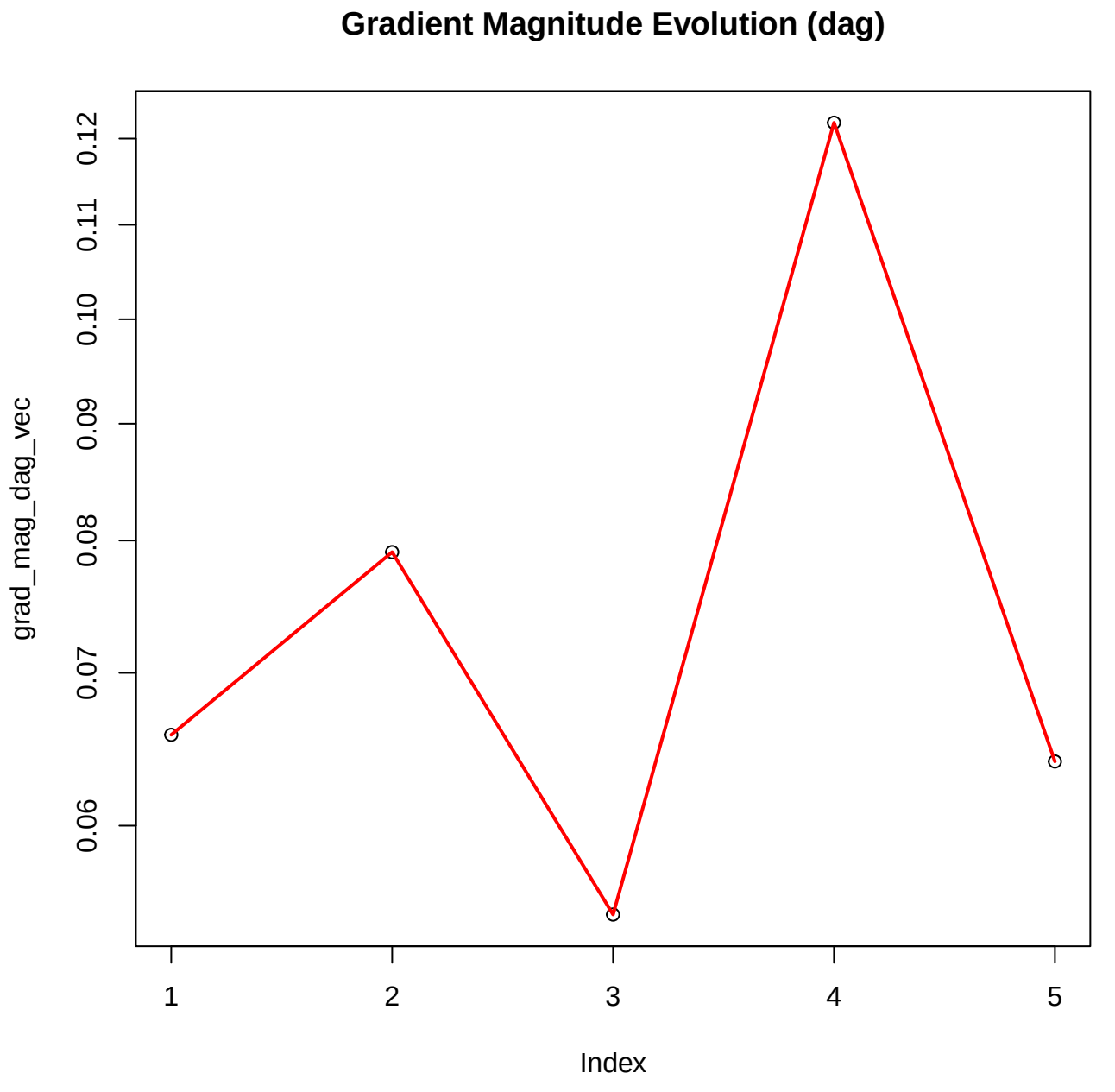

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.04
0.06
0.08
0.10
0.12
1
2
3
4
5
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.08
0.10
0.12
0.14
0.16
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.38
0.40
0.42
0.44
0.46
0.48
0.50
1
2
3
4
5
Index


Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.02
0.04
0.06
0.08
0.10
0.14
1
2
3
4
5
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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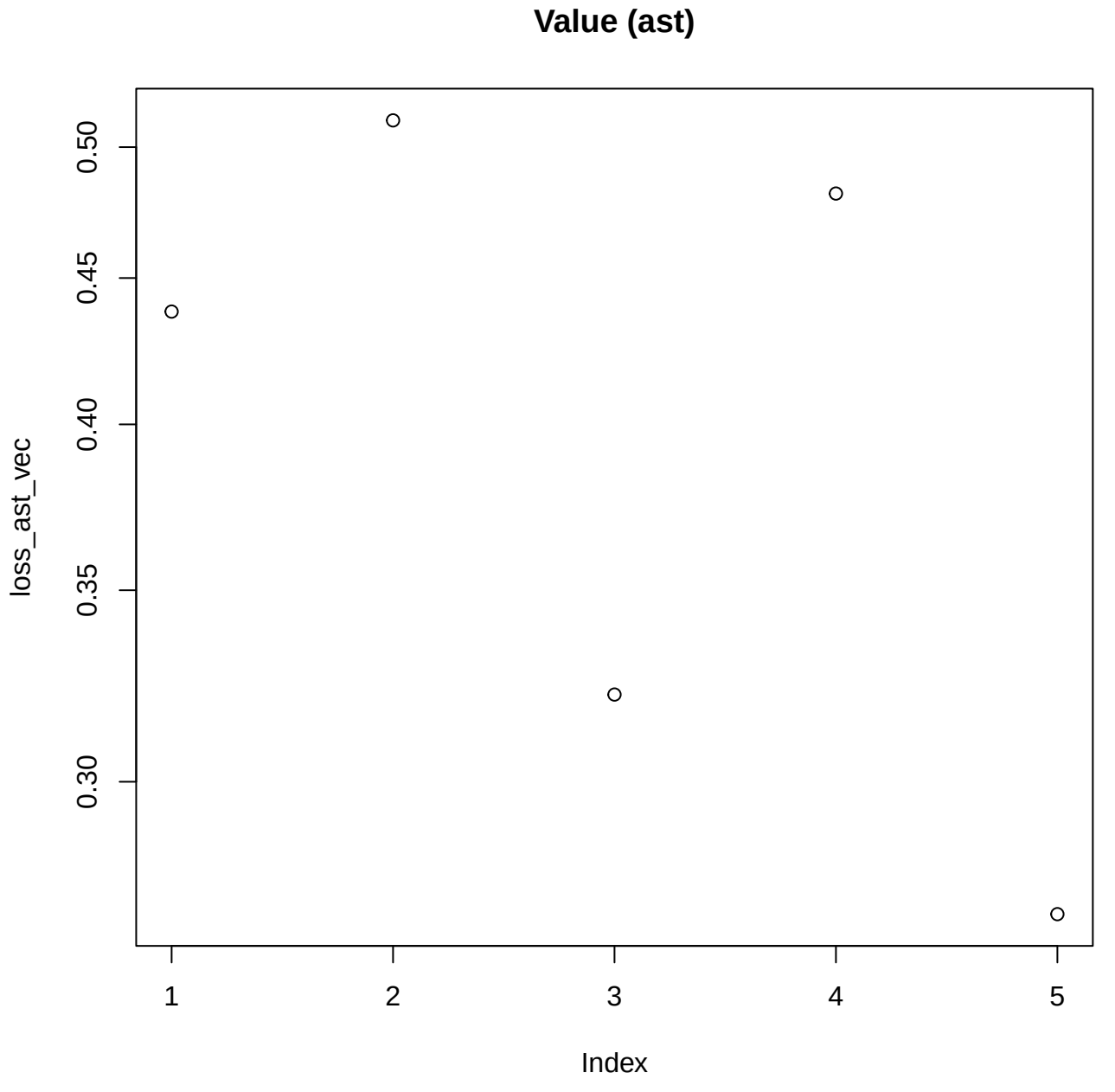
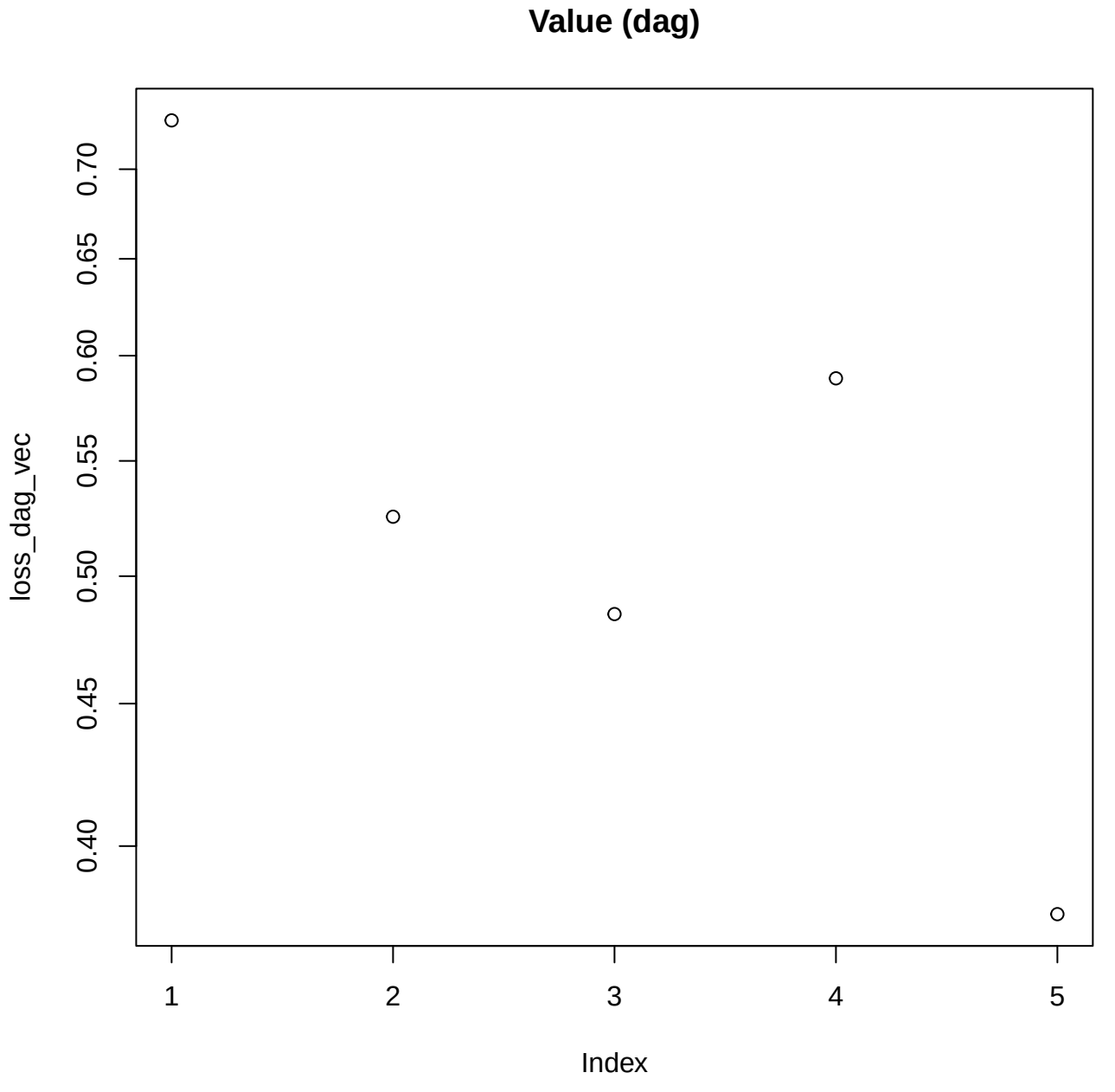
0.07
0.08
0.09
0.10
0.11
0.12
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.30
0.35
0.40
0.45
0.50
1
2
3
4
5
Index


Value (dag)
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0.55
0.60
0.65
0.70
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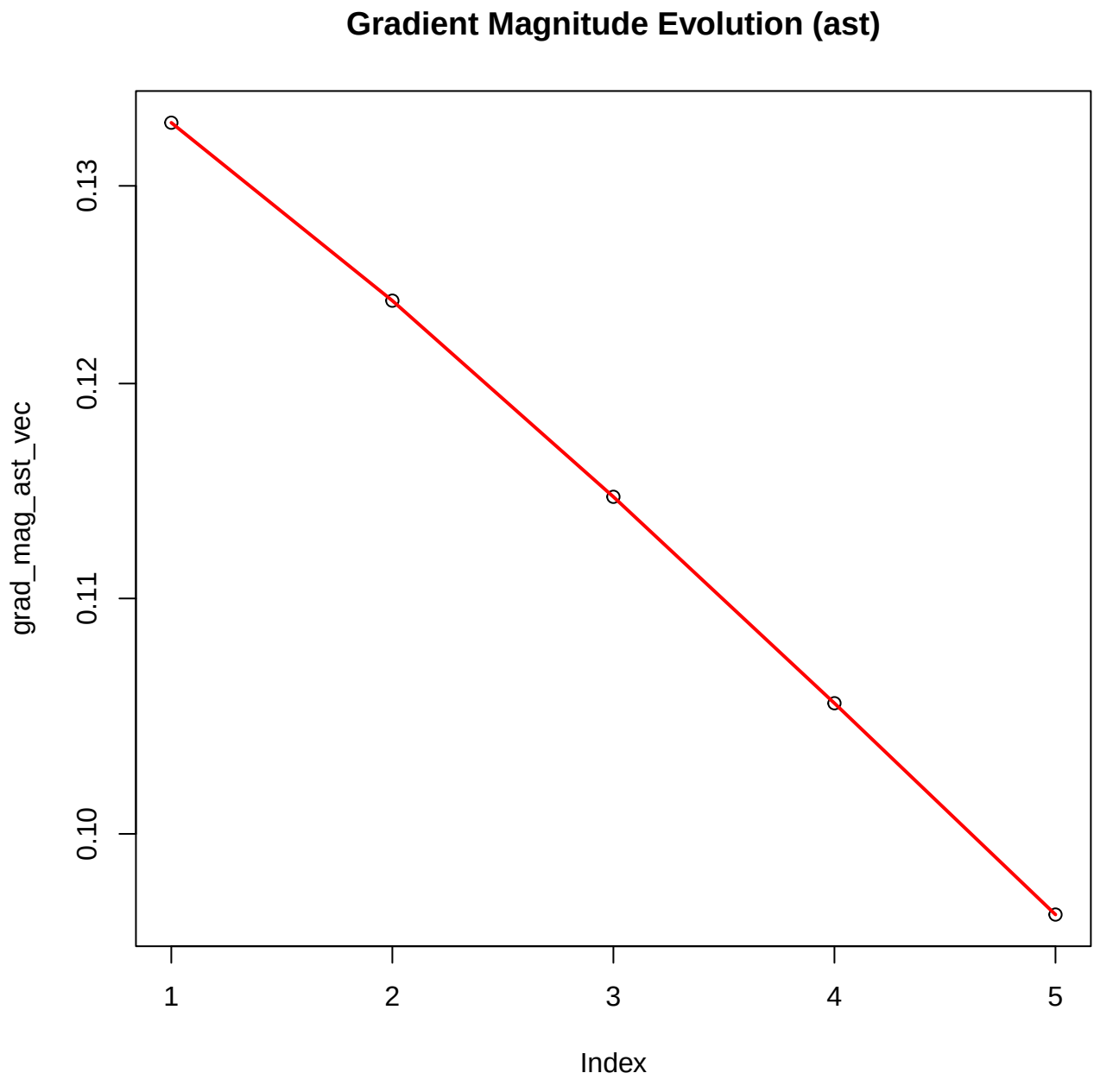
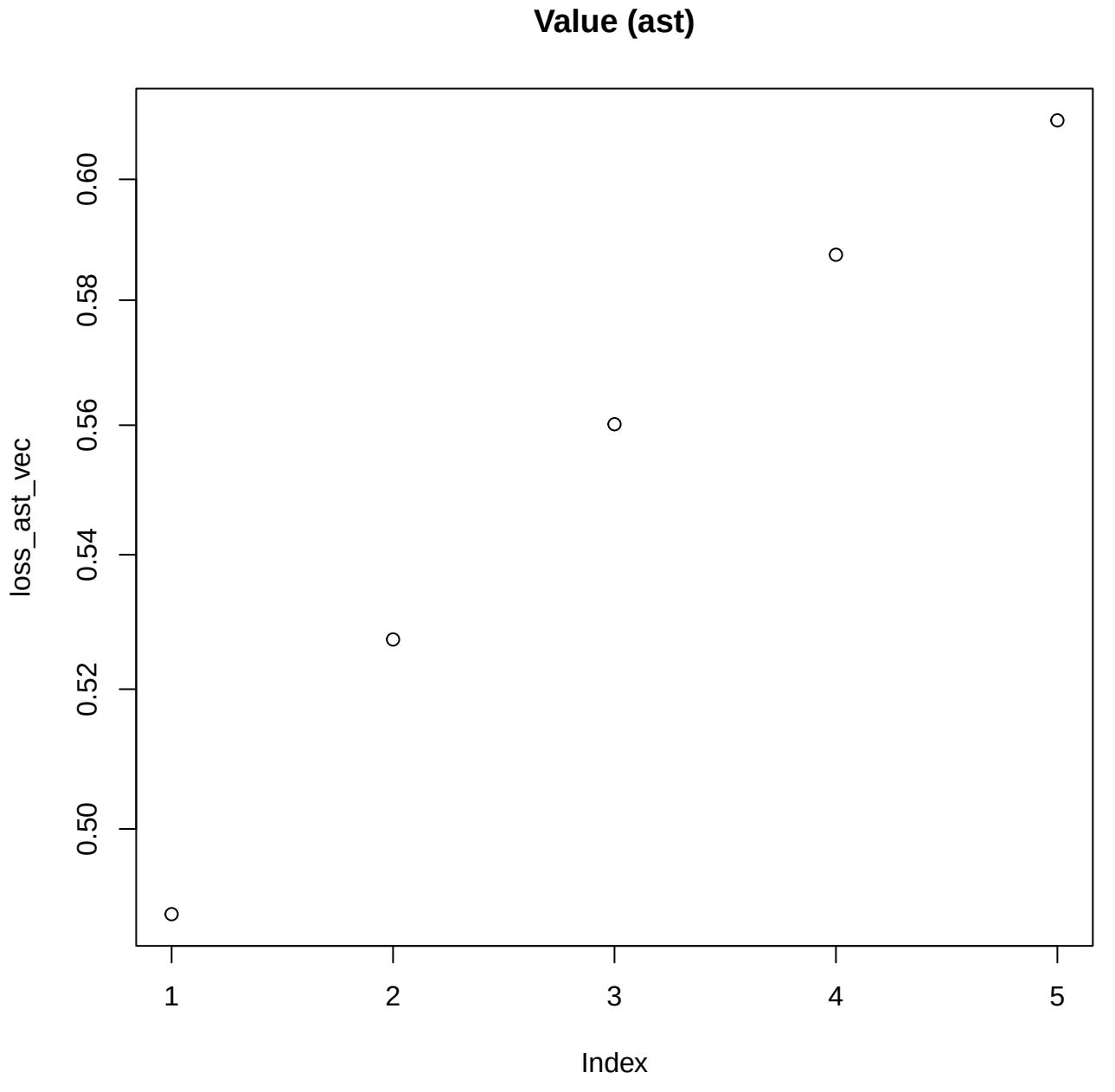
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.54
0.56
0.58
0.60
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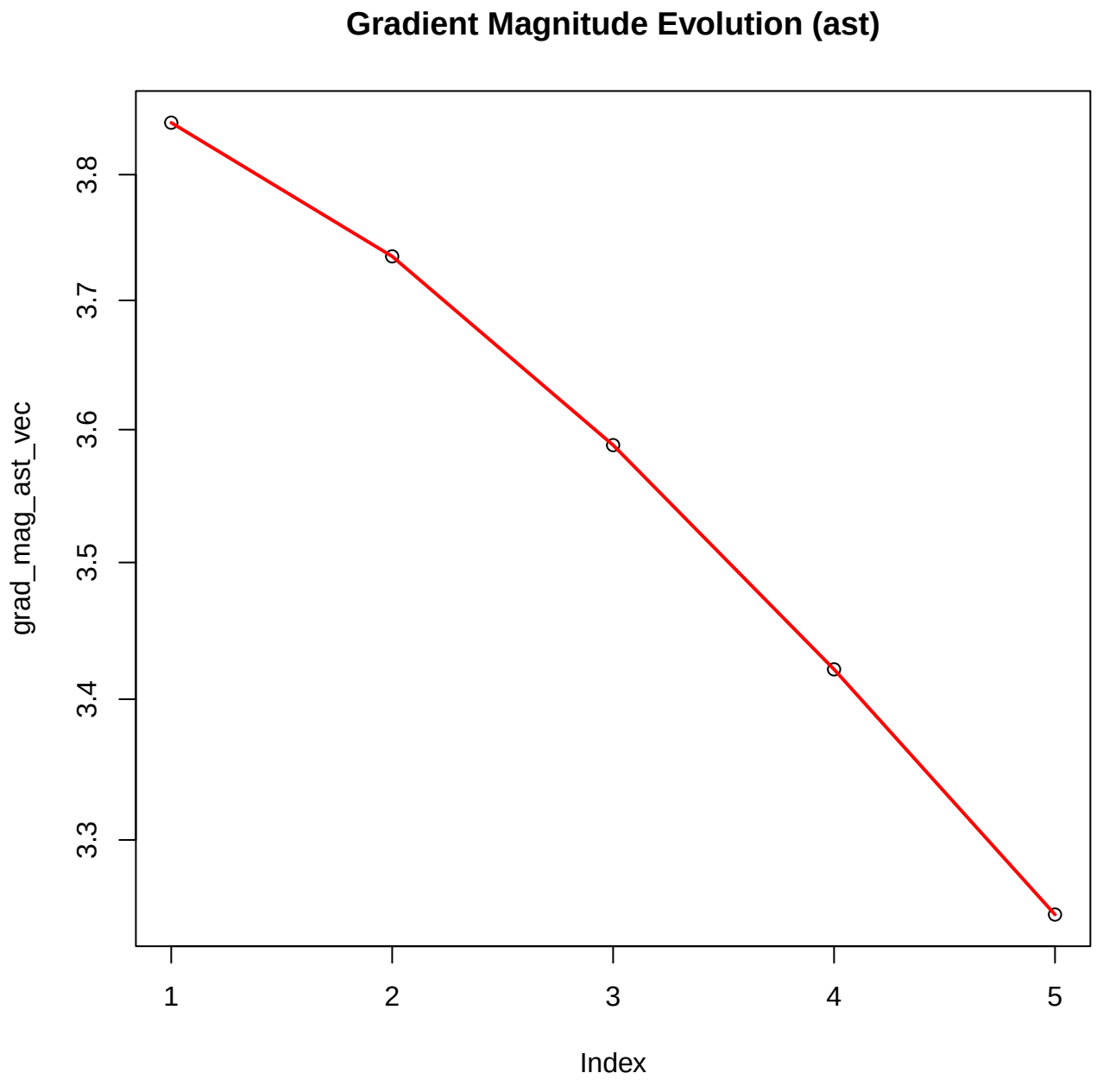
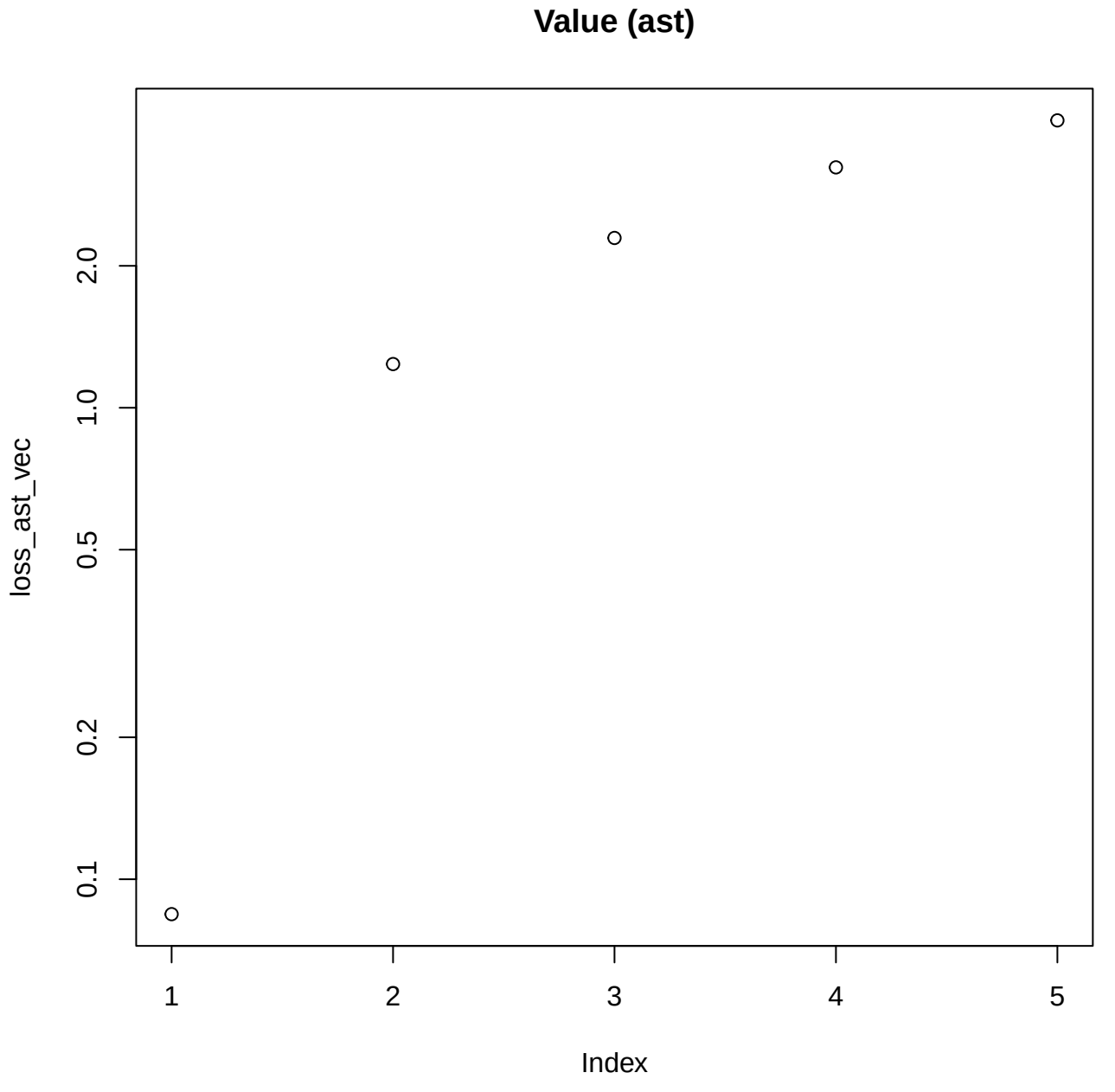
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
3.3
3.4
3.5
3.6
3.7
3.8
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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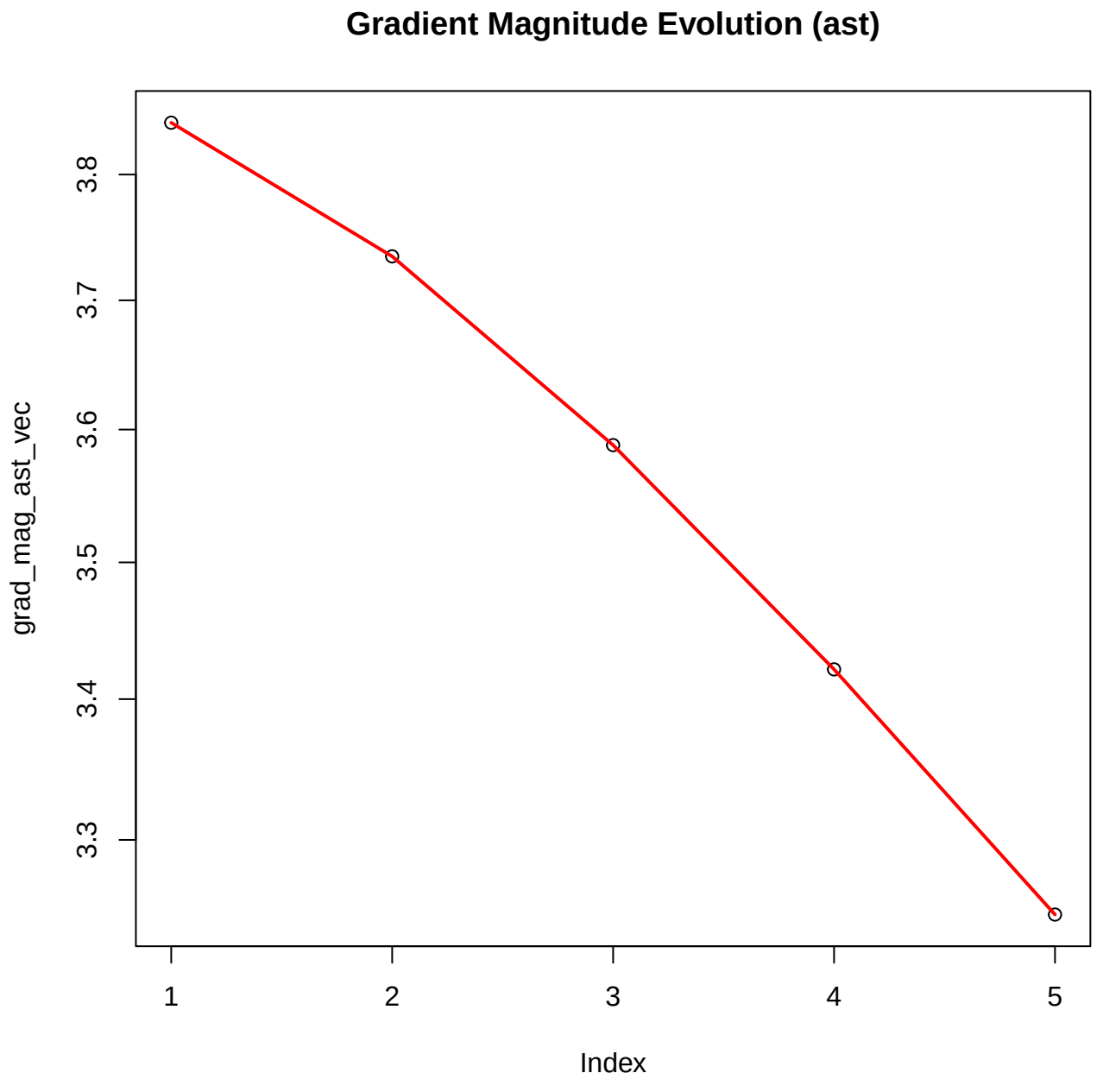
0.2
0.5
1.0
2.0
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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3.4
3.5
3.6
3.7
3.8
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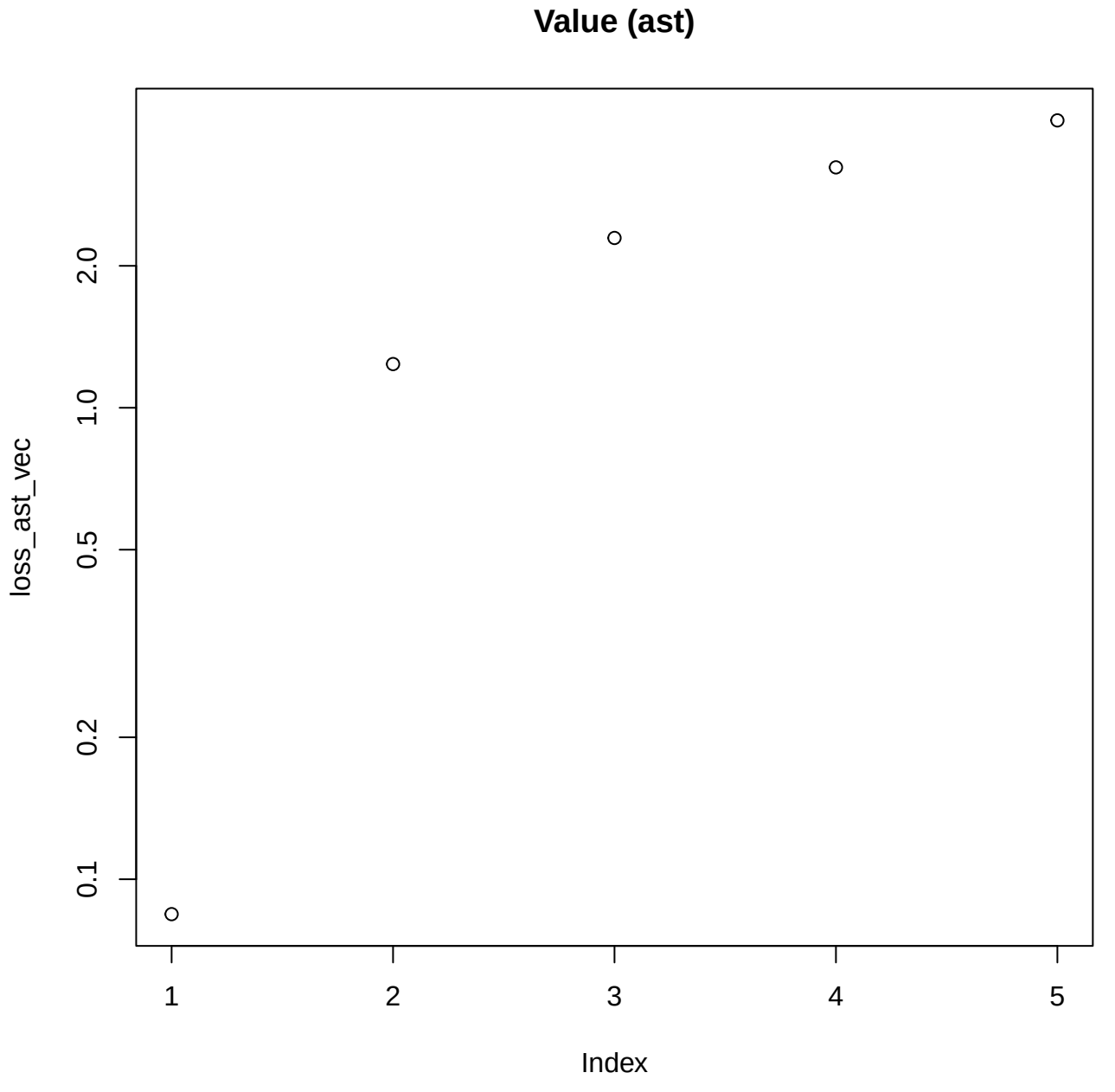
2
3
4
5
Index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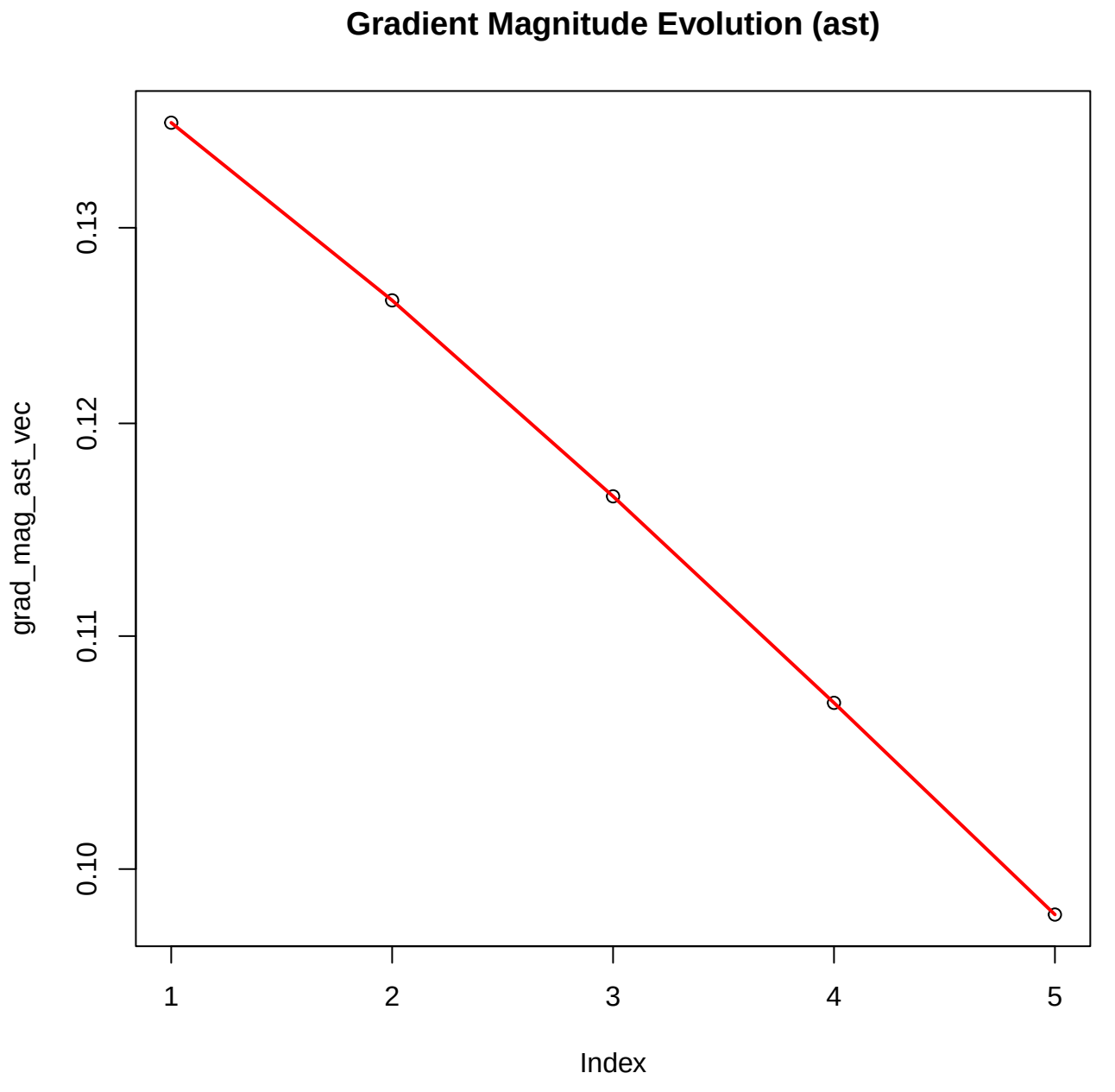
0.5
1.0
2.0
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.12
0.13
1
2
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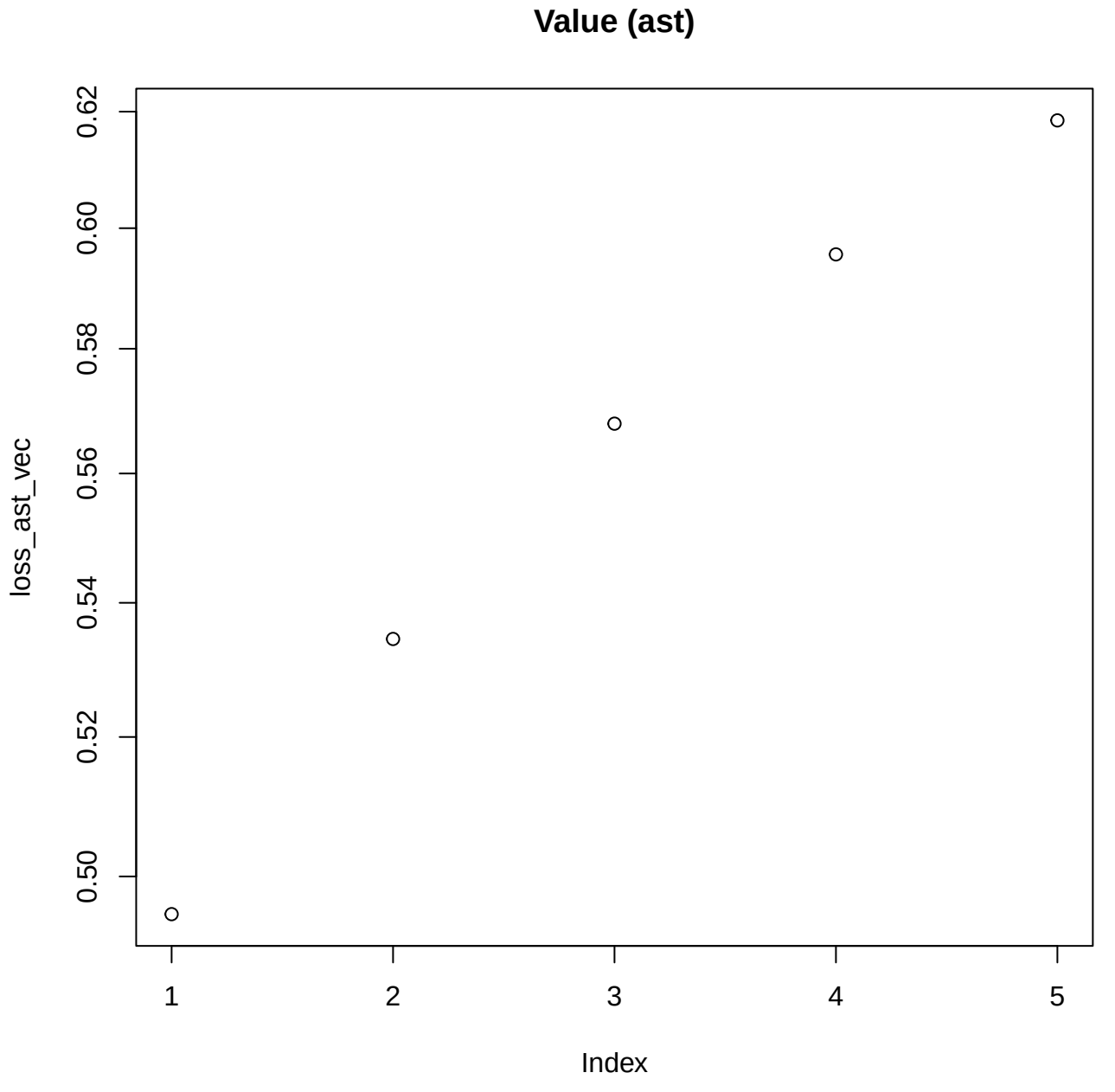
4
5
Index


Value (ast)
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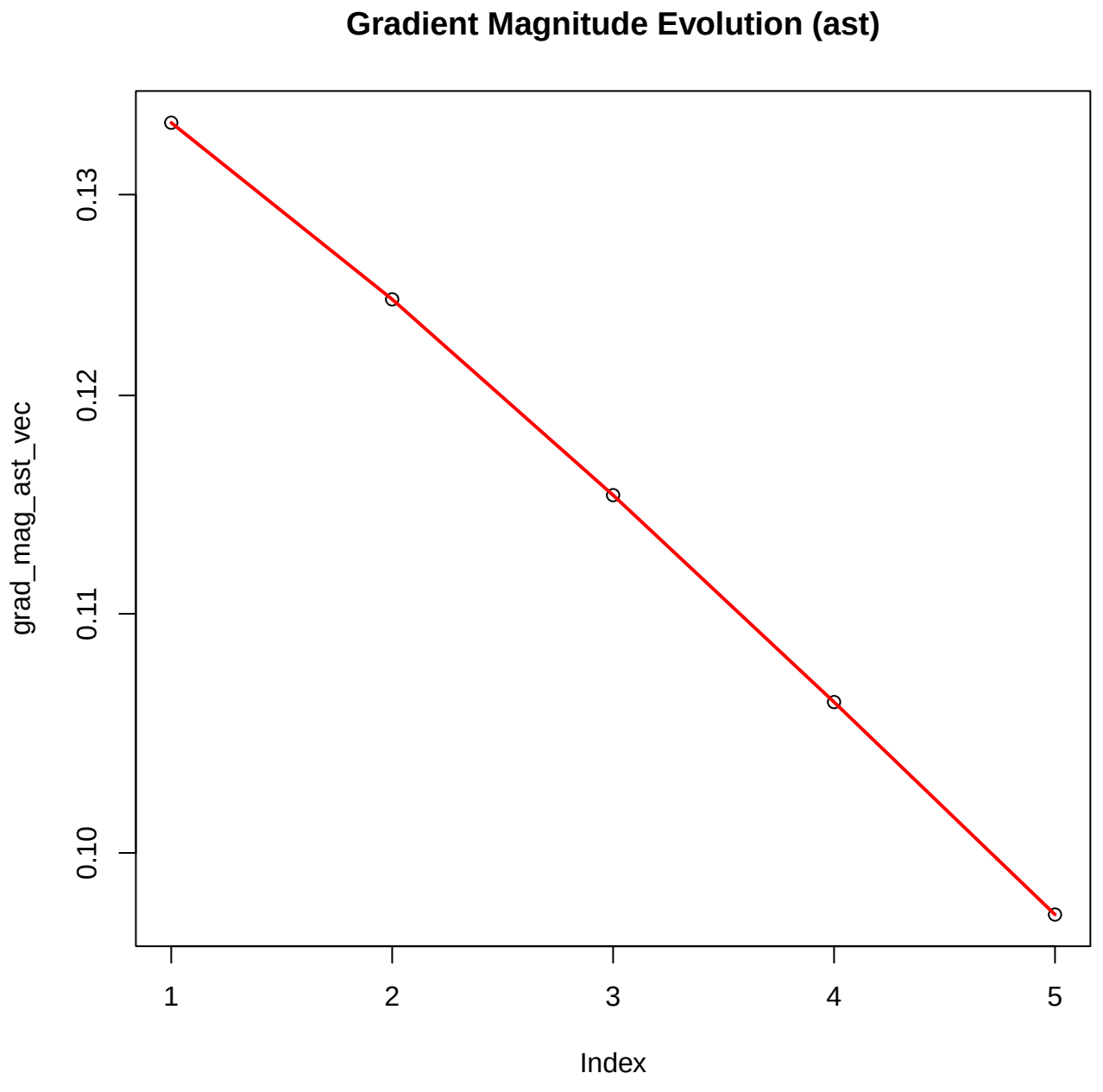
3
4
5
Index


Gradient Magnitude Evolution (ast)
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0.13
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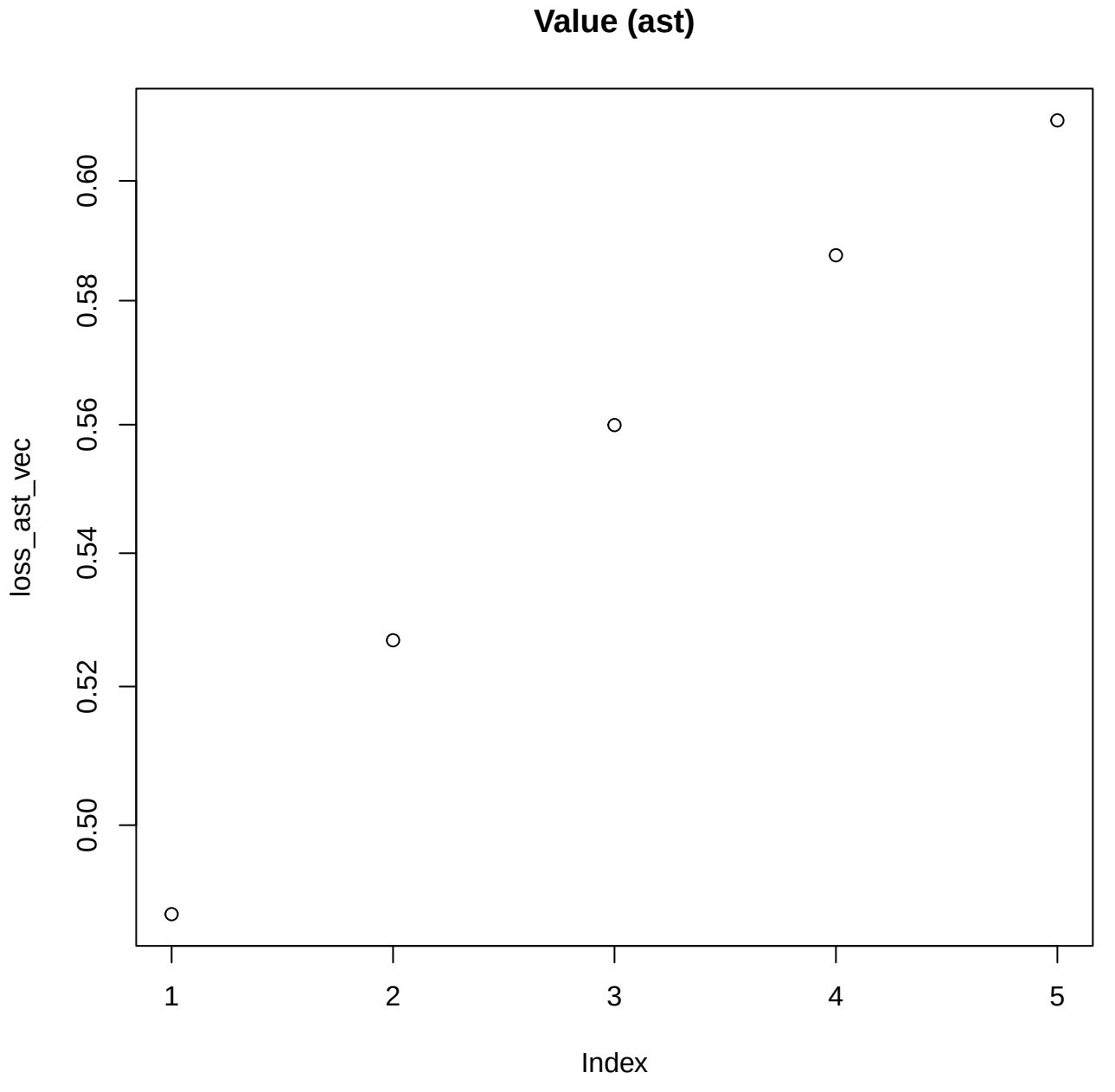
3
4
5
Index


Value (ast)
loss_ast_vec
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0.54
0.56
0.58
0.60
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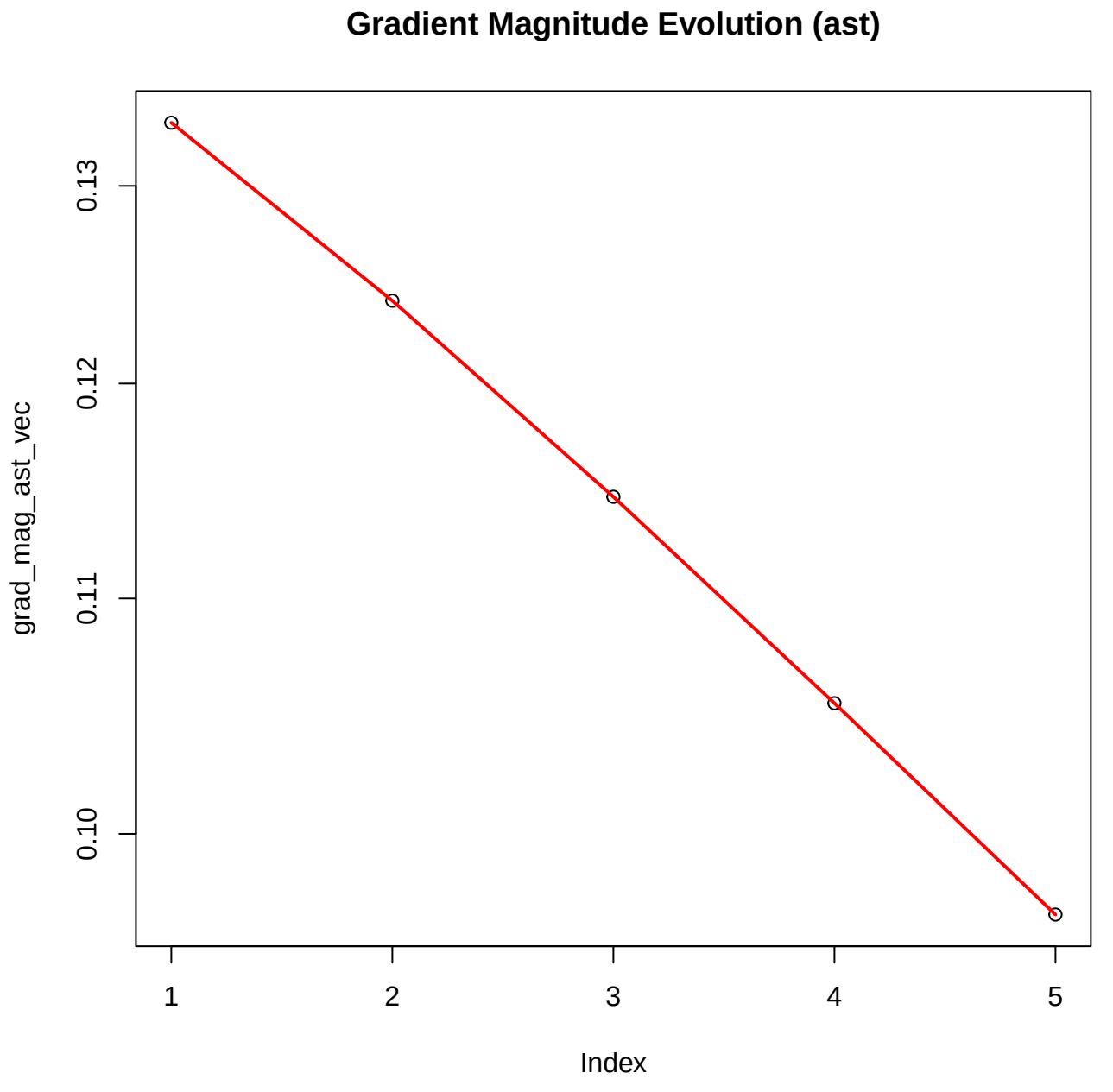
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.12
0.13
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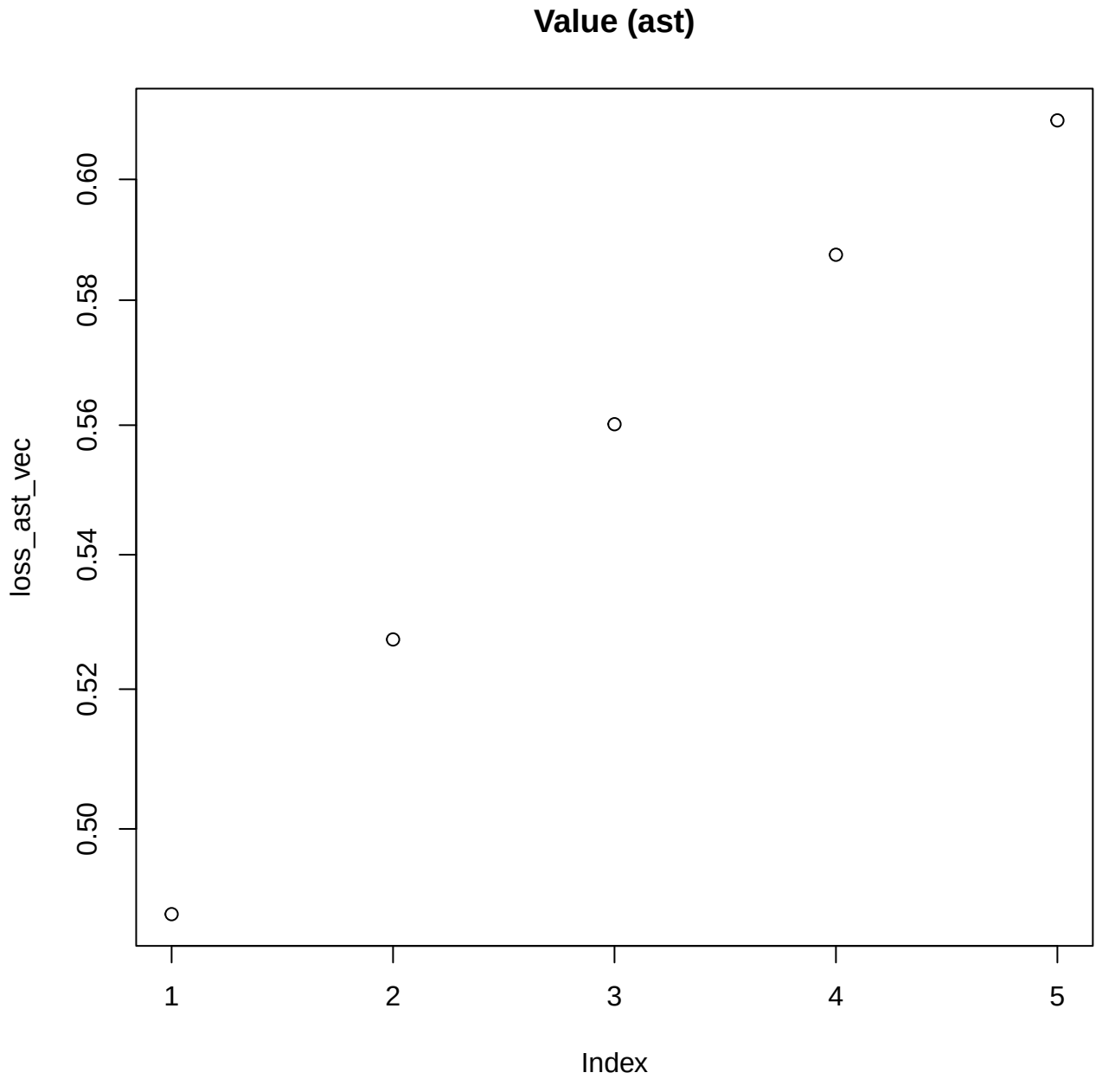
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.54
0.56
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0.60
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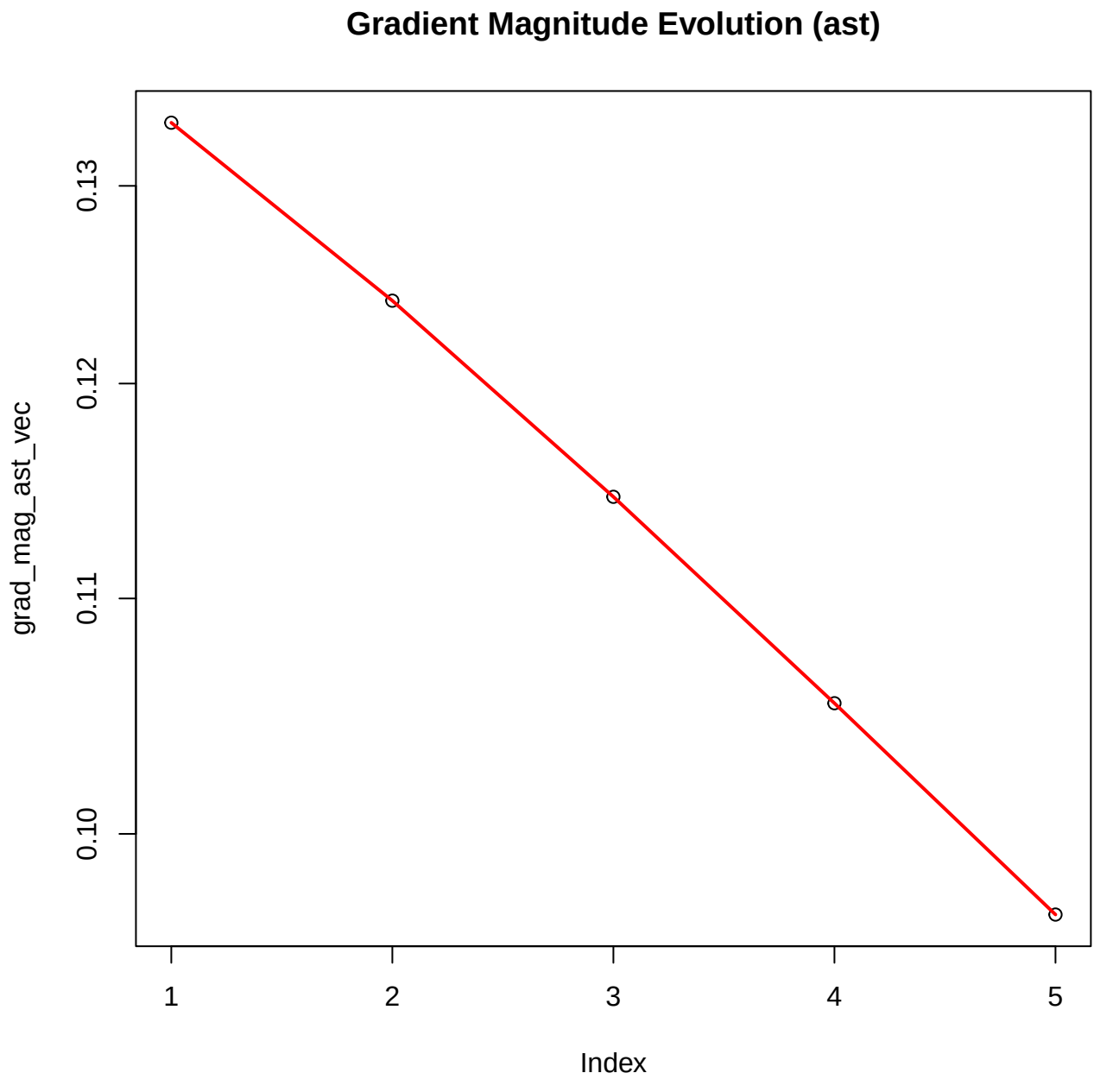
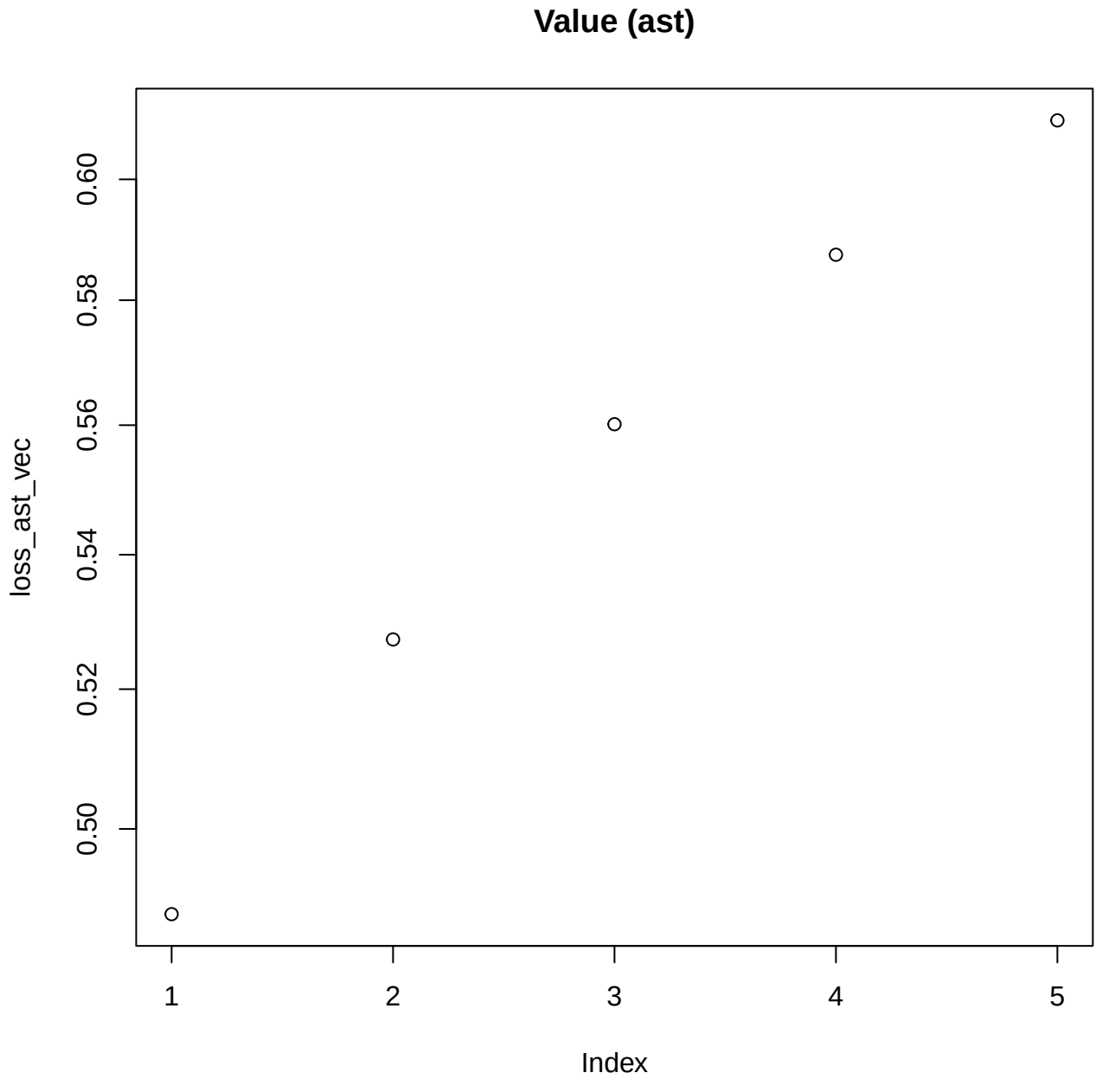
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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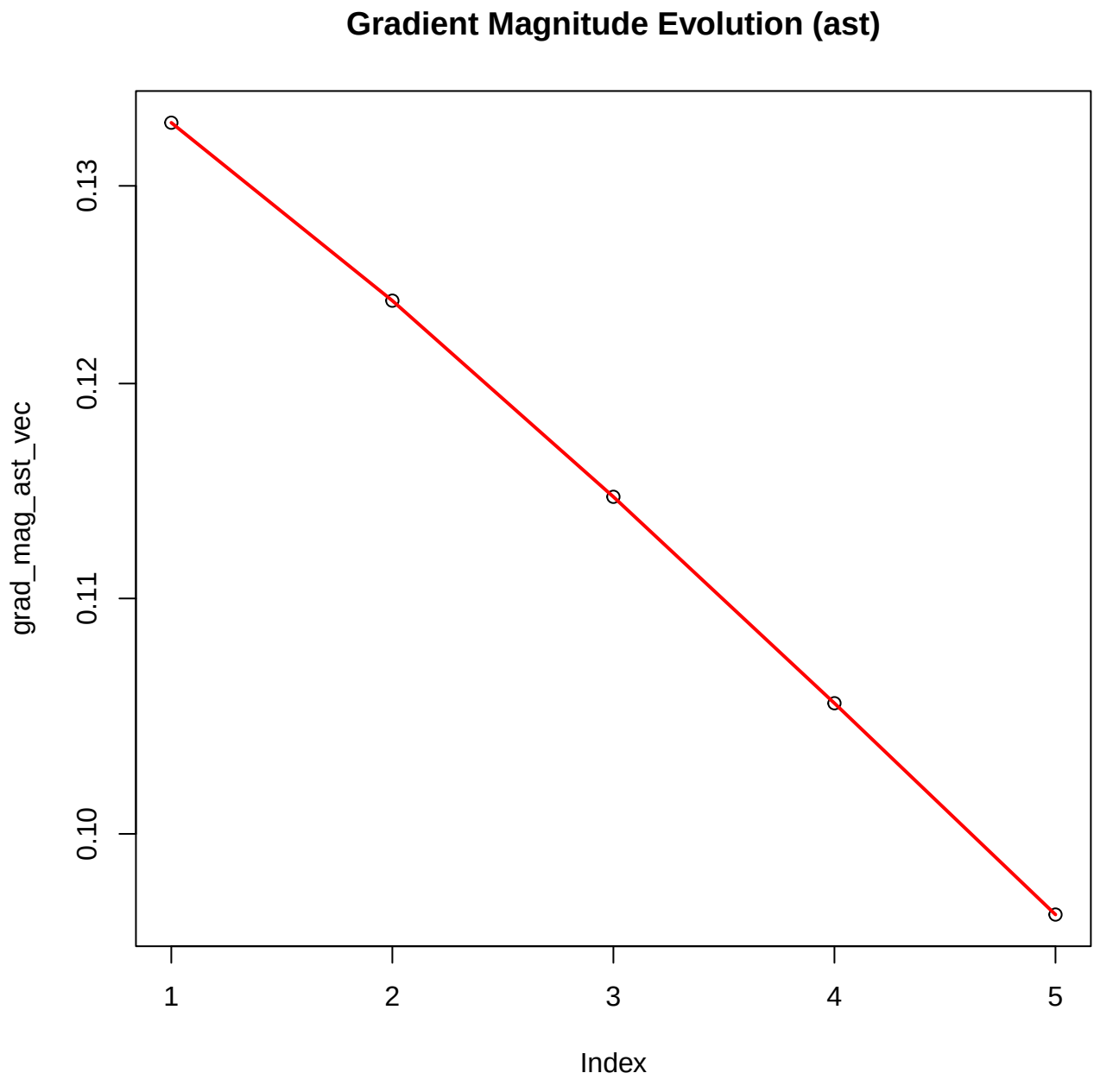
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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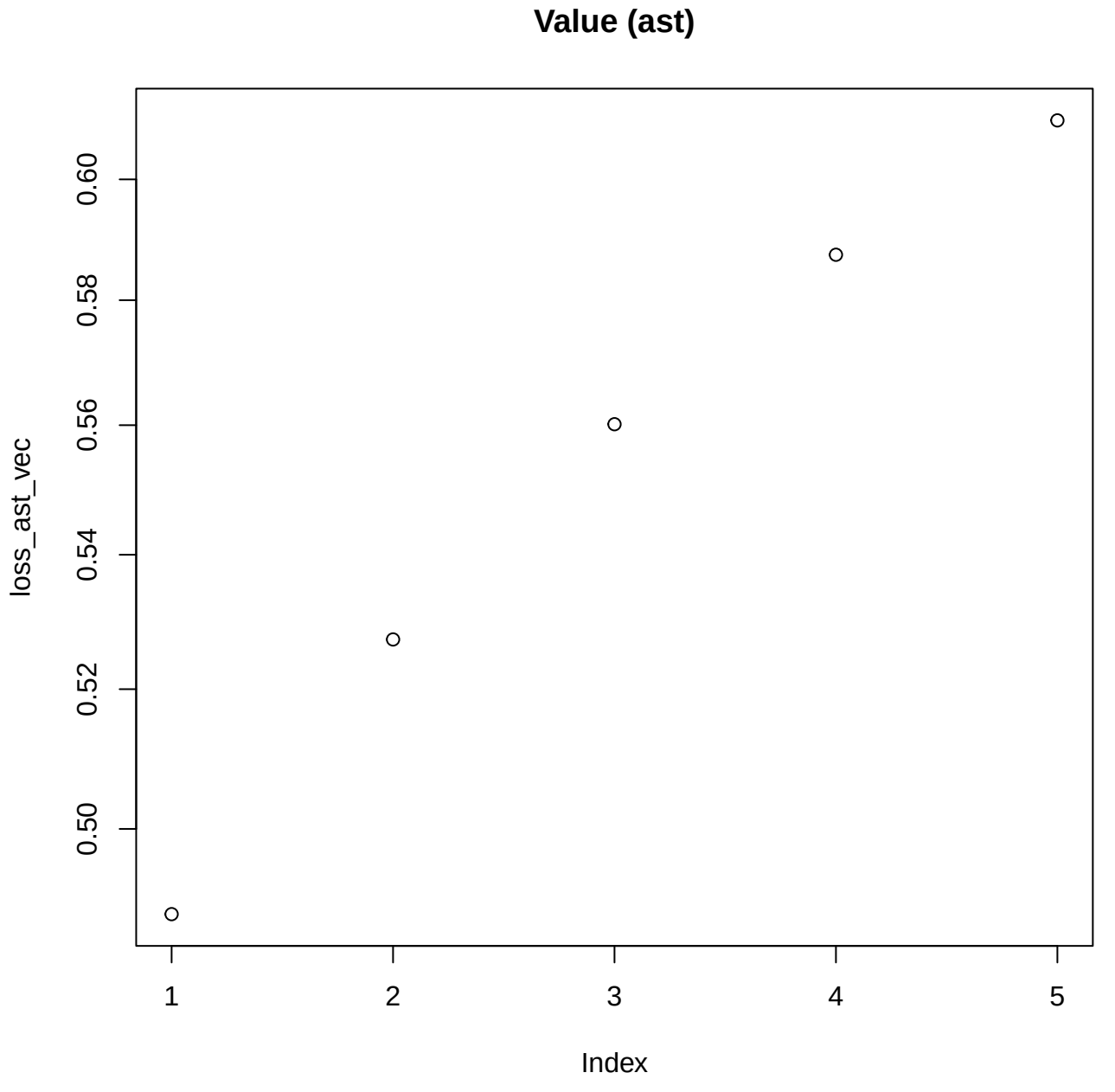
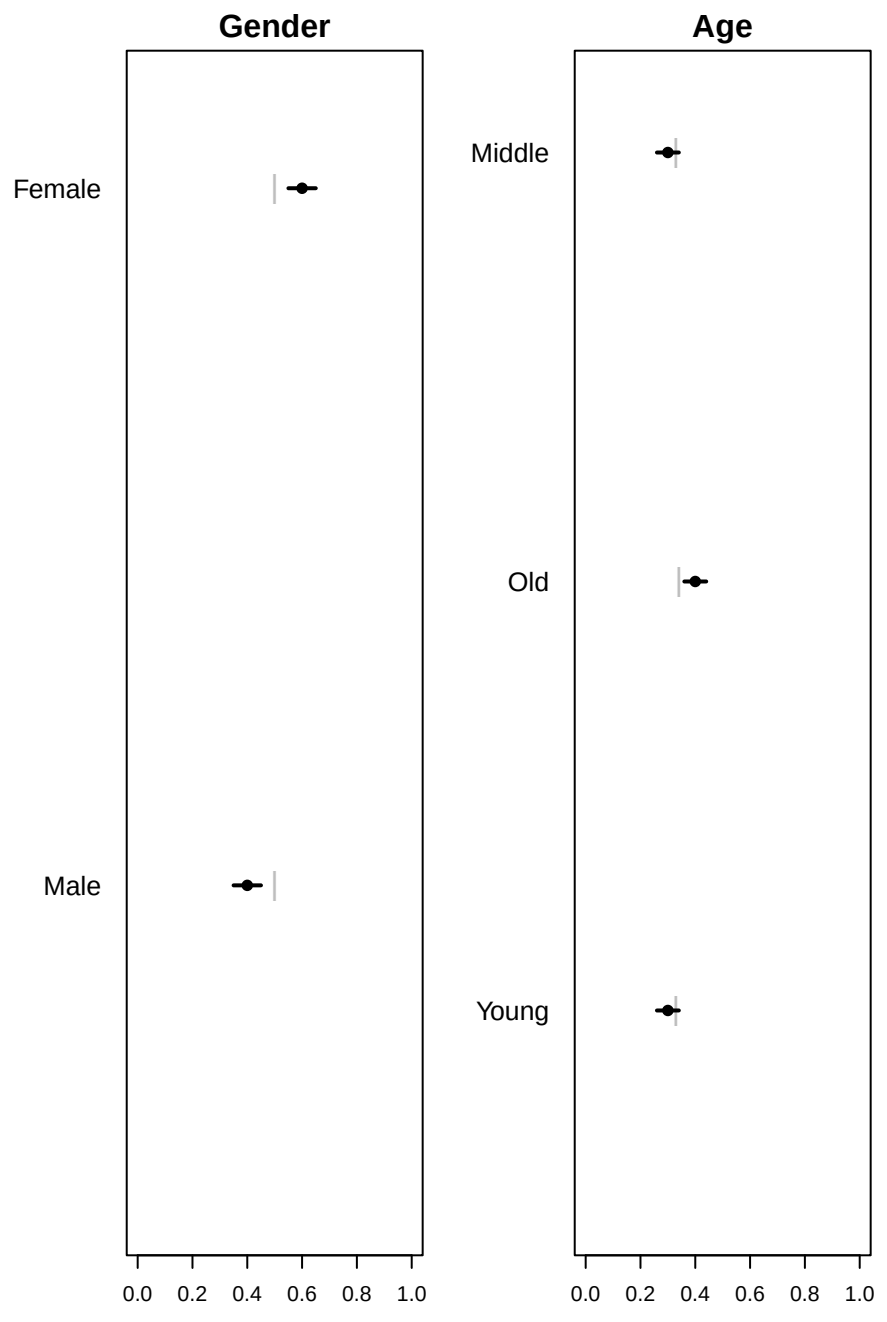
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index


Gender
Female
Male
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0.2
0.4
0.6
0.8
1.0
Age
Middle
Old
Young
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0.2
0.4
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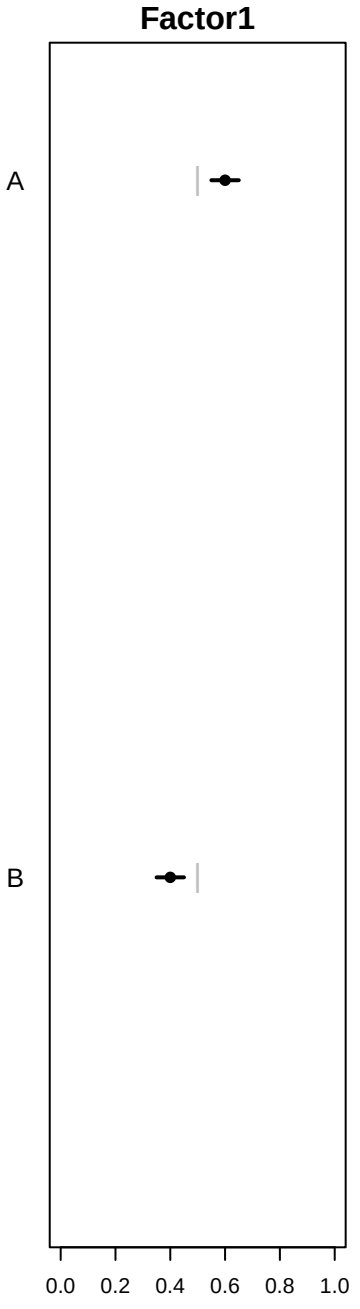
0.8
1.0


Factor1
A
B
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0.4
0.6
0.8
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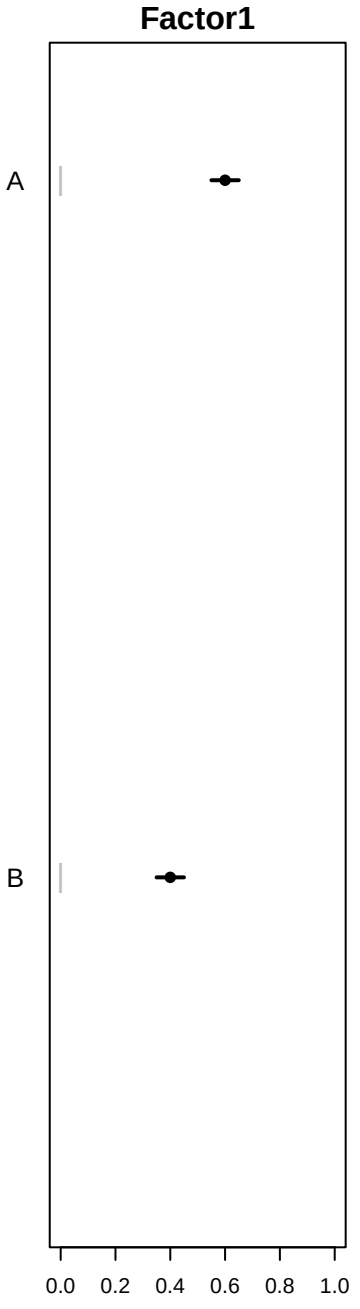
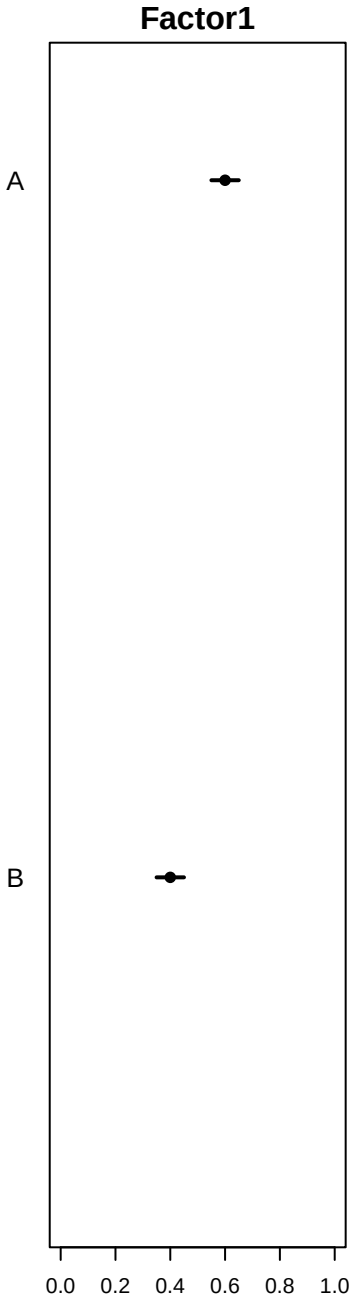

Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
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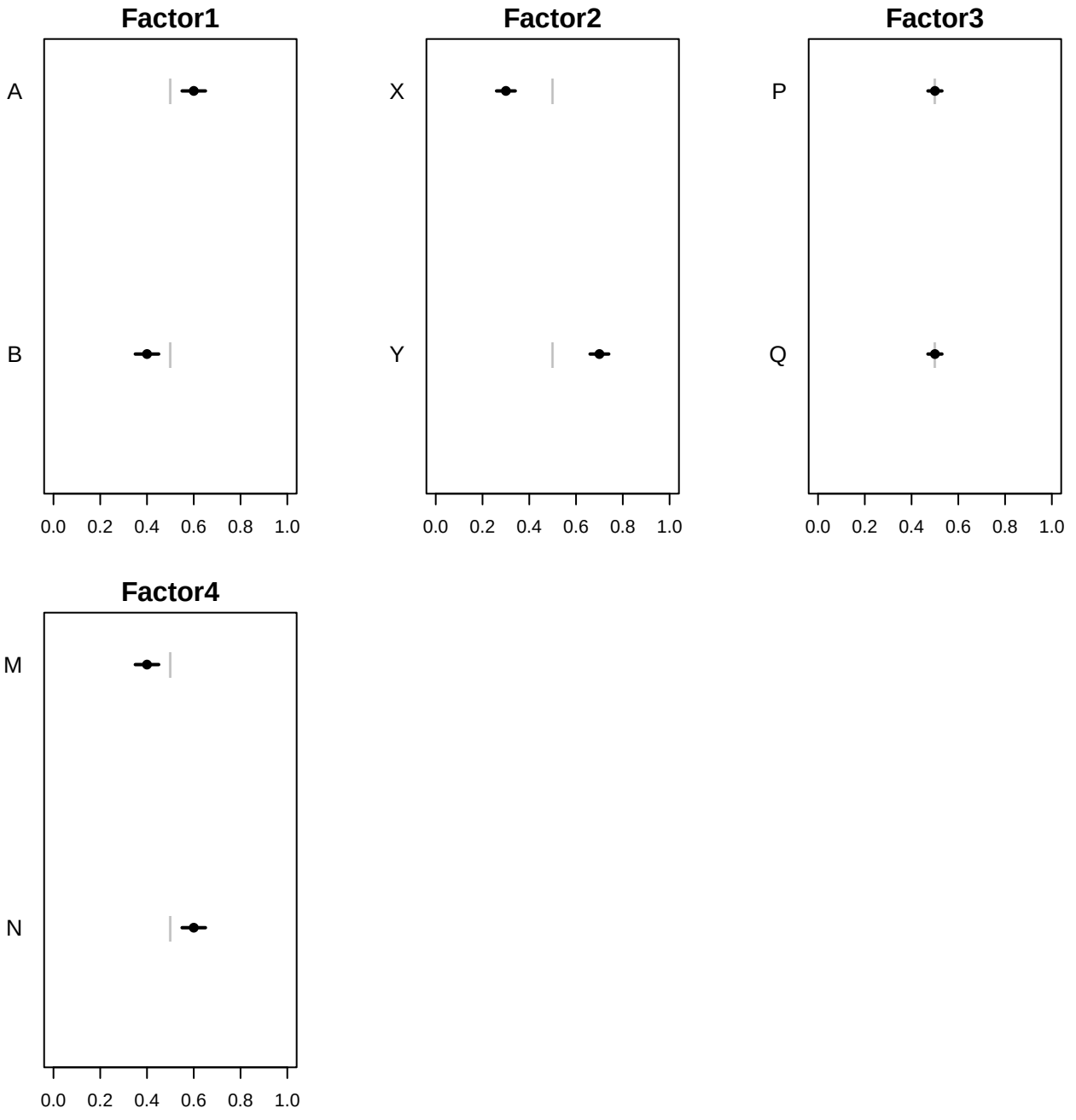
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
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0.2
0.4
0.6
0.8
1.0
Factor2
X
Y
0.0
0.2
0.4
0.6
0.8
1.0
Factor3
P
Q
0.0
0.2
0.4
0.6
0.8
1.0
Factor4
M
N
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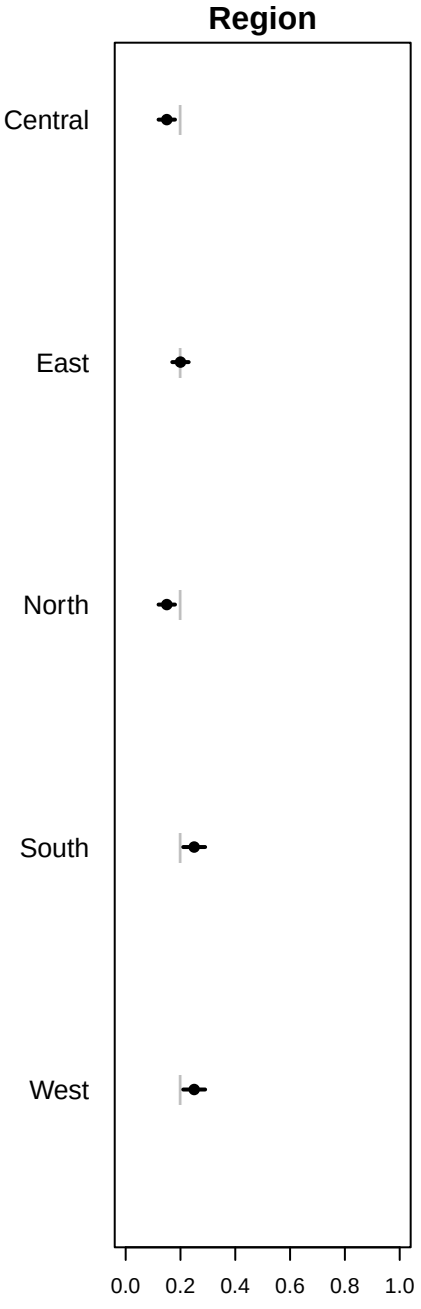
0.4
0.6
0.8
1.0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East
North
South
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0.8
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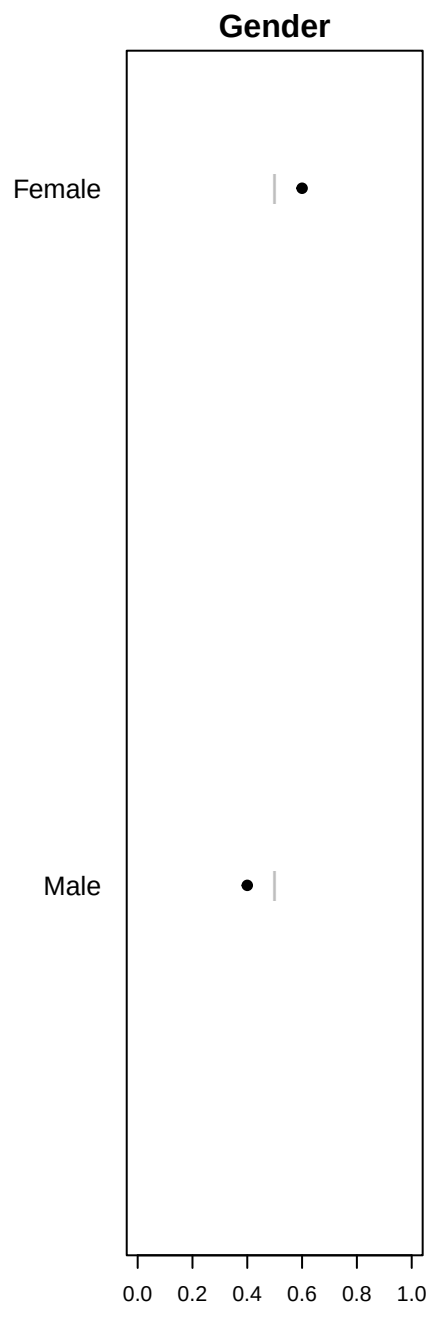
Gender
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Male
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0.4
0.6
0.8
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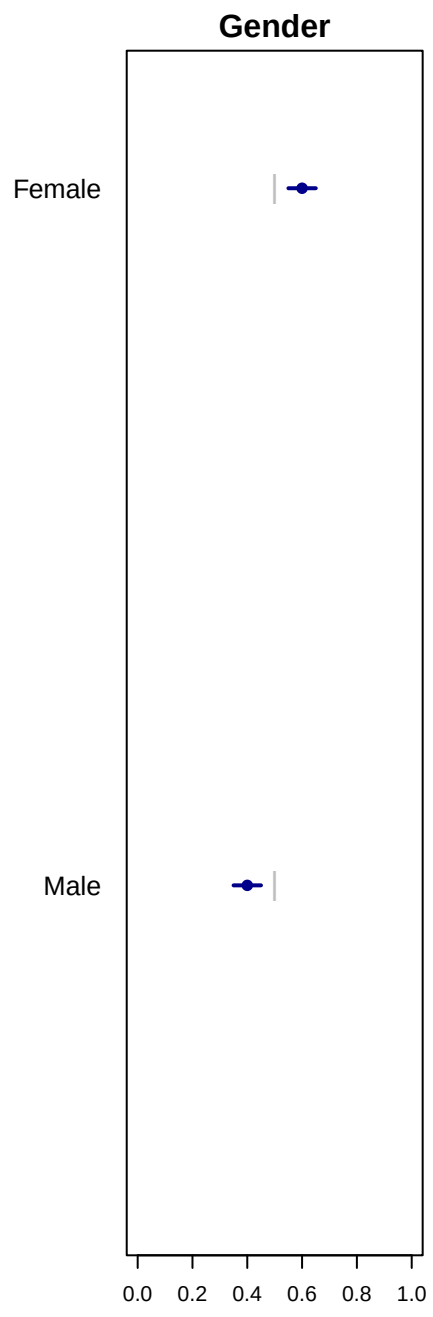
Gender
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Male
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0.4
0.6
0.8
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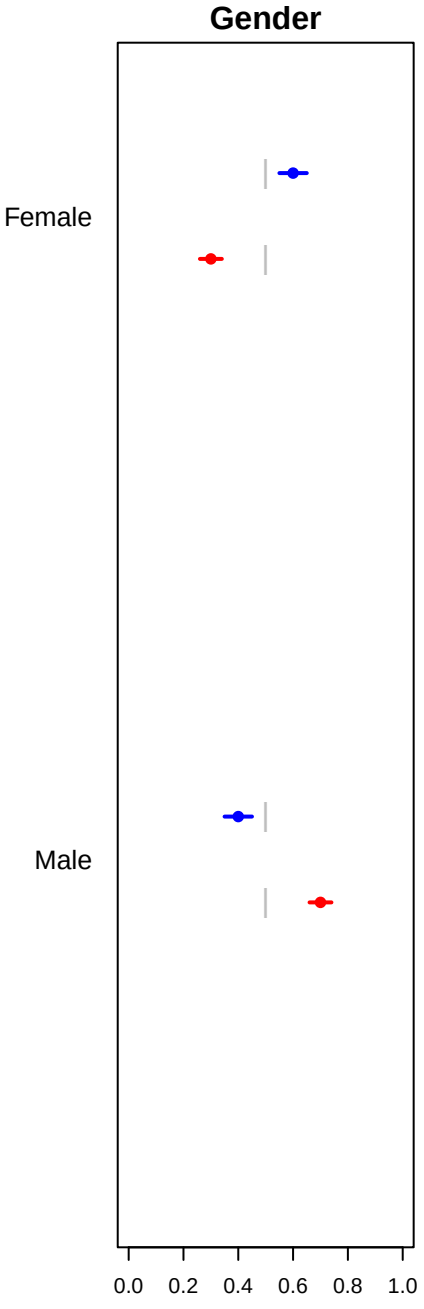
Gender
Female
Male
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0.4
0.6
0.8
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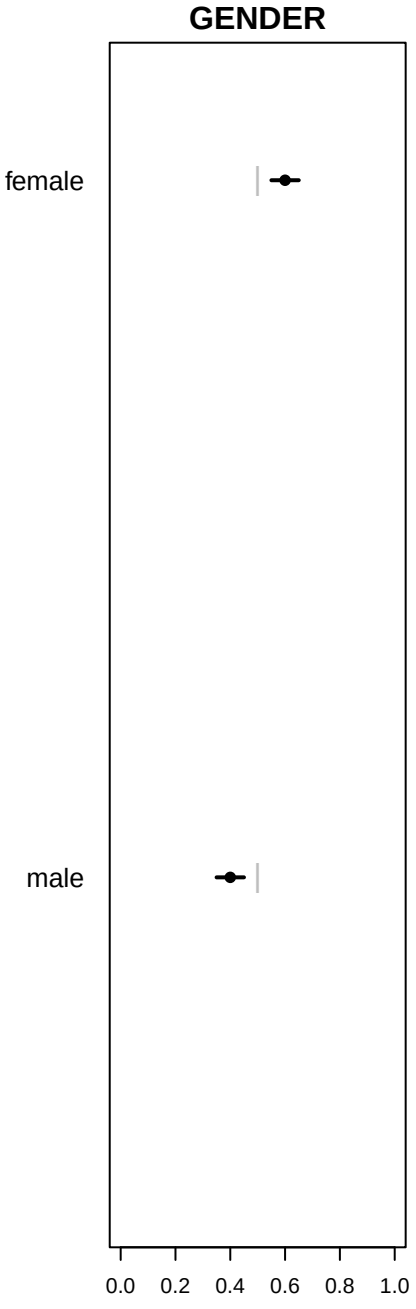
GENDER
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male
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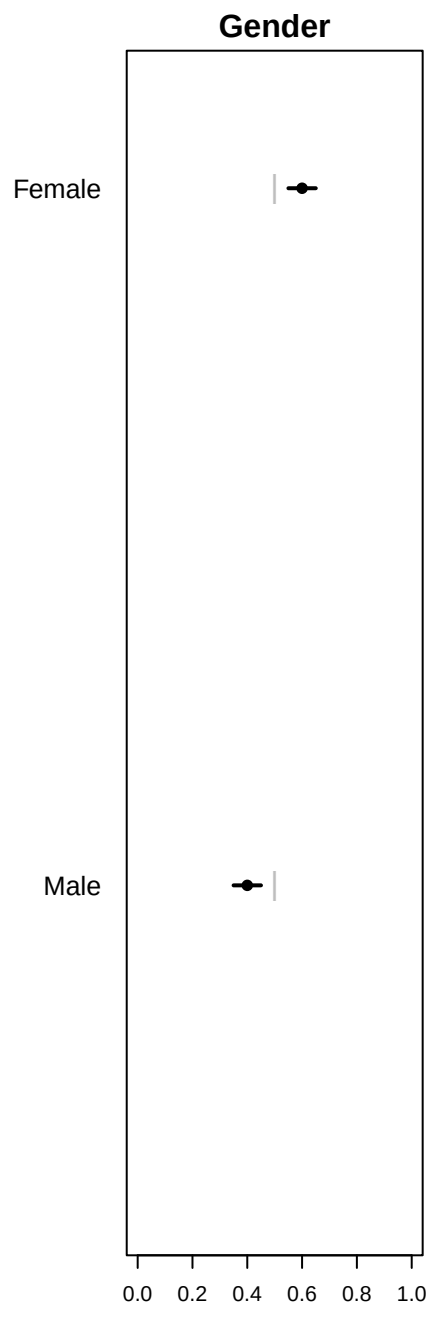
Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0